

Second Edition

**Eastern
Economy
Edition**

Textbook of Functional Analysis

A Problem-Oriented Approach



V.K. Krishnan

TEXTBOOK OF FUNCTIONAL ANALYSIS

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FUNCTIONAL ANALYSIS
A Problem-Oriented Approach

SECOND EDITION

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TEXTBOOK OF FUNCTIONAL ANALYSIS: A Problem-Oriented Approach, Second Edition
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Preface

Functional analysis is being increasingly recognised as a basic area of mathematics. This area is the natural platform for generalising many of the concepts of classical mathematics. The concepts and tools developed in this area are applied to various other branches of mathematics, especially to the areas of applied mathematics. Consequently, functional analysis is included as a core subject in the curriculum of the graduate courses in mathematics offered in all universities in India and abroad.

Many excellent textbooks on functional analysis are now available. These books are concerned with expounding the theoretical framework of the subject in detail. They provide statements and rigorous proofs of theorems. Even though most of these books provide problems of various levels of difficulty, they do not generally give the so-called solved problems. From the student's point of view, the study of proof of a theorem has a two-fold purpose. The first is, of course, to be convinced of the result. The second and perhaps more important, is to understand the techniques adopted in the proof. The second purpose is also served by the study of solutions of problems. Although many students are capable of reproducing the proofs of theorems, they find themselves incapable of solving a problem of even moderate difficulty level. Even some of the teachers find it difficult to solve such problems. A theory in any branch of mathematics is grasped properly only with a certain amount of drill involving exercises related to it. Many people ignore this aspect of learning and teaching.

This book does not use the lemma-theorem-proof-corollary style. Each section begins with an introduction in which the basic concepts are given briefly. It is followed by some exercises involving simple applications of these concepts. The exercises are intended mainly to motivate the concepts. The problems and solutions are given after this. The solutions are given in detail so that an average student can follow them. To make reading effortless, as far as possible, the solutions have been made self-contained.

As a consequence of this, ideas and techniques in one solution may be repeated in other solutions. Almost all the standard results are given as problems. This book, therefore, will serve the purpose of a textbook, rather than as a mere problem book. As the title suggests, it helps the students in learning the basics of functional analysis by way of problems and solutions.

The topics are chosen from the core of functional analysis which is commonly included in the M.Sc. syllabus of Indian universities. The first four chapters deal with the fundamentals of normed linear spaces, Banach spaces, duals and spectral notions. The remaining three chapters cover inner product spaces, Hilbert spaces, bounded operators and spectral theory in Hilbert spaces. Almost all the problems included in this book are given as exercises in the standard textbooks which are listed in the References at the end of the book. Some of the problems may be new.

In this new edition, several problems have been incorporated discussing the spectral properties of compact operators on Banach spaces as well as on Hilbert spaces. Besides, convergence of some additional problems based on the square root of a positive operator makes this book highly beneficial for the readers.

Just as in the case of any course in functional analysis, the reader is expected to possess a knowledge of the basics of linear algebra and real analysis, including metric spaces. A knowledge of Lebesgue integral and complex function theory is assumed occasionally.

A sincere attempt has been made to make the book free from errors. However, I do not claim that the goal has been achieved. Some errors may have still crept into it.

I am grateful to PHI Learning for persuading me to bring out the second edition. I am indebted to the referee for closely examining the manuscript and giving valuable suggestions for improvement. While preparing the final version, I have made use of a number of suggestions by the editor; I am thankful to him for the same. The typesetting has been done using the computer program LaTeX. All the technical help needed for this was rendered by V.N. Krishnachandran. My daughter V.K. Jayanthi has also assisted me in preparing the typescript. I sincerely express my gratitude to all those who have helped me in this endeavour.

Any suggestions for improving the content will be warmly appreciated.

V.K. Krishnan

Notation

For each symbol, the number of the page is mentioned on which the symbol is explained.

$A \geq 0$	A is positive, 302
$A \geq B$	$A - B$ is positive, 302
$A^2(D)$	A space of analytic functions on D , 223
$BL(X)$	Space of bounded linear maps from X to itself, 17
$BL(X, Y)$	Space of bounded linear maps from X to Y , 17
$BV[a, b]$	Space of scalar-valued functions of bounded variation on $[a, b]$, 10
$\mathbb{C}$	Complex field, 1
$C(T)$	Space of bounded continuous scalar-valued functions on T , 3
$C_0(T)$	Subspace of $C(T)$ consisting of functions vanishing at infinity, 3
$C_c(T)$	Subspace of $C(T)$ consisting of functions with compact support, 3
$C^n[a, b]$	Space of scalar-valued functions having continuous n -th derivatives on $[a, b]$, 4
$C^\infty[a, b]$	Space of infinitely differentiable scalar-valued functions on $[a, b]$, 4
c	Space of convergent sequences of scalars, 3
c_0	Subspace of c consisting of null sequences, 3
c_{00}	Subspace of c consisting of sequences having at most a finite number of non-zero terms, 3
$d(x, E)$	Distance between x and E , 8
$d(x, y)$	Distance between x and y , 3
E^a	Annihilator of a set E of vectors, 136
E^0	Interior of a set E , 46
$E^\perp$	Orthogonal complement of E , 212
$E \perp F$	E orthogonal to F , 212
$E_1 + E_2$	Sum of sets E_1 and E_2 , 7
e_i	The i -th unit vector, 2
$\ F\ $	Norm of linear map F , 17
$f _Z$	Restriction of f to Z , 55

H^p	A space of analytic functions, 82
I	Identity operator, 186
Im	Imaginary part, 53
J	Canonical mapping, 124
$\mathbb{K}$	Scalar field which may be the real field or complex field, 1
$\mathbb{K}^n$	Space of n -tuples of elements in $\mathbb{K}$, 2
$\bar{k}$	Complex conjugate of k , 210
(k_{ij})	Matrix with k_{ij} as the i -th row j -th column entry, 17
$L^p(T)$	Lebesgue space of p -integrable scalar-valued functions on T , 3
$L^\infty(T)$	Lebesgue space of essentially bounded measurable functions on T , 4
l^p	Space of p -summable scalar sequences, 3
l^∞	Space of bounded scalar sequences, 3
$l^2(S)$	Space of square-summable functions on S , 227
aM	Annihilator of a set M of functionals, 136
$(\overline{M})'$	Transpose of the conjugate of M , 303
M_A	Supremum of the numerical range of A , 328
m_A	Infimum of the numerical range of A , 328
$\mathbb{N}$	Set of natural numbers, 2
$\mathbb{R}$	Real field, 1
Re	Real part, 53
$R(F)$	Range of F , 17
$r_\sigma(T)$	Spectral radius of T , 197
$\text{span } S$	Span of the set S , 2
$\overline{\text{span } S}$	Closure of span of S , 245
$\sigma(A)$	Spectrum of A , 186
$\sigma_a(A)$	Set of approximate eigenvalues of A , 186
$\sigma_e(A)$	Eigenspectrum of A , 186
$\sigma_r(A)$	Residual spectrum of A , 198
T^*	Adjoint of T , 279
U	Open unit ball, 165
$U(a, r)$	Open ball with centre a and radius r , 6
u_x, u_y	Partial derivatives of u , 222
$V(x)$	Total variation of x , 10
$W(A)$	Numerical range of A , 327
$\xrightarrow{w}$	Weak convergence, 149
$\xrightarrow{w^*}$	Weak* convergence, 149
$(X, \mathbb{R})$	Associated real linear space, 1
X'	Dual space of X , 17
X''	Dual of X' , 124
$\tilde{X}$	Completion of X , 125
$\ x\ _p$	p -norm of x , 3
$\ x\ _\infty$	Sup norm of x , 3

$\hat{x}(n)$	Fourier coefficient of $x \in L^1[-\pi, \pi]$, 119
$\hat{x}(t)$	Fourier transform of $x \in L^1(\mathbb{R})$, 116
$X \times Y$	Cartesian product of X and Y , 4
X/Y	Quotient space, 4
$x \perp E, x \perp y$	x orthogonal to E, y , 212
$\ x + Y\ $	Quotient norm of $x + Y$, 5
$\langle x, y \rangle$	Inner product of x and y , 210
$\overline{Y}$	Closure of Y , 3
$\mathbb{Z}$	Set of all integers, 119
$Z(F)$	Zero space of F , 17
$\{0\}$	Zero-dimensional space consisting of 0 only, 2
$\emptyset$	Empty set, 8

1

Normed Linear Spaces

This chapter deals with the basic properties of normed linear spaces. In Section 1.1 we introduce linear spaces and norm. Linear maps are homomorphisms of linear spaces. In Section 1.2 we consider the continuity of linear maps. Finite dimensional normed linear spaces have many nice properties in common with Euclidean spaces. These are studied in Section 1.3. Hahn-Banach extension theorem is one of the fundamental theorems in functional analysis. Problems related to this are considered in Section 1.4.

1.1 Linear Spaces and Norm

A *linear space* consists of an additive abelian group X of *vectors*, a field K of *scalars* and a multiplication of vectors by scalars. The multiplication assigns a vector kx for each $k \in K$ and $x \in X$ in such a way that

$$\begin{aligned}k(x + y) &= kx + ky, \\(k + k')x &= kx + k'x, \\k(k'x) &= (kk')x, \\1x &= x,\end{aligned}$$

for all $k, k' \in K$ and $x, y \in X$. A linear space is also called a vector space.

In this book, *the scalar field* K (hereafter denoted by $\mathbb{K}$) is supposed to be either the field $\mathbb{R}$ of real numbers *or* the field $\mathbb{C}$ of complex numbers. When we consider two or more linear spaces in a context, it will be tacitly assumed that they have the same scalar field which may be either $\mathbb{R}$ or $\mathbb{C}$.

Let X be a linear space over the scalar field $\mathbb{C}$. The scalar multiplication, which maps (k, x) to kx , is defined for all $k \in \mathbb{C}, x \in X$. So it is defined for all $k \in \mathbb{R}$. Hence X is a linear space over $\mathbb{R}$ also. This linear space X over $\mathbb{R}$ is called the *associated real linear space*. The underlying abelian group X for the associated real linear space $(X, \mathbb{R})$ is the same as that for the original complex linear space $(X, \mathbb{C})$.

A subset Y of a linear space X is said to be a *subspace* if

$$ky_1 + y_2 \in Y \quad \text{for } k \in \mathbb{K}, \quad y_1, y_2 \in Y.$$

If S is a subset of X then the smallest subspace of X containing S is called the *span* of S .

We say that the vectors $x_1, \dots, x_n$ are *linearly independent* if the relation

$$k_1x_1 + \dots + k_nx_n = 0, \quad k_i \in \mathbb{K}$$

holds only if each k_i is 0. A set $S \subset X$ is said to be linearly independent if every finite subset of S is linearly independent. A linearly independent set S , whose span is equal to X , is called a *basis*. Every linear space has a basis. Any two bases have the same number of elements. The number of elements in a basis is called the *dimension* of the space.

A subset E of a linear space X is said to be *convex* if, whenever $x, y \in E$, we have

$$tx + (1 - t)y \in E \quad \text{for } 0 \leq t \leq 1.$$

The zero-dimensional space $X = \{0\}$ and the one-dimensional space $X = \mathbb{K}$ are two examples of linear spaces. Let T be a nonempty set. The set of all functions $x : T \rightarrow \mathbb{K}$ is a linear space if addition and scalar multiplication are defined pointwise as follows:

$$\begin{aligned} (x + y)(t) &= x(t) + y(t), \\ (kx)(t) &= kx(t) \end{aligned}$$

for $t \in T, k \in \mathbb{K}$.

A sequence is a function defined on the set $\mathbb{N}$ of natural numbers, and an n -tuple is a function defined on the set $\{1, \dots, n\}$. So *the space of sequences in $\mathbb{K}$ and the space $\mathbb{K}^n$ are considered as spaces of $\mathbb{K}$ -valued functions defined on $T = \mathbb{N}$ and $T = \{1, 2, \dots, n\}$, respectively*. These are, therefore, linear spaces. The i -th *unit vector* e_i in these spaces is given by

$$e_i(j) = \begin{cases} 0 & \text{if } j \neq i, \\ 1 & \text{if } j = i, \end{cases}$$

that is, $e_i = (0, \dots, 0, 1, 0, \dots)$, where 1 occurs at the i -th place and 0 elsewhere.

A *norm* on a linear space is a function which assigns a non-negative real number $\|x\|$ to each x in X having the following properties:

$$\begin{aligned} \|x\| &= 0 \quad \text{iff } x = 0, \\ \|kx\| &= |k| \|x\| \quad \text{for } k \in \mathbb{K}, x \in X, \\ \|x + y\| &\leq \|x\| + \|y\| \quad \text{for } x, y \in X. \end{aligned}$$

The inequality in the third line is called the *triangle inequality*. A linear space equipped with a norm is called a *normed linear space* or *normed space*.

A normed space becomes a metric space with the metric given by

$$d(x, y) = \|x - y\|, \quad x, y \in X.$$

Thus, a sequence $\{x_n\}$ converges to x if $\|x_n - x\| \rightarrow 0$ as $n \rightarrow \infty$.

While dealing with more than one normed space we may use, for convenience, the same symbol $\|\cdot\|$ to denote the norm in these spaces. If X is a complex normed space, then the same norm is given to the associated real linear space $(X, \mathbb{R})$.

If Y is a subspace of a normed linear space, then its closure $\bar{Y}$ is also a subspace.

The simplest example of a normed space is given by the one-dimensional space $X = \mathbb{K}$ with $\|x\| = |x|$.

The spaces $C(T), C_0(T), C_c(T)$. Let T be a metric space. The set of all bounded continuous $\mathbb{K}$ -valued functions on T is denoted by $C(T)$. This is a linear space w.r.t. pointwise addition and scalar multiplication of functions, and

$$\|x\|_\infty = \sup \{|x(t)| : t \in T\}, \quad x \in C(T),$$

defines a norm called *supnorm* on $C(T)$. By $C_0(T)$ we mean the set of all $x \in C(T)$ having the following property: For every $\epsilon > 0$ there is a compact subset S of T , depending on x and ϵ , such that

$$|x(t)| < \epsilon \quad \text{for all } t \notin S.$$

The set of all $x \in C(T)$ with the following property is denoted by $C_c(T)$: There is a compact subset S of T , depending on x , such that $x(t) = 0$ for all $t \notin S$.

The spaces $C_0(T)$ and $C_c(T)$ are subspaces of $C(T)$.

Let T be the set of natural numbers with the discrete metric. Then $C(T)$ is the set of all bounded sequences $\{x(n)\}$ in $\mathbb{K}$. This space is denoted by l^∞ , and the norm is given by

$$\|x\|_\infty = \sup_{n \geq 1} |x(n)|, \quad x \in l^\infty.$$

The subspace $C_0(T)$ becomes

$$c_0 = \{x \in l^\infty : \lim_{n \rightarrow \infty} x(n) = 0\},$$

and $C_c(T)$ becomes

$$c_{00} = \{x \in l^\infty : x(n) = 0 \quad \text{for all but a finite number of values of } n\}.$$

The set of all convergent scalar sequences is denoted by c .

The spaces $L^p(T), l^p$ with $1 \leq p \leq \infty$. Let T be a measure space with measure μ . If $1 \leq p < \infty$, then the set of all measurable functions x with $\int |x|^p d\mu < \infty$ is denoted by $L^p(T)$ or $L^p(\mu)$. This is a linear space and

$$\|x\|_p = \left(\int |x|^p d\mu \right)^{1/p}, \quad x \in L^p(T),$$

defines a norm on $L^p(T)$. For $1 < p < \infty$, the triangle inequality is proved using *Holder's inequality*, namely,

$$\int |xy| d\mu \leq \left(\int |x|^p d\mu \right)^{1/p} \left(\int |y|^q d\mu \right)^{1/q},$$

where $1/p + 1/q = 1$. The case $p = q = 2$ of Holder's inequality is called *Schwarz inequality*.

The set of all essentially bounded measurable functions is denoted by $L^\infty(T)$ or $L^\infty(\mu)$. This also is a linear space, and

$$\|x\|_\infty = \text{ess. sup } \{|x(t)| : t \in T\}, \quad x \in L^\infty(T),$$

defines a norm on $L^\infty(T)$. The spaces $L^p(T)$, with $1 \leq p \leq \infty$, are called *Lebesgue spaces*.

Let T be the set of all natural numbers with counting measure. If $1 \leq p < \infty$, then $L^p(T)$ becomes the set l^p of all scalar sequences $\{x(n)\}$ with

$$\|x\|_p = \left(\sum_{n=1}^{\infty} |x(n)|^p \right)^{1/p} < \infty.$$

The space $L^\infty(T)$ becomes the sequence space l^∞ given earlier.

If $T = \{1, \dots, n\}$ with counting measure, then $L^p(T)$ becomes $\mathbb{K}^n$ with norm $\|\cdot\|_p$ given by

$$\begin{aligned} \|x\|_p &= \left(\sum_{j=1}^n |x(j)|^p \right)^{1/p} & \text{if } 1 \leq p < \infty, \\ \|x\|_\infty &= \max_{1 \leq j \leq n} |x(j)| & \text{if } p = \infty. \end{aligned}$$

The spaces $C^n[a, b]$, $C^\infty[a, b]$. Let $[a, b]$ be a subinterval of $\mathbb{R}$. By $C^n[a, b]$ we mean the set of all scalar valued functions x on $[a, b]$ whose n -th derivative $x^{(n)}$ exists and belongs to $C[a, b]$. The set of all functions x such that $x \in C^n[a, b]$ for all n is denoted by $C^\infty[a, b]$.

Product of normed spaces. If X and Y are normed spaces, then $X \times Y$ is a linear space with addition and scalar multiplication defined coordinate-wise; and

$$\|(x, y)\| = \|x\| + \|y\|, \quad x \in X, y \in Y,$$

defines a norm on it.

Quotient space. Let Y be a closed subspace of a normed space X . The quotient space X/Y consists of the cosets

$$x + Y = \{x + y : y \in Y\}, \quad x \in X.$$

It is a linear space with addition and scalar multiplication defined as follows:

$$\begin{aligned}(x_1 + Y) + (x_2 + Y) &= (x_1 + x_2) + Y, \quad x_1, x_2 \in X, \\ k(x + Y) &= kx + Y, \quad k \in \mathbb{K}, x \in X.\end{aligned}$$

The norm on this space, called *quotient norm*, is defined by

$$\|x + Y\| = \inf \{\|x + y\| : y \in Y\}.$$

The norm $\|x + Y\|$ is the distance of x from the subspace Y .

EXERCISES

1. Let $x = (1, 2) \in \mathbb{R}^2$. Find $\|x\|_p$ when $p = 1, 2, 4, \infty$.
2. Let $X = L^p[0, 1]$ and $x(t) = t^2$. Find $\|x\|_p$ for $1 \leq p \leq \infty$. Show that $\lim_{p \rightarrow \infty} \|x\|_p = \|x\|_\infty$.
3. Let $X = L^p(0, \infty)$ and $x(t) = e^{-t}$. Find $\|x\|_p$ for $1 \leq p \leq \infty$. Show that $\lim_{p \rightarrow \infty} \|x\|_p = \|x\|_\infty$.
4. Let $X = Y = \mathbb{R}$, each having norm $\|\cdot\|_p, 1 \leq p < \infty$. Find the norm of an element (x, y) in the product space $X \times Y$ as given above.
5. Let $X = \mathbb{R}^2, Y = \mathbb{R}$, each having norm $\|\cdot\|_p, 1 \leq p < \infty$. Find the norm of an element (x, y) in the product space $X \times Y$. Find the limit of this norm as $p \rightarrow \infty$.
6. Let $X = \mathbb{R}^2$ with norm $\|\cdot\|_2$, and Y be the subspace of X consisting of all $x = (x(1), x(2))$ such that $x(2) = 2x(1)$. Find the norm of a general element $x + Y$ in X/Y .

PROBLEMS

PROBLEM 1.1.1 Let X be a normed linear space, $a \in X$ and k a non zero scalar. Prove that the mappings

$$x \mapsto x + a \quad \text{and} \quad x \mapsto kx$$

are homeomorphisms of X onto itself.

Solution For $a \in X$ and $0 \neq k \in \mathbb{K}$, let

$$\begin{aligned}T_a(x) &= x + a, \quad x \in X, \\ M_k(x) &= kx, \quad x \in X.\end{aligned}$$

Then,

$$\|T_a(x) - T_a(y)\| = \|(x + a) - (y + a)\| = \|x - y\|,$$

$$\|M_k(x) - M_k(y)\| = \|kx - ky\| = |k| \|x - y\|.$$

These show that T_a and M_k are uniformly continuous. Again,

$$T_{-a}T_a(x) = T_{-a}(x + a) = (x + a) - a = x,$$

$$M_{1/k}M_k(x) = M_{1/k}(kx) = \frac{1}{k}(kx) = x.$$

These imply that T_a and M_k have inverses, namely,

$$T_a^{-1} = T_{-a}, \quad M_k^{-1} = M_{1/k}.$$

These inverses are also of the same type. So they also are uniformly continuous. This proves the result. ■

PROBLEM 1.1.2 A maximal proper subspace of a linear space is called a *hyperspace*. If Y is a hyperspace in a normed linear space X , prove that the following are equivalent:

- (a) Y is closed.
- (b) Y is not dense in X .
- (c) Y is nowhere dense in X .

Solution A subset E of X is said to be dense in X if $\overline{E} = X$; it is said to be nowhere dense in X if $\overline{E}$ has no interior points.

If $\overline{Y}$ has no interior point, then $\overline{Y}$ cannot be equal to X ; that is, Y cannot be dense. Thus, (c) $\implies$ (b).

Since $Y \subset \overline{Y} \subset X$ and $\overline{Y}$ is a subspace, the maximality of Y implies that $\overline{Y} = Y$ or $\overline{Y} = X$. So, if $\overline{Y} \neq X$, then $\overline{Y} = Y$. Thus (b) $\implies$ (a). Suppose that a is an interior point of $\overline{Y}$. Then there is $r > 0$ such that

$$U(a, r) = \{x \in X : \|x - a\| < r\} \subset \overline{Y}.$$

Let x be any element of X . If $x \neq 0$, let

$$y = a + \frac{r}{2\|x\|}x.$$

Then, $\|y - a\| = (r/(2\|x\|))\|x\| < r$, and hence $y \in U(a, r) \subset \overline{Y}$. Since $\overline{Y}$ is a subspace we see that

$$x = 2\frac{\|x\|}{r}(y - a) \in \overline{Y}.$$

This is true even if $x = 0$. So $\overline{Y} = X$. Thus $\overline{Y} \neq Y$ since $Y \neq X$. It follows that if $\overline{Y} = Y$ then $\overline{Y}$ cannot have an interior point. Hence (a) $\implies$ (c). This completes the proof. ■

PROBLEM 1.1.3 Show that a normed space X is homeomorphic to the open ball

$$U(0, r) = \{x \in X : \|x\| < r\}, \quad r > 0.$$

Solution For $x, y \in X$ with $\|y\| < r$, let

$$\phi(x) = \frac{rx}{1 + \|x\|}, \quad \psi(y) = \frac{y}{r - \|y\|}.$$

If $y = \phi(x)$, then

$$\|y\| = \frac{r\|x\|}{1 + \|x\|} < r, \quad r - \|y\| = \frac{r}{1 + \|x\|}.$$

Hence $\psi(\phi(x)) = \psi(y) = x$. If $x = \psi(y)$, $\|y\| < r$ then $\|x\| = \|y\|/(r - \|y\|)$ and therefore $\phi(\psi(y)) = \phi(x) = y$. This shows that $\phi : X \rightarrow U(0, r)$ is a bijective mapping and $\psi = \phi^{-1}$. If $x_n \rightarrow x$, then $\|x_n\| \rightarrow \|x\|$, and hence

$$\phi(x_n) = \frac{rx_n}{1 + \|x_n\|} \rightarrow \frac{rx}{1 + \|x\|} = \phi(x).$$

Therefore, ϕ is continuous. Similarly, if $y_n \rightarrow y$ and $y_n, y \in U(0, r)$, then

$$\phi^{-1}(y_n) = \frac{y_n}{r - \|y_n\|} \rightarrow \frac{y}{r - \|y\|} = \phi^{-1}(y).$$

So ϕ^{-1} also is continuous. Thus, ϕ is a homeomorphism. ■

PROBLEM 1.1.4 Let E_1 and E_2 be subsets of a normed space X , and

$$E_1 + E_2 = \{x + y : x \in E_1, y \in E_2\}.$$

Prove the following:

- (a) If E_1 or E_2 is open then $E_1 + E_2$ is open.
- (b) If E_1 and E_2 are compact then $E_1 + E_2$ is compact.
- (c) If E_1 is compact and E_2 is closed then $E_1 + E_2$ is closed.
- (d) $E_1 + E_2$ need not be closed even though E_1 and E_2 are closed.

Solution (a) If $c \in E_1 + E_2$ then $c = a + b$ with $a \in E_1$ and $b \in E_2$. Suppose that E_1 is open. Then there is $r > 0$ such that

$$U(a, r) = \{x \in X : \|x - a\| < r\} \subset E_1.$$

Let $\|z - c\| < r$. If $x = z - b$ then $x - a = z - c$, $\|x - a\| < r$, and hence $x \in E_1$. Thus $z = x + b \in E_1 + E_2$. Since this is true whenever $\|z - c\| < r$, we see that $E_1 + E_2$ is open.

(b) We use the fact that a subset E of a metric space is compact if and only if every sequence in E has subsequence converging in E . Let $\{z_n\}$ be a sequence in $E_1 + E_2$. Then $z_n = x_n + y_n$ with $x_n \in E_1, y_n \in E_2$. Since E_1 is compact, $\{x_n\}$ has a subsequence $\{x_{n_i}\}$ converging to a point x in E_1 . Then, $\{y_{n_i} : i = 1, 2, \dots\}$ is a sequence in the compact set E_2 , where we have written n_i instead of n_i . So it has a subsequence $\{y_{n_{ij}} : j = 1, 2, \dots\}$

converging to a point y in E_2 . Clearly, $x_{nij} \rightarrow x$ as $j \rightarrow \infty$. Hence, as $n \rightarrow \infty$,

$$z_{nij} = x_{nij} + y_{nij} \rightarrow x + y.$$

Thus we get a subsequence $\{z_{nij}\}$ of $\{z_n\}$ converging to $x + y \in E_1 + E_2$. This proves that $\overline{E_1 + E_2}$ is compact.

(c) Let $z \in \overline{E_1 + E_2}$. Then there is a sequence $\{z_n\}$ such that

$$z_n \rightarrow z, \quad z_n = x_n + y_n, \quad x_n \in E_1, y_n \in E_2.$$

Since E_1 is compact $\{x_n\}$ has a subsequence $\{x_{n_i}\}$ converging to some x in E_1 . So, as $i \rightarrow \infty$,

$$y_{n_i} = z_{n_i} - x_{n_i} \rightarrow z - x.$$

Hence $z - x \in \overline{E_2} = E_2$ and, therefore, $z = x + (z - x) \in E_1 + E_2$. This proves that $E_1 + E_2$ is closed.

(d) Let $X = \mathbb{K}$ and consider the sets

$$E_1 = \{n + \frac{1}{n} : n \in \mathbb{N}\}, \quad E_2 = \{-n : n \in \mathbb{N}\}.$$

These sets have no limit points. Hence they are closed. Since

$$p + \frac{1}{n+p} = \left(n + p + \frac{1}{n+p}\right) + (-n) \in E_1 + E_2$$

for all $p, n \geq 1$, we see that p is a limit point of $E_1 + E_2$. If $p \in E_1 + E_2$, then

$$p = \left(n + \frac{1}{n}\right) + (-m),$$

that is, $n(m + p - n) = 1$. This is possible only if $n = 1, m = 2 - p$. So $p \notin E_1 + E_2$ if $p \geq 2$. Thus, $E_1 + E_2$ is not closed. ■

PROBLEM 1.1.5 Let E_1 and E_2 be subsets of a normed space X . If E_1 is compact, E_2 is closed and $E_1 \cap E_2 = \emptyset$, prove that there exists $r > 0$ such that

$$(E_1 + U(0, r)) \cap E_2 = \emptyset,$$

where $U(0, r) = \{x \in X : \|x\| < r\}$.

Solution Let

$$g(x) = d(x, E_2) = \inf \{\|x - y\| : y \in E_2\} \quad \text{for } x \in X$$

Then g is a real continuous function on X . Since E_1 is compact there is $x_1 \in E_1$ such that

$$g(x_1) \leq g(x) \quad \text{for all } x \in E_1. \quad (1)$$

Since $x_1 \notin E_2$ and E_2 is closed, we see that $d(x_1, E_2) > 0$, that is, $g(x_1) > 0$. Let $0 < r < g(x_1)$. Assume that $x \in (E_1 + U(0, r)) \cap E_2$. Then $x \in E_2$ and $x = y + z$ with $y \in E_1, z \in U(0, r)$. Therefore,

$$g(y) = d(y, E_2) \leq d(y, x) = \|x - y\| = \|z\| < r < g(x_1).$$

But $g(x_1) \leq g(y)$ by (1), since $y \in E_1$. This contradiction shows that

$$(E_1 + U(0, r)) \cap E_2 = \emptyset. \quad \blacksquare$$

PROBLEM 1.1.6 Let $0 < p < 1$ and

$$\|x\|_p = \left(\sum_{j=1}^n |x(j)|^p \right)^{1/p}, \quad x \in \mathbb{K}^n.$$

Show that $\|\cdot\|$ is not a norm on $\mathbb{K}^n$ if $n \geq 2$.

Solution The triangle inequality is not satisfied if $n \geq 2$. For example, let

$$x = (1, 0, 0, \dots, 0), \quad y = (0, 1, 0, \dots, 0).$$

Then $x + y = (1, 1, 0, \dots, 0)$,

$$\|x\|_p = (1^p)^{1/p} = 1 = \|y\|_p, \quad \|x + y\|_p = (1^p + 1^p)^{1/p} = 2^{1/p}.$$

Hence,

$$\|x\|_p + \|y\|_p = 2 < 2^{1/p} = \|x + y\|_p,$$

since $1/p > 1$. This proves the result. $\blacksquare$

PROBLEM 1.1.7 Let X be the linear space of all polynomials in one variable t with scalar coefficients. For $p \in X$ with

$$p(t) = a_0 + a_1 t + \dots + a_n t^n,$$

define

$$\|p\| = \sup \{|p(t)| : 0 \leq t \leq 1\}, \quad \|p\|_1 = |a_0| + |a_1| + \dots + |a_n|.$$

Prove that $\|\cdot\|$ and $\|\cdot\|_1$ are norms on X , and $\|p\| \leq \|p\|_1$ for every $p \in X$. Prove also that there is no constant α such that $\|p\|_1 \leq \alpha \|p\|$ for all $p \in X$.

Solution It can be seen that $\|\cdot\|$ is the sup norm on $C[0, 1]$ and that X is a subspace of $C[0, 1]$. An element p of X is uniquely determined by the finite sequence $\{a_j\}$ of its coefficients, and $\|p\|_1$ is just the l^1 norm of this sequence. Thus we see that $\|\cdot\|$ and $\|\cdot\|_1$ are norms on X .

If $p(t) = a_0 + a_1 t + \dots + a_n t^n$, then for $0 \leq t \leq 1$,

$$|p(t)| \leq |a_0| + |a_1|t + \dots + |a_n|t^n \leq |a_0| + |a_1| + \dots + |a_n| = \|p\|_1.$$

To prove the nonexistence of an α as stated in the problem, let

$$p(t) = 1 - t + t^2 - \dots - t^{2n-1}.$$

Now, for $0 \leq t \leq 1$, we have

$$p(t) = (1 - t) + t^2(1 - t) + \dots + t^{2n-2}(1 - t) \geq 0,$$

$$p(t) = 1 - t(1-t) - \dots - t^{2n-3}(1-t) - t^{2n-1} \leq 1.$$

Since $p(0) = 1$, we get

$$\|p\| = \sup \{|p(t)| : 0 \leq t \leq 1\} = 1.$$

Since $\|p\|_1 = 1 + 1 + \dots + 1 = 2n$ we have $\|p\|_1 > \alpha\|p\|$ if $2n > \alpha$. Thus for any α we can choose a polynomial $p(t)$ with $\|p\|_1 > \alpha\|p\|$. This completes the proof. ■

PROBLEM 1.1.8 Let $BV[a, b]$ denote the set of all functions of bounded variation on $[a, b]$. For $x \in [a, b]$, let

$$\|x\| = |x(a)| + V(x),$$

where $V(x)$ is the total variation of x . Prove that $\|\cdot\|$ is a norm on $BV[a, b]$.

Solution The total variation $V(x)$ is defined as the supremum of the sums

$$S(x, P) = \sum_{i=1}^n |x(t_i) - x(t_{i-1})|$$

corresponding to all possible partitions $P = (t_0, \dots, t_n)$ of $[a, b]$.

Obviously, $V(x) \geq 0$ and hence $\|x\| \geq 0$ for all $x \in BV[a, b]$. If $x(t) = 0$ for all $t \in [a, b]$, then $S(x, P) = 0$ for all P ; hence $\|x\| = 0$.

Let $\|x\| = 0$. Then $x(a) = 0$ and $V(x) = 0$. So $S(x, P) = 0$ for all P . Let $a < t < b$. If $P = (a, t, b)$, then

$$S(x, P) = |x(t) - x(a)| + |x(b) - x(t)|.$$

Hence, $|x(t) - x(a)| = |x(b) - x(t)| = 0$. Since $x(a) = 0$ we have $x(t) = 0$. Thus $x = 0$.

Let $x \in BV[a, b]$ and $k \in \mathbb{K}$. If $P = (t_0, \dots, t_n)$ is any partition of $[a, b]$, then

$$S(kx, P) = \sum_{i=1}^n |kx(t_i) - kx(t_{i-1})| = |k|S(x, P) \leq |k|V(x).$$

This implies that $kx \in BV[a, b]$ and $V(kx) \leq |k|V(x)$. Hence,

$$\|kx\| \leq |kx(a)| + |k|V(x) = |k|\|x\|.$$

If $k \neq 0$, change k to $1/k$ and x to kx in this. We get

$$\|x\| = \left\| \frac{1}{k} \cdot kx \right\| \leq \frac{1}{|k|} \|kx\| \leq \frac{1}{|k|} \cdot |k| \|x\| = \|x\|.$$

Hence $\|kx\| = |k|\|x\|$. This is true for $k = 0$ also.

Finally, let $x, y \in BV[a, b]$ and $z = x + y$. Let $P(t_0, \dots, t_n)$ be any partition of $[a, b]$. Since

$$|z(t_i) - z(t_{i-1})| = |x(t_i) + y(t_i) - x(t_{i-1}) - y(t_{i-1})|$$

$$\leq |x(t_i) - x(t_{i-1})| + |y(t_i) - y(t_{i-1})|$$

we get, on addition,

$$S(z, P) \leq S(x, P) + S(y, P) \leq V(x) + V(y).$$

This implies that $z \in BV[a, b]$ and $V(z) \leq V(x) + V(y)$. Since

$$|z(a)| \leq |x(a)| + |y(a)|,$$

we obtain $\|z\| \leq \|x\| + \|y\|$. This is the triangle inequality; and the proof is complete. ■

PROBLEM 1.1.9 A nonempty subset E of a linear space X is said to be *balanced* if $kx \in E$ whenever $x \in E$ and $k \in \mathbb{K}$ with $|k| \leq 1$. It is said to be *absorbing* if for every $x \in X$ there exists $r > 0$ with $r^{-1}x \in E$. Suppose that E is convex, balanced and absorbing, and also that no nonzero subspace of X is contained in E . For $x \in X$, let

$$\|x\| = \inf \{r > 0 : r^{-1}x \in E\}.$$

Prove that $\|\cdot\|$ is a norm on X , and that

$$\{x \in X : \|x\| < 1\} \subset E \subset \{x \in X : \|x\| \leq 1\}.$$

Show also that in any normed space X the norm is generated like this by some E .

Solution Given that $\|x\| = \inf S_x$, where

$$S_x = \{r > 0 : (1/r)x \in E\}. \quad (2)$$

Thus $S_x \subset (0, \infty)$, and hence $\|x\| \geq 0$ for all x . If $x \in E$ then $(-1)x \in E$ as E is balanced. So $0 = (1/2)x + (1/2)(-x) \in E$ as E is convex. Thus we see that

$$(1/r)0 \in E \quad \text{for all } r > 0, \\ S_0 = (0, \infty), \quad \|0\| = \inf S_0 = 0.$$

Let $x \neq 0$. Then $Y = \{kx : k \in \mathbb{K}\}$ is a nonzero subspace of X , and so Y is not contained in E . Hence there is $k_1 \in \mathbb{K}$ such that $k_1x \notin E$. Clearly, $k_1 \neq 0$. Suppose that $0 < r < 1/|k_1|$ and $r \in S_x$. Then $(1/r)x \in E$, and hence

$$k_1x = (k_1r)\left(\frac{1}{r}x\right) \in E$$

since $|k_1r| < 1$. This contradiction shows that

$$r \in S_x \implies r \geq 1/|k_1|.$$

Hence $\|x\| \geq 1/|k_1| > 0$. Thus we have proved that $\|x\| = 0 \Leftrightarrow x = 0$.

Let $k \neq 0$ and $r \in S_{kx}$. Then,

$$\frac{1}{r}kx \in E, \quad \frac{|k|}{r}x = \left(\frac{|k|}{k}\right)\left(\frac{kx}{r}\right) \in E.$$

Thus, $r/|k| \in S_x$, and hence $\|x\| \leq r/|k|$, that is, $|k|\|x\| \leq r$. This is true for all $r \in S_{kx}$. It follows that $|k|\|x\| \leq \|kx\|$. Changing k to $1/k$ and x to kx in this, we get

$$\left|\frac{1}{k}\right| \cdot \|kx\| \leq \left\|\frac{1}{k} \cdot kx\right\| = \|x\|, \quad \|kx\| \leq |k|\|x\|.$$

Therefore, $\|kx\| = |k|\|x\|$. This is true even if $k = 0$.

To prove the triangle inequality, let $x, y \in X$. Given $\epsilon > 0$ we can find r in S_x and s in S_y such that

$$r < \|x\| + \epsilon, \quad s < \|y\| + \epsilon.$$

Then,

$$\frac{1}{r}x \in E, \quad \frac{1}{s}y \in E, \quad \frac{1}{r+s}(x+y) = \frac{r}{r+s}\left(\frac{x}{r}\right) + \frac{s}{r+s}\left(\frac{y}{s}\right) \in E,$$

since E is convex. Hence,

$$r+s \in S_{x+y}, \quad \|x+y\| \leq r+s < \|x\| + \|y\| + 2\epsilon.$$

This implies the triangle inequality $\|x+y\| \leq \|x\| + \|y\|$ since $\epsilon > 0$ is arbitrary.

Thus, $\|\cdot\|$ is a norm on X . Let $\|x\| < 1$. Then there is $r \in S_x$ with $\|x\| < r < 1$. Hence,

$$\frac{1}{r}x \in E, \quad x = r\left(\frac{x}{r}\right) \in E,$$

as $0 < r < 1$. Therefore,

$$\{x \in X : \|x\| < 1\} \subset E.$$

If $x \in E$ then $x/1 \in E$, $1 \in S_x$ and hence $\|x\| \leq 1$. Thus,

$$E \subset \{x \in X : \|x\| \leq 1\}.$$

For the second part let X be a normed space, and

$$E = \{x \in X : \|x\| < 1\}.$$

Then E is convex, balanced and absorbing. Let $x \in X$ and S_x be defined by (2). If $r \in S_x$ then $r^{-1}x \in E$ and so $r^{-1}\|x\| < 1$. Thus $\|x\| < r$ for any $r \in S_x$. This implies that

$$\|x\| \leq \inf S_x. \quad (3)$$

If $\|x\| < \inf S_x$, then choose r with $\|x\| < r < \inf S_x$. Thus, $r^{-1}\|x\| < 1$, and so, $r^{-1}x \in E$; that is, $r \in S_x$. This is not possible since $r < \inf S_x$. This contradiction shows that equality holds in (3). This completes the proof. ■

PROBLEM 1.1.10 Let Y be a closed subspace of a normed space X . Show that $x_n + Y \rightarrow x + Y$ iff there is a sequence $\{y_n\}$ in Y such that $x_n + y_n \rightarrow x$ in X .

Solution Note that $\|x_n - x + Y\|$ is the greatest lower bound of the set

$$S_n = \{\|x_n - x + y\| : y \in Y\}.$$

So for $n = 1, 2, \dots$ there is y_n in Y such that

$$\|x_n - x + y_n\| < \|x_n - x + Y\| + \frac{1}{n}$$

Hence,

$$\|x_n - x + y_n\| < \|x_n - x + Y\| + \frac{1}{n} = \|(x_n + Y) - (x + Y)\| + \frac{1}{n} \rightarrow 0$$

as $n \rightarrow \infty$. Therefore, $x_n + y_n \rightarrow x$ in X .

Conversely, suppose that such a sequence $\{y_n\}$ exists in Y . Then,

$$\|(x_n + Y) - (x + Y)\| = \|x_n - x + Y\| \leq \|x_n - x + y_n\| \rightarrow 0$$

as $n \rightarrow \infty$. Hence, $x_n + Y \rightarrow x + Y$ in X/Y . This completes the proof. ■

PROBLEM 1.1.11 Let Y be a closed subspace of a normed space X and $F : X \rightarrow X/Y$ be the quotient map defined by $F(x) = x + Y$. Prove that F is continuous and maps open sets in X onto open sets in X/Y .

Solution Since

$$F(x_1) - F(x_2) = (x_1 + Y) - (x_2 + Y) = x_1 - x_2 + Y,$$

we get

$$\|F(x_1) - F(x_2)\| = \inf \{\|x_1 - x_2 + y\| : y \in Y\} \leq \|x_1 - x_2\|.$$

This shows that F is uniformly continuous.

Let V be an open set in X . Consider any element $F(x_1)$ in $F(V)$, where $x_1 \in V$. Since V is open there is $r > 0$ such that

$$\|x - x_1\| < r \implies x \in V. \quad (4)$$

Let

$$\|(x + Y) - F(x_1)\| < r. \quad (5)$$

Then

$$\|x - x_1 + Y\| = \|(x + Y) - (x_1 + Y)\| < r.$$

So by the definition of the quotient norm, we have $\|x - x_1 + y_1\| < r$ for some $y_1 \in Y$. Hence $x + y_1 \in V$ by (4). Thus, $F(x + y_1) \in F(V)$. But

$$F(x + y_1) = x + y_1 + Y = x + Y.$$

Therefore, $x + Y \in F(V)$ whenever (5) holds. This shows that $F(V)$ contains the open ball in X/Y with centre $F(x_1)$ and radius r . So $F(V)$ is open. This completes the proof. ■

PROBLEM 1.1.12 Prove that

- (a) l^p is separable for $1 \leq p < \infty$,
- (b) l^∞ is not separable.

Solution (a) Let M be a countable dense subset of $\mathbb{K}$. If $\mathbb{K} = \mathbb{R}$ we may take M as the set of all rationals, and if $\mathbb{K} = \mathbb{C}$ we may take M as the set of all complex numbers $\alpha + i\beta$ with α, β rationals. For $n = 1, 2, \dots$, let E_n be the set of y in l^p such that

$$y(j) \in M \quad \text{for } j \geq 1; \quad y(j) = 0 \quad \text{for } j > n.$$

Then E_n is countable since it is in one-one correspondence with M^n . Let $E = \bigcup_{n=1}^{\infty} E_n$. Then E is countable. We shall prove that E is dense in l^p .

Let $x \in l^p$ and $\epsilon > 0$ be given. There is n such that

$$\sum_{j=n+1}^{\infty} |x(j)|^p < \epsilon^p/2.$$

Fix n . For each j , with $1 \leq j \leq n$, choose $y(j)$ in M such that

$$|x(j) - y(j)| < (2n)^{-1/p} \epsilon,$$

and let $y(j) = 0$ for $j > n$. Then $y \in E_n \subset E$, and

$$\begin{aligned} \left(\|x - y\|_p \right)^p &= \sum_{j=1}^n |x(j) - y(j)|^p + \sum_{j=n+1}^{\infty} |x(j)|^p \\ &< n(2n)^{-1} \epsilon^p + 2^{-1} \epsilon^p = \epsilon^p. \end{aligned}$$

Thus, $\|x - y\|_p < \epsilon$. This proves that $x \in \overline{E}$. Therefore, $\overline{E} = l^p$.

(b) Suppose that $E \subset l^\infty$ and $\overline{E} = l^\infty$. Each real number t between 0 and 1 can be expressed as

$$t = \sum_{n=1}^{\infty} x(n) 2^{-n} \quad \text{with } x(n) = 0 \text{ or } 1 \quad \text{for all } n. \quad (6)$$

Hence, for each t in $(0, 1)$ we can find $h(t) = x$ in l^∞ satisfying (6); and then we can find $g(x)$ in E such that

$$\|g(x) - x\|_\infty < 1/4.$$

Let $f(t) = g(h(t))$, $0 < t < 1$. Suppose that $f(t_1) = f(t_2)$, $h(t_1) = x_1$ and $h(t_2) = x_2$. Then, $g(x_1) - g(x_2) = 0$, and therefore,

$$\|x_1 - x_2\| \leq \|x_1 - g(x_1)\|_\infty + \|g(x_1) - g(x_2)\|_\infty + \|g(x_2) - x_2\|_\infty < 1/2,$$

$$|x_1(n) - x_2(n)| \leq \|x_1 - x_2\|_\infty < 1/2.$$

But $x_1(n) - x_2(n) = 0, \pm 1$ for all n . So we must have $x_1(n) - x_2(n) = 0$ for all n ; hence,

$$t_1 = \sum_{n=1}^{\infty} x_1(n)2^{-n} = \sum_{n=1}^{\infty} x_2(n)2^{-n} = t_2.$$

This shows that $f : (0, 1) \rightarrow E$ is one-one. So E must be uncountable since $(0, 1)$ is uncountable. Thus we see that l^∞ cannot have a countable dense subset. This completes the proof. ■

PROBLEM 1.1.13 Show that if a normed space X is separable and Y is a closed subspace of X , then the quotient space X/Y is separable.

Solution Let E be a countable dense subset of X , and

$$F = \{y + Y : y \in E\}.$$

Then F is countable. Let $x + Y$ be any element of X/Y and $\epsilon > 0$ be given. There is y in E such that $\|y - x\| < \epsilon$. So by the definition of the quotient norm,

$$\|(x + Y) - (y + Y)\| = \|x - y + Y\| \leq \|x - y\| < \epsilon.$$

This implies that $\overline{F} = X/Y$. So X/Y is separable. ■

PROBLEM 1.1.14 If $x \in L^p(T)$, $x \in L^q(T)$ and $p < q$ show that $x \in L^r(T)$ for $p < r < q$.

Solution It is known that e^t is a convex function of t . So

$$e^{\lambda u + (1-\lambda)v} \leq \lambda e^u + (1-\lambda)e^v \quad \text{for } 0 < \lambda < 1.$$

If $t \in T$ and $0 < |x(t)| < \infty$ choose u, v and λ such that

$$e^u = |x(t)|^p, \quad e^v = |x(t)|^q, \quad r = \lambda p + (1-\lambda)q.$$

Thus,

$$\begin{aligned} |x(t)|^r &= |x(t)|^{p\lambda} \cdot |x(t)|^{(1-\lambda)q} = e^{\lambda u} \cdot e^{(1-\lambda)v} \\ &\leq \lambda e^u + (1-\lambda)e^v = \lambda |x(t)|^p + (1-\lambda)|x(t)|^q. \end{aligned}$$

Integrating, we obtain

$$\int |x|^r d\mu \leq \lambda \int |x|^p d\mu + (1-\lambda) \int |x|^q d\mu < \infty.$$

This proves that $x \in L^r(T)$. ■

PROBLEM 1.1.15 Prove that if $x \in l^r$ for some $r \geq 1$ then $x \in l^p$ for all $p > r$. Prove also that

$$\lim_{p \rightarrow \infty} \|x\|_p = \|x\|_\infty,$$

for any x in l^r or $\mathbb{K}^n$.

Solution If $x \in l^r$ then $\sum |x(n)|^r$ converges. So $|x(n)| < 1$ for all large n , say $n \geq n_0$. Thus, if $p > r$ then

$$|x(n)|^p \leq |x(n)|^r \quad \text{for } n \geq n_0.$$

Hence $\sum |x(n)|^p < \infty$. This proves the first part.

The second part is trivial if $x = 0$. So let $x \in l^r$ and $x \neq 0$. Then we have

$$\begin{aligned} \left(\frac{\|x\|_p}{\|x\|_\infty} \right)^p &= \sum_{n=1}^{\infty} \left(\frac{|x(n)|}{\|x\|_\infty} \right)^p \leq \sum_{n=1}^{\infty} \left(\frac{|x(n)|}{\|x\|_\infty} \right)^r = \left(\frac{\|x\|_r}{\|x\|_\infty} \right)^r, \\ \|x\|_p &\leq \left(\frac{\|x\|_r}{\|x\|_\infty} \right)^{r/p} \|x\|_\infty. \end{aligned}$$

Again,

$$(\|x\|_p)^p = \sum_{n=1}^{\infty} |x(n)|^p \geq |x(j)|^p.$$

Thus, $|x(j)| \leq \|x\|_p$ for $j \geq 1$ and so $\|x\|_\infty \leq \|x\|_p$. Therefore,

$$\|x\|_\infty \leq \|x\|_p \leq \left(\frac{\|x\|_r}{\|x\|_\infty} \right)^{r/p} \|x\|_\infty.$$

Letting $p \rightarrow \infty$ we get the desired result.

The case $x \in \mathbb{K}^n$ can be dealt with similarly. ■

PROBLEM 1.1.16 Show that $c_0 \subset l^p$ for $1 \leq p \leq \infty$. Is it true that $c_0 \subset l^p$?

Solution Let $x \in c_0$. Then $x(n) = 0$ for all large n ; hence $\sum |x(n)|^p < \infty$. Thus $x \in l^p$ if $1 \leq p < \infty$. Also, $x \in l^\infty$ since $\{x(n)\}$ is bounded.

Let $x(n) = n^{-1/p}$. Then $x(n) \rightarrow 0$ as $n \rightarrow \infty$; so $x \in c_0$. But

$$\sum |x(n)|^p = \sum \frac{1}{n} = \infty.$$

Hence $x \notin l^p$. On the other hand, $x \in l^\infty$ if $x \in c_0$ since convergent sequences are bounded. So the inclusion $c_0 \subset l^p$ is not true for $1 \leq p < \infty$, and it is true when $p = \infty$. ■

1.2 Continuity of Linear Maps

Let X and Y be normed spaces over the same scalar field $\mathbb{K}$. A map $F : X \rightarrow Y$ is said to be *linear* if

$$F(kx + y) = kF(x) + F(y) \quad \text{for } k \in \mathbb{K}, x, y \in X$$

A linear map $f : X \rightarrow \mathbb{K}$ is called a *linear functional*.

A map $F : X \rightarrow Y$ is said to be *bounded* if

$$\|F\| = \sup \{\|F(x)\| : x \in X, \|x\| \leq 1\} < \infty. \quad (1)$$

Let $F : X \rightarrow Y$ be linear. We often write, for convenience, Fx instead of $F(x)$. The *zero space*, also called *null space*, of F is

$$Z(F) = \{x \in X : F(x) = 0\}.$$

This is a subspace of X . Its *range*

$$R(F) = \{y \in Y : y = F(x), x \in X\}$$

is a subspace of Y . A linear map F is one-one iff

$$F(x) = 0 \Rightarrow x = 0, \quad \text{that is, } Z(F) = \{0\}.$$

If $F : X \rightarrow Y$ is linear and bijective then its inverse $F^{-1} : Y \rightarrow X$ is linear.

If $F : X \rightarrow Y$ is linear then the following statements are equivalent:

- (a) F is continuous at 0.
- (b) F is continuous on X .
- (c) F is bounded.

A linear map $F : X \rightarrow Y$ is bounded iff there is a constant $\alpha > 0$ such that

$$\|F(x)\| \leq \alpha \|x\| \quad \text{for all } x \in X.$$

We say that $F : X \rightarrow Y$ is *bounded below* if there is a constant $\beta > 0$ such that

$$\beta \|x\| \leq \|F(x)\| \quad \text{for all } x \in X.$$

The set of all bounded linear maps $F : X \rightarrow Y$ is denoted by $BL(X, Y)$. If $Y = X$ we write $BL(X)$ instead of $BL(X, X)$. If addition and scalar multiplication are defined pointwise then $BL(X, Y)$ becomes a linear space; and (1) defines a norm on $BL(X, Y)$. It can be seen that $\|F\|$ is the infimum of real numbers $\alpha > 0$ with the following property:

$$\|F(x)\| \leq \alpha \|x\| \quad \text{for all } x \in X.$$

Moreover,

$$\begin{aligned} \|F(x)\| &\leq \|F\| \|x\| \quad \text{for all } x \in X, \\ \|F\| &= \sup \{\|F(x)\| : x \in X, \|x\| = 1\} \quad \text{if } X \neq \{0\}. \end{aligned}$$

The set $BL(X, \mathbb{K})$ of all continuous linear functionals on X is denoted by X' and is called the *dual space* or *conjugate space* of X .

Linear maps are also called *linear operators* or *linear transformations*. Let (k_{ij}) be a $(m \times n)$ matrix with scalar entries, and

$$y(i) = \sum_{j=1}^n k_{ij} x(j), \quad 1 \leq i \leq m.$$

This defines a linear map $x \mapsto y$ from $\mathbb{K}^n$ to $\mathbb{K}^m$. Any linear map $F : \mathbb{K}^n \rightarrow \mathbb{K}^m$ is given by a matrix in this manner. If (k_{ij}) is an infinite matrix then the relation

$$y(i) = \sum_{j=1}^{\infty} k_{ij}x(j), \quad i = 1, 2, \dots$$

defines a linear transformation $x \mapsto y$ under certain conditions on k_{ij} .

The transformations $x \mapsto y$ given by

$$y(s) = \int_a^b k(s, t)x(t) dt$$

are linear. They are called integral transforms with kernel $k(s, t)$.

EXERCISES

1. Let X and Y be normed spaces and $F : X \rightarrow Y$ be linear. Show that $Z(F)$ and $R(F)$ are subspaces of X and Y respectively.
2. Let X and Y be normed spaces and $F : X \rightarrow Y$ be linear. Show that F is one-one iff $Z(F) = \{0\}$.
3. Let X and Y be normed spaces and $F : X \rightarrow Y$ be linear. Show that the following are equivalent: (i) F is continuous; (ii) F is continuous at 0; (iii) F is bounded.
4. Let X and Y be normed spaces and $F : X \rightarrow Y$ be linear. Show that

$$\|F(x)\| \leq \|F\| \|x\| \quad \text{for all } x \in X.$$

5. If $F : X \rightarrow Y$ is bijective and linear show that its inverse $F^{-1} : Y \rightarrow X$ is linear.
6. Let $X = \mathbb{R}^2$ with norm $\|\cdot\|_1$, and $A \in BL(X)$ be represented by the matrix $M = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. Show that $\|A\| = 6$.

PROBLEMS

PROBLEM 1.2.1 Let f be a linear functional on a linear space X and $f(x_0) \neq 0$ for some x_0 . Show that any x in X has a unique representation as $x = kx_0 + y$ with $k \in \mathbb{K}$ and $y \in Z(f)$, the zero space of f .

Solution If $x \in X$ let $k = f(x)/f(x_0)$ and $y = x - kx_0$. Then $x = kx_0 + y$ and

$$f(y) = f(x) - kf(x_0) = 0;$$

hence $y \in Z(f)$. Since $k \in \mathbb{K}$, this shows the existence of a representation. For uniqueness let

$$x = k_1x_0 + y_1 = k_2x_0 + y_2, \quad y_1, y_2 \in Z(f).$$

Then,

$$(k_1 - k_2)x_0 = y_2 - y_1 \in Z(f).$$

So $k_1 - k_2 = 0$ since $x_0 \notin Z(f)$. Hence, $k_1 = k_2$ and $y_1 = y_2$. This verifies uniqueness and completes the proof. ■

PROBLEM 1.2.2 Let f_1 and f_2 be two linear functionals on a linear space X . Show that if they have the same zero space then there is a nonzero scalar k such that $f_2(x) = kf_1(x)$ for all $x \in X$.

Solution If $f_1(x) = 0$ for all x , then $f_2(x) = 0$ for all x since $Z(f_2) = Z(f_1) = X$. We can take any $k \neq 0$ in this case. So suppose that $f_1(x_0) \neq 0$ for some x_0 . Then $f_2(x_0) \neq 0$. Let

$$k = \frac{f_2(x_0)}{f_1(x_0)}.$$

Then $k \neq 0$. If $x \in X$, let $\alpha = \frac{f_1(x)}{f_1(x_0)}$. Then,

$$f_1(x) = \alpha f_1(x_0) = f_1(\alpha x_0),$$

$$f_1(x - \alpha x_0) = 0,$$

$$x - \alpha x_0 \in Z(f_1) = Z(f_2).$$

Hence, $f_2(x - \alpha x_0) = 0$. Thus,

$$f_2(x) = \alpha f_2(x_0) = \alpha k f_1(x_0) = k f_1(x).$$

This proves the result. ■

PROBLEM 1.2.3 Let X and Y be normed spaces, $X \neq \{0\}$, and $F : X \rightarrow Y$ be a bounded linear map. Show that there is a sequence $\{x_n\}$ in X such that $\|x_n\| = 1$ for all n and $\|F(x_n)\| \rightarrow \|F\|$ as $n \rightarrow \infty$.

Solution By definition, $\|F\|$ is the supremum of the set

$$S = \{\|F(x)\| : x \in X, \|x\| \leq 1\}.$$

If $\|F\| = 0$ then S contains only 0. Let $\{x_n\}$ be any sequence in X with $\|x_n\| = 1$. Then $\|F(x_n)\| = 0$ for all n , and hence $\|F(x_n)\| \rightarrow 0 = \|F\|$ in this case.

Next suppose that $\|F\| \neq 0$. Then $\|F\| - \frac{1}{n}\|F\| < \|F\|$ for all $n \geq 1$. So there is $\alpha_n \in S$ such that

$$\|F\| - \frac{1}{n}\|F\| < \alpha_n \quad \text{for } n = 1, 2, \dots$$

Then $\alpha_n = \|F(z_n)\|$ with $z_n \in X, \|z_n\| \leq 1$. Since $\alpha_n > \|F\| - (1/n)\|F\| \geq 0$, we have $\alpha_n \neq 0$ and so $z_n \neq 0$, for any n . Let $x_n = z_n/\|z_n\|$. Then $\|x_n\| = 1$ for all $n \geq 1$ and so $\|F(x_n)\| \in S$. Thus $\|F(x_n)\| \leq \|F\|$. However,

$$\|F(x_n)\| = \frac{1}{\|z_n\|} \|F(z_n)\| = \frac{\alpha_n}{\|z_n\|} \geq \alpha_n.$$

Thus we have

$$\|F\| - \frac{1}{n}\|F\| < \alpha_n \leq \|F(x_n)\| \leq \|F\|$$

for all n . This implies that $\|F(x_n)\| \rightarrow \|F\|$ as $n \rightarrow \infty$. ■

PROBLEM 1.2.4 Let X be a normed space and f be a nonzero bounded linear functional on X . Show that

$$E = \{x \in X : f(x) = \|f\|\}$$

is a nonempty closed convex subset of X . Show also that

$$\inf \{\|x\| : x \in E\} = 1.$$

Solution Given that $f(x_0) \neq 0$ for some $x_0 \in X$. Let $k = \|f\|/f(x_0)$. Then,

$$f(kx_0) = kf(x_0) = \|f\|.$$

Hence $kx_0 \in E$; thus E is nonempty.

Let $x \in \overline{E}$. Thus there is a sequence $\{x_n\}$ in E such that $x_n \rightarrow x$. Then $f(x_n) = \|f\|$, and $f(x_n) \rightarrow f(x)$ by the continuity of f . Hence $f(x) = \|f\|$; that is, $x \in E$. Thus E is closed.

If $x, y \in E$ and $0 \leq t \leq 1$, then

$$f(tx + (1-t)y) = tf(x) + (1-t)f(y) = t\|f\| + (1-t)\|f\| = \|f\|.$$

Therefore, $tx + (1-t)y \in E$. This shows that E is convex.

If $x \in E$ then $\|f\| = f(x) \leq \|f\| \cdot \|x\|$. Hence, $1 \leq \|x\|$ for all $x \in E$, since $\|f\| > 0$. By Problem 1.2.3, there is a sequence $\{x_n\}$ in X such that $\|x_n\| = 1$ and $|f(x_n)| \rightarrow \|f\|$. Let

$$k_n = \frac{\|f\|}{f(x_n)}.$$

Then, $|k_n| \rightarrow 1$ and $\|f\| = k_n f(x_n) = f(k_n x_n)$. Thus $k_n x_n \in E$ and $\|k_n x_n\| = |k_n| \rightarrow 1$. This shows that $\inf \{\|x\| : x \in E\} = 1$. This completes the proof. ■

PROBLEM 1.2.5 For $x \in c_{00}$ let $f(x) = \sum_{n=1}^{\infty} x(n)$. Is f linear? Continuous?

Solution If $x \in c_{00}$ then the sum $\sum_{n=1}^{\infty} x(n)$ contains only a finite number of nonzero terms. Hence $f(x)$ exists for each $x \in c_{00}$. If $x, y \in c_{00}, k \in \mathbb{K}$, then

$$(kx + y)(n) = kx(n) + y(n) \quad \text{for all } n.$$

This gives, on addition, $f(kx + y) = kf(x) + f(y)$. Therefore, f is linear.

The norm on c_{00} is the sup norm $\|\cdot\|_{\infty}$. Consider $x_n \in c_{00}$ defined by

$$x_n(j) = \begin{cases} 1 & \text{for } 1 \leq j \leq n, \\ 0 & \text{for } j > n. \end{cases}$$

Then, $\|x_n\|_{\infty} = 1$, and

$$f(x_n) = \sum_{j=1}^{\infty} x_n(j) = n.$$

If f were continuous, there would exist $\alpha > 0$ such that

$$|f(x_n)| \leq \alpha \|x_n\|_{\infty} = \alpha,$$

that is, $n \leq \alpha$ for all n . This is possible only if $\alpha = \infty$. This implies that f is not continuous. ■

PROBLEM 1.2.6 Let c be the space of convergent sequences and

$$f_1(x) = \lim_{n \rightarrow \infty} x(n), \quad x \in c;$$

$$f_2(x) = \sum_{n=1}^{\infty} \frac{x(n)}{n^2}, \quad x \in l^{\infty}.$$

Show that f_1 and f_2 are bounded linear functionals on c and l^{∞} , respectively. Find their norms.

Solution If $x, y \in c, k \in \mathbb{K}$, then

$$(kx + y)(n) = kx(n) + y(n)$$

for all n . Letting $n \rightarrow \infty$, we get

$$f_1(kx + y) = kf_1(x) + f_1(y).$$

Thus, $f_1 : c \rightarrow \mathbb{K}$ is linear.

For $x \in c$ we have

$$\begin{aligned} |x(n)| &\leq \|x\|_{\infty} \quad \text{for all } n, \\ |x(n)| &\rightarrow |f_1(x)| \quad \text{as } n \rightarrow \infty. \end{aligned}$$

Hence, $|f_1(x)| \leq \|x\|_\infty$. This shows that f_1 is bounded and $\|f_1\| \leq 1$. On the other hand, if $x(n) = 1$ for all n , then $x \in c$, $f_1(x) = 1$ and $\|x\|_\infty = 1$. Thus,

$$1 = f_1(x) \leq \|f_1\| \|x\|_\infty = \|f_1\|.$$

It follows that $\|f_1\| = 1$.

If $x \in l^\infty$ then $|x(n)| \leq \|x\|_\infty$ for all n ; hence

$$\sum_{n=1}^{\infty} \left| \frac{x(n)}{n^2} \right| \leq \|x\|_\infty \sum_{n=1}^{\infty} \frac{1}{n^2} = \|x\|_\infty \frac{\pi^2}{6} \quad (2)$$

This shows that $f_2(x)$ exists for each x in l^∞ . Since

$$\frac{(kx + y)(n)}{n^2} = \frac{kx(n)}{n^2} + \frac{y(n)}{n^2},$$

we get, on addition,

$$f_2(kx + y) = kf_2(x) + f_2(y),$$

for $x, y \in l^\infty$, $k \in \mathbb{K}$. Thus, $f_2 : l^\infty \rightarrow \mathbb{K}$ is linear. Moreover, it follows from (2) that

$$|f_2(x)| \leq \frac{\pi^2}{6} \|x\|_\infty, \quad x \in l^\infty.$$

Hence f_2 is bounded and $\|f_2\| \leq \pi^2/6$. On the other hand, if $x(n) = 1$ for all n , then $x \in l^\infty$, $f_2(x) = \sum 1/n^2 = \pi^2/6$, and $\|x\|_\infty = 1$. Thus,

$$\frac{\pi^2}{6} = f_2(x) \leq \|f_2\| \|x\|_\infty = \|f_2\|.$$

It follows that $\|f_2\| = \pi^2/6$. ■

PROBLEM 1.2.7 Find the norm of the linear functional f defined on $C[-1, 1]$ by

$$f(x) = \int_{-1}^0 x(t) dt - \int_0^1 x(t) dt, \quad x \in C[-1, 1].$$

Solution It can be easily verified that f is linear. Since the norm on $C[-1, 1]$ is the sup norm, we have, for all $x \in C[-1, 1]$,

$$|x(t)| \leq \|x\|_\infty \quad \text{for } t \in [-1, 1],$$

$$|f(x)| \leq \int_{-1}^0 |x(t)| dt + \int_0^1 |x(t)| dt \leq \|x\|_\infty + \|x\|_\infty = 2\|x\|_\infty.$$

Hence, $\|f\| \leq 2$. For $n = 1, 2, \dots$, let

$$x_n(t) = \begin{cases} 1, & -1 \leq t \leq -1/n \\ -nt, & -1/n < t < 1/n \\ -1, & 1/n \leq t \leq 1. \end{cases}$$

Then, $x_n \in C[-1, 1]$, $\|x_n\|_\infty = 1$, and

$$f(x_n) = \int_{-1}^{-1/n} dt + \int_{-1/n}^0 -nt dt - \int_0^{1/n} -nt dt - \int_{1/n}^1 (-1) dt = 2 - 1/n.$$

Therefore,

$$2 - 1/n = f(x_n) \leq \|f\| \cdot \|x_n\|_\infty = \|f\|.$$

This is true for all n . It follows that $\|f\| = 2$. ■

PROBLEM 1.2.8 Show that the zero space of a discontinuous linear functional on a normed space X is dense in X .

Solution Let f be a discontinuous linear functional on X . Then f is not bounded. So, for $n = 1, 2, \dots$ there is $x_n \in X$ such that $\|x_n\| \leq 1$ and $|f(x_n)| > n$. Let $x \in X$ and $f(x) \neq 0$. By Problem 1.2.1 there exist $k_n \in \mathbb{K}$ and $y_n \in Z(f)$ such that $x_n = k_n x + y_n$. Then,

$$f(x_n) = k_n f(x) + f(y_n) = k_n f(x).$$

Hence,

$$n\|x_n\| \leq n < |f(x_n)| = |k_n| |f(x)|. \quad (3)$$

In particular, $k_n \neq 0$. Let $z_n = -(1/k_n)y_n$. Then, $z_n \in Z(f)$ and

$$x_n = k_n x - k_n z_n = k_n(x - z_n),$$

$$|k_n| \|x - z_n\| = \|x_n\| < \frac{1}{n} |k_n| |f(x)|$$

by (3). Hence, $\|x - z_n\| < |f(x)|/n$. Thus $z_n \rightarrow x$. Since $z_n \in Z(f)$, this shows that $x \in \overline{Z(f)}$ if $f(x) \neq 0$. On the other hand, if $f(x) = 0$, then $x \in Z(f) \subset \overline{Z(f)}$. Thus, in any case, $x \in \overline{Z(f)}$; hence $\overline{Z(f)} = X$. This proves the result. ■

PROBLEM 1.2.9 Let X be a normed linear space over $\mathbb{K}$ and $S = \{x \in X : \|x\| \leq 1\}$. Suppose that $g : S \rightarrow \mathbb{K}$ is a mapping such that

$$g(kx + y) = kg(x) + g(y) \quad (4)$$

whenever x, y and $kx + y$ are in S and k is in $\mathbb{K}$. Show that g can be extended uniquely to a linear functional f on X . Show also that f is continuous iff g is continuous.

Solution Define $f : X \rightarrow \mathbb{K}$ by

$$f(0) = 0, \quad f(x) = \|x\| g\left(\frac{x}{\|x\|}\right) \quad \text{if } x \neq 0.$$

Choosing $x = y = 0$ and $k = 1$ in (4) we see that $g(0) = g(0) + g(0)$. Hence, $g(0) = 0 = f(0)$. If $x \in S$ and $x \neq 0$, then choosing $k = \frac{1}{\|x\|}$, $y = 0$ in (4), we get

$$\frac{1}{\|x\|}f(x) = g\left(\frac{1}{\|x\|}x + 0\right) = \frac{1}{\|x\|}g(x) + g(0).$$

Hence $f(x) = g(x)$ for all $x \in S$. Thus, f is an extension of g .

If x and kx are in S , then

$$g(kx) = g(kx + 0) = kg(x) + g(0) = kg(x).$$

Hence for $k \neq 0, x \neq 0$, we have

$$g\left(\frac{kx}{\|kx\|}\right) = g\left(\frac{k}{|k|} \cdot \frac{x}{\|x\|}\right) = \frac{k}{|k|}g\left(\frac{x}{\|x\|}\right).$$

This means that

$$\frac{1}{\|kx\|}f(kx) = \frac{k}{|k|} \frac{1}{\|x\|}f(x), \quad f(kx) = kf(x).$$

This is true when $k = 0$ or $x = 0$ also.

Choose $k \neq 0$ such that kx, ky and $k(x + y)$ are in S . For this k we have

$$f(k(x + y)) = g(k(x + y)) = g(kx) + g(ky) = f(kx) + f(ky),$$

and hence,

$$kf(x + y) = kf(x) + kf(y).$$

Thus, $f(x + y) = f(x) + f(y)$. It follows that $f : X \rightarrow \mathbb{K}$ is linear.

If f_1 is any linear extension of g to X , then $f_1(0) = 0$, and for $x \neq 0$,

$$f_1(x) = \|x\|f_1\left(\frac{x}{\|x\|}\right) = \|x\|g\left(\frac{x}{\|x\|}\right) = f(x),$$

since $x/\|x\|$ is in S . This proves the uniqueness.

If f is continuous on X , then its restriction g to S is continuous. For the converse, let g be continuous on S and $x_n \rightarrow x$ in X . Then $\|x_n - x\| < 1$ for all large n . For such n ,

$$f(x_n - x) = g(x_n - x) \rightarrow g(0) = 0,$$

$$f(x_n) - f(x) \rightarrow 0.$$

This shows that f is continuous and completes the proof. ■

PROBLEM 1.2.10 Show that even though $F : X \rightarrow Y$ is a continuous bijective linear map, its inverse need not be continuous.

Solution Let $X = l^\infty$ and $y = F(x)$, where

$$y(n) = \frac{1}{n}x(n), \quad n = 1, 2, \dots, x \in X.$$

Then $F : X \rightarrow X$ is linear. If $F(x) = 0$,

$$\frac{1}{n}x(n) = F(x)(n) = 0 \quad \text{for all } n;$$

hence $x = 0$. Thus F is one-one. Let $Y = R(F)$, the range of F . Then $F : X \rightarrow Y$ is bijective, and

$$|F(x)(n)| = \frac{1}{n}|x(n)| \leq \frac{1}{n}\|x\|_\infty \leq \|x\|_\infty$$

for all n . So, $\|F(x)\|_\infty \leq \|x\|_\infty$. Therefore, F is continuous, with $\|F\| \leq 1$.

Let $G = F^{-1}$. Suppose that G is continuous. Then there is $\alpha > 0$ such that $\|G(y)\|_\infty \leq \alpha\|y\|_\infty$ for all $y \in Y$. Let

$$x_n(j) = \begin{cases} j & \text{for } 1 \leq j \leq n, \\ 0 & \text{for } j > n. \end{cases}$$

Then, $x_n \in l^\infty$ and $\|x_n\|_\infty = n$. If $y_n = F(x_n)$, then

$$\begin{aligned} y_n(j) &= F(x_n)(j) = 1 && \text{for } 1 \leq j \leq n, \\ y_n(j) &= 0 && \text{for } j > n. \end{aligned}$$

Thus, $y_n \in Y$, $\|y_n\|_\infty = 1$ and $G(y_n) = x_n$. Therefore,

$$n = \|x_n\|_\infty = \|G(y_n)\|_\infty \leq \alpha\|y_n\|_\infty = \alpha.$$

This gives a contradiction since n can be arbitrarily large. This proves the result. ■

PROBLEM 1.2.11 Let $T : C[0, 1] \rightarrow C[0, 1]$ be defined by $y = T(x)$, where

$$y(t) = \int_0^t x(u) du.$$

Find $R(T)$ and $T^{-1} : R(T) \rightarrow C[0, 1]$. Is T^{-1} linear? Bounded?

Solution Let $y = T(x)$. Then $y(0) = 0$, y is differentiable and $y'(t) = x(t)$ for all $t \in [0, 1]$. Thus,

$$R(T) = \{y \in C^1[0, 1] : y(0) = 0\},$$

where $C^1[0, 1]$ is the space of all continuously differentiable functions on $[0, 1]$. If $T(x_1) = y = T(x_2)$, then $x_1(t) = y'(t) = x_2(t)$ for $0 \leq t \leq 1$. This shows that T is one-one. Since T is linear we see that T^{-1} also is linear.

Let $x_n(t) = nt^{n-1}$ and $y_n = T(x_n)$. Then $y_n(t) = t^n$. Since the norm on $C[0, 1]$ is the sup norm $\|\cdot\|_\infty$,

$$\|x_n\|_\infty = n, \quad \|y_n\|_\infty = 1.$$

If $S = T^{-1} : R(T) \rightarrow C[0, 1]$ is bounded, then there is $\alpha > 0$ such that

$$\|x_n\|_\infty = \|S(y_n)\|_\infty \leq \alpha \|y_n\|_\infty = \alpha,$$

that is $n \leq \alpha$ for all n . This is not possible; so T^{-1} is not bounded. ■

PROBLEM 1.2.12 Define $S, T : C[0, 1] \rightarrow C[0, 1]$ by

$$(Sx)(s) = s \int_0^1 x(t) dt, \quad (Tx)(s) = sx(s) \quad \text{for } 0 \leq s \leq 1.$$

Do S and T commute? Find $\|S\|, \|T\|, \|ST\|$ and $\|TS\|$.

Solution Let $x(t) = 1$ for $0 \leq t \leq 1$, $y = S(x)$ and $z = T(x)$. Then,

$$y(s) = s \int_0^1 x(t) dt = s, \quad 0 \leq s \leq 1;$$

$$z(s) = sx(s) = s, \quad 0 \leq s \leq 1.$$

Hence for $0 \leq s \leq 1$,

$$(TSx)(s) = (Ty)(s) = sy(s) = s^2,$$

$$(STx)(s) = (Sz)(s) = s \int_0^1 z(t) dt = s \int_0^1 t dt = \frac{s}{2}.$$

Therefore, $ST \neq TS$.

Since the norm in $C[0, 1]$ is the sup norm $\|\cdot\|_\infty$, we have $|x(t)| \leq \|x\|$ for all t , and hence for $x \in C[0, 1]$, $0 \leq s \leq 1$,

$$|S(x)(s)| \leq s \int_0^1 |x(t)| dt \leq s\|x\| \leq \|x\|,$$

$$|Tx(s)| = s|x(s)| \leq s\|x\| \leq \|x\|.$$

Thus, $\|Sx\| \leq \|x\|, \|Tx\| \leq \|x\|$, and hence

$$\|TSx\| \leq \|Sx\| \leq \|x\|.$$

It follows that $\|S\| \leq 1, \|T\| \leq 1$ and $\|TS\| \leq 1$. Again, if $z = T(x)$ and $x \in C[0, 1]$ then, for $0 \leq s \leq 1$,

$$|z(s)| = s|x(s)| \leq s\|x\|,$$

$$|STx(s)| = |Sz(s)| \leq s \int_0^1 |z(t)| dt \leq \int_0^1 t\|x\| dt = \frac{1}{2}\|x\|;$$

$$\|STx\| \leq \frac{1}{2}\|x\|.$$

Thus, $\|ST\| \leq \frac{1}{2}$. If $x(t) = 1$ for all t , then $\|x\| = 1$, and

$$\begin{aligned}(Sx)(s) &= s, & (Tx)(s) &= s, \\ (TSx)(s) &= s^2, & (STx)(s) &= \frac{s}{2},\end{aligned}$$

as seen above. So,

$$\|Sx\| = \|Tx\| = \|TSx\| = 1 = \|x\|, \quad \|STx\| = \frac{1}{2} = \frac{1}{2}\|x\|.$$

It follows that

$$\|S\| \geq 1, \quad \|T\| \geq 1, \quad \|TS\| \geq 1, \quad \|ST\| \geq \frac{1}{2}.$$

Therefore, $\|S\| = 1$, $\|T\| = 1$, $\|TS\| = 1$, $\|ST\| = \frac{1}{2}$. ■

PROBLEM 1.2.13 Let $X = \mathbb{R}^2$ with the Euclidean norm $\|\cdot\|_2$ and $A \in BL(X)$ be represented by the matrix $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Show that $\|A\| = \frac{1}{2}(\sqrt{\alpha} + \sqrt{\beta})$, where

$$\alpha = a^2 + b^2 + c^2 + d^2 + 2(ad - bc)\beta = a^2 + b^2 + c^2 + d^2 - 2(ad - bc).$$

Solution Let $x \in \mathbb{R}^2$, $\|x\| = 1$ and $A(x) = y$. Then, $x = (\cos \theta, \sin \theta)$ for some real θ and

$$y(1) = a \cos \theta + b \sin \theta, \quad y(2) = c \cos \theta + d \sin \theta.$$

Hence,

$$\begin{aligned}\|Ax\|^2 &= (y(1))^2 + (y(2))^2 \\ &= (a^2 + c^2) \cos^2 \theta + (b^2 + d^2) \sin^2 \theta + 2(ab + cd) \cos \theta \sin \theta \\ &= \frac{a^2 + c^2 + b^2 + d^2}{2} + \frac{a^2 + c^2 - b^2 - d^2}{2} \cos(2\theta) + (ab + cd) \sin(2\theta) \\ &= \frac{\alpha + \beta}{4} + r \cos(2\theta - \phi)\end{aligned}\tag{5}$$

where α, β are given by (1.2.13) and

$$r \cos \phi = \frac{1}{2}(a^2 + c^2 - b^2 - d^2), \quad r \sin \phi = ab + cd.$$

So,

$$4r^2 = (a^2 + c^2 - b^2 - d^2)^2 + 4(ab + cd)^2$$

$$\begin{aligned}
&= (a^2 + c^2 + b^2 + d^2)^2 - 4(a^2 + c^2)(b^2 + d^2) + 4(ab + cd)^2 \\
&= (a^2 + c^2 + b^2 + d^2)^2 - 4(ad - bc)^2 \\
&= \alpha\beta.
\end{aligned}$$

Hence from (5),

$$\|Ax\|^2 = \frac{\alpha + \beta}{4} + \frac{\sqrt{\alpha\beta}}{2} \cos(2\theta - \phi) \leq \left(\frac{\sqrt{\alpha} + \sqrt{\beta}}{2} \right)^2,$$

where equality holds when $\theta = (1/2)\phi$. It follows that

$$\|A\| = \sup_{\|x\|=1} \|Ax\| = \frac{\sqrt{\alpha} + \sqrt{\beta}}{2}.$$

This completes the proof. ■

PROBLEM 1.2.14 Let X be the normed space of all bounded real-valued functions on $\mathbb{R}$ with norm defined by

$$\|x\| = \sup \{|x(t)|, t \in \mathbb{R}\}, \quad x \in X.$$

Let $T : X \rightarrow X$ be defined by

$$Tx(t) = x(t - \Delta)$$

where $\Delta > 0$ is a constant. Is T linear? Bounded?

Solution Given that $Tx(t) = x(s)$, where $s = t - \Delta$. If $z = kx + y$, where $k \in \mathbb{K}$, $x, y \in X$, then

$$\begin{aligned}
z(s) &= kx(s) + y(s), \\
Tz(t) &= kTx(t) + Ty(t), \quad T(z) = kT(x) + T(y).
\end{aligned}$$

Thus T is linear. If $y = T(x)$, then for all $t \in \mathbb{R}$,

$$|y(t)| = |x(t - \Delta)| \leq \|x\|, \quad |x(t)| = |y(t + \Delta)| \leq \|y\|.$$

It follows that $\|y\| = \|x\|$. Therefore, T is bounded and $\|T\| = 1$. ■

PROBLEM 1.2.15 Let X and Y be normed linear spaces over $\mathbb{K} = \mathbb{R}$. Suppose that $F : X \rightarrow Y$ is a mapping such that

$$F(x + y) = F(x) + F(y) \quad \text{for all } x, y \in X.$$

If, further, F is bounded on some ball

$$U(a, r) = \{x \in X : \|x - a\| < r\}, \quad r > 0,$$

show that F is continuous and linear. Show also that this result may be false when $\mathbb{K} = \mathbb{C}$.

Solution Since

$$F((n+1)x) = F(nx+x) = F(nx) + F(x),$$

it follows by induction that $F(nx) = nF(x)$ for all positive integers n and $x \in X$. Since $F(0) = F(0+0) = F(0) + F(0)$, we get $F(0) = 0$. So,

$$0 = F(x + (-x)) = F(x) + F(-x),$$

$$F(-x) = -F(x).$$

It follows that $F(nx) = nF(x)$ for all integers n and $x \in X$. If m and n are integers and $m > 0$ then, for all $x \in X$,

$$nF(x) = F(nx) = F\left(m \frac{n}{m}x\right) = mF\left(\frac{n}{m}x\right).$$

Thus $F(px) = pF(x)$ for all rational numbers p and $x \in X$.

Since F is bounded on $U(a, r)$, there is $M > 0$ such that $\|F(x)\| \leq M$ for all x in $U(a, r)$. Let $x \in X, x \neq 0$. Choose a rational number p such that

$$\frac{r}{2\|x\|} < p < \frac{r}{\|x\|}.$$

Then $\|px\| < r$. Thus, $px + a$ and a are in $U(a, r)$. Hence,

$$\|F(px)\| = \|F(px + a) - F(a)\| \leq \|F(px + a)\| + \|F(a)\|.$$

Therefore, $\|pF(x)\| = \|F(px)\| \leq 2M$, and so,

$$\|F(x)\| \leq \frac{1}{p}2M < \frac{2\|x\|}{r}2M.$$

Thus, $\|F(x)\| \leq \alpha\|x\|$ for some constant α . This is true even when $x = 0$. Hence,

$$\|F(x) - F(y)\| = \|F(x - y)\| \leq \alpha\|x - y\|$$

for all x, y in X . This shows that F is uniformly continuous on X .

Given $k \in \mathbb{R}$ we can find a sequence $\{p_n\}$ of rational numbers such that $p_n \rightarrow k$. So $p_n x \rightarrow kx$, and hence $F(p_n x) \rightarrow F(kx)$ by the continuity of F . But,

$$F(p_n x) = p_n F(x) \rightarrow kF(x).$$

Therefore, $F(kx) = kF(x)$. Thus F is linear.

Finally, if $X = Y = \mathbb{K} = \mathbb{C}$ and $F(z) = \bar{z}$, the complex conjugate of z , then F is not linear even though it has the properties stated in the problem. This completes the proof. ■

PROBLEM 1.2.16 Let X and Y be normed spaces and $F : X \rightarrow Y$ be linear. Show that the following statements are equivalent:

- (a) F is continuous.
- (b) F sends Cauchy sequences in X to Cauchy sequences in Y .
- (c) F sends convergent sequences in X to convergent sequences in Y .

Solution First suppose that F is continuous. If $\{x_n\}$ is a Cauchy sequence in X then

$$\|F(x_n) - F(x_m)\| = \|F(x_n - x_m)\| \leq \|F\| \|x_n - x_m\|.$$

This shows that $\{F(x_n)\}$ is a Cauchy sequence in Y .

If $x_n \rightarrow x$ in X , then

$$\|F(x_n) - F(x)\| = \|F(x_n - x)\| \leq \|F\| \|x_n - x\|.$$

This shows that $F(x_n) \rightarrow F(x)$ in Y . Thus we have proved that if (a) is true then (b) and (c) are true.

Next suppose that F is not continuous. Then, for $n = 1, 2, \dots$, there is x_n in X such that $\|x_n\| \leq 1$ and $\|F(x_n)\| > n$. Thus, $x_n \neq 0$. Let

$$y_n = \frac{x_n}{\|x_n\|}, \quad z_n = \frac{1}{\sqrt{n}} y_n.$$

Then, $\|y_n\| = 1$ and $\|z_n\| = 1/\sqrt{n} \rightarrow 0$. Hence, $z_n \rightarrow 0$ in X , but

$$\|F(z_n)\| = \frac{1}{\sqrt{n}} \|F(y_n)\| = \frac{1}{\sqrt{n} \|x_n\|} \|F(x_n)\| > \sqrt{n}.$$

Thus, $\{F(z_n)\}$ is not bounded. So $\{F(z_n)\}$ is neither a convergent nor a Cauchy sequence even though $z_n \rightarrow 0$. This shows that if (a) is false then (b) and (c) are false. This completes the proof. ■

PROBLEM 1.2.17 Let (k_{ij}) be a column-finite infinite matrix whose entries k_{ij} are scalars. For x in c_{00} , let $F(x) = y$, where

$$y(i) = \sum_{j=1}^{\infty} k_{ij} x(j), \quad i = 1, 2, \dots$$

Let $X = c_{00}$ with $\|\cdot\|_1$ and $Y = c_{00}$ with $\|\cdot\|_{\infty}$. Prove that $F : X \rightarrow Y$ is linear. Prove also that if $|k_{ij}| \leq \alpha$ for all $i, j \geq 1$ and some α in $\mathbb{R}$, then F is continuous.

Solution If $x \in c_{00}$, then

$$\sum_{j=1}^{\infty} k_{ij} x(j) = \sum_{j=1}^n k_{ij} x(j)$$

for some n , depending on x . Hence this sum exists for each $x \in c_{00}$. Given that each column of (k_{ij}) contains only a finite number of nonzero entries. Hence we can find m depending on n , such that

$$k_{ij} = 0 \quad \text{for } i > m, 1 \leq j \leq n.$$

Hence, if $y = F(x)$ then

$$y(i) = \sum_{j=1}^n k_{ij}x(j) = 0 \quad \text{for } i > m.$$

This shows that $y \in c_{00}$ and F maps c_{00} into itself.

If $z = kx + y$ where $k \in \mathbb{K}$ and $x, y \in c_{00}$, then

$$\begin{aligned} k_{ij}z(j) &= k_{ij}(kx(j) + y(j)), \\ \sum_{j=1}^{\infty} k_{ij}z(j) &= k \sum_{j=1}^{\infty} k_{ij}x(j) + \sum_{j=1}^{\infty} k_{ij}y(j), \\ F(z)(i) &= kF(x)(i) + F(y)(i), \quad i = 1, 2, \dots \end{aligned}$$

Therefore, $F(z) = kF(x) + F(y)$. Thus F is linear.

If $|k_{ij}| \leq \alpha$ for all i, j , then

$$\left| \sum_{j=1}^{\infty} k_{ij}x(j) \right| \leq \sum_{j=1}^{\infty} |k_{ij}| |x(j)| \leq \alpha \sum_{j=1}^{\infty} |x(j)|.$$

That is, $|F(x)(i)| \leq \alpha \|x\|_1$ for $i = 1, 2, \dots$. Hence,

$$\|F(x)\|_{\infty} = \sup \{|F(x)(i)| : i \geq 1\} \leq \alpha \|x\|_1.$$

This shows that F is continuous and $\|F\| \leq \alpha$. ■

PROBLEM 1.2.18 Let E be a measurable subset of $\mathbb{R}$ and

$$X = \{x \in L^2(E) : tx(t) \in L^2(E)\}.$$

Define $F : X \rightarrow L^2(E)$ by

$$F(x)(t) = tx(t), \quad t \in E, x \in X.$$

Prove that if $E = [a, b]$ then F is continuous and if $E = \mathbb{R}$ then F is discontinuous.

Solution If $x, y \in X$ and $k \in \mathbb{K}$, let $z = kx + y$. Then $F(x), F(y)$ are in $L^2(E)$, and

$$\begin{aligned} tz(t) &= t(kx(t) + y(t)) = ktx(t) + ty(t), \\ F(z)(t) &= kF(x)(t) + F(y)(t), \quad t \in E. \end{aligned}$$

Thus $z \in X$ and

$$F(z) = kF(x) + F(y) \in L^2(E).$$

This shows that X is a subspace of $L^2(E)$ and that F is linear.

Let $E = [a, b]$ and α be such that $|t| \leq \alpha$ for $t \in E$. If $x \in X$ and $y = F(x)$, then

$$\begin{aligned} |y(t)| &= |tx(t)| \leq \alpha|x(t)|, \\ \|y\|_2^2 &= \int_a^b |y(t)|^2 dt \leq \alpha^2 \int_a^b |x(t)|^2 dt = \alpha^2 \|x\|_2^2, \\ \|F(x)\|_2 &= \|y\|_2 \leq \alpha \|x\|_2. \end{aligned}$$

This shows that F is continuous.

Next let $E = \mathbb{R}$. Define x_n on $\mathbb{R}$ for $n = 1, 2, \dots$ by

$$x_n(t) = \begin{cases} t^{n-1}e^{-t/2} & \text{for } t > 0, \\ 0 & \text{for } t \leq 0. \end{cases}$$

Then

$$\begin{aligned} \int_{\mathbb{R}} |x_n(t)|^2 dt &= \int_0^\infty t^{2n-2} e^{-t} dt = \Gamma(2n-1) < \infty, \\ \int_{\mathbb{R}} |tx_n(t)|^2 dt &= \int_0^\infty t^{2n} e^{-t} dt = \Gamma(2n+1) < \infty. \end{aligned}$$

Therefore, $x_n \in L^2(E)$ and $tx_n(t) \in L^2(E)$. Thus, $x_n \in X$ for $n = 1, 2, \dots$. Further,

$$\begin{aligned} \|F(x_n)\|_2^2 &= \int_{\mathbb{R}} |F(x_n)(t)|^2 dt \\ &= \int_{\mathbb{R}} |tx_n(t)|^2 dt = \Gamma(2n+1), \\ \|F(x_n)\|_2 &= (2n(2n-1)\Gamma(2n-1))^{1/2} = (2n(2n-1))^{1/2} \|x_n\|. \end{aligned}$$

This holds for all n . Hence F cannot be bounded. ■

PROBLEM 1.2.19 Given a $m \times n$ matrix (k_{ij}) , define $F : \mathbb{K}^n \rightarrow \mathbb{K}^m$ by

$$F(x)(i) = \sum_{j=1}^n k_{ij}x(j), \quad 1 \leq i \leq m.$$

Let

$$\beta = \max_{1 \leq i \leq m} \sum_{j=1}^n |k_{ij}|, \quad \gamma = \max_{1 \leq j \leq n} \sum_{i=1}^m |k_{ij}|.$$

Suppose that $\mathbb{K}^n$ and $\mathbb{K}^m$ have norms $\|\cdot\|_p$ with the same p , $1 < p < \infty$. Prove that

$$\|F\| \leq \gamma^{1/p} \beta^{1/q},$$

where $1/p + 1/q = 1$. Deduce that if $n = m$ and (k_{ij}) is a diagonal matrix then

$$\|F\| = \max_{1 \leq i \leq m} |k_{ii}|.$$

Solution If $x \in \mathbb{K}^n$ then, for $1 \leq i \leq m$, we have

$$\begin{aligned} |F(x)(i)| &\leq \sum_{j=1}^n |k_{ij}| |x(j)| = \sum_{j=1}^n |k_{ij}|^{1/p} |x(j)| |k_{ij}|^{1/q} \\ &\leq \left(\sum_{j=1}^n |k_{ij}| |x(j)|^p \right)^{1/p} \left(\sum_{j=1}^n |k_{ij}| \right)^{1/q} \\ &\leq \left(\sum_{j=1}^n |k_{ij}| |x(j)|^p \right)^{1/p} \beta^{1/q}, \end{aligned}$$

using Holder's inequality. Hence,

$$\begin{aligned} \|F(x)\|_p^p &= \sum_{i=1}^m |F(x)(i)|^p \leq \sum_{i=1}^m \sum_{j=1}^n |k_{ij}| |x(j)|^p \beta^{p/q} \\ &= \beta^{p/q} \sum_{j=1}^n |x(j)|^p \sum_{i=1}^m |k_{ij}| \leq \beta^{p/q} \sum_{j=1}^n |x(j)|^p \gamma; \\ \|F(x)\|_p &\leq \beta^{1/q} \|x\|_p \gamma^{1/p}. \end{aligned}$$

This shows that $\|F\| \leq \beta^{1/q} \gamma^{1/p}$.

If $n = m$ and $k_{ij} = 0$ for $i \neq j$, then

$$\beta = \max_{1 \leq i \leq m} |k_{ii}| = \gamma.$$

So $\|F\| \leq \beta$ by what has been proved above. Moreover, $\beta = |k_{rr}|$ for some r . Let $\beta \neq 0$, $x(j) = 0$ for $j \neq r$ and $x(r) = \beta/k_{rr}$. Then, $|x(r)| = 1$, $\|x\|_p = 1$, and

$$\begin{aligned} F(x)(i) &= k_{ir} x(r) = 0 \quad \text{for } i \neq r, \\ F(x)(r) &= k_{rr} x(r) = \beta. \end{aligned}$$

Thus, $\|F(x)\|_p = \beta$, and so

$$\beta \leq \|F\| \cdot \|x\|_p = \|F\|.$$

Since this is true when $\beta = 0$ also, we get $\|F\| = \beta$. This completes the proof. ■

PROBLEM 1.2.20 Let $1 \leq p < \infty$ and $F : l^p \rightarrow l^p$ be a bounded linear map. Prove that F is represented by an infinite matrix (k_{ij}) in the sense that

$$F(x)(i) = \sum_{j=1}^{\infty} k_{ij} x(j), \quad i = 1, 2, \dots, x \in l^p.$$

Solution Let e_j denote the j -th unit vector, that is, $e_j = (0, \dots, 1, 0, \dots)$, where 1 occurs at the j -th place and 0 elsewhere. Let $F(e_j)(i) = k_{ij}$ for $i, j = 1, 2, \dots$. For $x \in l^p$, let

$$x_n = \sum_{j=1}^n x(j)e_j = (x(1), \dots, x(n), 0, 0, \dots).$$

Then, $x(j) - x_n(j) = 0$ for $j \leq n$, and $x(j) - x_n(j) = x(j)$ for $j > n$. Hence,

$$\|x - x_n\|_p^p = \sum_{j=n+1}^{\infty} |x(j)|^p \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

since $x \in l^p$. Thus, $x_n \rightarrow x$ in l^p . So $F(x_n) \rightarrow F(x)$ in l^p . Since

$$\begin{aligned} |F(x_n)(i) - F(x)(i)|^p &\leq \sum_{j=1}^{\infty} |F(x_n)(j) - F(x)(j)|^p \\ &= \|F(x_n) - F(x)\|_p^p \\ &\rightarrow 0 \quad \text{as } n \rightarrow \infty, \end{aligned}$$

we see that $F(x_n)(i) \rightarrow F(x)(i)$ as $n \rightarrow \infty$, for $i = 1, 2, \dots$. However,

$$\begin{aligned} F(x_n) &= \sum_{j=1}^n x(j)F(e_j), \\ F(x_n)(i) &= \sum_{j=1}^n x(j)k_{ij}. \end{aligned}$$

Therefore,

$$F(x)(i) = \lim_{n \rightarrow \infty} F(x_n)(i) = \lim_{n \rightarrow \infty} \sum_{j=1}^n k_{ij}x(j).$$

This proves the result. ■

PROBLEM 1.2.21 Let $F : c_0 \rightarrow c_0$ be a bounded linear map. Prove that F is represented by an infinite matrix (k_{ij}) in the sense that

$$F(x)(i) = \sum_{j=1}^{\infty} k_{ij}x(j), \quad i = 1, 2, \dots, x \in c_0.$$

Solution We proceed as in Problem 1.2.20. Let $k_{ij} = F(e_j)(i)$ for $i \geq 1, j \geq 1$. For $x \in c_0$, let

$$x_n = \sum_{j=1}^n x(j)e_j.$$

Since $x(j) - x_n(j) = 0$ for $j \leq n$ and $x(j) - x_n(j) = x(j)$ for $j > n$, we get

$$\|x - x_n\|_\infty = \sup_{j>n} |x(j)|.$$

This tends to 0 as $n \rightarrow \infty$ since $x \in c_0$. Therefore, $x_n \rightarrow x$ in c_0 and so $F(x_n) \rightarrow F(x)$ in c_0 . Thus,

$$|F(x_n)(i) - F(x)(i)| \leq \|F(x_n) - F(x)\|_\infty \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Since

$$F(x_n)(i) = \sum_{j=1}^n x(j)k_{ij},$$

we get

$$F(x)(i) = \sum_{j=1}^{\infty} k_{ij}x(j).$$

This completes the proof. ■

PROBLEM 1.2.22 Show that there are bounded linear maps $F : c \rightarrow c$ which cannot be represented by an infinite matrix (k_{ij}) in the sense that, for each $x \in c$,

$$F(x)(i) = \sum_{j=1}^{\infty} k_{ij}x(j), \quad i = 1, 2, \dots,$$

the series being convergent for all i, x .

Solution For $x \in c$ let $f(x) = \lim_{n \rightarrow \infty} x(n)$, and

$$F(x) = (f(x), f(x), \dots)$$

That is, $F(x)(i) = f(x)$ for all $i \geq 1$. Then f is a bounded linear functional on c by Problem 1.2.6. So F is linear. It is continuous since

$$\|F(x)\|_\infty = \sup_{i \geq 1} |F(x)(i)| = |f(x)| \leq \|f\| \cdot \|x\|_\infty.$$

Suppose that, for $x \in c$, $i = 1, 2, \dots$,

$$F(x)(i) = \sum_{j=1}^{\infty} k_{ij}x(j).$$

If $x = e_m$, this gives

$$F(x)(i) = k_{im}.$$

But $F(x)(i) = f(x) = \lim_{n \rightarrow \infty} e_m(n) = 0$. Hence, $k_{im} = 0$ for all $i, m \geq 1$. So $F(x)(i) = 0$ for all i, x . However, if $x = (1, 1, 1, \dots)$, then $x \in c$ and

$$F(x)(i) = f(x) = \lim_{n \rightarrow \infty} x(n) = 1.$$

This contradiction shows that there is no matrix (k_{ij}) representing F . ■

1.3 Equivalent Norms, Riesz Lemma, Finite Dimensional Spaces

Two norms $\|\cdot\|_1$ and $\|\cdot\|_2$ on a linear space X are said to be equivalent if there exist constants $\alpha > 0, \beta > 0$ such that

$$\alpha\|x\|_1 \leq \|x\|_2 \leq \beta\|x\|_1 \quad \text{for all } x \in X.$$

In this case,

- (i) $x_n \rightarrow x$ w.r.t. $\|\cdot\|_1$ iff $x_n \rightarrow x$ w.r.t. $\|\cdot\|_2$,
- (ii) $\{x_n\}$ is Cauchy w.r.t. $\|\cdot\|_1$ iff $\{x_n\}$ is Cauchy w.r.t. $\|\cdot\|_2$,
- (iii) $E \subset X$ is open w.r.t. $\|\cdot\|_1$ iff E is open w.r.t. $\|\cdot\|_2$.

The Heine-Borel theorem states that if $Y = \mathbb{K}^n$ with the Euclidean norm $\|\cdot\|_2$, then closed bounded subsets of Y are compact. Finite dimensional spaces have many interesting properties. Finite dimensional subspaces of a normed space are closed. (See Problem 1.3.6.) Any two norms on a finite dimensional linear space are equivalent. (See Problem 1.3.9.)

Riesz's Lemma states that if Y is a closed proper subspace of a normed space X and $0 < r < 1$ then there is $x_r \in X$ such that $\|x_r\| = 1$ and $r \leq d(x_r, Y) \leq 1$.

EXERCISES

1. Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be equivalent norms on a linear space X . Show that
 - (a) $x_n \rightarrow x$ w.r.t. $\|\cdot\|_1$ iff $x_n \rightarrow x$ w.r.t. $\|\cdot\|_2$.
 - (b) a subset E of X is open w.r.t. $\|\cdot\|_1$ iff it is open w.r.t. $\|\cdot\|_2$.
2. Let $X = \mathbb{R}^2$ with norm $\|\cdot\|_\infty$ and Y be the subspace of X consisting of all $y = (y(1), y(2)) \in \mathbb{R}^2$ with $y(1) + y(2) = 0$. Show that if $x_1 = (1, 1)$ then

$$d(x_1, Y) = 1, \quad \|x_1\| = 1.$$

PROBLEMS

PROBLEM 1.3.1 Let $X_1, \dots, X_n$ be normed spaces. Show that the following give equivalent norms on $X_1 \times \dots \times X_n$:

$$\|(x_1, \dots, x_n)\|_1 = \sum_{i=1}^n \|x_i\|, \quad x_i \in X_i;$$

$$\|(x_1, \dots, x_n)\|_2 = \left(\sum_{i=1}^n \|x_i\|^2 \right)^{1/2}, \quad x_i \in X_i;$$

$$\|(x_1, \dots, x_n)\|_\infty = \max_{1 \leq i \leq n} \|x_i\|, \quad x_i \in X_i.$$

Solution Let y denote $(x_1, \dots, x_n) \in X_1 \times \dots \times X_n$ with $x_i \in X_i$. It is clear that

$$\begin{aligned} \left(\sum_{i=1}^n \|x_i\| \right)^2 &\geq \sum_{i=1}^n \|x_i\|^2, \\ \frac{1}{n} \sum_{i=1}^n \|x_i\|^2 &\geq \left(\frac{1}{n} \sum_{i=1}^n \|x_i\| \right)^2. \end{aligned}$$

Hence,

$$\begin{aligned} \frac{1}{n} \left(\sum_{i=1}^n \|x_i\| \right)^2 &\leq \sum_{i=1}^n \|x_i\|^2 \leq \left(\sum_{i=1}^n \|x_i\| \right)^2, \\ \frac{1}{\sqrt{n}} \|y\|_1 &\leq \|y\|_2 \leq \|y\|_1, \quad \|y\|_2 \leq \|y\|_1 \leq \sqrt{n} \|y\|_2. \end{aligned}$$

Thus, $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent.

Since $\max_{1 \leq i \leq n} \|x_i\| = \|x_j\|$ for some j , we see that

$$\begin{aligned} \|x_j\| &\leq \sum_{i=1}^n \|x_i\| \leq n \|x_j\|, \\ \|y\|_\infty &\leq \|y\|_1 \leq n \|y\|_\infty. \end{aligned}$$

This shows that $\|\cdot\|_1$ and $\|\cdot\|_\infty$ are equivalent. ■

PROBLEM 1.3.2 If $1 \leq p < \infty$ and $x \in \mathbb{K}^n$, let

$$\|x\|_p = \left(\sum_{j=1}^n |x(j)|^p \right)^{1/p}.$$

Prove that, for all $x \in \mathbb{K}^n$,

$$n^{-1/2} \|x\|_2 \leq \|x\|_p \leq \|x\|_2 \quad \text{if } p > 2 \quad (1)$$

$$\|x\|_2 \leq \|x\|_p \leq n^{1/2} \|x\|_2 \quad \text{if } 1 \leq p \leq 2. \quad (2)$$

Solution If $p > r$ and $r/p + 1/q = 1$ then, by Holder's inequality,

$$\sum_{j=1}^n |a_j b_j| \leq \left(\sum_{j=1}^n |a_j|^{p/r} \right)^{r/p} \left(\sum_{j=1}^n |b_j|^q \right)^{1/q}.$$

This with $a_j = |x(j)|^r$, $b_j = 1$ gives

$$\sum_{j=1}^n |x(j)|^r \leq \left(\sum_{j=1}^n |x(j)|^p \right)^{r/p} n^{1/q}.$$

Thus,

$$\begin{aligned}\|x\|_r &\leq \|x\|_p n^{1/qr} = \|x\|_p n^{1/r-1/p}; \\ n^{-1/r} \|x\|_r &\leq n^{-1/p} \|x\|_p.\end{aligned}$$

This holds if $p > r$. Hence, if $p > 2$,

$$n^{-1/2} \|x\|_2 \leq n^{-1/p} \|x\|_p \leq \|x\|_p,$$

and if $1 \leq p < 2$,

$$\begin{aligned}n^{-1/p} \|x\|_p &\leq n^{-1/2} \|x\|_2, \\ \|x\|_p &\leq n^{1/p-1/2} \|x\|_2 \leq n^{1/2} \|x\|_2,\end{aligned}$$

since $1/p \leq 1$. Thus we get the first inequality in (1) and the second inequality in (2).

If $p > 1$, $a > 0$ and $\phi(t) = (t+a)^p - t^p$, then

$$\phi'(t) = p(t+a)^{p-1} - pt^{p-1} > 0.$$

Hence $\phi(t) \geq \phi(0) = a^p$ for $t \geq 0$. Thus,

$$(t+a)^p \geq t^p + a^p \quad \text{if } a \geq 0, t \geq 0, p > 1.$$

Using this, it follows by induction on n that if $p > 1$,

$$\left(\sum_{j=1}^n |a_j| \right)^p \geq \sum_{j=1}^n |a_j|^p.$$

This, with $a_j = |x(j)|^r$ and p changed to p/r , shows that if $p > r$,

$$\left(\sum_{j=1}^n |x(j)|^r \right)^{p/r} \geq \sum_{j=1}^n |x(j)|^p = (\|x\|_p)^p.$$

Thus, $\|x\|_r \geq \|x\|_p$ if $p > r$. This readily gives the second inequality in (1) and the first inequality in (2). This completes the proof. ■

PROBLEM 1.3.3 For $x \in \mathbb{K}^n$, let

$$\|x\|_2 = \left(\sum_{j=1}^n |x(j)|^2 \right)^{1/2}$$

and $\|\cdot\|$ be any other norm on $\mathbb{K}^n$. Find constants $\alpha > 0, \beta > 0$ such that

$$\alpha \|x\|_2 \leq \|x\| \leq \beta \|x\|_2 \quad \text{for all } x \in \mathbb{K}^n.$$

Can we choose α and β which do not depend on the general norm $\|\cdot\|$?

Solution Let $S = \{x \in \mathbb{K}^n : \|x\|_2 = 1\}$. Then S is a compact subset of $\mathbb{K}^n$ by the Heine-Borel theorem. From the triangle inequality of norm we see that the real function ϕ given by $\phi(x) = \|x\|$ is continuous. Hence there exist x_1, x_2 in S such that

$$\phi(x_1) \leq \phi(x) \leq \phi(x_2) \quad \text{for all } x \in S.$$

Let $\alpha = \phi(x_1) = \|x_1\|$ and $\beta = \phi(x_2) = \|x_2\|$. Since $\|x_1\|_2 = \|x_2\|_2 = 1$, we see that $x_1 \neq 0, x_2 \neq 0$. Hence, $\alpha > 0, \beta > 0$. Let $x \in \mathbb{K}^n, x \neq 0$. Then, $(1/\|x\|_2)x \in S$, and hence

$$\alpha \leq \phi\left(\frac{1}{\|x\|_2}x\right) \leq \beta,$$

$$\alpha \leq \frac{\|x\|}{\|x\|_2} \leq \beta,$$

$$\alpha\|x\|_2 \leq \|x\| \leq \beta\|x\|_2.$$

This is true even when $x = 0$. This finishes the first part.

For the second part note that for any $\lambda > 0$,

$$\|x\| = \lambda\|x\|_2, \quad x \in \mathbb{K}^n,$$

defines a norm on $\mathbb{K}^n$. So if α and β exist as above, then

$$\alpha\|x\|_2 \leq \lambda\|x\|_2 \leq \beta\|x\|_2 \quad \text{for all } x;$$

$$\alpha \leq \lambda \leq \beta.$$

So α, β depend on λ . This shows that we cannot choose constants α, β independent of $\|\cdot\|$. ■

PROBLEM 1.3.4 Let $1 \leq r < p \leq \infty$. Show that $\|\cdot\|_p$ and $\|\cdot\|_r$ are not equivalent on c_{00} .

Solution For $n = 1, 2, \dots$, let

$$x_n(j) = \begin{cases} j^{-1/r} & \text{for } 1 \leq j \leq n, \\ 0 & \text{for } j > n. \end{cases}$$

Then $x_n \in c_{00}$,

$$(\|x_n\|_r)^r = \sum_{j=1}^n j^{-1},$$

$$(\|x_n\|_p)^p = \sum_{j=1}^n j^{-p/r} \quad \text{if } p < \infty,$$

and $\|x_n\|_\infty = 1$. Hence, $\|x_n\|_p \leq \lambda$ for all n , where

$$\lambda = \sum_{j=1}^{\infty} j^{-p/r} \quad \text{if } p < \infty,$$

and $\lambda = 1$ if $p = \infty$. If $\|\cdot\|_p$ and $\|\cdot\|_r$ are equivalent on c_{00} , then there exist $\alpha > 0, \beta > 0$ such that

$$\alpha\|x_n\|_p \leq \|x_n\|_r \leq \beta\|x_n\|_p \quad \text{for all } n,$$

hence,

$$\left(\sum_{j=1}^n \frac{1}{j}\right)^{1/r} = \|x_n\|_r \leq \beta\lambda \quad \text{for all } n.$$

This cannot be true since the left member of this tends to ∞ as $n \rightarrow \infty$, and λ is finite. This completes the proof. ■

PROBLEM 1.3.5 Show that two norms on a linear space are equivalent iff every Cauchy sequence w.r.t. one of the norms is a Cauchy sequence w.r.t. the other norm.

Solution Suppose that two norms $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent on a linear space X . Then there exist $\alpha > 0$ and $\beta > 0$ such that

$$\alpha\|x\|_1 \leq \|x\|_2 \leq \beta\|x\|_1 \quad \text{for all } x \in X.$$

So for a sequence $\{x_n\}$ in X , we have

$$\alpha\|x_n - x_m\|_1 \leq \|x_n - x_m\|_2 \leq \beta\|x_n - x_m\|_1 \quad (3)$$

for all $n, m \geq 1$. The second inequality in (3) shows that if $\{x_n\}$ is Cauchy w.r.t. $\|\cdot\|_1$, then it is Cauchy w.r.t. $\|\cdot\|_2$. The first inequality in (3) shows that if $\{x_n\}$ is Cauchy w.r.t. $\|\cdot\|_2$ then it is Cauchy w.r.t. $\|\cdot\|_1$.

For the converse, suppose that the norms are not equivalent. Then, either (i) we cannot find α such that $\alpha\|x\|_1 \leq \|x\|_2$ for all x , or (ii) we cannot find β such that $\|x\|_2 \leq \beta\|x\|_1$ for all x . In case (i), for $n = 1, 2, \dots$, there exist x_n in X such that

$$\frac{1}{n}\|x_n\|_1 > \|x_n\|_2.$$

Let $y_n = \frac{1}{\sqrt{n}} \frac{1}{\|x_n\|_2} x_n$. Then,

$$\|y_n\|_2 = \frac{1}{\sqrt{n}} \rightarrow 0,$$

$$\|y_n\|_1 = \frac{1}{\sqrt{n}} \frac{1}{\|x_n\|_2} \|x_n\|_1 > \sqrt{n} \rightarrow \infty.$$

So $\{y_n\}$ is Cauchy w.r.t. $\|\cdot\|_2$ but not w.r.t. $\|\cdot\|_1$. Similarly, in case (ii) there is a sequence $\{z_n\}$ which is Cauchy w.r.t. $\|\cdot\|_1$, but not w.r.t. $\|\cdot\|_2$. This completes the proof. ■

PROBLEM 1.3.6 Suppose that X_1 and X_2 are subspaces of a normed space X , where X_1 is closed and X_2 is finite dimensional. Show that $X_1 + X_2$ is closed in X . Deduce that finite dimensional subspaces of X are closed.

Solution Suppose that Y is closed in X and

$$Z = Y + \{kx_0 : k \in \mathbb{K}\} = \text{span}\{Y, x_0\}.$$

We shall prove that Z is closed in X . If $x_0 \in Y$ then $Z = Y$. So let $x_0 \notin Y$, and let $c = d(x_0, Y)$. Then $c > 0$ since Y is closed. If $k \neq 0$, then

$$\|y + kx_0\| = \|k(x_0 + \frac{1}{k}y)\| \geq |k|c \quad \text{for } y \in Y.$$

This is true even if $k = 0$. Let $\{z_n\}$ be a sequence in Z , and $z_n \rightarrow z$. Then, $z_n = y_n + k_n x_0, y_n \in Y$. Hence,

$$\|z_n - z_m\| = \|(y_n - y_m) + (k_n - k_m)x_0\| \geq |k_n - k_m|c.$$

Since $\{z_n\}$ converges, it is a Cauchy sequence. As $c > 0$, this shows that $\{k_n\}$ is a Cauchy sequence in $\mathbb{K}$. Hence, $\{k_n\}$ converges. If $k_n \rightarrow k$, then $y_n = z_n - k_n x_0 \rightarrow z - kx_0$. Since $y_n \in Y$ and Y is closed we see that $z - kx_0 \in Y$. Hence, $z = y + kx_0$, where $y = z - kx_0 \in Y$; so $z \in Z$. This proves that Z is closed.

If $X_2 = \text{span}\{x_1, \dots, x_n\}$, then repeated application of the above result with Y replaced by

$$X_1, X_1 + \text{span}\{x_1\}, X_1 + \text{span}\{x_1, x_2\}, \dots$$

shows that

$$X_1 + \text{span}\{x_1, \dots, x_n\} = X_1 + X_2$$

is closed. This proves the first part.

If $X_1 = \{0\}$, then X_1 is closed and $X_1 + X_2 = X_2$. Therefore, X_2 is closed if it is finite dimensional. This completes the proof. ■

PROBLEM 1.3.7 Suppose that $u_1, \dots, u_n$ are linearly independent vectors in a normed space X . Prove that there is $c > 0$ such that for all scalars k_i we have

$$\left\| \sum_{i=1}^n k_i u_i \right\| \geq c |k_j| \quad \text{for } 1 \leq j \leq n.$$

Solution If $n = 1$ we can take $c = \|u_1\|$. If $n > 1$, let

$$Y_i = \text{span}\{u_1, \dots, u_{i-1}, u_{i+1}, \dots, u_n\},$$

$$c_i = d(u_i, Y_i), \quad c = \min_{1 \leq i \leq n} c_i.$$

Then Y_i , being finite dimensional, is closed, and $u_i \notin Y_i$. So $c_i > 0$ and $c > 0$. If $k_j \neq 0$, let

$$y_j = - \sum_{i \neq j} \left(\frac{k_i}{k_j} \right) u_i.$$

Then, $y_j \in Y_j$, and hence

$$\left\| \sum_{i=1}^n k_i u_i \right\| = \|k_j u_j - k_j y_j\| \geq |k_j| |c_j| \geq |k_j| c.$$

This is true even when $k_j = 0$. This proves the result. ■

PROBLEM 1.3.8 Suppose that X and Y are normed spaces and X is finite dimensional. Prove that every linear map from X to Y is continuous.

Solution We shall first prove that if f is a linear functional on a normed space X and if its zero space $Z(f)$ is closed in X , then f is continuous. If f is not continuous then, for $n = 1, 2, \dots$, there exist x_n in X such that $\|x_n\| \leq 1$ and $|f(x_n)| > n$ for all n . Then, $f(x_n) \neq 0$ for any n . Let

$$z_n = x_1 - \frac{f(x_1)}{f(x_n)} x_n.$$

Then,

$$\begin{aligned} f(z_n) &= f(x_1) - \frac{f(x_1)}{f(x_n)} f(x_n) = 0, \\ \|z_n - x_1\| &= \left\| \frac{f(x_1)}{f(x_n)} x_n \right\| < \frac{|f(x_1)|}{n} \rightarrow 0. \end{aligned}$$

Thus, $z_n \in Z(f)$ and $z_n \rightarrow x_1$. So $x_1 \in \overline{Z(f)}$. But, $x_1 \notin Z(f)$ since $f(x_1) \neq 0$. Thus, we see that $Z(f)$ is not closed if f is not continuous.

Next let X be a normed space of finite dimension n , and $\{v_1, \dots, v_n\}$ be a basis for X . For $i = 1, \dots, n$, let $f_i : X \rightarrow \mathbb{K}$ be defined by

$$f_i \left(\sum_{j=1}^n k_j v_j \right) = k_i.$$

Then each f_i is a linear functional on X . Since $Z(f_i) \subset X$ we see that $Z(f_i)$ is finite dimensional and hence closed. So each f_i is continuous.

Let $F : X \rightarrow Y$ be linear. If

$$x = \sum_{j=1}^n k_j v_j,$$

then

$$|k_i| = |f_i(x)| \leq \|f_i\| \cdot \|x\|,$$

$$\begin{aligned}\|F(x)\| &= \left\| \sum_{i=1}^n k_i F(v_i) \right\| \leq \sum_{i=1}^n |k_i| \|F(v_i)\| \\ &\leq \sum_{i=1}^n \|f_i\| \|x\| \|F(v_i)\| = \alpha \|x\|,\end{aligned}$$

where

$$\alpha = \sum_{i=1}^n \|f_i\| \|F(v_i)\|.$$

This proves that F is continuous. ■

PROBLEM 1.3.9 Show that any two norms on a finite dimensional linear space are equivalent.

Solution Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be two norms on a finite dimensional linear space X . Let X_1 denote X with the norm $\|\cdot\|_1$ and X_2 denote X with the norm $\|\cdot\|_2$. Consider the identity map I on X given by $I(x) = x$. It is linear. So $I : X_1 \rightarrow X_2$ is continuous by Problem 1.3.8 since X_1 is finite dimensional. Hence there is $\alpha > 0$ such that $\|Ix\|_2 \leq \alpha\|x\|_1$ for all $x \in X_1$, that is,

$$\|x\|_2 \leq \alpha\|x\|_1, \quad x \in X. \quad (4)$$

Similarly, since $I^{-1} = I$ is a linear map from X_2 to X_1 , and hence continuous, there is $\beta > 0$ such that $\|Ix\|_1 \leq \beta\|x\|_2$ for all $x \in X_2$, that is,

$$\|x\|_1 \leq \beta\|x\|_2, \quad x \in X.$$

Combining this with (4), we get

$$\frac{1}{\beta}\|x\|_1 \leq \|x\|_2 \leq \alpha\|x\|_1 \quad \text{for all } x \in X.$$

This proves the result. ■

PROBLEM 1.3.10 If X is an infinite dimensional normed space, prove the following:

- (a) There is a one-one linear map $F : X \rightarrow X$ which is not continuous.
- (b) There is a linear functional on X which is not continuous.
- (c) There is a subspace of X which is not closed.
- (d) For any normed space $Y \neq \{0\}$ there is a linear map $F : X \rightarrow Y$ which is not continuous.

Solution Let B be a basis for X . Since B is an infinite set, we can choose an infinite sequence $\{u_n\}$ of distinct elements from B .

(a) Let $F(u_n) = nu_n$ and $F(u) = u$ for all u in B not belonging to $\{u_n\}$. Extend F to X by

$$F(x) = \sum_i k_i F(v_i) \quad \text{for } x = \sum_i k_i v_i, \quad v_i \in B.$$

This gives a linear map $F : X \rightarrow X$. If it is continuous, then $\|F(x)\| \leq \alpha\|x\|$ for x in X and some constant α . So,

$$n\|u_n\| = \|F(u_n)\| \leq \alpha\|u_n\| \quad \text{for all } n.$$

This shows that F is not continuous.

To show that F is one-one, let

$$F(x) = 0, \quad x = \sum k_i v_i, \quad v_i \in B, k_i \in X.$$

Then,

$$\sum_i k_i F(v_i) = F(x) = 0.$$

But $F(v_i) = \alpha_i v_i$, where $\alpha_i = n$ if $v_i = u_n$, and $\alpha_i = 1$ otherwise. Hence,

$$\sum k_i \alpha_i v_i = 0.$$

This implies that $k_i \alpha_i = 0$ for all i and so $k_i = 0$ since $\alpha_i \neq 0$. Therefore, $x = 0$ if $F(x) = 0$. So F is one-one.

(b) Define f on B by

$$\begin{aligned} f(u_n) &= n\|u_n\| \\ f(u) &= 1 \quad \text{for } u \in B, u \neq u_n; \end{aligned}$$

and extend f to X by

$$f(x) = \sum_i k_i f(v_i) \quad \text{for } x = \sum_i k_i v_i, \quad v_i \in B.$$

This gives a linear functional f on X . Since $|f(u_n)| = n\|u_n\|$ for all n , we see, as before, that it is not continuous.

(c) In the proof of Problem 1.3.8 we found that if the zero space of a linear functional f is closed, then f is continuous. So choose a linear functional f on X which is not continuous, and let Z be its zero space. Then Z is not closed.

(d) Let $y_0 \in Y, y_0 \neq 0$. Define F on B by

$$F(u_n) = n\|u_n\|y_0, \quad F(u) = y_0 \quad \text{if } u \in B, u \neq u_n,$$

and extend F to X as in (a). Then $F : X \rightarrow Y$ is linear but not continuous, as in (a), since

$$\|F(u_n)\| = n\|u_n\| \|y_0\| \quad \text{for all } n.$$

This completes the proof. ■

PROBLEM 1.3.11 Find subspaces of l^∞ and l^2 which are not closed.

Solution It is clear that c_{00} is a subspace of l^2 as well l^∞ . We shall prove that it is not closed in these spaces. For this let

$$x_n(j) = \begin{cases} \frac{1}{j} & \text{for } 1 \leq j \leq n, \\ 0 & \text{for } j > n, \end{cases}$$

$$x(j) = \frac{1}{j} \quad \text{for all } j \geq 1.$$

Then $x_n \in c_{00}$ for all n , and $x \in l^2 \subset l^\infty$. Since

$$x - x_n = (0, \dots, 0, \frac{1}{n+1}, \frac{1}{n+2}, \dots),$$

we see that

$$\|x - x_n\|_\infty = \frac{1}{n+1} \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

$$\|x - x_n\|_2^2 = \sum_{j=n+1}^{\infty} \frac{1}{j^2} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Hence, $x_n \rightarrow x$ in l^∞ and l^2 . Therefore, x belongs to the closure of c_{00} in l^∞ and in l^2 . However, $x \notin c_{00}$. So c_{00} is not closed in l^∞ and l^2 . ■

PROBLEM 1.3.12 Let X and Y be normed spaces, Y be finite dimensional and $F : X \rightarrow Y$ be linear and onto. Show that F is an open map; that is, F maps open subsets of X onto open subsets of Y . Show also that F is continuous iff its zero space $Z(F)$ is closed in X .

Solution Suppose that Y has dimension n and $\{v_1, \dots, v_n\}$ is a basis for Y . Let $v_i = F(u_i)$, $i = 1, 2, \dots, n$. Define $G : Y \rightarrow X$ by

$$G(y) = \sum_{i=1}^n k_i u_i \quad \text{for } y = \sum_{i=1}^n k_i v_i.$$

It is clear that G is well-defined and linear. It is continuous by Problem 1.3.8.

Let V be an open subset of X and $y_0 \in F(V)$. Then $y_0 = F(x_0)$ where $x_0 \in V$. Since V is open, there is $r > 0$ such that

$$U(x_0, r) = \{x \in X : \|x - x_0\| < r\} \subset V.$$

Let $\delta = r/\|G\|$. Suppose that $\|y - y_0\| < \delta$. Let

$$y = \sum_{i=1}^n k_i v_i, \quad y_0 = \sum_{i=1}^n a_i v_i$$

$$x = G(y), \quad x_1 = G(y_0).$$

Then,

$$x = \sum_{i=1}^n k_i u_i, \quad x_1 = \sum_{i=1}^n a_i u_i$$

$$F(x) = y, \quad F(x_1) = y_0.$$

Thus, $x - x_1 = G(y - y_0)$, and so

$$\|x - x_1\| = \|G(y - y_0)\| \leq \|G\| \|y - y_0\| < \|G\| \delta = r.$$

This shows that $x - x_1 + x_0 \in U(x_0, r) \subset V$. Since

$$F(x - x_1 + x_0) = F(x) - F(x_1) + F(x_0) = y,$$

we see that $y \in F(V)$ whenever $\|y - y_0\| < \delta$. This shows that y_0 is an interior point of $F(V)$. Thus $F(V)$ is open whenever V is open in X .

If F is continuous, then $Z(F) = F^{-1}\{0\}$ is closed since $\{0\}$ is closed in Y . Conversely, suppose that $Z = Z(F)$ is closed in X . Let

$$\tilde{F}(x + Z) = F(x), \quad x \in X.$$

Since F is onto this defines a bijective linear map $\tilde{F} : X/Z \rightarrow Y$. Hence X/Z is of dimension n . Since Z is closed, X/Z is a normed space under the quotient norm. Since X/Z is finite dimensional, $\tilde{F}$ is continuous by Problem 1.3.8. Hence

$$\|F(x)\| = \|\tilde{F}(x + Z)\| \leq \|\tilde{F}\| \cdot \|x + Z\| \leq \|\tilde{F}\| \|x\|,$$

since $\|x + Z\| \leq \|x\|$. This shows that F is continuous. ■

PROBLEM 1.3.13 Suppose that E is a subset of a normed space X . Show that if $\text{span } E \neq X$ then E^0 , the interior of E , is empty. Show also that if X is a finite dimensional normed space over $\mathbb{K} = \mathbb{R}$, E is a convex subset of X and $0 \in E$ then $\text{span } E = X$ iff E^0 is nonempty.

Solution Let $\text{span } E = Y$ and a be an interior point of E . Then there is $r > 0$ such that

$$\{x \in X : \|x - a\| < r\} \subset E \subset Y.$$

Let $x \in X$ and $x \neq 0$. As in the solution of Problem 1.1.2, we see that

$$y = a + \frac{r}{2\|x\|}x \in Y,$$

and hence $x \in Y$. Thus, $Y = X$ if E^0 is nonempty. This proves the first part.

For the second part, suppose that X is a real normed space of finite dimension n , E is a convex subset of X and $0 \in E$. If $E^0 \neq \emptyset$ then $\text{span } E =$

X by the first part. Conversely, suppose that $\text{span } E = X$. Then E contains n linearly independent elements $u_1, \dots, u_n$.

If $t_i \geq 0$ and $t_1 + \dots + t_n \leq 1$, then

$$\sum_{i=1}^n t_i u_i = t_1 u_1 + \dots + t_n u_n + (1 - \sum_{i=1}^n t_i) \cdot 0 \in E$$

since E is convex and $0 \in E$. Let

$$a = \frac{1}{2n} \sum_{i=1}^n u_i.$$

Then $a \in E$. By Problem 1.3.7, there exists $c > 0$ such that

$$\left\| \sum_{i=1}^n k_i u_i \right\| \geq c |k_j| \quad \text{for } 1 \leq j \leq n, \quad k_i \in \mathbb{K}.$$

Let $\delta = c/2n$ and $\|x - a\| < \delta$. If $x = \sum k_i u_i$, then

$$\|x - a\| = \left\| \sum_{i=1}^n \left(k_i - \frac{1}{2n}\right) u_i \right\| \geq c \left| k_j - \frac{1}{2n} \right|$$

for $1 \leq j \leq n$. Therefore,

$$c \left| k_j - \frac{1}{2n} \right| \leq \|x - a\| < \frac{c}{2n},$$

$$\left| k_j - \frac{1}{2n} \right| < \frac{1}{2n},$$

for $1 \leq j \leq n$. Since $k_j \in \mathbb{K} = \mathbb{R}$, this implies that $0 < k_j < 1/n$ for $1 \leq j \leq n$. So $\sum k_i < 1$, and hence $x \in E$ as observed above. Since this is true whenever $\|x - a\| < \delta$, we see that $a \in E^0$. Thus, E^0 is nonempty if $\text{span } E = X$. This completes the proof. ■

PROBLEM 1.3.14 Let $\{Y_n\}$ be a sequence of closed subspaces of X such that $Y_n \subset Y_{n+1}$ and $Y_n \neq Y_{n+1}$ for $n = 1, 2, \dots$. Show that there is a sequence $\{y_n\}$ in X with $y_n \in Y_n$, $\|y_n\| = 1$ and $d(y_{n+1}, Y_n) \geq 1/2$ for all n .

Solution Apply the Riesz lemma with $r = 1/2$ to the proper closed subspace Y_n of Y_{n+1} . There is $y_{n+1} \in Y_{n+1}$ such that $\|y_{n+1}\| = 1$ and $d(y_{n+1}, Y_n) \geq 1/2$. This proves the result. ■

PROBLEM 1.3.15 If the ball

$$U(0, r) = \{x \in X : \|x\| < r\}, \quad r > 0$$

in a normed space X is totally bounded, show that X is finite dimensional.

Solution If $U(0, r)$ is totally bounded, $r > 0$, then there exist a finite number of open balls $U(x_i, \frac{r}{4})$, $i = 1, \dots, n$, whose union contains $U(0, r)$. Suppose that $Y = \text{span}\{x_1, \dots, x_n\}$. Then Y , being finite dimensional, is closed. If $Y \neq X$, then by Riesz lemma there is x_0 in X such that $\|x_0\| = 1$ and $d(x_0, Y) \geq 3/4$. Then $(r/2)x_0 \in U(0, r)$, and hence $(r/2)x_0 \in U(x_i, r/4)$ for some i . So,

$$\left\| \frac{r}{2}x_0 - x_i \right\| < \frac{r}{4},$$

that is, $\|x_0 - (2/r)x_i\| < 1/2$. This gives a contradiction since $d(x_0, Y) \geq \frac{3}{4}$, and $(2/r)x_i \in Y$. This shows that $Y = X$. This proves the result. ■

PROBLEM 1.3.16 Show that any two normed spaces having the same finite dimension are linearly homeomorphic. Deduce the following:

- (a) A closed bounded subset of a finite dimensional space is compact.
- (b) A normed space is finite dimensional iff it is locally compact.

Solution Let X and Y be two normed spaces with the same finite dimension n , and having bases $\{x_1, \dots, x_n\}$ and $\{y_1, \dots, y_n\}$, respectively. Setting

$$F(x_i) = y_i, \quad i = 1, \dots, n$$

we get a linear map $F : X \rightarrow Y$. In fact, F is given by

$$F\left(\sum_{i=1}^n k_i x_i\right) = \sum_{i=1}^n k_i y_i, \quad k_i \in \mathbb{K}.$$

This map is clearly bijective. It is continuous by Problem 1.3.8. Thus $F : X \rightarrow Y$ is a surjective linear homeomorphism. This proves the first part.

(a) Let X be a normed space of finite dimension n and S be a closed bounded subset of X . Let $Y = \mathbb{K}^n$ with the Euclidean norm $\|\cdot\|_2$. There is a linear homeomorphism $F : X \rightarrow Y$. Let $F(S) = T$. Then T is closed since F^{-1} is continuous and S is closed. Since S is bounded and F is a bounded linear map we see that T is bounded. Hence T is compact by the Heine-Borel theorem. So $S = F^{-1}(T)$ is compact since F^{-1} is continuous.

(b) Let X be a finite dimensional normed space and $x_0 \in X$. By (a)

$$S = \{x \in X : \|x - x_0\| \leq 1\}$$

is a compact neighbourhood of x_0 . This shows that X is locally compact.

Conversely, suppose X is a locally compact normed space. Then there is a compact set E containing 0 in its interior. In that case, E is totally bounded and, moreover, there is $r > 0$ such that if

$$U(0, r) = \{x \in X : \|x\| < r\}$$

then $U(0, r) \subset E$. So $U(0, r)$ is totally bounded, and hence X is finite dimensional by Problem 1.3.15. This completes the proof. ■

PROBLEM 1.3.17 Let Y be a finite dimensional proper subspace of a normed space X . Prove that there is x_1 in X such that $\|x_1\| = 1$ and $d(x_1, Y) = 1$; that is, Riesz lemma holds with $r = 1$.

Solution Choose $x \in X$ with $x \notin Y$. Let

$$d = d(x, Y) = \inf \{\|x - y\| : y \in Y\}.$$

Let $r = \|x\| + d + 1$, and

$$S = \{y \in Y : \|y\| \leq r\}.$$

Then S is a closed bounded subset of the finite dimensional space Y . Hence S is compact. The function ϕ defined by

$$\phi(y) = \|y - x\|, \quad y \in Y,$$

is real and continuous on Y . Hence there is $y_0 \in S$ such that $\phi(y_0) \leq \phi(y)$ for all y in S . Again, there is $y_1 \in Y$ such that $\|x - y_1\| < d + 1$. Then,

$$\|y_1\| \leq \|y_1 - x\| + \|x\| < d + 1 + \|x\| = r.$$

Hence $y_1 \in S$, and so

$$\phi(y_0) \leq \phi(y_1) = \|y_1 - x\| < d + 1. \quad (5)$$

Let $y \in Y$. If $y \notin S$, then

$$\|x\| + d + 1 = r < \|y\| \leq \|y - x\| + \|x\|.$$

Combining this with (5), we get

$$\phi(y_0) < d + 1 < \|y - x\|.$$

On the other hand, if $y \in S$, then

$$\phi(y_0) \leq \phi(y) = \|y - x\|.$$

Thus,

$$\|y_0 - x\| = \phi(y_0) \leq \|y - x\| \quad \text{for all } y \in Y.$$

Hence,

$$\|x - y_0\| = \inf \{\|x - y\| : y \in Y\} = d.$$

If $d = 0$, then $x = y_0 \in Y$, which is not true. Therefore, $d > 0$. Let

$$x_1 = \frac{1}{d}(x - y_0).$$

Then, $\|x_1\| = 1$, and for all $y \in Y$, we have

$$\|x_1 - y\| = \left\| \frac{1}{d}(x - z) \right\| \geq 1,$$

since $z = y_0 + dy \in Y$. This implies that $d(x_1, Y) = 1$ since $\|x_1 - 0\| = 1$ and $0 \in Y$. This completes the proof. ■

PROBLEM 1.3.18 Let $1 \leq p < \infty$ and

$$E = \{x \in l^p : |x(n)| \leq n^{-2/p} \text{ for all } n \geq 1\}.$$

Show that:

- (a) E is a compact convex subset of l^p .
- (b) E is not contained in any finite dimensional subspace of l^p .
- (c) E^0 , the interior of E , is empty.

[If $p = 2$, E is called the *Hilbert cube*.]

Solution (a) If $x, y \in E$ and $0 < t < 1$ let $z = tx + (1-t)y$. Then,

$$z(n) = tx(n) + (1-t)y(n),$$

$$|z(n)| \leq t|x(n)| + (1-t)|y(n)| \leq tn^{-2/p} + (1-t)n^{-2/p} = n^{-2/p}.$$

This shows that $z \in E$. Thus E is convex.

To prove that E is compact it is enough if we prove that E is a complete metric space and that E is totally bounded.

First let $x_n \in E$ and $x_n \rightarrow x$ in l^p . Then,

$$|x_n(j) - x(j)|^p \leq \sum_{j=1}^{\infty} |x_n(j) - x(j)|^p = \|x_n - x\|_p^p,$$

$$\begin{aligned} |x(j)| &\leq |x(j) - x_n(j)| + |x_n(j)| \\ &\leq \|x_n - x\|_p + j^{-2/p}. \end{aligned}$$

Letting $n \rightarrow \infty$ we see that $|x(j)| \leq j^{-2/p}$ for all $j \geq 1$. Thus $x \in E$. This shows that E is closed in l^p . But l^p is complete. (See Problem 2.1.7 below.) Hence E is complete.

To show that E is totally bounded, let $\epsilon > 0$ be given. Find N such that

$$\sum_{j=N+1}^{\infty} \frac{1}{j^2} < \left(\frac{\epsilon}{2}\right)^p.$$

Fix N , and let

$$Y = \text{span}\{e_1, \dots, e_N\} = \{x \in l^p : x(j) = 0 \text{ for } j > N\}.$$

Then Y is a finite dimensional subspace of l^p . So Y is closed in l^p . If $x \in E$, then

$$\|x\|_p^p = \sum_{j=1}^{\infty} |x(j)|^p \leq \sum_{j=1}^{\infty} \frac{1}{j^2}.$$

Thus E is bounded. Hence $E \cap Y$, being closed and bounded, is compact. So there exist a finite number of points $y_1, \dots, y_m$ in $E \cap Y$ such that the open

balls $U(y_i, \epsilon/2)$, $1 \leq i \leq m$, cover $E \cap Y$. Let x be any element of E . Let $y(j) = x(j)$ for $1 \leq j \leq N$ and $y(j) = 0$ for $j > N$. Then $y \in E \cap Y$, and hence $y \in U(y_i, \epsilon/2)$ for some i . Since

$$\|x - y\|_p^p = \sum_{j=N+1}^{\infty} |x(j)|^p < \left(\frac{\epsilon}{2}\right)^p,$$

we see that

$$\|x - y_i\|_p \leq \|x - y\|_p + \|y - y_i\|_p < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Thus the open balls $U(y_i, \epsilon)$, $1 \leq i \leq m$ cover E . This proves that E is totally bounded.

(b) Let $x_n = n^{-2/p}e_n$, where e_n is the n -th unit vector. Then $x_n \in E$. Suppose that $k_1x_1 + \cdots + k_nx_n = 0$. Then

$$\sum_{j=1}^n k_j j^{-2/p} e_j = 0.$$

So, $k_j j^{-2/p} = 0$; $k_j = 0$. This shows that E contains an infinite number of linearly independent elements $x_1, x_2, \dots$. Therefore, no finite dimensional space can contain E .

(c) Suppose that E has an interior point a . Then E contains some open ball $U(a, \epsilon)$, $\epsilon > 0$. Choose N such that

$$\sum_{j=N+1}^{\infty} j^{-3/2} < \left(\frac{\epsilon}{2}\right)^p.$$

Let $x(j) = a(j)$ for $1 \leq j \leq N$ and $x(j) = j^{-3/2p}$ for $j > N$. Since $j^{-3/2p} > j^{-2/p}$ if $j > 1$, we see that $x \notin E$, and for $j > N$,

$$|x(j) - a(j)| \leq |x(j)| + |a(j)| \leq j^{-3/2p} + j^{-2/p} < 2j^{-3/2p}.$$

Hence,

$$\|x - a\|_p^p = \sum_{j=N+1}^{\infty} |x(j) - a(j)|^p < \sum_{j=N+1}^{\infty} 2^p j^{-3/2} < \epsilon^p.$$

Thus, $\|x - a\|_p < \epsilon$, and so $x \in U(a, \epsilon) \subset E$. This contradiction shows that E^0 is empty. This completes the proof. ■

PROBLEM 1.3.19 Let $X = \{x \in C[0, 1] : x(0) = 0\}$ and

$$Y = \{x \in X : \int_0^1 x(t) dt = 0\}.$$

Prove that Y is a closed proper subspace and that there is no x_1 in X such that $\|x_1\| = 1$ and $d(x_1, Y) = 1$; that is, in this case, the Riesz lemma does not hold, with $r = 1$.

Solution Let

$$f(x) = \int_0^1 x(t) dt, \quad x \in X.$$

Then f is a linear functional on X , and

$$|f(x)| \leq \int_0^1 |x(t)| dt \leq \|x\|_\infty.$$

Hence, f is continuous with $\|f\| \leq 1$, and so $Y = \{x \in X : f(x) = 0\}$ is a closed subspace of X . It is proper since, for example, x defined by $x(t) = t, 0 \leq t \leq 1$, is in X but not in Y .

For the second part, consider any point x_1 in X with $\|x_1\| = 1$. We shall prove that $d(x_1, Y) < 1$. Since $x_1(0) = 0$, the continuity of x_1 implies that there is $\delta > 0$ such that

$$|x_1(t)| < \frac{1}{2} \quad \text{for } 0 \leq t < \delta.$$

Since the norm in X is the sup norm $\|\cdot\|_\infty$, we have $|x_1(t)| \leq \|x_1\|_\infty = 1$ for all t . Hence,

$$\begin{aligned} |f(x_1)| &\leq \int_0^1 |x_1(t)| dt = \int_0^\delta |x_1(t)| dt + \int_\delta^1 |x_1(t)| dt \\ &\leq \frac{1}{2}\delta + 1 - \delta. \end{aligned}$$

Thus, $|f(x_1)| < 1$. We shall first find x_2 in X such that $f(x_2) = |f(x_1)|$ and $\|x_2\|_\infty < 1$. Let $c = |f(x_1)|$. Then $c < 1$. If $c < 1/2$, take $x_2(t) = 2ct$ for $0 \leq t \leq 1$. Then,

$$f(x_2) = \int_0^1 x_2(t) dt = \int_0^1 2ct dt = c,$$

and $\|x_2\|_\infty = 2c < 1$. Next suppose that $\frac{1}{2} \leq c < 1$. Let $r = 2(1-c)/(1+c)$. Then, $0 < r < 1$. Let

$$x_2(t) = \begin{cases} (c+1)/(2r) t & \text{for } 0 \leq t \leq r, \\ (c+1)/2 & \text{for } r < t \leq 1. \end{cases}$$

Then x_2 is continuous, $x_2(0) = 0$, and hence $x_2 \in X$. Further,

$$f(x_2) = \int_0^1 x_2(t) dt = \int_0^r \frac{c+1}{2r} t dt + \int_r^1 \frac{c+1}{2} dt = c.$$

Clearly, $|x_2(t)| \leq (c+1)/2$ for $0 \leq t \leq 1$, and hence $\|x_2\|_\infty \leq (c+1)/2 < 1$.

Having found x_2 like this, let $y_1 = x_1 - kx_2$, where $k = f(x_1)/c$. Then, $y_1 \in X$ and

$$f(y_1) = f(x_1) - kf(x_2) = f(x_1) - kc = 0.$$

Thus, $y_1 \in Y$. Hence,

$$d(x_1, Y) \leq d(x_1, y_1) = \|x_1 - y_1\|_\infty = \|kx_2\|_\infty.$$

Since $|k| = |f(x_1)|/c = 1$, we get $d(x_1, Y) \leq \|x_2\|_\infty < 1$. This completes the proof. ■

1.4 Hahn-Banach Theorems

Hahn-Banach theorems are concerned with the existence of bounded linear functionals having prescribed properties.

The *Hahn-Banach separation theorem* states that if E_1 and E_2 are two disjoint nonempty convex subsets of a normed linear space X and if E_1 is open, then there is a bounded linear functional f on X and a real number α such that

$$\operatorname{Re} f(x_1) < \alpha \leq \operatorname{Re} f(x_2)$$

for all x_1 in E_1 and x_2 in E_2 .

The *Hahn-Banach extension theorem* is this: Let Y be a subspace of a normed linear space X and $g \in Y'$, the dual of Y . Then there is $f \in X'$, the dual of X , such that f is an extension of g and $\|f\| = \|g\|$.

The extension stated above need not be unique. *Taylor-Foguel theorem* states that a norm-preserving extension of $g \in Y'$ to an element $f \in X'$ is unique iff X' is strictly convex. A normed space X is said to be *strictly convex* if $\|(x+y)/2\| < 1$ whenever $x, y \in X$, $\|x\| = \|y\| = 1$ and $x \neq y$.

Let X be a normed space over the scalar field $\mathbb{K} = \mathbb{C}$. Let $f \in X'$, $u = \operatorname{Re} f$ and $v = \operatorname{Im} f$. Then u and v are real-linear functionals on the associated real linear space $(X, \mathbb{R})$,

$$v(x) = -u(ix) \quad \text{and} \quad \|u\| = \|f\| = \|v\|.$$

Let E be a subset of a normed space X and b be a boundary point of E . If f is a nonzero functional such that

$$\operatorname{Re} f(x) \leq \operatorname{Re} f(b) \quad \text{for all } x \in E,$$

then f is called a *support functional* of E at b , and the real hyperplane

$$H = \{x : \operatorname{Re} f(x) = \operatorname{Re} f(b)\} = \{b + x : \operatorname{Re} f(x) = 0\}$$

is called the *support hyperplane* of E at b .

EXERCISES

1. Let X be a complex normed space. Let $f \in X'$, $u = \operatorname{Re} f$ and $v = \operatorname{Im} f$. Show that:

- (a) u and v are real-linear functionals on the associated real linear space.
 (b) $v(x) = -u(ix)$ for all $x \in X$.
 (c) $\|u\| = \|f\| = \|v\|$.
2. Show that $\mathbb{R}^2$ with norm $\|\cdot\|_1$ is not strictly convex.
 3. Show that $\mathbb{R}^2$ with norm $\|\cdot\|_\infty$ is not strictly convex.
 4. Let ϕ and α be real numbers. Show that

$$\cos(\theta - \phi) \leq \cos(\alpha - \phi) \quad \text{for all } \theta \in \mathbb{R}$$

iff $\phi - \alpha = 2n\pi$ for some integer n . [Hint: Use the inequality with $\theta = \phi$.]

5. Let $X = \mathbb{R}^2$ with norm $\|\cdot\|_2$, $E = \{x \in \mathbb{R}^2 : \|x\| = 1\}$ be the unit circle and $b = (\cos \alpha, \sin \alpha)$. If a nonzero functional f in X' is a support functional of E at b show that $f(1, 0) = r \cos \alpha$, $f(0, 1) = r \sin \alpha$, $r > 0$. Deduce that the support hyperplane of E at b is

$$H = \{x \in \mathbb{R}^2 : x(1) \cos \alpha + x(2) \sin \alpha = 1\}$$

which is the tangent at b to E .

[Hint: Let $f(1, 0) = r \cos \phi$, $f(0, 1) = r \sin \phi$, where $r > 0$. If $f(x) \leq f(b)$ for all $x \in E$ then

$$r \cos \theta \cos \phi + r \sin \theta \sin \phi \leq r \cos \alpha \cos \phi + r \sin \alpha \sin \phi$$

for all $\theta \in \mathbb{R}$. Use the previous problem.]

PROBLEMS

PROBLEM 1.4.1 Let Y be a subspace of a normed space X . Prove that if $a \in X$, $a \notin \overline{Y}$ then there is $f \in X'$ such that f is zero everywhere on Y , $f(a) = d(a, Y)$ and $\|f\| = 1$.

Solution Let $\lambda = d(a, Y)$. Then $\lambda > 0$ since $a \notin \overline{Y}$. Let

$$Z = \text{span}\{Y, a\} = \{y + ka : y \in Y, k \in \mathbb{K}\}.$$

If $y + ka = y_1 + k_1 a$, where $y, y_1 \in Y$ and $k, k_1 \in \mathbb{K}$, then $(k - k_1)a = y_1 - y \in Y$. It follows that $k_1 = k$ since $a \notin Y$. Thus we can define $g : Z \rightarrow \mathbb{K}$ by

$$g(y + ka) = k\lambda, \quad y \in Y, \quad k \in \mathbb{K}.$$

Then g is linear, $g(a) = \lambda$ and $g(y) = 0$ for all $y \in Y$. If $k \neq 0$, then

$$\|y + ka\| = |k| \left\| a + \frac{1}{k}y \right\| \geq |k|\lambda,$$

$$|g(y + ka)| = |k|\lambda \leq \|y + ka\|.$$

This is true even when $k = 0$. It follows that $g \in Z'$ and $\|g\| \leq 1$.

On the other hand, there is a sequence $\{y_n\}$ in Y such that $\|a - y_n\| \rightarrow \lambda$. Since $g(a - y_n) = g(a) = \lambda$, we get

$$\lambda = |g(a - y_n)| \leq \|g\| \cdot \|a - y_n\| \rightarrow \|g\| \lambda;$$

hence $1 \leq \|g\|$. Thus, $\|g\| = 1$. By the Hahn-Banach extension theorem, there is $f \in X'$ such that $\|f\| = \|g\| = 1$ and $f|_Z = g$. Therefore,

$$\begin{aligned} f(a) &= g(a) = \lambda = d(a, Y); \\ f(y) &= g(y) = 0 \quad \text{for all } y \in Y. \end{aligned}$$

This completes the proof. ■

PROBLEM 1.4.2 Prove that if a is a nonzero element in a normed space X , then there is a bounded linear functional f on X such that

$$f(a) = \|a\|, \quad \|f\| = 1.$$

Solution Let $a \in X$, $a \neq 0$ and $Y = \{0\}$. Then Y is a subspace of X , $\overline{Y} = Y$, $a \notin \overline{Y}$ and $d(a, Y) = \|a\|$. So, by Problem 1.4.1, there exists $f \in X'$ such that

$$f(a) = d(a, Y) = \|a\|, \quad \|f\| = 1. \quad \blacksquare$$

PROBLEM 1.4.3 Let E be a subset of a normed space X , $Y = \text{span } E$ and $a \in X$. Show that $a \in \overline{Y}$ iff

$$f(a) = 0 \quad \text{whenever } f \in X' \text{ and } f \text{ is zero everywhere on } E.$$

Solution Let $a \in \overline{Y}$. Suppose that $f \in X'$ and f is zero everywhere on E . Then $Z(f)$, the zero space of f , contains E . But $Z(f)$ is a closed subspace of X . Therefore, we see that $\overline{Y} \subset Z(f)$. Thus f is zero on $\overline{Y}$ and so $f(a) = 0$.

For the converse, let $a \notin \overline{Y}$. Then by Problem 1.4.1 there is $f \in X'$ such that

$$Y \subset Z(f), \quad f(a) = d(a, Y), \quad \|f\| = 1.$$

But $d(a, Y) \neq 0$ since $a \notin \overline{Y}$. Hence there is $f \in X'$ such that

$$E \subset Y \subset Z(f), \quad f(a) \neq 0.$$

This completes the proof. ■

PROBLEM 1.4.4 For any element x in a normed space $X \neq \{0\}$ show that

$$\|x\| = \sup \{|f(x)| : f \in X', \|f\| = 1\}.$$

Deduce that if x is such that $f(x) = 0$ for all f in X' then $x = 0$.

Solution Let $a \in X$, $a \neq 0$. By Problem 1.4.2 there is f in X' such that $f(a) = \|a\|$ and $\|f\| = 1$. Thus X' contains elements f with $\|f\| = 1$, and hence the set

$$S = \{|f(x)| : f \in X', \|f\| = 1\}$$

is nonempty. If $x = 0$, then S contains 0 only, and so $\sup S = 0 = \|x\|$.

Let $x \neq 0$. Correspondingly there is f in X' with $f(x) = \|x\|$ and $\|f\| = 1$. This shows that S contains $\|x\|$. On the other hand, if $f \in X'$ and $\|f\| = 1$, then

$$|f(x)| \leq \|f\| \cdot \|x\| = \|x\|.$$

This shows that $\|x\|$ is an upper bound of S . Thus $\sup S = \|x\|$.

If $f(x) = 0$ for all f in X' then S contains 0 only. In this case, $\|x\| = \sup S = 0$, and hence, $x = 0$. ■

PROBLEM 1.4.5 Let Y be subspace of a normed space X , $g \in Y'$ and $f \in X'$ be a Hahn-Banach extension of g . Prove that

- (a) For every $a \notin \overline{Y}$ there is $h_a \in X'$ such that h_a is zero everywhere on Y and $h_a(a) \neq 0$.
- (b) $f + h_a$ is a continuous linear extension of g to X , although it may not have the same norm as g .

Solution (a) If $a \notin \overline{Y}$ then by Problem 1.4.1 there is $h_a \in X'$ such that $h_a(y) = 0$ for all y in Y , and $h_a(a) = d(a, Y) \neq 0$.

(b) Since

$$(f + h_a)(y) = f(y) + 0 = g(y) \quad \text{for } y \in Y,$$

we see that $f + h_a$ is an extension of g . It is clearly linear and continuous. The property of h_a stated in (a) is possessed by $h = kh_a$ for any nonzero $k \in \mathbb{K}$. If $\|f + h\| = \|g\|$ in all such cases, we must have

$$\|kh_a\| = \|h\| \leq \|h + f\| + \| - f\| = \|g\| + \|g\|.$$

Thus, $|k| \|h_a\| \leq 2\|g\|$. This cannot be true for large k . It follows that $\|f + h_a\| \neq \|g\|$ for some choice of h_a . This completes the proof. ■

PROBLEM 1.4.6 Let T be a nonempty set, and X be the complex normed linear space of all bounded complex-valued functions x on T with the sup norm $\|\cdot\|_\infty$. Let $t \in T$ and

$$H = \{x \in X : x(t) \in \mathbb{R}\}.$$

Show that

- (a) H is a (real) hyperspace in the associated real linear space $(X, \mathbb{R})$, and
- (b) $H \cap iH = \{x \in X : x(t) = 0\}$.

Solution (a) If $x, y \in H$ and $\alpha \in \mathbb{R}$, then

$$\begin{aligned}(x + y)(t) &= x(t) + y(t), \\ (\alpha x)(t) &= \alpha x(t)\end{aligned}$$

are real. Thus, $x + y \in H$, $\alpha x \in H$. If $x_0(s) = i$ for all $s \in T$ then $x_0 \in X$, but $x_0 \notin H$. This proves that H is a proper subspace of the associated real linear space $(X, \mathbb{R})$.

To show that H is maximal, let H_1 be a subspace of $(X, \mathbb{R})$ with $H \subset H_1 \subset X$ and $H \neq H_1$. Let $x_1 \in H_1, x_1 \notin H$. Then,

$$x_1(t) = \alpha_1 + i\beta_1, \quad \beta_1 \neq 0.$$

Let x be any element of X and $x(t) = \alpha + i\beta$. Then,

$$(\beta_1 x - \beta x_1)(t) = \beta_1(\alpha + i\beta) - \beta(\alpha_1 + i\beta_1) \in \mathbb{R}.$$

This shows that $y = \beta_1 x - \beta x_1 \in H \subset H_1$. Hence,

$$x = \beta_1^{-1}(y + \beta x_1) \in H_1.$$

Since this is true for any $x \in X$, we get $H_1 = X$. Therefore, H is a hyper-space.

(b) Let $x \in H \cap iH$. Then $x \in H$ and $x = iy, y \in H$. So $x(t), y(t) \in \mathbb{R}$, and hence

$$(1 + i)x(t) = x(t) - y(t) \in \mathbb{R}.$$

This implies that $x(t) = 0$. On the other hand, if $x(t) = 0$ then $x \in H$ and

$$-ix(t) = 0 \in \mathbb{R}, \quad -ix \in H, \quad x \in iH;$$

hence $x \in H \cap iH$. This completes the proof. ■

PROBLEM 1.4.7 Suppose that E_1 and E_2 are disjoint nonempty convex subsets of a normed space X , where E_1 is compact and E_2 is closed. Show that there exists f in X' and real numbers α_1, α_2 such that

$$\operatorname{Re} f(x_1) < \alpha_1 < \alpha_2 < \operatorname{Re} f(x_2)$$

for all x_1 in E_1 and x_2 in E_2 .

Solution By Problem 1.1.5 there exists $r > 0$ such that

$$(E_1 + U(0, r)) \cap E_2 = \emptyset.$$

Let $E_3 = E_1 + U(0, r)$. Then E_3 is open by Problem 1.1.4(a). Let $z_1, z_2 \in E_3$ and $0 \leq t \leq 1$. Then,

$$z_i = x_i + y_i, \quad x_i \in E_1, \quad y_i \in U(0, r), \quad i = 1, 2.$$

Hence, $tz_1 + (1-t)z_2 = x + y$, where

$$x = tx_1 + (1-t)x_2, \quad y = ty_1 + (1-t)y_2.$$

Since E_1 is convex and

$$\|y\| \leq t\|y_1\| + (1-t)\|y_2\| < tr + (1-t)r = r,$$

we see that $x \in E_1, y \in U(0, r)$, and hence $x + y \in E_3$. This shows that E_3 is convex.

Thus, E_3, E_2 are disjoint nonempty convex subsets of X with E_3 open. So by the Hahn-Banach separation theorem, there exists f in X' and α in $\mathbb{R}$ such that

$$\operatorname{Re} f(x_3) < \alpha \leq \operatorname{Re} f(x_2)$$

for all x_3 in E_3, x_2 in E_2 . Let $u = \operatorname{Re} f$. Let $x_2 \in E_2$, and $x_3 \in E_3$. Then, $u(x_3) < u(x_2)$, and

$$y_1 = \epsilon(x_2 - x_3) \in U(0, r), \quad \epsilon > 0,$$

if ϵ is small enough. Then, $u(y_1) > 0$. Let

$$\alpha_1 = \alpha - u(y_1), \quad \alpha_2 = \alpha - \frac{1}{2}u(y_1).$$

If $x_1 \in E_1, x_2 \in E_2$, then $x_1 + y_1 \in E_3$, and hence

$$u(x_1 + y_1) < \alpha \leq u(x_2),$$

$$u(x_1) + u(y_1) < \alpha = \alpha_1 + u(y_1)$$

and $u(x_2) \geq \alpha > \alpha_2$. Thus, $u(x_1) < \alpha_1 < \alpha_2 < u(x_2)$. This is true for all $x_1 \in E_1, x_2 \in E_2$. ■

PROBLEM 1.4.8 Let X be a real linear space, and $p : X \rightarrow \mathbb{R}$ satisfy

$$p(x + y) \leq p(x) + p(y),$$

$$p(rx) = rp(y)$$

for all x, y in X and $r \geq 0$. Let Y be a subspace of X , and $g : Y \rightarrow \mathbb{R}$ be a linear map such that

$$g(y) \leq p(y) \quad \text{for all } y \in Y.$$

Let

$$a \in X, \quad a \notin Y, \quad Z = \operatorname{span}\{Y, a\},$$

$$\alpha = \sup \{g(y) - p(y - a) : y \in Y\},$$

and

$$h(y + ta) = g(y) + t\alpha, \quad y \in Y, t \in \mathbb{R}.$$

Show that this defines a linear map $h : Z \rightarrow \mathbb{R}$ such that

$$h|_Y = g \quad \text{and} \quad h(z) \leq p(z) \quad \text{for all } z \in Z.$$

Solution For all x, y in Y ,

$$\begin{aligned} g(x) + g(y) &= g(x + y) \leq p(x + y) \\ &= p(x + a + y - a) \leq p(x + a) + p(y - a); \\ g(y) - p(y - a) &\leq p(x + a) - g(x). \end{aligned}$$

This with a fixed x shows that

$$\alpha \leq p(x + a) - g(x) \quad (1)$$

for any $x \in Y$. Since $a \notin Y$ we see, as in the solution of Problem 1.4.1, that every z in Z can be uniquely expressed as

$$z = y + ta, \quad y \in Y, t \in \mathbb{R}.$$

So h is well-defined on Z by

$$h(y + ta) = g(y) + t\alpha, \quad y \in Y, t \in \mathbb{R}.$$

It is easily seen that h is linear and $h|_Y = g$.

Let $z = y + ta, y \in Y, t \in \mathbb{R}$. If $t = 0$, then

$$h(z) = g(y) \leq p(y) = p(z)$$

since $z = y$. If $t > 0$, then (1) with $x = (1/t)y$ gives

$$\alpha \leq p\left(\frac{1}{t}y + a\right) - g\left(\frac{1}{t}y\right) = p\left(\frac{1}{t}z\right) - g\left(\frac{1}{t}y\right) = \frac{1}{t}p(z) - \frac{1}{t}g(y).$$

Hence, $t\alpha + g(y) \leq p(z)$, that is, $h(z) \leq p(z)$. If $t < 0$ then, since $(-1/t)y \in Y$, the definition of α gives

$$\alpha \geq g\left(-\frac{1}{t}y\right) - p\left(-\frac{1}{t}y - a\right) = g\left(-\frac{1}{t}y\right) - p\left(-\frac{1}{t}z\right) = -\frac{1}{t}g(y) + \frac{1}{t}p(z).$$

Since $-t > 0$, we see that $-t\alpha \geq g(y) - p(z)$. Therefore, $h(z) = g(y) + t\alpha \leq p(z)$. This completes the proof. ■

PROBLEM 1.4.9 Show that, under the hypothesis of Problem 1.4.8, there exists a linear map $f : X \rightarrow \mathbb{R}$ such that $f|_Y = g$ and

$$-p(-x) \leq f(x) \leq p(x) \quad \text{for all } x \in X.$$

Solution Consider the collection $\mathcal{E}$ of all pairs (Z, h) , where Z is a subspace of X , containing Y , and $h : Z \rightarrow \mathbb{R}$ is a linear extension of g such that $h(z) \leq p(z)$ for all z in Z . Then $\mathcal{E}$ is nonempty since it contains (Y, g) .

For $(Z_1, h_1), (Z_2, h_2)$ in $\mathcal{E}$ write $(Z_1, h_1) \leq (Z_2, h_2)$ if

$$Z_1 \subset Z_2, \quad h_1 = h_2|_{Z_1}.$$

Then ' $\leq$ ' is a partial ordering on $\mathcal{E}$. Consider a nonempty chain

$$\mathcal{F} = \{(Z_i, h_i) : i \in I\}$$

in $\mathcal{E}$. Let $Z = \bigcup\{Z_i : i \in I\}$. Write

$$h(z) = h_i(z) \quad \text{for } z \in Z_i. \quad (2)$$

Suppose that $z \in Z_i \cap Z_j$. Since $\mathcal{F}$ is a chain, we have either

$$Z_i \subset Z_j, \quad h_i = h_j|_{Z_i} \quad (3)$$

or

$$Z_j \subset Z_i, \quad h_j = h_i|_{Z_j}. \quad (4)$$

In either case, $h_i(z) = h_j(z)$. Hence, (2) defines a map $h : Z \rightarrow \mathbb{R}$. Let $x, y \in Z$, $k \in \mathbb{K}$. Then, $x \in Z_i$, $y \in Z_j$ for some i, j in I , and (3) or (4) must be true. If (3) is true, then

$$x, y, kx + y \in Z_j \subset Z,$$

$$h(kx + y) = h_j(kx + y) = kh_j(x) + h_j(y) = kh(x) + h(y).$$

Similarly, if (4) is true, then the above statement holds with Z_j, h_j replaced by Z_i, h_i . Thus we see that Z is a subspace of X and $h : Z \rightarrow \mathbb{R}$ is linear. Clearly,

$$Y \subset Z, \quad h|_Y = g, \quad h(z) \leq p(z) \quad \text{for all } z \in Z.$$

Hence, $\mathcal{F}$ has an upper bound (Z, h) in $\mathcal{E}$. So by Zorn's lemma, $\mathcal{E}$ must have a maximal element (Z_0, h_0) .

Assume that $Z_0 \neq X$. Then there is $a \in X$ with $a \notin Z_0$. Apply Problem 1.4.8 with (Y, g) replaced by (Z_0, h_0) . We see that if $Z_1 = \text{span}\{Z_0, a\}$, then there is a linear map $h_1 : Z_1 \rightarrow \mathbb{R}$ such that

$$h_1|_{Z_0} = h_0, \quad h_1(z) \leq p(z) \quad \text{for all } z \in Z_1.$$

Thus, $(Z_0, h_0) \leq (Z_1, h_1)$ and $(Z_0, h_0) \neq (Z_1, h_1)$. This contradicts the maximality of (Z_0, h_0) . Hence, $Z_0 = X$. Denote h_0 by f . Then, $f : X \rightarrow \mathbb{R}$ is linear, $f|_Y = g$ and $f(x) \leq p(x)$ for all x in X . So,

$$-f(x) = f(-x) \leq p(-x),$$

$$-p(-x) \leq f(x) \leq p(x) \quad \text{for all } x \in X.$$

This completes the proof. ■

PROBLEM 1.4.10 Let Y be a subspace of a linear space X , and p be a seminorm on X ; that is, p is a mapping from X to $\mathbb{R}$ such that

$$p(x) \geq 0, \quad p(kx) = |k|p(x),$$

$$p(x + y) \leq p(x) + p(y)$$

for all x, y in X and $k \in \mathbb{K}$. If $g : Y \rightarrow \mathbb{K}$ is linear and $|g(y)| \leq p(y)$ for all y in Y , show that there is a linear map $f : X \rightarrow \mathbb{K}$ such that $f|_Y = g$ and $|f(x)| \leq g(x)$ for all x in X .

Solution First suppose that the scalar field $\mathbb{K}$ is $\mathbb{R}$. By Problem 1.4.9, there is a linear map $f : X \rightarrow \mathbb{R}$ such that $f|_Y = g$, and

$$-p(x) \leq f(x) \leq p(x) \quad \text{for all } x \in X.$$

Since $p(-x) = p(x)$, it follows that

$$|f(x)| \leq p(x) \quad \text{for all } x \in X.$$

Next suppose that $\mathbb{K} = \mathbb{C}$. Let $X_1 = (X, \mathbb{R})$ and $Y_1 = (Y, \mathbb{R})$ denote the associated real linear spaces corresponding to X and Y . It is clear that Y_1 is a subspace of X_1 . Let $v = \operatorname{Re} g$. Then v is linear on Y_1 , and

$$|v(y)| \leq |g(y)| \leq p(y),$$

$$g(y) = v(y) - iv(iy)$$

for all y in Y . Therefore, by the case $\mathbb{K} = \mathbb{R}$, there is a linear map $u : X \rightarrow \mathbb{R}$ such that $u|_{Y_1} = v$ and $|u(x)| \leq p(x)$ for all x in X . Let

$$f(x) = u(x) - iu(ix).$$

Then, $f : X \rightarrow \mathbb{C}$ is linear. If $y \in Y$, then $iy \in Y$, and hence

$$f(y) = u(y) - iu(iy) = v(y) - iv(iy) = g(y).$$

Thus, $f|_Y = g$. Let $x \in X$ and

$$k = \begin{cases} \frac{|f(x)|}{f(x)} & \text{if } f(x) \neq 0, \\ 1 & \text{if } f(x) = 0. \end{cases}$$

Then, $|k| = 1$ and $kf(x) = |f(x)|$. Therefore,

$$f(kx) = kf(x) = |f(x)|,$$

$$|f(x)| = \operatorname{Re} f(kx) = u(kx) \leq p(kx) = |k|p(x),$$

$$|f(x)| \leq p(x).$$

This completes the proof. ■

PROBLEM 1.4.11 A *Banach limit* is a linear functional f on l^∞ having the following three properties:

- (i) If $a = (1, 1, 1, \dots)$, then $f(a) = 1$.
- (ii) If $x \in l^\infty$ and $x(j) \geq 0$ for all j , then $f(x) \geq 0$.
- (iii) If $T : l^\infty \rightarrow l^\infty$ is defined by

$$T(x) = (x(2), x(3), \dots), \quad x \in l^\infty,$$

then $f(Tx) = f(x)$ for all $x \in l^\infty$.

Show that every Banach limit has value $1/2$ at $(1, 0, 1, 0, \dots)$.

Solution Let $b = (1, 0, 1, 0, \dots)$. Then

$$\begin{aligned} T(b) &= (0, 1, 0, 1, \dots), \\ b + T(b) &= (1, 1, 1, 1, \dots) = a. \end{aligned}$$

Hence,

$$1 = f(a) = f(b) + f(Tb) = f(b) + f(b) = 2f(b)$$

for any Banach limit f . This proves the result. ■

PROBLEM 1.4.12 (See Problems 1.2.20, 1.2.21, 1.2.22.) Prove that there exist bounded linear maps $F : l^\infty \rightarrow l^\infty$ which cannot be represented by an infinite matrix (k_{ij}) in the sense that, for each $x \in l^\infty$,

$$F(x)(i) = \sum_{j=1}^{\infty} k_{ij}x(j), \quad i = 1, 2, \dots,$$

the series being convergent for all i, x .

Solution For $x \in c$ let $g(x) = \lim_{n \rightarrow \infty} x(n)$. Then g is a bounded linear functional on c (see Problem 1.2.6). Let f be a Hahn-Banach extension of g to l^∞ . For $x \in l^\infty$, let

$$F(x) = (f(x), f(x), \dots),$$

that is, $F(x)(i) = f(x)$ for all $i \geq 1$. Then $F : l^\infty \rightarrow l^\infty$ is linear. It is bounded since

$$\|F(x)\|_\infty = |f(x)| \leq \|f\| \cdot \|x\|_\infty.$$

Suppose that, for each $x \in l^\infty$,

$$F(x)(i) = \sum_{j=1}^{\infty} k_{ij}x(j), \quad i = 1, 2, \dots$$

Then,

$$f(x) = \sum_{j=1}^{\infty} k_{ij}x(j), \quad i = 1, 2, \dots$$

If $x = e_m$, the m -th unit vector, then

$$f(x) = g(x) = \lim_{n \rightarrow \infty} x(n) = 0.$$

Hence, $0 = k_{im}$. This is true for all i, m . So $f(x) = 0$ for all $x \in l^\infty$. But if $x = (1, 1, 1, \dots)$, then $f(x) = g(x) = 1$. This contradiction shows that there is no matrix (k_{ij}) representing F . ■

PROBLEM 1.4.13 Let X be a normed space, $x_\alpha \in X$ and $k_\alpha \in \mathbb{K}$ for α in some set A . Show that there exists f in X' such that $f(x_\alpha) = k_\alpha$ iff there exists $M > 0$ such that

$$\left| \sum_{\alpha} a_{\alpha} k_{\alpha} \right| \leq M \left\| \sum_{\alpha} a_{\alpha} x_{\alpha} \right\|,$$

where the sums are finite and are taken over all possible choices of $a_{\alpha} \in \mathbb{K}$.

Solution First suppose that there exists f in X' such that $f(x_\alpha) = k_\alpha$ for all α . Then,

$$\sum a_{\alpha} k_{\alpha} = \sum a_{\alpha} f(x_{\alpha}) = f\left(\sum a_{\alpha} x_{\alpha}\right)$$

if the sums are finite. Hence,

$$\left| \sum a_{\alpha} k_{\alpha} \right| = \left| f\left(\sum a_{\alpha} x_{\alpha}\right) \right| \leq M \cdot \left\| \sum a_{\alpha} x_{\alpha} \right\|,$$

where $M = \|f\|$.

For the converse, suppose that such a constant M exists. Let $\text{span}\{x_\alpha : \alpha \in A\}$ be Y . Then, $\{x_\alpha : \alpha \in A\}$ contains a basis $E = \{x_\alpha : \alpha \in B\}$ for Y , so that $B \subset A$. Define $f : E \rightarrow \mathbb{K}$ by $f(x_\alpha) = k_\alpha$ for $\alpha \in B$. Extend f to Y by

$$f\left(\sum a_{\alpha} x_{\alpha}\right) = \sum a_{\alpha} f(x_{\alpha}), \quad x_{\alpha} \in E.$$

This gives a linear functional on Y . If

$$y = \sum a_{\alpha} x_{\alpha}, \quad x_{\alpha} \in E, \alpha \in B,$$

is any element of Y , then

$$f(y) = \sum a_{\alpha} f(x_{\alpha}) = \sum a_{\alpha} k_{\alpha},$$

and so

$$|f(y)| = \left| \sum a_{\alpha} k_{\alpha} \right| \leq M \left\| \sum a_{\alpha} x_{\alpha} \right\| = M \|y\|.$$

Thus, $f \in Y'$ with $\|f\| \leq M$. So f can be extended to a bounded linear functional on X .

Finally, if $\alpha \in A$, $\alpha \notin B$, then $x_\alpha \in Y$,

$$x_{\alpha} = \sum a_i x_i, \quad x_i \in E, i \in B.$$

Hence,

$$f(x_{\alpha}) = \sum a_i f(x_i) = \sum a_i k_i.$$

By our assumption,

$$\left| (-1)k_{\alpha} + \sum a_i k_i \right| \leq M \left\| (-1)x_{\alpha} + \sum a_i x_i \right\| = 0.$$

Thus,

$$k_\alpha = \sum a_i k_i = f(x_\alpha).$$

This completes the proof. ■

PROBLEM 1.4.14 Let X be a normed space, $x, y \in X$, $\|x\| = \|y\| = 1$ and $x \neq y$. Show that if X is strictly convex, then

$$\|tx + (1-t)y\| < 1 \quad (5)$$

for $0 < t < 1$. Show also that X is strictly convex if there is some t with $0 < t < 1$ such that (5) holds whenever $x, y \in X$ with $\|x\| = \|y\| = 1$ and $x \neq y$.

Solution Suppose $0 < t_0 < 1$, and

$$\|t_0x + (1-t_0)y\| < 1. \quad (6)$$

Let $z = t_0x + (1-t_0)y$. If $0 < t < t_0$ let $s = t/t_0$. Then, $0 < s < 1$ and

$$\begin{aligned} sz + (1-s)y &= st_0x + [s(1-t_0) + 1-s]y \\ &= tx + (1-t)y, \end{aligned}$$

$$\|tx + (1-t)y\| \leq \|sz\| + \|(1-s)y\| < 1$$

since $\|z\| < 1$ by (6) and $\|y\| = 1$. If $t_0 < t < 1$, let $s = (1-t)/(1-t_0)$. Then, $0 < s < 1$ and

$$\begin{aligned} sz + (1-s)x &= (st_0 + 1-s)x + s(1-t_0)y \\ &= tx + (1-t)y, \end{aligned}$$

$$\|tx + (1-t)y\| \leq \|sz\| + \|(1-s)x\| < 1,$$

as before. Thus we see that (5) holds for $0 < t < 1$ if (6) holds for some t_0 . If X is strictly convex, then (6) holds with $t_0 = 1/2$. This proves both parts of the problem. ■

PROBLEM 1.4.15 Show that a normed space X is strictly convex iff it has the following property: If x and y are nonzero elements in X such that $\|x + y\| = \|x\| + \|y\|$, then $x = cy$ for some real $c > 0$.

Solution Suppose that the condition is satisfied. To show that X is strictly convex let $x, y \in X$ and $\|x\| = \|y\| = \|(x+y)/2\| = 1$. Then,

$$\|x + y\| = 2 = \|x\| + \|y\|;$$

hence $x = cy$ for some $c > 0$. So,

$$1 = \|x\| = c\|y\| = c; \quad x = y.$$

Thus X is strictly convex.

Conversely, suppose that X is strictly convex. Let x, y be nonzero elements in X such that $\|x + y\| = \|x\| + \|y\|$. Let

$$a = \|x\|, \quad b = \|y\|, \quad x_1 = \frac{x}{a}, \quad y_1 = \frac{y}{b}.$$

Then, $a > 0, b > 0, \|x_1\| = \|y_1\| = 1$, and

$$\|ax_1 + by_1\| = \|x + y\| = \|x\| + \|y\| = a + b,$$

$$\left\| \frac{a}{a+b}x_1 + \frac{b}{a+b}y_1 \right\| = 1,$$

$$\|tx_1 + (1-t)y_1\| = 1,$$

where $t = a/(a+b)$. Since $0 < t < 1$, we see by Problem 1.4.14 that this is possible only if $x_1 = y_1$; that is, $x = cy$, where $c = a/b$. This completes the proof. ■

PROBLEM 1.4.16 Show that l^p is strictly convex iff $1 < p < \infty$.

Solution If $x = (1, 0, 0, \dots)$ and $y = (0, 1, 0, \dots)$, then $x, y \in l^1$ and $(x + y)/2 = (\frac{1}{2}, \frac{1}{2}, 0, \dots)$, so that

$$\|x\|_1 = \|y\|_1 = \left\| \frac{x+y}{2} \right\|_1 = 1.$$

Hence l^1 is not strictly convex. Let $z = (1, 1, 0, \dots)$. Then $x, z \in l^\infty$, $(x + z)/2 = (1, \frac{1}{2}, 0, \dots)$, and

$$\|x\|_\infty = \|z\|_\infty = \left\| \frac{x+z}{2} \right\|_\infty = 1.$$

Hence l^∞ is not strictly convex.

Let $1 < p < \infty$. We shall prove that l^p is strictly convex. Suppose that $x, y \in l^p$, and $\|x\|_p = \|y\|_p = \|(x+y)/2\|_p = 1$. We shall prove that this happens only when $x = y$. Let $1/p + 1/q = 1$, and

$$z(n) = (|x(n)| + |y(n)|)^{p-1}.$$

Then $z \neq 0$. Since $(p-1)q = p$, we see that

$$\|z\|_q^q = \sum_{n=1}^{\infty} (|x(n)| + |y(n)|)^p = \| |x| + |y| \|_p^p < \infty. \quad (7)$$

We shall first prove that

$$ab \leq \frac{1}{p}a^p + \frac{1}{q}b^q \quad \text{for } a \geq 0, b \geq 0 \quad (8)$$

and that equality holds in (8) only if $a^p = b^q$. We may suppose that $a > 0, b > 0$. The function $\phi(t) = t^p - pt$, $0 < t < \infty$, has a unique minimum at $t = 1$. Hence, $\phi(t) > \phi(1)$ for $0 < t < 1$ and for $1 < t < \infty$. This with $t = ab^{-q/p}$ proves our claim.

Taking $a = |x(n)|$, $b = z(n)/\|z\|_q$ in (8), we get

$$|x(n)| \frac{z(n)}{\|z\|_q} \leq \frac{1}{p} |x(n)|^p + \frac{1}{q} \left(\frac{z(n)}{\|z\|_q} \right)^q \quad (9)$$

$$\begin{aligned} \sum_{n=1}^{\infty} |x(n)| \frac{z(n)}{\|z\|_q} &\leq \frac{1}{p} \sum_{n=1}^{\infty} |x(n)|^p + \frac{1}{q} \left(\frac{1}{\|z\|_q} \right)^q \sum_{n=1}^{\infty} (z(n))^q \\ &= \frac{1}{p} (\|x\|_p)^p + \frac{1}{q} = 1. \end{aligned} \quad (10)$$

If equality holds in (10), then equality holds in (9) for all n , and hence

$$|x(n)|^p = \left(\frac{z(n)}{\|z\|_q} \right)^q \quad \text{for all } n \geq 1. \quad (11)$$

Similarly,

$$\sum_{n=1}^{\infty} |y(n)| \frac{z(n)}{\|z\|_q} \leq 1 \quad (12)$$

with equality only if

$$|y(n)|^p = \left(\frac{z(n)}{\|z\|_q} \right)^q \quad \text{for all } n \geq 1. \quad (13)$$

Since $(|x(n)| + |y(n)|)z(n) = (|x(n)| + |y(n)|)^p$, we get

$$(\|z\|_q)^q = \sum_{n=1}^{\infty} (|x(n)| + |y(n)|)z(n) \leq \|z\|_q + \|z\|_q,$$

where we have used (7), (10) and (12). Thus, $(\|z\|_q)^{q-1} \leq 2$. However, $(q-1)p = q$; so

$$\|z\|_q^q \leq 2^p \quad (14)$$

If equality holds in (14), we must have equality in (10) and (12); and hence (11) and (13) must be true. But,

$$\begin{aligned} 1 &= \sum_{n=1}^{\infty} \left| \frac{x(n) + y(n)}{2} \right|^p \leq \sum_{n=1}^{\infty} \left(\frac{|x(n)| + |y(n)|}{2} \right)^p \\ &= 2^{-p} \|z\|_q^q, \end{aligned}$$

by (7). It follows from (14) that equality holds in it. So, (11) and (13) must be true. Moreover, the two sums involved in the last step are equal, and hence the corresponding terms in them are equal. Thus, for all $n \geq 1$,

$$|x(n)|^p = \left(\frac{z(n)}{\|z\|_q} \right)^q = |y(n)|^p,$$

$$|x(n) + y(n)| = |x(n)| + |y(n)|.$$

If u and v are real or complex numbers, then

$$|u + v|^2 + |u - v|^2 = 2|u|^2 + 2|v|^2.$$

Hence,

$$\begin{aligned} 2|x(n)|^2 + 2|y(n)|^2 &= |x(n) + y(n)|^2 + |x(n) - y(n)|^2 \\ &= (|x(n)| + |y(n)|)^2 + |x(n) - y(n)|^2, \\ 2|x(n)|^2 + 2|y(n)|^2 &= (2|x(n)|)^2 + |x(n) - y(n)|^2; \\ |x(n) - y(n)|^2 &= 0. \end{aligned}$$

This is true for all $n \geq 1$, and hence $x = y$. This completes the proof. ■

PROBLEM 1.4.17 A normed space X is said to be *smooth* if for every nonzero x in X there is a unique f in X' with

$$f(x) = \|x\|, \quad \|f\| = 1. \quad (15)$$

Show that

- (a) if X is smooth and Y is a subspace of X then Y is smooth;
- (b) if X' is strictly convex then X is smooth.

Solution The existence of $f \in X'$, for each x , satisfying (15) is guaranteed by Problem 1.4.2. So X is smooth if f is unique for each $x \neq 0$.

Suppose that X is smooth. Let $a \in Y, a \neq 0$, and $f, g \in Y'$ be such that

$$f(a) = \|a\| = g(a), \quad \|f\| = \|g\| = 1. \quad (16)$$

Then there exist f_1, g_1 in X' which are Hahn-Banach extensions of f, g . Since X is smooth, (16) implies that $f_1 = g_1$. Hence, $f = g$. Thus Y is smooth. This proves (a).

To prove (b) suppose that X' is strictly convex. Let $a \in X, a \neq 0$, and $f, g \in X'$ be such that (16) holds. Then,

$$\begin{aligned} 2\|a\| &= f(a) + g(a) = (f + g)(a) \leq \|f + g\| \|a\|, \\ 2 &\leq \|f + g\| \leq \|f\| + \|g\| = 2. \end{aligned}$$

Hence, $\|(f + g)/2\| = 1$. Since X' is strictly convex this is possible only if $f = g$. Thus X is smooth. This completes the proof. ■

PROBLEM 1.4.18 Let X be a separable normed space with norm $\|\cdot\|$. Show that there exists an equivalent norm on X in which X is strictly convex.

Solution Let $\{x_n\}$ be a countable dense subset of $S = \{x \in X : \|x\| = 1\}$. For each n there is $f_n \in X'$, by Problem 1.4.2, such that

$$f_n(x_n) = \|x_n\| = 1, \quad \|f_n\| = 1.$$

Let $x \in X$ and $y(n) = 2^{-n/2} f_n(x)$. Since

$$|f_n(x)| \leq \|f_n\| \|x\| = \|x\|,$$

we see that

$$\sum_{n=1}^{\infty} |y(n)|^2 \leq \sum_{n=1}^{\infty} 2^{-n} \|x\|^2 = \|x\|^2.$$

This shows that $y \in l^2$ and $\|y\|_2 \leq \|x\|$. Thus, a map $F : X \rightarrow l^2$ can be defined by

$$F(x)(n) = 2^{-n/2} f_n(x), \quad n = 1, 2, \dots, x \in X.$$

The linearity of f_n shows that F is linear. Since $\|F(x)\|_2 \leq \|x\|$, we see that F is bounded with $\|F\| \leq 1$.

Let $F(x) = 0$, $x \neq 0$ and $a = x/\|x\|$. Then, $F(a) = 0$ and $\|a\| = 1$. Hence,

$$f_n(a) = 2^{n/2} F(a)(n) = 0 \quad \text{for all } n \geq 1.$$

Since $\{x_n\}$ is dense in S , there is x_m such that $\|x_m - a\| < \frac{1}{2}$, and hence

$$|f_m(x_m)| = |f_m(x_m - a)| \leq \|f_m\| \cdot \|x_m - a\| < \frac{1}{2}.$$

This is not possible since $f_m(x_m) = 1$. This shows that if $F(x) = 0$, then $x = 0$; that is, F is one-one.

For $x \in X$, let

$$\|x\|_0 = \|x\| + \|F(x)\|_2.$$

Clearly, $\|x\|_0 \geq 0$. If $\|x\|_0 = 0$ then $\|x\| = 0$ and so $x = 0$. Thus, $\|x\|_0 = 0$ iff $x = 0$. Again,

$$\|kx\|_0 = \|kx\| + \|F(kx)\|_2 = |k| \|x\| + |k| \|F(x)\|_2 = |k| \|x\|_0,$$

and

$$\begin{aligned} \|x + y\|_0 &= \|x + y\| + \|F(x + y)\|_2 \\ &\leq \|x\| + \|y\| + \|F(x)\|_2 + \|F(y)\|_2 = \|x\|_0 + \|y\|_0. \end{aligned}$$

Thus we see that $\|\cdot\|_0$ is a norm on X . Since $\|F(x)\|_2 \leq \|x\|$, we see that

$$\|x\| \leq \|x\|_0 \leq \|x\| + \|x\| = 2\|x\|.$$

Thus, $\|\cdot\|_0$ is equivalent to $\|\cdot\|$.

To show that X is strictly convex under $\|\cdot\|_0$, let

$$\|x\|_0 = \|y\|_0 = \|(x + y)/2\|_0 = 1.$$

Then,

$$\begin{aligned}\|x + y\| + \|F(x) + F(y)\|_2 &= \|x + y\|_0 = 2, \\ \|x\| + \|y\| + \|F(x)\|_2 + \|F(y)\|_2 &= \|x\|_0 + \|y\|_0 = 2.\end{aligned}$$

Because of triangle inequality for $\|\cdot\|$ and $\|\cdot\|_2$, this implies that

$$\|F(x) + F(y)\|_2 = \|F(x)\|_2 + \|F(y)\|_2. \quad (17)$$

Since $x \neq 0$, $y \neq 0$, we see that $F(x)$ and $F(y)$ are nonzero elements of l^2 satisfying (17). Since l^2 is strictly convex, we see by Problem 1.4.15 that $F(x) = cF(y) = F(cy)$ for some $c > 0$. Hence, $x = cy$. Since $\|x\|_0 = \|y\|_0$ we get $c = 1$ and $x = y$. This completes the proof. ■

2

Banach Spaces

The concept of completeness is very important in metric spaces. This chapter is devoted to normed linear spaces which are complete; these are called Banach spaces. Examples of Banach spaces are discussed in Section 2.1. As a result of Baire's theorem for complete metric spaces, we get the three fundamental theorems for Banach spaces, namely, the uniform boundedness principle, the open mapping theorem and the closed graph theorem. Problems involving applications of these fundamental theorems are discussed in Sections 2.2 and 2.3.

2.1 Complete Normed Spaces

A sequence $\{x_n\}$ in a normed linear space X is said to be a *Cauchy sequence* if for each $\epsilon > 0$ there is a positive integer N such that

$$\|x_n - x_m\| < \epsilon \quad \text{for all } n, m \geq N.$$

It can be seen that a Cauchy sequence is bounded and that a convergent sequence is a Cauchy sequence. On the other hand, if every Cauchy sequence in X converges to an element in X , then we say that X is complete. A complete normed space is called a *Banach space*. To avoid trivialities we assume that $X \neq \{0\}$.

The Lebesgue spaces L^p , $1 \leq p \leq \infty$, are Banach. Many of the spaces introduced in Section 1.1 are Banach. These are proved in the problems below.

Let X be a normed space. A sequence $\{x_n\}$ in X is called a *Schauder basis* for X if $\|x_n\| = 1$ for $n = 1, 2, \dots$, and each x in X can be expressed as

$$x = \sum_{n=1}^{\infty} k_n x_n$$

where the series converges in the norm of X , and the scalars k_n are uniquely determined by x .

EXERCISES

1. Show that a closed subspace of a Banach space is Banach.
2. Suppose that $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent norms on a linear space X . Show that X is Banach under $\|\cdot\|_1$ iff it is Banach under $\|\cdot\|_2$.

PROBLEMS

PROBLEM 2.1.1 Let X be a normed space and Y be a Banach space. Show that the space $BL(X, Y)$ of bounded linear maps from X to Y , with norm defined by

$$\|F\| = \sup \{\|F(x)\| : x \in X, \|x\| \leq 1\}, \quad F \in BL(X, Y),$$

is a Banach space. Deduce that the dual X' of any normed space X is Banach.

Solution To show that $BL(X, Y)$ is complete, let $\{F_n\}$ be a Cauchy sequence in $BL(X, Y)$. Then for each $\epsilon > 0$, there is a positive integer N such that $\|F_n - F_m\| < \epsilon$ for all $n, m \geq N$. So for each $x \in X$ and $n, m \geq N$, we have

$$\begin{aligned} \|F_n(x) - F_m(x)\| &= \|(F_n - F_m)(x)\| \\ &\leq \|F_n - F_m\| \|x\| \\ &\leq \epsilon \|x\|. \end{aligned} \tag{1}$$

This shows that $\{F_n(x)\}$ is a Cauchy sequence in Y for each $x \in X$. But Y is given to be a Banach space. Hence, for each $x \in X$,

$$F(x) = \lim_{n \rightarrow \infty} F_n(x) \tag{2}$$

exists in Y . The map $F : X \rightarrow Y$ defined by (2) is easily seen to be linear. Keeping $n \geq N$ fixed and letting $m \rightarrow \infty$ in (1), we get

$$\|F_n(x) - F(x)\| \leq \epsilon \|x\|, \quad x \in X.$$

This shows that $F_n - F$ is a bounded linear map and $\|F_n - F\| \leq \epsilon$ for $n \geq N$. Hence, $F = F_n - (F_n - F) \in BL(X, Y)$ and $F_n \rightarrow F$ in the norm of $BL(X, Y)$. This proves the first result.

Since $X' = BL(X, \mathbb{K})$ and $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$ is a Banach space, the second part follows from the first. ■

PROBLEM 2.1.2 Let X and Y be normed spaces and $X \neq \{0\}$. Show that if $BL(X, Y)$ is a Banach space, then Y is Banach.

Solution Choose a nonzero element a in X . By Problem 1.4.2, there $f \in X'$ such that $f(a) = \|a\|$ and $\|f\| = 1$. To show that Y is Banach, let $\{y_n\}$ be a Cauchy sequence in Y . Define $F_n : X \rightarrow Y$ by

$$F_n(x) = f(x)y_n, \quad x \in X.$$

Since f is linear, we see that F_n is linear. Since

$$\|F_n(x)\| = |f(x)| \|y_n\| \leq \|f\| \|y_n\| \|x\|, \quad x \in X,$$

we see that $F_n \in BL(X, Y)$. Further,

$$\begin{aligned} \|F_n(x) - F_m(x)\| &= \|f(x)(y_n - y_m)\| \\ &= |f(x)| \|y_n - y_m\| \\ &\leq \|f\| \|x\| \|y_n - y_m\| \end{aligned}$$

for all $x \in X$, $n, m \geq 1$. Hence,

$$\|F_n - F_m\| \leq \|f\| \|y_n - y_m\| = \|y_n - y_m\|.$$

This shows that $\{F_n\}$ is a Cauchy sequence in $BL(X, Y)$. So it converges to some F in the norm of $BL(X, Y)$. Therefore,

$$\|F_n(a) - F(a)\| \leq \|F_n - F\| \|a\| \rightarrow 0$$

Thus, $\|a\|y_n = f(a)y_n = F_n(a) \rightarrow F(a)$. This shows that $\{y_n\}$ converges to $(1/\|a\|)F(a)$ in Y . This completes the proof. ■

PROBLEM 2.1.3 If the normed spaces X and Y are linearly homeomorphic and if one of them is Banach, show that the other also is Banach. Deduce that all finite dimensional spaces are Banach.

Solution Suppose that X is Banach. To show that Y is Banach, let $\{y_n\}$ be a Cauchy sequence in Y . Let $F : X \rightarrow Y$ be a linear homeomorphism and $G = F^{-1}$. Then, $F \in BL(X, Y)$ and $G \in BL(Y, X)$. Let $x_n = G(y_n)$. Then,

$$\|x_n - x_m\| = \|G(y_n - y_m)\| \leq \|G\| \|y_n - y_m\|.$$

Since $\{y_n\}$ is Cauchy in Y , this shows that $\{x_n\}$ is Cauchy in X . So $\{x_n\}$ converges to some x in X . Hence, $F(x_n) \rightarrow F(x)$ in Y . But $F(x_n) = y_n$. So $\{y_n\}$ converges in Y . This proves the first part.

For the second part, let X be a normed space of finite dimension n . Let $Y = \mathbb{K}^n$ with the Euclidean norm $\|\cdot\|_2$. Then Y is a Banach space. By Problem 1.3.16, the spaces X and Y are linearly homeomorphic. This completes the proof. ■

PROBLEM 2.1.4 Let X and Y be Banach spaces. Show that the product space $X \times Y$, with the norm defined by

$$\|(x, y)\| = \|x\| + \|y\|, \quad (x, y) \in X \times Y,$$

is Banach.

Solution To show that $X \times Y$ is complete, let $\{(x_n, y_n)\}$ be a Cauchy sequence in $X \times Y$. Since

$$\begin{aligned}\|(x_n, y_n) - (x_m, y_m)\| &= \|(x_n - x_m, y_n - y_m)\| \\ &= \|x_n - x_m\| + \|y_n - y_m\|,\end{aligned}$$

we have

$$\begin{aligned}\|x_n - x_m\| &\leq \|(x_n, y_n) - (x_m, y_m)\|, \\ \|y_n - y_m\| &\leq \|(x_n, y_n) - (x_m, y_m)\|\end{aligned}$$

for all $n, m \geq 1$. This shows that $\{x_n\}$ and $\{y_n\}$ are Cauchy sequences in X and Y , respectively. Since X and Y are Banach, they converge. Let $x_n \rightarrow x$ and $y_n \rightarrow y$. Then,

$$\begin{aligned}\|(x_n, y_n) - (x, y)\| &= \|(x_n - x, y_n - y)\| \\ &= \|x_n - x\| + \|y_n - y\| \rightarrow 0,\end{aligned}$$

that is, $(x_n, y_n) \rightarrow (x, y)$ in $X \times Y$. This proves the result. ■

PROBLEM 2.1.5 If T is a metric space show that $C(T)$ and $C_0(T)$ are Banach spaces under the sup norm $\|\cdot\|_\infty$. Show also that $C_c(T)$ need not be a Banach space under this norm.

Solution Let $\{x_n\}$ be a Cauchy sequence in $C(T)$. Then for each $\epsilon > 0$, there is a positive integer N such that

$$|x_n(t) - x_m(t)| \leq \|x_n - x_m\|_\infty \leq \epsilon$$

for all $n, m \geq N$, $t \in T$. This shows that $\{x_n\}$ satisfies Cauchy's condition for uniform convergence on T . So $\{x_n\}$ converges uniformly to some x , that is, $\|x_n - x\|_\infty \rightarrow 0$ as $n \rightarrow \infty$. In particular, $x_n - x$ is bounded on T , and so x is bounded. Hence, $x \in C(T)$ since the limit of a uniformly convergent sequence of continuous functions is continuous. Thus, $x_n \rightarrow x$ in $C(T)$. This proves that $C(T)$ is complete.

Let y be any element in the closure of $C_0(T)$ in $C(T)$. Then there is a sequence $\{y_n\}$ in $C_0(T)$ such that $\|y_n - y\| \rightarrow 0$ as $n \rightarrow \infty$. So for each $\epsilon > 0$ there is N such that $\|y_N - y\|_\infty < \epsilon/2$. Since $y_N \in C_0(T)$, there is a compact set $S \subset T$ such that $|y_N(t)| < \epsilon/2$ for all $t \notin S$. Hence,

$$\begin{aligned}|y_n(t)| &\leq |y(t) - y_N(t)| + |y_N(t)| \\ &\leq \|y - y_N\|_\infty + |y_N(t)| \\ &< \epsilon/2 + \epsilon/2\end{aligned}$$

for all $t \notin S$. This shows that $y \in C_0(T)$. Thus, $C_0(T)$ is a closed subspace of $C(T)$, and is therefore complete.

If $C_c(T)$ is complete, it will be closed in $C(T)$. To show that this need not be true, take $T = \mathbb{N}$ with the discrete metric. Then, $C(T) = l^\infty$ and $C_c(T) = c_{00}$. Let

$$x_n(j) = \begin{cases} \frac{1}{j} & \text{for } 1 \leq j \leq n, \\ 0 & \text{for } j > n. \end{cases}$$

Then, $x_n \in c_{00} = C_c(T)$. Let $x(j) = 1/j$ for all $j \geq 1$. Then $x \in l^\infty = C(T)$, and

$$x_n - x = (0, \dots, 0, \frac{1}{n+1}, \frac{1}{n+2}, \dots),$$

$$\|x_n - x\| = \frac{1}{n+1} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Since $x \notin c_{00}$, we see that $c_{00} = C_c(T)$ is not closed. Thus $C_c(T)$ need not be complete. ■

PROBLEM 2.1.6 Show that l^∞ , c_0 and c are Banach spaces under the sup norm $\|\cdot\|_\infty$ while c_{00} is not a Banach space.

Solution If $T = \mathbb{N}$ with the discrete metric, then $C(T) = l^\infty$ and $C_0(T) = c_0$. So the completeness of l^∞ and c_0 follow from Problem 2.1.5. An independent proof is given below:

Completeness of l^∞ . Let $\{x_n\}$ be a Cauchy sequence in l^∞ . Since

$$|x_n(j) - x_m(j)| \leq \|x_n - x_m\|_\infty \quad \text{for } j = 1, 2, \dots,$$

we see that, for each fixed j , $\{x_n(j)\}$ is a Cauchy sequence of scalars. Hence $x(j) = \lim_{n \rightarrow \infty} x_n(j)$ exists in $\mathbb{K}$. Let

$$\|x_n - x_m\|_\infty < \epsilon \quad \text{for } n, m \geq N.$$

Then $|x_n(j) - x_m(j)| < \epsilon$ for $m, n \geq N$. Letting $m \rightarrow \infty$,

$$|x_n(j) - x(j)| \leq \epsilon \quad \text{for } n \geq N.$$

This is true for all $j \geq 1$. This, in particular, implies that $x_n - x \in l^\infty$, and $x \in l^\infty$. It also shows that

$$\|x_n - x\| \leq \epsilon \quad \text{for } n \geq N.$$

Hence, $x_n \rightarrow x$ in l^∞ . Thus l^∞ is complete.

Completeness of c_0 . Let $x \in \bar{c}_0$, the closure of c_0 in l^∞ . Then there exist $x_n \in c_0$ with $\|x_n - x\| \rightarrow 0$ as $n \rightarrow \infty$. So, for each j ,

$$\begin{aligned} |x(j)| &\leq |x(j) - x_N(j)| + |x_N(j)| \\ &\leq \|x - x_N\|_\infty + |x_N(j)| < \frac{\epsilon}{2} + |x_N(j)| \end{aligned}$$

if N were large enough. Since $x_N(j) \rightarrow 0$ as $j \rightarrow \infty$, this shows that $|x(j)| < \epsilon$ if $j \geq j_0$. Thus, $x \in c_0$. Hence, c_0 is a closed subspace of l^∞ and is, therefore, complete.

Completeness of c . Let $x \in \bar{c}$, the closure of c in l^∞ . Then there exist $x_n \in c$, with $\|x_n - x\|_\infty \rightarrow 0$ as $n \rightarrow \infty$. So for all i, j ,

$$\begin{aligned} |x(i) - x(j)| &\leq |x(i) - x_N(i)| + |x_N(i) - x_N(j)| + |x_N(j) - x(j)| \\ &\leq \|x - x_N\|_\infty + |x_N(i) - x_N(j)| + \|x_N - x\|_\infty \\ &< \frac{\epsilon}{3} + |x_N(i) - x_N(j)| + \frac{\epsilon}{3} \end{aligned}$$

if N were large enough. Since $\lim_{j \rightarrow \infty} x_N(j)$ exists, this shows that

$$|x(i) - x(j)| < \epsilon \quad \text{for } i, j \geq j_0.$$

Hence, $x \in c$. Thus, c is a closed subspace of l^∞ and is, therefore, complete.

The last part of the solution of Problem 2.1.5 shows that c_{00} is not complete. This completes the proof. ■

PROBLEM 2.1.7 Show that l^p is a Banach space for $1 \leq p \leq \infty$.

Solution Since l^∞ is Banach by Problem 2.1.6, we may suppose that $1 \leq p < \infty$. Let $\{x_n\}$ be a Cauchy sequence in l^p , $1 \leq p < \infty$. For each j ,

$$\begin{aligned} |x_n(j) - x_m(j)|^p &\leq \sum_{i=1}^{\infty} |x_n(i) - x_m(i)|^p, \\ |x_n(j) - x_m(j)| &\leq \|x_n - x_m\|_p. \end{aligned}$$

Thus, $\{x_n(j)\}$ is a Cauchy sequence of scalars for $j = 1, 2, \dots$. Hence, $x(j) = \lim_{n \rightarrow \infty} x_n(j)$ exists in $\mathbb{K}$. Let

$$\|x_n - x_m\|_p < \epsilon \quad \text{for } n, m \geq n_0.$$

Then for all $N \geq 1$,

$$\sum_{j=1}^N |x_n(j) - x_m(j)|^p \leq \sum_{j=1}^{\infty} |x_n(j) - x_m(j)|^p < \epsilon^p$$

for $n, m \geq n_0$. Letting $m \rightarrow \infty$, keeping n, N fixed, we get

$$\sum_{j=1}^N |x_n(j) - x(j)|^p \leq \epsilon^p$$

for $n \geq n_0$, $N \geq 1$. Letting $N \rightarrow \infty$, keeping n fixed, we obtain

$$\sum_{j=1}^{\infty} |x_n(j) - x(j)|^p \leq \epsilon^p, \quad n \geq n_0.$$

This, first of all, shows that $x_n - x \in l^p$, and hence $x \in l^p$. It also shows that

$$\|x_n - x\|_p < \epsilon \quad \text{for } n \geq n_0.$$

Thus, $x_n \rightarrow x$ in l^p . Therefore, l^p is complete. ■

PROBLEM 2.1.8 Show that a normed space $X \neq \{0\}$ is a Banach space iff

$$S = \{x \in X : \|x\| = 1\}$$

is complete.

Solution If X is complete then S is complete since it is closed.

Conversely, suppose that S is complete. Let $\{x_n\}$ be a Cauchy sequence in X . From the triangle inequality, we get

$$|\|x_n\| - \|x_m\|| \leq \|x_n - x_m\|. \quad (3)$$

Thus $\{\|x_n\|\}$ is a Cauchy sequence and hence convergent. If $\|x_n\| \rightarrow 0$, then $x_n \rightarrow 0$. So let $\|x_n\| \rightarrow \lambda > 0$. Then there is n_0 such that $\|x_n\| > \frac{1}{2}\lambda$ for $n \geq n_0$. For $n \geq n_0$, let $y_n = x_n/\|x_n\|$. Then, $\{y_n : n \geq n_0\}$ is a sequence in S and

$$\begin{aligned} \|x_n\|(y_n - y_m) &= (x_n - x_m) + (\|x_m\| - \|x_n\|)y_m, \\ \|x_n\| \|y_n - y_m\| &\leq \|x_n - x_m\| + |\|x_m\| - \|x_n\|| \cdot \|y_m\| \\ &\leq \|x_n - x_m\| + \|x_n - x_m\| \end{aligned}$$

by (3). Hence,

$$\|y_n - y_m\| \leq \frac{2}{\|x_n\|} \|x_n - x_m\| \leq \frac{4}{\lambda} \|x_n - x_m\|.$$

Thus $\{y_n\}$ is a Cauchy sequence in S . So it converges. If $y_n \rightarrow y$, then $x_n \rightarrow \lambda y$. This shows that X is complete. ■

PROBLEM 2.1.9 Show that a normed space $X \neq \{0\}$ contains sequences $\{x_n\}$, $\{y_n\}$ such that:

- (a) $\sum \|x_n\|^2 < \infty$, but the sequence of partial sums of $\sum x_n$ is not a Cauchy sequence.
- (b) $\sum \|y_n\| = \infty$, but $\sum y_n$ converges.

Solution Let $x_0 \in X$, $x_0 \neq 0$, $x_n = \frac{1}{n}x_0$ and $y_n = \frac{(-1)^{n-1}}{n}x_0$. Then,

$$\sum_{n=1}^{\infty} \|x_n\|^2 = \sum_{n=1}^{\infty} \frac{1}{n^2} \|x_0\|^2 < \infty.$$

If $s_n = x_1 + \cdots + x_n$, then

$$\begin{aligned} s_{2n} - s_n &= \left(\frac{1}{n+1} + \cdots + \frac{1}{2n} \right) x_0, \\ \|s_{2n} - s_n\| &= \left(\frac{1}{n+1} + \cdots + \frac{1}{2n} \right) \|x_0\| > \frac{1}{2} \|x_0\|. \end{aligned}$$

This shows that $\{s_n\}$ is not a Cauchy sequence. Further,

$$\sum \|y_n\| = \sum \frac{1}{n} \|x_0\| = \infty.$$

Since $y_1 + \cdots + y_n = \alpha_n x_0$, where

$$\alpha_n = 1 - \frac{1}{2} + \cdots + \frac{(-1)^{n-1}}{n} \rightarrow \log 2,$$

we see that $\sum y_n$ converges to $(\log 2)x_0$. This completes the proof. ■

PROBLEM 2.1.10 Let X be a normed space. If $x_n \in X$ and $\sum \|x_n\| < \infty$, we say that the series $\sum x_n$ is *absolutely convergent*. Show that if X is a Banach space then every absolutely convergent series in it converges.

Solution Suppose that X is a Banach space and $\sum x_n$ is an absolutely convergent series in X . Let

$$\begin{aligned} y_n &= x_1 + \cdots + x_n, \\ t_n &= \|x_1\| + \cdots + \|x_n\|. \end{aligned}$$

Then for $n \geq m \geq 1$,

$$\begin{aligned} \|y_n - y_m\| &= \|x_{m+1} + \cdots + x_n\| \\ &\leq \|x_{m+1}\| + \cdots + \|x_n\| \\ &= t_n - t_m. \end{aligned}$$

Since $\{t_n\}$ converges, it is a Cauchy sequence. It follows that $\{y_n\}$ is a Cauchy sequence in X and hence convergent in X . Thus, $\sum x_n$ converges. This proves the result. ■

PROBLEM 2.1.11 Let X be a normed space such that every absolutely convergent series in X converges. Show that X is a Banach space.

Solution To show that X is complete suppose that $\{x_n\}$ is a Cauchy sequence in X . Let n_1 be a positive integer such that

$$\|x_n - x_{n_1}\| < 1 \quad \text{for all } n \geq n_1.$$

Choose $n_2, n_3, \dots$, successively, such that $n_r > n_{r-1}$ and

$$\|x_n - x_{n_r}\| < \frac{1}{r^2} \quad \text{for all } n \geq n_r, \quad r = 2, 3, \dots$$

Then

$$\|x_{n_{r+1}} - x_{n_r}\| < \frac{1}{r^2} \quad \text{for } r = 1, 2, \dots$$

Let

$$z_r = x_{n_{r+1}} - x_{n_r}, \quad r = 1, 2, \dots$$

Then,

$$\sum_{r=1}^{\infty} \|z_r\| = \sum_{r=1}^{\infty} \|x_{n_{r+1}} - x_{n_r}\| \leq \sum_{r=1}^{\infty} \frac{1}{r^2} < \infty.$$

Thus, $\sum z_r$ is absolutely convergent. So it converges to some x in X . Hence,

$$x_{n_r} = x_{n_1} + z_1 + \dots + z_{r-1} \rightarrow x_{n_1} + x \quad \text{as } r \rightarrow \infty.$$

Since $\{x_n\}$ is a Cauchy sequence, given $\epsilon > 0$ we can find N such that

$$\|x_n - x_{n_r}\| < \epsilon/2 \quad \text{for } n \geq N, n_r \geq N.$$

So choosing a large r , we see that

$$\|x_n - x_{n_1} - x\| \leq \|x_n - x_{n_r}\| + \|x_{n_r} - x_{n_1} - x\| < \epsilon$$

for all $n \geq N$. This proves that $x_n \rightarrow x_{n_1} + x$ and completes the proof. ■

PROBLEM 2.1.12 If X is a Banach space and Y is a closed subspace of X , show that the quotient space X/Y , under the norm defined by

$$\|x + Y\| = \inf \{\|x + y\| : y \in Y\}, \quad x + Y \in X/Y,$$

is a Banach space.

Solution We shall prove that every absolutely convergent series in X/Y is convergent. The desired result will then follow from Problem 2.1.11. Let $\sum \|x_n + Y\| < \infty$. By the definition of the norm on X/Y , for each n we can find $y_n \in Y$ such that

$$\|x_n + y_n\| < \|x_n + Y\| + \frac{1}{2^n}.$$

So,

$$\sum_{n=1}^{\infty} \|x_n + y_n\| \leq \sum_{n=1}^{\infty} \|x_n + Y\| + 1 < \infty.$$

Since X is Banach this implies, by Problem 2.1.10, that $\sum (x_n + y_n)$ converges to some x in X . Let

$$z_n = (x_1 + Y) + \dots + (x_n + Y) = \sum_{r=1}^n (x_r + y_r) + Y,$$

and $z = x + Y$. Then,

$$\begin{aligned}\|z_n - z\| &= \left\| \sum_{r=1}^n (x_r + y_r) + Y - (x + Y) \right\| \\ &= \left\| \sum_{r=1}^n (x_r + y_r) - x + Y \right\| \\ &\leq \left\| \sum_{r=1}^n (x_r + y_r) - x + 0 \right\|.\end{aligned}$$

This tends to zero as $n \rightarrow \infty$. Hence, $z_n \rightarrow z$ in X/Y , that is, $\sum(x_n + Y)$ converges to z in X/Y . This completes the proof. ■

PROBLEM 2.1.13 Let $X = C^1[a, b]$, the space of all continuously differentiable functions on $[a, b]$. For $x \in X$, let

$$\begin{aligned}\|x\| &= \|x\|_\infty + \|x'\|_\infty, \\ \|x\|_1 &= |x(a)| + \|x'\|_\infty,\end{aligned}$$

where x' is the derivative of x . Show that X is Banach space with either of the norms. Show also that, for all x in X ,

$$\|x\|_1 \leq \|x\| \leq (b - a + 1)\|x\|_1,$$

the constant $b - a + 1$ being the best possible.

Solution Since $|x(a)| \leq \|x\|_\infty$, we have

$$\|x\|_1 \leq \|x\|_\infty + \|x'\|_\infty = \|x\|.$$

On the other hand,

$$x(t) - x(a) = (t - a)x'(c)$$

for some c with $a < c < t$. Hence,

$$\begin{aligned}|x(t)| &\leq |x(a)| + |t - a||x'(c)| \\ &\leq |x(a)| + (b - a)\|x'\|_\infty.\end{aligned}$$

This is true for all t in $[a, b]$. Therefore,

$$\begin{aligned}\|x\|_\infty &\leq |x(a)| + (b - a)\|x'\|_\infty, \\ \|x\|_\infty + \|x'\|_\infty &\leq |x(a)| + (b - a + 1)\|x'\|_\infty, \\ \|x\| &\leq (b - a + 1)(|x(a)| + \|x'\|_\infty) = (b - a + 1)\|x\|_1.\end{aligned}$$

Thus the two norms are equivalent.

Let $\{x_n\}$ be a Cauchy sequence w.r.t. $\|\cdot\|$. Since

$$\|x_n - x_m\| = \|x_n - x_m\|_\infty + \|x'_n - x'_m\|_\infty,$$

we see that $\{x_n\}$ and $\{x'_n\}$ are Cauchy sequences in $C[a, b]$ w.r.t. the sup norm $\|\cdot\|_\infty$. Since $C[a, b]$ is complete w.r.t. $\|\cdot\|_\infty$, there exist x, y in $C[a, b]$ such that $x_n \rightarrow x$ and $x'_n \rightarrow y$ w.r.t. $\|\cdot\|_\infty$. But convergence w.r.t. $\|\cdot\|_\infty$ is uniform convergence. Hence, $x_n \rightarrow x$ and $x'_n \rightarrow y$ uniformly on $[a, b]$. Consequently, x is differentiable on $[a, b]$ and $x' = \lim x'_n = y$. Thus $x' = y$ is continuous. Hence $x \in C^1[a, b]$, and

$$\begin{aligned} \|x_n - x\|_\infty &\rightarrow 0, \quad \|x'_n - x'\|_\infty \rightarrow 0; \\ \|x_n - x\| &= \|x_n - x\|_\infty + \|x'_n - x'\|_\infty \rightarrow 0. \end{aligned}$$

Therefore, $x_n \rightarrow x$ w.r.t. $\|\cdot\|$. This shows that X is complete w.r.t. $\|\cdot\|$. Since $\|\cdot\|_1$ is equivalent to $\|\cdot\|$, we see that X is complete w.r.t. $\|\cdot\|_1$ also.

Finally, to show that the constant $b - a + 1$ is the best possible, suppose that c is a constant such that

$$\|x\| \leq c\|x\|_1 \quad \text{for all } x \in X. \quad (4)$$

Consider the function $x(t) = t - a$, $a \leq t \leq b$. For this x in X , we have

$$\|x\|_\infty = b - a, \quad x'(t) = 1, \quad \|x'\|_\infty = 1.$$

Hence, $\|x\| = b - a + 1$ and $\|x\|_1 = 1$. So (4) implies that $b - a + 1 \leq c$. This completes the proof. ■

PROBLEM 2.1.14 Let $X = C^n[a, b]$, the set of scalar-valued functions whose n -th derivatives exist and are continuous on $[a, b]$. For x in X , let

$$\|x\| = \sum_{m=0}^n \|x^{(m)}\|_\infty,$$

where $x^{(0)} = x$ and $x^{(m)}$ is the m -th derivative of x , $m = 1, 2, \dots, n$. Let Y be the set of all infinitely differentiable functions in X . Show that X is a Banach space, and also that Y is not Banach in the induced norm.

Solution Let $\{x_i\}$ be a Cauchy sequence in X . Then,

$$\sum_{m=0}^n \|x_i^{(m)} - x_j^{(m)}\|_\infty = \|x_i - x_j\| \rightarrow 0 \quad \text{as } i, j \rightarrow \infty.$$

It follows that for $m = 0, 1, \dots, n$,

$$\|x_i^{(m)} - x_j^{(m)}\|_\infty \rightarrow 0 \quad \text{as } i, j \rightarrow \infty,$$

that is, $\{x_i^{(m)} : i = 1, 2, \dots\}$ is a Cauchy sequence in $C[a, b]$ with the norm $\|\cdot\|_\infty$. Since $C[a, b]$ is complete under $\|\cdot\|_\infty$, there is $y_m \in C[a, b]$ such that

$$\|x_i^{(m)} - y_m\|_\infty \rightarrow 0 \quad \text{as } i \rightarrow \infty,$$

that is, $x_i^{(m)} \rightarrow y_m$ uniformly as $i \rightarrow \infty$. Thus we see that if $1 \leq m \leq n$ and $z_i = x_i^{(m-1)}$, then $\{z_i\}$ and $\{z'_i\}$ converge uniformly to y_{m-1} and y_m . It follows that y_{m-1} is differentiable and $y'_{m-1} = y_m$. This is true for $m = 1, \dots, n$. If we denote y_0 by x , then $y_m = x^{(m)}$ exists in $C[a, b]$, and

$$\|x_i^{(m)} - x^{(m)}\|_\infty = \|x_i^{(m)} - y_m\|_\infty \rightarrow 0 \quad \text{as } i \rightarrow \infty,$$

$$\|x_i - x\| = \sum_{m=0}^n \|x_i^{(m)} - x^{(m)}\|_\infty \rightarrow 0 \quad \text{as } i \rightarrow \infty.$$

Thus, $x_i \rightarrow x$ in X . This shows that X is complete under $\|\cdot\|$.

To show that Y is not complete, let

$$F(x)(t) = \int_a^t x(s) ds, \quad a \leq t \leq b, x \in C[a, b].$$

This defines a linear map $F : C[a, b] \rightarrow C[a, b]$. If $x \in C^{m-1}[a, b]$, then $F(x) \in C^m[a, b]$. Further,

$$|F(x)(t)| \leq \int_a^t |x(s)| ds \leq \|x\|_\infty (t - a).$$

Therefore, $\|F(x)\|_\infty \leq (b - a)\|x\|_\infty$. Thus, F is bounded w.r.t. $\|\cdot\|_\infty$.

Choose $y \in C[a, b]$ such that y is not differentiable on $[a, b]$; for example, $y(t) = (t - a)^{1/2}$. There is a sequence $\{p_i\}$ of polynomials such that $p_i \rightarrow y$ uniformly on $[a, b]$ as $i \rightarrow \infty$, that is, $p_i \rightarrow y$ w.r.t. $\|\cdot\|_\infty$. The continuity of F implies that

$$F(p_i) \rightarrow F(y), \quad F^2(p_i) = FF(p_i) \rightarrow F^2(y), \dots$$

in $C[a, b]$ w.r.t. $\|\cdot\|_\infty$. Let $F^n(p_i) = P_i$ and $F^n(y) = x$. Then P_i is a polynomial, and

$$\begin{aligned} x' &= F^{n-1}(y), \quad x'' = F^{n-2}(y), \dots, \quad x^{(n)} = y, \\ P'_i &= F^{n-1}(p_i), \quad P''_i = F^{n-2}(p_i), \dots, \quad P_i^{(n)} = p_i. \end{aligned}$$

Further, $x \in C^n[a, b]$ and $x \notin C^{n+1}[a, b]$, as $x^{(n)} = y$. Therefore,

$$\|P_i - x\| = \sum_{m=0}^n \|P_i^{(m)} - x^{(m)}\|_\infty = \sum_{m=0}^n \|F^{n-m}(p_i) - F^{n-m}(y)\|_\infty \rightarrow 0$$

as $i \rightarrow \infty$. Thus, $P_i \rightarrow x$ w.r.t. the norm $\|\cdot\|$. But $P_i \in Y$ and $x \notin Y$. This shows that Y is not closed in X under $\|\cdot\|$. So Y cannot be complete under $\|\cdot\|$. This completes the proof. ■

PROBLEM 2.1.15 Let D be the open unit disk in $\mathbb{C}$ and $\overline{D}$ be its closure. Let X be the set of all continuous complex-valued functions on $\overline{D}$ which are analytic in D . For $x \in X$, let

$$\|x\| = \sup \{|x(e^{it})| : 0 \leq t \leq 2\pi\}.$$

Show that X is a Banach space.

Solution The maximum modulus principle for analytic functions implies that

$$\sup \{|x(u)| : u \in \overline{D}\} = \sup \{|x(u)| : |u| = 1\}.$$

Thus the norm given here is the same as the sup norm $\|\cdot\|_\infty$ on $\overline{D}$. Let $\{x_n\}$ be a sequence in X and $x_n \rightarrow x$ in $C(\overline{D})$ w.r.t. $\|\cdot\|_\infty$. Thus $x_n \rightarrow x$ uniformly on $\overline{D}$. So $x_n \rightarrow x$ uniformly on every compact subset of D , and each x_n is analytic in D . Hence x is analytic in D ; that is, $x \in X$. This shows that X is a closed subspace of $C(\overline{D})$. But $C(\overline{D})$ is a Banach space under $\|\cdot\|_\infty$. So X is Banach. This completes the proof. ■

PROBLEM 2.1.16 Let D be the open unit disk in $\mathbb{C}$. If $x : D \rightarrow \mathbb{C}$ is analytic, let

$$\|x\|_p = \sup_{0 < r < 1} \left\{ \frac{1}{2\pi} \int_{-\pi}^{\pi} |x(re^{it})|^p dt \right\}^{1/p}, \quad 1 \leq p < \infty,$$

$$\|x\|_\infty = \sup_{u \in D} |x(u)|.$$

Let $H^p(D)$, $1 \leq p \leq \infty$, denote the set of all analytic functions $x : D \rightarrow \mathbb{C}$ such that $\|x\|_p < \infty$. Show that $H^p(D)$ is a Banach space for $1 \leq p \leq \infty$.

Solution Let $1 \leq p < \infty$. For $0 \leq r \leq 1$, write

$$M_p(x, r) = \left\{ \int_{-\pi}^{\pi} |x(re^{it})|^p dt \right\}^{1/p}.$$

Consequently,

$$\|x\|_p = \sup_{0 < r < 1} (2\pi)^{-1/p} M_p(x, r).$$

Consider a fixed r , where $0 < r < 1$. If $x : D \rightarrow \mathbb{C}$ is analytic, let $x_r : [-\pi, \pi] \rightarrow \mathbb{C}$ be defined by

$$x_r(t) = x(re^{it}).$$

Then $M_p(x, r)$ is the L^p -norm of x_r w.r.t. the Lebesgue measure. Consequently, $M_p(x, r) = 0$ iff x is zero on the circle $|z| = r$,

$$M_p(kx, r) = |k| M_p(x, r), \quad (5)$$

$$M_p(x + y, r) \leq M_p(x, r) + M_p(y, r). \quad (6)$$

From (5) we see that

$$(2\pi)^{-1/p} M_p(kx, r) = |k| (2\pi)^{-1/p} M_p(x, r) \leq |k| \|x\|.$$

It follows that

$$\|kx\| \leq |k| \|x\|.$$

Hence if $k \neq 0$, then

$$\|x\| = \left\| \frac{1}{k} kx \right\| \leq \frac{1}{|k|} \|kx\|.$$

Thus, $|k| \|x\| \leq \|kx\|$. This being trivial when $k = 0$,

$$\|kx\| = |k| \|x\|. \quad (7)$$

Similarly, from (6),

$$(2\pi)^{-1/p} M_p(x + y, r) \leq \|x\| + \|y\|.$$

Therefore,

$$\|x + y\| \leq \|x\| + \|y\|. \quad (8)$$

From (7) and (8) one can see that $H^p(D)$ is a linear space and $\|\cdot\|_p$ is a norm on it. If $p = \infty$, the norm $\|\cdot\|_\infty$ is the same as the usual sup norm.

To prove that $H^p(D)$ is complete, let $\{x_n\}$ be a Cauchy sequence in it. Let $0 < r < 1$. Fix r and choose R with $r < R < 1$.

If f is analytic in D and $|z| < r$, then by Cauchy's integral formula,

$$f(z) = \frac{1}{2\pi} \int_{|w|=R} \frac{f(w)}{w-z} dw;$$

$$|f(z)| \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{|f(R e^{i\theta})|}{R-r} R d\theta = \frac{1}{2\pi} \frac{R}{R-r} M_1(f, R).$$

If $1 < p < \infty$ and $1/p + 1/q = 1$, then Holder's inequality gives

$$|f(z)| \leq \frac{R}{2\pi(R-r)} \left\{ \int_{-\pi}^{\pi} |f(R e^{i\theta})|^p d\theta \right\}^{1/p} \left\{ \int_{-\pi}^{\pi} 1^q d\theta \right\}^{1/q}$$

$$\leq \frac{R}{R-r} (2\pi)^{\frac{1}{q}-1} M_p(f, R).$$

Thus,

$$|f(z)| \leq \frac{R}{R-r} (2\pi)^{-1/p} M_p(f, R) \leq \frac{R}{R-r} \|f\|_p$$

for $|z| < r$, $1 \leq p < \infty$. On the other hand,

$$|f(z)| \leq \|f\|_\infty \quad \text{for } |z| \leq r.$$

Apply this result with $f = x_n - x_m$. We see that, for $|z| \leq r$,

$$|x_n(z) - x_m(z)| \leq \frac{R}{R-r} \|x_n - x_m\|_p, \quad 1 \leq p < \infty,$$

$$|x_n(z) - x_m(z)| \leq \|x_n - x_m\|_\infty.$$

It follows that $\{x_n\}$ satisfies the Cauchy criterion for uniform convergence for $|z| \leq r$. This is true for any r with $0 < r < 1$. Hence, $\{x_n\}$ converges uniformly on compact subsets of D to a function x which is analytic in D .

Suppose that $\|x_n - x_m\|_p < \epsilon$ for all $n, m \geq N$. If $p = \infty$, this gives

$$|x_n(z) - x_m(z)| \leq \|x_n - x_m\|_\infty < \epsilon \quad \text{for } n, m \geq N, z \in D.$$

Keep n fixed and let $m \rightarrow \infty$. We thus obtain

$$|x_n(z) - x(z)| \leq \epsilon \quad \text{for } n \geq N, z \in D,$$

$$\|x_n - x\|_\infty \leq \epsilon \quad \text{for } n \geq N.$$

Next let $1 \leq p < \infty$ and $0 < r < 1$. If $n, m \geq N$, then

$$M_p(x_n - x_m, r) \leq (2\pi)^{1/p} \|x_n - x_m\|_p < (2\pi)^{1/p} \epsilon,$$

$$\int_{-\pi}^{\pi} |x_n(re^{it}) - x_m(re^{it})|^p dt < 2\pi \epsilon^p.$$

Keep n fixed and let $m \rightarrow \infty$. Since

$$|x_n(z) - x_m(z)| \rightarrow |x_n(z) - x(z)|$$

uniformly for $|z| = r$, we get

$$\int_{-\pi}^{\pi} |x_n(re^{it}) - x(re^{it})|^p dt \leq (2\pi) \epsilon^p \quad \text{for } n \geq N,$$

$$M_p(x_n - x, r) \leq (2\pi)^{1/p} \epsilon \quad \text{for } n \geq N.$$

This is true whenever $0 < r < 1$, and N is independent of r . It follows that

$$\|x_n - x\|_p \leq \epsilon \quad \text{for } n \geq N.$$

This was proved for $p = \infty$ also. This, first of all, shows that $x_n - x \in H^p(D)$, and hence $x \in H^p(D)$. Moreover, this shows also that $x_n \rightarrow x$ in $H^p(D)$. This completes the proof. ■

PROBLEM 2.1.17 Let X be a Banach space and Y be any normed space. If $F : X \rightarrow Y$ is a linear homeomorphism of X onto $R(F)$ and $R(F)$ is dense in Y , show that $R(F) = Y$.

Solution Let $y \in Y$. Since $R(F)$ is dense in Y , there is a sequence $\{y_n\}$ in $R(F)$ such that $y_n \rightarrow y$. Let $x_n = F^{-1}(y_n)$. Then,

$$\|x_n - x_m\| = \|F^{-1}(y_n - y_m)\| \leq \|F^{-1}\| \|y_n - y_m\|.$$

Since $\{y_n\}$ is a Cauchy sequence, this shows that $\{x_n\}$ also is a Cauchy sequence. Hence, $\{x_n\}$ converges to some x in X , and $y_n = F(x_n) \rightarrow F(x)$. Thus,

$$y = \lim y_n = F(x) \in R(F).$$

This shows that $R(F) = Y$. ■

PROBLEM 2.1.18 Let X denote a Banach space and $BL(X)$ be the space of bounded linear operators on X . For $A \in BL(X)$, let

$$\exp A = \sum_{n=0}^{\infty} \frac{A^n}{n!}$$

Show that the series defining $\exp A$ is absolutely convergent. Show also that, if $A, B \in BL(X)$ and $AB = BA$, then

$$\exp(A + B) = (\exp A)(\exp B).$$

Solution Note that $BL(X) = BL(X, X)$ is a Banach space since X is Banach. Multiplication in it is the composition of mappings. Since

$$\|A(Bx)\| \leq \|A\| \|Bx\| \leq \|A\| \|B\| \|x\|,$$

we see that $\|AB\| \leq \|A\| \|B\|$ for all A, B in $BL(X)$. Hence, $\|A^n\| \leq \|A\|^n$,

$$\sum_{n=0}^{\infty} \left\| \frac{1}{n!} A^n \right\| \leq \sum_{n=0}^{\infty} \frac{1}{n!} \|A\|^n < \infty.$$

Thus the series defining $\exp A$ is absolutely convergent in $BL(X)$. So it is convergent in $BL(X)$ by Problem 2.1.10 since $BL(X)$ is Banach.

If $A_n \rightarrow A$ and $B_n \rightarrow B$ in $BL(X)$, then $A_n + B_n \rightarrow A + B$ in $BL(X)$, and

$$\begin{aligned} \|A_n B_n - AB\| &= \|A_n(B_n - B) + (A_n - A)B\| \\ &\leq \|A_n\| \cdot \|B_n - B\| + \|A_n - A\| \cdot \|B\| \rightarrow 0; \end{aligned}$$

hence $A_n B_n \rightarrow AB$ in $BL(X)$. If $A, B \in BL(X)$ and $AB = BA$, let

$$F_n = \sum_{i=0}^n \frac{A^i}{i!}, \quad G_n = \sum_{j=0}^n \frac{B^j}{j!}, \quad H_n = \sum_{r=0}^n \frac{(A+B)^r}{r!}.$$

Then,

$$F_n G_n = \sum_{i=0}^n \sum_{j=0}^n \frac{A^i}{i!} \frac{B^j}{j!}.$$

If $0 \leq r \leq n$, the sum of terms in this with $i + j = r$ is

$$\sum_{i=0}^r \frac{A^i}{i!} \frac{B^{r-i}}{(r-i)!} = \frac{1}{r!} (A+B)^r$$

since $AB = BA$. If $n+1 \leq r \leq 2n$, the corresponding sum is

$$\sum_{i=r-n}^n \frac{A^i}{i!} \frac{B^{r-i}}{(r-i)!}.$$

Hence, $F_n G_n = H_n + D_n$, where

$$\begin{aligned} D_n &= \sum_{r=n+1}^{2n} \sum_{i=r-n}^n \frac{A^i}{i!} \frac{B^{r-i}}{(r-i)!}, \\ \|D_n\| &\leq \sum_{r=n+1}^{2n} \sum_{i=r-n}^n \frac{\|A\|^i}{i!} \frac{\|B\|^{r-i}}{(r-i)!}. \end{aligned} \quad (9)$$

Similarly, if $\alpha = \|A\|$ and $\beta = \|B\|$, then

$$\sum_{i=0}^n \frac{\alpha^i}{i!} \sum_{j=0}^n \frac{\beta^j}{j!} - \sum_{r=0}^n \frac{(\alpha + \beta)^r}{r!} = \sum_{r=n+1}^{2n} \sum_{i=r-n}^n \frac{\alpha^i}{i!} \frac{\beta^{r-i}}{(r-i)!}. \quad (10)$$

The left-hand side of (10) tends to zero as $n \rightarrow \infty$. If $\alpha = \|A\|$, $\beta = \|B\|$, then the right-hand sides of (9) and (10) are the same. It follows that $\|D_n\| \rightarrow 0$ as $n \rightarrow \infty$. Hence, $F_n G_n - H_n \rightarrow 0$ in $BL(X)$. But $F_n \rightarrow \exp A$, $G_n \rightarrow \exp B$ and $H_n \rightarrow \exp(A+B)$. This proves that

$$\exp(A+B) = (\exp A)(\exp B). \quad \blacksquare$$

PROBLEM 2.1.19 If $e = (1, 1, \dots)$ and $e_1, e_2, \dots$ are the unit vectors, show that $\{e, e_1, e_2, \dots\}$ is a Schauder basis for c .

Solution If $x \in c$ and $\lim_{n \rightarrow \infty} x(n) = \lambda$, let

$$x_n = \lambda e + \sum_{j=1}^n (x(j) - \lambda) e_j.$$

Then, $x_n(j) = \lambda + x(j) - \lambda = x(j)$ for $1 \leq j \leq n$ and $x_n(j) = \lambda$ for $j > n$. It follows that

$$\|x - x_n\|_\infty = \sup_{j \geq 1} |x(j) - x_n(j)| = \sup_{j > n} |x(j) - \lambda|.$$

This tends to 0 as $n \rightarrow \infty$ since $\lambda = \lim_{j \rightarrow \infty} x(j)$. Thus, $x_n \rightarrow x$ in the norm of c . Hence

$$x = \lim x_n = \lambda e + \sum_{j=1}^{\infty} (x(j) - \lambda) e_j.$$

To show that such an expression of x is unique, let

$$x = ae + \sum_{n=1}^{\infty} a_n e_n$$

be another such expression, and let

$$y_n = ae + \sum_{i=1}^n a_i e_i, \quad n = 1, 2, \dots$$

Then, $y_n(j) = a + a_j$ for $1 \leq j \leq n$ and $y_n(j) = a$ for $j > n$. Hence, for $n \geq j$,

$$|a + a_j - x(j)| = |y_n(j) - x(j)| \leq \|y_n - x\|_{\infty} \rightarrow 0$$

as $n \rightarrow \infty$. Thus we see that $a + a_j - x(j) = 0$ for each $j \geq 1$. Hence, $y_n(j) = x(j)$ for $1 \leq j \leq n$. So, as before,

$$\|x - y_n\|_{\infty} = \sup_{j \geq 1} |x(j) - y_n(j)| = \sup_{j > n} |x(j) - a|.$$

Since $y_n \rightarrow x$ in c , this implies that

$$a = \lim_{j \rightarrow \infty} x(j) = \lambda,$$

and hence $a_j = x(j) - a = x(j) - \lambda$ for all $j \geq 1$. This shows uniqueness, and the proof is complete. ■

PROBLEM 2.1.20 For $t \in \mathbb{R}$ let $x_0(t) = t$, $x_1(t) = 1 - t$,

$$x_2(t) = \begin{cases} 2t & \text{if } 0 \leq t \leq \frac{1}{2} \\ 2 - 2t & \text{if } \frac{1}{2} < t \leq 1 \\ 0 & \text{if } t < 0 \text{ or } t > 1. \end{cases}$$

For $n \geq 2$, let

$$\begin{aligned} n &= 2^r + j, \quad 1 \leq j \leq 2^r, \quad r = 0, 1, 2, \dots, \\ x_n(t) &= x_2(2^r t - j + 1), \quad t \in \mathbb{R}. \end{aligned} \tag{11}$$

Show that $\{x_n\}$ is a Schauder basis for $C[0, 1]$.

Solution Let $t_0 = 1$, $t_1 = 0$, and

$$t_n = 2^{-r-1}(2j - 1) \quad \text{for } n = 2, 3, \dots,$$

where r, j are determined by (11). Then,

$$x_n(t) = x_2\left(2^r(t - t_n) + \frac{1}{2}\right). \tag{12}$$

Note that $x_2(u) = 0$ if $2u$ is an integer other than 1. So, (12) shows that $x_n(t) = 0$ if $2^{r+1}t - 2^{r+1}t_n + 1$ is an integer other than 1. If $2 \leq m < n$, let $m = 2^s + i$, where $1 \leq i \leq 2^s, 0 \leq s \leq r$. Then, $2^{r+1}t_m$ is an integer. This is true for $m = 0, 1$ also. Again, by (12), $x_n(t_n) = x_2(\frac{1}{2}) = 1$. Thus,

$$x_n(t_m) = \begin{cases} 0 & \text{for } 0 \leq m < n \\ 1 & \text{for } m = n. \end{cases} \quad (13)$$

It can be seen that this is true for $n = 0, 1$ also.

Let $x \in C[0, 1]$. Choose $a_0, a_1, a_2, \dots$ successively as follows:

$$a_0 = x(1) = x(t_0), \quad a_1 = x(0) = x(t_1),$$

$$a_n = x(t_n) - \sum_{i=0}^{n-1} a_i x_i(t_n), \quad n = 2, 3, \dots$$

Let

$$y_n(t) = \sum_{i=0}^n a_i x_i(t), \quad n = 0, 1, 2, \dots,$$

so that $a_n = x(t_n) - y_{n-1}(t_n)$. Then (13) shows that $y_n(t_n) = x(t_n)$ and $y_n(t_m) = y_{n-1}(t_m)$ for $0 \leq m < n$. This is true for $n = 0, 1$ also. It follows that

$$y_n(t_m) = x(t_m) \quad \text{for } 0 \leq m \leq n. \quad (14)$$

Let n be given by (11) and $p = 2^s(2i - 1) < 2^{r+1}$. Then,

$$\frac{p}{2^{r+1}} = \frac{2i - 1}{2^{r-s+1}} = t_m, \quad \text{where } m = 2^{r-s} + i.$$

It follows that the numbers

$$0, \frac{1}{2^{r+1}}, \frac{2}{2^{r+1}}, \dots, \frac{2j-1}{2^{r+1}}, \frac{2j}{2^{r+1}}, \frac{2j+2}{2^{r+1}}, \dots, 1 \quad (15)$$

are precisely the numbers $t_0, t_1, \dots, t_n$ written in increasing order. The $n+1$ numbers in (15) form a partition of $[0, 1]$ into n subintervals. We shall prove that $y_n(t)$ is a linear function of t (that is, of the form $at + b$) in each of these subintervals. If $n = 2$, then the corresponding numbers are $0, \frac{1}{2}, 1$. It can be easily verified that $y_2(t)$ is linear in $[0, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$.

Assume that our claim holds for $n-1$. Removing $t_n = 2^{-r-1}(2j-1)$ from (15), we get the corresponding numbers $t_0, \dots, t_{n-1}$ for $n-1$. Since $x_2(t) = 0$ for $t \leq 0$ and for $t \geq 1$, we see from (12) that $x_n(t) = 0$ for $t \leq (2j-2)2^{-r-1}$ and for $t \geq 2j2^{-r-1}$. Hence, $y_n(t) = y_{n-1}(t)$ is linear in each of the $n-2$ subintervals formed by (15) other than

$$J_1 = \left[\frac{2j-2}{2^{r+1}}, \frac{2j-1}{2^{r+1}} \right], \quad J_2 = \left[\frac{2j-1}{2^{r+1}}, \frac{2j}{2^{r+1}} \right]$$

Since $x_2(t)$ is linear in $[0, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$ we see from (12) that $x_n(t)$ is linear in J_1 and J_2 . By assumption, $y_{n-1}(t)$ also is linear in J_1 and J_2 . Therefore, $y_n(t)$ is linear in J_1 and J_2 . This verifies our claim.

Since x is uniformly continuous on $[0, 1]$, given $\epsilon > 0$, we can find $\delta > 0$ such that

$$|x(u) - x(v)| < \epsilon \quad \text{for } |u - v| < \delta, \quad u, v \in [0, 1].$$

Let $n > 2/\delta$ and consider the corresponding $n + 1$ numbers in (15). Let $t \in [0, 1]$. There exist two consecutive terms u, v in (15) such that $u \leq t \leq v$. Since $v - u \leq 2(2^{-r-1})$ and $2^{r+1} \geq n > 2/\delta$ we have $|v - u| < \delta$. Hence,

$$|x(t) - x(u)| < \epsilon, \quad |x(v) - x(u)| < \epsilon, \quad (16)$$

by the choice of δ .

Since $y_n(t)$ is of the form $at + b$ for $u \leq t \leq v$, we have

$$\begin{aligned} \left| \frac{y_n(t) - y_n(u)}{y_n(v) - y_n(u)} \right| &= \left| \frac{a(t - u)}{a(v - u)} \right| \leq 1, \\ |y_n(t) - y_n(u)| &\leq |y_n(v) - y_n(u)|. \end{aligned}$$

Since $y_n(u) = x(u)$, $y_n(v) = x(v)$ by (14), the last relation gives

$$|y_n(t) - x(u)| \leq |x(v) - x(u)|.$$

Hence,

$$|y_n(t) - x(t)| \leq |y_n(t) - x(u)| + |x(u) - x(t)| < 2\epsilon$$

by (16). This, being true for all $t \in [0, 1]$, shows that $\|y_n - x\|_\infty \leq 2\epsilon$. This holds for all $n > 2/\delta$. So $y_n \rightarrow x$ in $C[0, 1]$. Thus,

$$x = \lim_{n \rightarrow \infty} \sum_{i=0}^n a_i x_i = \sum_{i=0}^{\infty} a_i x_i \quad \text{in } C[0, 1].$$

Finally, suppose that

$$x = \sum_{i=0}^{\infty} b_i x_i$$

is another such representation of x . Let

$$z_n = \sum_{i=0}^n b_i x_i, \quad n = 0, 1, 2, \dots$$

It follows from (13) that $z_n(t_m) = z_m(t_m)$ for all $n > m \geq 0$. Since

$$|z_n(t_m) - x(t_m)| \leq \|z_n - x\|_\infty \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

we see that

$$z_m(t_m) = x(t_m) \quad \text{for } m = 0, 1, 2, \dots$$

So (14) implies that

$$z_m(t_m) = y_m(t_m) \quad \text{for } m = 0, 1, 2, \dots$$

This with $m = 0$ implies that $b_0 = a_0$. Suppose that $b_i = a_i$ for $0 \leq i \leq m-1$. Then, $z_{m-1}(t) = y_{m-1}(t)$ for all t , and so

$$z_m(t_m) - z_{m-1}(t_m) = y_m(t_m) - y_{m-1}(t_m).$$

Hence,

$$b_m = b_m x_m(t_m) = a_m x_m(t_m) = a_m.$$

This shows that $b_n = a_n$ for all $n \geq 0$. Thus, every $x \in C[0, 1]$ has a unique expression in the form

$$x = \sum_{i=0}^{\infty} a_i x_i.$$

This completes the proof. ■

PROBLEM 2.1.21 Show that if a normed space has a Schauder basis then it is separable.

Solution Let $\{x_n\}$ be a Schauder basis for a normed space X , and M be a countable dense subset of the scalar field $\mathbb{K}$. If $\mathbb{K} = \mathbb{R}$ we may take M as the set of all rationals, and if $\mathbb{K} = \mathbb{C}$ we may take M as the set of all complex numbers $\alpha + i\beta$ with α, β rationals. Let E_n be the set of all $x \in X$ expressible as

$$x = \sum_{i=1}^n m_i x_i, \quad m_i \in M.$$

The coefficients m_i are uniquely determined by x . Thus there is a one-one correspondence between E_n and M^n . Hence E_n is countable. Let

$$E = \bigcup_{n=1}^{\infty} E_n.$$

Then E is countable. We prove that E is dense in X .

Let $x \in X$. There exist k_n in $\mathbb{K}$, uniquely determined by x , such that

$$x = \sum_{n=1}^{\infty} k_n x_n.$$

Let $\epsilon > 0$ be given. Correspondingly, there is n such that

$$\left\| x - \sum_{j=1}^n k_j x_j \right\| < \epsilon/2.$$

Since M is dense in $\mathbb{K}$, there exist m_j in M such that

$$|k_j - m_j| < \frac{\epsilon}{2n} \quad \text{for } 1 \leq j \leq n.$$

Let $y = \sum_{j=1}^n m_j x_j$. Then, $y \in E_n \subset E$, and

$$\begin{aligned} x - y &= x - \sum_{j=1}^n k_j x_j + \sum_{j=1}^n (k_j - m_j) x_j, \\ \|x - y\| &\leq \left\| x - \sum_{j=1}^n k_j x_j \right\| + \sum_{j=1}^n |k_j - m_j| \|x_j\| \\ &< \frac{\epsilon}{2} + n \frac{\epsilon}{2n}, \end{aligned}$$

since $\|x_j\| = 1$ for all j . Therefore, $\|x - y\| < \epsilon$, $y \in E$. This proves that E is dense in X . Thus the proof is complete. ■

2.2 Uniform Boundedness Principle

Baire's theorem states that if E is a complete metric space then the intersection of a countable collection of dense open sets is dense in E . The fundamental theorems for Banach spaces are proved using this. The first of them is the *uniform boundedness principle* stated below:

Let X be a Banach space, Y be a normed space, $\mathcal{F}$ be a family of bounded linear maps from X to Y , and D be the set of all points x in X such that the set

$$\{\|F(x)\| : F \in \mathcal{F}\}$$

is unbounded. If D is not dense in X , then the set

$$\{\|F\| : F \in \mathcal{F}\}$$

is bounded. In particular, if D is empty, then $\{\|F\| : F \in \mathcal{F}\}$ is bounded.

In fact, the particular case implies the general case (see Problem 2.2.3).

One of the many interesting applications of the uniform boundedness principle is to show the existence of continuous functions on $[-\pi, \pi]$ whose Fourier series does not converge at a given t_0 in $[-\pi, \pi]$.

Let X be a complex normed linear space, and D be an open subset of the complex plane $\mathbb{C}$. A function $F : D \rightarrow X$ is said to be *analytic* on D if for every z_0 in D the limit

$$\lim_{z \rightarrow z_0} \frac{1}{z - z_0} (F(z) - F(z_0))$$

exists in X . A necessary and sufficient condition for F to be analytic on D is that $x' \circ F$ is analytic on D for every x' in X' .

A bounded linear map $F : c \rightarrow c$ is said to be *regular* or *permanent* if

$$\lim_{n \rightarrow \infty} F(x)(n) = \lim_{n \rightarrow \infty} x(n).$$

Not all such F can be represented by matrices (see Problem 1.2.22). However, an infinite matrix (k_{ij}) represents a regular bounded linear map $F : c \rightarrow c$ under the following conditions, known as *Toeplitz conditions*:

- (i) $\sup_{i \geq 1} \sum_{j=1}^{\infty} |k_{ij}| < \infty$;
- (ii) $\lim_{i \rightarrow \infty} k_{ij} = 0$ for $j = 1, 2, \dots$;
- (iii) $\lim_{i \rightarrow \infty} \sum_{j=1}^{\infty} k_{ij} = 1$.

EXERCISES

1. Let $X = \mathbb{R}$ and, for $n = 1, 2, \dots$,

$$f_n(t) = \begin{cases} n^2 t^{2n} (1 - t^2) & \text{for } |t| \leq 1 \\ 0 & \text{for } |t| > 1. \end{cases}$$

Show that $\{f_n : n \geq 1\}$ is a family of bounded continuous functions on X such that $\{f_n(t) : n \geq 1\}$ is bounded for each $t \in X$. Show also that $\{\|f_n\| : n \geq 1\}$ is not bounded where $\|f_n\| = \sup_{|t| \leq 1} |f_n(t)|$. Does this contradict the uniform boundedness principle?

2. Let X be a complex normed space and D be an open subset of the complex plane $\mathbb{C}$. Show that if $F : D \rightarrow X$ is analytic, as defined above, then $x' \circ F$ is analytic on D for every $x' \in X'$.

PROBLEMS

PROBLEM 2.2.1 Show that a Banach space cannot have a countably infinite basis.

Solution Suppose that X is an infinite dimensional Banach space having a countable basis $\{x_1, x_2, \dots\}$. Let

$$Y_n = \text{span}\{x_1, \dots, x_n\}, \quad n = 1, 2, \dots$$

and V_n be the complement of Y_n in X . Then,

$$\bigcup_{n=1}^{\infty} Y_n = X, \quad \bigcap_{n=1}^{\infty} V_n = \emptyset.$$

Hence, $\bigcap V_n$ cannot be dense in X . Since Y_n is finite dimensional, it is closed. Therefore, each V_n is open in X . As X is a complete metric space, Baire's

theorem implies that some V_n must be nondense in X , that is, $\overline{V}_N \neq X$ for some N .

Let $a \in X$ and $a \notin \overline{V}_N$. Then there is $r > 0$ such that the open ball $U(a, r)$ is disjoint from V_N , that is, $U(a, r) \subset Y_N$. Let x be any element in X . There is $\delta > 0$ such that $\delta\|x - a\| < r$. So,

$$y = a + \delta(x - a) \in U(a, r) \subset Y_N.$$

Since

$$x = \frac{y - a}{\delta} + a = \frac{1}{\delta}y + (1 - \frac{1}{\delta})a,$$

and $a \in Y_N$, we see that $x \in Y_N$. This shows $X = Y_N$. So X is finite dimensional. This contradiction proves the result. ■

PROBLEM 2.2.2 Show that c_{00} and the linear space of all polynomials in one variable are not Banach spaces under any norm.

Solution If e_n is the n -th unit vector, then $\{e_1, e_2, e_3, \dots\}$ is a basis for c_{00} . Let $p_n(t) = t^n$, $n = 0, 1, 2, \dots$. Then $\{p_0, p_1, p_2, \dots\}$ is a basis for the linear space of all polynomials in one variable t . But a Banach space cannot have a countably infinite basis by Problem 2.2.1. So c_{00} and the linear space of polynomials cannot be Banach spaces under any norm. ■

PROBLEM 2.2.3 Let X and Y be normed spaces, $\mathcal{F}$ be a family of bounded linear maps from X to Y , D be the set of all $x \in X$ such that the set

$$S_x = \{\|F(x)\| : F \in \mathcal{F}\} \quad (1)$$

is unbounded. Show that if D is not dense in X then it must be empty.

Solution Suppose that D is not dense in X . Then there is $a \in X$ with $a \notin \overline{D}$. So there is $r > 0$ such that the open ball $U(a, r)$ is disjoint from D . In particular, $a \notin D$. Let x be any element of X . We can find δ , with $0 < \delta < 1$, such that $\delta\|x - a\| < r$. Let $y = a + \delta(x - a)$. Then, $\|y - a\| = \delta\|x - a\| < r$, and so $y \notin D$. Thus we see that the sets S_a and S_y , as defined by (1) with $x = a, y$, are bounded. So there exist finite constants M_1, M_2 such that

$$\|F(a)\| \leq M_1, \quad \|F(y)\| \leq M_2$$

for all $F \in \mathcal{F}$. Since

$$F(y) = F(a) + \delta F(x) - \delta F(a),$$

we get

$$\delta\|F(x)\| = \|F(y)\| + (\delta - 1)\|F(a)\| \leq M_2 + (1 - \delta)M_1$$

for all $F \in \mathcal{F}$. This implies that $x \notin D$. Since this is true for any $x \in X$, we see that D is empty. ■

PROBLEM 2.2.4 Show that in the uniform boundedness principle on a family of bounded linear maps $F : X \rightarrow Y$, the assumption of completeness of X is essential.

Solution Let $X = c_{00}$ with the sup norm $\|\cdot\|_\infty$, $Y = \mathbb{K}$ and $F_n : X \rightarrow Y$ be defined by

$$F_n(x) = \sum_{j=1}^n x(j), \quad n = 1, 2, \dots$$

For each $x \in c_{00}$, there is m such that $x(j) = 0$ for all $j > m$. If $\min(m, n) = p$, then

$$F_n(x) = \sum_{j=1}^p x(j);$$

$$\|F_n(x)\| = |F_n(x)| \leq p\|x\|_\infty \leq m\|x\|_\infty.$$

Thus the set $\{\|F_n(x)\| : n = 1, 2, \dots\}$ is bounded for each x in X . So, if the uniform boundedness principle were true, then the set $\{\|F_n\| : n = 1, 2, \dots\}$ would be bounded.

Let $x(j) = 1$ for $1 \leq j \leq n$ and $x(j) = 0$ for $j > n$. Then, $x \in X$, $\|x\|_\infty = 1$ and $F_n(x) = n$. Hence,

$$n = |F_n(x)| \leq \|F_n\| \cdot \|x\|_\infty = \|F_n\|.$$

This shows that $\{\|F_n\| : n = 1, 2, \dots\}$ is unbounded. Thus, the uniform boundedness principle is false if $X = c_{00}$, which is not complete. ■

PROBLEM 2.2.5 Let X be a Banach space, Y be a normed space and $\{F_n\}$ be a sequence in $BL(X, Y)$ such that $\{F_n(x)\}$ converges in Y for each x in X . Show that if $F(x) = \lim_{n \rightarrow \infty} F_n(x)$, then $F \in BL(X, Y)$.

Solution If $x, y \in X$ and $k \in \mathbb{K}$, then

$$F_n(kx + y) = kF_n(x) + F_n(y), \quad n = 1, 2, \dots$$

Letting $n \rightarrow \infty$,

$$F(kx + y) = kF(x) + F(y).$$

This shows that F is linear. Apply the uniform boundedness principle to the family $\{F_n : n = 1, 2, \dots\}$. For each $x \in X$, the sequence $\{\|F_n(x)\|\}$ converges to $\|F(x)\|$, and is, therefore, bounded. So $\{\|F_n\| : n = 1, 2, \dots\}$ is bounded. If $\|F_n\| \leq \alpha$ for all n , then

$$\|F_n(x)\| \leq \|F_n\| \|x\| \leq \alpha\|x\|.$$

Letting $n \rightarrow \infty$,

$$\|F(x)\| \leq \alpha\|x\|$$

for each $x \in X$. So F is bounded. Thus, $F \in BL(X, Y)$. ■

PROBLEM 2.2.6 Suppose that E is a subset of a normed space X . Prove that E is bounded in X iff $x'(E)$ is a bounded set of scalars for every $x' \in X'$.

Solution Suppose that E is bounded in X . Let $\|x\| \leq \alpha$ for all x in E . If $x' \in X'$, then

$$|x'(x)| \leq \|x'\| \|x\| \leq \alpha \|x'\| \quad \text{for all } x \in E.$$

Therefore, $x'(E) = \{x'(x) : x \in E\}$ is a bounded set in $\mathbb{K}$ for any x' in X' .

Conversely, suppose that for each x' in X' the set $x'(E)$ is bounded in $\mathbb{K}$. For each $x \in X$, define $F : X' \rightarrow \mathbb{K}$ by

$$F_x(x') = x'(x), \quad x' \in X'.$$

Clearly, F_x is linear. Since

$$|F_x(x')| = |x'(x)| \leq \|x'\| \|x\|,$$

we see that F_x is bounded and $\|F_x\| \leq \|x\|$. On the other hand, we see by Problem 1.4.2 that if $x \neq 0$ then there is x' in X' such that $\|x'\| = 1$ and $x'(x) = \|x\|$, that is, such that

$$F_x(x') = \|x\|, \quad \|x'\| = 1.$$

It follows that $\|F_x\| = \|x\|$. This is true even if $x = 0$.

Since X' is a Banach space, we can apply the uniform boundedness principle to the family $\{F_x : x \in E\}$ of bounded linear maps from X' to $\mathbb{K}$. Since $x'(E)$ is bounded, we have

$$\|F_x(x')\| = |F_x(x')| = |x'(x)| \leq \beta$$

for all x in E and some β independent of x . Thus $\{F_x : x \in E\}$ is pointwise bounded. Hence, $\{\|F_x\| : x \in E\}$ is bounded; that is, $\{\|x\| : x \in E\}$ is bounded. This completes the proof. ■

PROBLEM 2.2.7 Let X and Y be normed spaces and $F : X \rightarrow Y$ be linear. Show that F is continuous iff $y' \circ F \in X'$ for every $y' \in Y'$.

Solution Since F is given to be linear, we see that $y' \circ F$ is linear for each $y' \in Y'$. If F is continuous then $y' \circ F$ is continuous.

Conversely, suppose that $y' \circ F$ is continuous for each $y' \in Y'$. Let $S = \{x \in X : \|x\| \leq 1\}$. Then $(y' \circ F)(S)$ is bounded in $\mathbb{K}$. Since this is true for all $y' \in Y'$, we see by Problem 2.2.6 that $E = F(S)$ is bounded in Y , that is, F is bounded. This completes the proof. ■

PROBLEM 2.2.8 Let $S : l^2 \rightarrow l^2$ be defined by

$$(Sx)(i) = x(i+2), i = 1, 2, \dots, x \in l^2,$$

and let $T_n = S^n$. Find

- (a) a bound for $\{\|T_n(x)\| : n \geq 1\}$
- (b) $\lim_{n \rightarrow \infty} \|T_n(x)\|$, and
- (c) $\lim_{n \rightarrow \infty} \|T_n\|$.

Solution (a) It follows by induction on n that

$$(S^n x)(i) = x(i + 2n), \quad i = 1, 2, \dots$$

Hence,

$$\|T_n x\|_2^2 = \|S^n x\|_2^2 = \sum_{i=1}^{\infty} |x(i + 2n)|^2 \leq \|x\|_2^2.$$

Thus, $\|T_n(x)\|_2 \leq \|x\|_2$ for all $n \geq 1$, that is, $\|x\|_2$ is an upper bound of $\{\|T_n x\|_2 : n = 1, 2, \dots\}$.

(b) If $x \in \ell^2$, then

$$\sum_{j=1}^{\infty} |x(j)|^2 < \infty,$$

and hence

$$\|T_n(x)\|_2^2 = \sum_{i=1}^{\infty} |x(i + 2n)|^2 = \sum_{j=2n+1}^{\infty} |x(j)|^2 \rightarrow 0$$

as $n \rightarrow \infty$. Thus,

$$\lim_{n \rightarrow \infty} \|T_n x\|_2 = 0, \quad x \in \ell^2.$$

(c) Since $\|T_n x\|_2 \leq \|x\|_2$ for all x in ℓ^2 , we have $\|T_n\| \leq 1$. It can be seen that

$$T_n(e_{2n+1}) = S^n(e_{2n+1}) = e_1,$$

where e_i is the i -th unit vector. Hence, $\|T_n(e_{2n+1})\|_2 = \|e_1\|_2 = 1$. It follows that $\|T_n\| = 1$ for any n . Thus, $\lim_{n \rightarrow \infty} \|T_n\| = 1$. ■

PROBLEM 2.2.9 Let $F_n \in BL(X, Y)$, where X is a Banach space and Y is any normed space. Show that the following statements are equivalent:

- (a) $\{\|F_n\| : n = 1, 2, \dots\}$ is bounded.
- (b) $\{\|F_n(x)\| : n = 1, 2, \dots\}$ is bounded for each $x \in X$.
- (c) $\{|g(F_n(x))| : n = 1, 2, \dots\}$ is bounded for each $x \in X$ and $g \in Y'$.

Solution Since X is Banach we can apply uniform boundedness principle to the family $\{F_n : n = 1, 2, \dots\}$. We get (b) $\implies$ (a).

If $\|F_n\| \leq \alpha$ for all n , then for x in X and g in Y' , we have

$$|g(F_n(x))| \leq \|g\| \|F_n(x)\| \leq \|g\| \|F_n\| \|x\| \leq \alpha \|g\| \|x\|.$$

This shows that (a) $\implies$ (c).

For a fixed x in X , let

$$E = \{F_n(x) : n = 1, 2, \dots\}.$$

If (c) is true, then $g(E)$ is bounded in $\mathbb{K}$ for each $g \in Y'$; hence E is bounded in Y by Problem 2.2.6. This shows that (c) $\implies$ (b). This completes the proof. ■

PROBLEM 2.2.10 Let X and Y be normed spaces, E be a bounded complete convex subset of X , and $\mathcal{F}$ be a set of continuous maps $F : E \rightarrow Y$ satisfying

$$F(rx + (1-r)y) = rF(x) + (1-r)F(y)$$

for $0 < r < 1$ and $x, y \in E$. Show that $\mathcal{F}$ is uniformly bounded on E iff it is pointwise bounded on E .

Solution If $\mathcal{F}$ is uniformly bounded on E , then it is obviously pointwise bounded on E . So suppose that $\mathcal{F}$ is pointwise bounded on E . This means that for each $x \in E$ there is a constant $M = M(x) < \infty$ such that

$$\|F(x)\| \leq M(x) \quad \text{for all } F \in \mathcal{F}.$$

For $F \in \mathcal{F}$, $n = 1, 2, \dots$, let

$$V_n(F) = \{x \in X : \|F(x)\| > n\},$$

$$V_n = \bigcup \{V_n(F) : F \in \mathcal{F}\},$$

$$E_n = E \cap V_n.$$

If $F \in \mathcal{F}$, then F is continuous, and hence the function $g : X \rightarrow \mathbb{R}$ defined by $g(x) = \|F(x)\|$ is continuous. It follows that $V_n(F)$ is open in X for all n, F . So V_n , being a union of open sets, is open. Thus E_n is an open subset of the metric space E .

Let $x \in E_n$. Then, $x \in E$ and $x \in V_n$; so, $x \in V_n(F)$ for some $F \in \mathcal{F}$. Thus,

$$\|F(x)\| > n \quad \text{for some } F.$$

But, $\|F(x)\| \leq M(x)$ for any F . Hence, $M(x) > n$ whenever $x \in E_n$. It follows that

$$\bigcap_{n=1}^{\infty} E_n = \emptyset.$$

Thus, $\cap E_n$ cannot be dense in E . But E is a complete metric space and each E_n is an open subset of E . Hence Baire's theorem implies that some E_n must be nondense in E ; that is, there is N with $\overline{E_N} \neq E$.

Let $a \in E$, $a \notin \overline{E_N}$. Then there is $r > 0$ such that the neighbourhood $E \cap U(a, r)$ of a is disjoint from E_N . Let $x \in E \cap U(a, r)$. Then $x \notin E_N$; hence,

$$x \notin V_N, \quad x \notin V_N(F) \quad \text{for any } F,$$

$$\|F(x)\| \leq N \quad \text{for all } F \in \mathcal{F}.$$

Since E is bounded, there is $R > r$ such that $E \subset U(a, R)$. Let $x \in E$, $F \in \mathcal{F}$ and

$$y = a + \frac{r}{R}(x - a) = \frac{r}{R}x + (1 - \frac{r}{R})a.$$

Then, $y \in E$ since E is convex, and

$$\begin{aligned}\|y - a\| &= \frac{r}{R}\|x - a\| < r, \\ F(y) &= \frac{r}{R}F(x) + (1 - \frac{r}{R})F(a).\end{aligned}$$

Hence,

$$\begin{aligned}\|\frac{r}{R}F(x)\| &\leq \|F(y)\| + \|(1 - \frac{r}{R})F(a)\| \\ &\leq N + (1 - \frac{r}{R})N,\end{aligned}$$

since y and a belong to $E \cap U(a, r)$. Thus,

$$\|F(x)\| \leq \frac{R}{r}N + (\frac{R}{r} - 1)N.$$

This is true for all $x \in E$ and $F \in \mathcal{F}$. Hence, $\mathcal{F}$ is uniformly bounded on E . This completes the proof. ■

PROBLEM 2.2.11 Let X be a complete metric space and $\mathcal{F}$ set of continuous complex-valued functions on X . Prove that either (i) there exists a dense subset D of X such that

$$\sup\{|F(x)| : F \in \mathcal{F}\} = \infty \quad \text{for each } x \in D,$$

or (ii) there exists a nonempty open ball U in X such that

$$\sup\{|F(x)| : F \in \mathcal{F}, x \in U\} < \infty.$$

Solution For $n = 1, 2, 3, \dots$, $F \in \mathcal{F}$, let

$$\begin{aligned}Y_n(F) &= \{x \in X : |F(x)| > n\}, \\ V_n &= \bigcup\{Y_n(F) : F \in \mathcal{F}\}, \\ D &= \bigcap_{n=1}^{\infty} V_n.\end{aligned}$$

Then each V_n is open in X , as in the proof of Problem 2.2.10. Since X is a complete metric space, Baire's theorem implies that either (a) D is dense in X or (b) some V_n is nondense in X , that is, $\overline{V_n} \neq X$.

Let D be dense in X , and

$$s(x) = \sup\{|F(x)| : F \in \mathcal{F}\}.$$

If $x \in V_n$, then $x \in Y_n(F)$ for some $F \in \mathcal{F}$, and hence

$$s(x) \geq |F(x)| > n.$$

Thus, if $x \in D$, then $x \in V_n$ for all n , and so

$$s(x) > n \quad \text{for all } n, \quad s(x) = \infty.$$

We have the first alternative.

Let $\bar{V}_N \neq X$. Then there is $a \in X$, with $a \notin \bar{V}_N$. Hence there is $r > 0$ such that the open ball $U(a, r)$ is disjoint from V_N . So if $x \in U(a, r)$, then $x \notin V_N$, and hence

$$x \notin V_N(F) \quad \text{for any } F \in \mathcal{F},$$

$$|F(x)| \leq N \quad \text{for all } F \in \mathcal{F}.$$

Thus, $|F(x)| \leq N$ for all $F \in \mathcal{F}$ and $x \in U = U(a, r)$. We have the second alternative. This completes the proof. ■

PROBLEM 2.2.12 Let $\{f_n\}$ be a sequence of complex-valued continuous functions on the real line which converges at every point. Prove that there is an interval I and a finite real number M such that $|f_n(x)| < M$ for all $x \in I$ and $n = 1, 2, 3, \dots$

Solution This follows from Problem 2.2.11 with $X = \mathbb{R}$ and $\mathcal{F} = \{f_n : n = 1, 2, \dots\}$. Since $\{f_n(x)\}$ converges, it is bounded for each x . So the alternative (i) of Problem 2.2.11 is not possible here. ■

PROBLEM 2.2.13 Let $\{k_n\}$ be a sequence of scalars. Prove that

(a) if $\sum k_n x(n)$ converges for every $x \in l^1$ then $\sup_{n \geq 1} |k_n| < \infty$;

(b) if $\sum k_n x(n)$ converges for every $x \in c_0$ then $\sum_{n=1}^{\infty} |k_n| < \infty$.

Solution (a) For $n = 1, 2, \dots$, let

$$f_n(x) = \sum_{j=1}^n k_j x(j), \quad x \in l^1.$$

Then, $f_n : l^1 \rightarrow \mathbb{K}$ is linear and

$$|f_n(x)| \leq \alpha_n \sum_{j=1}^n |x(j)| \leq \alpha_n \|x\|,$$

where $\alpha_n = \max_{1 \leq j \leq n} |k_j|$. So f_n is bounded. Since l^1 is a Banach space, we can apply the uniform boundedness principle to $\{f_n : n = 1, 2, \dots\}$. For each $x \in l^1$, the sequence $\{f_n(x)\}$ is convergent and hence bounded in $\mathbb{K}$. Consequently, $\{\|f_n\| : n = 1, 2, \dots\}$ is bounded. Let $\|f_n\| \leq \alpha$ for all n . If $x = e_j$, the j -th unit vector, then $\|x\|_1 = 1$ and $f_n(x) = k_j$ for $n \geq j$. Hence, if $n \geq j$, then

$$|k_j| = |f_n(x)| \leq \|f_n\| \cdot \|x\|_1 \leq \alpha.$$

Since this is true for any j , we get

$$\sup_{j \geq 1} |k_j| \leq \alpha < \infty.$$

(b) For $n = 1, 2, \dots$, let

$$f_n(x) = \sum_{j=1}^n k_j x(j), \quad x \in c_0.$$

Then, as before, $f_n : c_0 \rightarrow \mathbb{K}$ is a bounded linear map with

$$\|f_n\| \leq \sum_{j=1}^n |k_j|.$$

As c_0 is a Banach space, we can apply the uniform boundedness principle to $\{f_n : n \geq 1\}$. Since $\{f_n(x)\}$ is convergent and hence bounded, we see that $\{\|f_n\| : n \geq 1\}$ is bounded. Let $\|f_n\| \leq \alpha$ for all n . For each j choose k'_j such that

$$k_j k'_j = |k_j|, \quad |k'_j| = 1.$$

Let $x_n = (k'_1, \dots, k'_n, 0, 0, \dots)$. Then, $x_n \in c_0$, $\|x_n\|_\infty = 1$ and

$$f_n(x_n) = \sum_{j=1}^n k_j x_n(j) = \sum_{j=1}^n k_j k'_j = \sum_{j=1}^n |k_j|.$$

Hence,

$$\sum_{j=1}^n |k_j| = |f_n(x_n)| \leq \|f_n\| \cdot \|x_n\|_\infty = \|f_n\| \leq \alpha,$$

$$\sum_{j=1}^{\infty} |k_j| \leq \alpha < \infty.$$

This completes the proof. ■

PROBLEM 2.2.14 Let X be a Banach space, Y be a normed space and $F_{nm} \in BL(X, Y)$ for $n, m = 1, 2, \dots$. If for each $m \geq 1$, there is x_m in X such that

$$\sup_{n \geq 1} \|F_{nm}(x_m)\| = \infty,$$

prove that there exists x in X such that

$$\sup_{n \geq 1} \|F_{nm}(x)\| = \infty \quad \text{for } m = 1, 2, \dots$$

Solution For $N, n, m \geq 1$, let

$$V_N(F_{nm}) = \{x \in X : \|F_{nm}(x)\| > N\},$$

$$V_{N,m} = \bigcup_{n=1}^{\infty} V_N(F_{nm}),$$

$$D = \bigcap_{N=1}^{\infty} \bigcap_{m=1}^{\infty} V_{N,m};$$

$$s_m(x) = \sup_{n \geq 1} \|F_{nm}(x)\|, \quad x \in X.$$

Then, $s_m(x_m) = \infty$ for each m . Each $V_{N,m}$ is an open subset of X , as in the proof of Problem 2.2.10. Since X is complete, Baire's theorem implies that either (i) D is dense in X or (ii) $\bar{V}_{N,m} \neq X$ for some N, m .

Suppose that D is dense in X . Then D is nonempty. Let $a \in D$. Then, $a \in V_{N,m}$ for all $N, m \geq 1$. So for each N, m , there is n such that

$$a \in V_N(F_{nm});$$

that is, $\|F_{nm}(a)\| > N$. Hence, $s_m(a) \geq \|F_{nm}(a)\| > N$. Thus for each m , we have

$$s_m(a) > N \quad \text{for all } N;$$

that is, $s_m(a) = \infty$.

Next suppose that $\bar{V}_{N,m} \neq X$ for some N, m . Then there is an open ball $U(a, r)$, with $r > 0$, which is disjoint from $V_{N,m}$. So if $x \in U(a, r)$, then $x \notin V_{N,m}$, and hence $x \notin V_N(F_{nm})$ for any n , that is,

$$\|F_{nm}(x)\| \leq N \quad \text{for any } n.$$

Choose $t > 0$ such that $t\|x_m - a\| < r$, and let

$$y = a + t(x_m - a) = (1 - t)a + tx_m.$$

Then, for all $n \geq 1$,

$$F_{nm}(y) = (1 - t)F_{nm}(a) + tF_{nm}(x_m).$$

Hence,

$$\begin{aligned} \|tF_{nm}(x_m)\| &\leq \|F_{nm}(y)\| + \|(1 - t)F_{nm}(a)\| \\ &\leq N + |1 - t| \cdot N, \end{aligned}$$

since y and a belong to $U(a, r)$. Thus,

$$t\|F_{nm}(x_m)\| \leq N + (1 + t)N = (2 + t)N,$$

$$\|F_{nm}(x_m)\| \leq \frac{2 + t}{t}N,$$

for all $n \geq 1$. This implies that

$$s_m(x_m) \leq \frac{2+t}{t}N.$$

But $s_m(x_m) = \infty$ by hypothesis. This contradiction shows that alternative (ii) is impossible. This completes the proof. ■

PROBLEM 2.2.15 Let X, Y, Z be normed spaces, of which X or Y is Banach. For $F : X \times Y \rightarrow Z$, define $F_x : Y \rightarrow Z$ and $F^y : X \rightarrow Z$ by

$$F_x(y) = F(x, y) = F^y(x), \quad x \in X, y \in Y.$$

If $F_x \in BL(Y, Z)$ for all $x \in X$ and $F^y \in BL(X, Z)$ for all $y \in Y$, show that F is jointly continuous, and

$$\|F(x, y)\| \leq \alpha \|x\| \|y\| \quad \text{for } x \in X, y \in Y,$$

where α is a constant.

Solution Suppose that X is Banach. Let $E = \{y \in Y : \|y\| \leq 1\}$. Consider the family $\{F^y : y \in E\}$ of bounded linear maps from X to Z . For $x \in X, y \in E$,

$$\|F^y(x)\| = \|F(x, y)\| = \|F_x(y)\| \leq \|F_x\| \cdot \|y\| \leq \|F_x\|.$$

This shows that this family is pointwise bounded on X . Hence the uniform boundedness principle implies that $\{\|F^y\| : y \in E\}$ is a bounded set. Let $\|F^y\| \leq \alpha$ for all $y \in E$.

For $x \in X, y \in Y, y \neq 0$ let $z = y/\|y\|$. Then $z \in E$, and so,

$$\|F(x, z)\| = \|F^z(x)\| \leq \|F^z\| \cdot \|x\| \leq \alpha \|x\|.$$

However,

$$F(x, z) = F_x(z) = \frac{1}{\|y\|} F_x(y) = \frac{1}{\|y\|} F(x, y).$$

Hence,

$$\|F(x, y)\| = \|y\| \|F(x, z)\| \leq \alpha \|x\| \|y\|.$$

Since $F(x, 0) = F_x(0) = 0$, this is true when $y = 0$ also. Therefore,

$$\|F(x, y)\| \leq \alpha \|x\| \|y\| \quad \text{for all } x \in X, y \in Y.$$

To prove that F is jointly continuous, let $(x_n, y_n) \rightarrow (x, y)$ in $X \times Y$. Then, $x_n \rightarrow x$ in X and $y_n \rightarrow y$ in Y . Since

$$F(x_n, y_n) - F(x, y_n) = F^{y_n}(x_n - x) = F(x_n - x, y_n),$$

$$F(x, y_n) - F(x, y) = F_x(y_n - y) = F(x, y_n - y),$$

we get

$$\begin{aligned}\|F(x_n, y_n) - F(x, y)\| &\leq \|F(x_n - x, y_n)\| + \|F(x, y_n - y)\| \\ &\leq \alpha \|x_n - x\| \|y_n\| + \alpha \|x\| \|y_n - y\| \\ &\rightarrow 0 \quad \text{as } n \rightarrow \infty.\end{aligned}$$

Thus F is continuous at (x, y) .

If Y is Banach, we can consider the family $\{F_x : x \in X, \|x\| \leq 1\}$ of bounded linear maps from Y to Z and proceed as above. This completes the proof. ■

PROBLEM 2.2.16 Let X be a normed space over $\mathbb{C}$. If $F : \mathbb{C} \rightarrow X$ is such that $x' \circ F$ is bounded and analytic on $\mathbb{C}$ for every x' in X' , show that F is constant.

Solution Given that $x' \circ F$ is a bounded entire function. Hence it is constant. If $z_1, z_2 \in \mathbb{C}$ and $F(z_1) \neq F(z_2)$, let $a = F(z_1) - F(z_2)$. Then $a \in X, a \neq 0$. So there is x' in X' such that $x'(a) = \|a\| \neq 0$, by Problem 1.4.2. Since $x' \circ F$ is constant we have

$$(x' \circ F)(z_1) = (x' \circ F)(z_2),$$

that is,

$$x'(a) = x'(F(z_1)) - x'(F(z_2)) = 0.$$

This contradiction shows that $F(z_1) = F(z_2)$. This proves the result. ■

PROBLEM 2.2.17 (a) Let (k_{ij}) be an infinite matrix of scalar entries such that

$$\sup_{i \geq 1} \sum_{j=1}^{\infty} |k_{ij}| < \infty. \quad (2)$$

Show that (k_{ij}) represents a bounded linear map $F : l^\infty \rightarrow l^\infty$ in the sense that

$$F(x)(i) = \sum_{j=1}^{\infty} k_{ij} x(j), \quad i = 1, 2, \dots, \quad (3)$$

the series being convergent for all $i \geq 1$ and x in l^∞ .

(b) On the other hand, if an infinite matrix (k_{ij}) is such that (3) gives a map F from c_0 into l^∞ , show that (2) holds.

Solution (a) Let

$$\alpha_i = \sum_{j=1}^{\infty} |k_{ij}|, \quad \alpha = \sup_{i \geq 1} \alpha_i.$$

If $x \in l^\infty$, then

$$\sum_{j=1}^{\infty} |k_{ij} x(j)| \leq \sum_{j=1}^{\infty} |k_{ij}| \|x\|_\infty = \alpha_i \|x\|_\infty.$$

Thus the series in (3) converges for each $i \geq 1$ and

$$|F(x)(i)| \leq \alpha_i \|x\|_\infty \leq \alpha \|x\|_\infty.$$

Hence, $F(x) \in l^\infty$ and $\|F(x)\|_\infty \leq \alpha \|x\|_\infty$. Thus, (3) defines a map $F : l^\infty \rightarrow l^\infty$, which is linear and bounded.

(b) First, suppose $\alpha_i = \infty$ for some i . It is known that if $\sum a_n$ is a divergent series of non-negative terms and $\{s_n\}$ is its sequence of partial sums, then

$$\sum \frac{a_n}{s_n} = \infty, \quad \frac{1}{s_n} \rightarrow 0.$$

[See the solution of Problem 2.3.18.] So we can find $b_j \geq 0$ such that

$$\sum_{j=1}^{\infty} |k_{ij}| b_j = \infty, \quad b_j \rightarrow 0 \quad \text{as } j \rightarrow \infty.$$

Let

$$x(j) = \begin{cases} \frac{|k_{ij}|}{k_{ij}} b_j & \text{if } k_{ij} \neq 0, \\ b_j & \text{if } k_{ij} = 0. \end{cases}$$

Then, $|x(j)| = b_j \rightarrow 0$ as $j \rightarrow \infty$, and

$$F(x)(i) = \sum_{j=1}^{\infty} k_{ij} x(j) = \sum_{j=1}^{\infty} |k_{ij}| b_j = \infty.$$

So, $F(x)$ does not exist even though $x \in c_0$. This contradiction shows that $\alpha_i < \infty$ for all $i \geq 1$.

For $i = 1, 2, \dots$, let $f_i(x) = F(x)(i)$, $x \in c_0$. Then, $f_i : c_0 \rightarrow \mathbb{K}$ is linear, and

$$|f_i(x)| = |F(x)(i)| \leq \alpha_i \|x\|_\infty,$$

as before. Thus f_i is bounded. Since c_0 is a Banach space, we can apply the uniform boundedness principle to the family $\{f_i : i = 1, 2, \dots\}$. Since

$$|f_i(x)| = |F(x)(i)| \leq \|F(x)\|_\infty \quad \text{for all } i,$$

this family is pointwise bounded on c_0 . Hence, $\{\|f_i\| : i \geq 1\}$ is bounded. Let $\|f_i\| \leq \beta$ for all i . For fixed i , and $n = 1, 2, \dots$, let

$$x_n(j) = \begin{cases} \frac{|k_{ij}|}{k_{ij}} & \text{if } 1 \leq j \leq n, \quad k_{ij} \neq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Then, $x_n \in c_0$, $\|x_n\|_\infty \leq 1$, and

$$f_i(x_n) = \sum_{j=1}^n k_{ij} x_n(j) = \sum_{j=1}^n |k_{ij}|,$$

$$\sum_{j=1}^n |k_{ij}| = |f_i(x_n)| \leq \|f_i\| \cdot \|x_n\|_\infty \leq \|f_i\|.$$

Letting $n \rightarrow \infty$, we get

$$\alpha_i \leq \|f_i\| \leq \beta.$$

Since this is true for all $i \geq 1$, we get (2). This completes the proof. ■

PROBLEM 2.2.18 (See Problem 1.2.21.) Show that an infinite matrix (k_{ij}) represents a bounded linear map $F : c_0 \rightarrow c_0$ iff

$$\sup_{i \geq 1} \sum_{j=1}^{\infty} |k_{ij}| < \infty, \quad (4)$$

$$\lim_{i \rightarrow \infty} k_{ij} = 0 \quad \text{for } j = 1, 2, \dots \quad (5)$$

Solution Let

$$\alpha_i = \sum_{j=1}^{\infty} |k_{ij}|, \quad \alpha = \sup_{i \geq 1} \alpha_i$$

First suppose that conditions (4) and (5) hold. If (4) holds, then (k_{ij}) represents a bounded linear map $F : l^\infty \rightarrow l^\infty$ by Problem 2.2.17(a). So it is enough if we prove that $F(x) \in c_0$ for all $x \in c_0$. Let $x \in c_0$ and $\epsilon > 0$ be given. There is m such that $|x(j)| < \epsilon$ for all $j \geq m$. So for $i = 1, 2, \dots$,

$$\begin{aligned} \sum_{j=1}^{\infty} |k_{ij}x(j)| &\leq \sum_{j=1}^{m-1} |k_{ij}||x(j)| + \sum_{j=m}^{\infty} |k_{ij}|\epsilon \\ &\leq \|x\|_\infty \sum_{j=1}^{m-1} |k_{ij}| + \epsilon \alpha_i. \end{aligned}$$

By (5) the last sum tends to zero as $i \rightarrow \infty$. Hence,

$$|F(x)(i)| \leq \|x\|_\infty \cdot \epsilon + \epsilon \alpha \quad \text{for } i \geq i_0.$$

This shows that $F(x) \in c_0$.

Conversely, suppose that (k_{ij}) represents a bounded linear map $F : c_0 \rightarrow c_0$. Since $c_0 \subset l^\infty$, F maps c_0 into l^∞ . So (4) holds by Problem 2.2.17(b). If e_n is the n -th unit vector, then $F(e_n) \in c_0$ since $e_n \in c_0$. But $F(e_n)(i) = k_{in}$. Hence, $k_{in} \rightarrow 0$ as $i \rightarrow \infty$. Thus (5) holds. ■

PROBLEM 2.2.19 (See Problem 1.2.22.) Show that an infinite matrix (k_{ij}) represents a bounded linear map $F : c \rightarrow c$ iff

$$\sup_{i \geq 1} \sum_{j=1}^{\infty} |k_{ij}| < \infty, \quad (6)$$

$$\lim_{i \rightarrow \infty} \sum_{j=r}^{\infty} k_{ij} \quad \text{exists for } r = 1, 2, \dots \quad (7)$$

Solution First suppose the conditions (6) and (7) hold. Let

$$\alpha_i = \sum_{j=1}^{\infty} |k_{ij}|, \quad \alpha = \sup_{i \geq 1} \alpha_i,$$

$$\beta_i(r) = \sum_{j=r}^{\infty} k_{ij}, \quad \beta(r) = \lim_{i \rightarrow \infty} \beta_i(r).$$

If (6) holds, then Problem 2.2.17(a) shows that (k_{ij}) represents a bounded linear map $F : l^\infty \rightarrow l^\infty$. So it is enough if we prove that $F(x) \in c$ for all $x \in c$. Let $x \in c$, $\lambda = \lim_{j \rightarrow \infty} x(j)$ and $\epsilon > 0$ be given. There is m such that

$$|x(j) - \lambda| < \epsilon \quad \text{for all } j \geq m.$$

Fix m and let

$$u_i = \sum_{j=1}^{m-1} k_{ij}x(j) + \sum_{j=m}^{\infty} k_{ij}\lambda,$$

$$v_i = \sum_{j=m}^{\infty} k_{ij}(x(j) - \lambda).$$

Then, $u_i + v_i = F(x)(i)$, and

$$|v_i| \leq \sum_{j=m}^{\infty} |k_{ij}| \epsilon \leq \epsilon \alpha_i \leq \epsilon \alpha \quad \text{for all } i.$$

Since $k_{ij} = \beta_i(j) - \beta_i(j+1) \rightarrow \beta(j) - \beta(j+1)$, we see that $u_i \rightarrow u$ as $i \rightarrow \infty$, where

$$u = \sum_{j=1}^{m-1} (\beta(j) - \beta(j+1))x(j) + \lambda\beta(m).$$

So there exists i_0 such that

$$|u_i - u_r| < \epsilon \quad \text{for all } i \geq i_0, r \geq i_0.$$

Hence,

$$\begin{aligned} |F(x)(i) - F(x)(r)| &\leq |u_i - u_r| + |v_i - v_r| \\ &< \epsilon + |v_i| + |v_r| \\ &< \epsilon + \epsilon\alpha + \epsilon\alpha \end{aligned}$$

whenever $i \geq i_0$, $r \geq i_0$. This shows that $\lim_{i \rightarrow \infty} F(x)(i)$ exists; hence $F(x) \in c$.

Conversely, suppose that (k_{ij}) represents a bounded linear map $F : c \rightarrow c$. Since $c_0 \subset c \subset l^\infty$, F maps c_0 into l^∞ . So (6) holds by Problem 2.2.17(b). Let

$$x_r(j) = \begin{cases} 0 & \text{for } 1 \leq j < r \\ 1 & \text{for } j \geq r. \end{cases}$$

Then $x_r \in c$ and so $F(x_r) \in c$. But,

$$F(x_r)(i) = \sum_{j=r}^{\infty} k_{ij} = \beta_i(r).$$

Therefore, $\lim_{i \rightarrow \infty} \beta_i(r) = \lim_{i \rightarrow \infty} F(x_r)(i)$ exists. Thus (7) holds. This completes the proof. ■

PROBLEM 2.2.20 (See Problems 1.4.12, 1.2.20.) Show that an infinite matrix (k_{ij}) represents a bounded linear map $F : l^\infty \rightarrow l^\infty$ iff

$$\sup_{i \geq 1} \sum_{j=1}^{\infty} |k_{ij}| < \infty, \quad (8)$$

and that it represents a bounded linear map $F : l^1 \rightarrow l^1$ iff

$$\sup_{j \geq 1} \sum_{i=1}^{\infty} |k_{ij}| < \infty. \quad (9)$$

Solution The first statement follows from Problem 2.2.17. For the second, suppose that (9) holds. Let

$$\lambda_j = \sum_{i=1}^{\infty} |k_{ij}|, \quad \lambda = \sup_{j \geq 1} \lambda_j.$$

If $\lambda < \infty$ and $x \in l^1$, then

$$\begin{aligned} \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} |k_{ij}x(j)| &= \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} |k_{ij}| |x(j)| \\ &= \sum_{j=1}^{\infty} |x(j)| \lambda_j \leq \lambda \sum_{j=1}^{\infty} |x(j)|. \end{aligned}$$

Since this is finite, we see that

$$F(x)(i) = \sum_{j=1}^{\infty} k_{ij}x(j) \quad (10)$$

exists for $i = 1, 2, \dots$, and that

$$\sum_{i=1}^{\infty} |F(x)(i)| \leq \lambda \|x\|_1 < \infty. \quad (11)$$

Hence, $F(x) \in l^1$. Thus, (k_{ij}) represents a map $F : l^1 \rightarrow l^1$, as given by (10). Clearly, F is linear, and (11) shows that $\|F(x)\| \leq \lambda \|x\|_1$. So F is bounded.

Conversely, suppose that (k_{ij}) represents a bounded linear map $F : l^1 \rightarrow l^1$, as given by (10). Then, $F(e_j) \in l^1$ since $e_j \in l^1$. As $F(e_j)(i) = k_{ij}$,

$$\sum_{i=1}^{\infty} |k_{ij}| = \sum_{i=1}^{\infty} |F(e_j)(i)| = \|F(e_j)\|_1 \leq \|F\|.$$

Thus, if $y \in l^\infty$, then

$$\sum_{i=1}^{\infty} |k_{ij}y(i)| \leq \sum_{i=1}^{\infty} |k_{ij}| \|y\|_\infty \leq \|F\| \|y\|_\infty.$$

Hence,

$$G(y)(j) = \sum_{i=1}^{\infty} k_{ij}y(i) \tag{12}$$

exists for each $j \geq 1$, and $|G(y)(j)| \leq \|F\| \|y\|_\infty$. So (12) defines an element $G(y)$ in l^∞ for each y in l^∞ . Thus we see that if $k'_{ji} = k_{ij}$ then the infinite matrix (k'_{ij}) represents a map G from l^∞ into l^∞ . Therefore, by Problem 2.2.17(b),

$$\sup_{j \geq 1} \sum_{i=1}^{\infty} |k'_{ji}| < \infty.$$

Since this is the same as (9), the proof is complete. ■

2.3 Open Mapping Theorem and Closed Graph Theorem

In this section we deal with the remaining two fundamental theorems for Banach spaces, namely, the open mapping theorem and the closed graph theorem.

The *open mapping theorem* states that if X and Y are Banach spaces and F is a bounded linear map from X onto Y , then F is an open mapping; that is, F maps open subsets of X onto open subsets of Y . A converse of this is given by Problem 2.3.1.

The following result, which may be called *bounded inverse theorem*, is a special case of open mapping theorem: If $F \in BL(X, Y)$, where X and Y are Banach spaces, and F is bijective then $F^{-1} \in BL(Y, X)$.

Recall that if X and Y are normed spaces then $X \times Y$ is a normed space with

$$\|(x, y)\| = \|x\| + \|y\|.$$

The *graph* of a map $F : X \rightarrow Y$ is the set

$$\{(x, y) : x \in X, y = F(x)\}.$$

This subset of $X \times Y$ is closed iff

$$x_n \rightarrow x \quad \text{and} \quad F(x_n) \rightarrow y \quad \implies \quad y = F(x).$$

Hence if F is continuous, then its graph is closed. Even discontinuous linear maps can have a closed graph (see Problem 2.3.3 below). The *closed graph theorem* is the following: If X and Y are Banach spaces, $F : X \rightarrow Y$ is linear and the graph of F is a closed subset of $X \times Y$, then F is continuous.

It can be seen that if F is bijective and the graph of F is closed in $X \times Y$, then the graph of F^{-1} is closed in $Y \times X$. So the special case of the open mapping theorem is also a corollary of the closed graph theorem.

EXERCISES

1. Let $X = [0, 2\pi)$ and Y be the unit circle in $\mathbb{R}^2$. Define $F : X \rightarrow Y$ by $F(t) = (\cos t, \sin t)$ and let $E = [0, 1)$. Show that F is onto but $F(E)$ is not open in Y even though E is open in X . Does this contradict the open mapping theorem?
2. Show that the graph of a map $F : X \rightarrow Y$ is a closed subset of $X \times Y$ iff the following condition is satisfied:

$$x_n \rightarrow x \quad \text{and} \quad F(x_n) \rightarrow y \quad \implies \quad y = F(x).$$

3. Let X and Y be normed spaces, $F : X \rightarrow Y$ be linear and X be finite dimensional. Show that the graph of F is a finite dimensional subspace of $X \times Y$. Deduce, using the closed graph theorem, that F is continuous.

PROBLEMS

PROBLEM 2.3.1 Let X, Y be normed spaces, and $F \in BL(X, Y)$. Show that if F maps open subsets of X onto open subsets of Y then F is onto.

Solution Let $U = \{x \in X : \|x\| < 1\}$. Then U is open in X and hence $F(U)$ is open in Y . Since $0 = F(0) \in F(U)$, there is $r > 0$ such that

$$S = \{y \in Y : \|y\| < r\} \subset F(U).$$

Let $y \in Y$, $y \neq 0$ and $y_1 = \frac{r}{2\|y\|}y$. Then, $\|y_1\| = \frac{1}{2}r < r$, and so $y_1 \in F(U)$; that is, $y_1 = F(x_1)$ with $x_1 \in U$. Hence

$$y = \frac{2\|y\|}{r}y_1 = F(x_2) \in F(X),$$

where $x_2 = \frac{2\|y\|}{r}x_1$. This proves that F is onto. ■

PROBLEM 2.3.2 From the special case of open mapping theorem, deduce

- (a) the open mapping theorem, and
- (b) the closed graph theorem.

Solution (a) Suppose that X and Y are Banach spaces, $F \in BL(X, Y)$, and F is onto. Let $Z = Z(F)$, the zero space of F . Then Z is closed in X since F is continuous. Hence the quotient space X/Z is a Banach space under the quotient norm $\|\cdot\|$.

Define $\tilde{F} : X/Z \rightarrow Y$ as follows: Let $\tilde{F}(x + Z) = F(x)$. If $x + Z = x_1 + Z$, then

$$x - x_1 \in Z, \quad F(x - x_1) = 0, \quad F(x) = F(x_1).$$

Thus $\tilde{F}$ is a well-defined map. It is onto since F is onto. It is linear since F is linear, and is one-one since $Z = Z(F)$.

Let $x_n + Z \rightarrow x + Z$ in X/Z . Then using Problem 1.1.10, we see that there is a sequence $\{z_n\}$ in Z such that $x_n + z_n \rightarrow x$ in X . So, $F(x_n + z_n) \rightarrow F(x)$ by the continuity of F . Hence,

$$\tilde{F}(x_n + Z) = F(x_n) = F(x_n + z_n) \rightarrow F(x),$$

that is, $\tilde{F}(x_n + Z) \rightarrow \tilde{F}(x + Z)$. This shows that $\tilde{F}$ is continuous. So by the theorem we have assumed, its inverse $G : Y \rightarrow X/Z$ is a bounded linear map.

Let E be an open set in X . To show that $F(E)$ is open in Y , let $y_0 \in F(E)$. Then, $y_0 = F(x_0)$ with $x_0 \in E$. Since E is open, there is $r > 0$ such that

$$\|x - x_0\| < r \implies x \in E.$$

Let $\|y - y_0\| < r/\|G\|$. Since F is onto, $y = F(x)$ for some x . Hence,

$$\begin{aligned} (x + Z) - (x_0 + Z) &= G(y) - G(y_0) = G(y - y_0), \\ \|x - x_0 + Z\| &= \|G(y - y_0)\| \leq \|G\| \cdot \|y - y_0\| < r. \end{aligned}$$

So there is $z \in Z$ such that

$$\|x - x_0 + z\| < r,$$

by the definition of the quotient norm. Hence, $x + z \in E$, and, therefore,

$$y = F(x) = F(x + z) \in F(E).$$

This is true whenever $\|y - y_0\| < r/\|G\|$. Thus, $F(E)$ is open.

(b) Suppose that X and Y are Banach spaces, $F : X \rightarrow Y$ is linear, and the graph

$$W = \{(x, y) : x \in X, y = F(x)\}$$

is a closed subset of $X \times Y$. The linearity of F shows that W is a linear subspace of $X \times Y$. It is a Banach space since $X \times Y$ is Banach. Define $T : W \rightarrow X$ by

$$T(x, y) = x \quad \text{for } (x, y) \in W.$$

It can be seen that T is onto, one-one and linear. Since

$$\|T(x, y)\| = \|x\| \leq \|x\| + \|y\| = \|(x, y)\|,$$

we see that T is continuous with $\|T\| \leq 1$. So by the theorem we have assumed, its inverse $S : X \rightarrow W$ is a bounded linear map. If $x \in X$ and $y = F(x)$, then

$$(x, y) \in W, \quad T(x, y) = x, \quad S(x) = (x, y),$$

and hence

$$\begin{aligned} \|x\| + \|y\| &= \|(x, y)\| = \|S(x)\| \leq \|S\| \cdot \|x\|, \\ \|F(x)\| &= \|y\| \leq \|x\| + \|y\| \leq \|S\| \cdot \|x\|. \end{aligned}$$

This shows that F is continuous with $\|F\| \leq \|S\|$. This gives the result. ■

PROBLEM 2.3.3 Show that even a discontinuous linear map can have a closed graph. Does this contradict the closed graph theorem?

Solution Let $X = C^1[0, 1]$ and $Y = C[0, 1]$ where both have the sup norm $\|\cdot\|_\infty$. Let $F(x) = x'$, the derivative of x . Then, $F : X \rightarrow Y$ is linear. If $x_n(t) = t^n$, then $x'_n(t) = nt^{n-1}$, and so

$$\begin{aligned} \|x_n\|_\infty &= 1 \\ \|F(x_n)\|_\infty &= \|x'_n\|_\infty = n = n\|x_n\|_\infty. \end{aligned}$$

This shows that F is not continuous.

To show that the graph of F is closed, let $x_n \rightarrow x$ in X and $F(x_n) \rightarrow y$ in Y . But convergence with respect to the sup norm is uniform convergence. So, $x'_n = F(x_n) \rightarrow y$ uniformly. It follows that $x' = y$, that is, $y = F(x)$. Thus the graph of F is closed.

Here, $X = C^1[0, 1]$ is not a Banach space under the sup norm. So the hypothesis of the closed graph theorem is not satisfied. The closed graph theorem is not applicable. Therefore, there is no contradiction. ■

PROBLEM 2.3.4 Let $X = c_{00}$ and $F : X \rightarrow X$ be defined by

$$F(x)(j) = \frac{1}{j}x(j), \quad j = 1, 2, 3, \dots$$

Show that F is bijective, linear and bounded, but F^{-1} is unbounded. Does this contradict the bounded inverse theorem?

Solution Let $G : c_{00} \rightarrow c_{00}$ be defined by

$$G(x)(j) = jx(j), \quad j = 1, 2, 3, \dots$$

Then F and G are inverses of each other. Hence F must be bijective. It is clearly linear. Since the norm on $X = c_{00}$ is $\|\cdot\|_\infty$, we get

$$|F(x)(j)| = \frac{1}{j}|x(j)| \leq |x(j)| \leq \|x\|_\infty$$

for $j = 1, 2, \dots$. Therefore, $\|F(x)\|_\infty \leq \|x\|_\infty$. Thus F is bounded with $\|F\| \leq 1$.

Let $x_n(j) = 1$ for $1 \leq j \leq n$ and $x_n(j) = 0$ for $j > n$. Then, $x_n \in X$, $\|x_n\|_\infty = 1$,

$$G(x_n)(j) = j \quad \text{for } 1 \leq j \leq n, \quad G(x_n)(j) = 0 \quad \text{for } j > n.$$

Hence, $\|G(x_n)\|_\infty = n = n\|x_n\|_\infty$. This is true for any n . Thus, $G = F^{-1}$ is not bounded.

Here, $X = c_{00}$ is not a Banach space. So the bounded inverse theorem cannot be applied since its hypothesis is not satisfied. There is no contradiction. ■

PROBLEM 2.3.5 Let X and Y be Banach spaces and $F : X \rightarrow Y$ be a one-one bounded linear map. Show that its range $R(F)$ is closed in Y iff $F^{-1} : R(F) \rightarrow X$ is bounded.

Solution Let $Y_1 = R(F)$ be closed in Y . Then Y_1 is Banach, since Y is Banach. Moreover, $F : X \rightarrow Y_1$ is a bijective bounded linear map and X is Banach. Therefore, the bounded inverse theorem gives the continuity of F^{-1} .

Conversely, suppose that $F^{-1} : Y_1 \rightarrow X$ is a bounded linear map, where $Y_1 = R(F)$. Let $y \in \overline{Y_1}$. Then there is a sequence $\{y_n\}$ in Y_1 such that $y_n \rightarrow y$. Let $F^{-1}(y_n) = x_n$. Then,

$$\|x_n - x_m\| = \|F^{-1}(y_n - y_m)\| \leq \|F^{-1}\| \cdot \|y_n - y_m\|.$$

But $\{y_n\}$ is a Cauchy sequence since it converges in Y . So, this shows that $\{x_n\}$ is a Cauchy sequence in X . Hence, $\{x_n\}$ converges. If $x_n \rightarrow x$ then $F(x_n) \rightarrow F(x)$ by the continuity of F , that is, $y_n \rightarrow F(x)$. Thus, $y = F(x) \in Y_1$. This shows that Y_1 is closed in Y . ■

PROBLEM 2.3.6 Let X be a Banach space under two norms $\|\cdot\|_1$ and $\|\cdot\|_2$. Prove that if there exists $\alpha > 0$ such that $\|x\|_1 \leq \alpha\|x\|_2$ for each $x \in X$, then there exists $\beta > 0$ such that $\|x\|_2 \leq \beta\|x\|_1$ for each $x \in X$, that is, $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent.

Solution Let X_1 denote X with the norm $\|\cdot\|_1$ and X_2 denote X with the norm $\|\cdot\|_2$. Let $I : X_2 \rightarrow X_1$ be the identity map given by $I(x) = x$. It is a bijective linear map. If $\|x\|_1 \leq \alpha\|x\|_2$, then $\|Ix\|_1 \leq \alpha\|x\|_2$, and hence I is bounded. Since X_2 and X_1 are Banach, we can apply the bounded inverse theorem to conclude that I^{-1} is bounded. Hence there exists $\beta > 0$ such that $\|I^{-1}x\|_2 \leq \beta\|x\|_1$ for all $x \in X$, that is, $\|x\|_2 \leq \beta\|x\|_1$ for all $x \in X$. This completes the proof. ■

PROBLEM 2.3.7 Let X be a Banach space under the norms $\|\cdot\|_1$ and $\|\cdot\|_2$. If $\|x_n\|_1 \rightarrow 0$ implies that $\|x_n\|_2 \rightarrow 0$ show that there exist $\alpha > 0$, $\beta > 0$ such that

$$\alpha\|x\|_1 \leq \|x\|_2 \leq \beta\|x\|_1 \quad \text{for all } x \in X,$$

that is, the norms are equivalent.

Solution Suppose that no such β exists. Then for $n = 1, 2, \dots$ there is $x_n \in X$ such that $\|x_n\|_2 > n\|x_n\|_1$. Let

$$y_n = \frac{1}{\|x_n\|_2} x_n.$$

Then, $\|y_n\|_2 = 1$, and

$$\|y_n\|_1 = \frac{1}{\|x_n\|_2} \|x_n\|_1 < \frac{1}{n}.$$

Thus, $\|y_n\|_1 \rightarrow 0$ but $\|y_n\|_2$ does not tend to zero. This contradiction proves the existence of β . The existence of α now follows from Problem 2.3.6 with $\|\cdot\|_1$ and $\|\cdot\|_2$ interchanged. ■

PROBLEM 2.3.8 Let $X = C[0, 1]$. Consider two norms, the L^p -norm $\|\cdot\|_p$ and the sup norm $\|\cdot\|_\infty$, on X . Show that if $1 \leq p < \infty$ then

$$\|x\|_p \leq \|x\|_\infty \quad \text{for all } x \in X,$$

but there is no $\beta > 0$ such that

$$\|x\|_\infty \leq \beta \|x\|_p \quad \text{for all } x \in X.$$

Does this contradict 2.3.6?

Solution For $x \in C[0, 1]$, we have

$$\begin{aligned} (\|x\|_p)^p &= \int_0^1 |x(t)|^p dt \leq (\|x\|_\infty)^p, \\ \|x\|_p &\leq \|x\|_\infty. \end{aligned}$$

Let $y_n(t) = n - n^2 t$ for $0 \leq t \leq 1/n$ and $y_n(t) = 0$ for $1/n < t \leq 1$. Then, $y_n \in C[0, 1]$ and $0 \leq y_n(t) \leq n$. Let

$$x_n(t) = (y_n(t))^{1/p}, \quad 0 \leq t \leq 1.$$

Then, $x_n \in C[0, 1]$, $\|x_n\|_\infty = n^{1/p}$, and

$$(\|x_n\|_p)^p = \int_0^1 y_n(t) dt = \int_0^{1/n} (n - n^2 t) dt = \frac{1}{2}.$$

This is true for $n = 1, 2, \dots$. So there is no β such that $\|x_n\|_\infty \leq \beta \|x_n\|_p$ for all n . This does not contradict Problem 2.3.6 since X is not Banach under $\|\cdot\|_p$ if $1 \leq p < \infty$. ■

PROBLEM 2.3.9 Let X and Y be Banach spaces and $F \in BL(X, Y)$. Let $R(F)$ and $Z(F)$ be range space and zero space of F . Show that $R(F)$ is closed in Y iff $R(F)$ is linearly homeomorphic to $X/Z(F)$.

Solution Let $Z = Z(F)$ and $R(F)$ be closed in Y . Define $G : X/Z \rightarrow R(F)$ by

$$G(x + Z) = F(x), \quad x \in X.$$

This is linear and onto. If $G(x + Z) = 0$, then $F(x) = 0$, $x \in Z$ and $x + Z$ is the zero of X/Z . Thus G is one-one. For any x in X , there is a sequence $\{z_n\}$ in Z such that $\|x + z_n\| \rightarrow \|x + Z\|$. Since

$$\|G(x + Z)\| = \|F(x)\| = \|F(x + z_n)\| \leq \|F\| \cdot \|x + z_n\|,$$

for all n , letting $n \rightarrow \infty$, we get

$$\|G(x + Z)\| \leq \|F\| \cdot \|x + Z\|.$$

Therefore, G is continuous. Since X/Z and $R(F)$ are Banach, we see that G is open, that is, G^{-1} is continuous. Thus G is a linear homeomorphism.

Conversely, suppose that $R(F)$ is linearly homeomorphic to X/Z . But X/Z is Banach. So $R(F)$ is Banach. Thus $R(F)$ is complete and hence closed in Y . This completes the proof. ■

PROBLEM 2.3.10 Show that every separable Banach space is linearly homeomorphic to l^1/Z for some closed subspace Z of l^1 .

Solution Let X be a separable Banach space, E be a countable dense subset of X , and

$$E_1 = \{x \in E : \|x\| < 1\} = E \cap U(0, 1),$$

where $U(0, 1)$ is the open unit ball. Since E_1 is countable, we may suppose that $E_1 = \{u_n : n = 1, 2, \dots\}$. If $y \in l^1$, then

$$\sum_{n=1}^{\infty} \|y(n)u_n\| = \sum_{n=1}^{\infty} |y(n)| \|u_n\| \leq \sum_{n=1}^{\infty} |y(n)| < \infty.$$

Thus the series $\sum y(n)u_n$ converges absolutely in X . Since X is Banach, it converges in X . Let

$$F(y) = \sum_{n=1}^{\infty} y(n)u_n, \quad y \in l^1.$$

This defines a map $F : l^1 \rightarrow X$. It is clearly linear. Since

$$\|F(y)\| \leq \sum_{n=1}^{\infty} \|y(n)u_n\| \leq \sum_{n=1}^{\infty} |y(n)| = \|y\|_1,$$

F is bounded.

Let $x \in X$, $\|x\| < \frac{1}{2}$. There is $u \in E$ such that $\|x - u\| < \frac{1}{4}$. Then,

$$\|u\| \leq \|u - x\| + \|x\| < \frac{3}{4}.$$

Thus, $u \in E_1$ and so $u = u_{n_1}$ for some n_1 . Suppose that we have already found positive integers $n_1, \dots, n_r$ such that $n_1 < n_2 < \dots < n_r$, and

$$\|x - u_{n_1} - \frac{1}{2}u_{n_2} - \dots - \frac{1}{2^{r-1}}u_{n_r}\| < \frac{1}{2^{r+1}} \quad (1)$$

Let $x_r = \sum_{i=1}^r 2^{-i+1}u_{n_i}$. Then,

$$\|x - x_r\| < 2^{-r-1}; \quad \|2^r(x - x_r)\| < \frac{1}{2}.$$

If $u \in E$ and $\|2^r(x - x_r) - u\| < 1/4$, then $u \in E_1$, as before. Since every neighbourhood of $2^r(x - x_r)$ contains an infinite number of elements of E , there is $n_{r+1} > n_r$ such that

$$\|2^r(x - x_r) - u_{n_{r+1}}\| < \frac{1}{4},$$

that is,

$$\|x - x_r - \frac{1}{2^r}u_{n_{r+1}}\| < \frac{1}{2^{r+2}}.$$

This is the same as (1) with r replaced by $r+1$. The process can be continued indefinitely. Thus we get a strictly increasing sequence $\{n_r\}$ of positive integers such that (1) holds for all r .

Let $y(n_i) = 2^{-i+1}$, $i = 1, 2, \dots$ and $y(j) = 0$ for the remaining j . Then, $y \in l^1$,

$$F(y) = \sum_{j=1}^{\infty} y(j)u_j = \sum_{i=1}^{\infty} 2^{-i+1}u_{n_i} = \lim_{r \rightarrow \infty} x_r.$$

But $x_r \rightarrow x$ by (1). Hence, $F(y) = x$. This proves that $x \in R(F)$ whenever $\|x\| < 1/2$. Given $x \in X$, we can find $k > 0$ such that $k\|x\| < 1/2$, and so $kx \in R(F)$. Thus $x \in R(F)$ for all $x \in X$, and hence $R(F) = X$ is closed in X . So from Problem 2.3.9 (with X, Y replaced by l^1, X), we see that $X = R(F)$ is linearly homeomorphic to l^1/Z where $Z = Z(F)$. This completes the proof. ■

PROBLEM 2.3.11 Let Y and Z be closed subspaces of a Banach space X such that $X = Y + Z$. Show that there exists $\alpha > 0$ such that every x in X has a representation $x = y + z$, where $y \in Y$, $z \in Z$ and $\|y\| + \|z\| \leq \alpha\|x\|$.

Solution Since X is Banach and Y, Z are its closed subspaces, we see that Y and Z are Banach; hence $W = Y \times Z$ is a Banach space under the norm

$$\|(y, z)\| = \|y\| + \|z\|, \quad y \in Y, z \in Z.$$

Since the mappings $(y, z) \rightarrow y$ and $(y, z) \rightarrow z$ are linear, their sum $F : W \rightarrow X$ defined by

$$F(y, z) = y + z, \quad y \in Y, z \in Z,$$

is linear. It is bounded since

$$\|F(y, z)\| = \|y + z\| \leq \|y\| + \|z\| = \|(y, z)\|.$$

As $X = Y + Z$, we see that F is onto. Hence F maps open subsets of W onto open subsets of X . Let U be the open unit ball in W and $V = F(U)$. Then V is open and $0 = F(0, 0) \in V$. So there is $r > 0$ such that

$$\{x \in X : \|x\| < r\} \subset V.$$

Let $x \in X$. If $x \neq 0$, let $x_1 = \frac{r}{2\|x\|}x$. Then, $\|x_1\| = \frac{1}{2}r < r$, and hence $x_1 \in V = F(U)$. Therefore,

$$x_1 = F(y_1, z_1) = y_1 + z_1, \quad (y_1, z_1) \in U.$$

Hence,

$$x = \frac{2}{r}\|x\|x_1 = \frac{2\|x\|}{r}y_1 + \frac{2\|x\|}{r}z_1 = y + z,$$

where

$$y = \frac{2\|x\|}{r}y_1 \in Y, \quad z = \frac{2\|x\|}{r}z_1 \in Z.$$

Thus,

$$\|y\| + \|z\| = \frac{2\|x\|}{r}(\|y_1\| + \|z_1\|) = \frac{2\|x\|}{r}\|(y_1, z_1)\| \leq \frac{2\|x\|}{r},$$

since $(y_1, z_1) \in U$. This proves the result with $\alpha = 2/r$. ■

PROBLEM 2.3.12 The Fourier transform $\hat{x}$ in $L^1(\mathbb{R})$ is given by

$$\hat{x}(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-itu} x(u) du, \quad t \in \mathbb{R}.$$

Let $F(x) = \hat{x}$ for $x \in L^1(\mathbb{R})$. Show that F maps $L^1(\mathbb{R})$ onto a proper subset of $C_0(\mathbb{R})$.

Solution Here we use some basic properties of Fourier transforms.¹ It is known that $\hat{x} \in C_0(\mathbb{R})$ for all $x \in L^1(\mathbb{R})$. Clearly, F is linear and

$$|\hat{x}(t)| \leq \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} |e^{-itu} x(u)| du = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} |x(u)| du.$$

Hence,

$$\|F(x)\|_{\infty} = \|\hat{x}\|_{\infty} \leq \frac{1}{\sqrt{2\pi}}\|x\|_1, \quad x \in L^1(\mathbb{R}).$$

Thus, $F : L^1(\mathbb{R}) \rightarrow C_0(\mathbb{R})$ is a bounded linear map. It is known that if $x \in L^1(\mathbb{R})$ and $\hat{x}(t) = 0$ for all $t \in \mathbb{R}$, then x is the zero element of $L^1(\mathbb{R})$.

¹See Rudin, W., *Real and Complex Analysis* (1966), Ch. 9.

This shows that F is one-one. Suppose that it is onto. Since $L^1(\mathbb{R})$ and $C_0(\mathbb{R})$ are Banach its inverse $G : C_0(\mathbb{R}) \rightarrow L^1(\mathbb{R})$ must be a bounded linear map.

For $n = 1, 2, 3, \dots$, let

$$y_n(t) = \begin{cases} 1 & \text{for } |t| \leq n-1, \\ \frac{n+1-|t|}{2} & \text{for } n-1 < |t| \leq n+1 \\ 0 & \text{for } |t| > n+1. \end{cases}$$

Then, $y_n(-t) = y_n(t)$; hence,

$$\begin{aligned} \sqrt{2\pi} \hat{y}_n(t) &= \int_{-\infty}^{\infty} e^{-itu} y_n(u) du = 2 \int_0^{\infty} (\cos tu) y_n(u) du \\ &= 2 \int_0^{n-1} \cos tu du + 2 \int_{n-1}^{n+1} (\cos tu) \frac{n+1-u}{2} du \\ &= \frac{1}{t^2} [\cos(n-1)t - \cos(n+1)t]. \end{aligned}$$

Thus, $\hat{y}_n(t) = \frac{2}{\sqrt{2\pi}} x_n(t)$, where

$$x_n(t) = \frac{\sin t \sin nt}{t^2}.$$

Since

$$|x_n(t)| \leq \frac{1}{t^2}, \quad |x_n(t)| = \left| \frac{\sin t}{t} \right| \left| \frac{\sin nt}{nt} \right| \quad n \leq n,$$

we see that

$$\begin{aligned} \int_{-\infty}^{\infty} |x_n(t)| dt &\leq \int_{-\infty}^{-1} \frac{1}{t^2} dt + \int_{-1}^1 n dt + \int_1^{\infty} \frac{1}{t^2} dt \\ &= 1 + 2n + 1. \end{aligned}$$

Thus, $\hat{y}_n = (2/\sqrt{2\pi})x_n \in L^1(\mathbb{R})$. The Fourier inversion formula gives

$$\begin{aligned} y_n(t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{itu} \hat{y}_n(u) du, \\ y_n(-t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-itu} \frac{2}{\sqrt{2\pi}} x_n(u) du = \frac{2}{\sqrt{2\pi}} \hat{x}_n(t). \end{aligned}$$

Hence, $\hat{x}_n(t) = \sqrt{(\pi/2)} y_n(-t) = \sqrt{(\pi/2)} y_n(t)$. Since $\|y_n\|_{\infty} = 1$, we have $\|\hat{x}_n\|_{\infty} = \sqrt{(\pi/2)}$. On the other hand,

$$\|x_n\|_1 = \int_{-\infty}^{\infty} \left| \frac{\sin t}{t} \frac{\sin nt}{t} \right| dt \geq \int_0^{\pi/2} \left| \frac{\sin t}{t} \frac{\sin nt}{t} \right| dt$$

$$\geq \frac{2}{\pi} \int_0^{\pi/2} \left| \frac{\sin nt}{t} \right| dt = \frac{2}{\pi} \int_0^{n\pi/2} |\sin u| \frac{du}{u},$$

using the fact that $(\sin t)/t$ decreases from 1 to $(2/\pi)$ as t increases from 0 to $\pi/2$. Since

$$\int_0^\infty \left| \frac{\sin u}{u} \right| du = \infty,$$

we see that $\|x_n\|_1 \rightarrow \infty$ as $n \rightarrow \infty$. Thus, $\|G(\hat{x}_n)\|_1 \rightarrow \infty$. However, $\|\hat{x}_n\|_\infty = \sqrt{(\pi/2)}$. This shows that $G = F^{-1}$ is not bounded, and hence F is not onto. This completes the proof. ■

PROBLEM 2.3.13 Let X, Y, Z be Banach spaces, $G \in BL(X, Z)$ and $H \in BL(Y, Z)$. Suppose that for every x in X the equation $G(x) = H(y)$ has a unique solution y in Y . Show that the map $F : X \rightarrow Y$, defined by $F(x) = y$ when $G(x) = H(y)$, is in $BL(X, Y)$.

Solution By hypothesis, $F : X \rightarrow Y$ is well-defined. Let $F(x_1) = y_1$, $F(x_2) = y_2$. Then,

$$G(kx_1 + x_2) = kG(x_1) + G(x_2) = kH(y_1) + H(y_2) = H(ky_1 + y_2).$$

Hence,

$$F(kx_1 + x_2) = ky_1 + y_2 = kF(x_1) + F(x_2).$$

Thus F is linear. To prove that F is continuous, it is enough if we prove that the graph of F is closed. For this let $x_n \rightarrow x$ in X and $F(x_n) \rightarrow y$ in Y . Let $F(x_n) = y_n$. Then $G(x_n) = H(y_n)$. Since G and H are continuous, we have

$$G(x_n) \rightarrow G(x), \quad H(y_n) \rightarrow H(y).$$

Therefore, $G(x) = H(y)$, that is, $F(x) = y$. So the graph of F is closed. This completes the proof. ■

PROBLEM 2.3.14 Let X, Y, Z be Banach spaces and $\{G_\alpha : \alpha \in A\}$ be a family of bounded linear maps from Y to Z . Suppose that if $G_\alpha(y) = 0$ for all α in A , then $y = 0$. Show that if $F : X \rightarrow Y$ is linear and $G_\alpha \circ F \in BL(X, Z)$ for every α in A , then $F \in BL(X, Y)$.

Solution It is enough if we prove that the graph of F is closed. For this let $x_n \rightarrow x$ in X and $F(x_n) \rightarrow y$ in Y . Let $F(x_n) = y_n$. Then $y_n \rightarrow y$, and so $G_\alpha(y_n) \rightarrow G_\alpha(y)$ for all α . But,

$$G_\alpha(y_n) = G_\alpha(F(x_n)) = (G_\alpha \circ F)(x_n) \rightarrow (G_\alpha \circ F)(x).$$

Hence,

$$G_\alpha(y) = (G_\alpha \circ F)(x) = G_\alpha(F(x)),$$

that is, $G_\alpha(y - F(x)) = 0$. Since this is true for all α , we must have $y - F(x) = 0$, $y = F(x)$. This completes the proof. ■

PROBLEM 2.3.15 For $x \in L^1[-\pi, \pi]$, let

$$\hat{x}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-int} x(t) dt, \quad n = 0, \pm 1, \pm 2, \dots$$

For a set E of integers, let

$$C_E = \{x \in C[-\pi, \pi] : \hat{x}(n) = 0 \text{ for all } n \notin E\}.$$

Prove that C_E is a closed subspace of $C[-\pi, \pi]$. Prove also that if

$$\sum_{n=-\infty}^{\infty} |\hat{x}(n)| < \infty \quad \text{for every } x \in C_E, \quad (2)$$

then there is $\alpha > 0$ such that

$$\sum_{n=-\infty}^{\infty} |\hat{x}(n)| \leq \alpha \|x\|_{\infty} \quad \text{for every } x \in C_E.$$

Solution Let $x, y \in C_E, k \in \mathbb{K}$ and $z = kx + y$. Then for all $n \notin E$, we have

$$\hat{z}(n) = k\hat{x}(n) + \hat{y}(n) = 0,$$

and hence $z \in C_E$. This shows that C_E is a subspace.

Let $x \in \overline{C_E}$. Then there is a sequence $\{x_n\}$ in C_E such that $x_n \rightarrow x$ in $C[-\pi, \pi]$, that is, $\|x_n - x\|_{\infty} \rightarrow 0$. Since

$$\begin{aligned} \hat{x}(m) - \hat{y}(m) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-imt} [x(t) - y(t)] dt, \\ |\hat{x}(m) - \hat{y}(m)| &\leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |x(t) - y(t)| dt \\ &\leq \|x - y\|_{\infty}, \end{aligned} \quad (3)$$

we get $|\hat{x}_n(m) - \hat{x}(m)| \leq \|x_n - x\|_{\infty} \rightarrow 0$ as $n \rightarrow \infty$. But $\hat{x}_n(m) = 0$ for all $m \notin E$. It follows that $\hat{x}(m) = 0$ for all $m \notin E$, that is, $x \in C_E$. This shows that C_E is closed in $C[-\pi, \pi]$.

For the next part, let $\mathbb{Z}$ be the set of all integers and $l^1(\mathbb{Z})$ be the set of all functions $y : \mathbb{Z} \rightarrow \mathbb{K}$ such that

$$\sum_{m=-\infty}^{\infty} |y(m)| < \infty.$$

Like l^1 this space $l^1(\mathbb{Z})$ also is a Banach space under the norm

$$\|y\|_1 = \sum_{m=-\infty}^{\infty} |y(m)|, \quad y \in l^1(\mathbb{Z}).$$

If (2) holds, then $\hat{x} \in l^1(\mathbb{Z})$ for all $x \in C_E$. Let $F(x) = \hat{x}$. Then, $F : C_E \rightarrow l^1(\mathbb{Z})$ is linear. We shall prove that the graph of F is closed. Let $x_n \rightarrow x$ in C_E and $F(x_n) \rightarrow y$ in $l^1(\mathbb{Z})$. Then, using (3) and the definition of $\|\cdot\|_1$, for each $m \in \mathbb{Z}$ we see that

$$\begin{aligned} |y(m) - \hat{x}(m)| &\leq |y(m) - \hat{x}_n(m)| + |\hat{x}_n(m) - \hat{x}(m)| \\ &\leq \|y - \hat{x}_n\|_1 + \|x_n - x\|_\infty \\ &= \|y - F(x_n)\|_1 + \|x_n - x\|_\infty. \end{aligned}$$

Letting $n \rightarrow \infty$, we get $y(m) = \hat{x}(m)$ for all $m \in \mathbb{Z}$, that is, $y = \hat{x} = F(x)$. Thus the graph of F is closed. Since C_E and $l^1(\mathbb{Z})$ are Banach spaces, we conclude that F is bounded. Hence,

$$\|\hat{x}\|_1 = \|F(x)\|_1 \leq \|F\| \cdot \|x\|_\infty,$$

$$\sum_{n=-\infty}^{\infty} |\hat{x}(n)| = \|\hat{x}\|_1 \leq \|F\| \|x\|_\infty,$$

for all $x \in C_E$. This completes the proof. ■

PROBLEM 2.3.16 Let $\|\cdot\|$ be a complete norm on $C^1[a, b]$ such that if $\|x_n - x\| \rightarrow 0$ then $x_n(t) \rightarrow x(t)$ for every $t \in [a, b]$. Prove that $\|\cdot\|$ is equivalent to the sup norm $\|\cdot\|_\infty$.

Solution Let X_1 denote $C[a, b]$ with the given norm $\|\cdot\|$ and X_2 denote $C[a, b]$ with the sup norm $\|\cdot\|_\infty$. These are Banach spaces. Consider the identity map $I : X_2 \rightarrow X_1$. This is linear. To show that its graph is closed, let $x_n \rightarrow x$ in X_2 and $I(x_n) \rightarrow y$ in X_1 , that is, $\|x_n - x\|_\infty \rightarrow 0$ and $\|I(x_n) - y\| \rightarrow 0$ as $n \rightarrow \infty$. Thus,

$$\|x_n - y\| = \|I(x_n) - y\| \rightarrow 0,$$

and so $x_n(t) \rightarrow y(t)$ for each t . Since

$$|x_n(t) - x(t)| \leq \|x_n - x\|_\infty \rightarrow 0,$$

it follows that $y(t) = x(t)$ for each $t \in [a, b]$. Thus, $y = x = I(x)$. So the graph of I is closed. Hence I is continuous. Thus there is $\alpha > 0$ such that $\|Ix\| \leq \alpha\|x\|_\infty$ for all x . Similarly, or using Problem 2.3.6, we see that there is $\beta > 0$ such that $\|x\|_\infty \leq \beta\|x\|$ for all x . This proves the result. ■

PROBLEM 2.3.17 Let $\|\cdot\|$ be a complete norm on $C^1[a, b]$ such that if $\|x_n - x\| \rightarrow 0$ then $x_n(a) \rightarrow x(a)$ and $x'_n(t) \rightarrow x'(t)$ for each t in $[a, b]$. Prove that $\|\cdot\|$ is equivalent to $\|\cdot\|_1$ given by

$$\|x\|_1 = \|x\|_\infty + \|x'\|_\infty, \quad x \in C^1[a, b].$$

Deduce that the norm $\|\cdot\|_2$ given by

$$\|x\|_2 = |x(a)| + \|x'\|_\infty, \quad x \in C^1[a, b],$$

is equivalent to $\|\cdot\|_1$.

Solution Let X and X_1 denote the same space $C^1[a, b]$ with the norms $\|\cdot\|$ and $\|\cdot\|_1$ respectively. These are Banach spaces. Consider the identity map $I : X \rightarrow X_1$. It is linear. To show that its graph is closed, let $x_n \rightarrow x$ in X and $I(x_n) \rightarrow y$ in X_1 , that is, $\|x_n - x\| \rightarrow 0$ and $\|I(x_n) - y\|_1 \rightarrow 0$. Hence, $x_n(a) \rightarrow x(a)$ and $x'_n(t) \rightarrow x'(t)$ for each t . Moreover,

$$\|x_n - y\|_\infty + \|x'_n - y'\|_\infty = \|x_n - y\|_1 = \|I(x_n) - y\|_1 \rightarrow 0.$$

It follows that

$$\|x_n - y\|_\infty \rightarrow 0, \quad \|x'_n - y'\|_\infty \rightarrow 0.$$

So,

$$x_n(t) \rightarrow y(t) \quad \text{and} \quad x'_n(t) \rightarrow y'(t)$$

for each $t \in [a, b]$. It follows that $y(a) = x(a)$ and $y'(t) = x'(t)$. Hence,

$$y(t) - y(a) = \int_a^t y'(s) ds = \int_a^t x'(s) ds = x(t) - x(a).$$

So $y(t) = x(t)$, $y = x = I(x)$. Thus the graph of I is closed. Since X and X_1 are Banach, we see that I is continuous. So there is $\alpha > 0$ such that $\|Ix\|_1 \leq \alpha\|x\|$; $\|x\|_1 \leq \alpha\|x\|$. Similarly, or using Problem 2.3.6, we see that there is $\beta > 0$ such that $\|x\| \leq \beta\|x\|_1$. This proves that $\|\cdot\|$ and $\|\cdot\|_1$ are equivalent.

Finally, since $C^1[a, b]$ is Banach under $\|\cdot\|_2$, we can apply this result with $\|\cdot\|_2$ in place of $\|\cdot\|$. Hence $\|\cdot\|_2$ and $\|\cdot\|_1$ are equivalent. ■

PROBLEM 2.3.18 Let $1 < p < \infty$, $1/p + 1/q = 1$ and $\{k_n\}$ be a sequence in $\mathbb{K}$. If $\sum k_j x(j)$ converges for every x in l^p , show that $(k_n) \in l^q$.

Solution We shall first prove that if $a_n \geq 0$, $s_n = a_1 + \cdots + a_n$ and $\sum a_n$ diverges, then $\sum a_n s_n^{-p}$ converges when $p > 1$ and diverges when $p = 1$:

If $1 < p \leq 2$, then

$$\begin{aligned} (s_n)^{p-1} - (s_{n-1})^{p-1} &= (s_n - s_{n-1})(p-1)t^{p-2} \quad (s_{n-1} \leq t \leq s_n) \\ &\geq a_n(p-1)(s_n)^{p-2}. \end{aligned}$$

Hence,

$$(s_{n-1})^{-p+1} - (s_n)^{-p+1} \geq a_n(p-1)(s_n)^{-1}(s_{n-1})^{-p+1} \geq (p-1)a_n(s_n)^{-p}.$$

This shows that $\sum a_n s_n^{-p}$ converges if $1 < p \leq 2$. If $p > 2$, then $a_n s_n^{-p} \leq a_n s_n^{-2}$ if $s_n > 1$. So, $\sum a_n s_n^{-p}$ converges if $p > 2$ also.

For $p = 1$, we have

$$\log \left(\frac{s_n}{s_{n-1}} \right) = \log \left(1 + \frac{a_n}{s_{n-1}} \right) \leq \frac{a_n}{s_{n-1}} \leq \frac{2a_n}{s_n},$$

provided $s_n \leq 2s_{n-1}$. If $a_n/s_n > 1/2$ for infinitely many n , then $\sum(a_n/s_n)$ diverges. So suppose that $a_n/s_n \leq 1/2$ for $n \geq n_0$. Then,

$$2a_n \leq s_n, \quad a_n \leq s_{n-1}, \quad s_n \leq 2s_{n-1}$$

for $n \geq n_0$. Therefore,

$$\frac{a_n}{s_n} \geq \frac{1}{2} \log(s_n - s_{n-1}) \quad \text{for } n \geq n_0,$$

and $\sum(a_n/s_n)$ diverges in this case also. This proves our claim.

To prove that $(k_n) \in l^q$, assume that $\sum |k_n|^q = \infty$. Take $a_n = |k_n|^q$, and

$$x(n) = \begin{cases} \frac{a_n}{s_n k_n} & \text{if } k_n \neq 0 \\ 0 & \text{if } k_n = 0. \end{cases}$$

Then for $k_n \neq 0$,

$$|x(n)|^p = \left(\frac{|k_n|^{q-1}}{s_n} \right)^p = \frac{|k_n|^q}{s_n^p} = a_n s_n^{-p},$$

and $k_n x(n) = a_n/s_n$. If $k_n = 0$, then $x(n) = 0 = a_n$. Thus we see that $\sum |x(n)|^p = \sum a_n s_n^{-p}$ converges while $\sum k_n x(n) = \sum(a_n/s_n)$ diverges. We get a contradiction, thereby proving the result. ■

PROBLEM 2.3.19 (See Problem 2.2.17.) Let $1 \leq p \leq \infty$ and $1/p + 1/q = 1$. Suppose that an infinite matrix (k_{ij}) of scalar entries represents a map $F : l^p \rightarrow l^q$ in the sense that

$$F(x)(i) = \sum_{j=1}^{\infty} k_{ij} x(j), \quad i = 1, 2, \dots, \quad (4)$$

the series being convergent for all $i \geq 1$ and $x \in l^p$. Prove that F is a bounded linear map.

Solution We shall first prove that if $\sum k_n x(n)$ converges for every $x \in l^p$ then $(k_n) \in l^q$. If $1 < p < \infty$, then this is given by Problem 2.3.18. If $p = \infty$, then $q = 1$. Let

$$x(j) = \begin{cases} \frac{|k_j|}{k_j} & \text{if } k_j \neq 0, \\ 1 & \text{if } k_j = 0. \end{cases}$$

Then $x \in l^\infty$ with $\|x\|_\infty = 1$, and $k_j x(j) = |k_j|$ for all j . So, $\sum |k_j| = \sum k_j x(j)$ converges; that is, $(k_n) \in l^1$.

Next let $p = 1$. Then, $q = \infty$. Assume that (k_n) is unbounded. Then we can find $n_1, n_2, \dots$ such that $n_1 < n_2 < \dots$, and

$$k_{n_1} \neq 0, \quad |k_{n_r}| > 2^r |k_{n_1}| \quad \text{for } r = 2, 3, \dots$$

Let $x(n_r) = 2^{-r}$ for $r = 1, 2, \dots$ and $x(j) = 0$ for the remaining j . Then, $x \in l^1$ and

$$|k_{n_r}x(n_r)| > |k_{n_1}|, \quad r = 2, 3, \dots$$

So $k_jx(j)$ does not tend to zero as $j \rightarrow \infty$, and hence $\sum k_jx(j)$ does not converge even though $x \in l^1$. This contradiction shows that $\{k_n\}$ is bounded, that is, $(k_n) \in l^\infty$. This proves our claim.

Given that the series in (4) converges for all $x \in l^p$, $i \geq 1$. Hence, if

$$z_i(j) = k_{ij}, \quad j \geq 1, i \geq 1,$$

then $z_i \in l^q$ by what has been observed above. So for $i \geq 1$, $x \in l^p$,

$$|F(x)(i)| \leq \|z_i\|_q \|x\|_p. \quad (5)$$

This follows directly from (4) if $p = \infty$, $q = 1$ or $p = 1$, $q = \infty$. If $1 < p < \infty$, this is given by Holder's inequality.

It is clear that F defined by (4) is linear. To prove its continuity it is enough if we prove that its graph is closed, since l^p and l^q are Banach spaces. For this let $x_n \rightarrow x$ in l^p and $F(x_n) \rightarrow y$ in l^q . For $i = 1, 2, \dots$,

$$|F(x_n)(i) - F(x)(i)| = |F(x_n - x)(i)| \leq \|z_i\|_q \|x_n - x\|_p,$$

from (5). Hence, $F(x_n)(i) \rightarrow F(x)(i)$ as $n \rightarrow \infty$. But $F(x_n)(i) \rightarrow y(i)$ since $\|F(x_n) - y\|_q \rightarrow 0$. Thus $y(i) = F(x)(i)$ for all i , that is, $y = F(x)$. This shows that the graph of F is closed. This completes the proof. ■

PROBLEM 2.3.20 Let X and Y be Banach spaces. Let Z be a subspace of X and $F : Z \rightarrow Y$ be a linear map whose graph is closed in $X \times Y$. For $z \in Z$, let

$$\|z\|_F = (\|z\|^2 + \|F(z)\|^2)^{1/2}.$$

Show that Z is a Banach space and that $F \in BL(Z, Y)$ in this norm. [$\|\cdot\|_F$ is called the *norm of the graph* of F .]

Solution Clearly, $\|z\| \leq \|z\|_F$ and $\|F(z)\| \leq \|z\|_F$ for all z in Z . Let $\{z_n\}$ be a Cauchy sequence in Z w.r.t. the norm $\|\cdot\|_F$. Let $y_n = F(z_n)$. Then,

$$\|z_n - z_m\| \leq \|z_n - z_m\|_F,$$

$$\|F(z_n) - F(z_m)\| = \|F(z_n - z_m)\| \leq \|z_n - z_m\|_F.$$

This shows that $\{z_n\}$ and $\{y_n\}$ are Cauchy sequences in X and Y respectively. Since X and Y are Banach, there exist $z \in X$ and $y \in Y$ such that $z_n \rightarrow z$ and $y_n \rightarrow y$. So $(z_n, y_n) \rightarrow (z, y)$ in $X \times Y$. Since (z_n, y_n) belongs to the graph of F which is given as closed in $X \times Y$, we see that (z, y) belongs to the graph of F . Hence, $z \in Z$ and $y = F(z)$. Thus,

$$\begin{aligned} \|z_n - z\|_F^2 &= \|z_n - z\|^2 + \|F(z_n - z)\|^2 \\ &= \|z_n - z\|^2 + \|y_n - y\|^2 \rightarrow 0. \end{aligned}$$

Therefore, $z_n \rightarrow z$ in Z w.r.t. $\|\cdot\|_F$. So Z is a Banach space. Since $\|F(z)\| \leq \|z\|_F$, we see that F is continuous. This completes the proof. ■

3

Duality

In this chapter we study the relationship between a normed space and its dual. In Section 3.1 the duals of some standard normed spaces are found, and the concept of reflexivity is introduced. In Section 3.2 the concept of weak convergence is taken up.

3.1 Duals

The space $BL(X, \mathbb{K})$ of bounded linear maps on a normed space X is called the *dual* or *conjugate space* of X , and is denoted by X' . It is always a Banach space since $\mathbb{K}$ is complete. For $x' \in X'$, we have

$$\|x'\| = \sup \{|x'(x)| : x \in X, \|x\| \leq 1\}$$

and for $x \in X$ we have, by Problem 1.4.4,

$$\|x\| = \sup \{|x'(x)| : x' \in X', \|x'\| \leq 1\}.$$

Moreover,

$$\begin{aligned} x'_1 = x'_2 &\iff x'_1(x) = x'_2(x) \quad \text{for all } x \in X, \\ x_1 = x_2 &\iff x'(x_1) = x'(x_2) \quad \text{for all } x' \in X'. \end{aligned}$$

If X' is separable then X is separable (see Problem 3.1.5).

For $x \in X$, define $f_x : X' \rightarrow \mathbb{K}$ by

$$f_x(x') = x'(x), \quad x' \in X'.$$

Then f_x is a bounded linear functional on X' and $\|f_x\| = \|x\|$. Define J on X by

$$J(x) = f_x, \quad x \in X.$$

Then, $J : X \rightarrow X''$ is a linear isometry, called the *canonical mapping* of X into X'' . The canonical mapping need not be surjective. If it is surjective we

say that X is *reflexive*. In any case, the closure of the range of J is a Banach space $\tilde{X}$ such that X is linearly isometric to a dense subspace of $\tilde{X}$. Thus $\tilde{X}$ is the *completion* of X .

If X' is linearly isometric to a normed space Y , then we say that X' is equal to Y . In this sense the dual of $\mathbb{K}^n$ with $\|\cdot\|_p$ is $\mathbb{K}^n$ with $\|\cdot\|_q$, where $1/p + 1/q = 1$, $1 \leq p \leq \infty$, $1 \leq q \leq \infty$. The dual of l^p is l^q [see Problem 3.1.4] and the dual of L^p is L^q , where $1/p + 1/q = 1$, $1 \leq p < \infty$, $1 < q \leq \infty$. But the dual of l^∞ is not l^1 (see Problem 3.1.6).

EXERCISES

1. Let X be a normed space. For $x \in X$, define $f_x : X' \rightarrow \mathbb{K}$ by

$$f_x(x') = x'(x), \quad x' \in X'.$$

Show that $f_x \in X''$ and $\|f_x\| = \|x\|$.

2. Show that the canonical mapping is a linear isometry.

3. If $X = \mathbb{K}^n$ with norm $\|\cdot\|_p$, show that X' is linearly isometric to $\mathbb{K}^n$ with norm $\|\cdot\|_q$, where $1/p + 1/q = 1$.

PROBLEMS

PROBLEM 3.1.1 Show that the dual of l^1 is l^∞ .

Solution If $y \in l^\infty$ and $x \in l^1$, then

$$\sum_{j=1}^{\infty} |y(j)x(j)| \leq \|y\|_{\infty} \sum_{j=1}^{\infty} |x(j)| = \|y\|_{\infty} \|x\|_1 < \infty.$$

So,

$$f(x) = \sum_{j=1}^{\infty} y(j)x(j)$$

exists in $\mathbb{K}$ for each x in l^1 and f maps l^1 into the scalar field $\mathbb{K}$. It is clearly linear,

$$|f(x)| \leq \|y\|_{\infty} \|x\|_1, \quad x \in l^1,$$

and hence $\|f\| \leq \|y\|_{\infty}$. Since $f(e_i) = y(i)$ for all i , we have

$$|y(i)| = |f(e_i)| \leq \|f\| \cdot \|e_i\|_1 = \|f\|, \quad i = 1, 2, \dots$$

Hence, $\|y\|_{\infty} \leq \|f\|$. Thus to each y in l^∞ , we associate an element $f = f_y$ in $(l^1)'$ such that $\|f_y\| = \|y\|_{\infty}$. Let

$$F(y) = f_y, \quad y \in l^\infty,$$

where

$$f_y(x) = \sum_{j=1}^{\infty} y(j)x(j), \quad x \in l^1.$$

This gives a mapping $F : l^\infty \rightarrow (l^1)'$. It can be easily verified that F is linear. It is an isometry since $\|F(y)\| = \|f_y\| = \|y\|_\infty$.

To show that F is onto, let g be any bounded linear functional on l^1 . Let $y(j) = g(e_j)$ for $j \geq 1$. Then

$$|y(j)| = |g(e_j)| \leq \|g\| \|e_j\|_1 = \|g\|.$$

Therefore, $\|y\|_\infty \leq \|g\|$, $y \in l^\infty$.

For $x \in l^1$, let

$$x_n = x(1)e_1 + \cdots + x(n)e_n = (x(1), \dots, x(n), 0, 0, \dots).$$

Then,

$$x_n(j) - x(j) = \begin{cases} 0 & \text{for } 1 \leq j \leq n \\ -x(j) & \text{for } j > n; \end{cases}$$

$$\|x_n - x\|_1 = \sum_{j=n+1}^{\infty} |x(j)|.$$

This tends to zero as $n \rightarrow \infty$ since $x \in l^1$. This means that $x_n \rightarrow x$ in l^1 , and so $g(x_n) \rightarrow g(x)$. Hence,

$$\begin{aligned} g(x) &= \lim_{n \rightarrow \infty} g(x_n) = \lim_{n \rightarrow \infty} \sum_{j=1}^n x(j)g(e_j) \\ &= \sum_{j=1}^{\infty} x(j)y(j) = f_y(x). \end{aligned}$$

Thus, $g = f_y = F(y)$. This shows that F is onto and hence F is a linear isometry onto $(l^1)'$. So, l^∞ is $(l^1)'$. ■

PROBLEM 3.1.2 Show that the dual of c_{00} is l^1 and that the dual of c_0 also is l^1 .

Solution Let $X = c_{00}$ or c_0 and $y \in l^1$. If $x \in c_0$, then

$$\sum_{j=1}^{\infty} |y(j)x(j)| \leq \|x\|_\infty \sum_{j=1}^{\infty} |y(j)| = \|x\|_\infty \cdot \|y\|_1 < \infty.$$

So,

$$f(x) = \sum_{j=1}^{\infty} y(j)x(j)$$

exists in $\mathbb{K}$ for each x in $X = c_{00}$ or c_0 . It clearly gives a linear map $f : X \rightarrow \mathbb{K}$,

$$|f(x)| \leq \|y\|_1 \|x\|_\infty, \quad x \in X,$$

and hence $\|f\| \leq \|y\|_1$. Let

$$k_j = \frac{|y(j)|}{y(j)} \quad \text{if } y(j) \neq 0 \quad \text{and} \quad k_j = 0 \quad \text{if } y(j) = 0.$$

Then $k_j y(j) = |y(j)|$ and $|k_j| \leq 1$ for all j . Let

$$u_n = k_1 e_1 + \cdots + k_n e_n = (k_1, \dots, k_n, 0, 0, \dots).$$

Then $u_n \in c_{00} \subset c_0$, $\|u_n\|_\infty \leq 1$, and

$$\begin{aligned} f(u_n) &= \sum_{j=1}^n y(j) k_j = \sum_{j=1}^n |y(j)|, \\ \sum_{j=1}^n |y(j)| &= |f(u_n)| \leq \|f\| \|u_n\|_\infty \leq \|f\|. \end{aligned}$$

This is true for all n . Letting $n \rightarrow \infty$, we get

$$\|y\|_1 \leq \|f\|.$$

Thus to each y in l^1 we associate an element $f = f_y$ in X' such that $\|f_y\| = \|y\|_1$. Let

$$F(y) = f_y, \quad y \in l^1,$$

where

$$f_y(x) = \sum_{j=1}^{\infty} y(j) x(j), \quad x \in X.$$

This gives a mapping $F : l^1 \rightarrow X'$. It can be easily verified that F is linear. It is an isometry since $\|F(y)\| = \|f_y\| = \|y\|_1$.

To show that F is onto, let g be any bounded linear functional on $X = c_{00}$ or c_0 . Let $g(e_j) = y(j)$. As before, choose k_j and let $u_n = k_1 e_1 + \cdots + k_n e_n$. Then $u_n \in X$, $\|u_n\|_\infty \leq 1$, and

$$g(u_n) = \sum_{j=1}^n k_j g(e_j) = \sum_{j=1}^n k_j y(j) = \sum_{j=1}^n |y(j)|.$$

Hence,

$$\begin{aligned} \sum_{j=1}^n |y(j)| &= |g(u_n)| \leq \|g\|; \\ \sum_{j=1}^{\infty} |y(j)| &\leq \|g\| < \infty. \end{aligned}$$

Thus, $y \in l^1$.

For $x \in X$, let

$$x_n = x(1)e_1 + \cdots + x(n)e_n = (x(1), \dots, x(n), 0, 0, \dots).$$

Then, $x_n \in X$,

$$x(j) - x_n(j) = \begin{cases} 0 & \text{for } 1 \leq j \leq n \\ x(j) & \text{for } j > n. \end{cases}$$

So $\|x - x_n\|_\infty = \sup_{j>n} |x(j)|$. This tends to zero as $n \rightarrow \infty$ since $x \in c_0$. Hence, $x_n \rightarrow x$ in X and so $g(x_n) \rightarrow g(x)$. Therefore,

$$\begin{aligned} g(x) &= \lim_{n \rightarrow \infty} g(x_n) = \lim_{n \rightarrow \infty} \sum_{j=1}^n x(j)g(e_j) \\ &= \sum_{j=1}^{\infty} x(j)y(j) = f_y(x). \end{aligned}$$

Thus, $g = f_y = F(y)$. This shows that F is onto, and hence F is a linear isometry onto X' . So l^1 is X' when $X = c_{00}$ and also when $X = c_0$. This completes the proof. ■

PROBLEM 3.1.3 Show that the dual of c is l^1 .

Solution We proceed as in Problem 3.1.2. Let $y \in l^1$. If $x \in c$, then $x(j)$ tends to some finite limit λ as $j \rightarrow \infty$. Since $|\lambda| \leq \|x\|_\infty$, we have

$$\begin{aligned} |y(1)\lambda| + \sum_{j=1}^{\infty} |y(j+1)x(j)| &\leq \|x\|_\infty \left\{ |y(1)| + \sum_{j=1}^{\infty} |y(j+1)| \right\} \\ &= \|x\|_\infty \|y\|_1 < \infty. \end{aligned}$$

So,

$$f(x) = y(1)\lambda + \sum_{j=1}^{\infty} y(j+1)x(j)$$

exists in $\mathbb{K}$ for each x in c . The map $f : c \rightarrow \mathbb{K}$ defined by this is clearly linear; and $|f(x)| \leq \|y\|_1 \cdot \|x\|_\infty$ for all x in c . So, $f \in c'$ and $\|f\| \leq \|y\|_1$. As in the solution of Problem 3.1.2 choose k_j such that

$$k_j y(j) = |y(j)|, \quad |k_j| \leq 1 \tag{1}$$

for all j . Let $e = (1, 1, 1, \dots)$ and

$$u_n = k_1 e + \sum_{j=1}^n (k_{j+1} - k_1) e_j = (k_2, \dots, k_{n+1}, k_1, k_1, \dots).$$

Then, $\|u_n\|_\infty \leq 1$ and $\lim_{j \rightarrow \infty} u_n(j) = k_1$. Hence,

$$\begin{aligned} f(u_n) &= y(1)k_1 + \sum_{j=1}^n y(j+1)k_{j+1} + \sum_{j=n+1}^{\infty} y(j+1)k_1 \\ &= \sum_{j=1}^{n+1} |y(j)| + k_1 \sum_{j=n+2}^{\infty} y(j). \end{aligned}$$

Thus,

$$\begin{aligned} \sum_{j=1}^{n+1} |y(j)| &= \left| f(u_n) - k_1 \sum_{j=n+2}^{\infty} y(j) \right| \leq |f(u_n)| + |k_1| \sum_{j=n+2}^{\infty} |y(j)| \\ &\leq \|f\| \|u_n\| + |k_1| \sum_{j=n+2}^{\infty} |y(j)|. \end{aligned}$$

Let $n \rightarrow \infty$. Since $\|u_n\|_\infty \leq 1$ and the last sum tends to zero as $n \rightarrow \infty$, we get $\|y\|_1 \leq \|f\|$. Thus to each y in l^1 we associate an element $f = f_y$ in c' such that $\|f_y\| = \|y\|_1$. Let

$$F(y) = f_y, \quad y \in l^1,$$

where

$$f_y(x) = y(1) \lim_{j \rightarrow \infty} x(j) + \sum_{j=1}^{\infty} y(j+1)x(j), \quad x \in c.$$

This gives a linear isometry $F : l^1 \rightarrow c'$.

To show that F is onto, let $g \in c'$. Let $y(j) = g(e_{j-1})$ for $j = 2, 3, \dots$. Choose k_j as in (1) and let

$$v_n = k_2 e_1 + \dots + k_{n+1} e_n = (k_2, \dots, k_{n+1}, 0, 0, \dots).$$

Then, $v_n \in c$, $\|v_n\|_\infty \leq 1$, and

$$g(v_n) = \sum_{j=2}^{n+1} k_j g(e_{j-1}) = \sum_{j=2}^{n+1} k_j y(j) = \sum_{j=2}^{n+1} |y(j)|.$$

Hence,

$$\begin{aligned} \sum_{j=2}^{n+1} |y(j)| &= |g(v_n)| \leq \|g\| \cdot \|v_n\|_\infty \leq \|g\|; \\ \sum_{j=2}^{\infty} |y(j)| &\leq \|g\| < \infty. \end{aligned}$$

Thus $y \in l^1$ whatever be the value of $y(1)$.

If $x \in c$ and $\lim_{j \rightarrow \infty} x(j) = \lambda$, let

$$x_n = \lambda e + \sum_{j=1}^n (x(j) - \lambda) e_j = (x(1), \dots, x(n), \lambda, \lambda, \dots).$$

Then $x_n \in c$, and

$$x(j) - x_n(j) = \begin{cases} 0 & \text{for } 1 \leq j \leq n \\ x(j) - \lambda & \text{for } j > n. \end{cases}$$

Hence

$$\|x - x_n\|_\infty = \sup_{j > n} |x(j) - \lambda|.$$

This tends to zero as $n \rightarrow \infty$. Therefore, $x_n \rightarrow x$ in c , and so $g(x_n) \rightarrow g(x)$. Thus,

$$\begin{aligned} g(x) &= \lim_{n \rightarrow \infty} g(x_n) = \lim_{n \rightarrow \infty} \left\{ \lambda g(e) + \sum_{j=1}^n (x(j) - \lambda) g(e_j) \right\} \\ &= \lambda g(e) + \sum_{j=1}^{\infty} (x(j) - \lambda) y(j+1) \\ &= \lambda y(1) + \sum_{j=1}^{\infty} x(j) y(j+1), \end{aligned}$$

where

$$y(1) = g(e) - \sum_{j=1}^{\infty} y(j+1).$$

Hence, $g = f_y = F_y$ for some $y \in l^1$. Therefore, F is onto c' . This shows that c' is l^1 . ■

PROBLEM 3.1.4 Show that if $1 < p < \infty$ and $1/p + 1/q = 1$ then the dual of l^p is l^q .

Solution We proceed as in Problem 3.1.2. Let $X = l^p$ and $y \in l^q$. If $x \in l^p$, then

$$\sum_{j=1}^{\infty} |y(j)x(j)| \leq \|y\|_q \|x\|_p < \infty,$$

by Holder's inequality. So if

$$f(x) = \sum_{j=1}^{\infty} y(j)x(j), \quad x \in X,$$

then f maps X onto the scalar field $\mathbb{K}$. It is linear,

$$|f(x)| \leq \|y\|_q \|x\|_p, \quad x \in X,$$

and hence $\|f\| \leq \|y\|_q$. Let $y \neq 0$, and

$$k_j = \begin{cases} \frac{|y(j)|^q}{y(j)} & \text{if } y(j) \neq 0, \\ 0 & \text{if } y(j) = 0. \end{cases}$$

Then for $y(j) \neq 0$,

$$|k_j|^p = |y(j)|^{pq} \cdot |y(j)|^{-p} = |y(j)|^q,$$

since $pq = p + q$. Thus, $|k_j|^p = |y(j)|^q$ for all j , this being trivial when $y(j) = 0$. Let

$$u_n = k_1 e_1 + \cdots + k_n e_n = (k_1, \dots, k_n, 0, 0, \dots).$$

Then $u_n \in l^p$,

$$\|u_n\|_p^p = \sum_{j=1}^n |k_j|^p = \sum_{j=1}^n |y(j)|^q \leq \|y\|_q^q$$

$$f(u_n) = \sum_{j=1}^n y(j) k_j = \sum_{j=1}^n |y(j)|^q.$$

Hence,

$$\sum_{j=1}^n |y(j)|^q = |f(u_n)| \leq \|f\| \cdot \|u_n\| \leq \|f\| \cdot (\|y\|_q)^{q/p}.$$

This is true for all n . Letting $n \rightarrow \infty$, we get

$$\begin{aligned} \|y\|_q^q &\leq \|f\| (\|y\|_q)^{q/p} = \|f\| \|y\|_q^{q-1}, \\ \|y\|_q &\leq \|f\|. \end{aligned}$$

Since this is true even when $y = 0$, we see that to each y in l^q we can associate an element $f = f_y$ in X' such that $\|f_y\| = \|y\|_q$. Let

$$F(y) = f_y, \quad y \in l^q,$$

where

$$f_y(x) = \sum_{j=1}^{\infty} y(j)x(j), \quad x \in X.$$

This gives a mapping $F : l^q \rightarrow X'$. It can be easily verified that F is linear. It is an isometry since $\|F(y)\| = \|f_y\| = \|y\|_q$.

To show that F is onto, let g be any bounded linear functional on X and $y(j) = g(e_j)$ for $j \geq 1$. If $y(j) \neq 0$ for some j , choose k_j , as before, and let $u_n = k_1 e_1 + \cdots + k_n e_n$. Then $u_n \in l^p$,

$$\|u_n\|_p^p = \sum_{j=1}^n |y(j)|^q.$$

Hence,

$$\begin{aligned} g(u_n) &= \sum_{j=1}^n k_j g(e_j) = \sum_{j=1}^n k_j y(j) = \sum_{j=1}^n |y(j)|^q = \|u_n\|_p^p, \\ \|u_n\|_p^p &= |g(u_n)| \leq \|g\| \cdot \|u_n\|_p, \\ \|u_n\|_p^{p-1} &\leq \|g\|, \\ \sum_{j=1}^n |y(j)|^q &= \|u_n\|_p^{(p-1)q} \leq \|g\|^q. \end{aligned}$$

(In the last line we have used the fact that $p = (p-1)q$.) This is true for all n . Letting $n \rightarrow \infty$, we see that $y \in l^q$.

For $x \in X = l^p$, let

$$x_n = x(1)e_1 + \cdots + x(n)e_n = (x(1), \dots, x(n), 0, 0, \dots).$$

Then $x_n \in l^p$,

$$x(j) - x_n(j) = \begin{cases} 0 & \text{for } 1 \leq j \leq n, \\ x(j) & \text{for } j > n. \end{cases}$$

So,

$$\|x - x_n\|_p^p = \sum_{j=n+1}^{\infty} |x(j)|^p.$$

This tends to zero as $n \rightarrow \infty$ since $x \in l^p$. So $x_n \rightarrow x$ in l^p , and hence $g(x_n) \rightarrow g(x)$. Thus,

$$\begin{aligned} g(x) &= \lim_{n \rightarrow \infty} g(x_n) = \lim_{n \rightarrow \infty} \sum_{j=1}^n x(j)g(e_j) \\ &= \sum_{j=1}^{\infty} x(j)y(j) = f_y(x). \end{aligned}$$

Thus, $g = f_y = F(y)$ for some $y \in l^q$. This shows that X' is l^q . ■

PROBLEM 3.1.5 Suppose that the dual X' of a normed space X is separable. Show that X is separable.

Solution Let $S = \{x'_n : n \geq 1\}$ be a countable dense subset of X' . We can choose x_n in X such that

$$|x'_n(x_n)| \geq \frac{1}{2}\|x'_n\|, \quad \|x_n\| \leq 1 \quad \text{for all } n \quad (2)$$

Let $Y = \text{span}\{x_n\}$.

We shall first prove that $\overline{Y} = X$. Suppose that there is $a \in X$ with $a \notin \overline{Y}$. Then by Problem 1.4.1, there is $x' \in X'$ such that

$$x'(a) \neq 0, \quad \|x'\| = 1, \quad x'(y) = 0 \quad \text{for all } y \in Y.$$

Since S is dense in X' , there is x'_n in S such that $\|x' - x'_n\| < 1/4$. As $x_n \in Y$ we see that $x'(x_n) = 0$, and hence

$$|x'_n(x_n)| = |(x'_n - x')(x_n)| \leq \|x'_n - x'\| \|x_n\| < \frac{1}{4}$$

since $\|x_n\| \leq 1$. Thus, $\|x'_n\| < 1/2$ by (2). This implies that

$$\|x'\| \leq \|x' - x'_n\| + \|x'_n\| < \frac{1}{4} + \frac{1}{2}.$$

This gives a contradiction as $\|x'\| = 1$. Therefore, $\overline{Y} = X$.

We now proceed as in the solution of Problem 2.1.21. Let M be a countable dense subset of the scalar field $\mathbb{K}$. Let E_n be the set of all $x \in X$ which are expressible as

$$x = \sum_{i=1}^n m_i x_i, \quad \text{with } m_i \in M,$$

and let

$$E = \bigcup_{n=1}^{\infty} E_n.$$

Since there is a map from M^n onto E_n and M^n is countable, we see that each E_n is countable. Hence E is countable.

Let $x \in X$ and $\epsilon > 0$ be given. There is $y \in Y$ such $\|x - y\| < \epsilon/2$. Since $Y = \text{span}\{x_n\}$, we have

$$y = \sum_{i=1}^n k_i x_i, \quad k_i \in \mathbb{K}.$$

Since M is dense in $\mathbb{K}$ there exist m_i in M such that

$$|k_i - m_i| < \frac{\epsilon}{2n} \quad \text{for } 1 \leq i \leq n.$$

Let

$$z = \sum_{i=1}^n m_i x_i.$$

Then $z \in E_n \subset E$ and

$$\|y - z\| = \left\| \sum_{i=1}^n (k_i - m_i)x_i \right\| \leq \sum_{i=1}^n |k_i - m_i| \|x_i\| < n \left(\frac{\epsilon}{2n} \right)$$

since $\|x_i\| \leq 1$. So,

$$\|x - z\| \leq \|x - y\| + \|y - z\| < \epsilon.$$

This shows that $\overline{E} = X$. Since E is countable the proof is complete. ■

PROBLEM 3.1.6 Prove that the dual of l^∞ contains a proper subspace which is linearly isometric to l^1 .

Solution If $y \in l^1$ and $x \in l^\infty$ then

$$\sum_{j=1}^{\infty} |y(j)x(j)| \leq \|x\|_\infty \sum_{j=1}^{\infty} |y(j)| = \|x\|_\infty \|y\|_1.$$

Hence, if

$$f_y(x) = \sum_{j=1}^{\infty} y(j)x(j), \quad x \in l^\infty, \quad (3)$$

then f_y maps l^∞ into the scalar field $\mathbb{K}$. It is linear,

$$|f_y(x)| \leq \|y\|_1 \|x\|_\infty, \quad x \in l^\infty,$$

and hence $\|f_y\| \leq \|y\|_1$. Exactly as in the first part of the solution of Problem 3.1.2, we can see that $\|y\|_1 \leq \|f_y\|$. Thus to each $y \in l^1$ we associate an element $f_y \in (l^\infty)'$ such that $\|f_y\| = \|y\|_1$. Let

$$F(y) = f_y, \quad y \in l^1,$$

where f_y is given by (3). This gives a mapping $F : l^1 \rightarrow (l^\infty)'$. It can be easily verified to be a linear isometry. If F is onto, then $(l^\infty)'$ is l^1 . But l^∞ is not separable while l^1 is separable (see Problem 1.1.12). But X' cannot be separable if X is not separable by Problem 3.1.5. This contradiction completes the proof. ■

PROBLEM 3.1.7 If X and Y are normed spaces and $F \in BL(X, Y)$, show that

$$\|F\| = \sup \{|y'(F(x))| : x \in X, \|x\| \leq 1, y' \in Y', \|y'\| \leq 1\}.$$

Solution Let

$$S = \{|y'(F(x))| : x \in X, \|x\| \leq 1, y' \in Y', \|y'\| \leq 1\}.$$

Then for all α in S ,

$$\alpha = |y'(F(x))| \leq \|y'\| \|F(x)\| \leq \|y'\| \|F\| \|x\| \leq \|F\|.$$

This shows that $\|F\|$ is an upper bound of S . If $\|F\| = 0$, then $\sup S = 0 = \|F\|$. So let $\|F\| \neq 0$. By the definition of $\|F\|$, there is a sequence $\{x_n\}$ in X such that $\|x_n\| \leq 1$, and

$$\|F(x_n)\| \rightarrow \|F\| \quad \text{as } n \rightarrow \infty.$$

So $y_n = F(x_n) \neq 0$ for all large n . For such n we can find, by Problem 1.4.2, y'_n in Y' such that

$$y'_n(y_n) = \|y_n\|, \quad \|y'_n\| = 1.$$

Then,

$$\|y_n\| = y'_n(y_n) = y'_n(F(x_n)) = |y'_n(F(x_n))| \in S.$$

Since $\|y_n\| = \|F(x_n)\| \rightarrow \|F\|$, we see that $\|F\| = \sup S$. This completes the proof. ■

PROBLEM 3.1.8 Let X be a Banach space, Y be any normed space and $F : X \rightarrow Y$ be linear. Suppose also that if $\{x_n\}$ is a sequence in X such that $x_n \rightarrow 0$ and $\{F(x_n)\}$ is Cauchy, then $F(x_n) \rightarrow 0$ in Y . Prove that $F \in BL(X, Y)$.

Solution Let J be the canonical mapping of Y into Y'' , and $G = J \circ F$. Then $G : X \rightarrow Y''$ is linear. To show that the graph of G is closed, let $x_n \rightarrow x$ in X and $G(x_n) \rightarrow y''$ in Y'' . Let $u_n = x_n - x$. Then $u_n \rightarrow 0$, and

$$G(u_n) = G(x_n) - G(x) \rightarrow y'' - G(x).$$

So $\{G(u_n)\}$ is a Cauchy sequence. Since J is an isometry, we get

$$\|G(u_n - u_m)\| = \|J \circ F(u_n - u_m)\| = \|F(u_n - u_m)\|,$$

$$\|G(u_n) - G(u_m)\| = \|F(u_n) - F(u_m)\|.$$

Hence, $\{F(u_n)\}$ is a Cauchy sequence in Y . So $F(u_n) \rightarrow 0$ by hypothesis. Therefore,

$$G(u_n) = J \circ F(u_n) \rightarrow 0,$$

$$G(x_n) = G(u_n) + G(x) \rightarrow G(x).$$

Thus, $y'' = G(x)$. This shows that the graph of G is closed. Since X and Y'' are Banach, we see that G is continuous by the closed graph theorem. Hence,

$$\|F(x)\| = \|J \circ F(x)\| = \|G(x)\| \leq \|G\| \cdot \|x\|,$$

and so F is bounded. This completes the proof. ■

PROBLEM 3.1.9 Let X be a normed space, $E \subset X$ and $M \subset X'$. The *annihilator* of E is the set

$$E^a = \{x' \in X' : x'(x) = 0 \text{ for all } x \in E\};$$

the *annihilator* of M is the set

$${}^aM = \{x \in X : x'(x) = 0 \text{ for all } x' \in M\}.$$

Prove that E^a is a closed linear subspace of X' and that aM is a closed linear subspace of X .

Solution Let $x', y' \in E^a$, $k \in \mathbb{K}$ and $z' = kx' + y'$. Then for all $x \in E$, we have

$$z'(x) = kx'(x) + y'(x) = k \cdot 0 + 0 = 0.$$

Thus $z' \in E^a$. This shows that E^a is a subspace of X' .

Let $\{x'_n\}$ be a sequence in E^a and $x'_n \rightarrow x'$ in X' . Then,

$$|x'_n(x) - x'(x)| = |(x'_n - x')x| \leq \|x'_n - x'\| \cdot \|x\| \rightarrow 0$$

as $n \rightarrow \infty$. So, $x'_n(x) \rightarrow x'(x)$. If $x \in E$, then $x'_n(x) = 0$ for all n ; hence $x'(x) = 0$. Thus, $x' \in E^a$. This shows that E^a is closed in X' .

Similarly, if $x, y \in {}^aM$ and $k \in \mathbb{K}$, then

$$x'(kx + y) = kx'(x) + x'(y) = 0 \text{ for } x' \in M;$$

hence $kx + y \in {}^aM$. Thus aM is a subspace of X .

If $x_n \in {}^aM$ and $x_n \rightarrow x$ in X , then

$$x'(x_n) \rightarrow x'(x) \text{ for all } x'.$$

If $x' \in M$ then $x'(x_n) = 0$ for all n , and hence $x'(x) = 0$. Thus $x \in {}^aM$. So aM is closed. This completes the proof. ■

PROBLEM 3.1.10 Let Y be a closed subspace of a normed space X and Y^a be its annihilator. Prove that

- (a) Y' is linearly isometric to X'/Y^a ,
- (b) $(X/Y)'$ is linearly isometric to Y^a .

Solution (a) For $x' + Y^a \in X'/Y^a$, let

$$F(x' + Y^a) = x'|_Y, \text{ the restriction of } x' \text{ to } Y.$$

If $x'_1 + Y^a = x'_2 + Y^a$, then $x'_1 - x'_2 \in Y^a$, and hence

$$x'_1(x) = x'_2(x) \text{ for all } x \in Y,$$

$$x'_{1|Y} = x'_{2|Y}.$$

Since $x'|_Y$ is a continuous linear functional on Y , we see that F is a well-defined map from X'/Y^a to Y' . This map is easily seen to be linear. Each $y' \in Y'$ has a Hahn-Banach extension x' so that

$$\begin{aligned} F(x' + Y^a) &= x'|_Y = y', \\ \|y'\| &= \|x'\| \leq \|x' + Y^a\|. \end{aligned}$$

This shows that F is onto and

$$\|F(x' + Y^a)\| \leq \|x' + Y^a\|. \quad (4)$$

On the other hand, if $F(x' + Y^a) = y'$ and $z' \in Y^a$ then for all $y \in Y$,

$$|(x' + z')(y)| = |x'(y)| = |y'(y)| \leq \|y'\| \cdot \|y\|;$$

hence $\|x' + z'\| \leq \|y'\|$. Since this is true for all $z' \in Y^a$ we get

$$\|x' + Y^a\| \leq \|y'\| = \|F(x' + Y^a)\|. \quad (5)$$

Thus F is an isometry by (4) and (5).

(b) For $x' \in Y^a$, let

$$f(x + Y) = x'(x), \quad x \in X.$$

If $x_1 + Y = x_2 + Y$, then $x_1 - x_2 \in Y$, and hence

$$x'(x_1 - x_2) = 0, \quad x'(x_1) = x'(x_2).$$

This shows that $f : X/Y \rightarrow \mathbb{K}$ is a well-defined map. Clearly, it is linear.

Given $x + Y \in X/Y$, we can find a sequence $\{y_n\}$ in Y such that

$$\|x + y_n\| \rightarrow \|x + Y\|.$$

Since $x'(y_n) = 0$, we see that

$$|f(x + Y)| = |x'(x)| = |x'(x + y_n)| \leq \|x'\| \cdot \|x + y_n\|.$$

Letting $n \rightarrow \infty$,

$$|f(x + Y)| \leq \|x'\| \cdot \|x + Y\|.$$

This is true for all $x + Y$ in X/Y . Hence f is a bounded linear functional on X/Y and $\|f\| \leq \|x'\|$. On the other hand,

$$|x'(x)| = |f(x + Y)| \leq \|f\| \cdot \|x + Y\| \leq \|f\| \cdot \|x\|,$$

for all x . Hence $\|x'\| \leq \|f\|$. Thus to each $x' \in Y^a$ we associate an element f in $(X/Y)'$ such that

$$f(x + Y) = x'(x), \quad x \in X \quad \text{and} \quad \|f\| = \|x'\|.$$

Define $F : Y^a \rightarrow (X/Y)'$ by $F(x') = f$. It can be seen to be a linear isometry.

To see that F is onto, let $f \in (X/Y)'$. Define x' on X by

$$x'(x) = f(x + Y), \quad x \in X.$$

Clearly, x' is linear and

$$|x'(x)| = |f(x + Y)| \leq \|f\| \cdot \|x + Y\| \leq \|f\| \cdot \|x\|.$$

Therefore, $x' \in X'$ with $\|x'\| \leq \|f\|$. If $y \in Y$ then $y + Y$ is the zero of X/Y ; hence

$$x'(y) = f(y + Y) = 0.$$

Thus, $x' \in Y^a$. The definition of x' shows that $F(x') = f$. This proves that Y^a is linearly isometric to $(X/Y)^a$. ■

PROBLEM 3.1.11 Let X and Y be normed spaces and $F : X \rightarrow Y$ be linear. For $y' \in Y'$, define $F'(y') : X \rightarrow \mathbb{K}$ by

$$F'(y')(x) = y'(F(x)), \quad x \in X.$$

Prove the following:

- (a) $F'(y')$ is linear for each $y' \in Y'$.
- (b) $F'(y') \in X'$ for all $y' \in Y'$ iff $F \in BL(X, Y)$.
- (c) If $F \in BL(X, Y)$ then $F' \in BL(Y', X')$, and $\|F'\| = \|F\|$.

F' is called the *transpose* or *adjoint* of F .

Solution (a) For $x_1, x_2 \in X, k \in \mathbb{K}$, we have

$$F(kx_1 + x_2) = kF(x_1) + F(x_2);$$

hence

$$\begin{aligned} y'(F(kx_1 + x_2)) &= ky'(F(x_1) + F(x_2)), \\ (F'y')(kx_1 + x_2) &= k(F'y')(x_1) + (F'y')(x_2). \end{aligned}$$

Thus $F'y'$ is linear on X .

(b) Suppose that $F \in BL(X, Y)$ and $y' \in Y'$. Then for all x in X ,

$$|(F'y')(x)| = |y'F(x)| \leq \|y'\| \|F(x)\| \leq \|y'\| \|F\| \|x\|.$$

Hence $F'y'$ is bounded and

$$\|F'y'\| \leq \|y'\| \|F\|. \quad (6)$$

Conversely, suppose that $F'y' \in X'$ for all $y' \in Y'$. If $F(x) \neq 0$, there is y' in Y' such that

$$y'(F(x)) = \|F(x)\|, \quad \|y'\| = 1.$$

Hence,

$$\|F(x)\| = |y'F(x)| = |F'y'(x)| \leq \|F'y'\| \cdot \|x\|.$$

Thus F is bounded and

$$\|F\| \leq \|F'y'\|. \quad (7)$$

(c) Let $F \in BL(X, Y)$. Then F' maps Y' into X' by (b). It can be verified that $F' : Y' \rightarrow X'$ is linear. Since (6) holds for all $y' \in Y'$, we see that F' is bounded and $\|F'\| \leq \|F\|$. Since (7) holds for some y' with $\|y'\| = 1$, we get

$$\|F\| \leq \|F'\| \cdot \|y'\| = \|F'\|.$$

Hence, $F' \in BL(Y', X')$ and $\|F'\| = \|F\|$. This completes the proof. ■

PROBLEM 3.1.12 Let X and Y be normed spaces, $F \in BL(X, Y)$ and $Y_1 = R(F)$. Prove that

- (a) the annihilator of Y_1 is the zero space of the transpose F' of F ,
- (b) the annihilator of the zero space of F' is Y_1 ,
- (c) if the linear equation $F(x) = y$ has a solution, then

$$y'(y) = 0$$

for every solution y' of the equation

$$F'(y') = 0.$$

Solution (a) The annihilator of Y_1 is the set

$$Y_1^a = \{y' \in Y' : y'(y) = 0 \text{ for all } y \in Y_1\}.$$

Let $y' \in Y_1^a$. For all $x \in X$, we have

$$F'(y')(x) = y'(Fx) = 0,$$

since $F(x) \in R(F) \subset Y_1$. Thus, $F'(y')$ is the zero of X' , that is, $y' \in Z(F')$.

Conversely, let $y' \in Z(F')$. Then $F'(y') = 0$, that is, $F'(y')(x) = 0$ for all $x \in X$. Let y be any element of Y_1 . Then there is a sequence $\{y_n\}$ in $R(F)$ such that $y_n \rightarrow y$ in Y . If $y_n = F(x_n)$, then

$$y'(y_n) = y'(Fx_n) = F'y'(x_n) = 0 \quad \text{for all } n.$$

But $y'(y_n) \rightarrow y'(y)$. Hence $y'(y) = 0$. This shows that $y' \in Y_1^a$. It follows that $Y_1^a = Z(F')$.

(b) The annihilator of $Z(F')$ is the set

$$E = \{y \in Y : y'(y) = 0 \text{ for all } y' \in Z(F')\}.$$

Let $y = F(x) \in R(F)$. If $y' \in Z(F')$, then $F'(y') = 0$, and so $F'y'(x) = 0$ for all x , that is, $y'F(x) = 0$. Thus,

$$y'(y) = 0 \quad \text{for any } y' \in Z(F'),$$

that is, $y \in E$. Hence, $R(F) \subset E$. But E is a closed subspace of Y by Problem 3.1.9. Therefore, $Y_1 = \overline{R(F)} \subset E$.

On the other hand, let $z \notin Y_1$. Then by Problem 1.4.1 there is $y' \in Y'$ such that $y'(y) = 0$ for all y in Y_1 , $y'(z) \neq 0$ and $\|y'\| = 1$. Thus, $y' \in Y_1^a$ and hence $y' \in Z(F')$ by (a). It follows that $z \notin E$. Thus if $z \in E$, then $z \in Y_1$. Therefore, $E = Y_1$.

(c) Suppose that the equation $F(x) = y$ has a solution, that is, $y = F(x)$ for some x . Hence, $y \in R(F) \subset E$. Thus $y'(y) = 0$ for all $y' \in Z(F')$, that is, for all y' with $F'(y') = 0$. This completes the proof. ■

PROBLEM 3.1.13 Let X, Y, Z be normed spaces, $F \in BL(X, Y)$ and $G \in BL(Y, Z)$. Prove that $(G \circ F)' = F' \circ G'$.

Solution Note that

$$\begin{aligned} F' &\in BL(Y', X') & G' &\in BL(Z', Y') \\ G \circ F &\in BL(X, Z) & (G \circ F)' &\in BL(Z', X'). \end{aligned}$$

Let $z' \in Z'$ and $y' = G'(z')$. If $x \in X$ and $y = F(x)$, then

$$F'y'(x) = y'(Fx) = y'(y) = G'z'(y);$$

hence,

$$F'y'(x) = z'(Gy) = z'(G \circ Fx) = (G \circ F)'(z')(x).$$

This is true for all x in X . Therefore,

$$F'y' = (G \circ F)'(z'),$$

that is, $F' \circ G'(z') = (G \circ F)'(z')$. This is true for all $z' \in Z'$. Thus,

$$F' \circ G' = (G \circ F)'. \quad \blacksquare$$

PROBLEM 3.1.14 Let X and Y be normed spaces, and Y be reflexive. Define

$$T : BL(X, Y) \rightarrow BL(Y', X')$$

by $T(F) = F'$, the transpose of F . Prove that T is onto.

Solution Let $G \in BL(Y', X')$. Let $J_1 : X \rightarrow X''$ and $J_2 : Y \rightarrow Y''$ be the canonical mappings. Then J_2 is onto since Y is reflexive. Composing the maps

$$J_1 : X \rightarrow X'', \quad G' : X'' \rightarrow Y'', \quad J_2^{-1} : Y'' \rightarrow Y$$

we get a bounded linear map

$$F = J_2^{-1}G'J_1 : X \rightarrow Y.$$

If $y' \in Y'$ and $x \in X$ let

$$y = F(x), \quad x' = G(y'), \quad x'' = J_1 x.$$

Then,

$$J_2 y(y') = y'(y) = y'(F(x)) = F'y'(x).$$

But,

$$J_2 y = J_2 Fx = G'J_1 x = G'x''.$$

Therefore,

$$\begin{aligned} F'y'(x) &= J_2 y(y') = G'x''(y') = x''Gy' \\ &= x''x' = J_1 x x' = x'(x) \\ &= Gy'(x). \end{aligned}$$

Thus, $Gy' = F'y'$ for all $y' \in Y'$, that is, $G = F' = T(F)$. This completes the proof. ■

PROBLEM 3.1.15 Show that a closed subspace of a reflexive space is reflexive.

Solution Suppose that X is a reflexive space and Y is a closed subspace of X . Let $J : X \rightarrow X''$ and $J_1 : Y \rightarrow Y''$ be the canonical mappings. Then J is onto.

To prove that J_1 is onto, let $y'' \in Y''$. If $x' \in X'$ its restriction to Y , $x'|_Y$, belongs to Y' . Define $x'' : X' \rightarrow \mathbb{K}$ by

$$x''(x') = y''(x'|_Y), \quad x' \in X'.$$

We see that x'' is linear and that if $y' = x'|_Y$, then

$$\begin{aligned} \|y'\| &\leq \|x'\|, \\ |x''(x')| &= |y''(y')| \leq \|y''\| \|x'\|. \end{aligned}$$

Thus, $x'' \in X''$. Since J is onto, there is x_1 in X such that $Jx_1 = x''$.

Suppose that $x_1 \notin Y$. Since Y is closed, we see by Problem 1.4.1 that there is $x'_1 \in X'$ such that

$$x'_1(Y) = \{0\}, \quad x'_1(x_1) \neq 0, \quad \|x'_1\| = 1.$$

So if $y'_1 = x'_{1|Y}$, then $y'_1 = 0$, and hence

$$x''(x'_1) = y''(y'_1) = 0.$$

However, $x''(x'_1) = Jx_1(x'_1) = x'_1(x_1) \neq 0$. This contradiction shows that $x_1 \in Y$. Given y' in Y' choose some extension x' of y' to X' ; we get

$$y''(y') = x''(x') = Jx_1(x') = x'(x_1) = y'(x_1).$$

This is true for all $y' \in Y'$. Thus, $J_1x_1 = y''$. This completes the proof. ■

PROBLEM 3.1.16 Let X and Y be two linearly homeomorphic normed spaces. If X is reflexive show that Y also is reflexive.

Solution Let $F : X \rightarrow Y$ be a bijective linear continuous map such that its inverse $G = F^{-1} : Y \rightarrow X$ also is continuous. Since $(G \circ F)' = F' \circ G'$ and $(F \circ G)' = G' \circ F'$, we see that $F' : Y' \rightarrow X'$ is invertible. For the same reason, $F'' : X'' \rightarrow Y''$ is invertible.

To show that Y is reflexive let $y'' \in Y''$. Then, $y'' = F''(x'')$, $x'' \in X''$. Since X is reflexive, there is $x \in X$ such that

$$x''(x') = x'(x) \quad \text{for all } x' \in X'.$$

Let y' be any element of Y' and $x' = F'(y')$. Hence,

$$x''(F'y') = F'y'(x).$$

However,

$$x''(F'(y')) = F''x''(y') = y''(y'),$$

$$F'y'(x) = y'(F(x)).$$

It follows that $y''(y') = y'(F(x)) = y'(y)$. Hence, $y'' = J_1(y)$, where $J_1 : Y \rightarrow Y''$ is the canonical map and $y = F(x)$. ■

PROBLEM 3.1.17 Let Y be a dense subset of a normed space X . Show that Y' is linearly isometric to X' .

Solution Let $y' \in Y'$. For each $x \in X$, there is a sequence $\{y_n\}$ in Y such that $y_n \rightarrow x$. Since

$$|y'(y_n) - y'(y_m)| \leq \|y'\| \cdot \|y_n - y_m\|$$

and $\{y_n\}$ is a Cauchy sequence, we see that $\{y'(y_n)\}$ is a Cauchy sequence of scalars. So,

$$x'(x) = \lim_{n \rightarrow \infty} y'(y_n) \quad (8)$$

exists. If $\{z_n\}$ is another sequence in Y such that $z_n \rightarrow x$, then

$$y'(y_n) - y'(z_n) = y'(y_n - z_n) \rightarrow 0.$$

This shows that choosing any sequence $\{y_n\}$ in Y with $y_n \rightarrow x$ one can define $x'(x)$ unambiguously by (8). Thus, (8) gives a map $x' : X \rightarrow \mathbb{K}$. If $y \in Y$, we can take $y_n = y$ for all n . So $x'(y) = y'(y)$ for $y \in Y$, that is, $x'|_Y = y'$.

If $y_n \rightarrow x_1$ and $z_n \rightarrow x_2$, where $y_n, z_n \in Y$, then

$$ky_n + z_n \rightarrow kx_1 + x_2, \quad y'(ky_n + z_n) = ky'(y_n) + y'(z_n).$$

It follows that

$$x'(kx_1 + x_2) = kx'(x_1) + x'(x_2).$$

Thus x' is a linear functional on X . If $y_n \in Y$ and $y_n \rightarrow x$, then

$$|y'(y_n)| \leq \|y'\| \cdot \|y_n\|.$$

Letting $n \rightarrow \infty$, we see that $|x'(x)| \leq \|y'\| \|x\|$. Thus x' is bounded and, further, $\|x'\| \leq \|y'\|$. Since

$$|y'(y)| = |x'(y)| \leq \|x'\| \|y\|$$

for all $y \in Y$, we readily see that $\|y'\| \leq \|x'\|$. So, $\|x'\| = \|y'\|$.

Thus, to each $y' \in Y'$ we associate an element $x' \in X'$ such that $x'|_Y = y'$. Moreover, $\|x'\| = \|y'\|$. Let x'_1 be another continuous extension of y' . Then $x'_1(y_n) = y'(y_n)$ if $y_n \in Y$. So if $y_n \rightarrow x$, then

$$x'_1(x) = \lim y'(y_n) = x'(x).$$

This shows that each $y' \in Y'$ has a unique extension $x' \in X'$.

Write $F(y') = x'$ where $x' \in X'$, $x'|_Y = y'$. This defines a map $F : Y' \rightarrow X'$. It can be seen that F is linear. Moreover,

$$\|F(y')\| = \|x'\| = \|y'\|,$$

that is, F is an isometry. It is surjective, since for any $x' \in X'$ we have $F(y') = x'$ if $y' = x'|_Y$. This proves the result. ■

PROBLEM 3.1.18 Show that l^p is reflexive for $1 < p < \infty$.

Solution Let $X = l^p$ and $Y = l^q$ where $1/p + 1/q = 1$. Let $F(y) = f_y$, $y \in Y$, where

$$f_y(x) = \sum_{j=1}^{\infty} y(j)x(j), \quad x \in X.$$

that is,

$$F(y)(x) = \sum_{j=1}^{\infty} y(j)x(j), \quad x \in X, y \in Y.$$

In the solution of Problem 3.1.4 we found that $F : Y \rightarrow X'$ is a surjective linear isometry. If p is changed to q , then X and Y are interchanged. So, for the same reason, if

$$G(x)(y) = \sum_{j=1}^{\infty} x(j)y(j), \quad y \in Y, x \in X,$$

then $G : X \rightarrow Y'$ is a surjective linear isometry.

Let $x'' \in X''$ and $y' = x'' \circ F$. Then, $y' : Y \rightarrow \mathbb{K}$ is a bounded linear functional. So there is $x \in X$ such that $G(x) = y'$. Given x' in X' we can find $y \in Y$ such that $F(y) = x'$. Hence,

$$\begin{aligned} x'(x) &= F(y)(x) = \sum_{j=1}^{\infty} y(j)x(j), \\ y'(y) &= G(x)(y) = \sum_{j=1}^{\infty} x(j)y(j). \end{aligned}$$

Thus, $y'(y) = x'(x)$, and so

$$x''(x') = x''(F(y)) = y'(y) = x'(x).$$

Since this is true for any $x' \in X'$, we see that $x'' = J(x)$, where J is the canonical map. This proves that $X = l^p$ is reflexive. ■

PROBLEM 3.1.19 If a normed space X is reflexive, show that X' is reflexive.

Solution Let $J : X \rightarrow X''$ and $J_1 : X' \rightarrow X'''$ be the canonical mappings. We prove that J_1 is onto if J is onto. Let x''' be any element in X''' . Define $x' : X \rightarrow \mathbb{K}$ by

$$x'(x) = x'''(Jx), \quad x \in X.$$

Then $x' = x''' \circ J \in X'$. Let $x'' \in X''$. There exists $x \in X$ such that $Jx = x''$. So,

$$x'''(x'') = x'''(Jx) = x'(x).$$

But,

$$J_1 x'(x'') = x''(x') = Jx(x') = x'(x).$$

Hence,

$$J_1 x'(x'') = x'''(x'') \quad \text{for all } x'' \in X'',$$

that is, $J_1 x' = x'''$. This proves the result. ■

PROBLEM 3.1.20 Show by means of an example that a normed space need not be reflexive even though X' is reflexive.

Solution Let $1 < p < \infty$, $X = c_{00}$ with the norm $\|\cdot\|_p$ and $Y = l^p$. If $y \in Y$ and

$$y_n = \sum_{j=1}^n y(j)e_j, \quad n = 1, 2, \dots,$$

then $y_n \in X$ and

$$\|y - y_n\|_p^p = \sum_{j=n+1}^{\infty} |y(j)|^p \rightarrow 0 \quad \text{as } n \rightarrow \infty;$$

hence $y_n \rightarrow y$. This shows that X is dense in Y . Using Problem 3.1.17 with X, Y interchanged we see that X' is linearly isometric to Y' . But, $Y = l^p$ is reflexive by Problem 3.1.18. So Y' is reflexive by Problem 3.1.19. Hence X' is reflexive by Problem 3.1.16.

If X were reflexive then it would be isometric to X'' which is Banach. But X , being proper and dense in Y , is not complete. Therefore, X is not reflexive. This completes the proof. ■

PROBLEM 3.1.21 Show that a Banach space X is reflexive if its dual X' is reflexive.

Solution Suppose that X is a Banach space and X' is reflexive. To prove that X is reflexive, let $J : X \rightarrow X''$ be the canonical mapping, and $Y = J(X)$. Then Y is complete since X is Banach and J is an isometry. Thus Y is closed in X'' . But X'' is reflexive by Problem 3.1.19 since X' is reflexive. Hence Y is reflexive by Problem 3.1.15. Let J' be the transpose of $J : X \rightarrow Y$ and J'' be the transpose of J' . To show that $J(X) = X''$, let $x'' \in X''$ and $y'' = J''(x'')$. Then, $y'' \in Y''$. Since Y is reflexive, there is $y \in Y$ such that $y'' = J_1(y)$, where $J_1 : Y \rightarrow Y''$ is the canonical map. Since $Y = J(X)$, there is $x \in X$ such that $y = J(x)$.

Let $J_2 = J^{-1} : Y \rightarrow X$. Given $x' \in X'$, let $y' = J'_2(x')$. Then,

$$J'(y') = (J' \circ J'_2)(x') = (J_2 \circ J)'(x') = x'.$$

Since $J'' = (J')'$, we have

$$(J''x'')(y') = x''(J'y') = x''(x').$$

Since $J''x'' = y'' = J_1y$, we get

$$\begin{aligned} x''(x') &= (J''x'')(y') = (J_1y)(y') = y'(y) \\ &= (J'_2x')(y) = x'(J_2y) = x'(x). \end{aligned}$$

Thus, $x''(x') = x'(x)$ for all x' in X' . Hence $x'' = Jx$. This completes the proof. ■

PROBLEM 3.1.22 Show that l^1 , l^∞ , c and c_0 are not reflexive.

Solution Suppose that X is reflexive and separable. Then $J : X \rightarrow X''$, the canonical map, is a surjective isometry, and hence X'' is separable. So X' is separable by Problem 3.1.5.

Let $X = l^1$. Then X is separable and X' is (linearly isometric to) l^∞ . But l^∞ is not separable. Hence X' cannot be separable. So $X = l^1$ cannot be reflexive.

Let $X = c_0$ or c . If X is reflexive, then X' is reflexive. But X' is linearly isometric to l^1 . So l^1 must be reflexive by Problem 3.1.16. This contradiction shows that $X = c_0$ or c is not reflexive.

Let $X = l^\infty$. Then X is linearly isometric to $(l^1)'$. Hence if X is reflexive, then $(l^1)'$ is reflexive by Problem 3.1.16. Since l^1 is a Banach space this implies, by Problem 3.1.21, that l^1 is reflexive. This contradiction shows that $X = l^\infty$ is not reflexive. This completes the proof. ■

PROBLEM 3.1.23 A normed space X is said to be *uniformly convex* if for every $\epsilon > 0$, there exists $\delta > 0$ such that $\|(x + y)/2\| \leq 1 - \delta$ for all x, y in X , with

$$\|x\| \leq 1, \quad \|y\| \leq 1, \quad \|x - y\| \geq \epsilon.$$

Let X be a normed space and x' be a nonzero element in X' . Prove the following:

- (a) If X is reflexive then there exists at least one x in X such that

$$x'(x) = \|x'\|, \quad \|x\| = 1. \quad (9)$$

- (b) If X is strictly convex, then there exists at most one x in X satisfying (9).

- (c) If X is a uniformly convex Banach space, then there exists exactly one x in X satisfying (9).

Solution (a) Since $x' \neq 0$, there is f in X'' such that

$$f(x') = \|x'\|, \quad \|f\| = 1$$

by Problem 1.4.2. Since X is reflexive, there is x in X such that $J(x) = f$, that is, $f(x') = x'(x)$ for all x' . Hence,

$$x'(x) = f(x') = \|x'\|.$$

Further, $\|x\| = \|J(x)\| = \|f\| = 1$. This proves (a).

- (b) Let X be strictly convex, and

$$x'(x_1) = \|x'\| = x'(x_2) \quad \text{and} \quad \|x_1\| = 1 = \|x_2\|.$$

Let $x = (x_1 + x_2)/2$. Then,

$$\|x'\| = \frac{1}{2}(x'(x_1) + x'(x_2)) = x'(x).$$

So $\|x'\| = |x'(x)| \leq \|x'\| \cdot \|x\|$. It follows that

$$1 \leq \|x\| = \left\| \frac{x_1 + x_2}{2} \right\| \leq \frac{\|x_1\| + \|x_2\|}{2} = 1.$$

Therefore, $\|(x_1 + x_2)/2\| = 1$. Since X is strictly convex, this is possible only if $x_1 = x_2$. This proves (b).

(c) Suppose that X is a uniformly convex Banach space. There is a sequence $\{x_n\}$ in X such that

$$\|x_n\| = 1, \quad |x'(x_n)| \rightarrow \|x'\|.$$

Let

$$k_n = \begin{cases} \frac{|x'(x_n)|}{x'(x_n)} & \text{if } x'(x_n) \neq 0, \\ 1 & \text{if } x'(x_n) = 0, \end{cases}$$

and $y_n = k_n x_n$. Then, for all n ,

$$\begin{aligned} |k_n| &= 1, \quad \|y_n\| = |k_n| \|x_n\| = 1, \\ x'(y_n) &= k_n x'(x_n) = |x'(x_n)|. \end{aligned} \tag{10}$$

Thus, $x'(y_n) \geq 0$ and $x'(y_n) \rightarrow \|x'\|$ as $n \rightarrow \infty$.

Let $\epsilon > 0$ be given. By the uniform convexity there exists $\delta > 0$ such that $\|(x + y)/2\| \leq 1 - \delta$ for all x, y in X , with

$$\|x\| \leq 1, \quad \|y\| \leq 1, \quad \|x - y\| \geq \epsilon.$$

Since $\delta\|x'\| > 0$, there is N such that

$$|x'(y_n) - \|x'\|| < \frac{\delta}{2}\|x'\| \quad \text{for all } n \geq N.$$

Let $n, m \geq N$. Then,

$$\begin{aligned} \|x'\| &\leq |\|x'\| - x'(y_n)| + |x'(y_n)| < \frac{\delta}{2}\|x'\| + x'(y_n), \\ \|x'\| &< \frac{\delta}{2}\|x'\| + x'(y_m). \end{aligned}$$

Hence,

$$\begin{aligned} 2\|x'\| &< \delta\|x'\| + x'(y_n + y_m), \\ \|x'\| - \frac{\delta}{2}\|x'\| &< x'\left(\frac{y_n + y_m}{2}\right) \leq \|x'\| \left\| \frac{y_n + y_m}{2} \right\|, \end{aligned}$$

and so, $1 - \delta/2 < \|(y_n + y_m)/2\|$. It follows that

$$\|y_n - y_m\| < \epsilon$$

by the choice of δ . This is true for all $m, n \geq N$. Thus, $\{y_n\}$ is a Cauchy sequence in X . Since X is complete, it converges to some x in X . Hence, $x'(y_n) \rightarrow x'(x)$, that is,

$$x'(x) = \lim x'(y_n) = \|x'\|.$$

Since

$$\|x\| = \lim \|y_n\| = 1,$$

by (10), we see that x satisfies (9). Since a uniformly convex space is strictly convex, there cannot be more than one such x by (b). This completes the proof. ■

PROBLEM 3.1.24 As in Problem 1.4.17, a normed space is said to be smooth if for every nonzero $x \in X$ there is only one x' in X' , with

$$x'(x) = \|x\|, \quad \|x'\| = 1. \quad (11)$$

Prove that if X is reflexive and smooth, then X' is strictly convex.

Solution Let X be reflexive and smooth. To prove that X' is strictly convex, let x', y' be elements in X' , with

$$\|x'\| = \|y'\| = \left\| \frac{x' + y'}{2} \right\| = 1.$$

By Problem 1.4.2 there exists $x'' \in X''$ such that

$$x''(x' + y') = \|x' + y'\| = 2, \quad \|x''\| = 1. \quad (12)$$

Since X is reflexive, there is $x \in X$ such that

$$x''(z') = z'(x) \quad \text{for all } z' \in X',$$

and such that $\|x''\| = \|x\| = 1$. Let $u = x'(x), v = y'(x)$. Then,

$$u + v = (x' + y')(x) = x''(x' + y') = 2$$

by (12). But,

$$|u| \leq \|x'\| \|x\| = 1, \quad |v| \leq \|y'\| \|x\| = 1.$$

Since

$$|u + v|^2 + |u - v|^2 = 2|u|^2 + 2|v|^2,$$

we see that $4 + |u - v|^2 \leq 2 + 2$. It follows that $u = v = (u + v)/2 = 1$. Hence,

$$x'(x) = 1 = \|x\| = y'(x), \quad \|x'\| = \|y'\| = 1.$$

Since X is smooth and $x \neq 0$, this implies that $x' = y'$. This proves X' strictly convex. This proves the result. ■

3.2 Weak Convergence

Let X be a normed space. A sequence $\{x_n\}$ in X is said to be *weak convergent* (or to converge *weakly*) to x in X if

$$x'(x_n) \rightarrow x'(x) \quad \text{for each } x' \in X'.$$

In this case, we write $x_n \xrightarrow{w} x$ in X .

A sequence $\{x'_n\}$ in X' is said to be *weak* convergent* to x' in X' if

$$x'_n(x) \rightarrow x'(x) \quad \text{for each } x \in X.$$

In this case we write $x'_n \xrightarrow{w^*} x'$ in X' .

We say that $\{x_n\}$ is a *weak Cauchy sequence* if $\{x'(x_n)\}$ is a Cauchy sequence of scalars for each x' in X' . A normed space X is said to be *weak complete* if every weak Cauchy sequence in X converges weakly in X .

If $x_n \xrightarrow{w} x$ and $x_n \xrightarrow{w} y$, then $x = y$. If $x_n \xrightarrow{w} x$, $y_n \xrightarrow{w} y$ and $k \in \mathbb{K}$, then $kx_n + y_n \xrightarrow{w} kx + y$.

Analogous results hold for weak* limits.

EXERCISES

Let X be a normed space and X' be its dual.

1. Show that if $x_n \rightarrow x$ in the norm of X , then $x_n \xrightarrow{w} x$.
2. If $x_n \xrightarrow{w} x$ and $x_n \xrightarrow{w} y$, show that $x = y$.
3. If $x_n \xrightarrow{w} x$, $y_n \xrightarrow{w} y$ and $k \in \mathbb{K}$, show that $kx_n + y_n \xrightarrow{w} kx + y$.
4. If $x'_n \xrightarrow{w^*} x'$ and $x'_n \xrightarrow{w^*} y'$, show that $x' = y'$.
5. If $x'_n \xrightarrow{w^*} x'$, $y'_n \xrightarrow{w^*} y'$ and $k \in \mathbb{K}$, show that $kx'_n + y'_n \xrightarrow{w^*} kx' + y'$.
6. If $X = \mathbb{R}^2$ and $x_n \xrightarrow{w} x$, show that $x_n \rightarrow x$ in the norm of X .

PROBLEMS

PROBLEM 3.2.1 Let X be a normed space. Prove that if $x'_n \xrightarrow{w} x'$ in X' , then $x'_n \xrightarrow{w^*} x'$ in X' , but not conversely.

Solution Let $x'_n \xrightarrow{w} x'$. This means that $x''(x'_n) \rightarrow x''(x')$ for every $x'' \in X''$. Let $x \in X$ and $x'' = J(x)$, where J is the canonical mapping defined in Section 3.1. Then,

$$x''(x'_n) = x'_n(x), \quad x''(x') = x'(x).$$

Hence, $x'_n(x) \rightarrow x'(x)$. This is true for all x in X . So this means that $x'_n \xrightarrow{w^*} x'$ in X' .

To see that the converse is not true, let $X = c_0$ and x'_n be defined by

$$x'_n(x) = x(n), \quad x \in X, \quad n = 1, 2, \dots$$

Then, $x'_n(x) = x(n) \rightarrow 0$ as $n \rightarrow \infty$, for each $x \in X$. Thus $x'_n \xrightarrow{w^*} 0$. In the solution of Problem 3.1.2 we found that if g is any bounded linear functional on c_0 and $g(e_j) = y(j)$, then $y \in l^1$, and

$$\|y\|_1 = \sum_{j=1}^{\infty} |g(e_j)| \leq \|g\|.$$

Thus,

$$\sum_{j=1}^{\infty} |x'(e_j)| \leq \|x'\| < \infty \quad \text{for all } x' \in X'.$$

Hence we can define $f : X' \rightarrow \mathbb{K}$ by

$$f(x') = \sum_{j=1}^{\infty} (-1)^j x'(e_j), \quad x' \in X'.$$

Clearly, f is linear and

$$|f(x')| = \sum_{j=1}^{\infty} |x'(e_j)| \leq \|x'\|, \quad x' \in X'.$$

Therefore, $f \in X''$ with $\|f\| \leq 1$. Since

$$f(x'_n) = \sum_{j=1}^{\infty} (-1)^j x'_n(e_j) = \sum_{j=1}^{\infty} (-1)^j e_j(n) = (-1)^n,$$

we see that $\{f(x'_n)\}$ does not converge. This shows that $\{x'_n\}$ does not converge weakly even though it is weak* convergent. This completes the proof. ■

PROBLEM 3.2.2 Let X be a reflexive normed space. Show that $x'_n \xrightarrow{w} x'$ in X' iff $x'_n \xrightarrow{w^*} x'$ in X' .

Solution If $x'_n \xrightarrow{w} x'$ then $x'_n \xrightarrow{w^*} x'$ by Problem 3.2.1. For the converse, let X be reflexive and $x'_n \xrightarrow{w^*} x'$. To show that $x'_n \xrightarrow{w} x'$ in X' , let x'' be any bounded linear functional on X' . Then, $x'' = J(x)$ for some x in X , where J is the canonical mapping. Since $x'_n \xrightarrow{w^*} x'$, we have

$$x'_n(x) \rightarrow x'(x).$$

Since $x''(x'_n) = x'_n(x)$ and $x''(x) = x'(x)$, this means that

$$x''(x'_n) \rightarrow x''(x).$$

This is true for all x'' in X'' . So $x'_n \xrightarrow{w} x'$. This completes the proof. ■

PROBLEM 3.2.3 If $x_n \in C[a, b]$ and $x_n \xrightarrow{w} x$ in $C[a, b]$, show that $\{x_n\}$ converges pointwise on $[a, b]$. Show also that a sequence $\{x_n\}$ in $C[a, b]$ can be pointwise convergent without being weak convergent.

Solution Let $X = C[a, b]$ and $t_0 \in [a, b]$. Define $f : X \rightarrow \mathbb{K}$ by

$$f(x) = x(t_0), \quad x \in X.$$

Then f is linear, and

$$|f(x)| = |x(t_0)| \leq \|x\|_\infty, \quad x \in X.$$

Hence, f is bounded with $\|f\| \leq 1$. Thus, $f \in X'$, and so $f(x_n) \rightarrow f(x)$, that is, $x_n(t_0) \rightarrow x(t_0)$. Since this is true for any $t_0 \in [a, b]$, the first statement is proved.

For the second statement let, for $n = 1, 2, \dots$,

$$x_n(t) = n^3(b-a)^{-n}(t-a)^{n-1}(b-t), \quad a \leq t \leq b.$$

Then $x_n \in X$. Since

$$x_n(t) = n^3 \left(\frac{t-a}{b-a} \right)^{n-1} \left(\frac{b-t}{b-a} \right),$$

we readily see that if $a \leq t < b$ then $x_n(t) \rightarrow 0$ as $n \rightarrow \infty$. Hence, $x_n(t) \rightarrow 0$ pointwise on $[a, b]$ since $x_n(b) = 0$. Let

$$f(x) = \int_a^b x(t) dt, \quad x \in C[a, b].$$

Then $f \in X'$. So if $\{x_n\}$ is weak convergent, then $\{f(x_n)\}$ must be convergent. Since

$$x_n(t) = n^3(b-a)^{-n} \left((b-a)(t-a)^{n-1} - (t-a)^n \right),$$

we get

$$f(x_n) = n^3 \left(\frac{b-a}{n} - \frac{b-a}{n+1} \right) = (b-a) \frac{n^2}{n+1}.$$

So $f(x_n) \rightarrow \infty$ as $n \rightarrow \infty$. This contradiction shows that $\{x_n\}$ is not weak convergent even though it is pointwise convergent. This completes the proof. ■

PROBLEM 3.2.4 If X is a normed space and $x_n \xrightarrow{w} x$ in X , show that

$$\|x\| \leq \liminf_{n \rightarrow \infty} \|x_n\|.$$

Solution By Problem 1.4.2, there is f in X' such that

$$f(x) = \|x\|, \quad \|f\| = 1.$$

Since $x_n \xrightarrow{w} x$, we get

$$f(x_n) \rightarrow f(x) = \|x\|.$$

Since $|f(x_n)| \leq \|f\| \|x_n\| = \|x_n\|$,

$$\liminf_{n \rightarrow \infty} |f(x_n)| \leq \liminf_{n \rightarrow \infty} \|x_n\|.$$

But $|f(x_n)| \rightarrow |f(x)| = \|x\|$. Hence,

$$\|x\| \leq \liminf_{n \rightarrow \infty} \|x_n\|.$$

This completes the proof. ■

PROBLEM 3.2.5 Let X be a normed space. Show that if

$$\operatorname{Re} f(x_n) \rightarrow \operatorname{Re} f(x) \quad \text{for every } f \in X',$$

then

$$x_n \xrightarrow{w} x.$$

Solution If the scalar field $\mathbb{K}$ is the field of reals there is nothing to prove. So let $\mathbb{K} = \mathbb{C}$. Given $f \in X'$, let $g = -if$, $u = \operatorname{Re} f$ and $v = \operatorname{Im} f$. Then $g \in X'$,

$$g(x) = -i(u(x) + iv(x)) = v(x) - iu(x),$$

and hence $\operatorname{Re} g = v$. By our assumption,

$$\operatorname{Re} f(x_n) \rightarrow \operatorname{Re} f(x),$$

$$\operatorname{Re} g(x_n) \rightarrow \operatorname{Re} g(x).$$

Hence, $u(x_n) \rightarrow u(x)$ and $v(x_n) \rightarrow v(x)$. Therefore,

$$f(x_n) = u(x_n) + iv(x_n) \rightarrow u(x) + iv(x) = f(x).$$

This shows that $x_n \xrightarrow{w} x$. ■

PROBLEM 3.2.6 If $x_n \xrightarrow{w} x$ in a normed space X , show that $x \in \overline{Y}$, where $Y = \operatorname{span}\{x_n\}$.

Solution Suppose that $x \notin \overline{Y}$. Then by Problem 1.4.1 there exists f in X' such that $f(y) = 0$ for all y in Y and

$$f(x) = d(x, Y), \quad \|f\| = 1.$$

Hence, $f(x_n) = 0$ for all n while $f(x) > 0$. So $f(x_n)$ cannot tend to $f(x)$, and $\{x_n\}$ cannot converge weakly to x . ■

PROBLEM 3.2.7 Let X be a normed space. A subset of X is said to be *weakly closed* if it contains the limits of weak convergent sequences of its elements. Prove the following:

- (a) A subspace of X is closed iff it is weakly closed.
- (b) A subset of X may be closed without being weakly closed, that is, (a) may not be true for subsets which are not subspaces.
- (c) If $x_n \xrightarrow{w} x$ in X then there is a sequence $\{y_m\}$ of finite linear combinations of elements of $\{x_n\}$, which converges to x in the norm of X .

Solution (a) Let Y be a closed subspace of X and $x_n \xrightarrow{w} x$, where $x_n \in Y$ for all n . Let $Y_1 = \text{span}\{x_n\}$. Then $x \in \overline{Y}_1$ by Problem 3.2.6. But Y is a closed subspace containing Y_1 . Hence, $\overline{Y}_1 \subset Y$ and so $x \in Y$. This shows that Y is weakly closed.

For the converse, suppose that Y is weakly closed and $x \in \overline{Y}$. Then there is a sequence $\{x_n\}$ in Y such that $x_n \rightarrow x$ in the norm of X . So if $x' \in X'$, then $x'(x_n) \rightarrow x'(x)$ by the continuity of x' . Thus,

$$x'(x_n) \rightarrow x'(x) \quad \text{for all } x' \in X',$$

that is, $x_n \xrightarrow{w} x$. Hence, $x \in Y$ since Y is weakly closed. This proves that Y is closed in X .

(b) Consider the normed space $X = l^2$. Let E be the set consisting of the unit vectors $e_1, e_2, \dots$. Then E is closed since it has no limit point. Let x' be any element of X' , and $x'(e_j) = y(j)$ for $j = 1, 2, \dots$. Then $y \in l^2$, as in the second part of the solution of Problem 3.1.4 with $p = 2$. Hence, $y(n) \rightarrow 0$ as $n \rightarrow \infty$, that is, $x'(e_n) \rightarrow 0$ as $n \rightarrow \infty$. Since this is true for any $x' \in X'$ we see that $e_n \xrightarrow{w} 0$. Since $e_n \in E$ and $0 \notin E$, this shows that E is not weakly closed.

(c) Let $Y = \text{span}\{x_n\}$. Then, $x \in \overline{Y}$ by Problem 3.2.6. So there is a sequence $\{y_m\}$ with $y_m \in Y$ and $\|y_m - x\| \rightarrow 0$ as $m \rightarrow \infty$. This proves the result since y_m is a finite linear combination of $\{x_n\}$. ■

PROBLEM 3.2.8 Show that in any normed space a weak convergent sequence is a weak Cauchy sequence. Show also that the converse is false.

Solution Let $x_n \xrightarrow{w} x$. Then, $x'(x_n) \rightarrow x'(x)$ for each x' in X' . But a convergent sequence is Cauchy. Hence $\{x'(x_n)\}$ is a Cauchy sequence for each x' in X' . This means that $\{x_n\}$ is a weak Cauchy sequence.

To see that the converse statement need not be true, let X be a normed space and Y be a subspace of X , which is not closed. For example, we can take $X = c_0$ and $Y = c_{00}$. Let $x \in \overline{Y}$, $x \notin Y$. Then there is a sequence $\{y_n\}$ in Y with $y_n \rightarrow x$ in the norm of X .

Suppose that $\{y_n\}$ is weak convergent in Y . Then, $y_n \xrightarrow{w} y$ for some y in Y . Let $x' \in X'$ and $y' = x'|_Y$. Then, $y' \in Y'$, and so

$$x'(y_n) = y'(y_n) \rightarrow y'(y) = x'(y).$$

But, $x'(y_n) \rightarrow x'(x)$ since $y_n \rightarrow x$ in the norm of X . Thus, $x'(y) = x'(x)$. Since this is true for any $x' \in X'$, we see that $x = y \in Y$. This contradicts the choice of x . So $\{y_n\}$ is not weak convergent in the normed space Y . Let y' be any element in Y' . Then,

$$|y'(y_n) - y'(y_m)| = |y'(y_n - y_m)| \leq \|y'\| \|y_n - y_m\|.$$

Since $\{y_n\}$ is convergent in X , it is a Cauchy sequence. It follows that $\{y'(y_n)\}$ is a Cauchy sequence for any $y' \in Y'$, that is, $\{y_n\}$ is a weak Cauchy sequence in the normed space Y even though it is not weak convergent in Y . This completes the proof. ■

PROBLEM 3.2.9 Prove the following for any normed space:

- (a) A weak Cauchy sequence is bounded.
- (b) A weak convergent sequence is bounded.
- (c) A bounded sequence need not be a weak Cauchy sequence even in a Banach space.

Solution (a) Let $\{x_n\}$ be a weak Cauchy sequence in a normed space X , and let $f_n = J(x_n)$, where $J : X \rightarrow X''$ is the canonical mapping. Then, $\{x'(x_n)\}$ is a Cauchy sequence of scalars for each x' in X' . But a Cauchy sequence is bounded, and $x'(x_n) = f_n(x')$. Hence, $\{f_n(x')\}$ is a bounded sequence for each x' in X' . Since X' is a Banach space, we can apply the uniform boundedness principle to the family $\{f_n : n = 1, 2, \dots\}$ of bounded linear maps from X' to $\mathbb{K}$. Since it is pointwise bounded on X' we see that $\{\|f_n\| : n = 1, 2, \dots\}$ is bounded. But, $\|f_n\| = \|J(x_n)\| = \|x_n\|$. Thus, $\{x_n\}$ is a bounded sequence.

(b) This follows from (a) since a weak convergent sequence is a weak Cauchy sequence by Problem 3.2.8.

(c) Let $X = l^1$ and

$$f(x) = \sum_{j=1}^{\infty} y(j)x(j), \quad x \in X,$$

where $y(j) = (-1)^j$. Then, $y \in l^\infty$ and so $f \in X'$ by the first part of the solution of Problem 3.1.1. If e_n is the n -th unit vector then,

$$f(e_n) = \sum_{j=1}^{\infty} (-1)^j e_n(j) = (-1)^n.$$

So $\{f(e_n)\}$ is not a Cauchy sequence while $f \in X'$. Thus, $\{e_n\}$ is not a weak Cauchy sequence in X even though it is bounded. This completes the proof. ■

PROBLEM 3.2.10 Show that if X is a Banach space and $x'_n \xrightarrow{w^*} x'$ in X' , then $\{x'_n\}$ is a bounded sequence in X' . Show also that if X is not Banach then this result may be false.

Solution Let $x'_n \xrightarrow{w^*} x'$ in X' . Then, $x'_n(x) \rightarrow x'(x)$ for each $x \in X$. Thus, $\{x'_n(x)\}$ is a bounded sequence for each x in X . Since $\{x'_n : n = 1, 2, \dots\}$ is a family of bounded linear maps from X to $\mathbb{K}$ and X is a Banach space, the uniform boundedness principle implies that $\{\|x'_n\| : n = 1, 2, \dots\}$ is bounded, that is, $\{x'_n\}$ is a bounded sequence.

For the second part, let $X = c_{00}$ and

$$x'_n(x) = \frac{1}{\sqrt{n}} \sum_{j=1}^n x(j), \quad x \in c_{00},$$

$n = 1, 2, \dots$. It is clear that $x'_n : X \rightarrow \mathbb{K}$ is linear, and

$$|x'_n(x)| \leq \frac{1}{\sqrt{n}} \sum_{j=1}^n |x(j)| \leq \frac{1}{\sqrt{n}} n \|x\|_\infty, \quad x \in X.$$

This shows that $x'_n \in X'$ and $\|x'_n\| \leq \sqrt{n}$. If

$$x = e_1 + \dots + e_n = (1, \dots, 1, 0, 0, \dots),$$

then $x \in X$, $\|x\| = \|x\|_\infty = 1$, and

$$x'_n(x) = \frac{1}{\sqrt{n}} (1 + \dots + 1) = \sqrt{n}.$$

It follows that $\|x'_n\| = \sqrt{n}$. Hence, $\{x'_n\}$ is an unbounded sequence in X' . On the other hand, for each x in c_{00} there is an integer m such that $x(j) = 0$ for all $j > m$; hence,

$$x'_n(x) = \frac{1}{\sqrt{n}} \sum_{j=1}^m x(j) \quad \text{for all } n \geq m.$$

This shows that, for each fixed x in X ,

$$x'_n(x) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Hence $x'_n \xrightarrow{w^*} 0$ in X' even though $\{x'_n\}$ is unbounded in X' . This completes the proof. ■

PROBLEM 3.2.11 Let E be a subset of a normed space X such that every sequence in E contains a weak Cauchy subsequence. Show that E is bounded.

Solution Assume that E is not bounded. Then, for $n = 1, 2, \dots$, there is $y_n \in E$ with $\|y_n\| > n$. Let $\{y_{n_i}\}$ be any subsequence of $\{y_n\}$. Then, $\|y_{n_i}\| \rightarrow \infty$ as $i \rightarrow \infty$. If $x_i = y_{n_i}$ and $\{x_i\}$ is a weak Cauchy sequence, then $\{x_i\}$ must be a bounded sequence by Problem 3.2.9(a). This contradiction shows that $\{y_{n_i}\}$ cannot be a weak Cauchy sequence. Hence E must be bounded. ■

PROBLEM 3.2.12 Show that a reflexive normed space is weakly complete.

Solution Let X be reflexive. Let $\{x_n\}$ be a weak Cauchy sequence in X and $J(x_n) = f_n$, where $J : X \rightarrow X''$ is the canonical map. Then

$$f_n(x') = x'(x_n), \quad x' \in X', \quad n = 1, 2, \dots$$

Since $\{x'(x_n)\}$ is a Cauchy sequence in $\mathbb{K}$ and $\mathbb{K}$ is complete, we see that

$$f(x') = \lim_{n \rightarrow \infty} x'(x_n) = \lim_{n \rightarrow \infty} f_n(x')$$

exists for each $x' \in X'$. This defines a map $f : X' \rightarrow \mathbb{K}$. Since each f_n is linear we see that f is linear.

By Problem 3.2.9(a) we see that $\{x_n\}$ is a bounded sequence in X . Let $\|x_n\| \leq \alpha$ for all n . Then

$$|f_n(x')| = |x'(x_n)| \leq \|x'\| \|x_n\| \leq \alpha \|x'\|.$$

Letting $n \rightarrow \infty$,

$$|f(x')| \leq \alpha \|x'\|, \quad x' \in X'.$$

This shows that $f \in X''$. Since X is reflexive there exists x in X such that $J(x) = f$. That is, $f(x') = x'(x)$ for all $x' \in X'$. Hence,

$$x'(x_n) \rightarrow f(x') = x'(x) \quad \text{for all } x' \in X'.$$

This means that $x_n \xrightarrow{w} x$. This completes the proof. ■

PROBLEM 3.2.13 Let $X = c_{00}$ or c_0 . Show that $x_n \xrightarrow{w} x$ in X iff $\{x_n\}$ is bounded in X and $x_n(j) \rightarrow x(j)$ for each $j \geq 1$.

Solution Suppose that $x_n \xrightarrow{w} x$. Then, by Problem 3.2.9(b), we see that $\{x_n\}$ is bounded. For $j = 1, 2, \dots$, let

$$f_j(x) = x(j), \quad x \in X.$$

Then, $f_j \in X'$, and so $f_j(x_n) \rightarrow f_j(x)$ as $n \rightarrow \infty$, that is,

$$x_n(j) \rightarrow x(j) \quad \text{as } n \rightarrow \infty. \quad (1)$$

Conversely, suppose that $\{x_n\}$ is a bounded sequence in X such that (1) holds for some x in X and all $j \geq 1$. Then there is $\alpha > 0$ such that $\|x_n\|_\infty < \alpha$ for all $n \geq 1$. Let g be any bounded linear functional on X . To prove that $g(x_n) \rightarrow g(x)$ as $n \rightarrow \infty$ let $g(e_j) = y(j)$. From the second part of the solution of Problem 3.1.2, we see that $y \in l^1$, and

$$g(x) = \sum_{j=1}^{\infty} y(j)x(j), \quad x \in X.$$

Given $\epsilon > 0$ we can find m such that

$$\sum_{j=m+1}^{\infty} |y(j)| < \epsilon. \quad (2)$$

Fix m . Because of (1), we can find N such that

$$\sum_{j=1}^m |y(j)| |x_n(j) - x(j)| < \epsilon \quad \text{for all } n \geq N. \quad (3)$$

Thus, for $n \geq N$,

$$\begin{aligned} |g(x_n - x)| &= \left| \sum_{j=1}^{\infty} y(j)(x_n(j) - x(j)) \right| \\ &\leq \left(\sum_{j=1}^m + \sum_{j=m+1}^{\infty} \right) |y(j)| |x_n(j) - x(j)| \end{aligned} \quad (4)$$

In the last sum in (4), we have

$$\begin{aligned} |x_n(j) - x(j)| &\leq |x_n(j)| + |x(j)| \leq \|x_n\|_\infty + \|x\|_\infty \\ &< \alpha + \|x\|_\infty \quad \text{for all } n \geq 1. \end{aligned}$$

The other sum in (4) is less than ϵ by (3). Thus we get, for $n \geq N$,

$$\begin{aligned} |g(x_n) - g(x)| &= |g(x_n - x)| \\ &< \epsilon + \sum_{j=m+1}^{\infty} |y(j)|(\alpha + \|x\|_\infty) \\ &< \epsilon + (\alpha + \|x\|_\infty)\epsilon, \end{aligned}$$

by (2). Since ϵ is arbitrary this shows that $g(x_n) \rightarrow g(x)$ as $n \rightarrow \infty$. Hence $x_n \xrightarrow{w} x$ and the proof is complete. ■

PROBLEM 3.2.14 If $1 < p < \infty$, show that $x_n \xrightarrow{w} x$ in l^p iff $\{x_n\}$ is bounded in l^p and $x_n(j) \rightarrow x(j)$ for each $j \geq 1$. Show also that this result is not true when $p = 1$ or $p = \infty$.

Solution We proceed as in Problem 3.2.13. Let $X = l^p$, where $1 < p < \infty$. Suppose that $x_n \xrightarrow{w} x$ in X . Then $\{x_n\}$ is bounded by Problem 3.2.9(b). For $j = 1, 2, \dots$, let

$$f_j(x) = x(j), \quad x \in X.$$

Then $f_j \in X'$, and so $f_j(x_n) \rightarrow f_j(x)$ as $n \rightarrow \infty$, that is,

$$x_n(j) \rightarrow x(j) \quad \text{as } n \rightarrow \infty. \quad (5)$$

Conversely, suppose that $\{x_n\}$ is a bounded sequence in X such that (5) holds for some x in X and all $j \geq 1$. Then there is $\alpha > 0$ such that $\|x_n\|_p < \alpha$ for all $n \geq 1$. Let g be any bounded linear functional on X . To prove that $g(x_n) \rightarrow g(x)$ as $n \rightarrow \infty$, let $g(e_j) = y(j)$. From the second part of the solution of Problem 3.1.4, we see that $y \in l^q$, and

$$g(x) = \sum_{j=1}^{\infty} y(j)x(j), \quad x \in X.$$

Given $\epsilon > 0$ we can find m such that

$$\sum_{j=m+1}^{\infty} |y(j)|^q < \epsilon^q. \quad (6)$$

Fix m . We have

$$\begin{aligned} |g(x_n - x)| &= \left| \sum_{j=1}^{\infty} y(j)(x_n(j) - x(j)) \right| \\ &\leq \left(\sum_{j=1}^m + \sum_{j=m+1}^{\infty} \right) |y(j)||x_n(j) - x(j)|. \end{aligned} \quad (7)$$

Using Holder's inequality, we see that the last sum in (7) does not exceed

$$\left(\sum_{j=m+1}^{\infty} |y(j)|^q \right)^{1/q} \left(\sum_{j=m+1}^{\infty} |x_n(j) - x(j)|^p \right)^{1/p} \leq \epsilon \|x_n - x\|_p,$$

by (6). Because of (5), we can find N such that the finite sum in (7) is less than ϵ for all $n \geq N$. Since

$$\|x_n - x\|_p \leq \|x_n\|_p + \|x\|_p \leq \alpha + \|x\|_p \quad \text{for all } n,$$

we get

$$|g(x_n) - g(x)| = |g(x_n - x)| < \epsilon + \epsilon(\alpha + \|x\|_p)$$

for all $n \geq N$. Since ϵ is arbitrary this shows that $g(x_n) \rightarrow g(x)$ as $n \rightarrow \infty$. Thus, $x_n \xrightarrow{w} x$. This proves the first part.

Next let $p = 1$ or ∞ . Exactly as above we see that if $x_n \xrightarrow{w} x$, then $\{x_n\}$ is bounded and (5) holds for all $j \geq 1$. We prove that the converse is false when $p = 1$, and also when $p = \infty$.

Let $p = 1$, and

$$x_n(j) = \begin{cases} 1/n & \text{for } 1 \leq j \leq n \\ 1/j^2 & \text{for } j \geq n+1. \end{cases}$$

Then

$$\sum_{j=1}^{\infty} |x_n(j)| = n \left(\frac{1}{n} \right) + \sum_{j=n+1}^{\infty} \frac{1}{j^2}.$$

So, $x_n \in l^1$ and $\|x_n\|_1 < 1 + 1/n$. Thus, $\{x_n\}$ is a bounded sequence in l^1 , and

$$x_n(j) = \frac{1}{n} \quad \text{for all } n \geq j.$$

Hence (5) holds with $x = 0$. Let

$$f(x) = \sum_{j=1}^{\infty} x(j), \quad x \in l^1.$$

Then f is a bounded linear functional on l^1 , and

$$f(x_n) = n \left(\frac{1}{n} \right) + \sum_{j=n+1}^{\infty} \frac{1}{j^2} \rightarrow 1 \quad \text{as } n \rightarrow \infty.$$

This shows that $\{x_n\}$ does not converge weakly to 0, even though (5) holds with $x = 0$.

Finally, let $p = \infty$, and let

$$x_n(j) = \begin{cases} 1/n & \text{for } 1 \leq j \leq n \\ 1 & \text{for } j \geq n+1. \end{cases}$$

Then, $\|x_n\|_{\infty} = 1$ and $x_n(j) \rightarrow 0$ for each j . So, $\{x_n\}$ is a bounded sequence in l^{∞} , and (5) holds with $x = 0$. Let

$$g(x) = \lim_{j \rightarrow \infty} x(j), \quad x \in c.$$

Then g is a bounded linear functional on c . Let f be an extension of g to a bounded linear functional on l^{∞} . Then

$$f(x_n) = g(x_n) = 1 \quad \text{for all } n.$$

This shows that $\{x_n\}$ does not converge weakly to 0, even though (5) holds with $x = 0$. This completes the proof. ■

PROBLEM 3.2.15 Show that $x_n \xrightarrow{w} x$ in c iff $\{x_n\}$ is a bounded sequence in c such that

$$\begin{aligned} x_n(j) &\rightarrow x(j) \quad \text{as } n \rightarrow \infty \quad \text{for } j = 1, 2, \dots, \\ \lim_{j \rightarrow \infty} x_n(j) &\rightarrow \lim_{j \rightarrow \infty} x(j) \quad \text{as } n \rightarrow \infty. \end{aligned}$$

Solution We proceed as in Problem 3.2.13. Let $X = c$. Suppose that $x_n \xrightarrow{w} x$ in X . Then, $\{x_n\}$ is bounded by Problem 3.2.9(b). For $j = 1, 2, \dots$, let

$$f_j(x) = x(j), \quad x \in X.$$

Then $f_j \in X'$, and so $f_j(x_n) \rightarrow f_j(x)$ as $n \rightarrow \infty$, that is,

$$x_n(j) \rightarrow x(j) \quad \text{as } n \rightarrow \infty. \quad (8)$$

Similarly, if $f(x) = \lim_{j \rightarrow \infty} x(j)$, $x \in X$, then $f \in X'$, and hence $f(x_n) \rightarrow f(x)$ as $n \rightarrow \infty$, that is,

$$\lim_j x_n(j) \rightarrow \lim_j x(j) \quad \text{as } n \rightarrow \infty \quad (9)$$

Conversely, suppose that $\{x_n\}$ is a bounded sequence in X such that (8) and (9) hold for some x in X and all $j \geq 1$. Then there is $\alpha > 0$ such that $\|x_n\|_\infty < \alpha$ for all $n \geq 1$. Let g be any bounded linear functional on X . To prove that $g(x_n) \rightarrow g(x)$, let $g(e_j) = y(j+1)$. From the second part of the proof of Problem 3.1.3, we see that $y \in l^1$ and, for a proper choice of $y(1)$,

$$g(x) = y(1)f(x) + \sum_{j=1}^{\infty} x(j)y(j+1), \quad x \in X,$$

where

$$f(x) = \lim_{j \rightarrow \infty} x(j), \quad x \in X.$$

Given $\epsilon > 0$ we can find m such that

$$\sum_{j=m+1}^{\infty} |y(j)| < \epsilon. \quad (10)$$

Fix m . We have

$$\begin{aligned} |g(x_n - x)| &\leq |y(1)||f(x_n) - f(x)| \\ &\quad + \left(\sum_{j=1}^{m-1} + \sum_{j=m}^{\infty} \right) |y(j+1)||x_n(j) - x(j)|. \end{aligned} \quad (11)$$

In the last sum in (11) we have

$$\begin{aligned} |x_n(j) - x(j)| &\leq |x_n(j)| + |x(j)| \leq \|x_n\|_\infty + \|x\|_\infty \\ &< \alpha + \|x\|_\infty \quad \text{for all } n \geq 1. \end{aligned}$$

Because of (8), we can find N_1 such that the other sum in (11), is less than ϵ for all $n \geq N_1$. Because of (9) we can find $N \geq N_1$ such that

$$|y(1)||f(x_n) - f(x)| < \epsilon \quad \text{for all } n \geq N.$$

Thus, we see from (11) that, for all $n \geq N$,

$$\begin{aligned} |g(x_n) - g(x)| &= |g(x_n - x)| \\ &< \epsilon + \epsilon + (\alpha + \|x\|_\infty) \sum_{j=m}^{\infty} |y(j+1)| \\ &< 2\epsilon + (\alpha + \|x\|_\infty)\epsilon, \end{aligned}$$

by (10). Since ϵ is arbitrary, this shows that $g(x_n) \rightarrow g(x)$ as $n \rightarrow \infty$. Therefore, $x_n \xrightarrow{w} x$, and the proof is complete. ■

PROBLEM 3.2.16 Show that $x_n \xrightarrow{w} x$ in l^1 iff $x_n \rightarrow x$ in the norm of l^1 .

Solution If $x_n \rightarrow x$ in the norm of l^1 , then

$$x'(x_n) \rightarrow x'(x) \quad \text{for each } x',$$

by the continuity of x' , hence $x_n \xrightarrow{w} x$.

To prove the converse, we first suppose that $x_n \xrightarrow{w} 0$ in l^1 , and also that $\|x_n\| \geq \epsilon$ for some $\epsilon > 0$ and all $n \geq 1$. We find infinite sequences $\{m_r\}$ and $\{n_r\}$ of positive integers as follows: Let $n_0 = 1$. Since $x_1 \in l^1$, we can find $m_1 > 1$ such that

$$\sum_{j=m_1+1}^{\infty} |x_1(j)| < \epsilon/5.$$

If $x'_j(x) = x(j)$, then x'_j is a bounded linear functional on l^1 , and hence

$$x_n(j) = x'_j(x_n) \rightarrow 0 \quad \text{as } n \rightarrow \infty. \quad (12)$$

So we can find $n_1 > 1$ such that

$$\sum_{j=1}^{m_1} |x_{n_1}(j)| < \epsilon/5.$$

Having chosen n_{r-1}, m_r , we find $n_r > n_{r-1}$ such that

$$\sum_{j=1}^{m_r} |x_{n_r}(j)| < \epsilon/5; \quad (13)$$

this is possible because of (12). Then find $m_{r+1} > m_r$ such that

$$\sum_{j=m_{r+1}+1}^{\infty} |x_{n_r}(j)| < \epsilon/5; \quad (14)$$

this is possible since $x_{n_r} \in l^1$. Thus we get $\{m_r\}$ and $\{n_r\}$.

Choose $y(j)$ such that

$$y(j)x_{n_r}(j) = |x_{n_r}(j)| \quad \text{for } m_r < j \leq m_{r+1}, r = 0, 1, 2, \dots,$$

and $|y(j)| \leq 1$ for all $j \geq 1$ (where $m_0 = 0$). Let

$$f(x) = \sum_{j=1}^{\infty} y(j)x(j), \quad x \in l^1.$$

The first part of the solution of Problem 3.1.1 shows that f is a bounded linear functional on l^1 . So $f(x_n) \rightarrow 0$ as $n \rightarrow \infty$. Now

$$\begin{aligned} f(x_{n_r}) &= \left(\sum_{j=1}^{m_r} + \sum_{j=m_r+1}^{m_{r+1}} + \sum_{j=m_{r+1}+1}^{\infty} \right) y(j)x_{n_r}(j) \\ &= \Sigma_1 + \Sigma_2 + \Sigma_3, \end{aligned} \tag{15}$$

say. In the middle sum,

$$y(j)x_{n_r}(j) = |x_{n_r}(j)|,$$

while

$$|y(j)x_{n_r}(j)| \leq |x_{n_r}(j)| \quad \text{for all } j \geq 1.$$

Therefore,

$$\begin{aligned} \Sigma_2 &= \sum_{j=m_r+1}^{m_{r+1}} |x_{n_r}(j)|, \\ |\Sigma_1| + |\Sigma_3| &\leq \sum_{j=1}^{m_r} |x_{n_r}(j)| + \sum_{j=m_{r+1}+1}^{\infty} |x_{n_r}(j)| < 2\epsilon/5, \end{aligned}$$

by (13) and (14). Thus from (15) we get

$$\sum_{j=m_r+1}^{m_{r+1}} |x_{n_r}(j)| = \Sigma_2 = |f(x_{n_r}) - \Sigma_1 - \Sigma_3| < |f(x_{n_r})| + 2\epsilon/5.$$

Adding (13) and (14) to this, we obtain

$$\sum_{j=1}^{\infty} |x_{n_r}(j)| < |f(x_{n_r})| + 4\epsilon/5.$$

This is true for all r . Since $|f(x_{n_r})| < \epsilon/5$ for all large r , we see that

$$\|x_{n_r}\|_1 < \epsilon \quad \text{for large } r.$$

But $\|x_n\|_1 \geq \epsilon$, for all n , by our choice of $\{x_n\}$. This contradiction shows that there is no sequence $\{x_n\}$ of the type we considered.

To complete the proof suppose that $y_n \xrightarrow{w} y$ in l^1 and that $\|y_n - y\|$ does not tend to 0. Then, for some ϵ , there exist an infinite number of positive integers $n_1, n_2, \dots$ such that $\|y_{n_i} - y\| \geq \epsilon$ for all i . Let $x_i = y_{n_i} - y$. Then, $x_i \xrightarrow{w} 0$ in l^1 , and $\|x_i\| \geq \epsilon$. We proved that this gives a contradiction. This completes the proof. ■

PROBLEM 3.2.17 Let X be a Banach space, Y be any normed space and $F_n \in BL(X, Y)$ for $n = 1, 2, \dots$. If $\{F_n(x)\}$ converges weakly in Y for each x in X , show that $\{\|F_n\|\}$ is bounded.

Solution If $F_n(x) \xrightarrow{w} y$ in Y , then $\{F_n(x)\}$ is bounded in Y by Problem 3.2.9(b). Thus, $\{F_n(x) : n = 1, 2, \dots\}$ is bounded for each $x \in X$. Since X is Banach, the uniform boundedness principle implies that $\{\|F_n\| : n = 1, 2, \dots\}$ is bounded. ■

PROBLEM 3.2.18 Let $t_0 \in [a, b]$, and for $n = 1, 2, \dots$, let $y_n \in C[a, b]$ satisfy

$$\begin{aligned} y_n(t) &\geq 0 \quad \text{for all } t \in [a, b], \\ y_n(t) &= 0 \quad \text{for } |t - t_0| > 1/n, \\ \int_a^b y_n(t) dt &= 1. \end{aligned} \tag{16}$$

Let x'_n, x' be defined on $C[a, b]$ by

$$\begin{aligned} x'_n(x) &= \int_a^b y_n(t)x(t) dt, \quad x \in C[a, b], \\ x'(x) &= x(t_0), \quad x \in C[a, b]. \end{aligned}$$

Show that $x'_n \xrightarrow{w^*} x'$.

Solution Let $x \in C[a, b]$. We have to prove that $x'_n(x) \rightarrow x'(x)$ as $n \rightarrow \infty$. Because of (16), we have

$$\begin{aligned} x'_n(x) - x'(x) &= \int_a^b y_n(t)x(t) dt - x(t_0) \int_a^b y_n(t) dt \\ &= \int_a^b y_n(t)(x(t) - x(t_0)) dt. \end{aligned}$$

Let $\epsilon > 0$ be given. Since x is continuous, there exists $\delta > 0$ such that

$$|x(t) - x(t_0)| < \epsilon \quad \text{for } |t - t_0| < \delta, t \in [a, b].$$

Let $1/n < \delta$ and $t \in [a, b]$. Then,

$$|y_n(t)(x(t) - x(t_0))| \leq \epsilon y_n(t) \quad \text{if } |t - t_0| < \frac{1}{n},$$

$$|y_n(t)(x(t) - x(t_0))| = 0 \quad \text{if } |t - t_0| > 1/n.$$

Hence,

$$\begin{aligned} |x'_n(x) - x'(x)| &\leq \int_a^b |y_n(t)(x(t) - x(t_0))| dt \\ &\leq \int_a^b \epsilon y_n(t) dt = \epsilon. \end{aligned}$$

This is true whenever $1/n < \delta$. This completes the proof. ■

PROBLEM 3.2.19 Let X and Y be normed spaces and $F : X \rightarrow Y$ be linear. Prove that the following statements are equivalent:

- (a) F is continuous.
- (b) If $x_n \rightarrow x$ in X , then $F(x_n) \rightarrow F(x)$ in Y .
- (c) If $x_n \xrightarrow{w} x$ in X , then $F(x_n) \xrightarrow{w} F(x)$ in Y .
- (d) If $x_n \rightarrow x$ in X , then $F(x_n) \xrightarrow{w} F(x)$ in Y .

Solution The equivalence of (a) and (b) is well-known in any metric space.

(a) $\implies$ (c): Let F be continuous and $x_n \xrightarrow{w} x$ in X . If y' is any element in Y' and $x' = y' \circ F$, then $x' \in X'$. Hence $x'(x_n) \rightarrow x'(x)$, that is, $y'(F(x_n)) \rightarrow y'(F(x))$. Thus, $F(x_n) \xrightarrow{w} F(x)$ in Y .

(c) $\implies$ (a): Suppose that (a) is not true. Then, for $n = 1, 2, \dots$, there is $z_n \in X$ such that

$$\|F(z_n)\| > n \quad \text{and} \quad \|z_n\| \leq 1.$$

Let $x_n = \frac{1}{\sqrt{n}} z_n$. Then,

$$\|x_n\| = \frac{1}{\sqrt{n}} \|z_n\| \leq \frac{1}{\sqrt{n}} \rightarrow 0,$$

$$\|F(x_n)\| = \frac{1}{\sqrt{n}} \|F(z_n)\| > \sqrt{n}.$$

Thus, $\{F(x_n)\}$ is unbounded. By Problem 3.2.9(b), we see that it is not weak convergent. But $x_n \xrightarrow{w} 0$ since $\|x_n\| \rightarrow 0$. So (c) is not true when (a) is not true.

(a) $\implies$ (d): If (a) is true and $x_n \rightarrow x$ in X , then $F(x_n) \rightarrow F(x)$ by continuity, and hence $F(x_n) \xrightarrow{w} F(x)$.

(d) $\implies$ (a): Suppose that (a) is not true. Then, as above, there is a sequence $\{x_n\}$ in X such that $x_n \rightarrow 0$ but $\{F(x_n)\}$ does not converge weakly in Y . Thus (d) is false when (a) is false. This proves the equivalence of (a), (b), (c) and (d). ■

4

Bounded Linear Maps

This chapter is devoted to the study of bounded linear maps. In Section 4.1 we consider compact linear maps.

To a bounded linear map from X to X we associate certain scalars called spectral values. These are studied in Section 4.2. The spectral values arise, in a natural way, in the context of solution of linear equations such as linear differential, integral and algebraic equations. So they are often helpful in the solution of such equations.

4.1 Compact Operators

Let X and Y be normed spaces, and

$$U = \{x \in X : \|x\| < 1\},$$

be the open unit ball in X . A linear map $F : X \rightarrow Y$ is continuous if $F(U)$ is a bounded subset of Y . If $\overline{F(U)}$ is a compact subset of Y , we say that F is *completely continuous* or *compact*. The set of all compact linear maps from X to Y is denoted by $CL(X, Y)$. If $Y = X$ we write $CL(X)$ instead of $CL(X, X)$.

Since compact sets are always bounded we see that compact linear maps are also bounded linear maps. Thus, $CL(X, Y)$ is a subset of $BL(X, Y)$; it is, in general, a proper subset [see, for example, Problem 4.1.1].

A linear map $F : X \rightarrow Y$ is compact iff, for every bounded sequence $\{x_n\}$ in X , the sequence $\{F(x_n)\}$ in Y has a subsequence converging in Y . If $F : X \rightarrow Y$ is a bounded linear map and the range of F is a finite dimensional subspace of Y , then F is compact.

If $F : X \rightarrow Y$ and $G : X \rightarrow Y$ are compact then kF and $F + G$ are also compact. Thus, $CL(X, Y)$ is a subspace of $BL(X, Y)$. If Y is a Banach space then $CL(X, Y)$ is a closed subspace of the Banach space $BL(X, Y)$.

If $F : X \rightarrow Y$ and $G : Y \rightarrow Z$ are bounded linear maps and one of them is compact, then $G \circ F : X \rightarrow Z$ is compact.

EXERCISES

1. Show that compact linear maps are continuous, but not conversely.
2. Let X be a normed space and $F : X \rightarrow \mathbb{K}$ be defined by

$$F(x) = \begin{cases} \|x\| & \text{for } \|x\| \leq 1, \\ 0 & \text{for } \|x\| > 1. \end{cases}$$

Show that $\overline{F(U)}$ is compact but F is not continuous.

3. Let X and Y be normed spaces. Show that a linear map $F : X \rightarrow Y$ is compact iff for every bounded sequence $\{x_n\}$ in X the sequence $\{F(x_n)\}$ has a subsequence converging in Y .
4. Let X be a normed space. Show that every bounded linear functional on X is compact.
5. Show that if $F : X \rightarrow Y$ and $G : Y \rightarrow Z$ are bounded linear maps and one of them is compact then $G \circ F : X \rightarrow Z$ is compact.

PROBLEMS

PROBLEM 4.1.1 Suppose that X and Y are normed spaces and X is infinite dimensional. Suppose that $F : X \rightarrow Y$ is linear and bounded below; that is, there is $\alpha > 0$ such that

$$\alpha\|x\| \leq \|F(x)\| \quad \text{for all } x \in X. \quad (1)$$

Show that F is not compact. Deduce that the identity map on an infinite dimensional normed space is not compact.

Solution Suppose that F is compact and is bounded below. So there is $\alpha > 0$ satisfying (1). We shall first prove that the open unit ball U is totally bounded. Let $\epsilon > 0$ be given. For $b \in Y$, let

$$V(b) = \{y \in Y : \|y - b\| < \epsilon\alpha\}.$$

Then $\{V(b) : b \in F(U)\}$ is an open covering of $\overline{F(U)} = E$. Since E is compact, a finite number of them cover E . Hence there exist $b_1, \dots, b_n$ in $F(U)$ such that $V(b_1), \dots, V(b_n)$ cover E . Let $b_i = F(a_i)$, where $a_i \in U$, and

$$U(a_i, \epsilon) = \{x \in X : \|x - a_i\| < \epsilon\}.$$

Let $x \in U$. Then $F(x) \in F(U) \subset E$; hence $F(x) \in V(b_i)$ for some i . So

$$\|F(x) - b_i\| < \epsilon\alpha,$$

$$\alpha\|x - a_i\| \leq \|F(x - a_i)\| = \|F(x) - b_i\| < \epsilon\alpha,$$

using (1). Thus $\|x - a_i\| < \epsilon$ for some i , that is, $x \in U(a_i, \epsilon)$. It follows that $U(a_1, \epsilon), \dots, U(a_n, \epsilon)$ cover U . Thus U is totally bounded. By Problem 1.3.15 this is possible only if X is finite dimensional. So if X is infinite dimensional, F cannot be compact.

If $I : X \rightarrow X$ is the identity map, then $\|I(x)\| = \|x\|$, and hence I is bounded below. So if X is infinite dimensional then I cannot be compact. This completes the proof. ■

PROBLEM 4.1.2 Let X be a normed space, $z \in X$ and $f \in X'$. Show that $T : X \rightarrow X$ defined by

$$T(x) = f(x)z, \quad x \in X$$

is a compact linear map.

Solution The linearity of T follows from that of f . Let $\{x_n\}$ be a bounded sequence in X . Since f is a bounded linear functional, we see that $\{f(x_n)\}$ is a bounded sequence of scalars. So it has a convergent subsequence $\{f(x_{n_i})\}$. If $f(x_{n_i}) \rightarrow k$ as $i \rightarrow \infty$, then $T(x_{n_i}) \rightarrow kz$ as $i \rightarrow \infty$. Thus, $\{T(x_{n_i})\}$ is a convergent subsequence of $\{T(x_n)\}$. This shows that T is compact. ■

PROBLEM 4.1.3 Let $X = l^p$ where $1 \leq p \leq \infty$, and $\{\alpha_n\}$ be a sequence of scalars such that $\alpha_n \rightarrow 0$ as $n \rightarrow \infty$. Define $A : X \rightarrow X$ by

$$A(x)(n) = \alpha_n x(n), \quad n = 1, 2, \dots, x \in X.$$

Show that A is compact.

Solution It is clear that A is linear. For $n = 0, 1, 2, \dots$, let

$$\beta_n = \sup_{j \geq n+1} |\alpha_j|, \quad \beta = \beta_0.$$

Then $\beta_n \rightarrow 0$ as $n \rightarrow \infty$. If $1 \leq p < \infty$, then

$$\sum_{n=1}^{\infty} |A(x)(n)|^p = \sum_{n=1}^{\infty} |\alpha_n|^p |x(n)|^p \leq \beta^p \sum_{n=1}^{\infty} |x(n)|^p.$$

This shows that $A(x) \in l^p$ for all $x \in l^p$. If $p = \infty$ then

$$|A(x)(n)| = |\alpha_n| |x(n)| \leq \beta \|x\|_{\infty};$$

hence $A(x) \in l^{\infty}$ for all $x \in l^{\infty}$.

Define $A_n : X \rightarrow X$ for $n = 1, 2, \dots$ by

$$A_n(x) = \sum_{j=1}^n \alpha_j x(j) e_j = (\alpha_1 x(1), \dots, \alpha_n x(n), 0, 0, \dots).$$

Then A_n is linear. If $1 \leq p < \infty$, then

$$\begin{aligned} (\|A_n(x)\|_p)^p &= \sum_{j=1}^n |\alpha_j|^p |x(j)|^p \leq \beta^p (\|x\|_p)^p, \\ \|A_n(x)\|_p &\leq \beta \|x\|_p. \end{aligned}$$

If $p = \infty$, then

$$\begin{aligned} |A_n(x)(j)| &\leq \beta |x(j)| \leq \beta \|x\|_\infty, \\ \|A_n(x)\|_\infty &\leq \beta \|x\|_\infty. \end{aligned}$$

Thus $A_n : X \rightarrow X$ is a bounded linear map. Its range is of finite dimension. Hence each A_n is compact; that is, $A_n \in CL(X)$. It is clear that

$$A_n(x)(j) - A(x)(j) = \begin{cases} 0 & \text{for } 1 \leq j \leq n \\ -\alpha_j x(j) & \text{for } j \geq n+1. \end{cases}$$

Hence if $1 \leq p < \infty$, then

$$\begin{aligned} (\|A_n x - Ax\|_p)^p &= \sum_{j=n+1}^{\infty} |\alpha_j x(j)|^p \leq \beta_n^p \sum_{j=n+1}^{\infty} |x(j)|^p \leq \beta_n^p (\|x\|_p)^p, \\ \|A_n x - Ax\|_p &\leq \beta_n \|x\|_p; \end{aligned} \tag{2}$$

and if $p = \infty$ then

$$\begin{aligned} |A_n x(j) - Ax(j)| &\leq \beta_n |x(j)| \leq \beta_n \|x\|_\infty; \\ \|A_n x - Ax\|_\infty &\leq \beta_n \|x\|_\infty. \end{aligned}$$

Thus (2) holds for all x in l^p , $1 \leq p \leq \infty$. This shows that $A_n - A \in BL(X)$ and

$$\|A_n - A\| \leq \beta_n \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Hence $A_n \rightarrow A$ in $BL(X)$. Since X is a Banach space we see that $CL(X)$ is a closed subspace of $BL(X)$. Thus, $A \in CL(X)$. ■

PROBLEM 4.1.4 Does there exist a compact linear map $T : l^\infty \rightarrow l^\infty$, which is surjective?

Solution The answer is ‘No’. We shall prove that if X and Y are Banach spaces and there is a surjective compact linear map $T : X \rightarrow Y$ then Y is finite dimensional. If such a map exists then it is bounded and onto. Hence the open mapping theorem implies that $T(U)$ is an open subset of Y where U is the open unit ball in X . So there is $r > 0$ such that if

$$V = \{y \in Y : \|y\| < r\}$$

then $V \subset T(U)$. Since $\overline{T(U)}$ is compact by our assumption and $\overline{V} \subset \overline{T(U)}$ we see that

$$\overline{V} = \{y \in Y : \|y\| \leq r\}$$

is compact. It follows from Problem 1.3.15 that Y is finite dimensional.

In the present context $X = Y = l^\infty$ is Banach and infinite-dimensional. So there cannot exist a surjective compact linear map $T : l^\infty \rightarrow l^\infty$. ■

PROBLEM 4.1.5 If X, Y are normed spaces, $F \in BL(X, Y)$ and F is not compact, can the restriction of F to an infinite dimensional subspace of X be compact?

Solution The answer is ‘Yes’. It is possible that the restriction is compact. Let $X = l^2$. For $x \in l^2, j = 1, 2, \dots$ let

$$F(x)(2j) = 0, \quad F(x)(2j-1) = x(2j-1).$$

Thus $F(x) = (x(1), 0, x(3), 0, \dots) \in l^2$ for $x \in l^2$. This defines a linear map $F : l^2 \rightarrow l^2$. Since

$$|F(x)(i)| \leq |x(i)| \quad \text{for all } i \geq 1,$$

we get $\|F(x)\|_2 \leq \|x\|_2$. Thus F is bounded. It can be seen that the range of F is

$$Y = \{y \in l^2 : y(2j) = 0 \text{ for } j \geq 1\}.$$

If $y \in Y$ and $y_n \rightarrow x$ in l^2 then, for each i ,

$$y_n(i) \rightarrow x(i) \quad \text{as } n \rightarrow \infty.$$

Since $y_n(2j) = 0$ we get $x(2j) = 0$, and so $x \in Y$. This shows that Y is a closed subspace of $X = l^2$. Thus X and Y are Banach spaces. In the solution of Problem 4.1.4 we first proved that if X and Y are Banach spaces and there is a surjective compact linear map $T : X \rightarrow Y$, then Y is finite dimensional. Here $Y = \text{span}\{e_1, e_3, e_5, \dots\}$ is infinite dimensional and $F : X \rightarrow Y$ is a surjective bounded linear map. Hence F cannot be compact.

Let $Z = \{x \in l^2 : x(2j-1) = 0 \text{ for } j \geq 2\}$, and G be the restriction of F to Z . Then

$$Z = \text{span}\{e_1, e_2, e_4, e_6, \dots\}$$

is infinite dimensional, and for $x \in Z$,

$$G(x)(2j) = F(x)(2j) = 0,$$

$$G(x)(2j-1) = F(x)(2j-1) = x(2j-1) = 0 \quad \text{for } j \geq 2.$$

Hence $G(x)(i) = 0$ for all $i \geq 2, x \in Z$. So G is compact since its range $= \text{span}\{e_1\}$ is finite dimensional. This completes the proof. ■

PROBLEM 4.1.6 If X is an infinite dimensional normed space and $F \in CL(X)$ show that F is not invertible.

Solution Suppose that $F \in BL(X)$ and F has an inverse G in $BL(X)$. Then, for all x in X ,

$$\|x\| = \|GF(x)\| \leq \|G\| \|F(x)\|.$$

This shows that F is bounded below. So by Problem 4.1.1, F cannot be compact. This proves the result. ■

PROBLEM 4.1.7 Let $X = l^p, Y = l^q$, where

$$1 < p \leq \infty, \quad 1 \leq q < \infty, \quad \frac{1}{p} + \frac{1}{q} = 1,$$

and $F : X \rightarrow Y$ be given by

$$F(x)(i) = \sum_{j=1}^{\infty} k_{ij}x(j), \quad i = 1, 2, \dots, x \in l^p.$$

Show that if

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} |k_{ij}|^q < \infty$$

then $F \in CL(X, Y)$.

Solution It is clear that F is linear. Let

$$\sum_{j=1}^{\infty} |k_{ij}|^q = (\alpha_i)^q, \quad i = 1, 2, \dots$$

$$\sum_{i=n+1}^{\infty} \alpha_i^q = (\beta_n)^q, \quad n = 0, 1, 2, \dots$$

and $\beta = \beta_0$. Since $\sum \alpha_i^q$ converges, by hypothesis we see that $\beta_n \rightarrow 0$ as $n \rightarrow \infty$. If $1 < p < \infty$ and $x \in l^p$, then

$$\sum_{j=1}^{\infty} |k_{ij}x(j)| \leq \left(\sum_{j=1}^{\infty} |k_{ij}|^q \right)^{1/q} \left(\sum_{j=1}^{\infty} |x(j)|^p \right)^{1/p} = \alpha_i \|x\|_p, \quad (3)$$

by Holder's inequality. If $p = \infty$ then $q = 1$,

$$|k_{ij}x(j)| \leq |k_{ij}| \|x\|_{\infty} \quad \text{for all } j,$$

$$\sum_{j=1}^{\infty} |k_{ij}x(j)| \leq \alpha_i \|x\|_{\infty}.$$

This together with (3) first of all shows that $F(x)(i)$ exists for each $i \geq 1$ and that

$$|F(x)(i)| \leq \alpha_i \|x\|_p \quad \text{for } i \geq 1, 1 < p \leq \infty. \quad (4)$$

Thus

$$\sum_{i=1}^{\infty} |F(x)(i)|^q \leq (\|x\|_p)^q \sum_{i=1}^{\infty} \alpha_i^q = (\beta\|x\|_p)^q.$$

Therefore, $F(x)$ is well-defined, $F(x) \in l^q$ and

$$\|F(x)\|_q \leq \beta\|x\|_p \quad \text{for all } x \in l^p.$$

Thus $F \in BL(X, Y)$.

For $n = 1, 2, \dots$, define $F_n : X \rightarrow Y$ by

$$F_n(x) = \sum_{i=1}^n F(x)(i)e_i = (F(x)(1), \dots, F(x)(n), 0, 0, \dots).$$

Clearly, $F_n(x) \in l^q$, F_n is linear and

$$(\|F_n(x)\|_q)^q = \sum_{i=1}^n |F(x)(i)|^q \leq (\|x\|_p)^q \sum_{i=1}^n \alpha_i^q$$

by (4). Hence,

$$\|F_n(x)\|_q \leq \|x\|_p \left(\sum_{i=1}^n \alpha_i^q \right)^{1/q} = \beta\|x\|_p.$$

Thus F_n is bounded. Its range is of finite dimension. Hence each F_n is compact. Since

$$F(x)(i) - F_n(x)(i) = \begin{cases} 0 & \text{for } 1 \leq i \leq n \\ F(x)(i) & \text{for } i > n, \end{cases}$$

we get, using (4),

$$\begin{aligned} (\|F(x) - F_n(x)\|_q)^q &= \sum_{i=n+1}^{\infty} |F(x)(i)|^q \leq \sum_{i=n+1}^{\infty} \alpha_i^q (\|x\|_p)^q \\ &= \beta_n^q (\|x\|_p)^q. \end{aligned}$$

Therefore, $\|F(x) - F_n(x)\|_q \leq \beta_n\|x\|_p$. Thus $\|F - F_n\| \leq \beta_n \rightarrow 0$. So, $F_n \rightarrow F$ in $BL(X, Y)$. But $F_n \in CL(X, Y)$ and $CL(X, Y)$ is a closed subspace of $BL(X, Y)$ since $Y = l^q$ is Banach. It follows that $F \in CL(X, Y)$. This completes the proof. ■

PROBLEM 4.1.8 Let $X = l^1$, $Y = l^\infty$ and $F : X \rightarrow Y$ be given by

$$F(x)(i) = \sum_{j=1}^{\infty} k_{ij}x(j), \quad i = 1, 2, \dots, \quad x \in l^1.$$

Show that if

$$\alpha_i = \sup_{j \geq 1} |k_{ij}| < \infty$$

for each i , and $\alpha_i \rightarrow 0$ as $i \rightarrow \infty$, then $F \in CL(X, Y)$.

Solution We proceed as in Problem 4.1.7. The linearity of F is obvious. If $x \in l^1$, then for $i = 1, 2, \dots$,

$$\sum_{j=1}^{\infty} |k_{ij}x(j)| \leq \sum_{j=1}^{\infty} \alpha_i |x(j)| = \alpha_i \|x\|_1.$$

This shows that $F(x)(i)$ exists for each $i \geq 1$ and that

$$|F(x)(i)| \leq \alpha_i \|x\|_1, \quad i = 1, 2, \dots \quad (5)$$

Since $\alpha_i \rightarrow 0$, we see that $\alpha_i \leq \beta$ for all i , some β . Hence $F(x)$ is well-defined, $F(x) \in l^\infty$, and

$$\|F(x)\|_\infty \leq \beta \|x\|_1, \quad x \in l^1.$$

Thus, $F \in BL(X, Y)$.

For $n = 1, 2, \dots$, define $F_n : X \rightarrow Y$ by

$$F_n(x) = \sum_{i=1}^n F(x)(i)e_i = (F(x)(1), \dots, F(x)(n), 0, 0, \dots).$$

Clearly, $F_n(x) \in l^\infty$, F_n is linear and

$$\|F_n(x)\|_\infty \leq \beta \|x\|_1$$

by (5). Hence F_n is bounded. Its range is of finite dimension. So F_n is compact. Since

$$F(x)(i) - F_n(x)(i) = \begin{cases} 0 & \text{for } 1 \leq i \leq n \\ F(x)(i) & \text{for } i > n \end{cases}$$

we get, using (5),

$$\|F(x) - F_n(x)\|_\infty = \sup_{i \geq n+1} |F(x)(i)| \leq \beta_n \|x\|_1,$$

where $\beta_n = \sup_{i \geq n+1} \alpha_i$. Therefore,

$$\|F - F_n\| \leq \beta_n.$$

But $\beta_n \rightarrow 0$ since $\alpha_i \rightarrow 0$ as $i \rightarrow \infty$. So $F_n \rightarrow F$ in $BL(X, Y)$. However, $F_n \in CL(X, Y)$ and $CL(X, Y)$ is a closed subspace of $BL(X, Y)$ since $Y = l^\infty$ is a Banach space. Thus $F \in CL(X, Y)$. ■

PROBLEM 4.1.9 Let $X = l^2$ and W be the set of all operators A in $BL(X)$ which are given by

$$A(x)(i) = \sum_{j=1}^{\infty} k_{ij}x(j), \quad i = 1, 2, \dots, \quad x \in l^2,$$

where

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} |k_{ij}|^2 < \infty.$$

Show that W is a proper subspace of $CL(X)$.

Solution By Problem 4.1.7 with $p = q = 2$, we see that W is a subset of $CL(X)$. Let $A, B \in W$ and $k \in \mathbb{K}$. Suppose that A, B are represented by matrices $M_1 = (a_{ij})$ and $M_2 = (b_{ij})$, respectively. Let

$$k_{ij} = ka_{ij} + b_{ij}.$$

Then, for $x \in X$, $i = 1, 2, \dots$,

$$\begin{aligned} kAx(i) + Bx(i) &= k \sum_{j=1}^{\infty} a_{ij}x(j) + \sum_{j=1}^{\infty} b_{ij}x(j), \\ (kA + B)x(i) &= \sum_{j=1}^{\infty} k_{ij}x(j). \end{aligned}$$

Thus, $kA + B$ is represented by (k_{ij}) . Since

$$|k_{ij}|^2 \leq (|k| |a_{ij}| + |b_{ij}|)^2,$$

we get

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} |k_{ij}|^2 \leq |k|^2 \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} |a_{ij}|^2 + 2|k| \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} |a_{ij}| |b_{ij}| + \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} |b_{ij}|^2.$$

By Schwarz inequality we have

$$\sum \sum |a_{ij}| |b_{ij}| \leq \left(\sum \sum |a_{ij}|^2 \right)^{1/2} \left(\sum \sum |b_{ij}|^2 \right)^{1/2}.$$

It follows that

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} |k_{ij}|^2 < \infty;$$

hence $kA + B \in W$. Thus, W is a subspace of $CL(X)$.

Let $k_{ij} = 0$ if $i \neq j$ and $k_{ii} = i^{-1/2}$. Then,

$$A(x)(i) = \frac{1}{\sqrt{i}}x(i), \quad i = 1, 2, \dots, \quad x \in X.$$

Since $1/\sqrt{i} \rightarrow 0$ as $i \rightarrow \infty$ we see by Problem 4.1.3 that A is compact. But

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} |k_{ij}|^2 = \sum_{i=1}^{\infty} \frac{1}{i} = \infty.$$

Hence, $A \notin W$ even though $A \in CL(X)$. So W is a proper subspace of $CL(X)$. This completes the proof. ■

PROBLEM 4.1.10 Let X, Y be normed spaces and $F \in CL(X, Y)$. Does it follow that

(a) $F \in CL(X, R(F))$?

(b) $F \in CL(X, \overline{R(F)})$?

Solution (a) The answer is ‘No’. To see this, let $X = c_{00}$, $Y = l^\infty$ and $A : l^\infty \rightarrow l^\infty$ be defined by

$$A(x)(j) = \frac{1}{j}x(j), \quad j = 1, 2, \dots, x \in l^\infty.$$

Let F be the restriction of A to X . From Problem 4.1.3 we see that A is compact. Let $\{x_n\}$ be a bounded sequence in X . Then $\{A(x_n)\}$ has a subsequence converging in Y ; hence $\{F(x_n)\}$ has a subsequence converging in Y . This shows that $F \in CL(X, Y)$.

Let $x_n = e_1 + \dots + e_n = (1, \dots, 1, 0, \dots)$ and

$$y_n = F(x_n) = \left(1, \frac{1}{2}, \dots, \frac{1}{n}, 0, 0, \dots\right).$$

If $y = (1, \frac{1}{2}, \frac{1}{3}, \dots)$, then

$$\|y - y_n\|_\infty = \sup \left\{ \frac{1}{n+1}, \frac{1}{n+2}, \dots \right\} = \frac{1}{n+1},$$

and so $y_n \rightarrow y$ in l^∞ . Hence if some subsequence of $\{F(x_n)\}$ converges, then it must converge to y . But, $R(F) = c_{00}$ and $y \notin c_{00}$. Hence no subsequence of $\{F(x_n)\}$ converges in $c_{00} = R(F)$. This shows that $F : X \rightarrow R(F)$ is not compact.

(b) The answer is ‘Yes’. Consider any bounded sequence $\{x_n\}$ in X . Then, $\{F(x_n)\}$ has a subsequence converging in Y . Since $F(x_n) \in R(F)$, the limit of the subsequence lies in $\overline{R(F)}$, that is, some subsequence of $\{F(x_n)\}$ converges in $\overline{R(F)}$. This shows that $F : X \rightarrow \overline{R(F)}$ is compact. This completes the proof. ■

PROBLEM 4.1.11 Let X and Y be normed spaces and $F \in CL(X, Y)$. Prove that $R(F)$ is a separable normed space.

Solution For $n = 1, 2, \dots$ and $b \in Y$, let

$$V_n(b) = \left\{ y \in Y : \|y - b\| < \frac{1}{n} \right\}.$$

For each fixed n the collection $\{V_n(b) : b \in F(U)\}$ is an open covering of $\overline{F(U)}$. Since $\overline{F(U)}$ is compact a finite number of them cover $\overline{F(U)}$. Hence for each n there exists a finite number of points $b_{n1}, b_{n2}, \dots$ in $F(U)$ such that $\{V_n(b_{ni}) : i = 1, 2, \dots\}$ cover $\overline{F(U)}$.

Let S denote the set of all points rb_{ni} , where r is a positive integer, $n \geq 1$, $i \geq 1$. Since $b_{ni} \in F(U)$, we see that $rb_{ni} \in F(X) = R(F)$. Thus S is a countable subset of $R(F)$. We shall prove that S is dense in $R(F)$.

Let $y \in R(F)$ and $\epsilon > 0$ be given. Then, $y = F(x)$ for some x in X . Let r be an integer greater than $\|x\|$. Then, $(1/r)x \in U$, and hence

$$(1/r)y = F(x/r) \in F(U).$$

Let $n > r/\epsilon$. There exists i such that $y/r \in V_n(b_{ni})$, that is,

$$\left\| \frac{y}{r} - b_{ni} \right\| < \frac{1}{n} < \frac{\epsilon}{r}.$$

Thus, $\|y - z\| < \epsilon$, where $z = rb_{ni} \in S$. This completes the proof. ■

PROBLEM 4.1.12 Suppose that X is a normed space, $A \in BL(X)$ and $\{x_n\}$ is a bounded sequence in X such that $\{Ax_n - x_n\}$ converges in X . Show that if A^m is compact for some positive integer m , then $\{x_n\}$ has a convergent subsequence.

Solution Let

$$\begin{aligned} B &= A^{m-1} + \cdots + A + I, \\ y_n &= Ax_n - x_n = (A - I)(x_n), \\ z_n &= B(y_n). \end{aligned}$$

Then, $\{y_n\}$ converges and hence $\{z_n\}$ converges since B is continuous. Further,

$$\begin{aligned} z_n &= (A^{m-1} + \cdots + A + I)(A - I)(x_n) = (A^m - I)(x_n), \\ x_n &= A^m x_n - z_n. \end{aligned}$$

Let A^m be compact. Then $\{A^m(x_n)\}$ will have a subsequence $\{A^m(x_{n_j})\}$ converging in X . Since $\{z_{n_i}\}$ converges, we see that $\{x_{n_i}\}$ converges. This proves the result. ■

PROBLEM 4.1.13 Let $X = C[0, 1]$ and k be a continuous scalar-valued function on the closed unit square

$$S = \{(s, t) : 0 \leq s \leq 1, 0 \leq t \leq 1\}.$$

Let $A : X \rightarrow X$ be defined by

$$A(x)(s) = \int_0^1 k(s, t)x(t) dt, \quad 0 \leq s \leq 1, x \in X.$$

Show that A is compact.

Solution Since the closed unit square S is compact, the function k is bounded and uniformly continuous on it. So given $\epsilon > 0$ we can find $\delta > 0$ such that

$$|k(s_1, t) - k(s_2, t)| < \epsilon \quad \text{for } |s_1 - s_2| < \delta, 0 \leq t \leq 1.$$

Therefore, for each x in X and $|s_1 - s_2| < \delta$, we have

$$A(x)(s_1) - A(x)(s_2) = \int_0^1 \{k(s_1, t) - k(s_2, t)\} x(t) dt,$$

$$|Ax(s_1) - Ax(s_2)| \leq \int_0^1 |k(s_1, t) - k(s_2, t)| |x(t)| dt \leq \epsilon \|x\|_\infty. \quad (6)$$

This shows that $Ax(s)$ is uniformly continuous for $0 \leq s \leq 1$. Hence, $A(x) \in X$. It is clear by the linearity of integrals that A is linear. Since k is bounded, there is $\alpha > 0$ such that

$$|k(s, t)| \leq \alpha \quad \text{for } 0 \leq s \leq 1, 0 \leq t \leq 1.$$

Therefore,

$$|Ax(s)| \leq \int_0^1 |k(s, t)| |x(t)| dt \leq \alpha \|x\|_\infty, \quad 0 \leq s \leq 1; \quad (7)$$

$$\|Ax\|_\infty \leq \alpha \|x\|_\infty.$$

Thus, A is a bounded linear map.

To show that A is compact, let $\{x_n\}$ be a bounded sequence in X . We shall prove that $\{Ax_n\}$ has a convergent subsequence. Let $\|x_n\|_\infty \leq \beta$ for all n , and $y_n = Ax_n$. Then by (7),

$$|y_n(s)| \leq \alpha \|x_n\|_\infty \leq \alpha \beta, \quad n = 1, 2, \dots$$

Thus, $\{y_n : n = 1, 2, \dots\}$ is a pointwise bounded family of scalar-valued functions on $[0, 1]$. If $|s_1 - s_2| < \delta$ then, by (6),

$$\begin{aligned} |y_n(s_1) - y_n(s_2)| &= |Ax_n(s_1) - Ax_n(s_2)| \\ &\leq \epsilon \|x_n\|_\infty \leq \epsilon \beta. \end{aligned}$$

Since ϵ is arbitrary this shows that $\{y_n\}$ is an equicontinuous family. Hence it has a subsequence $\{y_{n_i}\}$ converging uniformly on $[0, 1]$.¹ Suppose that $y_{n_i}(s) \rightarrow y(s)$ uniformly for s in $[0, 1]$. Then, $\|y_{n_i} - y\|_\infty \rightarrow 0$ as $i \rightarrow \infty$, that is,

$$\|Ax_{n_i} - y\|_\infty \rightarrow 0 \quad \text{as } i \rightarrow \infty.$$

Thus, $\{Ax_{n_i}\}$ is a convergent subsequence of $\{Ax_n\}$. This shows that A is compact and completes the proof. ■

PROBLEM 4.1.14 For $x \in C[0, 1]$, let

$$A(x)(s) = \int_0^1 k(s, t)x(t) dt, \quad 0 \leq s \leq 1,$$

where the kernel $k(s, t)$ satisfies the following two conditions:

- (i) $\sup_{0 \leq s \leq 1} \int_0^1 |k(s, t)| dt < \infty.$
- (ii) $\lim_{\delta \rightarrow 0} \int_0^1 |k(s + \delta, t) - k(s, t)| dt = 0 \quad \text{for each fixed } s \in [0, 1].$

Show that A is a compact operator on $C[0, 1]$.

¹See Rudin, W., *Principles of Mathematical Analysis* (1976), Ch. 7.

Solution For $0 \leq s \leq 1$, define k_s on $[0, 1]$ by

$$k_s(t) = k(s, t), \quad 0 \leq t \leq 1.$$

Then for $x \in C[0, 1]$,

$$A(x)(s) = \int_0^1 k_s(t)x(t) dt, \quad 0 \leq s \leq 1.$$

From condition (i) we see that $k_s \in L^1[0, 1]$ and

$$\|k_s\|_1 = \int_0^1 |k_s(t)| dt = \int_0^1 |k(s, t)| dt \leq \alpha \quad (8)$$

for all s in $[0, 1]$ and some $\alpha < \infty$. Condition (ii) states that

$$\|k_{s+\delta} - k_s\|_1 = \int_0^1 |k_{s+\delta}(t) - k_s(t)| dt \rightarrow 0 \quad \text{as } \delta \rightarrow 0,$$

s fixed. Let $\phi : [0, 1] \rightarrow L^1[0, 1]$ be defined by $\phi(s) = k_s$. Then,

$$\|\phi(s+\delta) - \phi(s)\|_1 = \|k_{s+\delta} - k_s\|_1 \rightarrow 0 \quad \text{as } \delta \rightarrow 0,$$

s fixed. This means that ϕ is continuous on $[0, 1]$. Since $[0, 1]$ is compact it is uniformly continuous on $[0, 1]$. Hence given $\epsilon > 0$ we can find $\delta > 0$ such that if $|u - s| < \delta$, $u, s \in [0, 1]$, then

$$\|\phi(u) - \phi(s)\|_1 < \epsilon,$$

that is,

$$\int_0^1 |k_u(t) - k_s(t)| dt = \|k_u - k_s\|_1 < \epsilon. \quad (9)$$

We now proceed as in Problem 4.1.13. Let $X = C[0, 1]$. For $x \in X$ and $|u - s| < \delta$, we have, by (9),

$$|Ax(u) - Ax(s)| \leq \int_0^1 |k_u(t) - k_s(t)| |x(t)| dt \leq \epsilon \|x\|_\infty. \quad (10)$$

This shows that Ax is uniformly continuous on $[0, 1]$; so $Ax \in X$. Clearly, A is linear and, for $0 \leq s \leq 1$,

$$|Ax(s)| \leq \int_0^1 |k_s(t)| |x(t)| dt \leq \alpha \|x\|_\infty \quad (11)$$

by (8). Hence, $\|Ax\|_\infty \leq \alpha \|x\|_\infty$, $x \in X$. Thus, $A \in BL(X)$.

To prove that A is compact, let $\{x_n\}$ be a bounded sequence in X . Assume that $\|x_n\|_\infty \leq \beta$ for all n and let $y_n = Ax_n$. Then, by (11),

$$|y_n(s)| \leq \alpha \|x_n\|_\infty \leq \alpha \beta, \quad n = 1, 2, \dots$$

Thus, $\{y_n : n = 1, 2, \dots\}$ is a pointwise bounded family of scalar-valued functions on $[0, 1]$. If $|u - s| < \delta$ then, by (10),

$$|y_n(u) - y_n(s)| = |Ax_n(u) - Ax_n(s)| \leq \epsilon \|x_n\|_\infty \leq \epsilon \beta.$$

Since ϵ is arbitrary, this shows that $\{y_n\}$ is an equicontinuous family. The rest is exactly as in Problem 4.1.13. We get a convergent subsequence $\{A(x_{n_i})\}$. This shows that A is compact and completes the proof. ■

PROBLEM 4.1.15 Let $X = L^2[0, 1]$ and k be a continuous scalar-valued function on the closed unit square

$$S = \{(s, t) : 0 \leq s \leq 1, 0 \leq t \leq 1\}.$$

For $x \in X$, let

$$A(x)(s) = \int_0^1 k(s, t)x(t) dt, \quad 0 \leq s \leq 1.$$

Show that $A : X \rightarrow X$ is a compact linear map.

Solution Proceed as in Problem 4.1.13. By the uniform continuity of k , given $\epsilon > 0$ we can find $\delta > 0$ such that

$$|k(s_1, t) - k(s_2, t)| < \epsilon \quad \text{for } |s_1 - s_2| < \delta, \quad 0 \leq t \leq 1.$$

Hence for $x \in X$ and $|s_1 - s_2| < \delta$ we have

$$\begin{aligned} |Ax(s_1) - Ax(s_2)| &\leq \int_0^1 |k(s_1, t) - k(s_2, t)| |x(t)| dt \\ &\leq \left(\int_0^1 |k(s_1, t) - k(s_2, t)|^2 dt \right)^{1/2} \|x\|_2 \\ &\leq \epsilon \|x\|_2 \end{aligned} \tag{12}$$

using Schwarz inequality. Thus $A(x)$ is uniformly continuous and hence $A(x) \in X$ for each $x \in X$. It is clear that A is linear. Since k is bounded on S , there exists $\alpha > 0$ such that

$$|k(s, t)| \leq \alpha \quad \text{for } 0 \leq s \leq 1, \quad 0 \leq t \leq 1.$$

Hence by Schwarz inequality,

$$\begin{aligned} |Ax(s)| &\leq \int_0^1 |k(s, t)| |x(t)| dt \\ &\leq \left(\int_0^1 |k(s, t)|^2 dt \right)^{1/2} \|x\|_2 \leq \alpha \|x\|_2, \end{aligned} \tag{13}$$

$$\|A(x)\|_2 = \left(\int_0^1 |A(x)(s)|^2 ds \right)^{1/2} \leq \alpha \|x\|_2.$$

Thus, $A \in BL(X)$.

To show that A is compact, let $\{x_n\}$ be a bounded sequence in X . Let $\|x_n\|_2 \leq \beta$ and $y_n = Ax_n$. Then, by (13),

$$|y_n(s)| \leq \alpha \|x_n\|_2 \leq \alpha\beta, \quad n = 1, 2, \dots$$

Thus, $\{y_n : n = 1, 2, \dots\}$ is pointwise bounded on $[0, 1]$. Using (12) we see, as in Problem 4.1.13, that $\{y_n\}$ is equicontinuous. Hence it has a subsequence $\{y_{n_i}\}$ converging uniformly on $[0, 1]$. But uniform convergence implies convergence in the norm of $L^2[0, 1]$. Hence this subsequence $\{A(x_{n_i})\}$ of $\{A(x_n)\}$ converges in $L^2[0, 1]$. This shows that A is compact and completes the proof. ■

PROBLEM 4.1.16 Let $X = L^2[0, 1]$ and k be a scalar-valued measurable function on the closed unit square such that

$$\int_0^1 \int_0^1 |k(s, t)|^2 ds dt < \infty.$$

For $x \in X$, let

$$A(x)(s) = \int_0^1 k(s, t)x(t) dt, \quad 0 \leq s \leq 1.$$

Show that $A : X \rightarrow X$ is a compact linear map.

Solution Let S denote the closed unit square, namely,

$$\{(s, t) : 0 \leq s \leq 1, 0 \leq t \leq 1\}.$$

Then, $k \in L^2(S)$. The set of all continuous functions on S is known to be a dense subset of $L^2(S)$.² Hence there is a sequence $\{k_n\}$ of continuous functions on S such that $k_n \rightarrow k$ in the norm of $L^2(S)$. For $n = 1, 2, \dots$, let

$$A_n(x)(s) = \int_0^1 k_n(s, t)x(t) dt, \quad 0 \leq s \leq 1, x \in X.$$

Then $A_n \in CL(X)$ by Problem 4.1.15. For $0 \leq s \leq 1$, we have

$$\begin{aligned} A_n(x)(s) - A(x)(s) &= \int_0^1 \{k_n(s, t) - k(s, t)\} x(t) dt, \\ |A_n x(s) - Ax(s)| &\leq \left(\int_0^1 |k_n(s, t) - k(s, t)|^2 dt \right)^{1/2} \|x\|_2, \end{aligned}$$

²See Rudin, W., *Real And Complex Analysis* (1966), Ch. 3.

by Schwarz inequality. Hence,

$$\begin{aligned}
 \|A_n x - Ax\|_2^2 &= \int_0^1 |A_n x(s) - Ax(s)|^2 ds \\
 &\leq \int_0^1 ds \int_0^1 |k_n(s, t) - k(s, t)|^2 dt \cdot \|x\|_2^2 \\
 &= \|x\|_2^2 \cdot \|k_n - k\|_2^2, \\
 \|A_n x - Ax\|_2 &\leq \|k_n - k\|_2 \|x\|_2. \tag{14}
 \end{aligned}$$

This first of all shows that $A_n x - Ax \in L^2[0, 1]$. Hence, $Ax \in X$ for each $x \in X$. It is clear that A is linear. Since (14) implies that $A_n - A$ is bounded, we see that $A \in BL(X)$. From (14) we also see that

$$\|A_n - A\| \leq \|k_n - k\|_2 \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Therefore, $A_n \rightarrow A$ in $BL(X)$. But $A_n \in CL(X)$ and $CL(X)$ is a closed subspace of $BL(X)$ since X is Banach. Thus $A \in CL(X)$. This completes the proof. ■

PROBLEM 4.1.17 Let $X = L^2(0, \infty)$. For $x \in X$, let

$$A(x)(s) = \frac{1}{s} \int_0^s x(t) dt, \quad 0 < s < \infty.$$

Prove that

- (a) $A \in BL(X)$,
- (b) $A \notin CL(X)$.

Solution (a) Let $C_c(0, \infty)$ denote the set of all continuous functions on $(0, \infty)$ vanishing outside a compact set. This is known to be a dense subset of $L^2(0, \infty)$.³

Let $x \in C_c(0, \infty)$ and $y = A(|x|)$. Then

$$sy(s) = \int_0^s |x(t)| dt, \quad 0 < s < \infty. \tag{15}$$

So $sy(s)$ is differentiable on $(0, \infty)$ since $|x|$ is continuous on $(0, \infty)$, and

$$sy'(s) + y(s) = |x(s)|, \quad 0 < s < \infty. \tag{16}$$

Integration by parts gives

$$\int_0^\infty (y(s))^2 ds = \left[s(y(s))^2 \right]_0^\infty - \int_0^\infty s 2y(s)y'(s) ds \tag{17}$$

³See Rudin, W., *Real And Complex Analysis* (1966), Ch. 3.

Suppose that $x(t) = 0$ for all $t \notin [a, b]$, where $0 < a < b < \infty$. Then by (15)

$$\begin{aligned} y(s) &= 0 \quad \text{for } 0 < s < a, \\ y(s) &= \frac{1}{s} \int_a^b |x(t)| dt = \frac{1}{s} \alpha \quad \text{for } s > b, \end{aligned} \quad (18)$$

where α is a finite constant. Thus the integrated term in (17) vanishes at both ends. Hence, using (16) we get from (17),

$$\begin{aligned} \int_0^\infty (y(s))^2 ds &= -2 \int_0^\infty y(s) \{|x(s)| - y(s)\} ds \\ &= 2 \int_0^\infty (y(s))^2 ds - 2 \int_0^\infty |x(s)| y(s) ds, \\ \int_0^\infty (y(s))^2 ds &= 2 \int_0^\infty |x(s)| y(s) ds \\ &\leq 2 \left(\int_0^\infty |x(s)|^2 ds \right)^{1/2} \left(\int_0^\infty (y(s))^2 ds \right)^{1/2}, \end{aligned} \quad (19)$$

by Schwarz inequality. Because of (18), we see that

$$(\|y\|_2)^2 = \int_0^\infty (y(s))^2 ds < \infty.$$

Hence, (19) implies that $\|y\|_2 \leq 2\|x\|_2$. Thus we proved that $A(|x|) \in X$ for $x \in C_c(0, \infty)$ and that

$$\|A(|x|)\|_2 \leq 2\|x\|_2, \quad x \in C_c(0, \infty) \quad (20)$$

Next consider any x in X . Since $C_c(0, \infty)$ is dense in X there is a sequence $\{x_n\}$ in $C_c(0, \infty)$ such that $x_n \rightarrow x$ in X . Let

$$y_n = A(|x_n|), \quad y = A(|x|).$$

Then,

$$\begin{aligned} s(y_n(s) - y(s)) &= \int_0^s (|x_n(t)| - |x(t)|) dt, \\ s|y_n(s) - y(s)| &\leq \int_0^s ||x_n(t)| - |x(t)|| dt \leq \int_0^s |x_n(t) - x(t)| dt \\ &\leq \left(\int_0^s 1^2 dt \right)^{1/2} \left(\int_0^s |x_n(t) - x(t)|^2 dt \right)^{1/2} \\ &\leq s^{1/2} \|x_n - x\|_2. \end{aligned}$$

Hence, $y_n(s) \rightarrow y(s)$ as $n \rightarrow \infty$, for each fixed s in $(0, \infty)$. Since $y_n \geq 0$ we can apply Fatou's lemma and get

$$\int_0^\infty (y(s))^2 ds \leq \liminf_{n \rightarrow \infty} \int_0^\infty (y_n(s))^2 ds.$$

But,

$$\int_0^\infty (y_n(s))^2 ds = \|A(|x_n|)\|_2^2 \leq 4\|x_n\|_2^2$$

by (20), and $\|x_n\|_2 \rightarrow \|x\|_2$ since $x_n \rightarrow x$. Hence,

$$\liminf \int_0^\infty (y_n(s))^2 ds \leq 4\|x\|_2^2,$$

$$\int_0^\infty (y(s))^2 ds \leq 4\|x\|_2^2.$$

Since

$$|Ax(s)| \leq \frac{1}{s} \int_0^s |x(t)| dt = y(s),$$

we see that

$$\int_0^\infty |Ax(s)|^2 ds \leq 4\|x\|_2^2.$$

Hence

$$\begin{aligned} A(x) &\in X && \text{for all } x \in X \\ \|A(x)\|_2 &\leq 2\|x\|_2 && \text{for all } x \in X. \end{aligned}$$

This proves that $A \in BL(X)$.

(b) To prove that A is not compact, for $n = 1, 2, \dots$, let

$$x_n(t) = \begin{cases} t^{-1/2} & \text{for } 4^{n-1} \leq t \leq 4^n \\ 0 & \text{for } 0 < t < 4^{n-1}, t > 4^n. \end{cases}$$

Then,

$$\int_0^\infty (x_n(t))^2 dt = \int_{4^{n-1}}^{4^n} t^{-1} dt = \log 4.$$

Thus, $x_n \in X$ and $\|x_n\|_2 = (\log 4)^{1/2}$ for all $n \geq 1$. Let $y_n = A(x_n)$. Then for $s \geq 4^n$, we have

$$sy_n(s) = \int_0^s x_n(t) dt = \int_{4^{n-1}}^{4^n} t^{-1/2} dt = 2^n.$$

Thus, $y_n(s) = 2^n/s$ for $s \geq 4^n$. Let $n > m$. Then,

$$\begin{aligned}\|y_n - y_m\|_2^2 &\geq \int_{4^n}^{\infty} |y_n(s) - y_m(s)|^2 ds = \int_{4^n}^{\infty} \left(\frac{2^n - 2^m}{s}\right)^2 ds \\ &= (2^n - 2^m)^2 \frac{1}{4^n} \geq \frac{1}{4}.\end{aligned}$$

This implies that no subsequence of $\{y_n\}$ can converge. Hence, $\{A(x_n)\}$ has no convergent subsequence even though $\{x_n\}$ is a bounded sequence in X . Consequently, that A is not compact and completes the proof. ■

PROBLEM 4.1.18 For $x \in l^2$ and $n = 1, 2, \dots$, let

$$A(x)(n) = \frac{1}{n}\{x(1) + x(2) + \dots + x(n)\}.$$

Prove that $A : l^2 \rightarrow l^2$ is a bounded linear map. Prove also that A is not compact.

Solution Let $x \in c_{00}$ and $y = A(|x|)$. Then,

$$ny(n) = |x(1)| + \dots + |x(n)|.$$

Hence, for $n = 1, 2, \dots$,

$$\begin{aligned}n^2(y(n))^2 - (n-1)^2(y(n-1))^2 &= \{ny(n) - (n-1)y(n-1)\} \\ &\quad \{ny(n) + (n-1)y(n-1)\} \\ &\leq |x(n)| 2ny(n),\end{aligned}$$

where $y(0) = 0$. Thus,

$$\begin{aligned}n(y(n))^2 - \frac{(n-1)^2}{n}(y(n-1))^2 &\leq 2|x(n)|y(n), \\ \sum_{n=1}^N n(y(n))^2 - \sum_{n=2}^N \frac{(n-1)^2}{n}(y(n-1))^2 &\leq 2 \sum_{n=1}^N |x(n)|y(n).\end{aligned}$$

It follows that

$$\sum_{n=1}^{N-1} (y(n))^2 \left(n - \frac{n^2}{n+1}\right) + N(y(N))^2 \leq 2 \left(\sum_{n=1}^N |x(n)|^2\right)^{1/2} \left(\sum_{n=1}^N (y(n))^2\right)^{1/2} \quad (21)$$

by Schwarz inequality for sums. Since $x \in c_{00}$, there is m such that $x(j) = 0$ for all $j > m$. So for all $n > m$,

$$y(n) = \frac{1}{n} \sum_{j=1}^m |x(j)| = \frac{1}{n} \alpha,$$

where α is independent of n . This shows that

$$\sum_{n=1}^{\infty} (y(n))^2 < \infty,$$

and $N(y(N))^2 \rightarrow 0$ as $N \rightarrow \infty$. Since

$$n - \frac{n^2}{n+1} = \frac{n}{n+1} \geq \frac{1}{2},$$

letting $N \rightarrow \infty$ in (21), we get

$$\frac{1}{2} \sum_{n=1}^{\infty} (y(n))^2 \leq \sum_{n=1}^{\infty} \frac{n}{n+1} (y(n))^2 \leq 2\|x\|_2 \left(\sum_{n=1}^{\infty} (y(n))^2 \right)^{1/2}.$$

It follows that $\|y\|_2 \leq 4\|x\|_2$. Thus we have proved that $A(|x|) \in l^2$ for all x in c_{00} and

$$\|A(|x|)\|_2 \leq 4\|x\|_2, \quad x \in c_{00}. \quad (22)$$

Next consider any x in l^2 and let

$$x_m = \sum_{j=1}^m x(j)e_j = (x(1), \dots, x(m), 0, 0, \dots),$$

$$y_m = A(|x_m|), \quad y = A(|x|).$$

Then, $x_m \in c_{00}$,

$$\|x_m\|_2 = \left(\sum_{j=1}^m |x(j)|^2 \right)^{1/2} \leq \|x\|_2,$$

$$\|y_m\|_2 = \|A(|x_m|)\|_2 \leq 4\|x_m\|_2 \leq 4\|x\|_2 \quad (23)$$

by (22). Since $x_m(j) = x(j)$ for all $m \geq j$, we see that, for $m > i$,

$$y_m(i) = \frac{1}{i} \sum_{j=1}^i |x_m(j)| = \frac{1}{i} \sum_{j=1}^i |x(j)| = y(i).$$

So if $m > N$ then, by (23),

$$\sum_{i=1}^N (y(i))^2 = \sum_{i=1}^N (y_m(i))^2 \leq \|y_m\|_2^2 \leq 16\|x\|_2^2.$$

This is true for any N . Hence $y \in l^2$, and

$$\sum_{i=1}^{\infty} (y(i))^2 \leq 16\|x\|_2^2.$$

Since

$$|A(x)(n)| \leq \frac{1}{n} \sum_{j=1}^n |x(j)| = y(n),$$

we see that $A(x) \in l^2$ and

$$\|Ax\|_2 \leq 4\|x\|_2.$$

Since A is clearly linear, this shows that $A \in BL(l^2)$.

To prove that A is not compact, for $n = 1, 2, \dots$, let

$$x_n(j) = \begin{cases} \sqrt{(j+1)} - \sqrt{j} & \text{for } 4^n \leq j < 4^{n+1} \\ 0 & \text{for } j < 4^n, \quad j \geq 4^{n+1}. \end{cases}$$

Since

$$(\sqrt{(j+1)} + \sqrt{j})^2 = j + 1 + 2\sqrt{(j+1)j} + j > 4j,$$

we get

$$(\sqrt{(j+1)} - \sqrt{j})^2 = \frac{1}{(\sqrt{(j+1)} + \sqrt{j})^2} < \frac{1}{4j} < \frac{1}{4} \log \left(1 - \frac{1}{j}\right)^{-1}.$$

Hence,

$$\begin{aligned} \sum_{j=1}^{\infty} (x_n(j))^2 &= \sum_{j=4^n}^{4^{n+1}-1} (\sqrt{(j+1)} - \sqrt{j})^2 < \frac{1}{4} \sum_{j=4^n}^{4^{n+1}-1} \log \left(\frac{j}{j-1}\right) \\ &= \frac{1}{4} \log \left(\frac{4^{n+1}-1}{4^n-1}\right). \end{aligned}$$

This tends to $\frac{1}{4}(\log 4)$ as $n \rightarrow \infty$. Hence $x_n \in l^2$ and $\{\|x_n\|\}$ is bounded. Let $y_n = A(x_n)$. If $i \geq 4^{n+1}$, then

$$\begin{aligned} iy_n(i) &= \sum_{j=1}^i x_n(j) = \sum_{j=4^n}^{4^{n+1}-1} (\sqrt{j+1} - \sqrt{j}) = \sqrt{4^{n+1}} - \sqrt{4^n}, \\ y_n(i) &= (2^{n+1} - 2^n)/i = 2^n/i. \end{aligned}$$

Let $n > m$. Then,

$$\|y_n - y_m\|_2^2 = \sum_{i=1}^{\infty} |y_n(i) - y_m(i)|^2 \geq \sum_{i=4^{n+1}}^{\infty} \left| \frac{2^n}{i} - \frac{2^m}{i} \right|^2.$$

Thus,

$$\|y_n - y_m\|_2^2 \geq (2^n - 2^m)^2 \sum_{i=4^{n+1}}^{\infty} \left(\frac{1}{i} - \frac{1}{i+1} \right)$$

$$= (2^n - 2^m)^2 \frac{1}{4^{n+1}} \geq \left(\frac{1}{4}\right)^2.$$

This implies that no subsequence of $\{y_n\}$ can converge. Hence, $\{A(x_n)\}$ has no convergent subsequence even though $\{x_n\}$ is a bounded sequence in l^2 . Thus A is not compact. This completes the proof. ■

4.2 Spectrum of a Bounded Operator

Let X be a normed space. The space of all bounded linear maps from X to X is denoted by $BL(X)$. We define multiplication in it as follows: If $A, B \in BL(X)$, then

$$AB(x) = A(Bx), \quad x \in X.$$

Thus $BL(X)$ is a normed algebra, that is, a normed space which is also a ring. It has the identity element I given by $I(x) = x$, $x \in X$, and called *identity operator*. Moreover,

$$\|AB\| \leq \|A\| \cdot \|B\| \quad \text{for all } A, B \in BL(X).$$

Members of $BL(X)$ are sometimes called bounded operators on X .

Let $A \in BL(X)$. A scalar k is said to be a *regular value* of A if $A - kI$ is invertible in $BL(X)$, that is, the inverse $(A - kI)^{-1}$ exists and is in $BL(X)$. A scalar k which is not a regular value is called a *spectral value* of A . The *spectrum* of A , denoted by $s(A)$ or $\sigma(A)$, is the set of all $k \in \mathbb{K}$ which are not regular.

If $A - kI$ is one-one, then there is a nonzero x in X such that $Ax = kx$. In this case, k is said to be an *eigenvalue* of A ; a nonzero x with $Ax = kx$ is called an *eigenvector* corresponding to k . The zero space $Z(A - kI)$ is called the *eigenspace* corresponding to k . The set of all eigenvalues of A is called the *eigenspectrum* or *point spectrum*, and is denoted by $e(A)$ or $\sigma_e(A)$.

If $k \in \mathbb{K}$ and there is a sequence $\{x_n\}$ in X such that $\|Ax_n - kx_n\| \rightarrow 0$ as $n \rightarrow \infty$ and $\|x_n\| = 1$ for all n , then k is called an *approximate eigenvalue*. The set of all approximate eigenvalues of A is denoted by $a(A)$ or $\sigma_a(A)$. Thus k is an approximate eigenvalue of A iff $A - kI$ is not bounded below.

Every eigenvalue is a spectral value. In fact,

$$\sigma_e(A) \subset \sigma_a(A) \subset \sigma(A) \quad \text{for any } A \in BL(X).$$

Eigenvectors corresponding to distinct values are linearly independent.

Let X be Banach space. Then, $BL(X)$ is a Banach space. Let $A \in BL(X)$. If $\|A\| < |k|$, then $k \notin \sigma(A)$ (see Problem 4.2.1). The spectrum $\sigma(A)$ is a compact subset of $\mathbb{K}$.

A linear map $A : X \rightarrow X$ is said to be of *finite rank* if its range is finite dimensional. If X is a Banach space and $A \in BL(X)$ is of finite rank, then every spectral value is an eigenvalue, that is, $\sigma_e(A) = \sigma(A)$.

Let X be a Banach space and $A : X \rightarrow X$ be a compact linear map. In Problem 4.2.29 we establish the following: The eigenspectrum $\sigma_e(A)$ is countable (perhaps finite, or even empty), and has no limit point except possibly 0. Every nonzero spectral value of A is an eigenvalue. Consequently, the spectrum $\sigma(A)$ is countable and has no limit point except possibly 0. The eigenspace $Z(A - kI)$ corresponding to a nonzero eigenvalue k is finite dimensional.

Let X be a complex Banach space and $A \in BL(X)$. Then the spectrum $\sigma(A)$ is nonempty, and

$$\lim_{n \rightarrow \infty} \|A^n\|^{1/n} = \sup \{|k| : k \in \sigma(A)\}.$$

This formula is called *spectral radius formula*.

EXERCISES

1. Let X be a normed space. Show that

$$\|AB\| \leq \|A\| \|B\|$$

for all $A, B \in BL(X)$.

2. Let X be a normed space and $A \in BL(X)$. Show that k is an approximate eigenvalue, as defined above, iff $A - kI$ is not bounded below.
3. Let X be a normed space. Show that

$$\sigma_e(A) \subset \sigma_a(A) \subset \sigma(A)$$

for any $A \in BL(X)$.

4. Show that eigenvectors corresponding to distinct eigenvalues are linearly independent.
5. Let $X = \mathbb{R}^3$ and $A \in BL(X)$ be represented by the matrix

$$M = \begin{pmatrix} 1 & 1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Find $\sigma(A)$.

PROBLEMS

PROBLEM 4.2.1 Let X be a Banach space, $A \in BL(X)$ and that $\|A^m\| < 1$ for some positive integer m . Prove that $I - A$ is invertible in $BL(X)$. Deduce that if $\|A\| < |k|$ then $k \notin \sigma(A)$.

Solution Let $\|A^m\| = \lambda$ and $\|A\| = \alpha$. So if $n = rm + j$, then

$$\|A^n\| = \|(A^m)^r \cdot A^j\| \leq \|A^m\|^r \|A\|^j = \lambda^r \alpha^j, \quad r \geq 0, j \geq 0.$$

Hence,

$$\begin{aligned} \sum_{n=0}^{\infty} \|A^n\| &= \sum_{r=0}^{\infty} \sum_{j=0}^{m-1} \|A^{rm+j}\| \leq \sum_{r=0}^{\infty} \sum_{j=0}^{m-1} \lambda^r \alpha^j \\ &= \left(\sum_{j=0}^{m-1} \alpha^j \right) \sum_{r=0}^{\infty} \lambda^r < \infty \end{aligned}$$

since $0 \leq \lambda < 1$. Thus, $\sum A^n$ converges absolutely in $BL(X)$. Since $BL(X)$ is a Banach space, it converges in $BL(X)$. Let

$$B = \sum_{n=0}^{\infty} A^n, \quad B_n = I + A + \cdots + A^{n-1}.$$

Then $B_n \rightarrow B$ in $BL(X)$,

$$(I - A)B_n = I - A^n = B_n(I - A).$$

Since $A^n = B_{n+1} - B_n \rightarrow 0$, we see that

$$(I - A)B = I = B(I - A).$$

This proves the first part.

Let $|k| > \|A\|$. Applying the above result with A replaced by $(1/k)A$ and $m = 1$, we see that

$$A - kI = -k(I - (1/k)A)$$

is invertible in $BL(X)$; hence $k \notin \sigma(A)$. This completes the proof. ■

PROBLEM 4.2.2 Let $X = l^p$ where $1 \leq p < \infty$, and e_i be the i -th unit vector. Suppose that an element T in $BL(X)$ is such that $T(e_i) = e_{i+1}$ for $i = 1, 2, \dots$. Find the eigenvalues and spectrum of T .

Solution Given x in l^p , let

$$x_n = \sum_{j=1}^n x(j)e_j = (x(1), \dots, x(n), 0, 0, \dots).$$

Then,

$$\|x - x_n\|_p^p = \sum_{j=n+1}^{\infty} |x(j)|^p \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Hence, $x_n \rightarrow x$ in l^p . So $T(x_n) \rightarrow T(x)$ in l^p . Since

$$\begin{aligned} |Tx_n(i) - Tx(i)|^p &\leq \sum_{j=1}^{\infty} |Tx_n(j) - Tx(j)|^p \\ &= \|Tx_n - Tx\|_p^p, \end{aligned}$$

we see that

$$Tx_n(i) \rightarrow Tx(i) \quad \text{as } n \rightarrow \infty$$

for each $i \geq 1$. Since

$$Tx_n = \sum_{j=1}^n x(j)T(e_j) = \sum_{j=1}^n x(j)e_{j+1},$$

we get

$$\begin{aligned} Tx_n(1) &= 0, \\ Tx_n(i) &= x(i-1) \quad \text{for } 2 \leq i \leq n+1. \end{aligned}$$

Hence,

$$\begin{aligned} Tx(1) &= 0, \\ Tx(i) &= x(i-1) \quad \text{for } i \geq 2. \end{aligned}$$

Thus,

$$\|Tx\|_p^p = \sum_{i=2}^{\infty} |x(i-1)|^p = \|x\|_p^p.$$

This shows that $\|T\| = 1$. It is known that X is Banach and if $|k| > \|T\|$, then $k \notin \sigma(T)$. So,

$$k \in \sigma(T) \implies |k| \leq 1. \quad (1)$$

On the other hand, let $|k| \leq 1$ and $(T - kI)(x) = e_1$ for some $x \in l^p$. Then,

$$\begin{aligned} Tx(1) - kx(1) &= 1, \\ Tx(i) - kx(i) &= 0 \quad \text{for } i \geq 2, \\ kx(1) = -1, \quad kx(i) &= Tx(i) = x(i-1) \quad \text{for } i = 2, 3, \dots \end{aligned}$$

It follows that

$$k^i x(i) = k^{i-1} x(i-1) = kx(1) = -1$$

for $i = 2, 3, \dots$. However, if $x \in l^p$, then $x(i) \rightarrow 0$ as $i \rightarrow \infty$, and so

$$k^i x(i) \rightarrow 0 \quad \text{as } i \rightarrow \infty$$

if $|k| \leq 1$. This contradiction shows that if $|k| \leq 1$, then $(T - kI)(x) \neq e_1$ for x in l^p . So $T - kI$ is not onto and hence it has no inverse if $|k| \leq 1$. It follows that

$$|k| \leq 1 \implies k \in \sigma(T).$$

This together with (1) shows that

$$\sigma(T) = \{k \in \mathbb{K} : |k| \leq 1\}.$$

Next suppose that $x \in l^p$ and $(T - kI)(x) = 0$. Then, as before,

$$Tx(i) - kx(i) = 0, \quad i = 1, 2, \dots,$$

$$k^i x(i) = k^{i-1} Tx(i) = k^{i-1} x(i-1) = kx(1) = 0, \quad i = 2, 3, \dots$$

Hence, if $k \neq 0$, then $x(i) = 0$ for all $i \geq 1$. If $k = 0$, then

$$x(i-1) = Tx(i) = kx(i) = 0, \quad i = 2, 3, \dots;$$

hence $x(i) = 0$ for $i \geq 1$. We have proved that $(T - kI)(x) = 0$ only if $x = 0$. Hence k is not an eigenvalue. This is true for $k \in \mathbb{K}$. So $\sigma_e(T)$ is empty. ■

PROBLEM 4.2.3 If $P \in BL(X)$ and $P^2 = P$, then P is said to be *idempotent* or a *projection*. If P is a projection on a normed space X , and $P \neq 0$, $P \neq I$, find the eigenvalues and spectrum of P .

Solution Since $P \neq 0$ and $P \neq I$, there exist x_1, x_2 in X such that $Px_1 \neq 0$ and $Px_2 \neq I(x_2) = x_2$. Let $y_1 = P(x_1)$ and $y_2 = P(x_2) - x_2$. Then,

$$Py_1 = P^2 x_1 = Px_1 = y_1,$$

$$Py_1 = 1 \cdot y_1, \quad y_1 \neq 0.$$

Hence 1 is an eigenvalue. Again,

$$Py_2 = P^2 x_2 - Px_2 = Px_2 - Px_2 = 0,$$

$$Py_2 = 0 \cdot y_2, \quad y_2 \neq 0.$$

So 0 is an eigenvalue.

Next let $k \neq 0$, $k \neq 1$ and $Q = \frac{1}{k(1-k)}P - \frac{1}{k}I$. Then, $Q \in BL(X)$ and

$$\begin{aligned} PQ = QP &= \frac{1}{k(1-k)}P^2 - \frac{1}{k}P = \frac{1}{k(1-k)}P - \frac{1}{k}P \\ &= \frac{1}{1-k}P = kQ + I. \end{aligned}$$

Hence, $(P - kI)Q = I = Q(P - kI)$. Thus, $(P - kI)^{-1} = Q$ exists in $BL(X)$, and so k is not a spectral value. We thus find that

$$\{0, 1\} \subset \sigma_e(P) \subset \sigma(P) \subset \{0, 1\}.$$

Therefore,

$$\sigma_e(P) = \sigma(P) = \{0, 1\}. \quad \blacksquare$$

PROBLEM 4.2.4 Let X be a normed space and $T \in BL(X)$. Let Y be the eigenspace corresponding to an eigenvalue λ of T . Find the spectrum of the restriction of T to Y .

Solution Let $S = T|_Y$, the restriction of T to Y . Then for all y in Y , we have

$$S(y) = T(y) = \lambda y.$$

This first of all shows that λ is an eigenvalue of S . If $k \neq \lambda$, let $R = 1/(\lambda - k)I$, where I is the identity on Y . Then, $R \in BL(Y)$ and, for $y \in Y$,

$$RSy - kRy = \frac{1}{\lambda - k}Sy - \frac{k}{\lambda - k}y = \frac{1}{\lambda - k}\lambda y - \frac{k}{\lambda - k}y = y.$$

Hence, $RS - kR = I$. Similarly, $SR - kR = I$. This shows that $(S - kI)^{-1} = R$ exists in $BL(Y)$. So k is not a spectral value. Hence if $k \in \sigma(S)$, then $k = \lambda$. Thus we get

$$\begin{aligned}\{\lambda\} &\subset \sigma_e(S) \subset \sigma(S) \subset \{\lambda\}, \\ \sigma_e(S) &= \sigma(S) = \{\lambda\}. \quad \blacksquare\end{aligned}$$

PROBLEM 4.2.5 Let $X = l^p$ where $1 \leq p \leq \infty$, and E be a compact subset of the scalar field. Show that there is A in $BL(X)$ whose spectrum is E .

Solution Choose a countable dense subset $\{k_n\}$ of E . Then $\{k_n\}$ is a bounded sequence since E is compact. Suppose that $|k_n| \leq \alpha$ for all n . Define A on X by

$$(Ax)(j) = k_j x(j), \quad j = 1, 2, \dots, \quad x \in X.$$

For $1 \leq p < \infty$,

$$\begin{aligned}\|Ax\|_p^p &= \sum_{j=1}^{\infty} |k_j x(j)|^p \leq \alpha^p \|x\|_p^p, \\ \|Ax\|_p &\leq \alpha \|x\|_p,\end{aligned}$$

and, for $p = \infty$,

$$\begin{aligned}|Ax(j)| &\leq \alpha |x(j)| \leq \alpha \|x\|_{\infty}, \\ \|A(x)\|_{\infty} &\leq \alpha \|x\|_{\infty}.\end{aligned}$$

It follows that A is a bounded linear map from X to X .

Let $k \in E$. Suppose that $k \notin \sigma_a(A)$. Then $A - kI$ is bounded below, that is, there is $\beta > 0$ such that

$$\beta \|x\|_p \leq \|Ax - kx\|_p \quad \text{for all } x \in X. \quad (2)$$

We can choose k_n such that $|k_n - k| < \beta$. Since

$$(Ae_n)(j) - ke_n(j) = k_j e_n(j) - ke_n(j) = \begin{cases} 0 & \text{if } j \neq n \\ k_n - k & \text{if } j = n \end{cases}$$

we see that

$$\|Ae_n - ke_n\|_p = |k_n - k| < \beta.$$

This contradicts (2) when $x = e_n$. It follows that $k \in \sigma_a(A)$. Thus,

$$E \subset \sigma_a(A). \quad (3)$$

Next suppose that $k \notin E$. Define B on X by

$$(Bx)(n) = k'_n x(n), \quad n = 1, 2, \dots,$$

where

$$k'_n = \frac{1}{k_n - k}.$$

Then $\{k'_n\}$ is bounded since E is compact and $k \notin E$. So, as before, B is a bounded linear map from X to X . Further, since

$$ABx(n) = k_n k'_n x(n) = BAx(n),$$

$$\begin{aligned} x(n) + kBx(n) &= (1 + k k'_n)x(n) \\ &= \left(1 + \frac{k}{k_n - k}\right)x(n) = k_n k'_n x(n), \end{aligned}$$

we see that $ABx = BAx = x + kBx$. Hence,

$$(A - kI)B(x) = x = B(A - kI)(x).$$

Hence, $(A - kI)^{-1} = B$ exists in $BL(X)$. Thus, k is not a spectral value of A . This proves that $\sigma(A) \subset E$. Since $\sigma_a(A) \subset \sigma(A)$, it follows from (3) that $\sigma_a(A) = \sigma(A) = E$. This completes the proof. ■

PROBLEM 4.2.6 Let $X = l^p$, where $1 \leq p \leq \infty$, and $\{\alpha_n\}$ be dense in $[0, 1]$. Define A in $BL(X)$ by

$$(Ax)(n) = \alpha_n x(n), \quad n = 1, 2, \dots, x \in X.$$

Find the eigenvalues and spectrum of A .

Solution Let $k \in \mathbb{K}$ and $Ax = kx$. Then,

$$\alpha_n x(n) - kx(n) = (Ax)(n) - kx(n) = 0.$$

Hence, for each n , $\alpha_n - k = 0$ or $x(n) = 0$. So if $k = \alpha_j$, we can have $x = e_j$. If $k \neq \alpha_n$ for any n , then we must have $x = 0$. This shows that the set of eigenvalues is

$$\sigma_e(A) = \{\alpha_n : n = 1, 2, \dots\}.$$

Let $[0, 1] = E$. Then E is a compact subset of $\mathbb{K}$ whether $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. So, as in Problem 4.2.5, we have $\sigma(A) = E = [0, 1]$. ■

PROBLEM 4.2.7 Let $X = C[0, 1]$ and $v \in X$. Define T on X by $T(x) = vx$. Find the spectrum and eigenvalues of T .

Solution Since v is continuous on $[0, 1]$, its range is a compact interval $[a, b]$. First suppose that $k \in [a, b]$. Then, $k = v(t_0)$ for some t_0 in $[0, 1]$. Let $\epsilon > 0$ be given. By the continuity of v , there exists $\delta > 0$ such that

$$|v(t) - v(t_0)| < \epsilon \quad \text{for } |t - t_0| \leq \delta, \quad t \in [0, 1]. \quad (4)$$

Let

$$y(t) = \begin{cases} 1 - \frac{|t - t_0|}{\delta} & \text{for } |t - t_0| \leq \delta \\ 0 & \text{for } |t - t_0| > \delta, \end{cases}$$

and x be the restriction of y to $[0, 1]$. Then y is continuous on $\mathbb{R}$ and

$$|x(t)| \leq 1 = x(t_0) \quad \text{for all } t \in [0, 1]. \quad (5)$$

Hence $x \in X$ and $\|x\|_\infty = 1$. Since $(Tx)(t) = v(t)x(t)$, we have

$$(Tx)(t) - kx(t) = v(t)x(t) - kx(t) = (v(t) - v(t_0))x(t).$$

Since $x(t) = 0$ for $|t - t_0| > \delta$, we see from (4) and (5) that

$$|v(t) - v(t_0)||x(t)| < \epsilon \quad \text{for } t \in [0, 1].$$

Therefore,

$$\|Tx - kx\|_\infty \leq \epsilon. \quad (6)$$

Thus we see that for each $\epsilon > 0$ there exists x in X with $\|x\|_\infty = 1$ and satisfying (6). This means that k is an approximate eigenvalue of T .

Next suppose that $k \notin [a, b]$ = the range of v . Let

$$u(t) = \frac{1}{v(t) - k}.$$

Then $u \in C[0, 1] = X$. Let $S(x) = ux$, $x \in X$. Since $u(t)v(t) - ku(t) = 1$, we see that

$$x + kSx = x + kux = uvx.$$

Hence

$$STx = uvx = x + kSx = TSx,$$

$$S(T - kI)x = x = (T - kI)Sx.$$

Thus, $(T - kI)^{-1} = S$ exists in $BL(X)$. Hence, $k \notin \sigma(T)$. So if $k \in \sigma(T)$, then $k \in [a, b]$. We have proved that

$$[a, b] \subset \sigma_a(T) \quad \text{and} \quad \sigma(T) \subset [a, b].$$

Since $\sigma_a(T) \subset \sigma(T)$, we get

$$\sigma(T) = [a, b] = \text{the range of } v.$$

To find the eigenvalues of T , let $T(x) = kx$ for some $x \neq 0$ and

$$V = \{t \in [0, 1] : x(t) \neq 0\}.$$

Then V is a nonempty open subset of $[0, 1]$ and

$$(v(t) - k)x(t) = (Tx)(t) - kx(t) = 0$$

for all t in $[0, 1]$. Hence, $v(t) = k$ for all t in a nonempty open set.

On the other hand, suppose that $v(t) = k$ for $\alpha < t < \beta$, where $0 \leq \alpha, \beta \leq 1$. Let

$$x(t) = \begin{cases} (t - \alpha)(t - \beta) & \text{for } \alpha \leq t \leq \beta \\ 0 & \text{for other } t. \end{cases}$$

Then $x \in X$, $x \neq 0$, and

$$(Tx)(t) - kx(t) = (v(t) - k)x(t) = 0 \quad \text{for } t \in [0, 1].$$

Hence, $Tx = kx$; k is an eigenvalue. Thus the set $\sigma_e(T)$ of all eigenvalues is the set of all k in $[a, b]$ such that

$$v(t) = k \quad \text{for } t \text{ in a nonempty open set.} \quad \blacksquare$$

PROBLEM 4.2.8 Let $X = C[0, 1]$. Show that there is T in $BL(X)$ whose spectrum is a given interval $[a, b]$.

Solution Choose v in X , whose range is $[a, b]$; for instance, we can choose

$$v(t) = a + (b - a)t, \quad 0 \leq t \leq 1.$$

Define T on X by $T(x) = vx$. Then $\sigma(T) = [a, b]$ by Problem 4.2.7. $\blacksquare$

PROBLEM 4.2.9 Let $X = l^p$ where $1 \leq p \leq \infty$. If $T \in BL(X)$ is defined by

$$(Tx)(j) = x(j + 1) \quad \text{for } j = 1, 2, \dots, \quad x \in X,$$

find the spectrum of T .

Solution If $1 \leq p < \infty$, then

$$\begin{aligned}\|T(x)\|_p^p &= \sum_{j=1}^{\infty} |Tx(j)|^p = \sum_{j=1}^{\infty} |x(j+1)|^p \leq \|x\|_p^p, \\ \|Tx\|_p &\leq \|x\|_p, \quad x \in X.\end{aligned}$$

If $p = \infty$, then

$$\begin{aligned}|Tx(j)| &= |x(j+1)| \leq \|x\|_{\infty}, \quad j = 1, 2, \dots, \\ \|T(x)\|_{\infty} &\leq \|x\|_{\infty}.\end{aligned}$$

Thus, $\|T\| \leq 1$ in either case. Since X is Banach, we see that if $|k| > 1$ then $k \notin \sigma(T)$. So

$$\sigma(T) \subset \{k : |k| \leq 1\}. \quad (7)$$

Next let $|k| < 1$ and $x = (1, k, k^2, \dots)$. Then, $x \in X$, $x \neq 0$ and

$$kx(j) = k^j = x(j+1) = T(x)(j)$$

for $j = 1, 2, \dots$. Hence, $T(x) = kx$. So k is an eigenvalue. So,

$$\{k : |k| < 1\} \subset \sigma_e(T). \quad (8)$$

Since $\sigma_e(T) \subset \sigma(T)$ and $\sigma(T)$ is closed, we see from (7) and (8) that

$$\sigma(T) = \{k \in \mathbb{K} : |k| \leq 1\}. \quad \blacksquare$$

PROBLEM 4.2.10 Show that the spectrum of a bounded operator on a real Banach space can be empty.

Solution Let $X = \mathbb{R}^2$, $\mathbb{K} = \mathbb{R}$ and A be represented by the 2×2 matrix

$$M = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}.$$

where $a, b \in \mathbb{R}$ and $b \neq 0$. Then, $A - kI$ is represented by the matrix

$$M - kI = \begin{pmatrix} a - k & b \\ -b & a - k \end{pmatrix}.$$

Since $\det(M - kI) = (a - k)^2 + b^2 \geq b^2 > 0$ for all k in $\mathbb{R}$, we see that $M - kI$ is nonsingular. So $M - kI$ is invertible. Thus, $A - kI$ is invertible for all k in $\mathbb{R}$. Hence $\sigma(A)$ is empty. This proves the result. $\blacksquare$

PROBLEM 4.2.11 Show that the spectrum of a bounded operator on a normed space X can be a nonempty noncompact subset of the scalar field.

Solution Let $X = c_{00}$ and $A : X \rightarrow X$ be defined by

$$(Ax)(1) = 0, \quad (Ax)(i) = x(i-1) \quad \text{for } i = 2, 3, \dots$$

Then $A \in BL(X)$. Suppose that $k \in \mathbb{K}$ and $k \notin \sigma(A)$. So $(A - kI)^{-1} = B$ exists in $BL(X)$. Let

$$y(1) = 1 - k, \quad y(2) = 1, \quad y(i) = 0 \quad \text{for } i > 2,$$

so that $y \in X$. Let $x = B(y)$. Then $x \in X$ and

$$y = (A - kI)B(y) = (A - kI)(x) = Ax - kx.$$

Therefore,

$$1 - k = y(1) = (Ax)(1) - kx(1) = -kx(1), \quad (9)$$

$$1 = y(2) = (Ax)(2) - kx(2) = x(1) - kx(2), \quad (10)$$

$$0 = y(i) = (Ax)(i) - kx(i) = x(i-1) - kx(i)$$

for $i > 2$. Multiplying (10) by k and adding to (9), we get

$$1 = -k^2 x(2).$$

This implies that $x(2) \neq 0$. Since $x \in X = c_{00}$ there is m such that $x(m) \neq 0$ and $x(j) = 0$ for all $j > m$. Then $m \geq 2$. So

$$0 = y(m+1) = x(m) - kx(m+1) = x(m).$$

This contradiction shows that $(A - kI)^{-1} = B$ does not exist in $BL(X)$. It follows that $k \in \sigma(A)$ for all k in $\mathbb{K}$. Hence, $\sigma(A) = \mathbb{K}$ is noncompact. This proves the result. ■

PROBLEM 4.2.12 An operator A is said to be *nilpotent* if there is a positive integer m such that $A^m = 0$. Find the spectrum of a nilpotent operator A on a complex Banach space X .

Solution If X is a complex Banach space and $A \in BL(X)$, then the spectral radius formula

$$\lim_{n \rightarrow \infty} \|A^n\|^{1/n} = \sup \{|k| : k \in \sigma(A)\}$$

is valid. If A is nilpotent then $A^m = 0$ for some m , and hence $A^n = 0$ for all $n \geq m$. So

$$\sup \{|k| : k \in \sigma(A)\} = 0.$$

This implies that $\sigma(A)$ does not contain a nonzero element. But the spectrum is nonempty since X is a complex Banach space. It follows that $\sigma(A) = \{0\}$. ■

PROBLEM 4.2.13 Let X be a complex Banach space. For $T \in BL(X)$, let

$$r_\sigma(T) = \sup \{|k| : k \in \sigma(T)\}$$

be the spectral radius of T . If $S, T \in BL(X)$, prove the following:

(a) If $ST = TS$, then

$$r_\sigma(ST) \leq r_\sigma(S)r_\sigma(T).$$

(b) The commutativity $ST = TS$ cannot be dropped in (a).

(c) Strict inequality can occur in (a).

Solution (a) The spectral radius formula states that

$$\|T^n\|^{1/n} \longrightarrow r_\sigma(T) \quad \text{as } n \rightarrow \infty.$$

Since $ST = TS$, we have

$$(ST)^n = S^n T^n \quad \text{for all } n \geq 1.$$

Since $\|AB\| \leq \|A\| \cdot \|B\|$, we get

$$\|(ST)^n\| \leq \|S^n\| \cdot \|T^n\|.$$

Taking n -th roots and letting $n \rightarrow \infty$, we obtain

$$r_\sigma(ST) \leq r_\sigma(S) \cdot r_\sigma(T).$$

(b) Let $X = \mathbb{C}^2$, $\mathbb{K} = \mathbb{C}$ and S, T be the operators represented by the matrices

$$M_1 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$$

respectively. Since $S - kI$ is represented by the matrix $M_1 - kI$ and $k \in \sigma(S)$ iff $S - kI$ is not invertible, we see that $k \in \sigma(S)$ iff

$$\det(M_1 - kI) = \begin{vmatrix} -k & 0 \\ 1 & -k \end{vmatrix} = 0,$$

that is, $k = 0$. Hence, $\sigma(S) = \{0\}$ and $r_\sigma(S) = 0$. Similarly, since ST is represented by

$$M_1 M_2 = \begin{pmatrix} 0 & 0 \\ 1 & 2 \end{pmatrix},$$

we see that $\sigma(ST)$ is given by

$$\begin{vmatrix} -k & 0 \\ 1 & 2 - k \end{vmatrix} = 0, \quad k = 0, 2.$$

Hence, $\sigma(ST) = \{0, 2\}$ and $r_\sigma(ST) = 2$. Thus,

$$r_\sigma(ST) = 2 > 0 = r_\sigma(S)r_\sigma(T).$$

(c) Choose $T \in BL(X)$ such that $T^2 = T$, $T \neq 0$ and $T \neq I$. Then, $\sigma(T) = \{0, 1\}$ by Problem 4.2.3. So $r_\sigma(T) = 1$. Let $S = I - T$. Then, $S^2 = S$, $S \neq 0$ and $S \neq I$. So $r_\sigma(S) = 1$. But $ST = 0 = TS$; hence $r_\sigma(ST) = 0$. Thus, $r_\sigma(ST) < r_\sigma(S) \cdot r_\sigma(T)$. This completes the proof. ■

PROBLEM 4.2.14 Let $A \in BL(X)$. The *residual spectrum* $\sigma_r(A)$ of A is the set of all scalars k such that the range of $A - kI$ is not dense in X . Show that if X is Banach and $A \in BL(X)$, then

$$\sigma(A) = \sigma_a(A) \cup \sigma_r(A).$$

Solution First suppose that $k \notin \sigma(A)$. Then $A - kI$ is invertible. So the range of $A - kI$ is equal to X and hence dense in X . So $k \notin \sigma_r(A)$. Thus, $\sigma_r(A) \subset \sigma(A)$. Since $\sigma_a(A) \subset \sigma(A)$, we get

$$\sigma_r(A) \cup \sigma_a(A) \subset \sigma(A). \quad (11)$$

Next suppose that $k \notin \sigma_r(A)$ and $k \notin \sigma_a(A)$. Then the range of $A - kI$ is dense in X and $A - kI$ is bounded below. Then $A - kI$ is one-one since it is bounded below. To show that it is onto, let $y \in X$. Then there is sequence $\{x_n\}$ in X such that $(A - kI)(x_n) \rightarrow y$. Let $y_n = (A - kI)(x_n)$. For some $\alpha > 0$, we have

$$\alpha \|x_n - x_m\| \leq \|(A - kI)(x_n - x_m)\| = \|y_n - y_m\|.$$

Since $y_n \rightarrow y$ this shows that $\{x_n\}$ is a Cauchy sequence in X . Since X is Banach, there is x in X such that $x_n \rightarrow x$. Hence,

$$y_n = (A - kI)(x_n) \longrightarrow (A - kI)(x).$$

Thus, $y = (A - kI)(x)$. This shows that $A - kI$ is onto. Since X is Banach, the open mapping theorem implies that $A - kI$ is invertible in $BL(X)$. Hence $k \notin \sigma(A)$ if $k \notin \sigma_r(A)$ and $k \notin \sigma_a(A)$. Therefore, if $k \in \sigma(A)$, then $k \in \sigma_r(A)$ or $k \in \sigma_a(A)$. This combined with (11) proves the result. ■

PROBLEM 4.2.15 If X is a normed space, and $A \in BL(X)$ is invertible, show that

$$\sigma(A^{-1}) = \{k \in \mathbb{K} : \frac{1}{k} \in \sigma(A)\}.$$

Solution Since $A^{-1} - 0 \cdot I = A^{-1}$ is invertible we see that $0 \notin \sigma(A^{-1})$. If $k \neq 0$, then

$$A^{-1} - kI = A^{-1} - kA^{-1}A = -kA^{-1}(A - \frac{1}{k}I).$$

Hence, $A^{-1} - kI$ is invertible iff $A - \frac{1}{k}I$ is invertible. It follows that

$$k \in \sigma(A^{-1}) \iff \frac{1}{k} \in \sigma(A).$$

This proves the result. ■

PROBLEM 4.2.16 If $A, B \in BL(X)$, where X is a Banach space. Show that $\sigma(AB)$ differs from $\sigma(BA)$ at most by $\{0\}$.

Solution Suppose that $k \notin \sigma(AB)$ and $k \neq 0$. Let $F = AB - kI$ and $G = BA - kI$. Then F is invertible in $BL(X)$, and

$$\begin{aligned} FA &= ABA - kA = AG, \\ GB &= BAB - kB = BF. \end{aligned}$$

Hence,

$$\begin{aligned} A &= F^{-1}AG, \quad B = GBF^{-1}; \\ kI &= BA - G = GBF^{-1}F^{-1}AG - G; \\ I &= GS = TG, \end{aligned}$$

where

$$\begin{aligned} S &= \frac{1}{k}(BF^{-1}F^{-1}AG - I), \\ T &= \frac{1}{k}(GBF^{-1}F^{-1}A - I). \end{aligned}$$

But $GS = I$ implies that G is onto and $TG = I$ implies that G is one-one. Therefore, G is bijective. Since $G \in BL(X)$ and X is Banach, the open mapping theorem implies that G is invertible in $BL(X)$. So $k \notin \sigma(BA)$.

We have proved that if $k \in \sigma(BA)$ and $k \neq 0$ then $k \in \sigma(BA)$. This proves the result. ■

PROBLEM 4.2.17 Let $A \in BL(X)$, where X is a Banach space. Prove the following:

- (a) The approximate eigenspectrum $\sigma_a(A)$ is a compact subset of $\mathbb{K}$.
- (b) Limits of eigenvalues of A lie in $\sigma_a(A)$.
- (c) The boundary of $\sigma(A)$ in $\mathbb{K}$ is contained in $\sigma_a(A)$.
- (d) If $\mathbb{K} = \mathbb{C}$ then $\sigma_a(A) \neq \emptyset$.

Solution (a) It is known that $k \notin \sigma_a(A)$ iff $A - kI$ is bounded below. Let $k \notin \sigma_a(A)$. Then there exists $\alpha > 0$ such that

$$\alpha\|x\| \leq \|Ax - kx\| \quad \text{for all } x \in X.$$

Let $|k' - k| < \frac{1}{2}\alpha$. Then,

$$\begin{aligned} \alpha\|x\| &\leq \|Ax - kx\| \leq \|Ax - k'x\| + \|k'x - kx\| \\ &\leq \|Ax - k'x\| + \frac{1}{2}\alpha\|x\|. \end{aligned}$$

Hence, $\frac{1}{2}\alpha\|x\| \leq \|Ax - k'x\|$ for all $x \in X$. Thus, $A - k'I$ is bounded below if $|k' - k| < \frac{1}{2}\alpha$, that is,

$$k' \notin \sigma_a(A) \quad \text{if } |k' - k| < \frac{1}{2}\alpha.$$

This shows that the complement of $\sigma_a(A)$ is open. So $\sigma_a(A)$ is closed. Since $\sigma_a(A) \subset \sigma(A)$ and $\sigma(A)$ is compact, we see that $\sigma_a(A)$ is compact.

(b) Let $k_n \in \sigma_e(A)$ and $k_n \rightarrow k$. Then, $k_n \in \sigma_a(A)$ since $\sigma_e(A) \subset \sigma_a(A)$. Hence, $k \in \sigma_a(A)$ since $\sigma_a(A)$ is closed.

(c) Suppose that $k \notin \sigma_a(A)$ and that k is a boundary point of $\sigma(A)$. Then $A - kI$ is bounded below, that is, there exists $\alpha > 0$ such that

$$\alpha\|x\| \leq \|Ax - kx\| \quad \text{for all } x.$$

Let y be any element of X . Since k is a boundary point, for $n = 1, 2, \dots$, we can find $k_n \notin \sigma(A)$, with $|k - k_n| < \alpha/n$. Let $x_n = (A - k_n I)^{-1}(y)$. Then, $y = Ax_n - k_n x_n$ and $\alpha\|x_n\| \leq \|Ax_n - kx_n\|$. Thus,

$$\begin{aligned} \alpha\|x_n\| &\leq \|y + k_n x_n - kx_n\| \leq \|y\| + |k_n - k|\|x_n\| \\ &\leq \|y\| + \frac{\alpha}{n}\|x_n\|. \end{aligned}$$

Hence, $\alpha(1 - 1/n)\|x_n\| \leq \|y\|$, and so

$$\|y - (Ax_n - kx_n)\| = \|kx_n - k_n x_n\| \leq \frac{\alpha}{n}\|x_n\| \leq \frac{1}{n-1}\|y\|.$$

for $n > 1$. This shows that y belongs to the closure of the range of $A - kI$. This is true for any $y \in X$. Hence the range of $A - kI$ is dense in X , that is, $k \notin \sigma_r(A)$. Since $k \notin \sigma_a(A)$, we see by Problem 4.2.14 that $k \notin \sigma(A)$. But $\sigma(A)$, being closed, contains its boundary points. So k cannot be a boundary point of $\sigma(A)$. This contradiction shows that $k \in \sigma_a(A)$ if k is a boundary point of $\sigma(A)$.

(d) Let $\mathbb{K} = \mathbb{C}$. If $\sigma_a(A) = \emptyset$, then $\sigma(A)$ has no boundary point by (c), and hence $\sigma(A)$ must be open. This is not possible since $\sigma(A)$ is a nonempty compact subset of $\mathbb{K}$. This completes the proof. ■

PROBLEM 4.2.18 Let X be a Banach space and $A \in BL(X)$, $A \neq 0$. Prove that A is of finite rank iff there exist linearly independent sets $\{x_1, \dots, x_n\}$ and $\{x'_1, \dots, x'_n\}$ in X and X' respectively such that

$$A(x) = \sum_{i=1}^n x'_i(x)x_i, \quad x \in X.$$

Deduce that the nonzero eigenvalues of A are the nonzero roots of the characteristic polynomial of the matrix (k_{ij}) , where $k_{ij} = x'_i(x_j)$, $i, j = 1, \dots, n$.

Solution Suppose that there exist $\{x_1, \dots, x_n\}$ and $\{x'_1, \dots, x'_n\}$ as mentioned in the problem. Then any element in the range $R(A)$ is of the form

$$y = A(x) = \sum_{i=1}^n k_i x_i,$$

where $k_i = x'_i(x)$. Thus, $y \in \text{span}\{x_1, \dots, x_n\}$, and so $R(A)$ is finite dimensional, that is, A is of finite rank.

Conversely, suppose that A is of finite rank. Then, $Y = R(A)$ has a finite basis. Let $\{x_1, \dots, x_n\}$ be a basis for Y . Each element y in Y can be uniquely expressed as $y = k_1x_1 + \dots + k_nx_n$. So we can define $f_j : Y \rightarrow \mathbb{K}$ by

$$f_j(y) = k_j \quad \text{if } y = \sum_{i=1}^n k_i x_i.$$

It is clear that f_j is linear. Since Y is finite dimensional, it is continuous by Problem 1.3.8. Let $x'_j = f_j \circ A$. Then, $x'_j \in X'$. Since $Ax \in Y$ for $x \in X$, we have $f_j(Ax) = k_j$ if

$$Ax = \sum_{i=1}^n k_i x_i.$$

Hence, for all $x \in X$,

$$A(x) = \sum_{i=1}^n k_i x_i = \sum_{i=1}^n f_i(Ax)x_i = \sum_{i=1}^n x'_i(x)x_i.$$

To prove that $x'_1, \dots, x'_n$ are linearly independent elements in X' , let $k_1x'_1 + \dots + k_nx'_n = 0$. Then,

$$\sum_{j=1}^n k_j x'_j(x) = 0 \quad \text{for all } x \in X.$$

By the definition of f_j we see that

$$f_j(x_i) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j. \end{cases}$$

Since there exist $z_1, \dots, z_n$ in X such that $A(z_i) = x_i$, we get

$$\begin{aligned} 0 &= \sum_{j=1}^n k_j x'_j(z_i) = \sum_{j=1}^n k_j f_j(Az_i) \\ &= \sum_{j=1}^n k_j f_j(x_i) = k_i. \end{aligned}$$

This is true for $i = 1, \dots, n$; so $x'_1, \dots, x'_n$ are linearly independent.

For the second part, let $k_{ij} = x'_i(x_j)$ and B be the restriction of A to $Y = R(A)$. Then,

$$B(x_j) = A(x_j) = \sum_{i=1}^n x'_i(x_j)x_i = \sum_{i=1}^n k_{ij}x_i.$$

This means that the matrix (k_{ij}) represents B w.r.t. the basis $\{x_1, \dots, x_n\}$. If $k \neq 0$ and $Ax = kx$, then $x \in Y$. From this it follows that the nonzero eigenvalues of A are precisely the nonzero eigenvalues of B . If $M = (k_{ij})$, then $B - kI$ is represented by the matrix $M - kI$. So the eigenvalues k of B are the roots of the characteristic equation

$$\det(M - kI) = 0.$$

Hence the nonzero eigenvalues of B are the nonzero roots of the characteristic polynomial of the matrix (k_{ij}) . This completes the proof. ■

PROBLEM 4.2.19 Let $X = l^p$, where $1 \leq p \leq \infty$, and $T \in BL(X)$ be given by

$$T(x) = (0, x(1), \frac{x(2)}{2}, \dots), \quad x \in X.$$

Find the eigenvalues and the spectrum of T .

Solution Given that T is defined by

$$(Tx)(1) = 0$$

$$(Tx)(n) = \frac{x(n-1)}{n-1} \quad \text{for } n = 2, 3, \dots, \quad x \in X.$$

It is clear that T is linear. Since $|(Tx)(n)| \leq |x(n-1)|$ for $n \geq 2$, we can see that

$$\|Tx\|_p \leq \|x\|_p, \quad x \in X.$$

Thus $T \in BL(X)$. Since

$$T(T^{r-1}x)(1) = 0,$$

$$T(T^{r-1}x)(n) = \frac{(T^{r-1}x)(n-1)}{n-1}, \quad n \geq 2,$$

we see by induction on r that

$$(T^r x)(n) = 0 \quad \text{for } 1 \leq n \leq r$$

$$(T^r x)(n) = \frac{x(n-r)}{(n-1)(n-2) \cdots (n-r)} \quad \text{for } n \geq r+1.$$

Since

$$(n-1)(n-2) \cdots (n-r) = \binom{n-1}{r} r! \geq r!,$$

we get

$$|(T^r x)(n)| \leq \frac{|x(n-r)|}{r!} \quad \text{for } n \geq r+1. \quad (12)$$

If $1 \leq p < \infty$, then

$$\|T^r(x)\|_p^p = \sum_{n=1}^{\infty} |T^r(x)(n)|^p \leq \sum_{n=r+1}^{\infty} \left(\frac{|x(n-r)|}{r!} \right)^p = \left(\frac{\|x\|_p}{r!} \right)^p.$$

Hence, $\|T^r x\| \leq (1/r!) \|x\|_p$. Because of (12) this is true when $p = \infty$ also. Since this is true for all x in X , we get

$$\|T^r\| \leq \frac{1}{r!}, \quad r = 1, 2, \dots$$

Let $k \neq 0$. Then,

$$T - kI = -k \left(-\frac{1}{k}T + I \right) x = -k(I - A),$$

where $A = (1/k)T$. As

$$\|A^m\| = \left\| \frac{1}{k^m} T^m \right\| = \left(\frac{1}{|k|} \right)^m \|T^m\| \leq \left(\frac{1}{|k|} \right)^m \frac{1}{m!} < 1$$

for some m , we see by Problem 4.2.1 that $I - A$ is invertible in $BL(X)$. Thus, $T - kI$ is invertible, and k is not a spectral value.

Since $(Tx)(1) = 0$, we see that there is no x in X with $Tx = (1, 0, 0, \dots)$. Hence T is not onto, and $T = T - 0 \cdot I$ is not invertible. This shows that 0 is a spectral value of T .

If $(Tx) = 0$, then

$$x(n) = (n+1)(Tx)(n+1) = 0 \quad \text{for all } n;$$

hence $x = 0$. This shows that 0 is not an eigenvalue of T .

Thus we find that

$$\sigma(T) = \{0\}, \quad \sigma_e(T) = \emptyset. \quad \blacksquare$$

PROBLEM 4.2.20 Let X be a Banach space, and $A, B \in BL(X)$. Show that if B is compact, the spectra of A and $A + B$ are the same except for eigenvalues.

Solution Let $k \notin \sigma_e(A)$ and $k \notin \sigma(A + B)$. Then, $T = (A + B - kI)^{-1}$ exists in $BL(X)$. So,

$$(A + B - kI)T = I, \quad BT - I = -(A - kI)T.$$

Suppose that $(BT - I)(x) = 0$. Then, $(A - kI)T(x) = 0$; and so $Tx = 0$ since $k \notin \sigma_e(A)$. Hence, $x = T^{-1}T(x) = 0$. This shows that 1 is not an eigenvalue of BT . But BT is compact since B is compact and T is bounded. So 1 is not a spectral value of BT . Thus, $BT - I$ is invertible. Hence,

$$A - kI = (I - BT)T^{-1}$$

is invertible, that is, $k \notin \sigma(A)$. It follows that

$$k \in \sigma(A), \quad k \notin \sigma_e(A) \implies k \in \sigma(A + B).$$

Similarly, changing A to $A + B$ and B to $-B$, we see that

$$k \in \sigma(A + B) \quad \text{and} \quad k \notin \sigma_e(A + B) \implies k \in \sigma(A).$$

This is precisely what is required to be proved. $\blacksquare$

PROBLEM 4.2.21 Let $X = l^p$, where $1 \leq p \leq \infty$, and $\{k_n\}$ be a bounded sequence of scalars. For $x \in X$, define $A(x)$ by

$$A(x)(j) = k_j x(j), \quad j = 1, 2, \dots$$

Show that $A \in BL(X)$. Find the eigenvalues of A and show that A is compact iff $k_n \rightarrow 0$.

Solution Let $|k_n| \leq \alpha$ for all n . If $1 \leq p < \infty$, then

$$\sum_{j=1}^{\infty} |Ax(j)|^p \leq \sum_{j=1}^{\infty} \alpha^p |x(j)|^p;$$

hence $A(x) \in l^p$ and

$$\|A(x)\|_p \leq \alpha \|x\|_p \quad \text{for all } x \in X. \quad (13)$$

If $p = \infty$, then

$$|A(x)(j)| \leq \alpha |x(j)| \leq \alpha \|x\|_{\infty} \quad \text{for } j \geq 1;$$

hence $A(x) \in l^{\infty}$ and (13) holds with $p = \infty$. Since A is obviously linear, we see that $A \in BL(X)$.

As in the proof of Problem 4.2.5 we see that

$$\sigma_e(A) = \{k_j : j = 1, 2, \dots\}.$$

If $k_n \rightarrow 0$, then A is compact by Problem 4.1.3.

Conversely, suppose that A is compact. To show that $k_n \rightarrow 0$, let $\epsilon > 0$ be given. If $k \in \sigma_e(A)$, $|k| \geq \epsilon$ and $k = k_{n_1} = k_{n_2} = \dots$, then $e_{n_1}, e_{n_2}, \dots$ are eigenvectors corresponding to k . But the eigenspace corresponding to k is finite dimensional, and $e_{n_1}, e_{n_2}, \dots$ are linearly independent. Hence there can be only a finite number of values of n with $k = k_n$. Since the eigenspectrum $\sigma_e(A)$ is a bounded set having no limit point other than 0, there can be at the most a finite number of elements $k \in \sigma_e(A)$ with $|k| \geq \epsilon$. It follows that $|k_n| \geq \epsilon$ for at the most a finite number of values of n . Therefore, $k_n \rightarrow 0$. This completes the proof. ■

PROBLEM 4.2.22 Let $X = l^p$, where $1 \leq p \leq \infty$, and $\{k_n\}$ be a bounded sequence of scalars. For $x \in X$, define $A(x)$ by

$$A(x)(1) = 0, \quad A(x)(i) = k_{i-1} x(i-1), \quad i \geq 2.$$

Show that $A \in BL(X)$. Show also that A is compact iff $k_n \rightarrow 0$.

Solution Define B, S , and T on X as follows:

$$\begin{aligned}(Bx)(i) &= k_i x(i), \quad i = 1, 2, \dots; \\ (Sx)(1) &= 0, \quad (Sx)(i) = x(i-1) \quad \text{for } i \geq 2; \\ (Tx)(i) &= x(i+1), \quad i = 1, 2, \dots\end{aligned}$$

Then $B \in BL(X)$ by Problem 4.2.21. For $x \in l^p$, $1 \leq p \leq \infty$, we have

$$\|Sx\|_p = \|x\|_p, \quad \|Tx\|_p \leq \|x\|_p.$$

Hence $S, T \in BL(X)$. If $x \in X$ and $y = Bx$, then

$$\begin{aligned}(Sy)(1) &= 0 = (Ax)(1), \\ (Sy)(i) &= y(i-1) = k_{i-1}x(i-1) = (Ax)(i)\end{aligned}$$

for $i \geq 2$. Therefore, $SBx = Sy = Ax$, that is, $SB = A$. This shows that $A \in BL(X)$, and that if B is compact then A is compact. Again, if $x \in X$ and $z = Ax$, then

$$(Tz)(i) = z(i+1) = k_i x(i) = (Bx)(i)$$

for $i \geq 1$. Hence, $TAx = Tz = Bx$, that is, $TA = B$. This shows that if A is compact then B is compact. Thus we see that A is compact iff B is compact. But B is compact iff $k_n \rightarrow 0$ by Problem 4.2.21. This completes the proof. ■

PROBLEM 4.2.23 Let $A : C[0, 1] \rightarrow C[0, 1]$ be defined by

$$A(x) = vx, \quad x \in C[0, 1],$$

where v is a fixed nonzero element of $C[0, 1]$. Show that A is not compact.

Solution Let $X = C[0, 1]$. By Problem 4.2.7 we see that $\sigma(A)$ is the range of v . If the range contains only one element c , then $c \neq 0$ since v is not identically zero. In this case, $A = cI$. Hence, A is not compact by Problem 4.1.1, since $c \neq 0$ and X is infinite dimensional.

If the range of v contains two distinct elements a and b , then it contains the whole interval $[a, b]$ since the continuity of v implies that its range is connected. Hence all numbers in $[a, b]$ are limit points of $\sigma(A)$. If A is compact then $\sigma(A)$ cannot have a limit point except possibly 0, since X is a Banach space. This shows that A is not compact in any case. ■

PROBLEM 4.2.24 Let X be a Banach space. If $A \in BL(X)$ and A' is the transpose of A , show that $\sigma(A) = \sigma_a(A) \cup \sigma_e(A')$.

Solution Let $\sigma_r(A)$ be the set of all k in $\mathbb{K}$ such that $R(A - kI)$ is not dense in X . Then by Problem 4.2.14,

$$\sigma(A) = \sigma_a(A) \cup \sigma_r(A).$$

Let $k \in \sigma_r(A)$ and $Y = R(A - kI)$. If $x_1 \in X$ and $x_1 \notin \overline{Y}$, then there is x' in X' such that

$$x'(x_1) = d(x_1, Y) > 0 \quad \text{and} \quad x'(y) = 0 \quad \text{for all } y \in Y.$$

So for all x in X ,

$$(A'x')(x) - kx'(x) = x'(Ax) - kx'(x) = x'(Ax - kx) = 0,$$

since $y = Ax - kx \in Y$. Thus,

$$A'x' - kx' = 0.$$

Since $x'(x_1) \neq 0$, x' is not the zero element in X' . This shows that k is an eigenvalue of A' .

On the other hand, suppose that k is an eigenvalue of A' . Then, $A'x' - kx' = 0$ for some nonzero $x' \in X'$. So for all $x \in X$,

$$0 = (A'x')(x) - kx'(x) = x'(Ax - kx).$$

Hence, $Ax - kx \in Z(x')$, the zero space of x' . Since this is true for all $x \in X$, we see that $R(A - kI) \subset Z(x')$. But $Z(x')$ is a closed proper subspace of X , since x' is not zero. Therefore,

$$\overline{R(A - kI)} \subset Z(x') \neq X;$$

hence $k \in \sigma_r(A)$.

Thus we have proved that $\sigma_r(A) = \sigma_e(A')$. This proves the result. ■

PROBLEM 4.2.25 Let X be a normed space and $A \in BL(X)$. Suppose that Y and Z are closed subspaces of X such that $Z \subset Y$, $Z \neq Y$ and $(A - I)(Y) \subset Z$. Prove that there exists $y \in Y$ with $\|y\| = 1$ and $\|A(y) - A(z)\| \geq 1/2$ for all $z \in Z$.

Solution By applying Riesz lemma to Y and its proper closed subspace Z , we get $y \in Y$ with $\|y\| = 1$ and $d(y, Z) \geq 1/2$. If $B = A - I$, then, for all $z \in Z$, we have $B(y), B(z) \in Z$ and hence,

$$\|A(y) - A(z)\| = \|y - [B(z) + z - B(y)]\| \geq d(y, Z) \geq \frac{1}{2}. \quad \blacksquare$$

PROBLEM 4.2.26 Let A be a compact operator on a Banach space X . Prove that its eigenspectrum $\sigma_e(A)$ is countable and has no limit point except possibly 0.

Solution For $t > 0$, let $E_t = \{k \in \sigma_e(A) : |k| > t\}$. To prove that E_t is finite for each $t > 0$, assume, on the contrary, that E_t is an infinite set for some $t > 0$. Then, there exist an infinite number of distinct values $k_n \in E_t$. Let $x_n \in X$ with $A(x_n) = k_n x_n$ and $x_n \neq 0$. Then, $\{x_n : n \geq 1\}$ is a linearly independent set since, the vectors corresponding to distinct eigenvalues are linearly independent.

Let $Y_n = \text{span}\{x_1, \dots, x_n\}$. Then, Y_n being finite dimensional, is closed. Further, $Y_n \subset Y_{n+1}$ and $Y_n \neq Y_{n+1}$. If $y = \sum_{i=1}^n k'_i x_i \in Y$, then

$$A(y) - k_n y = \sum_{i=1}^n k'_i k_i x_i - k_n \sum_{i=1}^n k'_i x_i = \sum_{i=1}^{n-1} k'_i (k_i - k_n) x_i \in Y_{n-1}.$$

It follows that $((1/k_n)A - I)(Y_n) \subset Y_{n-1}$. This is true for all $n > 1$. Apply Problem 4.2.25 with A replaced by $(1/k_n)A$. We see that there exists $y_n \in Y_n$ with $\|y_n\| = 1$ for each n and

$$\|A(y_n) - A(y_m)\| \geq \frac{|k_n|}{2} \quad \text{for all } n > m > 1.$$

Since $k_n \in E_t$, we get $\|A(y_n) - A(y_m)\| \geq t/2$ for $n > m > 1$. This implies that $\{A(y_n)\}$ has no convergent subsequence even though $\{y_n\}$ is a bounded sequence. This contradicts the compactness of A . Hence, E_t is a finite set for each $t > 0$. This completes the proof. ■

PROBLEM 4.2.27 If A is a compact operator on a Banach space X , prove the following:

- (a) The zero space $Z(A - I)$ of $A - I$ is finite dimensional.
- (b) The range $R(A - I)$ of $A - I$ is closed.

Solution (a) Let $B = A - I$ and assume that $Z(B)$ is infinite dimensional. Let $\{x_n : n = 1, 2, \dots\}$ be an infinite linearly independent set, $Y_n = \text{span}\{x_1, \dots, x_n\}$. Then, Y_n , being finite dimensional, is closed in X . Since $B(x_n) = 0$ for all n , we see that $B(Y_n) = \{0\}$. Further, $Y_n \subset Y_{n+1}$, $Y_n \neq Y_{n+1}$ and $B(Y_{n+1}) = \{0\} \subset Y_n$. Hence, by Problem 4.2.25, there is a sequence $\{y_n\}$ in X such that

$$\|y_n\| = 1 \quad \text{and} \quad \|A(y_n) - A(y_m)\| \geq \frac{1}{2} \quad \text{for } n > m > 1.$$

This is not possible as A is compact. So, $Z(B)$ is finite dimensional.

(b) Let $B = A - I$, as before, and $Z = Z(B)$. Then, Z has a finite basis $\{z_1, \dots, z_n\}$. Using Problem 1.4.1, one can prove that there exist functionals x'_j in X' such that for $1 \leq j \leq n$. So, we have

$$x'_j(z_i) = 0 \quad \text{for } i \neq j, \quad 1 \leq i \leq n, \quad \text{and} \quad x'_j(z_j) = 1.$$

Let Y be the set of all $x \in X$ such that $x'_j(x) = 0$ for $1 \leq j \leq n$. Then, Y is closed and $Y \cap Z = \{0\}$. If $x \in X$ and $z = \sum_{i=1}^n x'_i(x) z_i$, then $z \in Z$ and

$x'_j(z) = x'_j(x)$ for $1 \leq j \leq n$. Hence, if $y = x - z$, then $x'_j(y) = 0$. This shows that $y \in Y$ and $x = y + z$. Thus, we have proved that $X = Y + Z$ where Y is a closed subspace of X and $Y \cap Z = \{0\}$.

Let C denote the restriction of B to Y . If $C(y) = 0$, $y \in Y$, then $B(y) = 0$ and so, $y \in Y \cap Z = \{0\}$. Thus, C is one to one. Further, $R(C) = R(B)$. Let $x \in \overline{R(C)}$. Then, there exists a sequence $\{y_n\}$ in Y such that $C(y_n) \rightarrow x$. First assume that $\{y_n\}$ is an unbounded sequence. Passing to a subsequence, if necessary, we may suppose that $\|y_n\| \rightarrow \infty$. Let $z_n = y_n/\|y_n\|$. Then, $\|z_n\| = 1$ and $C(z_n) \rightarrow 0$. Since A is compact, $\{z_n\}$ has a subsequence $\{z_{n_j}\}$ such that $\{A(z_{n_j})\}$ converges to some $z \in X$. Now,

$$z_{n_j} = A(z_{n_j}) - B(z_{n_j}) = A(z_{n_j}) - C(z_{n_j}) \rightarrow z - 0 \quad \text{as } j \rightarrow \infty.$$

Since $z_{n_j} \in Y$ and Y is closed, this, first of all, shows that $z \in Y$. Consequently, $C(z_{n_j}) \rightarrow C(z)$ as $j \rightarrow \infty$. Thus, $C(z) = 0$. Hence, $z = 0$ as C is one to one. This leads to a contradiction, since $\|z_n\| = 1$ for all n . So, we conclude that $\{y_n\}$ is a bounded sequence.

The compactness of A , therefore, implies the existence of a subsequence $\{A(y_{n_i})\}$ converging to some $y \in X$. As before,

$$y_{n_i} = A(y_{n_i}) - C(y_{n_i}) \rightarrow y - x \quad \text{as } i \rightarrow \infty.$$

Since $y_{n_i} \in Y$ and Y is closed, this, first of all, shows that $y - x \in Y$. Consequently, $C(y_{n_i}) \rightarrow C(y - x)$ as $i \rightarrow \infty$. But $C(y_n) \rightarrow x$ as $n \rightarrow \infty$. Thus, we see that $x = C(y - x) \in R(C)$. Hence, $R(C)$ is closed in X , that is, $R(B) = R(C)$ is closed. This finishes the proof. ■

PROBLEM 4.2.28 If A is a compact operator on a Banach space X , prove that $A - I$ is one to one if and only if it is onto.

Solution First, suppose that B is one to one. Let $B = A - I$, as before. For $n = 1, 2, \dots$, we have

$$(-1)^n B^n = (I - A)^n = I - \binom{n}{1} A + \dots + (-1)^n \binom{n}{n} A^n = I - T,$$

say. Since $A, A^2, \dots$ are all compact, we see that $T = I - (-1)^n B^n$ is a compact operator. Hence, by Problem 4.2.27(b), $R(T - I)$ is closed. Since $(-B)(x) = B(-x) \in R(B)$, we see that $R(B^n) = R((-1)^{n-1} B^n) = R(T - I)$ is closed for each n .

Since $B^{n+1}(x) = B^n(Bx)$, we see that $R(B^{n+1}) \subset R(B^n)$ for all n . Assume that $R(B^{n+1}) \neq R(B^n)$ for an infinite number of values of n . Then, by Problem 4.2.25, there is an infinite sequence $\{y_n\}$ in X such that $\|y_n\| = 1$ and $\|A(y_n) - A(y_m)\| \geq 1/2$ for all $n > m$. This contradicts the compactness of A . Consequently, $R(B^{m+1}) = R(B^m)$ for some m .

Let $y \in R(B^{m-1})$, where we take $R(B^0) = R(I) = X$. Then, $B(y) \in R(B^m) = R(B^{m+1})$. Thus, $B(y) = B(B^m(x))$. Since B is one to one, we get

$y = B^m(x)$ for some x . So, $y \in R(B^m)$. It follows that $R(B^m) = R(B^{m-1})$. Hence, $R(B) = X$ by induction, that is, B is onto.

Now, suppose that $B = A - I$ is onto. The continuity of B^n implies that $Z(B^n)$ is closed for all n . If $B^n(x) = 0$, then $B^{n+1}(x) = 0$. Thus, $Z(B^n) \subset Z(B^{n+1})$. Assume that $Z(B^n) \neq Z(B^{n+1})$ for an infinite number of values of n . As before, we get a contradiction to the compactness of A . Consequently, $Z(B^m) = Z(B^{m+1})$ for some m . Let $y \in Z(B^m)$. Since B is onto, there is x such that

$$\begin{aligned} y &= B(x), & B^{m+1}(x) &= B^m(y) = 0, \\ x &\in Z(B^{m+1}) = Z(B^m), & B^{m-1}(y) &= B^m(x) = 0, \end{aligned}$$

where we take $B^0 = I$, and hence, $y \in Z(B^{m-1})$. It follows that $Z(B^m) = Z(B^{m-1})$. Hence, $Z(B) = Z(I) = \{0\}$ by induction, that is, B is one to one. This completes the proof. ■

PROBLEM 4.2.29 If A is a compact operator on a Banach space X , prove the following:

- (a) Every nonzero spectral value of A is an eigenvalue.
- (b) The eigenspaces of A corresponding to nonzero eigenvalues are all finite dimensional.
- (c) The spectrum of A is countable and has no limit except 0.

Solution (a) Let k be a nonzero scalar and $B = (1/k)A - I$. It is clear that k is an eigenvalue of A if and only if B is one to one and k is a spectral value of A if and only if B is invertible in $BL(X)$. Since $(1/k)A$ is compact, Problem 4.2.28 shows that B is one to one if and only if it is onto. If it is one to one and onto, then it has an inverse B^{-1} which is linear. Since X is Banach, the open mapping theorem implies that if B^{-1} exists, then it belongs to $BL(X)$. This proves (a).

(b) Let k be a nonzero eigenvalue of A and $B = (1/k)A - I$. Then the eigenspace corresponding to k is $Z(B)$. Since $(1/k)A$ is compact, Problem 4.2.27(a) shows that $Z(B)$ is finite dimensional. This gives (b).

(c) This follows from (a) above and Problem 4.2.26. This completes the proof. ■

5

Hilbert Spaces

We next consider linear spaces in which a norm arises in a natural way out of an inner product. Such spaces, called inner product spaces, are perhaps the most natural generalization of Euclidean spaces. As before, the scalar field is denoted by $\mathbb{K}$ and is assumed to be either the field $\mathbb{R}$ of real numbers or the field $\mathbb{C}$ of complex numbers.

In Section 5.1 we deal with the elementary properties of inner product spaces. An inner product space which is complete, under the norm induced by the inner product, is called a Hilbert space. The inner product enables us to introduce the concept of orthogonality which, in turn, is intimately related to the idea of minimum distance. Section 5.2 is devoted to orthonormal sets and the Riesz-Fischer theorem. Another important theorem for Hilbert spaces, namely the Riesz representation theorem, is studied in Section 5.3.

5.1 Inner Product Spaces

Let X be a linear space over the scalar field $\mathbb{K}$ which may be $\mathbb{R}$ or $\mathbb{C}$. We shall denote the complex conjugate of an element k in $\mathbb{K}$ by $\bar{k}$. Thus, $\bar{k} = k$ iff $k \in \mathbb{R}$. An *inner product* on X is a function which assigns a scalar $\langle x, y \rangle$ for all x, y in X such that the following hold for all $x, y, z \in X$ and $k \in \mathbb{K}$:

$$\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle, \quad \langle kx, y \rangle = k\langle x, y \rangle,$$

$$\langle y, x \rangle = \overline{\langle x, y \rangle}, \quad \langle x, x \rangle \geq 0,$$

$$\langle x, x \rangle = 0 \iff x = 0.$$

An *inner product space* (i.p.s.) is a linear space with an inner product defined in it. The norm induced by an inner product is given by

$$\|x\| = \sqrt{\langle x, x \rangle}, \quad x \in X.$$

If this norm makes X a complete metric space, then X is said to be a *Hilbert space*.

When we deal with two or more inner product spaces, we may use the same symbol to denote the inner product. To avoid triviality it will be tacitly assumed that a Hilbert space is nonzero.

From the defining properties of an inner product one can readily see that

$$\begin{aligned}\langle x, y + z \rangle &= \langle x, y \rangle + \langle x, z \rangle \\ \langle x, ky \rangle &= \bar{k} \langle x, y \rangle, \\ \langle 0, y \rangle &= 0 = \langle x, 0 \rangle,\end{aligned}$$

for all $x, y, z \in X$ and $k \in \mathbb{K}$.

The space $\mathbb{K}^n$: For $x, y \in \mathbb{K}^n$, let

$$\langle x, y \rangle = \sum_{j=1}^n x(j) \overline{y(j)}.$$

This gives an inner product on $\mathbb{K}^n$ with the induced norm $\|\cdot\|_2$. This makes $\mathbb{K}^n$ a Hilbert space.

The space l^2 : For $x, y \in l^2$, let

$$\langle x, y \rangle = \sum_{j=1}^{\infty} x(j) \overline{y(j)}.$$

This gives an inner product on l^2 making l^2 a Hilbert space with the norm $\|\cdot\|_2$.

The space $L^2(T)$: If T is a measure space with measure μ and $x, y \in L^2(T)$, let

$$\langle x, y \rangle = \int_T x \bar{y} d\mu.$$

This makes $L^2(T)$ a Hilbert space with the norm $\|\cdot\|_2$.

The product of inner product spaces : If X and Y are inner product spaces then an inner product is defined on $X \times Y$ as follows:

$$\langle (x_1, y_1), (x_2, y_2) \rangle = \langle x_1, x_2 \rangle + \langle y_1, y_2 \rangle$$

for $x_1, x_2 \in X$ and $y_1, y_2 \in Y$.

Let X be an inner product space. Then

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2 \quad (1)$$

$$|\langle x, y \rangle| \leq \|x\| \cdot \|y\| \quad (2)$$

Formula (1) is known as *parallelogram law* and (2) is known as *Schwarz inequality*. If the scalar field $\mathbb{K}$ is $\mathbb{R}$, then

$$4\langle x, y \rangle = \|x + y\|^2 - \|x - y\|^2.$$

If $\mathbb{K}$ is $\mathbb{C}$, then

$$\begin{aligned} 4 \operatorname{Re} \langle x, y \rangle &= \|x + y\|^2 - \|x - y\|^2 \\ \operatorname{Im} \langle x, y \rangle &= \operatorname{Re} \langle x, iy \rangle \end{aligned}$$

These are known as *polarisation identities*.

Using Schwarz inequality, we can see that the inner product is a continuous function on $X \times X$. Thus, if $x_n \rightarrow x$ and $y_n \rightarrow y$, then

$$\langle x_n, z \rangle \rightarrow \langle x, z \rangle, \quad \langle z, y_n \rangle \rightarrow \langle z, y \rangle, \quad \langle x_n, y_n \rangle \rightarrow \langle x, y \rangle.$$

If $\langle x, y \rangle = 0$, we say that x and y are *orthogonal* and write $x \perp y$. If $\langle x, y \rangle = 0$ for all y in a set E , we write $x \perp E$. If $\langle x, y \rangle = 0$ for all $x \in E, y \in F$, we say that E and F are *orthogonal* and write $E \perp F$. The set of all x such that $x \perp E$ is called the *orthogonal complement* of E and is denoted by $E^\perp$.

Let E be a subset of X and $x \in X$. An element $y \in E$ is said to be a *best approximation* to x from E if

$$\|x - y\| \leq \|x - z\| \quad \text{for all } z \in E.$$

EXERCISES

1. Consider the inner product space $X = \mathbb{R}^2$ over the scalar field $\mathbb{R}$. Let $x = (1, 1)$. Find all $y \in \mathbb{R}^2$ such that $x \perp y$ and $\|x\| = \|y\|$.
2. Let $X = \mathbb{R}, Y = \mathbb{R}^2$ and $Z = \mathbb{R}^3$. Compare the inner products in Z and the product space $X \times Y$ as defined above.
3. Prove the parallelogram law.
4. Prove that

$$\operatorname{Re} \langle x, y \rangle = \frac{1}{4} \{ \|x + y\|^2 - \|x - y\|^2 \}$$

for all x, y in an inner product space.

5. Prove that if the scalar field is $\mathbb{C}$ then

$$\operatorname{Im} \langle x, y \rangle = \operatorname{Re} \langle x, iy \rangle$$

for all x, y .

6. Let $x_n \rightarrow x$ and $y_n \rightarrow y$. Prove that

$$\langle x_n, z \rangle \rightarrow \langle x, z \rangle, \quad \langle z, y_n \rangle \rightarrow \langle z, y \rangle, \quad \langle x_n, y_n \rangle \rightarrow \langle x, y \rangle$$

for all x, y .

PROBLEMS

PROBLEM 5.1.1 Let X be an inner product space and $x, y \in X$. Prove the following:

(a) $x \perp y$ iff

$$\|kx + y\|^2 = \|kx\|^2 + \|y\|^2 \quad \text{for all } k \in \mathbb{K}. \quad (3)$$

(b) Pythagoras theorem: If $x \perp y$ then

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2. \quad (4)$$

(c) If X is real, then (4) implies that $x \perp y$.

(d) If X is complex then (4) does not imply that $x \perp y$.

Solution (a) By the defining properties of inner product and norm, we have

$$\begin{aligned} \|kx + y\|^2 &= \langle kx + y, kx + y \rangle \\ &= \langle kx, kx + y \rangle + \langle y, kx + y \rangle \\ &= \langle kx, kx \rangle + \langle kx, y \rangle + \langle y, kx \rangle + \langle y, y \rangle \\ &= \|kx\|^2 + k\langle x, y \rangle + \bar{k}\overline{\langle x, y \rangle} + \|y\|^2. \end{aligned}$$

This shows that (3) holds iff

$$k\langle x, y \rangle + \bar{k}\overline{\langle x, y \rangle} = 0 \quad \text{for all } k \in \mathbb{K}. \quad (5)$$

If $x \perp y$ then $\langle x, y \rangle = 0$ and (5) holds. Conversely, if (5) holds then choosing $k = \langle x, y \rangle$ in it, we get

$$|\langle x, y \rangle|^2 + |\langle x, y \rangle|^2 = 0,$$

and hence $\langle x, y \rangle = 0$. Thus, $x \perp y$ iff (5) holds. This gives (a).

(b) Using (a) with $k = 1$ we get this.

(c) If $\mathbb{K} = \mathbb{R}$ and (3) holds with $k = 1$ then (5) holds with $k = 1$ and so

$$\langle x, y \rangle + \langle x, y \rangle = 0.$$

Thus, if $\mathbb{K} = \mathbb{R}$ and (4) holds then $\langle x, y \rangle = 0$. This gives (c).

(d) To see this let $X = \mathbb{K} = \mathbb{C}$. Then $\langle x, y \rangle = x\bar{y}$. Let $x = 1, y = i$, so that

$$\langle x, y \rangle + \overline{\langle x, y \rangle} = (-i) + i = 0,$$

that is, (5) holds with $k = 1$. Hence (4) holds even though $\langle x, y \rangle = -i \neq 0$. This completes the proof. ■

PROBLEM 5.1.2 Show that the following statements are equivalent in an inner product space X :

(a) $x \perp y$

(b) $\|x + ky\| = \|x - ky\|$ for all scalars k .

(c) $\|x + ky\| \geq \|x\|$ for all scalars k .

Solution For all $x, y \in X$ and scalars k , we have

$$\begin{aligned}
 \|x + ky\|^2 &= \langle x + ky, x + ky \rangle \\
 &= \langle x, x \rangle + \langle x, ky \rangle + \langle ky, x \rangle + \langle ky, ky \rangle \\
 &= \langle x, x \rangle + \bar{k}\langle x, y \rangle + k\overline{\langle x, y \rangle} + k\bar{k}\langle y, y \rangle \\
 &= \|x\|^2 + 2 \operatorname{Re} \bar{k}\langle x, y \rangle + |k|^2 \|y\|^2.
 \end{aligned} \tag{6}$$

Changing k to $-k$, we get

$$\|x - ky\|^2 = \|x\|^2 - 2 \operatorname{Re} \bar{k}\langle x, y \rangle + |k|^2 \|y\|^2. \tag{7}$$

Suppose that (a) holds. Then, $\langle x, y \rangle = 0$. Hence from (6) and (7),

$$\begin{aligned}
 \|x + ky\|^2 &= \|x - ky\|^2 \\
 \|x + ky\|^2 &= \|x\|^2 + |k|^2 \|y\|^2 \geq \|x\|^2
 \end{aligned}$$

This shows that (a) $\implies$ (b) and (a) $\implies$ (c).

On the other hand, suppose that (b) holds. Then from (6) and (7), we see that

$$\operatorname{Re} \bar{k}\langle x, y \rangle = 0 \quad \text{for all } k$$

Taking $k = \langle x, y \rangle$ we get

$$|\langle x, y \rangle|^2 = 0,$$

and so $\langle x, y \rangle = 0$, that is, (a) holds.

Finally, suppose that (c) holds. Then (6) shows that

$$2 \operatorname{Re} \bar{k}\langle x, y \rangle + |k|^2 \|y\|^2 \geq 0 \quad \text{for all } k$$

If $y = 0$, then $\langle x, y \rangle = 0$. If $y \neq 0$, take

$$k = \frac{-\langle x, y \rangle}{\|y\|^2}.$$

Then,

$$\begin{aligned}
 0 \leq 2 \operatorname{Re} \bar{k}\langle x, y \rangle + |k|^2 \|y\|^2 &= -2 \operatorname{Re} \frac{\overline{\langle x, y \rangle}}{\|y\|^2} \langle x, y \rangle \frac{|\langle x, y \rangle|^2}{\|y\|^4} \|y\|^2 \\
 &= -\frac{|\langle x, y \rangle|^2}{\|y\|^2}.
 \end{aligned}$$

This implies that $\langle x, y \rangle = 0$, that is, (a) holds. ■

PROBLEM 5.1.3 Prove that for all x, y, z in an inner product space X we have the Appolonius identity:

$$t\|x - y\|^2 + (1 - t)\|x - z\|^2 = \|x - u\|^2 + t(1 - t)\|y - z\|^2,$$

where

$$u = ty + (1 - t)z, \quad t \in \mathbb{R}.$$

Deduce that

$$\frac{1}{2}\|x - y\|^2 + \frac{1}{2}\|x - z\|^2 = \|x - (y + z)/2\|^2 + \frac{1}{4}\|y - z\|^2.$$

Solution For all $v, w \in X$, we have

$$\begin{aligned} \|v + w\|^2 &= \langle v + w, v + w \rangle = \langle v, v \rangle + \langle v, w \rangle + \langle w, v \rangle + \langle w, w \rangle \\ &= \|v\|^2 + 2 \operatorname{Re} \langle v, w \rangle + \|w\|^2. \end{aligned}$$

Hence,

$$\begin{aligned} \|x - y\|^2 &= \|x - u + u - y\|^2 \\ &= \|x - u\|^2 + 2 \operatorname{Re} \langle x - u, u - y \rangle + \|u - y\|^2, \end{aligned} \quad (8)$$

$$\|x - z\|^2 = \|x - u\|^2 + 2 \operatorname{Re} \langle x - u, u - z \rangle + \|u - z\|^2. \quad (9)$$

Multiply (8) by t and (9) by $1 - t$, and add.

$$\begin{aligned} t\|x - y\|^2 + (1 - t)\|x - z\|^2 &= \|x - u\|^2 + t\|u - y\|^2 \\ &\quad + (1 - t)\|u - z\|^2 + 2t \operatorname{Re} \langle x - u, u - y \rangle \\ &\quad + 2(1 - t) \operatorname{Re} \langle x - u, u - z \rangle \end{aligned} \quad (10)$$

Since

$$u - y = (1 - t)(z - y), \quad u - z = -t(z - y),$$

and t is real, we have

$$\begin{aligned} t\langle x - u, u - y \rangle &= t(1 - t)\langle x - u, z - y \rangle, \\ (1 - t)\langle x - u, u - z \rangle &= -t(1 - t)\langle x - u, z - y \rangle. \end{aligned}$$

So (10) gives

$$\begin{aligned} t\|x - y\|^2 + (1 - t)\|x - z\|^2 &= \|x - u\|^2 \\ &= t(1 - t)^2\|z - y\|^2 + (1 - t)t^2\|z - y\|^2 + 0 \\ &= t(1 - t)\|z - y\|^2. \end{aligned}$$

Thus we get the general formula. The second part is obtained from this with $t = \frac{1}{2}$. ■

PROBLEM 5.1.4 Let X be a normed space. Prove:

- If the norm obeys the parallelogram law, then X can be made an inner product space, that is, its norm is induced by an inner product.
- If every two dimensional subspace can be made an inner product space, then X can be made an inner product space.

Solution (a) Let $\mathbb{K}$ be the scalar field. If $\mathbb{K} = \mathbb{R}$, define

$$\langle x, y \rangle = \frac{1}{4} \{ \|x + y\|^2 - \|x - y\|^2 \},$$

and if $\mathbb{K} = \mathbb{C}$, define $\langle x, y \rangle$ by

$$\operatorname{Re} \langle x, y \rangle = \frac{1}{4} \{ \|x + y\|^2 - \|x - y\|^2 \} \quad (11)$$

$$\operatorname{Im} \langle x, y \rangle = \operatorname{Re} \langle x, iy \rangle.$$

It can be verified directly from these that

$$\langle x, 0 \rangle = 0 = \langle 0, y \rangle \quad (12)$$

$$\operatorname{Re} \langle y, x \rangle = \operatorname{Re} \langle x, y \rangle \quad (13)$$

irrespective of $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. If $\mathbb{K} = \mathbb{C}$, then

$$\begin{aligned} \operatorname{Im} \langle y, x \rangle &= \operatorname{Re} \langle y, ix \rangle = \frac{1}{4} \{ \|y + ix\|^2 - \|y - ix\|^2 \} \\ &= \frac{1}{4} \{ \|x - iy\|^2 - \|x + iy\|^2 \} \\ &= -\operatorname{Re} \langle x, iy \rangle = -\operatorname{Im} \langle x, y \rangle, \end{aligned}$$

and hence

$$\langle y, x \rangle = \overline{\langle x, y \rangle}.$$

If $\mathbb{K} = \mathbb{R}$, this is the same as (13). This with $y = x$ shows that $\langle x, x \rangle$ is real and so $\langle x, x \rangle = \|x\|^2$ by (11). It remains to prove that, for $x, y, z \in X$ and $k \in \mathbb{K}$,

$$\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle \quad (14)$$

$$\langle kx, z \rangle = k \langle x, z \rangle. \quad (15)$$

Using the parallelogram law, we get

$$\begin{aligned} \|(x + \frac{z}{2}) + (y + \frac{z}{2})\|^2 + \|(x + \frac{z}{2}) - (y + \frac{z}{2})\|^2 &= 2\|x + \frac{z}{2}\|^2 + 2\|y + \frac{z}{2}\|^2, \\ \|(x - \frac{z}{2}) + (y - \frac{z}{2})\|^2 + \|(x - \frac{z}{2}) - (y - \frac{z}{2})\|^2 &= 2\|x - \frac{z}{2}\|^2 + 2\|y - \frac{z}{2}\|^2 \end{aligned}$$

Subtract the last equation from the previous one and divide by 4. Since (11) holds whether $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, we get

$$\operatorname{Re} \langle x + y, z \rangle = 2 \operatorname{Re} \langle x, z/2 \rangle + 2 \operatorname{Re} \langle y, z/2 \rangle \quad (16)$$

In this taking $y = 0$ and $x = 0$ we obtain, respectively,

$$\operatorname{Re} \langle x, z \rangle = 2 \operatorname{Re} \langle x, z/2 \rangle,$$

$$\operatorname{Re} \langle y, z \rangle = 2 \operatorname{Re} \langle y, z/2 \rangle,$$

because of (12). Hence (16) gives

$$\operatorname{Re} \langle x + y, z \rangle = \operatorname{Re} \langle x, z \rangle + \operatorname{Re} \langle y, z \rangle.$$

If $\mathbb{K} = \mathbb{R}$ this is the same as (14). If $\mathbb{K} = \mathbb{C}$, changing z to iz in this, we get

$$\operatorname{Im} \langle x + y, z \rangle = \operatorname{Im} \langle x, z \rangle + \operatorname{Im} \langle y, z \rangle.$$

Thus we get (14) when $\mathbb{K} = \mathbb{C}$ also.

It follows from (14) by induction that

$$\langle nx, z \rangle = n \langle x, z \rangle \quad (17)$$

for all positive integers n . If $n = 0$, this is true by (12) and if $n = -m$ is a negative integer, then

$$\begin{aligned} \langle nx, z \rangle + \langle mx, z \rangle &= \langle nx + mx, z \rangle = \langle 0, z \rangle, \\ \langle nx, z \rangle + m \langle x, z \rangle &= 0. \end{aligned}$$

Hence (17) holds for all integers n . If $r = p/q$, where p and q are integers, then

$$q \langle rx, z \rangle = \langle qrx, z \rangle = \langle px, z \rangle = p \langle x, z \rangle,$$

and hence $\langle rx, z \rangle = r \langle x, z \rangle$ for all rational numbers r .

Let α be any real number. There is a sequence $\{r_n\}$ of rational numbers such that $r_n \rightarrow \alpha$. Hence, $r_n x \rightarrow \alpha x$, and the continuity of inner product gives

$$\langle r_n x, z \rangle \longrightarrow \langle \alpha x, z \rangle$$

But $\langle r_n x, z \rangle = r_n \langle x, z \rangle$. Hence,

$$\langle \alpha x, z \rangle = \lim_{n \rightarrow \infty} \langle r_n x, z \rangle = \alpha \langle x, z \rangle. \quad (18)$$

If $\mathbb{K} = \mathbb{R}$, then (15) is given by this.

Let $\mathbb{K} = \mathbb{C}$. Then

$$\begin{aligned} \operatorname{Re} \langle ix, z \rangle &= \operatorname{Re} \langle z, ix \rangle = \operatorname{Im} \langle z, x \rangle = -\operatorname{Im} \langle x, z \rangle, \\ \operatorname{Im} \langle ix, z \rangle &= \operatorname{Re} \langle ix, iz \rangle = \operatorname{Re} \langle x, z \rangle, \end{aligned}$$

where the last equality follows from (11). It follows that

$$\langle ix, z \rangle = i \langle x, z \rangle.$$

This together with (14) and (18) shows that for $\alpha, \beta \in \mathbb{R}$,

$$\langle (\alpha + i\beta)x, z \rangle = (\alpha + i\beta) \langle x, z \rangle.$$

This completes the proof of (15) and hence the proof of (a).

(b) If the dimension of X is less than two, then the result is trivial. Suppose that the dimension of X is at least two. Let x, y be any two elements in X . There is a two dimensional subspace Y containing x and y . By our assumption the norm in Y is induced by an inner product, and hence the parallelogram law holds in Y . Thus,

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2. \quad (19)$$

Since the norm in Y is the same as that given by X we see that (19) holds for all $x, y \in X$. Hence X can be made an inner product space by (a). This completes the proof. ■

PROBLEM 5.1.5 Let Y be a closed subspace of an inner product space X . Show that there is an inner product on the quotient space X/Y which induces the quotient norm

$$\|x + Y\| = \inf \{\|x + y\| : y \in Y\}, \quad x + Y \in X/Y$$

Solution Let $x_1 + Y$ and $x_2 + Y$ be any two elements of X/Y . There exist sequences $\{y_n\}$ and $\{z_n\}$ in Y such that

$$\|x_1 + y_n\| \rightarrow \|x_1 + Y\|, \quad \|x_2 + z_n\| \rightarrow \|x_2 + Y\|$$

as $n \rightarrow \infty$. Since $y_n \pm z_n \in Y$, we have

$$\begin{aligned} \|x_1 + x_2 + Y\| &\leq \|x_1 + x_2 + y_n + z_n\|, \\ \|x_1 - x_2 + Y\| &\leq \|x_1 - x_2 + y_n - z_n\|. \end{aligned}$$

Hence,

$$\begin{aligned} \|x_1 + x_2 + Y\|^2 + \|x_1 - x_2 + Y\|^2 &\leq \|(x_1 + y_n) + (x_2 + z_n)\|^2 \\ &\quad + \|(x_1 + y_n) - (x_2 + z_n)\|^2 \\ &= 2\|x_1 + y_n\|^2 + 2\|x_2 + z_n\|^2 \end{aligned}$$

by the parallelogram law in X . Letting $n \rightarrow \infty$, we get

$$\|x_1 + x_2 + Y\|^2 + \|x_1 - x_2 + Y\|^2 \leq 2\|x_1 + Y\|^2 + 2\|x_2 + Y\|^2. \quad (20)$$

Next let $\{u_n\}, \{v_n\}$ be sequences in Y such that

$$\begin{aligned} \|x_1 + x_2 + u_n\| &\rightarrow \|x_1 + x_2 + Y\|, \\ \|x_1 - x_2 + v_n\| &\rightarrow \|x_1 - x_2 + Y\| \end{aligned}$$

as $n \rightarrow \infty$. Since $\frac{1}{2}(u_n \pm v_n) \in Y$, we have

$$\begin{aligned} 2\|x_1 + Y\|^2 + 2\|x_2 + Y\|^2 &\leq 2\left\|x_1 + \frac{u_n + v_n}{2}\right\|^2 + 2\left\|x_2 + \frac{u_n - v_n}{2}\right\|^2 \\ &= \|x_1 + x_2 + u_n\|^2 + \|x_1 - x_2 + v_n\|^2 \end{aligned}$$

by the parallelogram law in X . Letting $n \rightarrow \infty$, we obtain

$$2\|x_1 + Y\|^2 + 2\|x_2 + Y\|^2 \leq \|x_1 + x_2 + Y\|^2 + \|x_1 - x_2 + Y\|^2. \quad (21)$$

Combining (20) and (21), we see that

$$\|(x_1 + Y) + (x_2 + Y)\|^2 + \|(x_1 + Y) - (x_2 + Y)\|^2 = 2\|x_1 + Y\|^2 + 2\|x_2 + Y\|^2.$$

Thus the quotient norm on X/Y satisfies the parallelogram law. Hence, by Problem 5.1.4, this norm is induced by an inner product. This completes the proof. ■

PROBLEM 5.1.6 (a) Let $\{u_1, \dots, u_n\}$ be a basis for a finite dimensional linear space X . Show that an inner product on X is completely determined by its values $k_{ij} = \langle u_i, u_j \rangle$. If $n = 2$ and X is real find conditions under which a 2×2 matrix (k_{ij}) can determine an inner product like this.

(b) Show that on any linear space an inner product can be introduced.

Solution (a) Every x in X is uniquely expressible as a linear combination of the basis elements. If

$$x = \sum_{i=1}^n a_i u_i, \quad y = \sum_{j=1}^n b_j u_j,$$

where $a_i, b_j \in \mathbb{K}$, then

$$\begin{aligned} \langle x, y \rangle &= \langle x, \sum_{j=1}^n b_j u_j \rangle = \sum_{j=1}^n \langle x, b_j u_j \rangle, \\ \langle x, b_j u_j \rangle &= \overline{b_j} \langle \sum_{i=1}^n a_i u_i, u_j \rangle = \overline{b_j} \sum_{i=1}^n a_i k_{ij}, \\ \langle x, y \rangle &= \sum_{j=1}^n \sum_{i=1}^n k_{ij} a_i \overline{b_j}. \end{aligned}$$

This shows that if k_{ij} are known then $\langle x, y \rangle$ can be found uniquely for all $x, y \in X$.

Let $n = 2$ and X be real. Given a 2×2 matrix (k_{ij}) define $\langle x, y \rangle$ by

$$\langle x, y \rangle = k_{11}a_1b_1 + k_{12}a_1b_2 + k_{21}a_2b_1 + k_{22}a_2b_2 \quad (22)$$

when

$$x = a_1u_1 + a_2u_2, \quad y = b_1u_1 + b_2u_2.$$

This ensures the first two properties, namely,

$$\begin{aligned} \langle x + y, z \rangle &= \langle x, z \rangle + \langle y, z \rangle, \\ \langle kx, y \rangle &= k\langle x, y \rangle. \end{aligned}$$

Since $\langle u_2, u_1 \rangle = \langle u_1, u_2 \rangle$, we must have $k_{21} = k_{12}$. On the other hand, if $k_{21} = k_{12}$ then $\langle y, x \rangle = \langle x, y \rangle$ for all x, y by (22). So let $k_{21} = k_{12}$. Then,

$$\langle x, x \rangle = k_{11}a_1^2 + 2k_{12}a_1a_2 + k_{22}a_2^2.$$

This must be positive unless $a_1 = a_2 = 0$. Taking $a_1 = 1, a_2 = 0$ and $a_1 = 0, a_2 = 1$, we get

$$k_{11} > 0 \quad \text{and} \quad k_{22} > 0,$$

respectively. Hence

$$\langle x, x \rangle = \left(\sqrt{k_{11}} a_1 + \frac{k_{12}}{\sqrt{k_{11}}} a_2 \right)^2 + \left(k_{22} - \frac{k_{12}^2}{k_{11}} \right) a_2^2.$$

So it is necessary and sufficient that

$$k_{22} - \frac{k_{12}^2}{k_{11}} > 0.$$

Thus the desired conditions are

$$k_{21} = k_{12}, \quad k_{11} > 0, \quad k_{22} > 0, \quad k_{11}k_{22} > k_{12}^2.$$

These are necessary and sufficient.

(b) Any linear space will have a basis. Let S be a basis for X . If

$$x = \sum_{i=1}^n a_i u_i, \quad y = \sum_{i=1}^n b_i u_i,$$

where $a_i, b_i \in \mathbb{K}$, $u_i \in S$, let

$$\langle x, y \rangle = \sum_{i=1}^n a_i \bar{b}_i.$$

This defines an inner product on X . This completes the proof. ■

PROBLEM 5.1.7 Prove that l^p with usual norm is not an inner product space unless $p = 2$.

Solution If the norm is induced by an inner product then it must obey the parallelogram law. If e_1, e_2 are the first two unit vectors in l^p , then

$$\begin{aligned} \|e_1\| &= \|e_2\| = 1, \\ e_1 + e_2 &= (1, 1, 0, \dots), \quad e_1 - e_2 = (1, -1, 0, \dots) \\ \|e_1 + e_2\| &= (1 + 1)^{1/p} = 2^{1/p}, \quad \|e_1 - e_2\| = (1 + 1)^{1/p} = 2^{1/p}. \end{aligned}$$

The parallelogram law gives

$$\|e_1 + e_2\|^2 + \|e_1 - e_2\|^2 = 2\|e_1\|^2 + 2\|e_2\|^2,$$

that is, $2^{2/p} + 2^{2/p} = 2 + 2$. This is possible only if $2/p = 1$. This proves the result. ■

PROBLEM 5.1.8 Let X be the real linear space consisting of all real polynomials of degree not exceeding 2, including the zero polynomial. Let $[a, b]$ be some finite interval and

$$\langle x, y \rangle = \int_a^b x(t)y(t) dt, \quad x, y \in X.$$

Let Y consist of all $x \in X$ such that $x(a) = 0$. Show that X is a complete inner product space and Y is a subspace of X . Do all x in X of degree 2 form a subspace of X ?

Solution Consider the Hilbert space $H = L^2[a, b]$ with $\mathbb{K} = \mathbb{R}$. The given X is a three dimensional subspace of H with the same inner product as in H . Since finite dimensional subspaces are closed we see that X is a closed subspace of H . As H is complete, X also is complete.

If $x, y \in Y$ and $k \in \mathbb{K}$, then

$$(kx + y)(a) = kx(a) + y(a) = 0,$$

and hence $kx + y \in Y$. So Y is a subspace of X .

Since the zero element in X is not a polynomial of degree 2, the answer to the last question is 'No'. ■

PROBLEM 5.1.9 Let $\{x_n\}$ be a sequence in an inner product space X . Prove that:

- (a) If $\langle x_n, x \rangle \rightarrow \langle x, x \rangle$ and $\|x_n\| \rightarrow \|x\|$, then $x_n \rightarrow x$.
- (b) If $x_n \xrightarrow{w} x$ and $\|x_n\| \rightarrow \|x\|$, then $x_n \rightarrow x$.

Solution (a) Since

$$\begin{aligned} \|x_n - x\|^2 &= \langle x_n - x, x_n - x \rangle \\ &= \langle x_n, x_n \rangle - \langle x_n, x \rangle - \langle x, x_n \rangle + \langle x, x \rangle, \\ \langle x_n, x_n \rangle &= \|x_n\|^2 \rightarrow \|x\|^2, \quad \langle x, x_n \rangle = \overline{\langle x_n, x \rangle} \rightarrow \langle x, x \rangle, \end{aligned}$$

we see that

$$\|x_n - x\|^2 \rightarrow \|x\|^2 - \langle x, x \rangle - \langle x, x \rangle + \langle x, x \rangle = 0.$$

That is, $x_n \rightarrow x$.

(b) Let $z \in X$. Define $f : X \rightarrow \mathbb{K}$ by

$$f(x) = \langle x, z \rangle, \quad x \in X.$$

Then f is a linear functional and

$$|f(x)| = |\langle x, z \rangle| \leq \|x\| \|z\|.$$

Hence $f \in X'$. So if $x_n \xrightarrow{w} x$, then $f(x_n) \rightarrow f(x)$, that is, $\langle x_n, z \rangle \rightarrow \langle x, z \rangle$. This is true for any z in X . Use this with $z = x$. The desired result follows from (a). ■

PROBLEM 5.1.10 Let D be a domain in $\mathbb{C}$ bounded by a simple smooth closed curve Γ . Let X be the set of all functions $u \in C(D \cup \Gamma)$ such that u has continuous and bounded partial derivatives u_x, u_y in D . For $u, v \in X$, let

$$\langle u, v \rangle = \iint_D (u_x \overline{v_x} + u_y \overline{v_y}) dx dy + \int_{\Gamma} u \overline{v} ds,$$

where ds is the arc length differential. Show that this defines an inner product on X .

Solution Since

$$(u + w)_x = u_x + w_x, \quad (u + w)_y = u_y + w_y, \quad (ku)_x = ku_x, \quad (ku)_y = ku_y,$$

we get

$$\begin{aligned} \langle u + w, v \rangle &= \iint_D (u_x \overline{v_x} + w_x \overline{v_x} + u_y \overline{v_y} + w_y \overline{v_y}) dx dy + \int_{\Gamma} (u \overline{v} + w \overline{v}) ds \\ &= \langle u, v \rangle + \langle w, v \rangle, \\ \langle ku, v \rangle &= \iint_D (ku_x \overline{v_x} + ku_y \overline{v_y}) dx dy + \int_{\Gamma} ku \overline{v} ds = k \langle u, v \rangle. \end{aligned}$$

It is clear that $\langle v, u \rangle = \overline{\langle u, v \rangle}$. Finally,

$$\langle u, u \rangle = \iint_D (|u_x|^2 + |u_y|^2) dx dy + \int_{\Gamma} |u|^2 ds$$

This is clearly nonnegative. It is equal to zero iff

$$|u_x|^2 + |u_y|^2 = 0 \quad \text{a.e. in } D, \quad |u| = 0 \quad \text{a.e. in } \Gamma.$$

Since u_x, u_y are continuous on D and u is continuous on Γ , this is possible only if

$$u_x = u_y = 0 \quad \text{on } D \quad \text{and} \quad u = 0 \quad \text{on } \Gamma.$$

Since D is connected, $u_x = u_y = 0$ on D only if $u = 0$ on D . Hence,

$$\langle u, u \rangle = 0 \iff u = 0 \quad \text{on } D \cup \Gamma.$$

This completes the proof. ■

PROBLEM 5.1.11 Let X and Y be two inner product spaces and $F : X \rightarrow Y$ be linear. Show that

$$\|F(x)\| = \|x\| \quad \text{for all } x \in X \tag{23}$$

iff

$$\langle F(x_1), F(x_2) \rangle = \langle x_1, x_2 \rangle \quad \text{for all } x_1, x_2 \in X. \tag{24}$$

Solution Since (24) with $x_1 = x_2 = x$ gives (23), it is enough if we prove that (23) $\implies$ (24).

Suppose that (23) holds. Let $y_1 = F(x_1)$ and $y_2 = F(x_2)$. Then,

$$y_1 \pm y_2 = F(x_1 \pm x_2),$$

$$\begin{aligned} 4 \operatorname{Re} \langle y_1, y_2 \rangle &= \|y_1 + y_2\|^2 - \|y_1 - y_2\|^2 \\ &= \|F(x_1 + x_2)\|^2 - \|F(x_1 - x_2)\|^2 \\ &= \|x_1 + x_2\|^2 - \|x_1 - x_2\|^2 = 4 \operatorname{Re} \langle x_1, x_2 \rangle \end{aligned} \quad (25)$$

This is true whether the scalar field $\mathbb{K}$ is $\mathbb{R}$ or $\mathbb{C}$. If $\mathbb{K} = \mathbb{R}$, then (25) gives (24). So let $\mathbb{K} = \mathbb{C}$. Then, $iy_2 = F(ix_2)$. So by (25),

$$\operatorname{Im} \langle y_1, y_2 \rangle = \operatorname{Re} \langle y_1, iy_2 \rangle = \operatorname{Re} \langle x_1, ix_2 \rangle = \operatorname{Im} \langle x_1, x_2 \rangle.$$

This combined with (25) gives

$$\langle F(x_1), F(x_2) \rangle = \langle y_1, y_2 \rangle = \langle x_1, x_2 \rangle.$$

This proves the result. $\blacksquare$

PROBLEM 5.1.12 Let D be a domain in $\mathbb{C}$ and $A^2(D)$ be the set of all functions f , analytic on D and lying in $L^2(D)$. Prove that

- (a) if $f_n \rightarrow f$ in $L^2(D)$ and $f_n \in A^2(D)$ then $f_n \rightarrow f$ uniformly on every compact subset of D ,
- (b) $A^2(D)$ is a Hilbert space under the inner product of $L^2(D)$.

Solution (a) Let $z_0 \in D$ and $r > 0$ be such that

$$E = \{z \in \mathbb{C} : |z - z_0| \leq r\} \subset D.$$

Let $f \in A^2(D)$. Then,

$$\iint_E f(z) \, dx \, dy = \int_0^r \int_{-\pi}^{\pi} f(z_0 + te^{i\theta}) t \, dt \, d\theta.$$

The mean value property gives

$$f(z_0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(z_0 + te^{i\theta}) \, d\theta \quad \text{for } 0 \leq t \leq r.$$

Therefore, if $d\mu = dx \, dy$, then

$$\iint_E f(z) \, d\mu = \int_0^r 2\pi f(z_0) t \, dt = \pi r^2 f(z_0). \quad (26)$$

But

$$\begin{aligned} \left| \int_E f \, d\mu \right| &\leq \left(\int_E |f|^2 \, d\mu \right)^{1/2} \left(\int_E 1 \, d\mu \right)^{1/2} \\ &\leq \left(\int_D |f|^2 \, d\mu \right)^{1/2} (\mu(E))^{1/2} \\ &= \|f\| (\pi r^2)^{1/2}, \end{aligned} \quad (27)$$

using Schwarz inequality for integrals. It follows from (26) and (27) that

$$|f(z_0)| \leq \frac{\|f\|}{\sqrt{\pi} r}. \quad (28)$$

This is true provided that $r > 0$ and

$$\|z - z_0\| \leq r \implies z \in D. \quad (29)$$

Let $\{f_n\}$ be a sequence in $A^2(D)$ and $f_n \rightarrow f$ in $L^2(D)$. Let S be a compact subset of D and ρ be the distance between S and the complement of D . Then, $\rho > 0$. Let $0 < r < \rho$. Suppose that z_0 is any element in S . Thus, (29) is true. So (28) with f replaced by $f_n - f_m$ gives

$$|f_n(z_0) - f_m(z_0)| \leq \frac{1}{\sqrt{\pi} r} \|f_n - f_m\|.$$

Since r is independent of n, m and $z_0 \in S$, this shows that $\{f_n\}$ satisfies the Cauchy criterion for uniform convergence on S . This proves (a).

(b) It is known that if $f_n \rightarrow f$ uniformly on compact subsets of D and each f_n is analytic on D , then f is analytic on D . So the limit function f in (a) belongs to $A^2(D)$. It follows from (a) that $A^2(D)$ is a closed subspace of $L^2(D)$. So $A^2(D)$ is a Hilbert space since $L^2(D)$ is a Hilbert space. This completes the proof. ■

PROBLEM 5.1.13 Let m be a positive integer, and X be the set of all scalar-valued functions x on $[a, b]$ such that its $(m-1)$ -th derivative $x^{(m-1)}$ is absolutely continuous on $[a, b]$ and the m -th derivative $x^{(m)}$ is in $L^2[a, b]$. For $x, y \in X$, let

$$\langle x, y \rangle = \sum_{i=0}^m \int_a^b x^{(i)}(t) \overline{y^{(i)}(t)} \, dt.$$

Show that

- (a) this defines an inner product which makes X a Hilbert space,
- (b) $C^m[a, b]$ is dense in X .

Solution (a) For all $x \in X$, we have

$$\langle x, x \rangle = \sum_{i=0}^m \int_a^b |x^{(i)}(t)|^2 dt = \sum_{i=0}^m \|x^{(i)}\|_2^2. \quad (30)$$

Hence, $\langle x, x \rangle \geq 0$ and $\langle x, x \rangle = 0$ only if $\|x\|_2 = \|x^{(0)}\|_2 = 0$, that is, $x = 0$ a.e. Since x is continuous, we see that $\langle x, x \rangle = 0$ iff $x = 0$. The remaining properties of the inner product follow easily from the definition.

To show that X is complete, let $\{x_n\}$ be a Cauchy sequence in X . From (30) we see that

$$\|x_n - x_r\|^2 \geq \|x_n^{(i)} - x_r^{(i)}\|_2^2, \quad 0 \leq i \leq m.$$

This implies that $\{x_n^{(i)}\}$ is a Cauchy sequence in $L^2[a, b]$. Since $L^2[a, b]$ is complete, there exists $y_i \in L^2[a, b]$ such that $x_n^{(i)} \rightarrow y_i$ in $L^2[a, b]$ for $0 \leq i \leq m$.

If $x \in L^2[a, b]$ then, by Schwarz inequality for integrals,

$$\int_a^b |x| dt \leq \left(\int_a^b |x|^2 dt \right)^{1/2} \left(\int_a^b 1 dt \right)^{1/2}. \quad (31)$$

This shows that $x \in L^1[a, b]$ and that convergence in $L^2[a, b]$ implies convergence in $L^1[a, b]$.

Since $\{x_n^{(i)}\}$ converges to y_i in $L^2[a, b]$, some subsequence of it converges a.e. to y_i . Hence, for each i , we can find c_i in $[a, b]$ such that some subsequence of $\{x_n^{(i)}(c_i)\}$ converges to $y_i(c_i)$. Define $z_m, z_{m-1}, \dots, z_0$ successively as follows:

$$\begin{aligned} z_m &= y_m \\ z_i(t) &= y_i(c_i) + \int_a^t z_{i+1}(s) ds - \int_a^{c_i} z_{i+1}(s) ds, \quad 0 \leq i \leq m-1. \end{aligned} \quad (32)$$

Since

$$x_n^{(i)}(t) = x_n^{(i)}(c_i) + \int_a^t x_n^{(i+1)}(s) ds - \int_a^{c_i} x_n^{(i+1)}(s) ds$$

for $0 \leq i \leq m-1$, we see from (32) that

$$\begin{aligned} |x_n^{(i)}(t) - x_n^{(i)}(c_i) - z_i(t) + y_i(c_i)| &\leq \int_a^t |x_n^{(i+1)} - z_{i+1}| + \int_a^{c_i} |x_n^{(i+1)} - z_{i+1}| \\ &\leq 2\|x_n^{(i+1)} - z_{i+1}\|_1. \end{aligned}$$

Suppose that $z_{i+1} = y_{i+1}$ a.e.. This is true when $i = m-1$. It follows that, as $n \rightarrow \infty$,

$$\|x_n^{(i+1)} - z_{i+1}\|_1 \rightarrow 0, \quad x_n^{(i)} - x_n^{(i)}(c_i) \rightarrow z_i(t) - y_i(c_i),$$

uniformly, and hence in the norm of $L^2[a, b]$. But there is a subsequence $\{x_{n_j}^{(i)}(c_i)\}$ of $\{x_n^{(i)}(c_i)\}$ converging to $y_i(c_i)$. Consequently, letting $n \rightarrow \infty$ through $\{n_j\}$, we see that

$$y_i - y_i(c_i) = z_i - y_i(c_i) \quad \text{a.e.}$$

It follows, successively, that $y_i = z_i$ a.e. for $i = m, m-1, \dots, 0$.

Further, (32) shows that z_i is absolutely continuous on $[a, b]$ and that $z'_i = z_{i+1}$ a.e., for $i \leq m-1$. Thus, $z'_i = z_{i+1}$ if z_{i+1} is continuous, that is, if $i+1 \leq m-1$. It follows that $z_0^{(i)} = z_i$ for $i = 0, 1, \dots, m-1$ and $z_0^{(m)} = z_m$ a.e. Let $x = z_0$. Then, $x \in X$ and

$$x^{(i)} = z_i = y_i \quad \text{a.e. for } i = 0, 1, \dots, m.$$

So by (30)

$$\begin{aligned} \|x_n - x\|^2 &= \sum_{i=0}^m \|x_n^{(i)} - x^{(i)}\|_2^2 \\ &= \sum_{i=0}^m \|x_n^{(i)} - y_i\|_2^2 \rightarrow 0 \quad \text{as } n \rightarrow \infty. \end{aligned}$$

This proves that X is complete.

(b) Let $x \in X$ and $\epsilon > 0$ be given. Since continuous functions are dense in $L^2[a, b]$ and $x^{(m)} \in L^2[a, b]$ there is y in $C[a, b]$ such that

$$\|x^{(m)} - y\|_2 < \epsilon.$$

Let $y_0 = y$ and

$$y_j(t) = x^{(m-j)}(a) + \int_a^t y_{j-1}(s) ds, \quad 1 \leq j \leq m.$$

Let $z = y_m$. Then $y'_j = y_{j-1}$, and so we see that

$$y = y_0 = y'_1 = y''_2 = \dots = y_m^{(m)} = z^{(m)}.$$

Thus, $z \in C^m[a, b]$, and

$$\|x^{(m)} - z^{(m)}\|_2 = \|x^{(m)} - y\|_2 < \epsilon. \quad (33)$$

Since

$$x^{(j-1)}(a) = y_{m-j+1}(a) = z^{(j-1)}(a),$$

we get

$$\left| x^{(j-1)}(t) - z^{(j-1)}(t) \right| = \left| \int_a^t x^{(j)}(s) ds - \int_a^t z^{(j)}(s) ds \right|$$

$$\begin{aligned} &\leq \int_a^b |x^{(j)}(s) - z^{(j)}(s)| ds \\ &\leq \|x^{(j)} - z^{(j)}\|_2 (b-a)^{1/2} \end{aligned}$$

using (31). It follows that

$$\|x^{(j-1)} - z^{(j-1)}\|_2^2 \leq \|x^{(j)} - z^{(j)}\|_2^2 (b-a)^2.$$

This is true for $1 \leq j \leq m$. Repeated application of this shows that

$$\|x^{(i)} - z^{(i)}\|_2 \leq \|x^{(m)} - z^{(m)}\|_2 (b-a)^{m-i} < \epsilon (b-a)^{m-i}$$

by (33). Hence from (30),

$$\|x - z\|^2 = \sum_{i=0}^m \|x^{(i)} - z^{(i)}\|_2^2 < \epsilon^2 \sum_{i=0}^m (b-a)^{2(m-i)}.$$

Since ϵ is arbitrary, this completes the proof. ■

PROBLEM 5.1.14 Let S be a nonempty set and $l^2(S)$ be the set of all scalar-valued functions x on S such that

- (i) $\{s \in S : x(s) \neq 0\}$ is a countable set, and
- (ii) $\sum_{s \in S} |x(s)|^2 < \infty$.

For $x, y \in l^2(S)$, let

$$\langle x, y \rangle = \sum_{s \in S} x(s) \overline{y(s)}. \quad (34)$$

Show that this makes $l^2(S)$ a Hilbert space.

Solution Let $x, y, z \in l^2(S)$. Since the union of three countable sets is countable, we see that there is a sequence $\{t_i\}$ of elements in S such that

$$x(t_i) = y(t_i) = z(t_i) = 0$$

for any s not in this sequence. By Schwarz inequality for sequences,

$$\sum_{i=1}^{\infty} |x(t_i) \overline{y(t_i)}| \leq \left(\sum_{i=1}^{\infty} |x(t_i)|^2 \right)^{1/2} \left(\sum_{i=1}^{\infty} |y(t_i)|^2 \right)^{1/2} < \infty.$$

This shows that

$$\sum_{s \in S} x(s) \overline{y(s)} = \sum_{i=1}^{\infty} x(t_i) \overline{y(t_i)}$$

exists in the scalar field $\mathbb{K}$. It is clear that $\langle y, x \rangle = \overline{\langle x, y \rangle}$ and that

$$\langle x, x \rangle > 0 \quad \text{if } x \neq 0.$$

Since

$$\sum_{i=1}^{\infty} (kx(t_i) + y(t_i))\overline{z(t_i)} = k \sum_{i=1}^{\infty} x(t_i)\overline{z(t_i)} + \sum_{i=1}^{\infty} y(t_i)\overline{z(t_i)}$$

we see that

$$\sum_{s \in S} (kx(s) + y(s))\overline{z(s)} = k \sum_{s \in S} x(s)\overline{z(s)} + \sum_{s \in S} y(s)\overline{z(s)}.$$

Thus,

$$\langle kx + y, z \rangle = k\langle x, z \rangle + \langle y, z \rangle.$$

Further,

$$kx(s) + y(s) = 0 \quad \text{when } s \neq t_i,$$

and

$$|kx(s) + y(s)|^2 = |kx(s)|^2 + kx(s)\overline{y(s)} + \overline{kx(s)}y(s) + |y(s)|^2.$$

This shows that

$$\sum_{s \in S} |kx(s) + y(s)|^2 < \infty.$$

Thus, $kx + y \in l^2(S)$. Since x, y, z are arbitrary we have proved that $l^2(S)$ is a linear space and that (34) gives an inner product on it.

To prove that $l^2(S)$ is complete, let $\{x_n\}$ be a Cauchy sequence in it. Let

$$\|x_n - x_m\| < \epsilon \quad \text{for all } n, m \geq n_0.$$

Since the union of a sequence of countable sets is countable, we see that there is a sequence $\{s_i\}$ of points in S such that $x_n(s) = 0$ for any $n \geq 1$ and any s not belonging to this sequence. Let j be some positive integer. For $N \geq j$ and $n, m \geq n_0$, we have

$$\begin{aligned} |x_n(s_j) - x_m(s_j)|^2 &\leq \sum_{i=1}^N |x_n(s_i) - x_m(s_i)|^2 \\ &\leq \sum_{s \in S} |x_n(s) - x_m(s)|^2 \\ &= \|x_n - x_m\|^2 < \epsilon^2. \end{aligned} \tag{35}$$

This first of all shows that $\{x_n(s_j) : n \geq 1\}$ is a Cauchy sequence of scalars. So,

$$\lim_{n \rightarrow \infty} x_n(s_j) = x(s_j)$$

exists for each $j \geq 1$. Let $x(s) = 0$ for any s not in $\{s_i\}$. Then,

$$x_n(s) \rightarrow x(s) \quad \text{for all } s \in S.$$

Keep n, N fixed and let $m \rightarrow \infty$ in (35). We get

$$\sum_{i=1}^N |x_n(s_i) - x(s_i)|^2 \leq \epsilon^2 \quad \text{for } n \geq n_0.$$

Letting $N \rightarrow \infty$ in this, we obtain

$$\sum_{i=1}^{\infty} |x_n(s_i) - x(s_i)|^2 \leq \epsilon^2 \quad \text{for } n \geq n_0.$$

Hence,

$$\sum_{s \in S} |x_n(s) - x(s)|^2 \leq \epsilon^2 \quad \text{for } n \geq n_0.$$

This shows that $x_n - x \in l^2(S)$ and that

$$\|x_n - x\| \leq \epsilon \quad \text{for } n \geq n_0.$$

Therefore, $x = x_n - (x_n - x) \in l^2(S)$ and $x_n \rightarrow x$ in $l^2(S)$. Thus, $l^2(S)$ is complete. This completes the proof. ■

PROBLEM 5.1.15 Let $X = C[a, b]$. For $x, y \in X$ let

$$\begin{aligned} \langle x, y \rangle_0 &= \int_a^b \int_a^b x(s) \overline{y(t)} \, ds \, dt, \\ \langle x, y \rangle &= \int_a^b \int_a^b \frac{\sin(s-t)}{s-t} x(s) \overline{y(t)} \, ds \, dt. \end{aligned}$$

Show that $\langle \cdot, \cdot \rangle_0$ is not an inner product on X and that $\langle \cdot, \cdot \rangle$ is an inner on X .

Solution We can choose a nonzero continuous function x on $[a, b]$ such that

$$\int_a^b x(s) \, ds = 0$$

For this x we have $\langle x, x \rangle_0 = 0$. This shows that $\langle \cdot, \cdot \rangle_0$ cannot be an inner product.

By the linearity of the integral, we readily see that

$$\begin{aligned} \langle x + y, z \rangle &= \langle x, z \rangle + \langle y, z \rangle, \\ \langle kx, y \rangle &= k \langle x, y \rangle, \end{aligned}$$

for $x, y, z \in X, k \in \mathbb{K}$. Also, $\langle y, x \rangle = \overline{\langle x, y \rangle}$.

It remains to prove that $\langle x, x \rangle \geq 0$ and that $\langle x, x \rangle = 0$ only if $x = 0$. Since

$$\int_{-1}^1 e^{i\theta u} \, du = 2 \frac{\sin \theta}{\theta},$$

we have

$$\begin{aligned}\langle x, x \rangle &= \int_a^b \int_a^b x(s) \overline{x(t)} \left(\frac{1}{2} \int_{-1}^1 e^{i(s-t)u} du \right) ds dt \\ &= \frac{1}{2} \int_{-1}^1 \left(\int_a^b x(s) e^{isu} ds \int_a^b \overline{x(t)} e^{-itu} dt \right) du \\ &= \frac{1}{2} \int_{-1}^1 |z(u)|^2 du,\end{aligned}$$

where

$$z(u) = \int_a^b x(s) e^{isu} ds.$$

This shows that $\langle x, x \rangle \geq 0$ for all $x \in X$ and that $\langle x, x \rangle = 0$ iff $z(u) = 0$ a.e. in $[-1, 1]$. Now,

$$|e^{isu} - e^{isv}| = \left| e^{isv} \frac{e^{is(u-v)} - 1}{s(u-v)} s(u-v) \right| \leq M|u-v|,$$

since

$$\left| \frac{e^{i\theta} - 1}{\theta} \right| = \left| e^{i\frac{\theta}{2}} \frac{2i \sin \frac{\theta}{2}}{\theta} \right| \leq 1 \quad \text{for all } \theta \in \mathbb{R}.$$

Hence,

$$\begin{aligned}|z(u) - z(v)| &= \left| \int_a^b x(s) (e^{isu} - e^{isv}) ds \right| \\ &\leq M|u-v| \int_a^b |x(s)| ds.\end{aligned}$$

This shows that z is uniformly continuous on $[-1, 1]$. So, $z(u) = 0$ a.e. in $[-1, 1]$ implies that $z(u) = 0$ for $-1 < u < 1$. Thus its derivatives $z'(u), z''(u), \dots$ are zero in $(-1, 1)$. Since

$$z^{(n)}(u) = \int_a^b x(s) (is)^n e^{isu} ds, \quad n = 0, 1, 2, \dots,$$

we see that

$$\left(\frac{1}{i} \right)^n z^{(n)}(0) = \int_a^b s^n x(s) ds = 0 \quad \text{for } n = 0, 1, 2, \dots$$

Hence if $p(t)$ is any polynomial in t , then

$$\int_a^b p(t)x(t) dt = 0.$$

If $x \in X$, then x is continuous on $[a, b]$. So there is a sequence $\{p_n\}$ of polynomials such that

$$p_n(t) \longrightarrow \overline{x(t)} \quad \text{uniformly on } [a, b]$$

Therefore,

$$p_n(t)x(t) \longrightarrow |x(t)|^2 \quad \text{uniformly}$$

Thus,

$$\int_a^b |x(t)|^2 dt = \lim_{n \rightarrow \infty} \int_a^b p_n(t)x(t) dt = 0.$$

This implies that $x(t) = 0$ a.e. Since x is continuous, we get $x(t) = 0$ for all t . This completes the proof. ■

PROBLEM 5.1.16 Let Y be a subspace of an inner product space X and $x \in X$. Show that

- (a) an element $y \in Y$ is a best approximation to x from Y iff $x - y \perp Y$,
- (b) if a best approximation to x from Y exists then it is unique.

Solution (a) Suppose that $x - y \perp Y$. Let $z \in Y$. Then, $y - z \in Y$ and so $(x - y) \perp (y - z)$. Hence,

$$\begin{aligned} \|x - z\|^2 &= \|(x - y) + (y - z)\|^2 \\ &= \|x - y\|^2 + \|y - z\|^2 \geq \|x - y\|^2, \end{aligned} \quad (36)$$

using Pythagoras theorem. Thus, y is a best approximation to x from Y .

Conversely, suppose that y is a best approximation to x from Y . Let $z \in Y$. If $z \neq 0$, let

$$k = \frac{\langle x - y, z \rangle}{\|z\|^2}.$$

Since $y + kz \in Y$, we have

$$\begin{aligned} \|x - y\|^2 &\leq \|x - y - kz\|^2 = \langle x - y - kz, x - y - kz \rangle \\ &= \langle x - y, x - y \rangle - \bar{k}\langle x - y, z \rangle - k\langle z, x - y \rangle + k\bar{k}\langle z, z \rangle \\ &= \|x - y\|^2 - \bar{k}k\|z\|^2 - k\bar{k}\|z\|^2 + k\bar{k}\|z\|^2. \end{aligned}$$

Therefore,

$$\|x - y\|^2 \leq \|x - y\|^2 - |k|^2\|z\|^2,$$

which shows that $|k|^2\|z\|^2 \leq 0$. This implies that $k = 0$. Thus, $\langle x - y, z \rangle = 0$. This is true if $z = 0$. Hence, $x - y \perp z$ for all $z \in Y$, that is, $x - y \perp Y$.

(b) If y is a best approximation to x from Y , then $x - y \perp Y$; hence (36) shows that

$$\|x - z\|^2 > \|x - y\|^2 \quad \text{for all } z \neq y.$$

So there cannot be another best approximation to x . This completes the proof. ■

PROBLEM 5.1.17 Show that the element x in $L^2[-\pi, \pi]$, for which $\|x\|_2$ is a minimum, subject to

$$\int_{-\pi}^{\pi} x(t)t \, dt = 1, \quad \int_{-\pi}^{\pi} x(t) \sin t \, dt = 2,$$

is given by

$$x(t) = \frac{1}{\pi(6 - \pi^2)} \left[\frac{9t}{2} + (3 - 2\pi^2) \sin t \right]$$

Solution Let $x_1(t) = t$, $x_2(t) = \sin t$ and $x = ax_1 + bx_2$, where

$$a = \frac{9}{2\pi(6 - \pi^2)}, \quad b = \frac{3 - 2\pi^2}{\pi(6 - \pi^2)}.$$

Then,

$$\langle x_1, x_1 \rangle = \int_{-\pi}^{\pi} x_1(t) \overline{x_1(t)} \, dt = \int_{-\pi}^{\pi} t^2 \, dt = \frac{2}{3}\pi^3,$$

$$\langle x_2, x_1 \rangle = \langle x_1, x_2 \rangle = \int_{-\pi}^{\pi} t \sin t \, dt = 2\pi,$$

$$\langle x_2, x_2 \rangle = \int_{-\pi}^{\pi} \sin^2 t \, dt = \pi.$$

Hence, using the values of a, b , we get

$$\langle x, x_1 \rangle = a\langle x_1, x_1 \rangle + b\langle x_2, x_1 \rangle = 1,$$

$$\langle x, x_2 \rangle = a\langle x_1, x_2 \rangle + b\langle x_2, x_2 \rangle = 2.$$

So the two given conditions

$$\int_{-\pi}^{\pi} x(t)t \, dt = 1, \quad \int_{-\pi}^{\pi} x(t) \sin t \, dt = 2$$

are satisfied by x . Let y be any element in $L^2[-\pi, \pi]$ satisfying these conditions, that is,

$$\langle y, x_1 \rangle = 1, \quad \langle y, x_2 \rangle = 2. \quad (37)$$

Then,

$$\langle y - x, x_1 \rangle = 0 = \langle y - x, x_2 \rangle.$$

Hence,

$$\langle x, y - x \rangle = a\langle x_1, y - x \rangle + b\langle x_2, y - x \rangle = 0,$$

$$\|y\|^2 = \|x + (y - x)\|^2 = \|x\|^2 + \|y - x\|^2$$

by Pythagoras theorem. This shows that $\|y\| \geq \|x\|$, that is, $\|x\|_2 \leq \|y\|_2$ for all y in $L^2[-\pi, \pi]$ satisfying (37). This proves the result. ■

PROBLEM 5.1.18 Let X be an inner product space, x_1 be a nonzero element in X , and c be a scalar. Show that the element x in X which minimizes $\langle x, x \rangle$ subject to $\langle x, x_1 \rangle = c$ is $cx_1/(\langle x_1, x_1 \rangle)$.

Solution Let $y = c_1 x_1$, where

$$c_1 = c / (\langle x_1, x_1 \rangle).$$

Then,

$$\langle y, x_1 \rangle = \langle c_1 x_1, x_1 \rangle = c_1 \langle x_1, x_1 \rangle = c.$$

If x is any element in X with $\langle x, x_1 \rangle = c$, then

$$\langle x - y, x_1 \rangle = \langle x, x_1 \rangle - \langle y, x_1 \rangle = 0.$$

Hence,

$$\langle x - y, y \rangle = \langle x - y, c_1 x_1 \rangle = 0,$$

$$\|x\|^2 = \|(x - y) + y\|^2 = \|x - y\|^2 + \|y\|^2 \geq \|y\|^2,$$

using Pythagoras theorem. Thus, $\|y\|^2 \leq \|x\|^2$ for all x with $\langle x, x_1 \rangle = c$. This proves the result. ■

PROBLEM 5.1.19 Let H be a Hilbert space and E be a nonempty closed convex subset of H . Show that for each x in H there is a unique y in E such that

$$\|x - y\| \leq \|x - z\| \quad \text{for all } z \in E.$$

Deduce that if F is a closed subspace of H then every x in H has a unique best approximation from F .

Solution Let $d = \inf \{\|x - z\| : z \in E\}$. Then there is a sequence $\{z_n\}$ in E such that $\|x - z_n\| \rightarrow d$ as $n \rightarrow \infty$. Since E is convex we have $\frac{1}{2}(z_n + z_m) \in E$ for all n, m . Hence,

$$\|x - \frac{z_n + z_m}{2}\| \geq d,$$

$$\|(x - z_n) + (x - z_m)\|^2 \geq 4d^2.$$

Add $\|(x - z_n) - (x - z_m)\|^2$ to this and use the parallelogram law. We get

$$2\|x - z_n\|^2 + 2\|x - z_m\|^2 \geq 4d^2 + \|z_m - z_n\|^2,$$

for all n, m . As n and m tend to ∞ , the left-hand side of this inequality tends to zero. This shows that $\{z_n\}$ is a Cauchy sequence in H . But H is complete. So $z_n \rightarrow y$ for some y in H . But E is closed and $z_n \in E$. Hence, $y \in E$. Thus,

$$\inf \{\|x - z\| : z \in E\} = d = \lim_{n \rightarrow \infty} \|x - z_n\| = \|x - y\|.$$

This means that $\|x - y\| \leq \|x - z\|$ for all $z \in E$.

For uniqueness suppose that there is another element $y_1 \in E$ such that

$$\|x - y_1\| \leq \|x - z\| \quad \text{for all } z \in E.$$

It follows that $\|x - y_1\| = \|x - y\|$. Hence,

$$\begin{aligned} \|(x - y) + (x - y_1)\|^2 + \|(x - y) - (x - y_1)\|^2 \\ = 2\|x - y\|^2 + 2\|x - y_1\|^2 = 4\|x - y\|^2 \end{aligned}$$

by the parallelogram law. Therefore,

$$\left\|x - \frac{y + y_1}{2}\right\|^2 + \left\|\frac{y_1 - y}{2}\right\|^2 = \|x - y\|^2 \leq \left\|x - \frac{y + y_1}{2}\right\|^2$$

since $(y + y_1)/2 \in E$. This shows that $y_1 = y$. This proves the first part.

The second part follows from the first part since a subspace is necessarily convex. This completes the proof. ■

PROBLEM 5.1.20 Let $X = \mathbb{R}^n$ with the Euclidean norm, and

$$E = \{x \in \mathbb{R}^n : \sum_{j=1}^n \alpha_j x(j) = 1\},$$

where $\alpha_1, \dots, \alpha_n$ are real numbers of which at least one is nonzero. Show that there is a unique vector of minimum norm in E , and find it.

Solution If

$$f(x) = \sum_{j=1}^n \alpha_j x(j), \quad x \in X,$$

then $f : X \rightarrow \mathbb{R}$ is a bounded linear map and $E = \{x \in X : f(x) = 1\}$. This shows that E is a closed convex subset of $X = \mathbb{R}^n$. So, by Problem 5.1.19, corresponding to $x = 0$ there is a unique y in E such that

$$\|0 - y\| \leq \|0 - z\| \quad \text{for all } z \in E,$$

that is, y is a vector of minimum norm in E .

To find y let c be given by

$$c \sum_{j=1}^n \alpha_j^2 = 1,$$

and let $y(j) = c\alpha_j$, $1 \leq j \leq n$. Then,

$$\sum_{j=1}^n \alpha_j y(j) = \sum_{j=1}^n c\alpha_j^2 = 1.$$

So $y \in E$. For any $z \in E$, we have

$$\langle y, z \rangle = \sum_{j=1}^n y(j)z(j) = c \sum_{j=1}^n \alpha_j z(j) = c.$$

In particular, $\langle y, y \rangle = c$ and so $\langle y, y - z \rangle = \langle y, y \rangle - \langle y, z \rangle = 0$. Hence by Pythagoras theorem,

$$\|z\|^2 = \|z - y + y\|^2 = \|z - y\|^2 + \|y\|^2 \geq \|y\|^2.$$

Thus, $\|y\| \leq \|z\|$ for all $z \in E$, that is, $y = c(\alpha_1, \dots, \alpha_n)$ is the desired vector of minimum norm. ■

PROBLEM 5.1.21 Suppose that E is a nonempty closed convex subset of a Hilbert space H and $x \in H$. Show that an element y in E is a best approximation to x from E iff

$$\operatorname{Re} \langle x - y, z \rangle \leq \operatorname{Re} \langle x - y, y \rangle$$

for all $z \in E$.

Solution Consider any two elements y, z in E . Let

$$y_t = (1 - t)y + tz = y + t(z - y), \quad 0 \leq t \leq 1.$$

Then, we have

$$\begin{aligned} \|x - y_t\|^2 &= \langle x - y + t(y - z), x - y + t(y - z) \rangle \\ &= \langle x - y, x - y \rangle + t\langle x - y, y - z \rangle \\ &\quad + t\langle y - z, x - y \rangle + t^2\langle y - z, y - z \rangle. \end{aligned}$$

Thus,

$$\begin{aligned} \|x - y_t\|^2 &= t^2\|y - z\|^2 + 2t \operatorname{Re} \langle x - y, y - z \rangle + \|x - y\|^2 \\ &= at^2 + 2bt + c, \end{aligned} \tag{38}$$

where

$$\begin{aligned} a &= \|y - z\|^2, \quad c = \|x - y\|^2, \\ b &= \operatorname{Re} \langle x - y, y - z \rangle = \operatorname{Re} \langle x - y, y \rangle - \operatorname{Re} \langle x - y, z \rangle. \end{aligned}$$

First suppose that y is a best approximation to x from E . We have to prove that $b \geq 0$. Suppose that $b < 0$. Then we can choose a small t with $0 < t < 1$ and such that $at + 2b < 0$. Thus for some t with $0 < t < 1$, we have $at^2 + 2bt < 0$, and so

$$\|x - y_t\|^2 = at^2 + 2bt + c < c = \|x - y\|^2 \tag{39}$$

from (38). But, $y_t \in E$ for $0 < t < 1$ since E is convex and $y, z \in E$. So (39) contradicts the fact that y is best approximation to x . Therefore, $b \geq 0$.

Conversely, suppose that $y \in E$ is such that

$$\operatorname{Re} \langle x - y, y - z \rangle \geq 0 \quad \text{for all } z \in E.$$

Then (38) with $t = 1$ shows that

$$\|x - z\|^2 = \|y - z\|^2 + 2 \operatorname{Re} \langle x - y, y - z \rangle + \|x - y\|^2 \geq \|x - y\|^2$$

for all $z \in E$. Thus, y is a best approximation to x from E . This completes the proof. ■

PROBLEM 5.1.22 Let H be a Hilbert space and Y be a closed proper subspace of H . Show that there exists x_1 in H such that $\|x_1\| = 1$ and $d(x_1, Y) = 1$.

Solution Let $x \in H$ and $x \notin Y$. By Problem 5.1.19 there is a best approximation, say y , to x from Y . Then, $x \neq y$ since $x \notin Y$. Let

$$x_1 = \frac{x - y}{\|x - y\|}.$$

So, $\|x_1\| = 1$. If $z \in Y$,

$$\|x_1 - z\| = \left\| \frac{x - y}{\|x - y\|} - z \right\| = \frac{1}{\|x - y\|} \|x - y_1\|,$$

where $y_1 = y + \|x - y\|z$. Since y is a best approximation to x from Y and $y_1 \in Y$, we must have

$$\|x_1 - z\| = \frac{\|x - y_1\|}{\|x - y\|} \geq 1$$

Since this is true for all $z \in Y$, we get $d(x_1, Y) \geq 1$. But, $\|x_1 - 0\| = 1$, and $0 \in Y$. It follows that $d(x_1, Y) = 1$. ■

5.2 Orthonormal Sets

Let X be an inner product space. A nonempty subset S of X is said to be an *orthogonal* set if $\langle x, y \rangle = 0$ for any two distinct elements x, y in S . An orthogonal set in which every element is of norm 1 is said to be *orthonormal*. A maximal orthonormal set in X is called an *orthonormal basis* or a *total orthonormal set* or a *complete orthonormal set*. In other words, an orthonormal set is total if it is not properly contained in another orthonormal set.

A subset of an orthonormal set is orthonormal. An orthogonal set of nonzero vectors is linearly independent.

The unit vectors $e_1, e_2, \dots$ form an orthonormal set in l^2 . The sets

$$\left\{ \frac{1}{\sqrt{2\pi}} e^{int} : n = 0, \pm 1, \pm 2, \dots \right\}$$

$$\left\{ \frac{1}{\sqrt{2\pi}}, \frac{1}{\sqrt{\pi}} \cos nt, \frac{1}{\sqrt{\pi}} \sin nt : n = 1, 2, \dots \right\}$$

are orthonormal in $L^2[-\pi, \pi]$.

Given linearly independent vectors $x_1, x_2, \dots$, we can construct an orthonormal sequence $u_1, u_2, \dots$ such that

$$\text{span}\{u_1, \dots, u_n\} = \text{span}\{x_1, \dots, x_n\}, \quad n = 1, 2, \dots$$

The procedure of constructing $u_1, u_2, \dots$, known as *Gram-Schmidt orthonormalisation process*, is given below:

Let $y_1 = x_1$ and $u_1 = y_1/\|y_1\|$. Having found $u_1, \dots, u_{n-1}$, let

$$y_n = x_n - \sum_{i=1}^{n-1} \langle x_n, u_i \rangle u_i, \quad u_n = \frac{y_n}{\|y_n\|}.$$

The procedure is illustrated in Problems 5.2.3 and 5.2.4.

If $\{u_n\}$ is a countable orthonormal set then, for each $x \in X$,

$$\sum_{n \geq 1} |\langle x, u_n \rangle|^2 \leq \|x\|^2.$$

This is known as *Bessel's inequality*. If E is an orthonormal set, then for each x in X , the set

$$E_x = \{u \in E : \langle x, u \rangle \neq 0\}$$

is countable.

The Riesz-Fischer theorem states that if $\{u_n\}$ is an orthonormal sequence in a Hilbert space H and $\{k_n\}$ is sequence of scalars such that $\sum |k_n|^2 < \infty$, then $\sum k_n u_n$ converges in H .

If E is an orthonormal basis in Hilbert space H then, for each x in H , we have

- (i) *Fourier expansion*: $x = \sum_{u \in E} \langle x, u \rangle u$.
- (ii) *Parseval's formula*: $\|x\|^2 = \sum_{u \in E} |\langle x, u \rangle|^2$.
- (iii) $\langle x, u \rangle = 0$ for all $u \in E \implies x = 0$.

If E is only an orthonormal set but not a basis, then there is $x \neq 0$ such that $\langle x, u \rangle = 0$ for all $u \in E$. Thus (a), (b) and (c) are false in that case.

Every nonzero Hilbert space has an orthonormal basis. To avoid trivialities, we shall assume that a Hilbert space contains nonzero vectors. A Hilbert space is separable iff it has a countable orthonormal basis see Problems 5.2.11 and 5.2.13).

Let $\{u_n\}$ be an orthonormal sequence in a Hilbert space H and F be the closure of $\text{span}\{u_n\}$. Then the unique best approximation to any x in H from F is given by

$$y = \sum \langle x, u_n \rangle u_n.$$

EXERCISES

1. Consider the inner product space $X = \mathbb{R}^3$. Let $x_1 = (1, 1, 0)$, $x_2 = (1, 0, 0)$ and $x_3 = (1, 1, 1)$. Show that x_1, x_2, x_3 are linearly independent. Orthonormalise x_1, x_2, x_3 using Gram-Schmidt process.

2. Suppose that E is an orthonormal set in an inner product space X , and x is any element in X . For $n = 1, 2, \dots$, let

$$E_{x,n} = \{u \in E : |\langle x, u \rangle| > \frac{1}{n}\}.$$

Show that $E_{x,n}$ contains at the most a finite number of elements. [Hint: Use Bessel's inequality.]

3. Let E and x be as in the previous problem. Show that

$$E_x = \{u \in E : \langle x, u \rangle \neq 0\}$$

is a countable set. [Hint: $E_x = \bigcup_{n=1}^{\infty} E_{x,n}$, where $E_{x,n}$ is as above.]

PROBLEMS

PROBLEM 5.2.1 Let $\{x_1, \dots, x_n\}$ be an orthogonal set of nonzero vectors in an inner product space X . Show that if the only vector in X which is orthogonal to each x_i is the zero vector, then $x_1, \dots, x_n$ form an algebraic basis for X .

Solution Let $k_1x_1 + \dots + k_nx_n = 0$. Taking inner product with x_i , we get

$$k_i \langle x_i, x_i \rangle = 0$$

But $\langle x_i, x_i \rangle \neq 0$ since $x_i \neq 0$. Hence, $k_i = 0$. This shows that $x_1, \dots, x_n$ are linearly independent.

Let x be any element in X , and $y = c_1x_1 + \dots + c_nx_n$, where c_i are given by

$$c_i \langle x_i, x_i \rangle = \langle x, x_i \rangle, \quad 1 \leq i \leq n.$$

Since $\langle x_i, x_j \rangle = 0$ for $i \neq j$, we see that

$$\langle y, x_j \rangle = c_j \langle x_j, x_j \rangle = \langle x, x_j \rangle.$$

Therefore, $\langle x - y, x_j \rangle = 0$. This is true for $1 \leq j \leq n$. So we must have $x - y = 0$. Thus,

$$x = y \in \text{span}\{x_1, \dots, x_n\}.$$

This proves the result. ■

PROBLEM 5.2.2 Let X be the real space $C[-1, 1]$ of all real-valued continuous functions on $[-1, 1]$ with the inner product given by

$$\langle x, y \rangle = \int_{-1}^1 x(t)y(t) dt, \quad x, y \in X.$$

If Y is the set of all odd functions in X , find its orthogonal complement $Y^\perp$.

Solution A function x is odd if $x(-t) = -x(t)$ for all t ; it is even if $x(-t) = x(t)$ for all t . The orthogonal complement of Y is

$$Y^\perp = \{x \in X : \langle x, y \rangle = 0 \text{ for all } y \in Y\}.$$

Let x be any even function and $y \in Y$. Then, $x(t)y(t)$ is an odd function of t , and hence

$$\langle x, y \rangle = \int_{-1}^1 x(t)y(t) dt = 0$$

Thus $\langle x, y \rangle = 0$ for all $y \in Y$. Therefore, $x \in Y^\perp$.

On the other hand, let $x \in Y^\perp$, and $y(t) = x(t) - x(-t)$. Then, $y(-t) = x(-t) - x(t) = -y(t)$, and so $y \in Y$. Hence, $\langle x, y \rangle = 0$. Since

$$(y(t))^2 = (x(t) - x(-t))y(t) = x(t)y(t) + x(-t)y(-t),$$

we get

$$\begin{aligned} \int_{-1}^1 (y(t))^2 dt &= \int_{-1}^1 x(t)y(t) dt + \int_{-1}^1 x(-t)y(-t) dt \\ &= \langle x, y \rangle + \langle x, y \rangle = 0. \end{aligned}$$

Since y is continuous this implies that $y(t) = 0$ for all t , that is, $x(-t) = x(t)$. So x is an even function if $x \in Y^\perp$. Thus, $Y^\perp$ is the set of all even functions. ■

PROBLEM 5.2.3 Let $x_n(t) = t^n$ for $t \in [-1, 1]$, $n = 0, 1, 2, \dots$. Orthonormalise the first three terms of $\{x_n\}$, where the inner product is given by

$$\langle x, y \rangle = \int_{-1}^1 x(t)y(t) dt$$

Solution The first three terms u_0, u_1, u_2 of the desired orthonormal sequence are determined as follows: Since $x_0(t) = 1$ for all t ,

$$\|x_0\|^2 = \int_{-1}^1 1 dt = 2.$$

So we take

$$u_0 = \frac{x_0}{\|x_0\|} = \frac{1}{\sqrt{2}}.$$

Let

$$y_1 = x_1 - \langle x_1, u_0 \rangle u_0.$$

Then, $\langle y_1, u_0 \rangle = 0$. Since

$$\langle x_1, u_0 \rangle = \int_{-1}^1 t \frac{1}{\sqrt{2}} dt = 0,$$

we have $y_1 = x_1$, and so

$$\|y_1\|^2 = \int_{-1}^1 t^2 dt = \frac{2}{3}.$$

Take

$$u_1 = \frac{y_1}{\|y_1\|} = \sqrt{(3/2)} x_1.$$

Let

$$y_2 = x_2 - \langle x_2, u_1 \rangle u_1 - \langle x_2, u_0 \rangle u_0.$$

Then, $\langle y_2, u_1 \rangle = 0 = \langle y_2, u_0 \rangle$. Since

$$\begin{aligned} \langle x_2, u_1 \rangle &= \int_{-1}^1 t^2 \sqrt{\frac{3}{2}} t dt = 0, \\ \langle x_2, u_0 \rangle &= \int_{-1}^1 t^2 \sqrt{\frac{1}{2}} dt = \frac{\sqrt{2}}{3}, \end{aligned}$$

we get

$$\begin{aligned} y_2 &= x_2 - \frac{\sqrt{2}}{3} u_0 = x_2 - \frac{1}{3}, \\ \|y_2\|^2 &= \int_{-1}^1 (y_2(t))^2 dt = \int_{-1}^1 (t^2 - 1/3)^2 dt = \frac{8}{45}. \end{aligned}$$

Take

$$u_2 = \frac{y_2}{\|y_2\|} = \sqrt{\frac{45}{8}} \left(x_2 - \frac{1}{3} \right).$$

Thus,

$$u_0(t) = \sqrt{\frac{1}{2}}, \quad u_1(t) = \sqrt{\frac{3}{2}} t, \quad u_2(t) = \sqrt{\frac{5}{2}} \left(\frac{3t^2 - 1}{2} \right). \quad \blacksquare$$

PROBLEM 5.2.4 Let $x_1(t) = t^2$, $x_2(t) = t$ and $x_3(t) = 1$. Orthonormalise x_1, x_2, x_3 in this order, on the interval $[-1, 1]$ with respect to the inner product

$$\langle x, y \rangle = \int_{-1}^1 x(t)y(t) dt.$$

Solution We find orthonormal functions u_1, u_2, u_3 , as in Problem 5.2.3. Since

$$\|x_1\|^2 = \int_{-1}^1 (x_1(t))^2 dt = \int_{-1}^1 t^4 dt = \frac{2}{5},$$

we take

$$u_1 = \frac{x_1}{\|x_1\|} = \sqrt{(5/2)} x_1.$$

Let

$$y_2 = x_2 - \langle x_2, u_1 \rangle u_1.$$

Then, $\langle y_2, u_1 \rangle = 0$. Since

$$\langle x_2, u_1 \rangle = \int_{-1}^1 t \sqrt{5/2} t^2 dt = 0,$$

we have $y_2 = x_2$,

$$\|y_2\|^2 = \int_{-1}^1 (x_2(t))^2 dt = \frac{2}{3}.$$

So we take

$$u_2 = \frac{y_2}{\|y_2\|} = \sqrt{(3/2)} x_2.$$

Let

$$y_3 = x_3 - \langle x_3, u_2 \rangle u_2 - \langle x_3, u_1 \rangle u_1.$$

Then, $\langle y_3, u_2 \rangle = 0 = \langle y_3, u_1 \rangle$. Since

$$\langle x_3, u_2 \rangle = \int_{-1}^1 \sqrt{(3/2)} x_2(t) dt = 0,$$

$$\langle x_3, u_1 \rangle = \int_{-1}^1 \sqrt{(5/2)} x_1(t) dt = \sqrt{(5/2)} \frac{2}{3},$$

we get

$$y_3 = x_3 - \frac{2}{3} \sqrt{(5/2)} u_1 = x_3 - \frac{5}{3} x_1,$$

$$\|y_3\|^2 = \int_{-1}^1 (y_3(t))^2 dt = \int_{-1}^1 \left(1 - \frac{5}{3} t^2\right)^2 dt = \frac{8}{9}.$$

Take

$$u_3 = \frac{y_3}{\|y_3\|} = \sqrt{(9/8)} \left(x_3 - \frac{5}{3} x_1\right).$$

Thus,

$$u_1(t) = \sqrt{(5/2)} t^2, \quad u_2(t) = \sqrt{(3/2)} t, \quad u_3(t) = \sqrt{(1/2)} \frac{3 - 5t^2}{2}. \quad \blacksquare$$

PROBLEM 5.2.5 Let H be a Hilbert space and $\{x_n\}$ be an orthogonal sequence in H . Show that $\sum x_n$ converges in H iff $\sum \|x_n\|^2 < \infty$.

Solution Let

$$s_n = x_1 + \cdots + x_n, \quad t_n = \|x_1\|^2 + \cdots + \|x_n\|^2.$$

Then, for $n > m$,

$$\begin{aligned} \|s_n - s_m\|^2 &= \left\langle \sum_{i=m+1}^n x_i, \sum_{j=m+1}^n x_j \right\rangle \\ &= \sum_i \sum_j \langle x_i, x_j \rangle = \sum_{i=m+1}^n \langle x_i, x_i \rangle \end{aligned}$$

by orthogonality. Thus,

$$\|s_n - s_m\|^2 = \sum_{i=m+1}^n \|x_i\|^2 = t_n - t_m.$$

This shows that $\{s_n\}$ is a Cauchy sequence in H iff $\{t_n\}$ is a Cauchy sequence in $\mathbb{R}$. Since H and $\mathbb{R}$ are complete, we see that $\{s_n\}$ converges in H iff $\{t_n\}$ converges in $\mathbb{R}$. This proves the result. ■

PROBLEM 5.2.6 Let H be the Hilbert space l^2 or $L^2[-\pi, \pi]$. In either case, find an infinite orthonormal set in l^2 , which is (i) an orthonormal basis, (ii) not an orthonormal basis.

Solution Consider the case $H = l^2$. Let e_n be the n -th unit vector in l^2 . Then, $\langle e_i, e_j \rangle = e_i(j) = 0$ or 1 according as $i \neq j$ or $i = j$. So $\{e_n\}$ is an orthonormal set in l^2 . Since $\langle x, e_n \rangle = x(n)$, we see that $\langle x, e_n \rangle = 0$ for all n only if $x(n) = 0$ for all n , that is $x = 0$. This implies that $\{e_n\}$ is a total orthonormal set in l^2 .

Removing some vectors from $\{e_n\}$, we find an infinite orthonormal set which is not total.

Next, let $H = L^2[-\pi, \pi]$. For integers n, m , we have

$$\int_{-\pi}^{\pi} e^{int} e^{-imt} dt = \begin{cases} 0 & \text{if } n \neq m \\ 2\pi & \text{if } n = m. \end{cases}$$

So let

$$u_n(t) = (2\pi)^{-1/2} e^{int}, \quad n = 0, \pm 1, \pm 2, \dots$$

Then, $\{u_n : n = 0, \pm 1, \pm 2, \dots\}$ is orthonormal in $L^2[-\pi, \pi]$. To show that it is an orthonormal basis, we use Fejer's theorem which states that if x is a 2π -periodic continuous function and

$$\hat{x}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} x(t) e^{-int} dt = \frac{1}{\sqrt{2\pi}} \langle x, u_n \rangle,$$

$$s_N(t) = \sum_{-N}^N \hat{x}(n) e^{int}, \quad N = 0, 1, 2, \dots,$$

then $(1/N)(s_0 + s_1 + \dots + s_{N-1}) \rightarrow x$ uniformly.

Suppose that $u \in L^2[-\pi, \pi]$ and $\langle u, u_n \rangle = 0$ for all integers n , that is, $\hat{u}(n) = 0$ for all n . Let

$$x(t) = \int_{-\pi}^t u(s) ds, \quad -\pi \leq t \leq \pi.$$

Then x is absolutely continuous, $x' = u$ a.e. and

$$x(\pi) = \int_{-\pi}^{\pi} u(s) ds = 2\pi \hat{u}(0) = 0 = \hat{x}(-\pi).$$

If $n \neq 0$, then integration by parts gives

$$\begin{aligned} 2\pi \hat{x}(n) &= \int_{-\pi}^{\pi} x(t) e^{-int} dt \\ &= \frac{1}{-in} \left[x(\pi) e^{-in\pi} - x(-\pi) e^{in\pi} \right] + \frac{2\pi}{in} \hat{u}(n) = 0. \end{aligned}$$

Thus, $\hat{x}(n) = 0$ for all $n \neq 0$, and

$$s_N(t) = \hat{x}(0), \quad \frac{1}{N}(s_0 + \dots + s_{N-1}) = \hat{x}(0)$$

for all N . It follows from Fejer's theorem that $x(t) = \hat{x}(0)$ for all t . Hence $u = x' = 0$ a.e.. Thus, $\langle u, u_n \rangle = 0$ for all n only if u is the zero element of $L^2[-\pi, \pi]$. Therefore, $\{u_n\}$ is total.

Removing some vectors from $\{u_n\}$, we get an orthonormal set which is not total. ■

PROBLEM 5.2.7 Let $\{u_n\}$ be an orthonormal sequence in a Hilbert space H , $Y = \text{span}\{u_n\}$ and $x \in H$. Show that the following are equivalent:

- (a) $x \in \overline{Y}$.
- (b) $x = \sum_{n=1}^{\infty} \langle x, u_n \rangle u_n$.
- (c) $\|x\|^2 = \sum_{n=1}^{\infty} |\langle x, u_n \rangle|^2$.

Solution Let

$$y_n = \sum_{i=1}^n \langle x, u_i \rangle u_i, \quad n = 1, 2, \dots \quad (1)$$

Then $y_n \in Y$. By Bessel's inequality we have

$$\sum_{n=1}^{\infty} |\langle x, u_n \rangle|^2 \leq \|x\|^2 < \infty.$$

Hence the Riesz-Fischer theorem shows that

$$y = \sum_{n=1}^{\infty} \langle x, u_n \rangle u_n$$

exists in H . Thus, $y_n \rightarrow y$ in H and so $y \in \overline{Y}$. Further, by the orthonormality of $\{u_n\}$, we see that

$$\langle y_n, u_j \rangle = \langle x, u_j \rangle \quad \text{for all } n \geq j,$$

and hence, $\langle y, u_j \rangle = \langle x, u_j \rangle$, on letting $n \rightarrow \infty$. Hence, $\langle y - x, u_j \rangle = 0$ for $j = 1, 2, \dots$. If $z = k_1 u_1 + \dots + k_n u_n$ is any element in Y , then

$$\langle y - x, z \rangle = \sum_{j=1}^n \langle y - x, k_j u_j \rangle = 0.$$

Let z be any element in $\overline{Y}$. Then there is a sequence $\{z_n\}$ in Y such that $z_n \rightarrow z$. So, $\langle y - x, z_n \rangle = 0$ and

$$\langle y - x, z_n \rangle \rightarrow \langle y - x, z \rangle.$$

Therefore, $\langle y - x, z \rangle = 0$ for all $z \in \overline{Y}$.

First suppose that (a) holds. Then $y - x \in Y$, and hence

$$\langle y - x, y - x \rangle = 0, \quad x = y = \sum_{n=1}^{\infty} \langle x, u_n \rangle u_n.$$

So (b) holds.

Next suppose that (b) holds. Then, $x = y \in \overline{Y}$. So (a) holds, and from (1) we see that

$$\langle y_n, x \rangle = \sum_{i=1}^n \langle x, u_i \rangle \langle u_i, x \rangle = \sum_{i=1}^n |\langle x, u_i \rangle|^2. \quad (2)$$

Since $\langle y_n, x \rangle \rightarrow \langle y, x \rangle$, (2) shows that (c) holds.

Finally, suppose that (c) holds. By the orthonormality of $\{u_n\}$, we get

$$\langle y_n, y_n \rangle = \sum_{i=1}^n |\langle x, u_i \rangle|^2.$$

Hence using (2),

$$\begin{aligned} \|x - y_n\|^2 &= \langle x - y_n, x - y_n \rangle \\ &= \langle x, x \rangle - \langle x, y_n \rangle - \langle y_n, x \rangle + \langle y_n, y_n \rangle \\ &= \|x\|^2 - \sum_{i=1}^n |\langle x, u_i \rangle|^2. \end{aligned}$$

This tends to zero if (c) holds. Hence, $y_n \rightarrow x$. So $x = y \in \overline{Y}$. So (b) and (a) hold. This completes the proof. ■

PROBLEM 5.2.8 Let H be a Hilbert space, and $\{u_n\}$ and $\{v_n\}$ be orthonormal sequences in H . Put

$$Y_1 = \text{span}\{u_n\}, \quad Y_2 = \text{span}\{v_n\}.$$

Show that $\overline{Y}_1 = \overline{Y}_2$ iff

$$u_n = \sum_{m=1}^{\infty} k_{nm} v_m, \quad n = 1, 2, \dots, \quad (3)$$

$$v_n = \sum_{m=1}^{\infty} \bar{k}_{mn} u_m, \quad n = 1, 2, \dots, \quad (4)$$

where $k_{nm} = \langle u_n, v_m \rangle$.

Solution Suppose that $\overline{Y}_1 = \overline{Y}_2$. Then $v_n \in \overline{Y}_1$ and $u_n \in \overline{Y}_2$. So using Problem 5.2.7, we get

$$\begin{aligned} v_n &= \sum_{m=1}^{\infty} \langle v_n, u_m \rangle u_m = \sum_{m=1}^{\infty} \bar{k}_{mn} u_m, \\ u_n &= \sum_{m=1}^{\infty} \langle u_n, v_m \rangle v_m = \sum_{m=1}^{\infty} k_{nm} v_m, \end{aligned}$$

Thus, (3) and (4) hold if $\overline{Y}_1 = \overline{Y}_2$.

Conversely, suppose that (3) and (4) hold. Since

$$\sum_{m=1}^N k_{nm} v_m \in Y_2 \quad \text{for } N \geq 1,$$

(3) shows that $u_n \in \overline{Y}_2$. Similarly, $v_n \in \overline{Y}_1$ from (4). Let x be any element in $\overline{Y}_1$. Then,

$$x = \sum_{m=1}^{\infty} \langle x, u_m \rangle u_m$$

by Problem 5.2.7. Since each $u_m \in \overline{Y}_2$, the partial sums of this series belong to $\overline{Y}_2$. Hence $x \in \overline{Y}_2$. Thus, $\overline{Y}_1 \subset \overline{Y}_2$. Similarly, $\overline{Y}_2 \subset \overline{Y}_1$. This completes the proof. ■

PROBLEM 5.2.9 Suppose that $x_1, x_2, \dots$ are linearly independent vectors in a Hilbert space H . Let

$$\begin{aligned} Y_n &= \text{span}\{x_1, \dots, x_n\}, \quad n = 1, 2, \dots, \\ Y &= \overline{\text{span}}\{x_1, x_2, \dots\}, \end{aligned}$$

and $x \in H$. If y_n is the best approximation to x from Y_n , show that $\{y_n\}$ converges to the unique best approximation to x from Y .

Solution Let $\{u_n\}$ be the orthonormal sequence constructed out of $\{x_n\}$ so that

$$Y_n = \text{span}\{x_1, \dots, x_n\} = \text{span}\{u_1, \dots, u_n\}.$$

As in the solution, of Problem 5.2.7, we see that

$$z = \sum_{n=1}^{\infty} \langle x, u_n \rangle u_n$$

exists in H . Let

$$z_n = \sum_{i=1}^n \langle x, u_i \rangle u_i.$$

Then $\langle z_n, u_i \rangle = \langle x, u_i \rangle$ for $n \geq i$, $z_n \in Y_n \subset Y$ and $z_n \rightarrow z$. Hence, $z \in Y$ and

$$\langle x - z_n, u_i \rangle = 0 \quad \text{for } 1 \leq i \leq n. \quad (5)$$

Since $Y_n = \text{span}\{u_1, \dots, u_n\}$, (5) shows that $x - z_n \perp Y_n$. Thus we see by Problem 5.1.16 that z_n is the best approximation to x from Y_n , that is, $z_n = y_n$.

Letting $n \rightarrow \infty$, keeping i fixed in (5), we get

$$\langle x - z, u_i \rangle = 0 \quad \text{for } i = 1, 2, \dots$$

Since $x_n \in \text{span}\{u_1, \dots, u_n\}$, this shows that $\langle x - z, x_n \rangle = 0$ for all $n \geq 1$. Let $Z = \text{span}\{x_n\}$. Then $x - z \perp Z$. If y is any element in Y , there is a sequence $\{w_n\}$ in Z such that $w_n \rightarrow y$. So,

$$\langle x - z, y \rangle = \lim_{n \rightarrow \infty} \langle x - z, w_n \rangle = 0.$$

Thus, $x - z \perp Y$. Hence, by Problem 5.1.16, z is the best approximation to x from Y . Since $y_n = z_n \rightarrow z$, this completes the proof. ■

PROBLEM 5.2.10 Prove that a Hilbert space is finite dimensional iff every total orthonormal set in it is an algebraic basis.

Solution Let H be a finite dimensional Hilbert space and E be a total orthonormal set in H . Then E is linearly independent set since an orthogonal set of nonzero vectors is linearly independent [see the solution of Problem 5.2.1]. Hence E must be a finite set $\{u_1, \dots, u_n\}$. Since E is total, the only vector which is orthogonal to each u_i is the zero vector. So E is an algebraic basis by Problem 5.2.1.

On the other hand, suppose that H is infinite dimensional. We shall prove that if E is any orthonormal set, then E cannot be an algebraic basis, that is, $\text{span } E \neq H$. This is true if E is a finite set. So let E contain an infinite sequence $\{u_n\}$ of distinct elements. Since $\sum n^{-2}$ converges, the Riesz-Fischer theorem implies that

$$x = \sum_{n=1}^{\infty} \frac{1}{n} u_n$$

exists in H . Moreover, $\langle x, u_n \rangle = 1/n$ for all n . If $x \in \text{span } E$, then

$$x = k_1 v_1 + \cdots + k_m v_m,$$

where $v_i \in E$ and k_i are scalars. Then, $\langle x, u \rangle = 0$ for any $u \in E$ other than $v_1, \dots, v_m$. Thus the set

$$\{u \in E : \langle x, u \rangle \neq 0\}$$

is the finite set $\{v_1, \dots, v_m\}$. But this set contains each u_n . This contradiction shows that $x \notin \text{span } E$. Hence $\text{span } E \neq H$. ■

PROBLEM 5.2.11 Let H be a separable Hilbert space. Prove that:

(a) Every orthonormal set in H is countable.

(b) H possesses a Schauder basis.

Solution (a) Let E be an orthonormal set in H and F be a countable dense subset of H . For each x in E , there is y in F such that $\|x - y\| < \frac{1}{2}$. Thus we can define a mapping $x \mapsto y$ from E into F . If distinct elements x_1, x_2 in E are mapped to the same y in F , then

$$\|x_1 - x_2\| \leq \|x_1 - y\| + \|y - x_2\| < \frac{1}{2} + \frac{1}{2} = 1.$$

But, $\langle x_1 - x_2, x_1 - x_2 \rangle = 1 + 1$ by the orthonormality of E . Hence,

$$\|x_1 - x_2\| = \sqrt{2} > 1.$$

This contradiction shows that the mapping is one-one. Since F is countable, we see that E also is countable.

(b) Every Hilbert space H has an orthonormal basis E . If H is separable, then $E = \{u_n\}$ is countable by (a). Every x in H has the Fourier expansion

$$x = \sum_{n=1}^{\infty} k_n u_n, \quad (6)$$

where

$$k_n = \langle x, u_n \rangle. \quad (7)$$

If the series in (6) converges in H then the coefficients k_n are uniquely determined by (7). Therefore, $\{u_n\}$ is a Schauder basis for H . This completes the proof. ■

PROBLEM 5.2.12 Let E be an orthonormal basis for a Hilbert space H and S be any set having the same cardinality as E . Show that H is linearly isometric to the Hilbert space $l^2(S)$ defined in Problem 5.1.14.

Solution Let $\phi : S \longrightarrow E$ be a bijective mapping and $\psi : E \longrightarrow S$ be its inverse. For $x \in H$ define $y : S \longrightarrow \mathbb{K}$ by

$$y(s) = \langle x, \phi(s) \rangle. \quad (8)$$

Let

$$\begin{aligned} E_x &= \{u \in E : \langle x, u \rangle \neq 0\}, \\ S_x &= \{s \in S : \phi(s) \in E_x\} = \psi(E_x). \end{aligned}$$

Then E_x is countable and hence S_x also is countable. Moreover,

$$s \in S_x \Leftrightarrow \phi(s) \in E_x \Leftrightarrow \langle x, \phi(s) \rangle \neq 0 \Leftrightarrow y(s) \neq 0.$$

Thus, $\{s \in S : y(s) \neq 0\}$ is countable and

$$\sum_{s \in S} |y(s)|^2 = \sum_{s \in S} |\langle x, \phi(s) \rangle|^2 = \sum_{u \in E} |\langle x, u \rangle|^2 = \|x\|^2$$

by Parseval's formula. This shows that $y \in l^2(S)$ and $\|y\|^2 = \|x\|^2$.

For $x \in H$ write $T(x) = y$, where y is defined by (8). This gives a mapping $T : H \longrightarrow l^2(S)$ such that $\|T(x)\| = \|x\|$. If $x_1, x_2 \in H$ and k is a scalar, then

$$\langle kx_1 + x_2, \phi(s) \rangle = k\langle x_1, \phi(s) \rangle + \langle x_2, \phi(s) \rangle,$$

that is, $y(s) = ky_1(s) + y_2(s)$, where

$$y = T(kx_1 + x_2), \quad y_1 = T(x_1), \quad y_2 = T(x_2).$$

Thus we see that T is a linear isometry.

Finally, we show that T is surjective. For this let $y \in l^2(S)$. Then,

$$\{s \in S : y(s) \neq 0\}$$

is a countable set. Let it be $\{s_n\}$ and $u_n = \phi(s_n)$. Since

$$\sum_{n=1}^{\infty} |y(s_n)|^2 < \infty,$$

the Riesz-Fischer theorem implies that

$$x = \sum_{n=1}^{\infty} y(s_n)u_n$$

exists in H . For this x we have

$$\langle x, u_n \rangle = y(s_n), \quad \langle x, u \rangle = 0 \quad \text{for } u \neq u_n.$$

If $s \neq s_n$, then $y(s) = 0$ and $\phi(s) \neq \phi(s_n)$. It follows that

$$y(s) = \langle x, \phi(s) \rangle \quad \text{for all } s \in S.$$

Thus, $y = T(x)$. So T is onto. This completes the proof. ■

PROBLEM 5.2.13 Let H be a Hilbert space. The cardinal number of an orthonormal basis for H is called the *orthogonal dimension* of H .

- (a) Prove that two Hilbert spaces are linearly isometric iff they have the same orthogonal dimension.
- (b) If a Hilbert space has a countable orthonormal basis, show that it is separable.

Solution (a) Let H_1 and H_2 be Hilbert spaces. Let E_1, E_2 be total orthonormal sets in H_1, H_2 , respectively.

Suppose that H_1 and H_2 have the same orthogonal dimension. Then E_1 and E_2 have the same cardinality. So Problem 5.2.12 implies that H_1 and H_2 are linearly isometric to $l^2(S)$, where $S = E_1$. Thus, H_1 and H_2 are linearly isometric.

Conversely, suppose that there is a surjective linear map $T : H_1 \rightarrow H_2$ such that $\|T(x)\| = \|x\|$ for all $x \in H_1$. Then,

$$\langle Tx_1, Tx_2 \rangle = \langle x_1, x_2 \rangle \quad \text{for } x_1, x_2 \in H_1 \quad (9)$$

by Problem 5.1.11. Let $T(E_1) = E_3$. Then E_3 is an orthonormal set because of (9) since E_1 is orthonormal. Suppose that $\langle y, v \rangle = 0$ for all $v \in E_3$. If $x = T^{-1}(y)$, we get

$$\langle x, u \rangle = \langle Tx, Tu \rangle = \langle y, Tu \rangle = 0$$

for all $u \in E_1$. Hence, $x = 0$ since E_1 is total. Thus, $y = T(x) = 0$ if $\langle y, v \rangle = 0$ for all $v \in E_3$. So E_3 is a total orthonormal set in H_2 . Hence, E_3 and E_2 have the same cardinality. But E_3 and E_1 have the same cardinality since $T : E_1 \rightarrow E_3$ is a bijection. So E_1 and E_2 have the same cardinality. This proves (a).

(b) Suppose that a Hilbert space H has a countable orthonormal basis E . Then E has the same cardinality as S , where $S = \{1, \dots, n\}$ or $\{1, 2, \dots\}$. So by Problem 5.2.12, H is linearly isometric to $l^2(S)$ which is $\mathbb{K}^n$ or l^2 . Since $\mathbb{K}^n$ and l^2 are separable, we see that H is separable. This completes the proof. ■

PROBLEM 5.2.14 Let H be Hilbert space and $\{u_\alpha\}$ be an orthonormal set in H . Prove that the following are equivalent:

- (a) $\{u_\alpha\}$ is an orthonormal basis for H .
- (b) $H = \overline{\text{span}}\{u_\alpha\}$.
- (c) For all $x, y \in H$, we have

$$\langle x, y \rangle = \sum_{n=1}^{\infty} \langle x, u_n \rangle \langle u_n, y \rangle,$$

where

$$\{u_n : n = 1, 2, \dots\} = \{u_\alpha : \langle x, u_\alpha \rangle \neq 0 \text{ or } \langle y, u_\alpha \rangle \neq 0\}.$$

Solution Suppose that (a) holds. Let $x \in H$ and

$$\{u_\alpha : \langle x, u_\alpha \rangle \neq 0\} \subset \{u_n : n = 1, 2, \dots\}.$$

Then x has the Fourier expansion

$$x = \sum \langle x, u_\alpha \rangle u_\alpha = \sum_{n \geq 1} \langle x, u_n \rangle u_n.$$

Let

$$x_m = \sum_{n=1}^m \langle x, u_n \rangle u_n.$$

Then $x_m \in \text{span}\{u_n\} \subset \text{span}\{u_\alpha\}$, and $x_m \rightarrow x$. Hence, $x \in \overline{\text{span}}\{u_\alpha\}$. Since this is true for any x in H , we see that (b) holds.

Further, the Fourier expansions of two elements x and y in H can be written as

$$x = \sum_{n=1}^{\infty} \langle x, u_n \rangle u_n, \quad y = \sum_{n=1}^{\infty} \langle y, u_n \rangle u_n,$$

where $\{u_n\}$ is as stated in (c). Let x_m and y_m denote the m -th partial sums of these series. By the linearity of inner product and orthonormality of $\{u_n\}$, we get

$$\langle x_m, y_m \rangle = \sum_{n=1}^m \langle x, u_n \rangle \langle u_n, y \rangle.$$

By the continuity of the inner product, we obtain (c) on letting $m \rightarrow \infty$. Thus we have proved that (a) implies both (b) and (c).

Next suppose that (b) holds. Let $\langle x, u_\alpha \rangle = 0$ for all α . There is a sequence $\{y_n\}$ in $\text{span}\{u_\alpha\}$ such that $y_n \rightarrow x$. Since each y_n is a finite linear combination of u_α and $\langle x, u_\alpha \rangle = 0$ for all α we see that $\langle x, y_n \rangle = 0$ for each n . Hence, $\langle x, x \rangle = 0$, that is, $x = 0$. Thus, $\{u_\alpha\}$ is total; so (a) holds.

Finally, suppose that (c) holds and let $\langle x, u_\alpha \rangle = 0$ for all α . Taking $y = x$ in (c), we get

$$\langle x, x \rangle = \sum \langle x, u_n \rangle \langle u_n, x \rangle = 0;$$

hence $x = 0$. Thus $\{u_\alpha\}$ is total and so (a) holds. This proves the equivalence of (a), (b) and (c). ■

PROBLEM 5.2.15 Let Y be a closed subspace of a Hilbert space H and $\{u_\alpha\}$ be an orthonormal basis for Y . If $x \in H$, show that the unique best approximation to x from Y is given by

$$y = \sum \langle x, u_\alpha \rangle u_\alpha.$$

Solution The set $\{u_\alpha : \langle x, u_\alpha \rangle \neq 0\}$ is countable. Let it be $\{u_n : n = 1, 2, \dots\}$. By Bessel's inequality,

$$\sum_{n \geq 1} |\langle x, u_n \rangle|^2 \leq \|x\|^2 < \infty.$$

So the Riesz-Fischer theorem implies that

$$y = \sum \langle x, u_n \rangle u_n = \sum \langle x, u_\alpha \rangle u_\alpha \quad (10)$$

exists in H . Moreover,

$$\langle y, u_\alpha \rangle = \langle x, u_\alpha \rangle \quad \text{for all } \alpha.$$

Hence, $\langle x - y, u_\alpha \rangle = 0$ for all α . Let z be any element in Y . As $\{u_\alpha\}$ is an orthonormal basis for Y , using Problem 5.2.14 we see that $Y = \overline{\text{span}}\{u_\alpha\}$. So there is a sequence $\{z_n\}$ in $\text{span}\{u_\alpha\}$ such that $z_n \rightarrow z$. Since z_n is a finite linear combination of u_α and $\langle x - y, u_\alpha \rangle = 0$ for all α we see that $\langle x - y, z_n \rangle = 0$ for all n . Hence

$$\langle x - y, z \rangle = \lim_{n \rightarrow \infty} \langle x - y, z_n \rangle = 0.$$

Since this is true for any z in Y we see that $x - y \perp Y$. As (10) implies that

$$y \in \overline{\text{span}}\{u_n\} \subset Y$$

we see by Problem 5.1.16 that y is the best approximation to x from Y . ■

PROBLEM 5.2.16 Let H be a Hilbert space and $F_1, F_2, \dots$ be closed subspaces of H such that $F_n \perp F_m$ for $n \neq m$. Let

$$F = \overline{\text{span}}\{F_n : n = 1, 2, \dots\}$$

Show that for every $x \in F$ there exists unique $x_n \in F_n$, for $n = 1, 2, \dots$, such that $x = \sum_{n \geq 1} x_n$.

Solution Suppose that

$$x = \sum_{n=1}^{\infty} x_n, \quad x = \sum_{n=1}^{\infty} y_n$$

where $x_n, y_n \in F_n$. Let $z_n = x_n - y_n$. Then $z_n \in F_n$, and hence $\langle z_n, z_m \rangle = 0$ for $n \neq m$. Thus, the Pythagoras theorem gives

$$\|z_1 + \dots + z_n\|^2 = \|z_1\|^2 + \dots + \|z_n\|^2 \geq \|z_m\|^2 \quad (11)$$

if $n \geq m$. However,

$$z_1 + \dots + z_n = \sum_{j=1}^n (x_j - y_j) \longrightarrow x - x = 0$$

as $n \rightarrow \infty$. So letting $n \rightarrow \infty$ in (11), we get $z_m = 0$, that is, $x_m = y_m$. This is true for any m . This proves the uniqueness.

Since F_n is a closed subspace of a Hilbert space H , we see that each F_n is a Hilbert space. So F_n has an orthonormal basis

$$E_n = \{u_{n,\alpha} : \alpha \in A_n\}.$$

Let

$$E = \bigcup_{n=1}^{\infty} E_n.$$

Then E is an orthonormal set of vectors in $\text{span}\{F_n : n = 1, 2, \dots\}$. It is clear that $\text{span } E_n \subset \text{span } E$ and $\overline{\text{span}} E_n = F_n$. Hence, $F_n \subset \overline{\text{span}} E$. Thus,

$$F = \overline{\text{span}}\{F_n : n \geq 1\} \subset \overline{\text{span}}(E)$$

But $E \subset F$. So, E is an orthonormal basis for F . Hence, each x in F has a Fourier expansion

$$x = \sum_{u \in E} \langle x, u \rangle u = \sum_{n=1}^{\infty} \sum_{\alpha \in A_n} \langle x, u_{n,\alpha} \rangle u_{n,\alpha}.$$

Let

$$x_n = \sum_{\alpha \in A_n} \langle x, u_{n,\alpha} \rangle u_{n,\alpha}.$$

Then, $x_n \in F_n$ and $x = \sum x_n$. This completes the proof. ■

PROBLEM 5.2.17 Let $u_{0,0}(t) = 1$ for $0 \leq t \leq 1$,

$$u_{1,0}(t) = \begin{cases} 1 & \text{for } t \in [0, 1/2) \\ -1 & \text{for } t \in (1/2, 1] \\ 0 & \text{for } t = 1/2. \end{cases}$$

For $n = 1, 2, \dots, j = 1, 2, \dots, 2^n$, let

$$u_{n,j}(t) = \begin{cases} \sqrt{2^n} & \text{for } t \in \left[\frac{j-1}{2^n}, \frac{j-1/2}{2^n} \right) \\ -\sqrt{2^n} & \text{for } t \in \left(\frac{j-1/2}{2^n}, \frac{j}{2^n} \right] \\ 0 & \text{otherwise.} \end{cases}$$

Show that the functions $u_{n,j}$ defined above is an orthonormal basis for $L^2[0, 1]$.

Solution It can be directly verified that

$$\int_0^1 (u_{n,j}(t))^2 dt = 1$$

for all relevant n, j . Also,

$$\int_0^1 u_{0,0}(t)u_{1,0}(t) dt = 0.$$

If $x(t)$ is constant on $((j-1)/2^n, j/2^n)$, then

$$\int_0^1 x(t)u_{n,j}(t) dt = 0.$$

Hence,

$$\int_0^1 u_{0,0}(t)u_{n,j}(t) dt = 0 = \int_0^1 u_{1,0}(t)u_{n,j}(t) dt$$

for $n \geq 1, j \geq 1$. Let $n = m + r, r \geq 1$. Since $u_{m,i}$ is constant on

$$\left[\frac{(i-1)}{2^m}, \frac{(i-1/2)}{2^m}\right) = \left[\frac{2^r(i-1)}{2^n}, \frac{2^r(i-1/2)}{2^n}\right)$$

and on

$$\left(\frac{(i-1/2)}{2^m}, \frac{i}{2^m}\right] = \left(\frac{2^r(i-1/2)}{2^n}, \frac{2^r i}{2^n}\right],$$

we see that $u_{m,i}$ is constant on $((j-1)/2^n, j/2^n)$ for $1 \leq j \leq 2^n$. Hence,

$$\int_0^1 u_{m,i}(t)u_{n,j}(t) dt = 0.$$

This is true for $n = m, i \neq j$. Thus we have proved that $\{u_{n,j}\}$ is an orthonormal set.

To prove that $\{u_{n,j}\}$ is an orthonormal basis for $L^2[0, 1]$, let $x \in L^2[0, 1]$ and $\langle x, u_{n,j} \rangle = 0$ for all n, j . Let

$$y(s) = \int_0^s x(t) dt, \quad 0 \leq s \leq 1.$$

Then y is absolutely continuous and $y' = x$ a.e. Now,

$$y(1) = \int_0^1 x(t) dt = \langle x, u_{0,0} \rangle = 0,$$

$$\begin{aligned} 0 = \langle x, u_{1,0} \rangle &= \int_0^{1/2} x(t) dt - \int_{1/2}^1 x(t) dt = y(1/2) - [y(1) - y(1/2)] \\ &= 2y(1/2) - y(1). \end{aligned}$$

Thus, $y(1/2) = y(2/2) = 0$. Assume that

$$y\left(\frac{j}{2^n}\right) = 0 \quad (12)$$

for some positive integer n and all j with $1 \leq j \leq 2^n$. This is true when $n = 1$. Also,

$$y\left(\frac{2j}{2^{n+1}}\right) = 0 \quad \text{for } j = 1, 2, \dots, 2^n. \quad (13)$$

Again,

$$\begin{aligned} \langle x, u_{n,j} \rangle &= \int_{(j-1)/2^n}^{(j-\frac{1}{2})/2^n} x(t)\sqrt{2^n} dt - \int_{(j-\frac{1}{2})/2^n}^{j/2^n} x(t)\sqrt{2^n} dt \\ &= \sqrt{2^n} \left\{ y\left(\frac{j-1/2}{2^n}\right) - y\left(\frac{j-1}{2^n}\right) - y\left(\frac{j}{2^n}\right) + y\left(\frac{j-1/2}{2^n}\right) \right\} \\ &= \sqrt{2^n} 2y\left(\frac{j-1/2}{2^n}\right) \end{aligned}$$

by (12). Since $\langle x, u_{n,j} \rangle = 0$, we get

$$y\left(\frac{2j-1}{2^{n+1}}\right) = 0 \quad \text{for } j = 1, 2, \dots, 2^n. \quad (14)$$

Combining (13) and (14), we see that

$$y\left(\frac{i}{2^{n+1}}\right) = 0 \quad \text{for } i = 1, 2, \dots, 2^{n+1}.$$

Thus we have proved by induction that (12) holds for all positive integers n and all j with $1 \leq j \leq 2^n$. Since y is continuous on $[0, 1]$ and the numbers $j/2^n$ form a dense subset of $[0, 1]$, we see that $y(s) = 0$ for $0 \leq s \leq 1$. Hence, $x = y' = 0$ a.e. This shows that $\{u_{n,j}\}$ is total in $L^2[0, 1]$. This completes the proof. ■

PROBLEM 5.2.18 Let D be the open unit disk in $\mathbb{C}$ and

$$u_n(z) = \sqrt{(n/\pi)} z^{n-1} \quad \text{for } |z| < 1, n = 1, 2, \dots$$

Show that $\{u_n : n = 1, 2, \dots\}$ is an orthonormal basis for the Hilbert space $A^2(D)$ defined in Problem 5.1.12.

Solution The inner product in $A^2(D)$ is that of $L^2(D)$. Hence,

$$\langle u_n, u_m \rangle = \iint_D u_n(z) \overline{u_m(z)} dx dy$$

$$\begin{aligned}
 &= \iint_D \frac{\sqrt{nm}}{\pi} z^{n-1} (\bar{z})^{m-1} dx dy \\
 &= \frac{\sqrt{mn}}{\pi} \int_0^1 \int_{-\pi}^{\pi} \left(r e^{i\theta} \right)^{n-1} \left(r e^{-i\theta} \right)^{m-1} r dr d\theta \\
 &= \frac{\sqrt{mn}}{\pi} \int_0^1 r^{m+n-1} dr \int_{-\pi}^{\pi} e^{i(n-m)\theta} d\theta
 \end{aligned}$$

Since

$$\int_{-\pi}^{\pi} e^{i(n-m)\theta} d\theta = \begin{cases} 0 & \text{if } n \neq m \\ 2\pi & \text{if } n = m, \end{cases}$$

we get

$$\langle u_n, u_m \rangle = \begin{cases} 0 & \text{if } n \neq m \\ 1 & \text{if } n = m. \end{cases}$$

Thus, $\{u_n\}$ is an orthonormal set in $A^2(D)$.

For $f \in A^2(D)$, we have

$$\begin{aligned}
 \langle f, u_n \rangle &= \iint_D f(z) \overline{u_n(z)} dx dy \\
 &= \int_0^1 \int_{-\pi}^{\pi} f(re^{i\theta}) \left(\frac{n}{\pi} \right)^{1/2} (re^{-i\theta})^{n-1} r dr d\theta \\
 &= \left(\frac{n}{\pi} \right)^{1/2} \int_0^1 r^{2n-1} dr \int_{-\pi}^{\pi} f(re^{i\theta}) \frac{re^{i\theta}}{(re^{i\theta})^n} d\theta \\
 &= \left(\frac{n}{\pi} \right)^{1/2} \int_0^1 r^{2n-1} dr \int_{|z|=r} f(z) \frac{dz}{iz^n}.
 \end{aligned}$$

But f is analytic on D if $f \in A^2(D)$. It follows that

$$\begin{aligned}
 \langle f, u_n \rangle &= \left(\frac{n}{\pi} \right)^{1/2} \int_0^1 r^{2n-1} dr \frac{2\pi}{(n-1)!} f^{(n-1)}(0) \\
 &= \frac{f^{(n-1)}(0)}{(n-1)!} \left(\frac{\pi}{n} \right)^{1/2}.
 \end{aligned}$$

Thus, if $\langle f, u_n \rangle = 0$ for all $n \geq 1$, then $f^{(n-1)}(0) = 0$ for all $n \geq 1$, and hence

$$f(z) = \sum_{n=1}^{\infty} \frac{f^{(n-1)}(0)}{(n-1)!} z^{n-1} = 0, \quad z \in D.$$

This shows that $\{u_n\}$ is total and completes the proof. ■

PROBLEM 5.2.19 Let $x_n(t) = t^n$ for $n = 0, 1, 2, \dots$, and $\{u_n\}$ be the orthonormal sequence constructed out of $\{x_n\}$ using Gram-Schmidt process w.r.t. the inner product in $L^2[-1, 1]$. Show that

- (a) $\{u_n\}$ is an orthonormal basis for $L^2[-1, 1]$,
 (b) $L^2[-1, 1]$ is separable,
 (c) $u_n(t) = ((2n+1)/2)^{1/2} P_n(t)$, where $P_0(t) = 1$, and

$$P_n(t) = \frac{1}{2^n n!} \frac{d^n}{dt^n} \left((t^2 - 1)^n \right) \quad (15)$$

for $n \geq 1$. [P_n is called the *Legendre polynomial* of order n .]

Solution By construction, $\{u_n\}$ is an orthonormal sequence in $L^2[-1, 1]$ such that

$$Y_n = \text{span}\{x_0, x_1, \dots, x_n\} = \text{span}\{u_0, u_1, \dots, u_n\}.$$

Since $x_n(t) = t^n$, we see that Y_n consists of polynomials of degree not exceeding n . To prove that $\{u_n\}$ is an orthonormal basis for $L^2[-1, 1]$, let $x \in L^2[-1, 1]$ and $\langle x, u_n \rangle = 0$ for all $n \geq 0$. It follows that $\langle x, y \rangle = 0$ if $y \in Y_m$ for some m .

Let y be any continuous function on $[-1, 1]$. Then there is a sequence $\{y_n\}$ of polynomials converging uniformly to y on $[-1, 1]$. So, $y_n \rightarrow y$ in the norm of $L^2[-1, 1]$. Hence $\langle x, y_n \rangle \rightarrow \langle x, y \rangle$. But each polynomial y_n lies in some Y_m , and so $\langle x, y_n \rangle = 0$ for all n . Thus, $\langle x, y \rangle = 0$.

Since continuous functions are dense in $L^2[-1, 1]$, there is a sequence $\{z_n\}$ of continuous functions on $[-1, 1]$ such that $z_n \rightarrow x$ in the norm of $L^2[-1, 1]$. Hence, $\langle x, z_n \rangle \rightarrow \langle x, x \rangle$. But $\langle x, z_n \rangle = 0$ for all n . Thus, $\langle x, x \rangle = 0$ and $x = 0$. This proves (a); and (b) follows from Problem 5.2.13(b).

To prove (c) let $v_n(t) = ((2n+1)/2)^{1/2} P_n(t)$, where $P_n(t)$ is defined by (15), for $n = 0, 1, 2, \dots$. For a fixed $n \geq 1$, let

$$y(t) = y_0(t) = (t^2 - 1)^n$$

and $y^{(i)}$ denote the i -th derivative of y . Integrating by parts n times, we get

$$\begin{aligned} \int_{-1}^1 y^{(n)} y^{(n)} dt &= \left[y^{(n)} y^{(n-1)} - y^{(n+1)} y^{(n-2)} + \dots \right. \\ &\quad \left. + (-1)^{n-1} y^{(2n-1)} y^{(0)} \right]_{-1}^1 + (-1)^n \int_{-1}^1 y^{(2n)} y^{(0)} dt. \end{aligned}$$

It can be seen that $y^{(0)}, y^{(1)}, \dots, y^{(n-1)}$ contain $t^2 - 1$ as a factor; hence the integrated terms vanish at $t = 1$ and $t = -1$. Since

$$\begin{aligned} y^{(2n)} &= \frac{d^{2n}}{dt^{2n}} (t^2 - 1)^n = (2n)!, \\ y^{(n)} &= \frac{d^n}{dt^n} (t^2 - 1)^n = 2^n n! P_n(t), \end{aligned}$$

we get

$$\begin{aligned} \int_{-1}^1 (2^n n! P_n(t))^2 dt &= \int_{-1}^1 (2n)! (1-t^2)^n dt \\ &= (2n)! \int_0^1 (1-u)^n u^{-1/2} dt \\ &= (2n)! \frac{\Gamma(n+1)\Gamma(1/2)}{\Gamma(n+3/2)} = 2^{2n+1} \frac{(n!)^2}{2n+1}. \end{aligned}$$

Hence,

$$\int_{-1}^1 v_n^2 dt = \frac{2n+1}{2} \int_{-1}^1 P_n^2 dt = 1.$$

This is true for $n = 0$ also. Thus, $\|v_n\| = 1$ for $n = 0, 1, 2, \dots$

If $n > m \geq 1$, integration by parts m times gives, as before,

$$\begin{aligned} \int_{-1}^1 t^m y^{(n)} dt &= \left[t^m y^{(n-1)} - m t^{m-1} y^{(n-2)} + \dots + (-1)^m m! y^{(n-m-1)} \right]_{-1}^1 \\ &= 0. \end{aligned}$$

Hence,

$$\int_{-1}^1 t^m v_n dt = 0.$$

This is true when $m = 0$ also. It follows that, if z is any polynomial of degree m and $n > m \geq 0$,

$$\langle z, v_n \rangle = \int_{-1}^1 z(t) v_n(t) dt = 0.$$

Since $v_n \in Y_n = \text{span}\{u_0, \dots, u_n\}$, we see that

$$v_n = \sum_{j=1}^n \alpha_j u_j.$$

But, $\alpha_j = \langle v_n, u_j \rangle = 0$ for $j < n$. So $v_n = \alpha_n u_n$. As $\|v_n\| = \|u_n\| = 1$, we must have $|\alpha_n| = 1$.

Now, $\{u_n\}$ is constructed out of $\{x_n\}$ as follows: $u_n = y_n / \|y_n\|$, where

$$y_0 = x_0, \quad y_n = x_n - \sum_{i=1}^{n-1} \langle x_n, u_i \rangle u_i$$

Since $x_n(t) = t^n$, this shows that u_n is a polynomial with real coefficients and the leading coefficient in it is $1/\|y_n\|$ which is positive. The definition of v_n shows that v_n also is a polynomial with real coefficients with leading coefficient positive. It follows that α_n is real and positive. So, $\alpha_n = 1, v_n = u_n$. This completes the proof. ■

PROBLEM 5.2.20 Let $x_n(t) = t^n e^{t^2/2}$ for $n = 0, 1, 2, \dots$ and $\{u_n\}$ be the orthonormal sequence constructed out of $\{x_n\}$ using Gram-Schmidt process w.r.t. the inner product of the space $L^2(-\infty, \infty)$. Show that $\{u_n\}$ is an orthonormal basis for $L^2(-\infty, \infty)$. [$H_n(t) = (2^n n!)^{1/2} \pi^{1/4} e^{t^2/2} u_n(t)$ is a polynomial called *Hermite polynomial* of order n .]

Solution We proceed as in Problem 5.2.19. By construction, $\{u_n\}$ is an orthonormal sequence in $L^2(-\infty, \infty)$. To prove that it is an orthonormal basis, suppose that $x \in L^2(-\infty, \infty)$ and $\langle x, u_n \rangle = 0$ for $n \geq 0$. Let

$$Y_n = \text{span}\{u_0, \dots, u_n\} = \text{span}\{x_0, \dots, x_n\}.$$

Then, $\langle x, y \rangle = 0$ for any $y \in Y_n$. In particular, $\langle x, x_n \rangle = 0$ for all $n \geq 0$. For any real number u , we have

$$\begin{aligned} \int_{-\infty}^{\infty} \sum_{n=0}^{\infty} |x(t)| e^{-t^2/2} \frac{|-itu|^n}{n!} dt &= \int_{-\infty}^{\infty} |x(t)| e^{-t^2/2 + |tu|} dt \\ &\leq \left(\int_{-\infty}^{\infty} |x|^2 dt \right)^{1/2} \left(\int_{-\infty}^{\infty} e^{-t^2 + 2|tu|} dt \right)^{1/2}. \end{aligned}$$

Since this is finite for all real u , we can integrate the series

$$\sum_{n=0}^{\infty} x(t) e^{-t^2/2} \frac{(-itu)^n}{n!} = x(t) e^{-t^2/2} e^{-itu}$$

term by term. We obtain

$$\begin{aligned} \int_{-\infty}^{\infty} x(t) e^{-t^2/2} e^{-itu} dt &= \sum_{n=0}^{\infty} \frac{(-iu)^n}{n!} \int_{-\infty}^{\infty} x(t) e^{-t^2/2} t^n dt \\ &= \sum_{n=0}^{\infty} \frac{(-iu)^n}{n!} \langle x, x_n \rangle = 0. \end{aligned}$$

This is true for all real u . This means that the Fourier transform of the function $x(t) e^{-t^2/2}$ in $L^1(-\infty, \infty)$ is identically zero on $(-\infty, \infty)$. Hence $x(t) e^{-t^2/2} = 0$ a.e., that is, $x = 0$ a.e. This completes the proof. ■

PROBLEM 5.2.21 Let $x_n(t) = e^{-t^2/2} t^n$ for $n = 0, 1, 2, \dots$, and $\{u_n\}$ be the orthonormal sequence constructed out of $\{x_n\}$ using Gram-Schmidt process w.r.t. the inner product of the space $L^2(0, \infty)$. Show that $\{u_n\}$ is an orthonormal basis for $L^2(0, \infty)$. [$L_n(t) = e^{t^2/2} u_n(t)$ is a polynomial called *Laguerre polynomial* of order n .]

Solution We proceed as in Problem 5.2.20. Since $\{u_n\}$ is an orthonormal sequence in $L^2(0, \infty)$, it is enough if we prove that if $x \in L^2(0, \infty)$ and $\langle x, u_n \rangle = 0$ for all $n \geq 0$, then $x = 0$ a.e. Assume that $\langle x, u_n \rangle = 0$ for all n . Then $\langle x, x_n \rangle = 0$ for all $n \geq 0$, as in Problem 5.2.20. For real u, v and integers $m \geq 0$, we have

$$\begin{aligned} \int_0^\infty \sum_{n=0}^\infty \left| x(t) e^{-t/2} e^{-itu} t^m \frac{(-itv)^n}{n!} \right| dt &= \int_0^\infty |x(t)| e^{-t/2} t^m e^{|tv|} dt \\ &\leq \left(\int_0^\infty |x(t)|^2 dt \right)^{1/2} \left(\int_0^\infty t^{2m} e^{-t+2|v|t} dt \right)^{1/2}. \end{aligned}$$

This is finite if $2|v| < 1$. Consequently, if $2|v| < 1$, then we can integrate the series

$$\sum_{n=0}^\infty x(t) e^{-t/2} e^{-itu} t^m \frac{(-itv)^n}{n!} = x(t) e^{-t/2} e^{-it(u+v)} t^m$$

term by term. Thus, if

$$g_m(u) = \int_0^\infty x(t) e^{-t/2} e^{-itu} t^m dt,$$

then

$$\begin{aligned} g_m(u+v) &= \sum_{n=0}^\infty \int_0^\infty x(t) e^{-t/2} e^{-itu} t^m \frac{(-itv)^n}{n!} dt \\ &= \sum_{n=0}^\infty \frac{(-iv)^n}{n!} g_{m+n}(u) \end{aligned}$$

for $|v| < 1/2$. Hence, if $g_n(u) = 0$ for all $n \geq 0$, then $g_m(u+v) = 0$ for all $m \geq 0$ and $|v| < 1/2$. But,

$$g_n(0) = \int_0^\infty x(t) e^{-t/2} t^n dt = \langle x, x_n \rangle = 0$$

for all $n \geq 0$. From this we can deduce that $g_m(u) = 0$ for each real number u and $m \geq 0$. Since $g_0(u)$ is the Fourier transform of $x(t)e^{-t/2}$ over $(0, \infty)$, this implies that $x(t)e^{-t/2} = 0$ a.e. Hence, $x = 0$ a.e. This completes the proof. ■

5.3 Orthogonal Complements and Representation of Functionals

Let H be a Hilbert space. Let F be a closed subspace of H and $F^\perp$ be the set of all elements in H which are orthogonal to F . Then every x in H can be uniquely expressed as

$$x = y + z, \quad y \in F, z \in F^\perp.$$

In fact, y is the best approximation to x from F , and is the *projection* of x onto F .

Let $y \in H$ and define $f : H \rightarrow \mathbb{K}$ by

$$f(x) = \langle x, y \rangle, \quad x \in H. \quad (1)$$

Then f is a bounded linear functional on H with $\|f\| = \|y\|$ [see Problem 5.3.8 below]. On the other hand, for every bounded linear functional f on H there is a unique y in H such that (1) holds. This result is known as *Riesz representation theorem*, and the unique element y is called the *representer* of f .

As a consequence of the Riesz representation theorem, we have the following results for a Hilbert space H :

(i) $x_n \xrightarrow{w} x$ iff $\langle x_n, y \rangle \rightarrow \langle x, y \rangle$, as $n \rightarrow \infty$, for each $y \in H$.

(ii) The annihilator of a nonempty subset E of H is the set

$$\{x' \in H' : T(x') \in E^\perp\}$$

where $T(x')$ is the representer of x' .

EXERCISES

1. Let $H = \mathbb{R}^3$ and F be the set of all $x = (x(1), x(2), x(3))$ in H such that $x(3) = 0$. Find $F^\perp$. Verify that every $x \in H$ can be expressed as $x = y + z$, where $y \in F$ and $z \in F^\perp$.
2. Let $H = \mathbb{R}^2$ and F be the set of all $x = (x(1), x(2))$ in H such that $x(1) + 2x(2) = 0$. Find $F^\perp$. Verify that every $x \in H$ can be expressed as $x = y + z$, where $y \in F$ and $z \in F^\perp$.
3. Recall the definition of weak convergence given in Section 3.2. Let H be a Hilbert space. Use the Riesz representation theorem to prove that $x_n \rightarrow x$ iff $\langle x_n, y \rangle \rightarrow \langle x, y \rangle$ as $n \rightarrow \infty$, for each $y \in H$.
4. Recall the definition of annihilator of a subset of a normed space given in Problem 3.1.9. Let E be a nonempty subset of a Hilbert space H . Show that the annihilator of E is the set $\{x' \in H' : T(x') \in E^\perp\}$, where $T(x')$ is the representer of x' .
5. Let $H = \mathbb{R}^3$. Define $f \in H'$ by

$$f(x) = 2x(1) + x(2) - x(3), \quad x = (x(1), x(2), x(3)).$$

Find the representer of f .

PROBLEMS

PROBLEM 5.3.1 If M a nonempty subset of an inner product space X , show that its orthogonal complement $M^\perp$ is a closed subspace of X .

Solution If $x, y \in M^\perp$ and k is a scalar, then for all $z \in M$, we have

$$\langle kx + y, z \rangle = k\langle x, z \rangle + \langle y, z \rangle = 0,$$

since $\langle x, z \rangle = 0 = \langle y, z \rangle$. This shows that $kx + y \in M^\perp$. Thus $M^\perp$ is a subspace.

Let x be any element in the closure of the subspace $M^\perp$. Then there exist x_n in $M^\perp$ such that $x_n \rightarrow x$. So, $\langle x_n, z \rangle \rightarrow \langle x, z \rangle$ for any z . If $z \in M$, then $\langle x_n, z \rangle = 0$ for all n and hence $\langle x, z \rangle = 0$ for any $z \in M$. Thus, $x \in M^\perp$, and therefore, $M^\perp$ is closed. This finishes the proof. ■

PROBLEM 5.3.2 Let A and B be nonempty subsets of inner product space X , and $A \subset B$. Show that

- (a) $B^\perp \subset A^\perp$,
- (b) $A \subset A^{\perp\perp}$,
- (c) $A^{\perp\perp\perp} = A^\perp$.

Give examples of situations where we have equalities and strict inclusions in (a) and (b).

Solution Let $x \in B^\perp$ and $y \in A$. Then, $y \in B$ and so $\langle x, y \rangle = 0$. Thus, if $x \in B^\perp$, then $\langle x, y \rangle = 0$ for all $y \in A$, and hence $x \in A^\perp$. This proves (a).

Let $x \in A$ and $y \in A^\perp$. So, $\langle x, y \rangle = 0$. Thus, if $x \in A$ then $\langle x, y \rangle = 0$ for all $y \in A^\perp$, and hence $x \in A^{\perp\perp}$. This proves (b).

Using (a) we deduce from (b) that $A^{\perp\perp\perp} \subset A^\perp$. Using (b) with A replaced by $A^\perp$, we get the reverse inclusion. This gives (c).

For the second part, let B be a nonzero subspace of X and $A = \{0\}$. Since $\langle x, 0 \rangle = 0$ for any x , we see that $x \in A^\perp$ for any x . Thus $A^\perp = X$. If $x \in B$ and $x \in B^\perp$, then $\langle x, x \rangle = 0$ and so $x = 0$. Hence, if $x \neq 0$ and $x \in B$, then $x \notin B^\perp$. So, $B^\perp \subsetneq X = A^\perp$. Thus we have a situation where the inclusion in (a) is strict.

Next, as above, let B be a nonzero subspace of X and A be the set obtained by removing a nonzero element u from B . Then $A \subsetneq B$. If $x \in A^\perp$,

$$\langle x, u \rangle = -\langle x, -u \rangle = 0$$

since $-u \in A$. Thus we see that $\langle x, y \rangle = 0$ for all $y \in B$, that is, $x \in B^\perp$. Hence, $A^\perp \subset B^\perp$. So by (a), equality holds in (a). If $x \in B^\perp$, then $\langle x, u \rangle = 0$ since $u \in B$. Thus,

$$u \perp x \quad \text{for all } x \in B^\perp = A^\perp.$$

Hence, $u \in A^{\perp\perp}$ while $u \notin A$. So we have a situation where the inclusion in (b) is strict.

Finally, if $A = E^\perp$ for some $E \subset X$, then

$$A^{\perp\perp} = E^{\perp\perp\perp} = E^\perp = A$$

by (c). Thus equality holds in (b). ■

PROBLEM 5.3.3 Let Y and Z be arbitrary subspaces of an inner product space X . Prove that

$$Y^{\perp\perp} + Z^{\perp\perp} \subset (Y + Z)^{\perp\perp}.$$

Solution Since Y and Z are subspaces we see that if $y \in Y$, then

$$y = y + 0 \in Y + Z.$$

Thus, $Y \subset Y + Z$ and, similarly, $Z \subset Y + Z$. If $A \subset B$, then $B^\perp \subset A^\perp$, and hence $A^{\perp\perp} \subset B^{\perp\perp}$ by Problem 5.3.2. Therefore,

$$Y^{\perp\perp} \subset (Y + Z)^{\perp\perp}, \quad Z^{\perp\perp} \subset (Y + Z)^{\perp\perp}.$$

Since $M^\perp$ is a subspace for any nonempty M , we see that $(Y + Z)^{\perp\perp}$ is a subspace containing $Y^{\perp\perp}$ and $Z^{\perp\perp}$. Thus,

$$(Y + Z)^{\perp\perp} \supset Y^{\perp\perp} + Z^{\perp\perp}.$$

This completes the proof. ■

PROBLEM 5.3.4 If Y is a complete subspace of an inner product space X , prove that

(a) $X = Y \oplus Y^\perp$, that is, every x in X is uniquely expressible as

$$x = y + z \quad \text{with } y \in Y, z \in Y^\perp;$$

(b) $Y^{\perp\perp} = Y$.

Show also that the completeness of Y is essential in (a) and (b).

Solution (a) Let $x \in X$ and $d = \inf \{\|x - z\| : z \in Y\}$. Then there is sequence $\{z_n\}$ in Y such that $\|x - z_n\| \rightarrow d$ as $n \rightarrow \infty$. Since Y is convex we see, exactly as in the solution of Problem 5.1.19, that $\{z_n\}$ is a Cauchy sequence in Y . Since Y is complete, there is $y \in Y$ such that $z_n \rightarrow y$. Hence,

$$d = \lim_{n \rightarrow \infty} \|x - z_n\| = \|x - y\|.$$

This implies that y is a best approximation to x from Y . So $x - y \perp Y$ by Problem 5.1.16(a). Thus, if $z = x - y$, then

$$x = y + z, \quad y \in Y, z \in Y^\perp.$$

For uniqueness, suppose that $x = y_1 + z_1$ is another such expression. Then $y - y_1 = z_1 - z$. But $y - y_1 \in Y$ and $z_1 - z \in Y^\perp$, since Y and $Y^\perp$ are subspaces. Therefore,

$$0 = \langle y - y_1, z_1 - z \rangle = \langle y - y_1, y - y_1 \rangle.$$

So $y = y_1$ and $z = z_1$. This proves (a).

(b) By Problem 5.3.2, we have $Y \subset Y^{\perp\perp}$. For the reverse inclusion, let $x \in Y^{\perp\perp}$. Then

$$x = y + z, \quad y \in Y, z \in Y^\perp.$$

So $\langle y, z \rangle = 0$ and $\langle x, z \rangle = 0$. Hence,

$$\langle z, z \rangle = \langle x - y, z \rangle = 0.$$

Thus, $z = 0$ and $x = y \in Y$. This proves (b).

To prove that the assumption of completeness cannot be dropped, let $X = l^2$ and $Y = c_{00}$. Then Y is not complete since it is not closed. Let $x \in Y^\perp$. If e_j is the j -th unit vector, then $e_j \in Y$, and so

$$x(j) = \langle x, e_j \rangle = 0 \quad \text{for } j = 1, 2, \dots$$

Thus $x = 0$. This shows that $Y^\perp = \{0\}$. Hence,

$$Y^{\perp\perp} = \{0\}^\perp = X \neq Y,$$

and (b) is not true. If $x \in X$ and $x \notin Y$, then x cannot be expressed as $x = y + z$ with $y \in Y, z \in Y^\perp = \{0\}$. So (a) is not true. This completes the proof. ■

PROBLEM 5.3.5 Prove the following:

(a) If S is a nonempty subset of an inner product space X , and $Y = \overline{\text{span}} S$, then $Y^\perp = S^\perp$.

(b) If M is a nonempty subset of a Hilbert space H , then $M^{\perp\perp} = \overline{\text{span}} M$.

Solution (a) Let $\text{span } S = Z$. Then, $S \subset Z \subset Y$, and hence $Y^\perp \subset S^\perp$ by Problem 5.3.2(a). For the reverse inclusion, let $x \in S^\perp$. Then, $\langle x, u \rangle = 0$ for all $u \in S$. If $z \in Z$,

$$z = k_1 u_1 + \dots + k_n u_n, \quad u_i \in S,$$

and hence $\langle x, z \rangle = 0$. Let $y \in Y$. Then there is a sequence $\{z_n\}$ in Z such that $z_n \rightarrow y$. So $\langle x, z_n \rangle \rightarrow \langle x, y \rangle$. But $\langle x, z_n \rangle = 0$ for all n . Hence, $\langle x, y \rangle = 0$. This is true for any $y \in Y$. This means that $x \in Y^\perp$. Thus $S^\perp \subset Y^\perp$. This proves (a).

(b) Let $Y = \overline{\text{span}} M$. Then, $Y^\perp = M^\perp$ by (a). Since Y is a closed subspace of a Hilbert space we see that Y is complete. So $Y^{\perp\perp} = Y$ by Problem 5.3.4(b). Thus, $M^{\perp\perp} = Y^{\perp\perp} = Y$. This completes the proof. ■

PROBLEM 5.3.6 Suppose that F_1 and F_2 are closed subspaces of a Hilbert space H . Show that

(a) $(F_1 \cap F_2)^\perp$ equals the closure of $F_1^\perp + F_2^\perp$;

(b) if, further, $F_1 \perp F_2$, then $F_1 + F_2$ is a closed subspace.

Show also that $F_1 + F_2$ need not be closed even though F_1 and F_2 are closed subspaces.

Solution (a) Since $F_1 \cap F_2 \subset F_1$ and $F_1 \cap F_2 \subset F_2$, we have by Problem 5.3.2,

$$F_1^\perp \subset (F_1 \cap F_2)^\perp, \quad F_2^\perp \subset (F_1 \cap F_2)^\perp.$$

Thus, $(F_1 \cap F_2)^\perp$ is a closed subspace containing $F_1^\perp$ and $F_2^\perp$, and hence $F_1^\perp + F_2^\perp$. So if F is the closure of $F_1^\perp + F_2^\perp$, then

$$(F_1 \cap F_2)^\perp \supset F. \quad (2)$$

On the other hand,

$$\begin{aligned} F_1^\perp &\subset F, \quad F_2^\perp \subset F; \\ F_1^{\perp\perp} &\supset F^\perp, \quad F_2^{\perp\perp} \supset F^\perp. \end{aligned}$$

But F_1, F_2, F are closed subspaces of a Hilbert space H . So, by Problem 5.3.4(b),

$$F_1^{\perp\perp} = F_1, \quad F_2^{\perp\perp} = F_2, \quad F^{\perp\perp} = F.$$

Hence,

$$\begin{aligned} F_1 &\supset F^\perp, \quad F_2 \supset F^\perp, \quad F_1 \cap F_2 \supset F^\perp, \\ (F_1 \cap F_2)^\perp &\subset F^{\perp\perp} = F. \end{aligned}$$

This together with (2) proves (a).

(b) Let x be any element in the closure of $F_1 + F_2$. Then there is a sequence $\{x_n\}$ in $F_1 + F_2$ such that $x_n \rightarrow x$. Thus,

$$x_n = y_n + z_n, \quad y_n \in F_1, z_n \in F_2.$$

Since $F_1 \perp F_2$, we get

$$\begin{aligned} \|x_n - x_m\|^2 &= \|(y_n - y_m) + (z_n - z_m)\|^2 \\ &= \|y_n - y_m\|^2 + \|z_n - z_m\|^2 \end{aligned}$$

for all n, m . This shows that $\{y_n\}, \{z_n\}$ are Cauchy sequences in F_1, F_2 respectively. But F_1, F_2 are complete as they are closed in H . Hence there exist $y \in F_1, z \in F_2$ such that $y_n \rightarrow y, z_n \rightarrow z$. Hence, $x_n \rightarrow y + z$. So $x = y + z \in F_1 + F_2$. This shows that $F_1 + F_2$ is closed.

For the last part consider the Hilbert space $H = l^2$. Let

$$F_1 = \{x \in l^2 : x(2j) = 0, j \geq 1\},$$

$$F_2 = \{x \in l^2 : x(2j-1) = jx(2j), j \geq 1\}.$$

Then F_1 and F_2 are easily seen to be subspaces of l^2 . If $x_n \rightarrow x$ in l^2 , then $x_n(j) \rightarrow x(j)$ for each j . Using this we also see that F_1 and F_2 are closed. Let e_n be the n -th unit vector,

$$y_n = \sum_{j=1}^n \left(\frac{1}{2j-1} - \frac{1}{2} \right) e_{2j-1},$$

$$z_n = \sum_{j=1}^n \left(\frac{1}{2} e_{2j-1} + \frac{1}{2j} e_{2j} \right).$$

Then, $y_n \in F_1$,

$$\begin{aligned} z_n(2j-1) &= \frac{1}{2}, & z_n(2j) &= \frac{1}{2j} & \text{for } 1 \leq j \leq n, \\ z_n(2j-1) &= 0 = z_n(2j) & & & \text{for } j > n. \end{aligned}$$

Hence $z_n \in F_2$. Also,

$$y_n + z_n = \sum_{i=1}^{2n} \frac{1}{i} e_i$$

So, if $x(m) = \frac{1}{m}$ for all m , then $x \in l^2$ and $y_n + z_n \rightarrow x$. Hence, $x \in \overline{F_1 + F_2}$. Suppose that $x = y + z$, where $y \in F_1$ and $z \in F_2$. Then,

$$\begin{aligned} \frac{1}{2j} &= x(2j) = y(2j) + z(2j) = z(2j); \\ z(2j-1) &= jz(2j) = \frac{1}{2}. \end{aligned}$$

This is not possible since $z \in l^2$. Thus $F_1 + F_2$ is not closed. This completes the proof. ■

PROBLEM 5.3.7 Let Y be a closed subspace of a Hilbert space X . Show that X/Y is linearly isometric to $Y^\perp$.

Solution Let $Y^\perp = Z$. Define $T : Z \rightarrow X/Y$ by

$$T(z) = z + Y, \quad z \in Z.$$

The relation

$$k(z_1 + Y) + (z_2 + Y) = kz_1 + z_2 + Y$$

states that T is linear. If $z \in Z$ and $T(z)$ is the zero element of X/Y , then $z \in Y$, and hence $z = 0$ since $Y \cap Z = Y \cap Y^\perp = \{0\}$. This shows that T is one-one. Let $x + Y$ be any element of X/Y . Since

$$x = y + z, \quad y \in Y, z \in Y^\perp = Z,$$

we have

$$x + Y = y + z + Y = z + Y = T(z).$$

Thus T is onto.

Let $z \in Z$. For all $y \in Y$, we have $\langle y, z \rangle = 0$, and hence

$$\|y + z\|^2 = \|y\|^2 + \|z\|^2 \geq \|z\|^2.$$

Therefore,

$$\inf \{\|z + y\| : y \in Y\} = \|z\|,$$

that is, $\|z + Y\| = \|z\|$. This means that T is an isometry. Thus we get a linear isometry T between $Y^\perp$ and X/Y . This proves the result. ■

PROBLEM 5.3.8 If z is a fixed element of an inner product space X , show that

$$f(x) = \langle x, z \rangle, \quad x \in X$$

defines a bounded linear functional on X , of norm $\|z\|$. Show also that if the mapping $X \rightarrow X'$ given by $z \mapsto f$ is surjective then X must be a Hilbert space.

Solution The linearity of $f : X \rightarrow \mathbb{K}$ is given by

$$\langle kx + y, z \rangle = k\langle x, z \rangle + \langle y, z \rangle.$$

The Schwarz inequality gives

$$|f(x)| = |\langle x, z \rangle| \leq \|x\| \cdot \|z\|, \quad x \in X.$$

Hence, f is bounded and $\|f\| \leq \|z\|$. Since $f(z) = \langle z, z \rangle$, we get

$$\|z\|^2 = |f(z)| \leq \|f\| \cdot \|z\|.$$

This implies that $\|z\| \leq \|f\|$. Therefore, $\|f\| = \|z\|$. This proves the first part.

For the second part suppose, that the mapping $z \mapsto f$ is surjective. To prove that X is complete, let $\{z_n\}$ be a Cauchy sequence in X . Let $\|z_n - z_m\| < \epsilon$ for $n, m \geq N$. Then for $x \in X$,

$$|\langle x, z_n \rangle - \langle x, z_m \rangle| = |\langle x, z_n - z_m \rangle| \leq \|x\| \|z_n - z_m\| \leq \epsilon \|x\| \quad (3)$$

for $n, m \geq N$. This shows that, for each x in X , $\{\langle x, z_n \rangle\}$ is a Cauchy sequence of scalars. So it converges. Let

$$f(x) = \lim_{n \rightarrow \infty} \langle x, z_n \rangle, \quad x \in X.$$

Since

$$\langle kx + y, z_n \rangle = k\langle x, z_n \rangle + \langle y, z_n \rangle$$

we see that f is linear. Since a Cauchy sequence is bounded, we have $\|z_n\| \leq \alpha$ for some α and all n . So,

$$|\langle x, z_n \rangle| \leq \|x\| \cdot \|z_n\| \leq \alpha \|x\|.$$

Letting $n \rightarrow \infty$,

$$|f(x)| \leq \alpha \|x\|, \quad x \in X.$$

Thus, $f \in X'$. By our assumption there is z in X such that $f(x) = \langle x, z \rangle$ for all $x \in X$.

Keeping x and n fixed and letting $m \rightarrow \infty$ in (3), we get

$$|\langle x, z_n \rangle - f(x)| \leq \epsilon \|x\| \quad \text{for } n \geq N, x \in X.$$

Hence,

$$\begin{aligned} |\langle z_n, z_n \rangle - f(z_n)| &\leq \epsilon \|z_n\| \leq \alpha\epsilon, \quad n \geq N, \\ |\langle z, z_n \rangle - f(z)| &\leq \epsilon \|z\|, \quad n \geq N. \end{aligned}$$

Now

$$\begin{aligned} \langle z_n - z, z_n - z \rangle &= \langle z_n, z_n \rangle - \langle z_n, z \rangle - \langle z, z_n \rangle + \langle z, z \rangle \\ &= \langle z_n, z_n \rangle - f(z_n) - \langle z, z_n \rangle + f(z), \end{aligned}$$

and so

$$\begin{aligned} \|z_n - z\|^2 &\leq |\langle z_n, z_n \rangle - f(z_n)| + |\langle z, z_n \rangle - f(z)| \\ &\leq \alpha\epsilon + \epsilon \|z\| \end{aligned}$$

for all $n \geq N$. Since ϵ is arbitrary, this implies that $z_n \rightarrow z$ in the norm of X . Thus every Cauchy sequence in X converges; so X is Hilbert space. This completes the proof. ■

PROBLEM 5.3.9 Show that the Riesz representation theorem is not true in arbitrary inner product spaces.

Solution Let $H = l^2$ and $X = c_{00}$. Then X is a subspace of H . The inner product in l^2 makes X an inner product space. For $x \in X$,

$$f(x) = \sum_{n=1}^{\infty} \frac{1}{n} x(n)$$

exists in $\mathbb{K}$. This gives a linear functional f on X . Since

$$|f(x)| \leq \sum_{n=1}^{\infty} \frac{1}{n} |x(n)| \leq \left(\sum_{n=1}^{\infty} \frac{1}{n^2} \right)^{1/2} \|x\|_2,$$

using the Schwarz inequality for sums, we see that f is bounded. If the Riesz representation theorem is true in X then there is y in X such that $f(x) = \langle x, y \rangle$ for all x in X . If e_j is the j -th unit vector, then $e_j \in X$, and hence

$$y(j) = \langle y, e_j \rangle = \overline{f(e_j)} = \sum_{n=1}^{\infty} \frac{1}{n} e_j(n) = \frac{1}{j}.$$

Thus, $y(j) = 1/j$ for $j = 1, 2, \dots$. This y cannot belong to $c_{00} = X$. This contradiction proves the result. ■

PROBLEM 5.3.10 Let X be an inner product space. Prove that $x_n \rightarrow x$ iff $\langle x_n, y \rangle \rightarrow \langle x, y \rangle$ uniformly for y in X with $\|y\| = 1$.

Solution Suppose that $x_n \rightarrow x$. Then, for $y \in X$ with $\|y\| = 1$, we have

$$|\langle x_n, y \rangle - \langle x, y \rangle| = |\langle x_n - x, y \rangle| \leq \|x_n - x\| \|y\| = \|x_n - x\|.$$

This shows that $\langle x_n, y \rangle \rightarrow \langle x, y \rangle$ uniformly for y in X with $\|y\| = 1$.

Conversely, suppose that $\langle x_n, y \rangle \rightarrow \langle x, y \rangle$ uniformly for y in X with $\|y\| = 1$. Then given $\epsilon > 0$, we can find N such that

$$|\langle x_n, y \rangle - \langle x, y \rangle| < \epsilon \quad \text{for } n \geq N, \|y\| = 1. \quad (4)$$

For a fixed n , let

$$f_n(z) = \langle z, x_n - x \rangle, \quad z \in X.$$

Then f_n is a continuous linear functional on X with $\|f_n\| = \|x_n - x\|$, by Problem 5.3.8. If $n \geq N$ and $z \neq 0$, then

$$\begin{aligned} |f_n(z)| &= |\langle x_n - x, z \rangle| \\ &= \|z\| \left| \left\langle x_n - x, \frac{z}{\|z\|} \right\rangle \right| \\ &= \|z\| \left| \left\langle x_n, \frac{z}{\|z\|} \right\rangle - \left\langle x, \frac{z}{\|z\|} \right\rangle \right| < \|z\| \epsilon \end{aligned}$$

by (4). It follows that $\|f_n\| \leq \epsilon$ for all $n \geq N$. That is,

$$\|x_n - x\| \leq \epsilon \quad \text{for all } n \geq N.$$

Thus, $x_n \rightarrow x$. This completes the proof. ■

PROBLEM 5.3.11 Let H be a Hilbert space. Prove that H' also is a Hilbert space and that H is reflexive.

Solution The Riesz representation theorem states that for each $x' \in H'$, there is a unique y in H such that $\|y\| = \|x'\|$, and

$$x'(x) = \langle x, y \rangle \quad \text{for all } x \in H.$$

This element y is the representer $T(x')$ of x' . Thus, $\|T(x')\| = \|x'\|$ and

$$x'(x) = \langle x, T(x') \rangle \quad \text{for } x \in H, x' \in H'.$$

If $x', y' \in H', k \in \mathbb{K}$, then for all $x \in H$,

$$\begin{aligned} \langle x, T(kx' + y') \rangle &= (kx' + y')(x) = kx'(x) + y'(x) \\ &= k\langle x, T(x') \rangle + \langle x, T(y') \rangle \\ &= \langle x, \bar{k}T(x') + T(y') \rangle. \end{aligned}$$

This shows that

$$T(kx' + y') = \bar{k}T(x') + T(y'). \quad (5)$$

For $x', y' \in H'$, write

$$\langle x', y' \rangle = \langle T(y'), T(x') \rangle. \quad (6)$$

Then using (5), we get

$$\begin{aligned} \langle kx' + y', z' \rangle &= \langle T(z'), T(kx' + y') \rangle = \langle T(z'), \bar{k}T(x') + T(y') \rangle \\ &= k\langle T(z'), T(x') \rangle + \langle T(z'), T(y') \rangle \\ &= k\langle x', z' \rangle + \langle y', z' \rangle. \end{aligned}$$

Clearly,

$$\begin{aligned} \langle y', x' \rangle &= \langle T(x'), T(y') \rangle = \overline{\langle x', y' \rangle}, \\ \langle x', x' \rangle &= \langle T(x'), T(x') \rangle = \|T(x')\|^2 = \|x'\|^2. \end{aligned}$$

Thus we see that (6) defines an inner product on H' which induces the norm on H' . Since X' is a Banach space for any normed space X , we see that H' is complete. Therefore, H' is a Hilbert space.

Let $f \in H''$. Since H' is a Hilbert space, by the Riesz representation theorem there is a unique $y' \in H'$ such that $\|y'\| = \|f\|$ and

$$f(x') = \langle x', y' \rangle \quad \text{for all } x' \in H'.$$

Let $T(y') = y$. Then for all $x' \in H'$,

$$f(x') = \langle x', y' \rangle = \langle T(y'), T(x') \rangle = \langle y, T(x') \rangle = x'(y) = (Jy)(x'),$$

where $J : H \rightarrow H''$ is the canonical mapping. Thus, $f = J(y)$. This means that H is reflexive. This completes the proof. ■

PROBLEM 5.3.12 Let Y be a subspace of a Hilbert space H and g be a continuous linear functional on Y . Using the Riesz representation theorem, prove that there is a unique $f \in H'$ such that $f|_Y = g$ and $\|f\| = \|g\|$.

Solution Let $y \in \bar{Y}$. Then there is a sequence $\{y_n\}$ in Y such that $y_n \rightarrow y$. So, $\{y_n\}$ is a Cauchy sequence, and

$$|g(y_n) - g(y_m)| = |g(y_n - y_m)| \leq \|g\| \|y_n - y_m\|.$$

This shows that $\{g(y_n)\}$ is a Cauchy sequence of scalars and hence convergent. If $\{z_n\}$ is another such sequence in Y such that $z_n \rightarrow y$, then

$$|g(y_n) - g(z_n)| = |g(y_n - z_n)| \leq \|g\| \|y_n - z_n\|,$$

and hence $\{g(y_n)\}$ and $\{g(z_n)\}$ converge to the same limit. So we can unambiguously define $h : \bar{Y} \rightarrow \mathbb{K}$ by

$$h(y) = \lim_{n \rightarrow \infty} g(y_n),$$

where $\{y_n\}$ is a sequence in Y such that $y_n \rightarrow y$.

If $y \in Y$, then we can take $y_n = y$ and so $h(y) = g(y)$. Thus, $h|_Y = g$. If $y_n \rightarrow y$ and $z_n \rightarrow z$, where $y_n, z_n \in Y$, then $ky_n + z_n \rightarrow ky + z$, and

$$g(ky_n + z_n) = kg(y_n) + g(z_n).$$

It follows that

$$h(ky + z) = kh(y) + h(z).$$

Thus h is a linear functional on $\bar{Y}$. If $y_n \rightarrow y$, then

$$|g(y_n)| \leq \|g\| \|y_n\| \rightarrow \|g\| \|y\|;$$

hence, $|h(y)| \leq \|g\| \|y\|$ for $y \in \bar{Y}$. Thus h is bounded and $\|h\| \leq \|g\|$. On the other hand, for $y \in Y$,

$$|g(y)| = |h(y)| \leq \|h\| \|y\|,$$

which implies that $\|g\| \leq \|h\|$. Thus, $\|h\| = \|g\|$.

Since H is a Hilbert space we see that $\bar{Y}$ is a Hilbert space. So by the Riesz representation theorem, there is $z \in \bar{Y}$ such that $\|z\| = \|h\|$, and

$$h(y) = \langle y, z \rangle, \quad y \in \bar{Y}.$$

Let

$$f(x) = \langle x, z \rangle, \quad x \in H.$$

Then, $f \in H'$, $\|f\| = \|z\| = \|h\| = \|g\|$, and

$$f(y) = h(y) = g(y) \quad \text{for } y \in Y.$$

This proves the existence of f .

For uniqueness let f_1 be another such extension of g and f be as determined above. If z_1 is the representer of f_1 , then

$$\|z_1\| = \|f_1\| = \|g\| = \|f\| = \|z\|,$$

$$f(y) = g(y) = f_1(y), \quad y \in Y.$$

Thus we see that

$$\{x \in H : f(x) = f_1(x)\}$$

is a closed subspace containing Y , and hence containing $\bar{Y}$. Since $z \in \bar{Y}$, we see that

$$f_1(z) = f(z) = \langle z, z \rangle = \|z\|^2 = \|z_1\|^2.$$

Hence,

$$\begin{aligned} \langle z - z_1, z - z_1 \rangle &= \langle z, z \rangle - \langle z, z_1 \rangle - \langle z_1, z \rangle + \langle z_1, z_1 \rangle \\ &= \|z\|^2 - f_1(z) - \overline{f_1(z)} + \|z_1\|^2 = 0 \end{aligned}$$

Thus, $z = z_1$, and $f = f_1$. This completes the proof. ■

PROBLEM 5.3.13 Let F be a closed subspace of H and $a \in H, a \notin F$. Show that there is a unique $f \in H'$ such that $f(a) = d(a, F)$, $f(x) = 0$ for all $x \in F$ and $\|f\| = 1$.

Solution There exist b, c such that

$$a = b + c, \quad b \in F, c \in F^\perp.$$

Since $a - b = c \in F^\perp$, we see that b is the best approximation to a from F . Further, $c \neq 0$ since $a \notin F$. Let

$$y = \frac{c}{\|c\|} \quad \text{and} \quad f(x) = \langle x, y \rangle, \quad x \in H.$$

Then, $f \in H'$ and $\|f\| = \|y\| = 1$. If $x \in F$, then $x \in H$, and hence

$$f(x) = \langle x, \frac{c}{\|c\|} \rangle = 0.$$

Thus, $f(x) = 0$ for all $x \in F$. Further,

$$f(a) = \langle a, \frac{c}{\|c\|} \rangle = \frac{1}{\|c\|} \langle b + c, c \rangle = \frac{1}{\|c\|} \langle c, c \rangle = \|c\|.$$

Since $d(a, F) = \|a - b\| = \|c\|$, we see that $f(a) = d(a, F)$. This completes the proof. ■

PROBLEM 5.3.14 Let $H = L^2[-1, 1]$ and

$$x_n(t) = \frac{\sin(\pi nt)}{|\sin(\pi nt)|} \quad \text{for } |t| \neq 0, 1/n, 2/n, \dots, 1.$$

Show that $x_n \xrightarrow{w} 0$ in H even though $x_n = \pm 1$ for almost all t in $[-1, 1]$.

Solution Since $\sin(\pi nt) = 0$ only if $\pi nt = m\pi$ for some integer m , we see that $x_n(t) = \pm 1$ if t does not belong to the set

$$\{0, \pm \frac{1}{n}, \pm \frac{2}{n}, \dots\}.$$

Since this set has measure zero, we see that $x_n = \pm 1$ a.e.

Let y be a continuous function on $[-1, 1]$. Since y is uniformly continuous, for each $\epsilon > 0$, there is $\delta > 0$ such that

$$|y(s) - y(t)| < \epsilon \quad \text{for } |s - t| < \delta, \quad s, t \in [-1, 1].$$

Let n be so large that $2/n < \delta$. If

$$\frac{m}{n} < t < \frac{m+1}{n}, \quad \alpha = \pi nt - m\pi,$$

then

$$0 < \alpha < \pi, \quad \sin(\pi nt) = (-1)^m \sin \alpha.$$

Hence,

$$x_n(t) = (-1)^m \quad \text{for } \frac{m}{n} < t < \frac{m+1}{n}.$$

Using the mean value theorem for integrals, we see that

$$\int_{m/n}^{(m+1)/n} x_n(t)y(t) dt = (-1)^m \frac{1}{n} y(t_m),$$

where $m/n \leq t_m \leq (m+1)/n$. Thus,

$$\begin{aligned} \int_{m/n}^{(m+2)/n} x_n(t)y(t) dt &= (-1)^m \frac{1}{n} y(t_m) + (-1)^{m+1} \frac{1}{n} y(t_{m+1}) \\ &= \frac{(-1)^m}{n} [y(t_m) - y(t_{m+1})]. \end{aligned}$$

Since $|t_m - t_{m+1}| \leq 2/n < \delta$, we have

$$\left| \int_{m/n}^{(m+2)/n} x_n(t)y(t) dt \right| = \frac{1}{n} |y(t_m) - y(t_{m+1})| < \frac{1}{n} \epsilon.$$

Use this with $m = -n, -n+2, \dots, n-2$, and add the resulting n inequalities. Then we obtain

$$\left| \int_{-1}^1 x_n(t)y(t) dt \right| \leq \sum_{r=0}^{n-1} \left| \int_{(-n+2r)/n}^{(-n+2r+2)/n} x_n y dt \right| < \epsilon.$$

Thus we see that $|\langle y, x_n \rangle| < \epsilon$ for $n > 2/\delta$. This shows that $\langle x_n, y \rangle \rightarrow 0$ as $n \rightarrow \infty$ if y is continuous on $[-1, 1]$.

Let $f \in H'$. There is $z \in H$ such that

$$f(x) = \langle x, z \rangle, \quad x \in H.$$

Since continuous functions are dense in $H = L^2[-1, 1]$, given $\epsilon > 0$ we can find a continuous function y in H such that $\|y - z\| < \epsilon$. So,

$$|\langle x_n, z \rangle| = |\langle x_n, z - y \rangle + \langle x_n, y \rangle| \leq \|x_n\| \|z - y\| + |\langle x_n, y \rangle|.$$

But,

$$\langle x_n, x_n \rangle = \int_{-1}^1 1 dt = 2.$$

Hence,

$$|f(x_n)| = |\langle x_n, z \rangle| \leq \sqrt{2} \epsilon + |\langle x_n, y \rangle| < (\sqrt{2} + 1) \epsilon$$

for all large n . This shows that $f(x_n) \rightarrow 0$ as $n \rightarrow \infty$. Since this is true for any $f \in H'$, the proof is complete. ■

PROBLEM 5.3.15 Using the Riesz-Fischer theorem, instead of Baire's theorem, prove the following special case of the uniform boundedness principle: Suppose that H is a Hilbert space, $E \subset H$, and for every y in H there is a constant $\alpha(y)$, depending on y , such that

$$|\langle x, y \rangle| \leq \alpha(y) \quad \text{for all } x \in E.$$

Then $\sup \{\|x\| : x \in E\} < \infty$. Deduce that a weakly convergent sequence in H is bounded.

Solution Assume that E is unbounded. We construct an orthonormal sequence $\{u_n\}$ as follows: Choose $x_1 \in E$ with $\|x_1\| > 1$. Write

$$y_1 = x_1, \quad u_1 = \frac{y_1}{\|y_1\|}.$$

Having found an orthonormal set $\{u_1, \dots, u_{n-1}\}, n \geq 2$, in H we choose $x_n \in E$ such that

$$\|x_n\| > n^2 \left(\sum_{j=1}^{n-1} \alpha(u_j) \right)^2,$$

and let

$$y_n = x_n - \sum_{j=1}^{n-1} \langle x_n, u_j \rangle u_j.$$

Since

$$\alpha(u_1) \geq |\langle x_1, u_1 \rangle| = \left| \left\langle x_1, \frac{x_1}{\|x_1\|} \right\rangle \right| > 1,$$

we see that

$$\sum_{j=1}^{n-1} \alpha(u_j) > 1.$$

Hence, $\|x_n\| > n^2$. By orthonormality of $u_1, \dots, u_{n-1}$, we have

$$\begin{aligned} \|x_n - y_n\|^2 &= \left\| \sum_{j=1}^{n-1} \langle x_n, u_j \rangle u_j \right\|^2 = \sum_{j=1}^{n-1} |\langle x_n, u_j \rangle|^2 \\ &\leq \sum_{j=1}^{n-1} (\alpha(u_j))^2 \leq \left(\sum_{j=1}^{n-1} \alpha(u_j) \right)^2 \\ &< \frac{1}{n^2} \|x_n\| < \frac{1}{n^2} \|x_n\|^2. \end{aligned}$$

Therefore,

$$\begin{aligned} \|x_n\| &\leq \|x_n - y_n\| + \|y_n\| < \frac{1}{n} \|x_n\| + \|y_n\|; \\ \|y_n\| &> \left(1 - \frac{1}{n} \right) \|x_n\|. \end{aligned} \tag{7}$$

This shows, in particular, that $y_n \neq 0$. Let

$$u_n = \frac{y_n}{\|y_n\|}.$$

Since

$$\langle y_n, u_j \rangle = \langle x_n, u_j \rangle - \langle x_n, u_j \rangle = 0$$

for $1 \leq j \leq n-1$, it follows that $\{u_1, \dots, u_n\}$ is an orthonormal set in H . Since E is unbounded, this process can be continued indefinitely.

Let $\lambda_n = \|y_n\|^{1/2}$. Then, by (7),

$$\lambda_n^2 > \left(1 - \frac{1}{n}\right) \|x_n\| > \left(1 - \frac{1}{n}\right) n^2. \quad (8)$$

So $\sum \lambda_n^{-2}$ converges. Hence,

$$z = \sum_{n=1}^{\infty} \lambda_n^{-1} u_n$$

exists in H by the Riesz-Fischer theorem. For this z we have $\langle z, u_n \rangle = \lambda_n^{-1}$. Moreover,

$$\langle y_n, z \rangle = \langle x_n, z \rangle - \sum_{j=1}^{n-1} \langle x_n, u_j \rangle \langle u_j, z \rangle. \quad (9)$$

Since $\lambda_1^2 = \|y_1\| = \|x_1\| > 1$, we see from (8) that $\lambda_n > 1$ for all n , and hence from (9),

$$\begin{aligned} |\langle y_n, z \rangle| &\leq |\langle x_n, z \rangle| + \sum_{j=1}^{n-1} |\langle x_n, u_j \rangle| \lambda_j^{-1} \leq \alpha(z) + \sum_{j=1}^{n-1} \alpha(u_j) \\ &< \alpha(z) + \frac{1}{n} \|x_n\|^{1/2} < \alpha(z) + \frac{1}{\sqrt{n(n-1)}} \lambda_n \end{aligned}$$

using (8). However,

$$\langle z, y_n \rangle = \langle z, \lambda_n^2 u_n \rangle = \lambda_n^2 \langle z, u_n \rangle = \lambda_n.$$

Hence,

$$\begin{aligned} \lambda_n &= |\langle z, y_n \rangle| < \alpha(z) + \frac{1}{\sqrt{n(n-1)}} \lambda_n, \\ \alpha(z) &> \left(1 - \frac{1}{\sqrt{n(n-1)}}\right) \lambda_n > \left(1 - \frac{1}{n-1}\right) \left(1 - \frac{1}{n}\right)^{1/2} n \end{aligned}$$

by (8). This gives a contradiction for large n . It follows that E is bounded.

Finally, if $x_n \xrightarrow{w} x$ then $\langle x_n, y \rangle \rightarrow \langle x, y \rangle$ for each $y \in H$, and so $\{\langle x_n, y \rangle\}$ is bounded for each y . Applying the above result to the set

$$E = \{x_n : n = 1, 2, \dots\},$$

we see that $\{x_n\}$ is a bounded sequence if it converges weakly. This completes the proof. ■

PROBLEM 5.3.16 Let H be a Hilbert space and $\{x_n\}$ be a sequence in H such that $\{\langle x_n, y \rangle\}$ converges for every $y \in H$. Show that there is a unique $x \in H$ such that $x_n \xrightarrow{w} x$.

Solution For each $y \in H$ the sequence $\{\langle x_n, y \rangle\}$ converges and hence bounded. So for each y there is a constant $\alpha(y)$ such that $|\langle x_n, y \rangle| \leq \alpha(y)$ for all $n \geq 1$. Applying Problem 5.3.15 we see that the set

$$E = \{x_n : n = 1, 2, \dots\}$$

is bounded. Suppose that $\|x_n\| \leq \beta$ for all n . Let

$$f(y) = \lim_{n \rightarrow \infty} \langle y, x_n \rangle, \quad y \in H.$$

Then f is defined and linear on H . Since

$$|\langle y, x_n \rangle| \leq \|y\| \|x_n\| \leq \beta \|y\|,$$

we get, letting $n \rightarrow \infty$,

$$|f(y)| \leq \beta \|y\|, \quad y \in H.$$

Therefore, $f \in H'$. So by the Riesz representation theorem, there is $x \in H$ such that

$$f(y) = \langle y, x \rangle \quad \text{for all } y \in H.$$

Hence, for all $y \in H$,

$$\langle y, x_n \rangle \rightarrow f(y) = \langle y, x \rangle, \quad \langle x_n, y \rangle \rightarrow \langle x, y \rangle.$$

If g is any element in H' , there is $y \in H$ such that $g(x) = \langle x, y \rangle$ for all x . Hence,

$$g(x_n) = \langle x_n, y \rangle \rightarrow \langle x, y \rangle = g(x).$$

This shows that $x_n \xrightarrow{w} x$.

Suppose that there is another $x_0 \in H$ such that $\langle x_n, y \rangle \rightarrow \langle x_0, y \rangle$ for all y . Then $\langle x, y \rangle = \langle x_0, y \rangle$ for all $y \in H$. Thus, $x = x_0$. This completes the proof. ■

PROBLEM 5.3.17 Let H be a Hilbert space and $\{u_\alpha\}$ be an orthonormal basis for H . Show that a sequence $\{x_n\}$ in H converges weakly to x iff $\{x_n\}$ is bounded and

$$\langle x_n, u_\alpha \rangle \rightarrow \langle x, u_\alpha \rangle \quad \text{for all } \alpha.$$

Solution If $x_n \xrightarrow{w} x$ then

$$\langle x_n, u_\alpha \rangle \rightarrow \langle x, u_\alpha \rangle \quad \text{for each } \alpha; \tag{10}$$

and $\{x_n\}$ is bounded by Problem 5.3.15.

Conversely, suppose that $\{x_n\}$ is bounded and (10) holds for some $x \in H$. Let Y be the set of all $y \in H$ such that $\langle x_n, y \rangle \rightarrow \langle x, y \rangle$. Then $u_\alpha \in Y$ for each α . Since

$$\langle x_n, ky_1 + y_2 \rangle = \bar{k}\langle x_n, y_1 \rangle + \langle x_n, y_2 \rangle$$

$$\rightarrow \bar{k}\langle x, y_1 \rangle + \langle x, y_2 \rangle = \langle x, ky_1 + y_2 \rangle,$$

we see that if $y_1, y_2 \in Y$ then $ky_1 + y_2 \in Y$. Thus Y is a subspace.

Let $y \in \bar{Y}$. Since $\{x_n\}$ is bounded, there is $\lambda > 0$ such that

$$\|x_n - x\| < \lambda \quad \text{for all } n \geq 1.$$

Given $\epsilon > 0$, find $y_1 \in Y$ such that

$$\|y - y_1\| < \frac{\epsilon}{2\lambda}$$

and then find N such that

$$|\langle x_n, y_1 \rangle - \langle x, y_1 \rangle| < \frac{\epsilon}{2} \quad \text{for all } n \geq N.$$

Then, for $n \geq N$,

$$\begin{aligned} |\langle x_n, y \rangle - \langle x, y \rangle| &= |\langle x_n - x, y \rangle| \\ &\leq |\langle x_n - x, y - y_1 \rangle| + |\langle x_n - x, y_1 \rangle| \\ &\leq \|x_n - x\| \|y - y_1\| + |\langle x_n, y_1 \rangle - \langle x, y_1 \rangle| \\ &\leq \lambda \|y - y_1\| + \epsilon/2 < \epsilon. \end{aligned}$$

This shows that $\langle x_n, y \rangle \rightarrow \langle x, y \rangle$. So, $y \in Y$. Thus Y is a closed subspace containing each u_α . But $\overline{\text{span}}\{u_\alpha\} = H$ since $\{u_\alpha\}$ is an orthonormal basis for H . It follows that $Y = H$, and the proof is complete. ■

PROBLEM 5.3.18 Let H be a Hilbert space and $\{x_n\}$ be a bounded sequence in H such that, for each j , the sequence $\{\langle x_j, x_n \rangle\}$ converges. Show that $\{x_n\}$ converges weakly in H .

Solution Let Y be the set of all $x \in H$ such that the sequence $\{\langle x, x_n \rangle\}$ converges. Given that $x_j \in Y$ for all j . Since

$$\langle kx + y, x_n \rangle = k\langle x, x_n \rangle + \langle y, x_n \rangle,$$

we see that if $x, y \in Y$ then $kx + y \in Y$. Thus Y is a subspace.

Let $y \in \bar{Y}$. Since $\{x_n\}$ is bounded there is $\alpha > 0$ such that $\|x_n\| < \alpha$ for all n . Given $\epsilon > 0$ we can find $y_1 \in Y$ such that $\|y - y_1\| < \epsilon/(4\alpha)$. Since $\{\langle y_1, x_n \rangle\}$ converges, we can find N such that, for all $n, m \geq N$,

$$|\langle y_1, x_n \rangle - \langle y_1, x_m \rangle| < \epsilon/2.$$

Hence for all $n, m \geq N$,

$$\begin{aligned} |\langle y, x_n \rangle - \langle y, x_m \rangle| &= |\langle y, x_n - x_m \rangle| \\ &\leq |\langle y - y_1, x_n - x_m \rangle| + |\langle y_1, x_n - x_m \rangle| \end{aligned}$$

$$\begin{aligned} &\leq \|y - y_1\| \|x_n - x_m\| + \frac{\epsilon}{2} \\ &< \frac{\epsilon}{4\alpha}(2\alpha) + \frac{\epsilon}{2}. \end{aligned}$$

This implies that $\{\langle y, x_n \rangle\}$ is a Cauchy sequence of scalars and hence convergent. So $y \in Y$. Thus, Y is a closed subspace.

If x is any element in H , then $x = y + z$, where $y \in Y, z \in Y^\perp$. So,

$$\langle x, x_n \rangle = \langle y, x_n \rangle + \langle z, x_n \rangle = \langle y, x_n \rangle,$$

since $x_n \in Y$ and $z \in Y^\perp$. Hence, $\{\langle x, x_n \rangle\}$ converges for any $x \in H$. Therefore, $\{x_n\}$ converges weakly in H by Problem 5.3.16. ■

PROBLEM 5.3.19 Prove that any bounded sequence in a Hilbert space contains a weakly convergent subsequence.

Solution Let x_n be a bounded sequence in a Hilbert space H . Then there exists $\alpha > 0$ such that $\|x_n\| < \alpha$ for all n . Thus,

$$|\langle x_1, x_n \rangle| \leq \|x_1\| \|x_n\| < \alpha^2.$$

This means that $\{\langle x_1, x_n \rangle\}$ is a bounded sequence of scalars. It has a convergent subsequence $\{\langle x_1, x_{n_i} \rangle\}$. Let S_1 denote the subsequence $\{x_{n_i}\}$ of $\{x_n\}$, and write x_{1i} instead of x_{n_i} . Since S_1 is bounded we can apply the same argument with S_1 instead of $\{x_n\}$ and x_2 instead of x_1 . In general, we can find a subsequence $S_j = \{x_{j i} : i \geq 1\}$ out of S_{j-1} and x_j exactly as S_1 is found out of $\{x_n\}$ and x_1 , so that the sequence

$$\{\langle x_j, x_{j i} \rangle : i = 1, 2, \dots\}$$

converges.

Consider the diagonal sequence $S = \{x_{nn}\}$. Let $m < n$. Then S_n is a subsequence of S_m . Since x_{nn} is the n -th term of this subsequence S_n , we have $x_{nn} = x_{mr}$, where $r \geq n$. Since S_m is a subsequence of S_1 and $m < n \leq r$, we have

$$x_{mm} = x_{1i}, \quad x_{mr} = x_{1j},$$

where $i < j$. Since $x_{nn} = x_{mr}$ this shows that S is a subsequence of S_1 , and hence of $\{x_n\}$. In an exactly similar manner, we can show that $\{x_{nn} : n \geq m\}$ is a subsequence of S_m . Hence, $\{\langle x_m, x_{nn} \rangle : n \geq m\}$ converges. It follows that $\{\langle x_m, x_{nn} \rangle : n \geq 1\}$ converges. This is true for any m . Therefore, $\{\langle x_{jj}, x_{nn} \rangle\}$ converges for each $j \geq 1$. Applying Problem 5.3.18 to the subsequence $\{x_{nn}\}$, we get the desired result. ■

6

Operators on Hilbert Spaces

This chapter and the next are devoted to bounded linear maps, also called bounded operators, on Hilbert spaces. In Section 6.1 we introduce the adjoint of an operator. In Section 6.2 three important classes of operators, namely, self-adjoint, unitary and normal, are studied.

6.1 Bounded Operators and Adjoints

Let X and Y be linear spaces. A linear mapping $\phi : X \times Y \longrightarrow \mathbb{K}$ is called a *sesquilinear functional* or *sesquilinear form* if it is linear in the first variable and conjugate linear in the second variable, that is,

$$\begin{aligned}\phi(kx_1 + x_2, y) &= k\phi(x_1, y) + \phi(x_2, y), \\ \phi(x, ky_1 + y_2) &= \bar{k}\phi(x, y_1) + \phi(x, y_2).\end{aligned}$$

If X and Y are normed spaces, then ϕ is said to be *bounded* if

$$\|\phi\| = \sup \{ |\phi(x, y)| : \|x\| \leq 1, \|y\| \leq 1 \} < \infty.$$

Let $X = Y$. If $\phi(y, x) = \overline{\phi(x, y)}$, then ϕ is said to be *symmetric*; if $\phi(x, x) \geq 0$, then ϕ is said to be *positive*.

Let ϕ be a sesquilinear functional on $X \times Y$, and

$$\begin{aligned}f_1(x) &= \phi(x, y_0), & x \in X, \\ f_2(y) &= \phi(x_0, y), & y \in Y,\end{aligned}$$

where $x_0 \in X$ and $y_0 \in Y$ are fixed. Then, f_1 and f_2 are linear functionals on X and Y , respectively.

Let X be an inner product space, and $\phi(x, y) = \langle x, y \rangle$ for $x, y \in X$. Then ϕ is a bounded sesquilinear functional on X , that is, on $X \times X$. It is symmetric and positive.

Let H_1 and H_2 be Hilbert spaces, and $T : H_1 \rightarrow H_2$ be a bounded linear map. Then there is a unique bounded linear map $T^* : H_2 \rightarrow H_1$ such that

$$\langle Tx, y \rangle = \langle x, T^*y \rangle \quad \text{for all } x \in H_1, y \in H_2,$$

(see Problem 6.1.8). The operator T^* is called the *adjoint* of T .

Let $S : H_1 \rightarrow H_2$ and $T : H_1 \rightarrow H_2$ be bounded linear maps. Then,

$$(S + T)^* = S^* + T^*, \quad (kT)^* = \bar{k}T^*, \quad (T^*)^* = T,$$

$$\|T^*\| = \|T\|, \quad \|T^*T\| = \|TT^*\| = \|T\|^2,$$

$$T^*T = 0 \iff T = 0.$$

Let $H_1 = H_2$. Then, $(ST)^* = T^*S^*$, $I^* = I$ and $0^* = 0$, where I is the identity on H_1 . If T^{-1} exists, then $(T^*)^{-1}$ exists and is equal to $(T^{-1})^*$.

EXERCISES

1. Which of the following functionals are sesquilinear?

$$\phi_1(s, t) = st + t, \quad s, t \in \mathbb{R},$$

$$\phi_2(s, t) = 2st, \quad s, t \in \mathbb{R}.$$

2. For $x = (s, t) \in \mathbb{R}^2$, $u \in \mathbb{R}$, let

$$\phi(x, u) = (2s + t)u.$$

Show that ϕ is a bounded sesquilinear functional on $\mathbb{R}^2 \times \mathbb{R}$.

3. For $x = (s, t) \in \mathbb{R}^2$, $y = (u, v) \in \mathbb{R}^2$, let

$$\phi(x, y) = (s + 2t)v + (2s - t)u.$$

Show that ϕ is a bounded sesquilinear functional on $\mathbb{R}^2$.

4. Let H_1, H_2 be Hilbert spaces and $T \in BL(H_1, H_2)$. Show that

$$(kT)^* = \bar{k}T^*, \quad (T^*)^* = T, \quad \|T^*\| = \|T\|,$$

$$\|T^*T\| = \|TT^*\| = \|T\|^2,$$

and also that $T^*T = 0$ iff $T = 0$.

5. Let H_1, H_2 be Hilbert spaces and $S, T \in BL(H_1, H_2)$. Show that

$$(S + T)^* = S^* + T^*$$

and that if $H_2 = H_1$, then

$$(ST)^* = T^*S^*.$$

6. Let $H = \mathbb{R}^2$ and A be defined by

$$A(e_1) = e_1, \quad A(e_2) = 2e_2,$$

where $e_1 = (1, 0)$, $e_2 = (0, 1)$. Let $x_1 = e_1$ and $x_2 = e_1 + e_2$. Find the matrix M representing A w.r.t. $\{x_1, x_2\}$. Show that $A^* = A$ even though M is not symmetric. [Compare with Problem 6.1.10.]

PROBLEMS

PROBLEM 6.1.1 Consider the infinite matrix

$$\begin{bmatrix} a_1 & b_1 & 0 & 0 & 0 & \cdots \\ c_2 & a_2 & b_2 & 0 & 0 & \cdots \\ 0 & c_3 & a_3 & b_3 & 0 & \cdots \\ 0 & 0 & c_4 & a_4 & b_4 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$$

If

$$\beta = \sup \{|c_n| + |a_n| + |b_n| : n = 1, 2, \dots\} < \infty,$$

$$\gamma = \sup \{|b_{n-1}| + |a_n| + |c_{n+1}| : n = 1, 2, \dots\} < \infty,$$

where $b_0 = 0 = c_1$, show that the above matrix represents a bounded operator A on l^2 with respect to the standard orthonormal basis, and that $\|A\| \leq (\beta\gamma)^{1/2}$. [Such matrices are called *Jacobi matrices*.]

Solution We say that an infinite matrix (k_{ij}) represents A w.r.t. the standard orthonormal basis $\{e_n\}$ if

$$A(x)(i) = \sum_{j=1}^{\infty} k_{ij}x(j), \quad i = 1, 2, \dots, x \in l^2.$$

So,

$$|A(x)(i)| \leq \sum_{j=1}^{\infty} |k_{ij}|^{1/2} |k_{ij}|^{1/2} |x(j)| \leq \left(\sum_{j=1}^{\infty} |k_{ij}| \right)^{1/2} \left(\sum_{j=1}^{\infty} |k_{ij}| |x(j)|^2 \right)^{1/2}$$

by the Schwarz inequality for sums. Here, for $i = 1, 2, \dots$,

$$\begin{aligned} k_{ii} &= a_i, & k_{i,i+1} &= b_i, & k_{i,i-1} &= c_i, \\ k_{ij} &= 0 & \text{for } j < i-1, j > i+1. \end{aligned}$$

Hence,

$$\sum_{j=1}^{\infty} |k_{ij}| = |c_i| + |a_i| + |b_i| \leq \beta \quad \text{for } i \geq 1,$$

$$\sum_{i=1}^{\infty} |k_{ij}| = |b_{j-1}| + |a_j| + |c_{j+1}| \leq \gamma \quad \text{for } j \geq 1;$$

$$|A(x)(i)|^2 \leq \beta \sum_{j=1}^{\infty} |k_{ij}| |x(j)|^2.$$

Therefore,

$$\begin{aligned} \|Ax\|^2 &= \sum_{i=1}^{\infty} |A(x)(i)|^2 \leq \beta \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} |k_{ij}| |x(j)|^2 \\ &= \beta \sum_{j=1}^{\infty} |x(j)|^2 \sum_{i=1}^{\infty} |k_{ij}| \leq \beta \gamma \|x\|^2. \end{aligned}$$

Thus, $\|Ax\| \leq (\beta\gamma)^{1/2} \|x\|$ for all $x \in l^2$. This implies that A is bounded, and that $\|A\| \leq (\beta\gamma)^{1/2}$. ■

PROBLEM 6.1.2 Let $\phi : X \times X \longrightarrow \mathbb{K}$ be a sesquilinear functional on a linear space X over $\mathbb{K}$, and

$$q(x) = \phi(x, x), \quad x \in X$$

be the *associated quadratic functional*. Prove the following:

- (a) $2\phi(x, y) + 2\phi(y, x) = q(x + y) - q(x - y)$.
- (b) If $\mathbb{K} = \mathbb{C}$, then

$$4\phi(x, y) = q(x + y) - q(x - y) + iq(x + iy) - iq(x - iy).$$

- (c) If $\mathbb{K} = \mathbb{C}$ or $\mathbb{R}$ and ϕ is symmetric, then

$$4 \operatorname{Re} \phi(x, y) = q(x + y) - q(x - y).$$

Solution (a) A sesquilinear functional is linear in the first variable and conjugate linear in the second variable. So,

$$\begin{aligned} q(x + y) &= \phi(x + y, x + y) = \phi(x, x + y) + \phi(y, x + y) \\ &= \phi(x, x) + \phi(x, y) + \phi(y, x) + \phi(y, y), \\ q(x - y) &= \phi(x, x) + \phi(x, -y) + \phi(-y, x) + \phi(-y, -y) \\ &= \phi(x, x) - \phi(x, y) - \phi(y, x) + \phi(y, y). \end{aligned}$$

Hence,

$$q(x + y) - q(x - y) = 2\phi(x, y) + 2\phi(y, x).$$

- (b) If $\mathbb{K} = \mathbb{C}$, then changing y to iy in (a), we get

$$\begin{aligned} q(x + iy) - q(x - iy) &= 2\phi(x, iy) + \phi(iy, x) \\ &= -2i\phi(x, y) + 2i\phi(y, x) \end{aligned}$$

Hence, using (a),

$$\begin{aligned} q(x+y) - q(x-y) + iq(x+iy) - iq(x-iy) \\ = 2\phi(x, y) + 2\phi(y, x) + 2\phi(x, y) - 2\phi(y, x) \\ = 4\phi(x, y). \end{aligned}$$

(c) If ϕ is symmetric, then $\phi(y, x) = \overline{\phi(x, y)}$, and hence (a) gives

$$q(x+y) - q(x-y) = 2\phi(x, y) + 2\overline{\phi(x, y)} = 4 \operatorname{Re} \phi(x, y).$$

This completes the proof. ■

PROBLEM 6.1.3 Let $\phi : X \times X \rightarrow \mathbb{K}$ be a bounded sesquilinear functional on an inner product space X , and $q(x) = \phi(x, x)$, $x \in X$. Let

$$\|q\| = \sup \{|q(x)| : \|x\| \leq 1\}.$$

If the scalar field $\mathbb{K}$ is $\mathbb{C}$, prove that

$$\|q\| \leq \|\phi\| \leq 2\|q\| \quad (1)$$

and that the bounds are the best possible. Show also that the second inequality of (1) may be false if $\mathbb{K} = \mathbb{R}$.

Solution Since

$$\|\phi\| = \sup \{|\phi(x, y)| : \|x\| \leq 1, \|y\| \leq 1\},$$

we see that, for $\|x\| \leq 1$,

$$|q(x)| = |\phi(x, x)| \leq \|\phi\|.$$

This shows that $\|q\| \leq \|\phi\|$.

If r is real, then

$$q(rx) = \phi(rx, rx) = r^2\phi(x, x) = r^2q(x).$$

So if $x \neq 0$, then

$$q(x) = q\left(\|x\| \frac{x}{\|x\|}\right) = \|x\|^2 q\left(\frac{x}{\|x\|}\right).$$

It follows that

$$|q(x)| \leq \|x\|^2 \|q\|.$$

This is true even when $x = 0$ since $q(0) = 0$. Hence,

$$\begin{aligned} |q(x+y)| + |q(x-y)| &\leq \|q\| \{\|x+y\|^2 + \|x-y\|^2\} \\ &= \|q\| \{2\|x\|^2 + 2\|y\|^2\}, \\ |q(x+iy)| + |q(x-iy)| &\leq \|q\| \{2\|x\|^2 + 2\|y\|^2\}. \end{aligned}$$

Thus, if $\|x\| \leq 1, \|y\| \leq 1$ then, by Problem 6.1.2,

$$\begin{aligned} 4|\phi(x, y)| &\leq |q(x+y)| + |q(x-y)| + |q(x+iy)| + |q(x-iy)| \\ &\leq \|q\|(4\|x\|^2 + 4\|y\|^2), \\ |\phi(x, y)| &\leq \|q\|(\|x\|^2 + \|y\|^2) \leq 2\|q\|. \end{aligned}$$

This shows that $\|\phi\| \leq 2\|q\|$. This finishes the first part.

For the rest, consider any inner product space X containing at least two orthonormal vectors u_1, u_2 . For $x, y \in X$, let

$$\begin{aligned} \phi_1(x, y) &= \langle x, y \rangle, \quad \phi_2(x, y) = \langle x, u_1 \rangle \langle u_2, y \rangle, \\ \phi_3(x, y) &= \langle x, u_1 \rangle \langle u_2, y \rangle - \langle x, u_2 \rangle \langle u_1, y \rangle, \\ q_j(x) &= \phi_j(x, x), \quad j = 1, 2, 3. \end{aligned}$$

Then, ϕ_1, ϕ_2, ϕ_3 are sesquilinear on $X \times X$. Since $q_1(x) = \|x\|^2$ and

$$|\phi_1(x, y)| \leq \|x\| \|y\|,$$

we see that $\|q_1\| = 1 = \|\phi_1\|$. Again, since

$$\begin{aligned} |\phi_2(x, y)| &\leq \|x\| \|u_1\| \|u_2\| \|y\| = \|x\| \|y\|, \\ \phi_2(u_1, u_2) &= \langle u_1, u_1 \rangle \langle u_2, u_2 \rangle = 1, \end{aligned}$$

we see that $\|\phi_2\| = 1$. Further,

$$|q_2(x)| = |\langle x, u_1 \rangle \langle u_2, x \rangle| \leq \frac{1}{2} [|\langle x, u_1 \rangle|^2 + |\langle u_2, x \rangle|^2] \leq \frac{1}{2} \|x\|^2,$$

where the last inequality is obtained from Bessel's inequality. This shows that $\|q_2\| \leq 1/2$. If $\mathbb{K} = \mathbb{C}$, then by the first part,

$$\|q_2\| \geq \frac{1}{2} \|\phi_2\| = \frac{1}{2}.$$

Therefore, $\|q_2\| = 1/2$, and $\|\phi_2\| = 2\|q_2\|$.

Thus we have proved that the bounds $\|\phi\| = \|q\|$ and $\|\phi\| = 2\|q\|$ in the first part are attained by ϕ_1 and ϕ_2 .

Finally, if $\mathbb{K} = \mathbb{R}$, then

$$q_3(x) = \langle x, u_1 \rangle \langle u_2, x \rangle - \langle x, u_2 \rangle \langle u_1, x \rangle = 0$$

for all x . So $\|q_3\| = 0$ if $\mathbb{K} = \mathbb{R}$. Since $\phi_3(u_1, u_2) = 1$, we have $\|\phi_3\| \geq 1$. So the inequality $\|\phi_3\| \leq 2\|q_3\|$ is not true when $\mathbb{K} = \mathbb{R}$. This completes the proof. ■

PROBLEM 6.1.4 Let X, ϕ, q be as in Problem 6.1.3 where the scalar field $\mathbb{K}$ is $\mathbb{R}$ or $\mathbb{C}$. If ϕ is symmetric, prove that $\|\phi\| = \|q\|$.

Solution Since ϕ is symmetric we have by Problem 6.1.2(c),

$$4 \operatorname{Re} \phi(x, y) = q(x + y) - q(x - y).$$

Since $|q(x)| \leq \|q\| \|x\|^2$ for all x , as in the solution of Problem 6.1.3, we get

$$\begin{aligned} 4|\operatorname{Re} \phi(x, y)| &\leq |q(x + y)| + |q(x - y)| \\ &\leq \|q\| (\|x + y\|^2 + \|x - y\|^2) \\ &= \|q\| (2\|x\|^2 + 2\|y\|^2) \end{aligned} \quad (2)$$

for all x, y . Let $\|x\| \leq 1, \|y\| \leq 1$, and

$$k = \begin{cases} \frac{|\phi(x, y)|}{\phi(x, y)}, & \text{if } \phi(x, y) \neq 0, \\ 1 & \text{if } \phi(x, y) = 0. \end{cases}$$

Then, $|k| = 1$ and

$$|\phi(x, y)| = k\phi(x, y) = \phi(kx, y).$$

So $\operatorname{Re} \phi(kx, y) = |\phi(x, y)|$ and hence, using (2), we get

$$4|\phi(x, y)| = 4 \operatorname{Re} \phi(kx, y) \leq 2\|q\| (\|kx\|^2 + \|y\|^2) \leq 2\|q\| (1 + 1).$$

Hence,

$$|\phi(x, y)| \leq \|q\| \quad \text{for } \|x\| \leq 1, \|y\| \leq 1.$$

Thus, $\|\phi\| \leq \|q\|$. On the other hand, since

$$|q(x)| = |\phi(x, x)| \leq \|\phi\| \quad \text{for } \|x\| \leq 1,$$

we must have $\|q\| \leq \|\phi\|$. This proves the result. ■

PROBLEM 6.1.5 Let X be a linear space, $\phi : X \times X \rightarrow \mathbb{K}$ be a symmetric and positive sesquilinear functional, and $q(x) = \phi(x, x)$. Prove that

- (a) $|\phi(x, y)|^2 \leq q(x)q(y)$ for all $x, y \in X$;
- (b) $p(x) = \sqrt{q(x)}$ defines a seminorm on X , that is, for all $x, y \in X, k \in \mathbb{K}$,

$$p(x) \geq 0, \quad (3)$$

$$p(kx) = |k|p(x), \quad (4)$$

$$p(x + y) \leq p(x) + p(y); \quad (5)$$

- (c) $\{x \in X : q(x) = 0\} = \{x \in X : \phi(x, y) = 0 \text{ for all } y \in X\}.$

Solution (a) Since ϕ is positive and symmetric, we have, for all $x, y \in X$ and $k \in \mathbb{K}$,

$$\begin{aligned} 0 &\leq \phi(kx + y, kx + y) = k\phi(x, kx + y) + \phi(y, kx + y) \\ &= k\{\bar{k}\phi(x, x) + \phi(x, y)\} + \bar{k}\phi(y, x) + \phi(y, y) \\ &= |k|^2 q(x) + k\phi(x, y) + \bar{k}\overline{\phi(x, y)} + q(y) \end{aligned} \quad (6)$$

If $q(x) \neq 0$, let

$$k = -\frac{\overline{\phi(x, y)}}{q(x)}.$$

We get from (6),

$$0 \leq \frac{|\phi(x, y)|^2}{q(x)} - \frac{|\phi(x, y)|^2}{q(x)} - \frac{|\phi(x, y)|^2}{q(x)} + q(y).$$

It follows that

$$|\phi(x, y)|^2 \leq q(x)q(y). \quad (7)$$

Similarly, considering $\phi(x + ky, x + ky)$, we again get (7) if $q(y) \neq 0$. So suppose that $q(x) = 0 = q(y)$. Then (6) shows that

$$k\phi(x, y) + \bar{k}\overline{\phi(x, y)} \geq 0 \quad \text{for all } k \in \mathbb{K}. \quad (8)$$

Choosing $k = 1$ and $k = -1$, we obtain from this relation,

$$\operatorname{Re} \phi(x, y) = 0.$$

If $\mathbb{K} = \mathbb{R}$, then $\phi(x, y) = 0$. If $\mathbb{K} = \mathbb{C}$, we can choose $k = i$ and $k = -i$ in (8) and get

$$\operatorname{Im} \phi(x, y) = 0.$$

Hence, $\phi(x, y) = 0$ in this case also. Thus, (7) holds in all cases.

(b) Since $p(x)$ is taken as the non-negative square root of $q(x)$, we have (3). Since

$$q(kx) = \phi(kx, kx) = k\bar{k}\phi(x, x) = |k|^2 q(x)$$

we get (4). Taking $k = 1$ in (6), we get

$$\begin{aligned} q(x + y) &= \phi(x + y, x + y) = q(x) + 2\operatorname{Re} \phi(x, y) + q(y) \\ &\leq q(x) + 2|\phi(x, y)| + q(y) \\ &\leq q(x) + 2(q(x)q(y))^{1/2} + q(y) \end{aligned}$$

using (a). Hence,

$$(p(x + y))^2 \leq (p(x))^2 + 2p(x)p(y) + (p(y))^2$$

which gives (5).

(c) If $q(x) = 0$, then by (a),

$$\phi(x, y) = 0 \quad \text{for all } y \in X.$$

If this holds then, in particular, $\phi(x, x) = 0$, that is, $q(x) = 0$. This completes the proof. ■

PROBLEM 6.1.6 Let X and Y be normed spaces and $\phi : X \times Y \longrightarrow \mathbb{K}$ be sesquilinear. For $x \in X, y \in Y$, let

$$f^y(x) = \phi(x, y) = \overline{f_x(y)}.$$

Prove that

- (a) if ϕ is bounded then it is continuous on $X \times Y$;
- (b) if ϕ is bounded then $f^y \in X'$ for each $y \in Y$ and $f_x \in Y'$ for each $x \in X$;
- (c) if $f^y \in X', f_x \in Y'$ for all $y \in Y, x \in X$ and if X or Y is Banach, then ϕ is bounded.

Solution (a) Let ϕ be bounded. If $x \neq 0, y \neq 0$, then

$$\begin{aligned} \phi(x, y) &= \|x\| \|y\| \phi\left(\frac{x}{\|x\|}, \frac{y}{\|y\|}\right), \\ |\phi(x, y)| &\leq \|x\| \|y\| \|\phi\|. \end{aligned} \tag{9}$$

Since $\phi(x, y) = 0$ when $x = 0$ or $y = 0$, (9) is true for all $x \in X, y \in Y$.

Let $x_n \rightarrow x$ in X and $y_n \rightarrow y$ in Y . Then,

$$\begin{aligned} \phi(x_n, y_n) - \phi(x, y) &= \phi(x_n, y_n) - \phi(x_n, y) + \phi(x_n, y) - \phi(x, y) \\ &= \phi(x_n, y_n - y) + \phi(x_n - x, y), \\ |\phi(x_n, y_n) - \phi(x, y)| &\leq \|\phi\| \|x_n\| \|y_n - y\| + \|\phi\| \|x_n - x\| \|y\|, \end{aligned} \tag{10}$$

where we have made use of (9). Since

$$\|x_n\| \rightarrow \|x\|, \|y_n - y\| \rightarrow 0, \|x_n - x\| \rightarrow 0$$

as $n \rightarrow \infty$, the right-hand side of (10) tends to zero as $n \rightarrow \infty$. Hence,

$$\phi(x_n, y_n) \rightarrow \phi(x, y)$$

whenever $(x_n, y_n) \rightarrow (x, y)$ in $X \times Y$. This proves (a).

(b) It is easily seen that $f^y : X \longrightarrow \mathbb{K}$ and $f_x : Y \longrightarrow \mathbb{K}$ are linear. Since

$$\begin{aligned} |f^y(x)| &= |\phi(x, y)| \leq \|\phi\| \|x\| \|y\|, \\ |f_x(y)| &= |\phi(x, y)| \leq \|\phi\| \|x\| \|y\|, \end{aligned}$$

we have

$$\|f^y\| \leq \|\phi\| \|y\|, \quad \|f_x\| \leq \|\phi\| \|x\|.$$

Thus, f^y and f_x are bounded. This proves (b).

(c) Suppose that X is Banach. Consider the family $\{f^y : y \in Y, \|y\| \leq 1\}$ of bounded linear maps from X to $\mathbb{K}$. For each $x \in X$,

$$|f^y(x)| = |\phi(x, y)| = |f_x(y)| \leq \|f_x\| \|y\| \leq \|f_x\|$$

for all $y \in Y$ with $\|y\| \leq 1$. Since X is Banach, we can apply the uniform boundedness principle to this family and see that $\{\|f^y\| : \|y\| \leq 1\}$ is bounded. If $\|f^y\| \leq \alpha$ for $\|y\| \leq 1$ then, for $\|x\| \leq 1, \|y\| \leq 1$,

$$|\phi(x, y)| = |f^y(x)| \leq \|f^y\| \|x\| \leq \alpha.$$

Hence, ϕ is bounded. If Y is Banach, a similar argument applied to the family $\{f_x : x \in X, \|x\| \leq 1\}$ yields the same result. This completes the proof. ■

PROBLEM 6.1.7 Let H_1, H_2 be Hilbert spaces and $\phi_1 : H_1 \times H_2 \rightarrow \mathbb{K}$ be a bounded sesquilinear functional. Show that there is a unique $S \in BL(H_1, H_2)$ such that

$$\phi(x, y) = \langle Sx, y \rangle \quad \text{for } x \in H_1, y \in H_2.$$

Show also that $\|S\| = \|\phi\|$.

Solution For a fixed x in H_1 , let

$$f_x(y) = \overline{\phi(x, y)}, \quad y \in H_2.$$

Then, $f_x \in H_2'$ by Problem 6.1.6(b). So by the Riesz representation theorem, there is a unique $z \in H_2$ such that

$$f_x(y) = \langle y, z \rangle \quad \text{for all } y \in H_2. \quad (11)$$

Since z is uniquely determined by x in H_1 , we can define a map $S : H_1 \rightarrow H_2$ by

$$S(x) = z, \quad x \in H_1,$$

where z is given by (11). Thus,

$$\begin{aligned} f_x(y) &= \langle y, z \rangle = \langle y, Sx \rangle, \\ \phi(x, y) &= \overline{f_x(y)} = \overline{\langle Sx, y \rangle} \quad \text{for } x \in H_1, y \in H_2. \end{aligned} \quad (12)$$

To show that $S : H_1 \rightarrow H_2$ is linear, let $S(x_1) = z_1, S(x_2) = z_2$ and $S(kx_1 + x_2) = z$. Then from (12) we have, for all $y \in H_2$,

$$\begin{aligned} \phi(x_1, y) &= \langle z_1, y \rangle, \quad \phi(x_2, y) = \langle z_2, y \rangle, \\ \phi(kx_1 + x_2, y) &= k\phi(x_1, y) + \phi(x_2, y) \\ &= k\langle z_1, y \rangle + \langle z_2, y \rangle = \langle kz_1 + z_2, y \rangle. \end{aligned}$$

Hence,

$$\langle z, y \rangle = \phi(kx_1 + x_2, y) = \langle kz_1 + z_2, y \rangle.$$

The validity of this for all $y \in H_2$ implies that $z = kz_1 + z_2$, that is,

$$S(kx_1 + x_2) = kS(x_1) + S(x_2).$$

Thus, S is linear.

As ϕ is bounded, we have

$$|\phi(x, y)| \leq \|\phi\| \|x\| \|y\| \quad \text{for all } x \in H_1, y \in H_2.$$

Since (12) with $y = Sx$ shows that $\phi(x, Sx) = \langle Sx, Sx \rangle$, we get

$$\begin{aligned} \|Sx\|^2 &= |\phi(x, Sx)| \leq \|\phi\| \|x\| \|Sx\|; \\ \|Sx\| &\leq \|\phi\| \|x\|. \end{aligned}$$

This is true for all $x \in H_1$. Hence, S is bounded and $\|S\| \leq \|\phi\|$. On the other hand, if $\|x\| \leq 1, \|y\| \leq 1$, then by (12),

$$|\phi(x, y)| \leq \|Sx\| \|y\| \leq \|S\| \|x\| \|y\| \leq \|S\|.$$

Therefore, $\|\phi\| \leq \|S\|$. Thus $\|S\| = \|\phi\|$.

Finally, if S_1 is another such element in $BL(H_1, H_2)$, then

$$\langle S_1x, y \rangle = \phi(x, y) = \langle Sx, y \rangle$$

for all $x \in H_1, y \in H_2$. The validity of this for all $y \in H_2$ implies that $S_1x = Sx$. Thus, $S_1 = S$. This completes the proof. ■

PROBLEM 6.1.8 Let H_1 and H_2 be Hilbert spaces and $T : H_1 \rightarrow H_2$ be a bounded linear map. Show that there exists a unique bounded linear map $T^* : H_2 \rightarrow H_1$ such that $\|T^*\| = \|T\|$, and

$$\langle Tx, y \rangle = \langle x, T^*y \rangle \quad \text{for all } x \in H_1, y \in H_2.$$

[The map T^* is called the *adjoint* of T .]

Solution Define $\phi : H_2 \times H_1 \rightarrow \mathbb{K}$ by

$$\phi(x, y) = \langle x, Ty \rangle \quad \text{for } x \in H_2, y \in H_1.$$

Since

$$\begin{aligned} \langle kx_1 + x_2, Ty \rangle &= k\langle x_1, Ty \rangle + \langle x_2, Ty \rangle, \\ \langle x, kTy_1 + Ty_2 \rangle &= \bar{k}\langle x, Ty_1 \rangle + \langle x, Ty_2 \rangle, \end{aligned}$$

we see that

$$\phi(kx_1 + x_2, y) = k\phi(x_1, y) + \phi(x_2, y),$$

$$\phi(x, ky_1 + y_2) = \langle x, kTy_1 + Ty_2 \rangle = \bar{k}\phi(x, y_1) + \phi(x, y_2).$$

Thus ϕ is a sesquilinear functional on $H_2 \times H_1$. By Schwarz inequality, we have

$$|\phi(x, y)| = |\langle x, Ty \rangle| \leq \|x\| \|Ty\| \leq \|x\| \|T\| \|y\|$$

for $x \in H_2, y \in H_1$. This shows that ϕ is bounded and $\|\phi\| \leq \|T\|$. On the other hand, taking $x = Ty$, we have

$$\phi(Ty, y) = \langle Ty, Ty \rangle = \|Ty\|^2 \quad \text{for } y \in H_1.$$

Hence,

$$\begin{aligned} \|Ty\|^2 &= |\phi(Ty, y)| \leq \|\phi\| \|Ty\| \|y\|, \\ \|Ty\| &\leq \|\phi\| \|y\|. \end{aligned}$$

Therefore, $\|T\| \leq \|\phi\|$. Thus $\|\phi\| = \|T\|$.

Use Problem 6.1.7, with H_1, H_2 interchanged. We see that there exists S in $BL(H_2, H_1)$ such that $\|S\| = \|\phi\|$ and

$$\phi(x, y) = \langle Sx, y \rangle \quad \text{for } x \in H_2, y \in H_1.$$

Hence,

$$\begin{aligned} \langle Ty, x \rangle &= \overline{\phi(x, y)} = \langle y, Sx \rangle, \quad x \in H_2, y \in H_1, \\ \langle Tx, y \rangle &= \langle x, Sy \rangle, \quad y \in H_2, x \in H_1. \end{aligned}$$

Since $\|S\| = \|\phi\| = \|T\|$, we can take $T^* = S$.

Finally, to prove uniqueness, suppose that $BL(H_2, H_1)$ contains two elements S and S_1 such that

$$\langle x, Sy \rangle = \langle Tx, y \rangle = \langle x, S_1y \rangle$$

for all $x \in H_1, y \in H_2$. Choosing $x = Sy - S_1y$, we get

$$\langle Sy - S_1y, Sy - S_1y \rangle = \langle x, Sy - S_1y \rangle = 0,$$

and hence $Sy - S_1y = 0$ for all $y \in H_2$. This completes the proof. ■

PROBLEM 6.1.9 Show that, in general, a bounded linear operator on an inner product space need not have an adjoint. [This implies that completeness is essential in Problem 6.1.8.]

Solution Let $X = c_{00}$ with the inner product of l^2 . If $x \in X$, then

$$\sum_{j=1}^{\infty} \left| \frac{x(j)}{j} \right| \leq \left(\sum_{j=1}^{\infty} |x(j)|^2 \right)^{1/2} \left(\sum_{j=1}^{\infty} \frac{1}{j^2} \right)^{1/2} = \|x\| \left(\frac{\pi^2}{6} \right)^{1/2}.$$

This shows that

$$f(x) = \sum_{j=1}^{\infty} \frac{x(j)}{j}, \quad x \in X$$

defines a bounded linear functional on X . Let

$$A(x) = (f(x), 0, 0, \dots), \quad x \in X.$$

Then $A : X \rightarrow X$ is a bounded linear map. Suppose that there is $A^* \in BL(X)$ such that

$$\langle Ax, y \rangle = \langle x, A^*y \rangle \quad \text{for all } x, y \in X.$$

Let e_n be the n -th unit vector. Then $f(e_n) = 1/n$ and

$$\begin{aligned} A(e_n) &= \left(\frac{1}{n}, 0, 0, \dots\right), \\ \langle Ae_n, e_1 \rangle &= \frac{1}{n}, \\ \langle e_n, A^*e_1 \rangle &= \langle Ae_n, e_1 \rangle = \frac{1}{n}. \end{aligned}$$

But $\langle A^*e_1, e_n \rangle = (A^*e_1)(n)$. Hence, $(A^*e_1)(n) = 1/n$. But then A^*e_1 cannot belong to $X = c_{00}$. This contradiction shows that A^* does not exist. ■

PROBLEM 6.1.10 Suppose that a linear operator $T : \mathbb{K}^n \rightarrow \mathbb{K}^m$ is given by

$$(Tx)(i) = \sum_{j=1}^n k_{ij}x(j), \quad i = 1, \dots, m, \quad x \in \mathbb{K}^n.$$

Show that $T^* : \mathbb{K}^m \rightarrow \mathbb{K}^n$ is given by

$$(T^*y)(j) = \sum_{i=1}^m \overline{k_{ij}}y(i), \quad j = 1, \dots, n, \quad y \in \mathbb{K}^m.$$

Solution Let $u_1, \dots, u_n$ be the unit vectors in $\mathbb{K}^n$ and $v_1, \dots, v_m$ be the unit vectors in $\mathbb{K}^m$. Given $y \in \mathbb{K}^m$ let $T^*(y) = x$. Then,

$$x(j) = \langle x, u_j \rangle = \langle T^*y, u_j \rangle.$$

Hence,

$$\langle T(u_j), y \rangle = \langle u_j, T^*y \rangle = \overline{x(j)} \quad (13)$$

But

$$T(u_j)(i) = \sum_{r=1}^n k_{ir}u_j(r) = k_{ij}$$

for $i = 1, \dots, m$. Since

$$\langle T(u_j), y \rangle = \sum_{i=1}^m (Tu_j)(i) \overline{y(i)} = \sum_{i=1}^m k_{ij} \overline{y(i)},$$

we get, from (13),

$$x(j) = \langle y, Tu_j \rangle = \sum_{i=1}^m \overline{k_{ij}} y(i);$$

that is,

$$(T^*y)(j) = \sum_{i=1}^m \overline{k_{ij}} y(i)$$

for $j = 1, \dots, m$. This proves the result. ■

PROBLEM 6.1.11 Let H be a separable Hilbert space and $\{u_n\}$ be an orthonormal basis for H . Suppose that A and B in $BL(H)$ are represented with respect to $\{u_n\}$ by the matrices (a_{ij}) and (b_{ij}) respectively. Show that

(a) the rows and columns of these matrices are square-summable,

(b) AB and A^* are represented by (c_{ij}) and (d_{ij}) , where

$$c_{ij} = \sum_{k=1}^{\infty} a_{ik} b_{kj}, \quad d_{ij} = \overline{a_{ji}}.$$

Solution A sequence $\{k_n\}$ of scalars is said to be square-summable if $\sum |k_n|^2 < \infty$.

Since $\{u_n\}$ is an orthonormal basis, we have the Fourier expansion

$$A(u_j) = \sum_{i=1}^{\infty} \langle Au_j, u_i \rangle u_i, \quad j = 1, 2, \dots$$

We say that the matrix (a_{ij}) represents A w.r.t. $\{u_n\}$ if

$$A(u_j) = \sum_{i=1}^{\infty} a_{ij} u_i, \quad j = 1, 2, \dots$$

This will be the case iff

$$\langle Au_j, u_i \rangle = a_{ij}, \quad i \geq 1, j \geq 1.$$

So,

$$\sum_{i=1}^{\infty} |a_{ij}|^2 = \sum_{i=1}^{\infty} |\langle Au_j, u_i \rangle|^2 \leq \|Au_j\|^2 < \infty$$

by Bessel's inequality. Since

$$a_{ij} = \langle Au_j, u_i \rangle = \langle u_j, A^* u_i \rangle,$$

we have

$$\sum_{j=1}^{\infty} |a_{ij}|^2 = \sum_{j=1}^{\infty} |\langle A^* u_i, u_j \rangle|^2 \leq \|A^* u_i\|^2 < \infty.$$

Similarly,

$$\sum_{i=1}^{\infty} |b_{ij}|^2 < \infty, \quad \sum_{j=1}^{\infty} |b_{ij}|^2 < \infty.$$

This proves (a).

Now,

$$x = \sum_{n=1}^{\infty} \langle x, u_n \rangle u_n, \quad y = \sum_{n=1}^{\infty} \langle y, u_n \rangle u_n.$$

As in Problem 5.2.14(c), the linearity and continuity of the inner product implies that

$$\langle x, y \rangle = \sum_{n=1}^{\infty} \langle x, u_n \rangle \langle u_n, y \rangle.$$

So,

$$\begin{aligned} \langle Bu_j, A^* u_i \rangle &= \sum_{k=1}^{\infty} \langle Bu_j, u_k \rangle \langle u_k, A^* u_i \rangle \\ &= \sum_{k=1}^{\infty} \langle Bu_j, u_k \rangle \langle Au_k, u_i \rangle = \sum_{k=1}^{\infty} b_{kj} a_{ik}. \end{aligned}$$

Hence, if AB is represented by (c_{ij}) , then

$$c_{ij} = \langle ABu_j, u_i \rangle = \langle Bu_j, A^* u_i \rangle = \sum_{k=1}^{\infty} b_{kj} a_{ik}.$$

Finally, if (d_{ij}) represents A^* , then

$$d_{ij} = \langle A^* u_j, u_i \rangle = \langle u_j, Au_i \rangle = \overline{\langle Au_i, u_j \rangle} = \overline{a_{ji}}.$$

This completes the proof. ■

PROBLEM 6.1.12 Let $k(s, t)$, $0 \leq s \leq 1$, $0 \leq t \leq 1$ be a measurable function on $[0, 1] \times [0, 1]$ and assume that

$$\begin{aligned} \sup_{0 \leq s \leq 1} \int_0^1 |k(s, t)| dt &= \beta < \infty, \\ \sup_{0 \leq t \leq 1} \int_0^1 |k(s, t)| ds &= \gamma < \infty. \end{aligned}$$

For $x \in L^2[0, 1]$, let

$$(Ax)(s) = \int_0^1 k(s, t)x(t) dt, \quad 0 \leq s \leq 1.$$

Show that this defines a bounded operator A on $L^2[0, 1]$ with $\|A\| \leq (\beta\gamma)^{1/2}$, and that, for $x \in L^2[0, 1]$,

$$(A^*x)(s) = \int_0^1 \overline{k(t, s)}x(t) dt, \quad 0 \leq s \leq 1.$$

Solution For $0 \leq s \leq 1$, we have

$$\begin{aligned} |(Ax)(s)| &\leq \int_0^1 |k(s, t)| |x(t)| dt = \int_0^1 |k(s, t)|^{1/2} |k(s, t)|^{1/2} |x(t)| dt \\ &\leq \left(\int_0^1 |k(s, t)| dt \right)^{1/2} \left(\int_0^1 |k(s, t)| |x(t)|^2 dt \right)^{1/2} \\ &\leq \beta^{1/2} \left(\int_0^1 |k(s, t)| |x(t)|^2 dt \right)^{1/2}. \end{aligned}$$

Hence,

$$\begin{aligned} \|Ax\|^2 &= \int_0^1 |(Ax)(s)|^2 ds \leq \int_0^1 \left\{ \beta \int_0^1 |k(s, t)| |x(t)|^2 dt \right\} ds \\ &= \beta \int_0^1 |x(t)|^2 dt \int_0^1 |k(s, t)| ds \end{aligned}$$

A change of order of integration is permissible here since the integrand is non-negative. So,

$$\|Ax\|^2 \leq \beta \int_0^1 |x(t)|^2 dt \gamma = \beta \gamma \|x\|^2.$$

Thus, $\|Ax\| \leq (\beta \gamma)^{1/2} \|x\|$ for all $x \in L^2[0, 1]$. This first of all shows that $Ax \in L^2[0, 1]$ for $x \in L^2[0, 1]$. It also shows that A is bounded and $\|A\| \leq (\beta \gamma)^{1/2}$. Since A is easily seen to be linear, it is a bounded linear operator on $H = L^2[0, 1]$.

For the second part, let

$$k_1(s, t) = \overline{k(t, s)}, \quad k_2(s, t) = |k(s, t)|,$$

and, for $x \in L^2[0, 1]$,

$$B_i(x)(s) = \int_0^1 k_i(s, t) x(t) dt, \quad 0 \leq s \leq 1, \quad i = 1, 2.$$

By the same argument as above, we see that B_1 and B_2 are bounded linear operators on $H = L^2[0, 1]$. For $x, y \in L^2[0, 1]$, we have

$$\langle x, B_1 y \rangle = \int_0^1 x(s) \overline{B_1(y)(s)} ds = \int_0^1 x(s) ds \int_0^1 \overline{k_1(s, t) y(t)} dt.$$

We wish to change the order of integration. Since $\overline{k_1(s, t)} = k_2(t, s)$, we have

$$\int_0^1 dt \int_0^1 |x(s)| \overline{k_1(s, t) y(t)} ds = \int_0^1 |y(t)| dt \int_0^1 k_2(t, s) |x(s)| ds$$

$$\begin{aligned}
&= \int_0^1 |y(t)| B_2(|x|)(t) dt \\
&= \langle B_2(|x|), |y| \rangle
\end{aligned}$$

This is finite since $B_2(|x|)$ and $|y|$ are in $L^2[0, 1]$. So we can apply Fubini's theorem to change the order of integration. Thus,

$$\begin{aligned}
\langle x, B_1 y \rangle &= \int_0^1 \overline{y(t)} dt \int_0^1 \overline{k_1(s, t)} x(s) ds \\
&= \int_0^1 \overline{y(t)} dt \int_0^1 k(t, s) x(s) ds \\
&= \int_0^1 \overline{y(t)} dt A(x)(t) = \langle Ax, y \rangle.
\end{aligned}$$

This shows that $B_1 = A^*$ and completes the proof. ■

PROBLEM 6.1.13 Let H be a separable Hilbert space and $\{u_n\}$ be an infinite orthonormal basis for H . If $A \in BL(H)$ is such that $A(u_n) = u_{n+1}$ for $n = 1, 2, \dots$, show that $A^*(u_1) = 0$ and $A^*(u_n) = u_{n-1}$ for $n = 2, 3, \dots$. Show also that $A^*A = I \neq AA^*$, where I is the identity on H .

Solution For $m = 1, 2, \dots$, we have

$$\langle u_{n+1}, u_m \rangle = \langle Au_n, u_m \rangle = \langle u_n, A^*u_m \rangle.$$

Since $\{u_n\}$ is an orthonormal basis, every y in H has the Fourier expansion

$$y = \sum_{n=1}^{\infty} \langle y, u_n \rangle u_n. \quad (14)$$

Hence,

$$A^*(u_m) = \sum_{n=1}^{\infty} \langle A^*u_m, u_n \rangle u_n = \sum_{n=1}^{\infty} \langle u_m, u_{n+1} \rangle u_n.$$

Since $\{u_n\}$ is orthonormal, we see that if $m = 1$ then each term in the last sum is zero, and if $m > 1$, then the only nonzero term is

$$\langle u_m, u_m \rangle u_{m-1} = u_{m-1}.$$

Hence,

$$A^*(u_1) = 0, \quad A^*(u_m) = u_{m-1} \quad \text{for } m \geq 2.$$

For the second part, we have

$$AA^*(u_1) = A(0) = 0 \neq I(u_1),$$

and so $AA^* \neq I$. On the other hand,

$$\langle A^*Ax, u_n \rangle = \langle Ax, Au_n \rangle = \langle Ax, u_{n+1} \rangle$$

$$= \langle x, A^* u_{n+1} \rangle = \langle x, u_n \rangle$$

for $n = 1, 2, \dots$. Therefore, using (14)

$$\begin{aligned} A^* A x &= \sum_{n=1}^{\infty} \langle A^* A x, u_n \rangle u_n \\ &= \sum_{n=1}^{\infty} \langle x, u_n \rangle u_n = x. \end{aligned}$$

Thus, $A^* A = I$, and the proof is complete. ■

PROBLEM 6.1.14 Let $H = \mathbb{K}^n$ and $A \in BL(H)$. Let $x_1, \dots, x_n$ be an algebraic basis for H and A be represented by the matrix (k_{ij}) w.r.t. $x_1, \dots, x_n$. Show that $(\overline{k_{ji}})$ may not represent A^* w.r.t. $x_1, \dots, x_n$. (See Problem 6.1.10.)

Solution We say that a matrix $M = (k_{ij})$ represents A w.r.t. $x_1, \dots, x_n$ if

$$A(x_j) = \sum_{i=1}^n k_{ij} x_i, \quad j = 1, \dots, n.$$

Let $t_{ij} = \overline{k_{ji}}$ and B be represented by $M_1 = (t_{ij})$. Then,

$$B(x_j) = \sum_{i=1}^n t_{ij} x_i = \sum_{i=1}^n \overline{k_{ji}} x_i.$$

Thus,

$$\begin{aligned} \langle Ax_j, x_r \rangle &= \left\langle \sum_{i=1}^n k_{ij} x_i, x_r \right\rangle = \sum_{i=1}^n k_{ij} \langle x_i, x_r \rangle, \\ \langle x_j, Bx_r \rangle &= \left\langle x_j, \sum_{i=1}^n \overline{k_{ri}} x_i \right\rangle = \sum_{i=1}^n k_{ri} \langle x_j, x_i \rangle. \end{aligned}$$

Let $c_{ij} = \langle x_j, x_i \rangle$ and G be the matrix (c_{ij}) . Then,

$$\langle Ax_j, x_r \rangle = \sum_{i=1}^n k_{ij} c_{ri} = \sum_{i=1}^n c_{ri} k_{ij};$$

this is the (r, j) -th element of GM . Further,

$$\langle x_j, Bx_r \rangle = \sum_{i=1}^n k_{ri} c_{ij},$$

which is the (r, j) -th element of MG . Thus we see that if

$$\langle Ax_j, x_r \rangle = \langle x_j, Bx_r \rangle$$

for all j, r , then $GM = MG$, which need not be true. In fact, if $GM = MG$ for all M , then G must be of the form cI , where I is the identity matrix and c is a scalar. This completes the proof. ■

PROBLEM 6.1.15 Let H_1 and H_2 be Hilbert spaces and $T : H_1 \rightarrow H_2$ be a bounded linear operator.

- (a) If $M_1 \subset H_1$ and $T(M_1) \subset M_2$, show that $M_1^\perp \supset T^*(M_2^\perp)$.
- (b) If M_1 and M_2 are closed subspaces of H_1 and H_2 respectively, show that $T(M_1) \subset M_2$ iff $M_1^\perp \supset T^*(M_2^\perp)$.
- (c) If $Z(A)$ and $R(A)$ denote the zero space and range of an operator A , prove that

$$R(T^*) \subset Z(T)^\perp, \quad (15)$$

$$Z(T) = R(T^*)^\perp. \quad (16)$$

Solution (a) Let $x \in T^*(M_2^\perp)$ and $u \in M_1$. Then, $x = T^*(y)$ for some $y \in M_2^\perp$, and

$$T(u) \in T(M_1) \subset M_2.$$

Hence, $\langle T(u), y \rangle = 0$. Therefore,

$$\langle u, x \rangle = \langle u, T^*y \rangle = \langle Tu, y \rangle = 0.$$

Thus we see that if $x \in T^*(M_2)$ then $\langle x, u \rangle = 0$ for every $u \in M_1$, that is, $x \in M_1^\perp$. This proves (a).

(b) Let M_1 and M_2 be closed subspaces of H_1 and H_2 . If $T(M_1) \subset M_2$, then

$$M_1^\perp \supset T^*(M_2^\perp) \quad (17)$$

by (a). Conversely, suppose that (17) holds. Apply (a) with T replaced by T^* and M_1, M_2 replaced by $E_1 = M_2^\perp, E_2 = M_1^\perp$, respectively. Since (17) becomes

$$T^*(E_1) \subset E_2,$$

we get, by (a),

$$E_1^\perp \supset T^{**}(E_2^\perp).$$

But

$$E_1^\perp = M_2^{\perp\perp} = M_2, \quad E_2^\perp = M_1^{\perp\perp} = M_1, \quad T^{**} = T.$$

Thus, $M_2 \supset T(M_1)$. This proves (b).

(c) Let $M_1 = Z(T)$ and $M_2 = \{0\} \subset H_2$. Then, $T(M_1) = \{0\} \subset M_2$. So by (b),

$$M_1^\perp \supset T^*(M_2^\perp) \quad (18)$$

However, $M_2^\perp = \{0\}^\perp = H_2$, and so

$$T^*(M_2) = T^*(H_2) = R(T^*).$$

So (18) gives (15).

Let $y \in R(T)^\perp$. Then, since $Tx \in R(T)$ for all $x \in H_1$, we have

$$\langle x, T^*y \rangle = \langle Tx, y \rangle = 0.$$

Therefore, $\langle x, T^*y \rangle = 0$ for any x in H_1 . This implies that $T^*y = 0$, that is, $y \in Z(T^*)$. Thus we obtain

$$R(T)^\perp \subset Z(T^*). \quad (19)$$

From (15),

$$R(T^*)^\perp \supset Z(T)^{\perp\perp} = Z(T) \quad (20)$$

since $Z(T)$ is a closed subspace. Replacing T by T^* in (19), we get

$$R(T^*)^\perp \subset Z(T^{**}) = Z(T). \quad (21)$$

Combining (20) and (21) we get (16). This completes the proof. ■

PROBLEM 6.1.16 Show that a bounded linear map $T : H \rightarrow H$ on a Hilbert space H has a finite dimensional range iff T can be represented in the form

$$T(x) = \sum_{j=1}^n \langle x, u_j \rangle v_j, \quad x \in H, \quad (22)$$

where $u_1, \dots, u_n, v_1, \dots, v_n$ are fixed vectors in H .

Solution Suppose that there exist vectors $u_1, \dots, u_n, v_1, \dots, v_n$ in H such that (22) holds. Then $T(x) \in \text{span}\{v_1, \dots, v_n\}$ for each $x \in H$. Hence $R(T)$, the range of T , is finite dimensional in this case.

Conversely, suppose that $R(T)$ is finite dimensional. Let $\{v_1, \dots, v_n\}$ be an orthonormal basis for $R(T)$. Then,

$$T(x) = \sum_{j=1}^n \langle Tx, v_j \rangle v_j, \quad x \in H.$$

Since $\langle Tx, v_j \rangle = \langle x, T^*v_j \rangle$, we can take $u_j = T^*v_j$ and get (22). This proves the result. ■

PROBLEM 6.1.17 Let H be a Hilbert space, and $\phi : H \times H \rightarrow \mathbb{K}$ be a sesquilinear functional on H . Suppose that there exist constants $\alpha, \beta > 0$ such that

$$|\phi(x, y)| \leq \alpha \|x\| \|y\| \quad (23)$$

$$\beta \|x\|^2 \leq \text{Re } \phi(x, x) \quad (24)$$

for all $x, y \in H$. Show that there exists a unique $S \in BL(H)$ such that

$$\langle x, y \rangle = \phi(x, Sy) \quad \text{for all } x, y \in H.$$

Show also that S^{-1} exists in $BL(H)$, $\|S\| \leq 1/\beta$ and $\|S^{-1}\| \leq \alpha$. [This result is known as the *Lax-Milgram theorem*].

Solution From (23) we see that ϕ is bounded and $\|\phi\| \leq \alpha$. So by Problem 6.1.7 there is $A \in BL(H)$ such that

$$\phi(x, y) = \langle Ax, y \rangle \quad \text{for } x, y \in H. \quad (25)$$

Moreover, $\|A\| = \|\phi\| \leq \alpha$. Using (24) and (25), we get

$$\beta\|x\|^2 \leq \operatorname{Re} \phi(x, x) \leq |\phi(x, x)| = |\langle Ax, x \rangle| \leq \|Ax\| \|x\|.$$

Hence,

$$\beta\|x\| \leq \|Ax\| \quad \text{for all } x \in H. \quad (26)$$

Since $\beta > 0$ this shows that $A(x) = 0$ only if $x = 0$, and hence A is one-one.

The range $R(A)$ is a subspace of H . Let $y \in \overline{R(A)}$. Then there is a sequence $\{y_n\}$ in $R(A)$ such that $y_n \rightarrow y$, and $y_n = A(x_n)$, $x_n \in H$. So from (26),

$$\beta\|x_n - x_m\| \leq \|A(x_n - x_m)\| = \|y_n - y_m\|.$$

Since $\beta > 0$, this shows that $\{x_n\}$ is a Cauchy sequence in H and hence convergent. If $x_n \rightarrow x$, then $y_n = A(x_n) \rightarrow A(x)$. Therefore, $y = A(x) \in R(A)$. Thus $R(A)$ is a closed subspace of H .

Let x be any element of H . Then, $x = x_1 + x_2$, where $x_1 \in R(A)$ and $x_2 \in R(A)^\perp$. Using (24) and (25), we get

$$\beta\|x_2\|^2 \leq \operatorname{Re} \phi(x_2, x_2) = \operatorname{Re} \langle Ax_2, x_2 \rangle = 0$$

since $Ax_2 \in R(A)$ and $x_2 \in R(A)^\perp$. This shows that $x_2 = 0$, and hence $x = x_1 \in R(A)$. Thus we see that $R(A) = H$, that is, $A : H \rightarrow H$ is a bijective bounded linear map. From (26) we see that

$$\beta\|A^{-1}x\| \leq \|AA^{-1}x\| = \|x\|, \quad x \in H.$$

Hence, A^{-1} is bounded and $\|A^{-1}\| \leq 1/\beta$. Thus, A^* and $(A^{-1})^*$ exist in $BL(H)$, and

$$A^*(A^{-1})^* = (A^{-1}A)^* = I^* = I,$$

$$(A^{-1})^*A^* = (AA^{-1})^* = I^* = I.$$

Let $S = (A^{-1})^*$. Then $S^{-1} = A^*$ exists in $BL(H)$ and

$$\|S\| = \|(A^{-1})^*\| = \|A^{-1}\| \leq \frac{1}{\beta},$$

$$\|S^{-1}\| = \|A^*\| = \|A\| = \|\phi\| \leq \alpha$$

and from (25),

$$\begin{aligned} \phi(x, Sy) &= \langle Ax, Sy \rangle = \langle Ax, (A^{-1})^*y \rangle \\ &= \langle A^{-1}Ax, y \rangle = \langle x, y \rangle \end{aligned}$$

for all $x, y \in H$. This completes the proof. ■

PROBLEM 6.1.18 Let H be a Hilbert space and $A : H \rightarrow H$ be a linear map. Suppose that there is a linear map $B : H \rightarrow H$ such that

$$\langle Ax, y \rangle = \langle x, By \rangle \quad \text{for all } x, y \in H.$$

Show that A and B are bounded and $B = A^*$.

Solution Let

$$E = \{A(x) : \|x\| \leq 1\}.$$

Let $z \in E$. Then, $z = A(x)$, where $\|x\| \leq 1$. So for every $y \in H$, we have

$$\begin{aligned} |\langle z, y \rangle| &= |\langle Ax, y \rangle| = |\langle x, By \rangle| \\ &\leq \|x\| \|By\| \leq \|By\|. \end{aligned}$$

Thus, $|\langle z, y \rangle| \leq \|By\|$ for all $z \in E$. Hence Problem 5.3.15 shows that E is a bounded set, that is, A is bounded. So A has a unique adjoint A^* in $BL(H)$ such that $\langle Ax, y \rangle = \langle x, A^*y \rangle$ for all $x, y \in H$. Therefore,

$$\langle x, By \rangle = \langle Ax, y \rangle = \langle x, A^*y \rangle \quad \text{for all } x, y \in H.$$

It follows that $B(y) = A^*(y)$ for all $y \in H$. Thus, $B = A^*$ and $B \in BL(H)$. This completes the proof. ■

PROBLEM 6.1.19 Let $A \in BL(H)$, where H is a Hilbert space. Show that A is bounded below iff A^* is onto.

Solution Suppose that A^* is onto. To show that A is bounded below, let

$$E = \{x \in H : \|Ax\| \leq 1\}.$$

Let y be any element in H . There is $z \in H$ such that $A^*(z) = y$. So for all $x \in E$,

$$\begin{aligned} |\langle x, y \rangle| &= |\langle x, A^*z \rangle| = |\langle Ax, z \rangle| \\ &\leq \|Ax\| \|z\| \leq \|z\| \end{aligned}$$

If $\alpha(y) = \|z\|$, where z is some element with $A^*(z) = y$, then

$$|\langle x, y \rangle| \leq \alpha(y) \quad \text{for all } x \in E.$$

Hence, Problem 5.3.15 shows that E is a bounded set. So there is $\lambda > 0$ such that $\|x\| \leq \lambda$ for all $x \in E$. If $x \in H, x \neq 0$, let

$$k = \frac{\lambda + 1}{\|x\|}.$$

Then $\|kx\| = \lambda + 1 > \lambda$; hence,

$$kx \notin E, \quad \|A(kx)\| > 1,$$

$$\|A(x)\| > \frac{1}{k} = \frac{\|x\|}{\lambda + 1}.$$

Therefore,

$$\frac{1}{\lambda + 1} \|x\| \leq \|A(x)\|$$

which is true even when $x = 0$. Thus, A is bounded below.

Conversely, suppose that A is bounded below. Then there is $\beta > 0$ such that

$$\beta \|x\| \leq \|Ax\| \quad \text{for } x \in H.$$

So $A(x) = 0$ only if $x = 0$, that is, A is one-one. Let $Y = R(A)$, the range of A . Then Y is a subspace of H , $A : H \rightarrow Y$ is bijective, and its inverse $A^{-1} : Y \rightarrow H$ is linear.

Let z be any element of H and

$$g(y) = \langle A^{-1}y, z \rangle, \quad y \in Y.$$

Then g is a linear functional on Y . Since

$$\beta \|A^{-1}y\| \leq \|AA^{-1}y\| = \|y\| \quad \text{for } y \in Y,$$

we have

$$|g(y)| = |\langle A^{-1}y, z \rangle| \leq \|A^{-1}y\| \|z\| \leq \frac{1}{\beta} \|y\| \|z\|,$$

for all $y \in Y$. Thus g is bounded. By Problem 5.3.12, there is $f \in H'$ such that

$$f|_Y = g \quad \text{and} \quad \|f\| = \|g\|.$$

By the Riesz representation theorem, there is $y \in H$ such that

$$f(x) = \langle x, y \rangle \quad \text{for } x \in H.$$

Hence, for all $x \in H$,

$$\begin{aligned} \langle x, A^*y \rangle &= \langle Ax, y \rangle = f(Ax) \\ &= g(Ax) = \langle A^{-1}Ax, z \rangle = \langle x, z \rangle. \end{aligned}$$

Since this is true for all $x \in H$, we must have $A^*(y) = z$. This is true for any $z \in H$. Hence A^* is onto, and the proof is complete. ■

PROBLEM 6.1.20 (*The bounded inverse theorem.*) Suppose that H_1 and H_2 are Hilbert spaces, $T \in BL(H_1, H_2)$ and T is bijective. Show that its inverse T^{-1} belongs to $BL(H_2, H_1)$.

Solution By Problem 6.1.8, there is $T^* \in BL(H_2, H_1)$ such that

$$\langle Tx, y \rangle = \langle x, T^*y \rangle \quad \text{for all } x \in H_1, y \in H_2.$$

Let

$$E = \{y \in H_2 : \|T^*y\| \leq 1\}.$$

Let y be any element in H_2 . Since T is bijective, there is a unique $u \in H_1$ such that $T(u) = y$. So for all $z \in E$,

$$|\langle z, y \rangle| = |\langle z, Tu \rangle| = |\langle T^*z, u \rangle| \leq \|T^*z\| \|u\| \leq \|u\|.$$

Hence, if $\alpha(y) = \|T^{-1}(y)\| = \|u\|$, then

$$|\langle z, y \rangle| \leq \alpha(y) \quad \text{for all } z \in E.$$

This is true for any $y \in H_2$. It follows from Problem 5.3.15 that the subset E of H_2 is bounded. Let $\|y\| \leq \lambda$ for all $y \in E$.

Let y be any element of H_2 . If $y \neq 0$, let $k = (\lambda + 1)/\|y\|$. Then, $\|ky\| = \lambda + 1 > \lambda$, and so $ky \notin E$. Hence,

$$\|T^*(ky)\| > 1, \quad \|T^*y\| > \frac{1}{k} = \frac{1}{\lambda + 1} \|y\|.$$

Thus,

$$\|y\| \leq (\lambda + 1) \|T^*y\| \quad \text{for all } y \in H_2, \quad (27)$$

this being trivial when $y = 0$.

Let $R(T^*)$ be the range of T^* and $x \in \overline{R(T^*)}$. Then there is a sequence $\{x_n\}$ in H_1 such that $x_n \rightarrow x$ and $x_n = T^*(y_n)$, $y_n \in H_2$. By (27),

$$\|y_n - y_m\| \leq (\lambda + 1) \|T^*(y_n - y_m)\| = (\lambda + 1) \|x_n - x_m\|.$$

This shows that $\{y_n\}$ is a Cauchy sequence in H_2 and hence it converges. If $y_n \rightarrow y$ in H_2 , then $x_n = T^*(y_n) \rightarrow T^*(y)$ in H_1 . So $x = T^*(y) \in R(T^*)$. This shows that $R(T^*)$ is a closed subspace of H_1 . Hence,

$$R(T^*)^{\perp\perp} = R(T^*).$$

However, $R(T^*)^{\perp} = Z(T)$ by Problem 6.1.15(c) and $Z(T) = \{0\}$ since T is one-one. Therefore,

$$R(T^*) = Z(T)^{\perp} = \{0\}^{\perp} = H_1,$$

that is, T^* is onto. Since (27) implies that T^* is one-one, we see that there is a linear map $S : H_1 \rightarrow H_2$ such that

$$T^*S(x) = x \quad \text{for all } x \in H_1.$$

By (27),

$$\|Sx\| \leq (\lambda + 1) \|T^*Sx\| = (\lambda + 1) \|x\|.$$

Thus, S is bounded, and hence has a unique adjoint $S^* \in BL(H_2, H_1)$. Let $x \in H_1$, $y \in H_2$ and $z = T^{-1}(y)$. Then,

$$\langle Sx, y \rangle = \langle Sx, Tz \rangle = \langle T^*Sx, z \rangle = \langle x, z \rangle = \langle x, T^{-1}y \rangle.$$

Thus,

$$\langle x, T^{-1}y \rangle = \langle Sx, y \rangle = \langle x, S^*y \rangle$$

for all $x \in H_1$, $y \in H_2$. This implies that $T^{-1}(y) = S^*(y)$ for all $y \in H_2$, that is, $T^{-1} = S^*$. Since $S^* \in BL(H_2, H_1)$ the proof is complete. ■

PROBLEM 6.1.21 Let H_1 and H_2 be Hilbert spaces, $T : H_1 \longrightarrow H_2$ be a linear map, and

$$G(T) = \{(x, Tx) : x \in H_1\}$$

be the graph of T . Show that T is continuous iff $G(T)$ is closed in $H_1 \times H_2$.

Solution This is closed graph theorem for Hilbert spaces. We shall prove this using Hilbert space techniques. The graph $G(T)$ is a subspace of the product $H_1 \times H_2$, since T is linear. It is closed iff the following condition is satisfied:

$$x_n \rightarrow x \quad \text{and} \quad T(x_n) \rightarrow y \implies y = T(x).$$

This condition is satisfied if T is continuous.

For the converse suppose that $H_3 = G(T)$ is closed in $H_1 \times H_2$. Then H_3 is a Hilbert space since $H_1 \times H_2$ is a Hilbert space. Let

$$S(x, Tx) = x, \quad x \in H_1.$$

This defines a map $S : H_3 \longrightarrow H_1$. It can be seen that S is linear, one-one and onto. Since

$$\|(x, Tx)\|^2 = \|x\|^2 + \|Tx\|^2 \geq \|x\|^2 = \|S(x, Tx)\|^2,$$

we see that S is bounded. Thus, $S \in BL(H_3, H_1)$ and S is bijective. Hence, $S^{-1} \in BL(H_1, H_3)$ by Problem 6.1.20. So if $\|S^{-1}\| = \alpha$, then

$$\|(x, Tx)\| = \|S^{-1}(x)\| \leq \alpha\|x\|.$$

Hence,

$$\begin{aligned} \|x\|^2 + \|Tx\|^2 &= \|(x, Tx)\|^2 \leq \alpha^2 \|x\|^2, \\ \|Tx\|^2 &\leq (\alpha^2 - 1)\|x\|^2. \end{aligned}$$

This shows that T is bounded, and the proof is completed. ■

6.2 Self-Adjoint, Unitary and Normal Operators

Let H be a Hilbert space, $A \in BL(H)$ and A^* be the adjoint of A . We say that A is *self-adjoint* or *Hermitian* if $A^* = A$. If A^* commutes with A , that is, $A^*A = AA^*$ then A is said to be *normal*. If $A^*A = I = AA^*$, where I is the identity operator, then A is said to be *unitary*.

If A is self-adjoint or unitary, then it is normal. If A is self-adjoint, then

$$\langle Ax, x \rangle = \langle x, Ax \rangle = \overline{\langle Ax, x \rangle},$$

and hence $\langle Ax, x \rangle$ is real for all x . If A is self-adjoint and $\langle Ax, x \rangle \geq 0$ for all $x \in H$, then A is said to be *positive*, and we write $A \geq 0$. If A and B are self-adjoint and $A - B \geq 0$ we write $A \geq B$ or $B \leq A$.

Let $T : \mathbb{K}^n \rightarrow \mathbb{K}^n$ be represented by the matrix $M = (k_{ij})$ in the sense that

$$(Tx)(i) = \sum_{j=1}^n k_{ij}x(j), \quad i = 1, \dots, n, \quad x \in \mathbb{K}^n.$$

Then T^* is given by

$$(T^*y)(i) = \sum_{j=1}^n \overline{k_{ji}} y(j),$$

as shown in Problem 6.1.10. Thus T^* is represented by the conjugate transpose $(\overline{M})'$ of M . So T is self-adjoint iff $(\overline{M})' = M$ and T is unitary iff $(\overline{M})'$ is the inverse of M .

Let $A \in BL(H)$, $B = \frac{1}{2}(A + A^*)$ and $C = \frac{1}{2i}(A - A^*)$. Then, $A = B + C$ with $B^* = B$ and $C^* = -C$. If the scalar field $\mathbb{K}$ is $\mathbb{C}$ and

$$B = \frac{1}{2}(A + A^*), \quad C = \frac{1}{2i}(A - A^*),$$

then B and C are self-adjoint and $A = B + iC$.

EXERCISES

1. Let $A \in BL(H)$ where H is a complex Hilbert space, and $B = c(A - A^*)$ where c is a scalar. If A is not self-adjoint find all c such that B is self-adjoint.
2. Let A, c and B be as above. Show that B is normal for all c .
3. Let $H = \mathbb{C}^3$, $\mathbb{K} = \mathbb{C}$, and $A \in BL(H)$ be defined by the matrix

$$M = \begin{pmatrix} 1 & a & 0 \\ i & 1 & 0 \\ 0 & 0 & i \end{pmatrix}.$$

Find values of a , if any, such that A is normal.

PROBLEMS

PROBLEM 6.2.1 Let $H = \mathbb{R}^2$, $\mathbb{K} = \mathbb{R}$ and $A \in BL(H)$. Suppose that A is represented by the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ w.r.t. the standard basis $e_1 = (1, 0)$, $e_2 = (0, 1)$. Show that

- (a) A is self-adjoint iff $b = c$;
- (b) A is positive iff

$$b = c, \quad a \geq 0, \quad d \geq 0, \quad ad \geq b^2;$$

(c) A is unitary iff, for some θ with $0 \leq \theta < 2\pi$,

$$a = d = \cos \theta, \quad c = -b = \sin \theta,$$

or else,

$$a = -d = \cos \theta, \quad c = b = \sin \theta;$$

(d) A is normal iff

$$b = c \quad \text{or else} \quad b = -c, a = d.$$

Solution If A is represented by $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then A^* is represented by

$$\begin{pmatrix} \bar{a} & \bar{c} \\ \bar{b} & \bar{d} \end{pmatrix} = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$$

since the scalar field $\mathbb{K}$ is $\mathbb{R}$. So the matrices representing A and A^* are equal iff $b = c$. Thus, $A = A^*$ iff $b = c$. This gives (a).

We say that A is positive if it is self-adjoint and $\langle Ax, x \rangle \geq 0$ for all $x \in H$. If $y = A(x)$, then

$$\begin{aligned} y(1) &= ax(1) + bx(2), \quad y(2) = cx(1) + dx(2), \\ \langle Ax, x \rangle &= y(1)x(1) + y(2)x(2) \\ &= a(x(1))^2 + (b+c)x(1)x(2) + d(x(2))^2. \end{aligned}$$

So A is positive iff $b = c$ and

$$a(x(1))^2 + 2bx(1)x(2) + d(x(2))^2 \geq 0 \quad \text{for all } x \in \mathbb{R}^2. \quad (1)$$

If this holds, then taking $x = (1, t)$ and $x = (t, 1)$, we get

$$a + 2bt + dt^2 \geq 0, \quad at^2 + 2bt + d \geq 0$$

for all t . This is possible only if

$$a \geq 0, \quad d \geq 0, \quad b^2 \leq ad.$$

On the other hand, if this holds then it can be verified that (1) holds. This proves (b).

For (c) note that $A^*A = I = AA^*$ iff

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a & c \\ b & d \end{pmatrix}$$

That is,

$$\begin{pmatrix} a^2 + c^2 & ab + cd \\ ab + cd & b^2 + d^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a^2 + b^2 & ac + bd \\ ac + bd & c^2 + d^2 \end{pmatrix}$$

Thus A is unitary iff

$$a^2 + c^2 = b^2 + d^2 = a^2 + b^2 = c^2 + d^2 = 1, \quad (2)$$

$$ab + cd = ac + bd = 0. \quad (3)$$

If these equalities hold, then there is θ such that

$$a = \cos \theta, \quad c = \sin \theta, \quad 0 \leq \theta < 2\pi.$$

So,

$$b^2 = 1 - a^2 = \sin^2 \theta, \quad d^2 = 1 - c^2 = \cos^2 \theta,$$

$$b = \pm \sin \theta, \quad d = \mp \cos \theta.$$

On the other hand, if

$$a = \cos \theta, \quad c = \sin \theta, \quad b = \pm \sin \theta, \quad d = \mp \cos \theta$$

then (2) and (3) hold, and so A is unitary. This proves (c).

Finally, $A^*A = AA^*$ iff

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a & c \\ b & d \end{pmatrix}$$

That is,

$$\begin{pmatrix} a^2 + c^2 & ab + cd \\ ab + cd & b^2 + d^2 \end{pmatrix} = \begin{pmatrix} a^2 + b^2 & ac + bd \\ ac + bd & c^2 + d^2 \end{pmatrix}$$

This holds iff $b^2 = c^2$ and $ab + cd = ac + bd$. That is,

$$(b - c)(b + c) = 0 = (b - c)(a - d).$$

This gives the condition

$$b = c; \text{ or else } b = -c \text{ and } a = d.$$

This completes the proof. ■

PROBLEM 6.2.2 Let H be a Hilbert space and $T \in BL(H)$. If T is nonzero and self-adjoint, show that T^n is nonzero and self-adjoint.

Solution If A and B are self-adjoint and $AB = BA$, then

$$(AB)^* = B^*A^* = BA = AB,$$

and hence AB is self-adjoint. From this it follows that if T is self-adjoint, then T^n is self-adjoint for all $n \geq 1$.

If T is self-adjoint and $T^2(x) = 0$, then

$$\langle Tx, Tx \rangle = \langle T^*Tx, x \rangle = \langle T^2x, x \rangle = 0,$$

and so $T(x) = 0$. Hence if T is nonzero and self-adjoint, then T^2 also is nonzero and self-adjoint. It follows by induction that T^n is nonzero and self-adjoint when $n = 2^j, j = 1, 2, 3, \dots$. If $2^j < n < 2^{j+1}$, let $m = 2^j$ and $r = n - m$. Then,

$$\langle T^n x, T^{m-r} x \rangle = \langle T^{n-1} x, T^* T^{m-r} x \rangle = \langle T^{n-1} x, T^{m-r+1} x \rangle.$$

Applying this r times, we get

$$\langle T^n x, T^{m-r} x \rangle = \langle T^{n-r} x, T^{m-r+r} x \rangle = \langle T^m x, T^m x \rangle.$$

Hence, if $T^n(x) = 0$, then $T^m(x) = 0$. Thus, if T^m is nonzero, then T^n is nonzero for $n = m + r, 1 \leq r < m$. This completes the proof. ■

PROBLEM 6.2.3 Let A be a self-adjoint operator on a Hilbert space. Show that $\langle Ax, x \rangle = 0$ for all $x \in H$ iff $A = 0$.

Solution If $A = 0$ then $\langle Ax, x \rangle = \langle 0, x \rangle = 0$ for all $x \in H$. Conversely, suppose that $\langle Ay, y \rangle = 0$ for all $y \in H$. If $y = x + Ax$ then $A(y) = A(x) + A^2(x)$, and so

$$\begin{aligned} 0 &= \langle Ay, y \rangle = \langle Ax + A^2x, x + Ax \rangle \\ &= \langle Ax, x \rangle + \langle Ax, Ax \rangle + \langle A^2x, x \rangle + \langle A^2x, Ax \rangle. \end{aligned}$$

The first term in the last line is zero. The last term also is zero since it is $\langle Az, z \rangle$, where $z = Ax$. Since A is self-adjoint, the third term is $\langle Ax, Ax \rangle$. Thus we get

$$0 = 2\langle Ax, Ax \rangle \quad \text{for all } x \in H.$$

Hence, $Ax = 0$ for all $x \in H$, that is, $A = 0$. This proves the result. ■

PROBLEM 6.2.4 Let $A \in BL(H)$, where H is a Hilbert space. Show that

- (a) if the scalar field $\mathbb{K}$ is $\mathbb{C}$, then A is self-adjoint iff $\langle Ax, x \rangle$ is real for all $x \in H$,
- (b) if $\mathbb{K}$ is $\mathbb{R}$ then A is self-adjoint iff $\langle Ax, y \rangle = \langle Ay, x \rangle$ for all $x, y \in H$.

Solution (a) If A is self-adjoint, then

$$\langle Ax, x \rangle = \langle x, Ax \rangle = \overline{\langle Ax, x \rangle};$$

hence $\langle Ax, x \rangle$ is real for any $x \in H$.

Conversely, suppose that $\mathbb{K} = \mathbb{C}$ and $\langle Ax, x \rangle$ is real for all $x \in H$. Then,

$$\langle A^*x, x \rangle = \langle x, Ax \rangle = \langle Ax, x \rangle \quad \text{for all } x.$$

So if $B = i(A^* - A)$, then

$$\langle Bx, x \rangle = i\langle A^*x - Ax, x \rangle = 0 \quad \text{for all } x.$$

Now, $B^* = -i(A - A^*) = B$, that is, B is self-adjoint. Hence $B = 0$ by Problem 6.2.3 with A replaced by B . Thus $i(A^* - A) = 0$; that is, $A^* = A$ and A is self-adjoint.

(b) If A is self-adjoint then for all $x, y \in H$,

$$\langle Ax, y \rangle = \langle x, Ay \rangle = \overline{\langle Ay, x \rangle} = \langle Ay, x \rangle$$

since $\mathbb{K} = \mathbb{R}$.

Conversely, suppose that this condition is satisfied. Then, for all x, y ,

$$\langle Ax, y \rangle = \langle Ay, x \rangle = \langle y, A^*x \rangle = \langle A^*x, y \rangle.$$

This with $y = Ax - A^*x$ shows that

$$\langle Ax - A^*x, Ax - A^*x \rangle = \langle Ax - A^*x, y \rangle = 0.$$

Therefore, $Ax - A^*x = 0$ for all x , that is, $A = A^*$. Thus A is self-adjoint. This completes the proof. ■

PROBLEM 6.2.5 Let H be a Hilbert space. Prove the following:

- (a) The set of all self-adjoint operators on H forms a closed real-linear subspace of $BL(H)$.
- (b) If A and B are self-adjoint then AB is self-adjoint iff $AB = BA$.
- (c) If A and B are self-adjoint then $AB = 0$ iff the range of A is orthogonal to the range of B .

Solution (a) Let W be the set of all self-adjoint operators in $BL(H)$. If $A, B \in W$ and $k \in \mathbb{R}$, then

$$(kA + B)^* = \bar{k}A^* + B^* = kA + B;$$

hence $kA + B \in W$. Thus W is a real-linear subspace of $BL(H)$.

If A is any element in the closure of W in $BL(H)$, then there is a sequence $\{A_n\}$ in W such that $A_n \rightarrow A$ in the norm of $BL(H)$. Since

$$\|A_n^* - A^*\| = \|(A_n - A)^*\| = \|A_n - A\|,$$

we get

$$\begin{aligned} \|A - A^*\| &\leq \|A - A_n\| + \|A_n - A_n^*\| + \|A_n^* - A^*\| \\ &= \|A - A_n\| + 0 + \|A_n - A\|. \end{aligned}$$

This tends to zero as $n \rightarrow \infty$. So we must have $A = A^*$, that is, $A \in W$. Thus W is closed.

(b) Since $(AB)^* = B^*A^* = BA$ we see that AB is self-adjoint iff $AB = BA$.

(c) If $AB = 0$, then for all x, y ,

$$\langle Ax, By \rangle = \langle x, A^*By \rangle = \langle x, AB y \rangle = 0,$$

and this implies that the ranges $R(A)$ and $R(B)$ are orthogonal. On the other hand, if $A(x) \perp B(y)$ for all x, y , then

$$0 = \langle Ax, By \rangle = \langle x, A^*By \rangle = \langle x, AB y \rangle.$$

This, with $x = AB(y)$, shows that $AB(y) = 0$ for all y , that is, $AB = 0$. This completes the proof. ■

PROBLEM 6.2.6 Let W be the space of all self-adjoint operators in $BL(H)$. For $A, B \in W$ write $A \leq B$ if

$$\langle Ax, x \rangle \leq \langle Bx, x \rangle \quad \text{for all } x \in H.$$

Prove the following:

- (a) This defines a partial order on W .
- (b) If $A \leq B$ then $A + C \leq B + C$ for all $C \in W$ and $tA \leq tB$ for all real $t \geq 0$.
- (c) $0 \leq A \leq B$ does not imply that $A^2 \leq B^2$.

Solution (a) That $A \leq A$ is trivial. If $A \leq B$ and $B \leq C$, then

$$\langle Ax, x \rangle \leq \langle Bx, x \rangle, \quad \langle Bx, x \rangle \leq \langle Cx, x \rangle.$$

Hence, $\langle Ax, x \rangle \leq \langle Cx, x \rangle$ for all $x \in H$. Therefore, $A \leq C$.

If $A \leq B$ and $B \leq A$ then, for all $x \in H$,

$$\begin{aligned} \langle Ax, x \rangle &\leq \langle Bx, x \rangle \leq \langle Ax, x \rangle; \\ \langle Ax, x \rangle &= \langle Bx, x \rangle, \quad \langle (A - B)x, x \rangle = 0. \end{aligned}$$

Since $A - B$ is self-adjoint this implies, by Problem 6.2.3 that $A - B = 0$, that is, $A = B$. Thus, ' $\leq$ ' is a partial order on W .

(b) If $A \leq B$, then

$$\langle Ax, x \rangle \leq \langle Bx, x \rangle \quad \text{for all } x \in H.$$

So if $C \in W$ and $t \geq 0$ then, for all $x \in H$,

$$\begin{aligned} \langle Ax, x \rangle + \langle Cx, x \rangle &\leq \langle Bx, x \rangle + \langle Cx, x \rangle, \\ \langle (A + C)x, x \rangle &\leq \langle (B + C)x, x \rangle, \\ t\langle Ax, x \rangle &\leq t\langle Bx, x \rangle, \quad \langle tAx, x \rangle \leq \langle tBx, x \rangle. \end{aligned}$$

Hence,

$$A + C \leq B + C, \quad tA \leq tB.$$

(c) Let $H = \mathbb{R}^2$, $\mathbb{K} = \mathbb{R}$, and A, B be operators represented by the matrices

$$M_1 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

respectively. We see by Problem 6.2.1 that A and B are self-adjoint. If $A(x) = y$ and $B(x) = z$, then

$$\begin{aligned} y(1) &= x(1) + x(2) = y(2), \\ z(1) &= 2x(1) + x(2), \quad z(2) = x(1) + x(2). \end{aligned}$$

Hence,

$$\begin{aligned} \langle Ax, x \rangle &= y(1)x(1) + y(2)x(2) = (x(1) + x(2))^2, \\ \langle Bx, x \rangle &= z(1)x(1) + z(2)x(2) \\ &= (x(1))^2 + (x(1) + x(2))^2. \end{aligned}$$

Thus,

$$0 \leq \langle Ax, x \rangle \leq \langle Bx, x \rangle \quad \text{for all } x.$$

Therefore, $0 \leq A \leq B$.

If $x = (-1, 2)$, then $A(x) = (1, 1)$ and $B(x) = (0, 1)$, and so

$$\begin{aligned} \langle A^2x, x \rangle &= \langle Ax, Ax \rangle = \|Ax\|^2 = 2, \\ \langle B^2x, x \rangle &= \langle Bx, Bx \rangle = \|Bx\|^2 = 1, \\ \langle A^2x, x \rangle &> \langle B^2x, x \rangle. \end{aligned}$$

So it is not true that $A^2 \leq B^2$. This completes the proof. ■

PROBLEM 6.2.7 Let A and B be bounded self-adjoint operators on a Hilbert space H . Show that

- (a) A^2 is self-adjoint and positive;
- (b) if B is positive and $AB = BA$, then A^2B is self-adjoint and positive.

Solution (a) Since $(AA)^* = A^*A^* = AA$, we see that A^2 is self-adjoint. For all $x \in H$, we have

$$\langle A^2x, x \rangle = \langle Ax, A^*x \rangle = \langle Ax, Ax \rangle \geq 0;$$

hence A^2 is positive.

(b) Since $(A^2B)^* = BA^2 = A^2B$, we see that A^2B is self-adjoint. For all $x \in H$, we have

$$\begin{aligned} \langle A^2Bx, x \rangle &= \langle ABx, A^*x \rangle = \langle BAx, Ax \rangle \\ &= \langle By, y \rangle \geq 0, \end{aligned}$$

where $y = Ax$. Thus A^2B is positive if B is positive. ■

PROBLEM 6.2.8 Let H be a Hilbert space and A be a bounded self-adjoint positive operator on H . Show that $I + A$ is invertible in $BL(H)$. Deduce that if $T \in BL(H)$ then $I + T^*T$ is invertible in $BL(H)$.

Solution Let A be self-adjoint and positive and $B = I + A$. Then,

$$\begin{aligned}\langle Bx, Bx \rangle &= \langle x + Ax, x + Ax \rangle \\ &= \langle x, x \rangle + \langle x, Ax \rangle + \langle Ax, x \rangle + \langle Ax, Ax \rangle.\end{aligned}$$

Hence,

$$\|Bx\|^2 = \|x\|^2 + \langle Ax, x \rangle + \langle Ax, x \rangle + \|Ax\|^2 \geq \|x\|^2.$$

Thus, $\|x\| \leq \|Bx\|$ for all $x \in H$. So B is bounded below and is one-one. Hence B^* is onto by Problem 6.1.19. Since $B^* = I^* + A^* = B$, we see that B is bijective and hence has a linear inverse B^{-1} . Since $\|x\| \leq \|Bx\|$ for all x , we get

$$\|B^{-1}x\| \leq \|B(B^{-1}x)\| = \|x\| \quad \text{for all } x;$$

hence $B^{-1} \in BL(H)$. Thus, $I + A = B$ is invertible in $BL(H)$.

If T is any element in $BL(H)$, then T^*T is self-adjoint, and

$$\langle T^*Tx, x \rangle = \langle Tx, Tx \rangle \geq 0.$$

Therefore, T^*T is positive. Thus we can take $A = T^*T$ in the result already proved. ■

PROBLEM 6.2.9 Suppose that A and B are bounded self-adjoint operators on a Hilbert space H . Show that if A and B are positive and $AB = BA$, then AB is positive.

Solution We have to prove that $\langle ABx, x \rangle \geq 0$ for all $x \in H$. Since this is trivial when $A = 0$, we suppose $A \neq 0$. Let

$$A_1 = \frac{1}{\|A\|}A, \quad A_{n+1} = A_n - A_n^2$$

for $n = 1, 2, \dots$. Then each A_n is self-adjoint. For all $x \in H$,

$$0 \leq \frac{1}{\|A\|} \langle Ax, x \rangle \leq \frac{1}{\|A\|} \|Ax\| \|x\| \leq \|x\|^2.$$

Thus,

$$0 \leq \langle A_1x, x \rangle \leq \langle x, x \rangle = \langle Ix, x \rangle.$$

This shows that $A_1 \geq 0$ and $I - A_1 \geq 0$.

Assume that

$$A_n \geq 0, \quad I - A_n \geq 0 \tag{4}$$

for some n . Using Problem 6.2.7, we see that

$$A_n^2 \geq 0, \quad A_n^2(I - A_n) \geq 0, \quad (I - A_n)^2 A_n \geq 0.$$

Hence,

$$I - A_{n+1} = (I - A_n) + A_n^2 \geq 0,$$

$$\begin{aligned} 0 &\leq A_n^2(I - A_n) + (I - A_n)^2 A_n \\ &= A_n - A_n^2 = A_{n+1}. \end{aligned}$$

Thus we have proved by induction that (4) holds for all $n \geq 1$.

Since A_j is self-adjoint,

$$\langle A_j^2 x, x \rangle = \langle A_j x, A_j x \rangle = \|A_j x\|^2$$

for all $x \in H$. Since $A_n^2 = A_n - A_{n+1}$, we get

$$A_1^2 + \cdots + A_n^2 = A_1 - A_{n+1}, \quad n \geq 1. \quad (5)$$

Hence,

$$\begin{aligned} \sum_{j=1}^n \|A_j x\|^2 &= \sum_{j=1}^n \langle A_j^2 x, x \rangle = \left\langle \sum_{j=1}^n A_j^2 x, x \right\rangle \\ &= \langle A_1 x - A_{n+1} x, x \rangle = \langle A_1 x, x \rangle - \langle A_{n+1} x, x \rangle \\ &\leq \langle A_1 x, x \rangle \end{aligned}$$

since $A_{n+1} \geq 0$. The validity of this for all n implies that

$$\sum_{j=1}^{\infty} \|A_j x\|^2 \leq \langle A_1 x, x \rangle < \infty.$$

It follows that $\|A_n x\| \rightarrow 0$ as $n \rightarrow \infty$. This is true for all $x \in H$.

Since $AB = BA$, it follows from the definition that $A_n B = B A_n$. Hence,

$$\langle A_j^2 Bx, x \rangle = \langle A_j Bx, A_j x \rangle = \langle B A_j x, A_j x \rangle = \langle B y_j, y_j \rangle,$$

where $y_j = A_j(x)$, $x \in H$. Since

$$A_1 B = \sum_{j=1}^n A_j^2 B + A_{n+1} B$$

by (5), we have

$$\begin{aligned} \langle A_1 Bx, x \rangle &= \left\langle \sum_{j=1}^n A_j^2 Bx + A_{n+1} Bx, x \right\rangle \\ &= \sum_{j=1}^n \langle A_j^2 Bx, x \rangle + \langle A_{n+1} Bx, x \rangle \\ &= \sum_{j=1}^n \langle B y_j, y_j \rangle + \langle A_{n+1} Bx, x \rangle. \end{aligned}$$

But,

$$|\langle A_{n+1}Bx, x \rangle| \leq \|A_{n+1}Bx\| \|x\| \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Therefore,

$$\begin{aligned} \langle A_1Bx, x \rangle &= \lim_{n \rightarrow \infty} \sum_{j=1}^n \langle By_j, y_j \rangle \geq 0; \\ \frac{1}{\|A\|} \langle ABx, x \rangle &\geq 0. \end{aligned}$$

This proves the desired result. ■

PROBLEM 6.2.10 Let $\{A_n\}$ be a sequence of bounded self-adjoint operators on a Hilbert space H such that $A_n \leq A_{n+1} \leq T$, where T is a self-adjoint operator in $BL(H)$. Suppose that any A_j commutes with T and every A_n . Prove that there is a self-adjoint operator $A \in BL(H)$ such that $A \leq T$ and $A_n x \rightarrow Ax$ for all $x \in H$.

Solution Let $B_n = T - A_n$. Then, $B_n \geq 0$ and each B_j commutes with every B_n . Further, if $m < n$, then $B_m - B_n = A_n - A_m \geq 0$. Hence, by Problem 6.2.9, $B_m(B_m - B_n)$ and $(B_m - B_n)B_n$ are positive. Thus,

$$\begin{aligned} B_m^2 - B_m B_n &\geq 0, \quad B_m B_n - B_n^2 \geq 0, \\ B_m^2 &\geq B_m B_n \geq B_n^2 \quad \text{for } n > m. \end{aligned} \quad (6)$$

This implies that if $\alpha_n = \langle B_n^2 x, x \rangle$, then $\alpha_m \geq \alpha_n \geq 0$ for $n > m$. Hence, the sequence $\{\alpha_n\}$ converges. Now,

$$\begin{aligned} \|B_m(x) - B_n(x)\|^2 &= \langle (B_m - B_n)(x), (B_m - B_n)(x) \rangle \\ &= \langle (B_m - B_n)^2(x), x \rangle, \end{aligned}$$

since $B_m - B_n$ is self-adjoint. Further, $(B_m - B_n)^2 = B_m^2 - 2B_m B_n + B_n^2$, and $-2B_m B_n \leq -2B_n^2$ from (6); hence, $(B_m - B_n)^2 \leq B_m^2 - B_n^2$. Thus, for $n > m$,

$$\|B_m(x) - B_n(x)\|^2 \leq \langle (B_m^2 - B_n^2)(x), x \rangle = \alpha_m - \alpha_n = |\alpha_n - \alpha_m|.$$

Since $\{\alpha_n\}$ converges, and hence satisfies Cauchy's condition. This implies that $\{B_n(x)\}$ is a sequence in H satisfying Cauchy's condition. Since H is complete, we see that $B(x) = \lim_{n \rightarrow \infty} B_n(x)$ exists in H . Since each B_n is linear and positive, we can see that B is linear and positive. Thus, each B_n is a bounded linear map and the sequence $\{B_n(x)\}$ is bounded for each $x \in H$. So, the uniform boundedness theorem implies that there is a constant M such that $\|B_n\| \leq M$ for all n . Hence,

$$\|B_n(x)\| \leq \|B_n\| \|x\| \leq M \|x\| \quad \text{for all } n.$$

Taking limits, we get $\|B(x)\| \leq M\|x\|$. Thus, $B \in BL(H)$. Further, $\langle B_n x, y \rangle = \langle x, B_n y \rangle$. Letting $n \rightarrow \infty$, we get $\langle Bx, y \rangle = \langle x, By \rangle$, that is, B is self-adjoint. Now, $A_n x = Tx - B_n x \rightarrow Tx - Bx$. So, if $A = T - B$, then $A_n x \rightarrow Ax$, where $A \in BL(H)$ and is self-adjoint. Finally, $A \leq T$ as $B \geq 0$. This completes the proof. ■

PROBLEM 6.2.11 Let A be a positive self-adjoint operator in $BL(H)$. Prove that there is a unique positive self-adjoint operator $B \in BL(H)$ such that $B^2 = A$. Also prove that B commutes with every T in $BL(H)$ which commutes with A . [We say that B is the positive square root of A .]

Solution Existence: If $A = 0$, we can take $B = 0$. Suppose that $A \neq 0$. First consider the special case in which $A \leq I$. Define a sequence $\{B_n\}$ by $B_0 = 0$ and

$$B_{n+1} = B_n + \frac{1}{2}[A - B_n^2] \quad \text{for } n = 0, 1, 2, \dots \quad (7)$$

Since $B_0 = 0$, we get $B_1 = \frac{1}{2}A$, $B_2 = A - \frac{1}{8}A^2$ and so on. Thus, each B_n is a polynomial in A with real coefficients. Hence, if $TA = AT$, then $TB_n = B_n T$ for all $n \geq 0$. We see also that each B_n is self-adjoint, since A is self-adjoint and the coefficients in the polynomial for B_n are real.

We have

$$2I - 2B_{n+1} = 2I - 2B_n - [A - B_n^2] = (I - A) + (I - B_n)^2.$$

Since $A \leq I$ in the present case and $I - B_n$ is self-adjoint, we see that $(I - A)$ and $(I - B_n)^2$ are positive. So, $B_{n+1} \leq I$. Since $B_0 \leq I$, we see that $B_n \leq I$ for all $n \geq 0$. Again, by (7),

$$B_n = B_{n-1} + \frac{1}{2}[A - B_{n-1}^2] \quad \text{for } n > 0,$$

Hence, 4pt

$$\begin{aligned} 2[B_{n+1} - B_n] &= 2[B_n - B_{n-1}] - [B_n^2 - B_{n-1}^2] \\ &= [B_n - B_{n-1}][(I - B_n) + (I - B_{n-1})] \end{aligned}$$

for $n > 0$. Suppose that $B_n \geq B_{n-1}$. Now, $I - B_n$ and $I - B_{n-1}$ are positive. Hence, $B_{n+1} \geq B_n$, by considering Problem 6.2.9. Since $B_1 \geq B_0 = 0$, we see by induction that

$$0 \leq B_n \leq B_{n+1} \leq I \quad \text{for all } n.$$

It follows from Problem 6.2.10 that there is a self-adjoint operator $B \in BL(H)$ such that $B_n x \rightarrow Bx$ for each $x \in H$. Now, $\langle B_n x, x \rangle \geq 0$, as $B_n \geq 0$. Hence, $\langle Bx, x \rangle \geq 0$, that is, B is positive. We noted that if $TA = AT$, then $TB_n = B_n T$ for all n . Taking limits, we get $TB = BT$. Further, from (7), we get $A(x) - B_n^2(x) = 2B_{n+1}(x) - 2B_n(x) \rightarrow 0$. Hence, $B^2 = A$, that is, $B = A^{1/2}$.

Now, next consider the general case. We may suppose that $A \neq 0$. Let $T = (1/\|A\|)A$. Then, $T \geq 0$ and

$$\begin{aligned}\langle Ax, x \rangle &= |\langle Ax, x \rangle| \leq \|Ax\| \|x\| \leq \|A\| \langle x, x \rangle, \\ \langle Tx, x \rangle &= \frac{1}{\|A\|} \langle Ax, x \rangle \leq \langle x, x \rangle.\end{aligned}$$

Thus, $T \leq I$ and hence, $S = T^{1/2}$ exists. Let $B = \|A\|^{1/2}S$. Then, $B^2 = \|A\|S^2 = \|A\|T = A$. Thus, $A^{1/2}$ exists in any case.

Uniqueness: Suppose that C is self-adjoint, positive and $C^2 = A$. We have $CA = CC^2 = C^2C = AC$. Thus, C commutes with A . So, it commutes with B by the construction of B . Let x be any element in H and $y = Bx - Cx$. Then,

$$By + Cy = B^2x - BCx + CBx - C^2x = 0,$$

since $B^2 = A = C^2$ and $BC = CB$. Thus, $\langle By, y \rangle + \langle Cy, y \rangle = 0$. Since B and C are positive, this implies that $\langle By, y \rangle = 0 = \langle Cy, y \rangle$. We have assumed that C is self-adjoint and positive. So, it has a positive square root D . We get

$$0 = \langle Cy, y \rangle = \langle D^2y, y \rangle = \langle Dy, Dy \rangle = \|Dy\|^2.$$

Thus, $Dy = 0$ and so, $Cy = D(Dy) = 0$. Similarly, using the fact that $\langle By, y \rangle = 0$, we get $By = 0$. Since $y = Bx - Cx$, we get

$$\langle y, y \rangle = \langle Bx, y \rangle - \langle Cx, y \rangle = \langle x, By \rangle - \langle x, Cy \rangle = 0 - 0.$$

Hence, $y = 0$, that is, $Bx = Cx$ for any x . This completes the proof. ■

PROBLEM 6.2.12 Let H be a Hilbert space, $A \in BL(H)$ and A be an isometry, that is, $\|Ax\| = \|x\|$ for all $x \in H$. Show that:

- (a) $A^*A = I$.
- (b) A is unitary if H is finite dimensional.
- (c) AA^* is idempotent, self-adjoint and positive.
- (d) $R(AA^*) = Z(A^*A - AA^*)$.

Solution (a) Since A is an isometry we have, for all $x \in H$,

$$\langle A^*Ax, x \rangle = \langle Ax, Ax \rangle = \langle x, x \rangle.$$

So if $B = A^*A - I$, then

$$\langle Bx, x \rangle = \langle A^*Ax - x, x \rangle = 0.$$

Since $B^* = (A^*A)^* - I^* = A^*A - I = B$, we see by Problem 6.2.3 that $B = 0$, that is, $A^*A = I$.

(b) If $A(x) = 0$ then $\|x\| = \|Ax\| = 0$, $x = 0$. Hence, A is one-one. So it is also onto if H is finite dimensional. Therefore, there is a linear map $A^{-1} : H \rightarrow H$. Thus,

$$AA^* = AA^*AA^{-1} = AIA^{-1} = I = A^*A$$

by (a). Hence A is unitary.

(c) Let $P = AA^*$. Then, using (a),

$$P^2 = AA^*AA^* = AIA^* = P,$$

$$P^* = (AA^*)^* = AA^* = P,$$

$$\langle Px, x \rangle = \langle AA^*x, x \rangle = \langle A^*x, A^*x \rangle \geq 0.$$

Thus P is idempotent, self-adjoint and positive.

(d) Let $P = AA^*$ and $y \in R(P)$. Then $y = P(x)$. Since $A^*A = I$, we have

$$(A^*A - AA^*)(y) = y - P(y) = P(x) - P^2(x) = 0.$$

So $y \in Z(A^*A - AA^*)$. On the other hand, if $y \in Z(A^*A - AA^*)$, then

$$0 = (A^*A - AA^*)(y) = y - P(y),$$

$$y = P(y) \in R(P).$$

Thus, $R(P) = Z(A^*A - AA^*)$, and the proof is complete. ■

PROBLEM 6.2.13 Let H be a Hilbert space and $T : H \rightarrow H$ be an isometric linear map. Show that the following statements are equivalent:

- (a) T is unitary.
- (b) T is onto.
- (c) The range of T is dense in H .

Solution If T is unitary then, for all $x \in H$,

$$x = TT^*(x) \in R(T),$$

and hence T is onto. Thus, (a) implies (b). That (b) implies (c) is trivial.

Suppose that the range $R(T)$ is dense in H . Let $y \in H$. Then there is a sequence $\{y_n\}$ in $R(T)$ such that $y_n \rightarrow y$. Let $y_n = T(x_n)$. Then,

$$\|x_n - x_m\| = \|T(x_n - x_m)\| = \|y_n - y_m\|.$$

Thus, $\{x_n\}$ is a Cauchy sequence in H . Hence, $\{x_n\}$ converges to some $x \in H$. So, $y_n = T(x_n) \rightarrow T(x)$. Thus, $y = T(x) \in R(T)$. This shows that T is onto. Thus (c) implies (b).

If T is onto then T is bijective, and hence there is a linear map $S : H \rightarrow H$ such that $ST = TS = I$. Since $T^*T = I$ by Problem 6.2.12, we have

$$TT^* = TT^*TS = TIS = I = T^*T.$$

Therefore, T is unitary. Thus (b) implies (a). This completes the proof. ■

PROBLEM 6.2.14 Let $T \in BL(H)$, where H is a Hilbert space. Show that T is unitary iff it transforms every total orthonormal set $\{u_i : i \in L\}$ onto a total orthonormal set $\{T(u_i) : i \in L\}$.

Solution First suppose that T is unitary. Let $\{u_i : i \in L\}$ be a total orthonormal set in H and $v_i = T(u_i)$. For all $i, j \in L$, we have

$$\langle v_i, v_j \rangle = \langle Tu_i, Tu_j \rangle = \langle u_i, T^*Tu_j \rangle = \langle u_i, u_j \rangle.$$

This shows that $\{v_i\}$ is an orthonormal set. To show that it is total, suppose that $\langle x, v_i \rangle = 0$ for all $i \in L$. Then,

$$\langle T^*x, u_i \rangle = \langle x, Tu_i \rangle = \langle x, v_i \rangle = 0$$

for all $i \in L$. Hence, $T^*(x) = 0$ since $\{u_i\}$ is total. So $x = TT^*(x) = 0$. This shows that $\{v_i\}$ is total.

For the converse, consider a total orthonormal set $\{u_i : i \in L\}$. Let $v_i = T(u_i)$ and suppose that $\{v_i : i \in L\}$ is a total orthonormal set. For any $x \in H$, we have the Fourier expansion

$$x = \sum_{i \in L} \langle x, u_i \rangle u_i.$$

This sum contains only a countable number of nonzero terms. Hence the linearity and continuity of T show that

$$T(x) = \sum_i \langle x, u_i \rangle T(u_i) = \sum_i \langle x, u_i \rangle v_i,$$

and so $\langle Tx, v_j \rangle = \langle x, u_j \rangle$ for all $j \in L$. Thus,

$$\langle x, T^*v_j \rangle = \langle Tx, v_j \rangle = \langle x, u_j \rangle. \quad (8)$$

Since this is true for all $x \in H$, we see that $T^*(v_j) = u_j$ for any $j \in L$. Hence,

$$\begin{aligned} \langle TT^*x, v_j \rangle &= \langle T^*x, T^*v_j \rangle = \langle T^*x, u_j \rangle \\ &= \langle x, Tu_j \rangle = \langle x, v_j \rangle. \end{aligned}$$

Since $\{v_i\}$ is total, the validity of this for all $j \in L$ implies that $TT^*x = x$. This is true for any $x \in H$, that is, $TT^* = I$.

Again, from (8),

$$\langle x, u_j \rangle = \langle Tx, v_j \rangle = \langle Tx, Tu_j \rangle = \langle T^*Tx, u_j \rangle.$$

Since $\{u_i\}$ is total, the validity of this for all j implies that $x = T^*Tx$ for any $x \in H$. Hence,

$$T^*T = I = TT^*.$$

Thus we see that if T transforms some total orthonormal set $\{u_i\}$ onto a total orthonormal set $\{T(u_i)\}$, then T is unitary. ■

PROBLEM 6.2.15 Let H be a Hilbert space and W be the set of all unitary operators on H . Show that the multiplication in $BL(H)$ makes W a group. Show also that W is a closed subset of $BL(H)$. Is W a subspace of $BL(H)$?

Solution Given that W is the set of all $T \in BL(H)$ such that

$$TT^* = T^*T = I.$$

If A, B are in W , then

$$\begin{aligned} AB(AB)^* &= ABB^*A^* = AIA^* = I, \\ (AB)^*AB &= B^*A^*AB = B^*IB = I. \end{aligned}$$

Hence $AB \in W$. Multiplication in $BL(H)$ is associative. Since $I^* = I$, we see that $I \in W$. Since $(A^*)^* = A$, we see that $A^* \in W$ if $A \in W$. If $A \in W$ then A is invertible in $BL(H)$ and $A^{-1} = A^* \in W$. This shows that W is a group under the multiplication in $BL(H)$.

For the second part let $A \in \overline{W}$. Then there is a sequence $\{A_n\}$ in W such that $A_n \rightarrow A$ in the norm of $BL(H)$. So for all n , we have

$$\begin{aligned} I - AA^* &= A_nA_n^* - A_nA^* + A_nA^* - AA^* \\ &= A_n(A_n^* - A^*) + (A_n - A)A^*. \end{aligned}$$

Hence,

$$\begin{aligned} \|I - AA^*\| &\leq \|A_n(A_n^* - A^*)\| + \|(A_n - A)A^*\| \\ &\leq \|A_n\| \|A_n^* - A^*\| + \|A_n - A\| \|A^*\|. \end{aligned}$$

This tends to 0 as $n \rightarrow \infty$ since $\|A_n - A\| \rightarrow 0$. This is possible only if $I = AA^*$. Similarly,

$$\begin{aligned} \|I - A^*A\| &= \|A_n^*A_n - A_n^*A + A_n^*A - A^*A\| \\ &\leq \|A_n^*\| \|A_n - A\| + \|A_n^* - A^*\| \|A\| \\ &= \|A_n\| \|A_n - A\| + \|A_n - A\| \|A\| \rightarrow 0 \end{aligned}$$

as $n \rightarrow \infty$. This implies that $I = A^*A$. Hence, $A \in W$. Thus W is closed. Since the zero operator is not unitary, $0 \notin W$. So W cannot be a subspace. This completes the proof. ■

PROBLEM 6.2.16 A rigid motion of $\mathbb{R}^2$ is a map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, not necessarily linear, such that $\|T(x) - T(y)\| = \|x - y\|$ for all $x, y \in \mathbb{R}^2$. Prove:

- If T is a rigid motion of $\mathbb{R}^2$ such that $T(0) = 0$, then T is a linear unitary operator on $\mathbb{R}^2$.
- Every rigid motion of $\mathbb{R}^2$ is a translation of $\mathbb{R}^2$ followed by a rotation, possibly followed by a reflection in the x -axis.

Solution (a) Let T be a rigid motion such that $T(0) = 0$. Let $x \neq 0$. Then $x = (r \cos \theta, r \sin \theta)$, where $r > 0$. Let $e_1 = (1, 0)$, $e_2 = (0, 1)$,

$$y_1 = T(e_1), \quad y_2 = T(e_2), \quad y = T(x).$$

Since

$$\|T(z)\| = \|T(z) - T(0)\| = \|z - 0\| = \|z\|$$

for all $z \in \mathbb{R}^2$, we see that $\|y_1\| = \|y_2\| = 1$ and $\|y\| = r$. Hence we can write

$$y_1 = (\cos \alpha, \sin \alpha), \quad y_2 = (\cos \beta, \sin \beta), \quad y = (r \cos \phi, r \sin \phi).$$

Since

$$\|y - y_1\| = \|T(x) - T(e_1)\| = \|x - e_1\|,$$

we get

$$\begin{aligned} (r \cos \phi - \cos \alpha)^2 + (r \sin \phi - \sin \alpha)^2 &= (r \cos \theta - 1)^2 + (r \sin \theta)^2, \\ r^2 + 1 - 2r(\cos \phi \cos \alpha + \sin \phi \sin \alpha) &= r^2 + 1 - 2r \cos \theta. \end{aligned}$$

That is,

$$\cos(\phi - \alpha) = \cos \theta. \quad (9)$$

Similarly, since $\|y - y_2\| = \|x - e_2\|$, we obtain

$$r^2 + 1 - 2r(\cos \phi \cos \beta + \sin \phi \sin \beta) = (r \cos \theta)^2 + (r \sin \theta - 1)^2,$$

that is,

$$\cos(\phi - \beta) = \sin \theta. \quad (10)$$

Applying (9) with $x = e_2$, $y = y_2$, we get

$$\cos(\beta - \alpha) = \cos(\pi/2) = 0 \quad (11)$$

Thus, $\sin(\beta - \alpha) = \pm 1$.

From (9), (10) and (11), we have

$$\begin{aligned} \sin \theta &= \cos(\phi - \beta) \\ &= \cos(\phi - \alpha) \cos(\alpha - \beta) - \sin(\phi - \alpha) \sin(\alpha - \beta) \\ &= \sin(\phi - \alpha) \sin(\beta - \alpha). \end{aligned} \quad (12)$$

First suppose that $\sin(\beta - \alpha) = 1$. Then, $\sin \theta = \sin(\phi - \alpha)$ by (12). Hence, by (9),

$$\begin{aligned} \cos \alpha \cos \theta - \sin \alpha \sin \theta &= \cos \alpha \cos(\phi - \alpha) - \sin \alpha \sin(\phi - \alpha) = \cos \phi, \\ \sin \alpha \cos \theta + \cos \alpha \sin \theta &= \sin \alpha \cos(\phi - \alpha) + \cos \alpha \sin(\phi - \alpha) = \sin \phi. \end{aligned}$$

These give

$$\begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} r \cos \theta \\ r \sin \theta \end{pmatrix} = \begin{pmatrix} r \cos \phi \\ r \sin \phi \end{pmatrix}.$$

So the transform $y = T(x)$ is represented by the matrix

$$M = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}.$$

Next suppose that $\sin(\beta - \alpha) = -1$. Then, $\sin \theta = -\sin(\phi - \alpha)$ by (12). Hence, as before,

$$\cos \alpha \cos \theta + \sin \alpha \sin \theta = \cos \phi,$$

$$\sin \alpha \cos \theta - \cos \alpha \sin \theta = \sin \phi.$$

These give

$$\begin{pmatrix} \cos \alpha & \sin \alpha \\ \sin \alpha & -\cos \alpha \end{pmatrix} \begin{pmatrix} r \cos \theta \\ r \sin \theta \end{pmatrix} = \begin{pmatrix} r \cos \phi \\ r \sin \phi \end{pmatrix}.$$

The transform $y = T(x)$ is given by the matrix

$$M_1 = \begin{pmatrix} \cos \alpha & \sin \alpha \\ \sin \alpha & -\cos \alpha \end{pmatrix}$$

Thus we see that in both cases T is a linear map. If $\sin(\beta - \alpha) = 1$, T is represented by M , and in the other case T is represented by M_1 . Since $\|T(x)\| = \|x\|$ and $H = \mathbb{R}^2$ is finite dimensional, we see by Problem 6.2.12(b) that T is an isometry. This proves (a).

(b) If $T(0) = a$ let $T_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by

$$T_1(x) = T(x) - a, \quad x \in \mathbb{R}^2.$$

Then $T_1(0) = 0$ and

$$\|T_1(x) - T_1(y)\| = \|T(x) - T(y)\| = \|x - y\|.$$

Hence, T_1 is a rigid motion with $T_1(0) = 0$, and $T(x) = a + T_1(x)$. So T_1 is a linear isometry by (a). Thus it is represented by M or M_1 . The transformation represented by M maps $(r \cos \theta, r \sin \theta)$ to $(r \cos(\theta + \alpha), r \sin(\theta + \alpha))$; hence it is a rotation through an angle α . The transformation represented by M_1 maps $(r \cos \theta, r \sin \theta)$ to $(r \cos(\theta - \alpha), -r \sin(\theta - \alpha))$; hence it is a rotation through $-\alpha$ followed by reflection in the x -axis. If $b = T_1^{-1}(a)$, then $T(x) = T_1(b) + T_1(x) = T_1(x + b)$. Thus, T is a translation through b followed by T_1 . This completes the proof. ■

PROBLEM 6.2.17 Let H be a separable Hilbert space and $A \in BL(H)$. Suppose that the matrix (k_{ij}) represents A with respect to an orthonormal basis $\{u_n\}$. Show that

- (a) if H is infinite dimensional then A is unitary iff the rows, as well as the columns, of the matrix (k_{ij}) form an orthonormal basis for ℓ^2 ,
- (b) if H is finite dimensional and A is unitary then $|\det(k_{ij})| = 1$.

Solution (a) By Problem 6.1.11 we see that

- (i) the rows and columns of (k_{ij}) are elements of l^2 ,
- (ii) A^* is represented by $(\overline{k_{ji}})$, and
- (iii) AA^*, A^*A are represented by $(k_{ij})(\overline{k_{ji}}), (\overline{k_{ji}})(k_{ij})$, respectively.

Hence A is unitary iff

$$(k_{ij})(\overline{k_{ji}}) = (\overline{k_{ji}})(k_{ij}) = I, \text{ the unit matrix.}$$

But

$$(k_{ij})(\overline{k_{ji}}) = \text{the unit matrix} \quad (13)$$

means that the rows of (k_{ij}) are orthonormal, and

$$(\overline{k_{ji}})(k_{ij}) = \text{the unit matrix} \quad (14)$$

means that the columns of (k_{ij}) are orthonormal.

If $(k_n) \in l^2$ and (k_n) is orthogonal to all the columns, then

$$\sum_{i=1}^{\infty} \overline{k_i} k_{ij} = 0 \quad \text{for } j \geq 1;$$

hence,

$$(\overline{k_i})(k_{ij})(\overline{k_{ji}}) = 0.$$

So $(\overline{k_i}) = 0$ if (13) holds. This shows that the columns form a total orthonormal set if (13) holds. Similarly, if (14) holds, then the rows form a total orthonormal set. This finishes the first part.

(b) If H is of finite dimension n , then (k_{ij}) is a $n \times n$ square matrix. Let $\det(k_{ij}) = c$. Then,

$$\det(\overline{k_{ji}}) = \det(\overline{k_{ij}}) = \bar{c}.$$

Hence,

$$\det(k_{ij}) \cdot \det(\overline{k_{ji}}) = c\bar{c} = |c|^2.$$

But if A is unitary, then

$$\begin{aligned} \det(k_{ij}) \cdot \det(\overline{k_{ji}}) &= \det((k_{ij})(\overline{k_{ji}})) \\ &= \det(I) = 1. \end{aligned}$$

Thus, $|c| = 1$. This completes the proof. ■

PROBLEM 6.2.18 Let $A \in BL(H)$, where H is a Hilbert space. Prove the following:

(a) A is normal iff

$$\|Ax\| = \|A^*x\| \quad \text{for all } x \in H.$$

(b) If α, β are scalars such that $|\alpha| = |\beta|$ then $\alpha A + \beta A^*$ is normal.

(c) If A is normal then $\|A^2\| = \|A\|^2$.

(d) If A is normal then $\overline{R(A)} = \overline{R(A^*)}$.

Solution (a) For all $x \in H$, we have

$$\begin{aligned}\langle A^*Ax, x \rangle &= \langle Ax, Ax \rangle = \|Ax\|^2, \\ \langle AA^*x, x \rangle &= \langle A^*x, A^*x \rangle = \|A^*x\|^2.\end{aligned}$$

Hence, $\|Ax\| = \|A^*x\|$ iff

$$\langle A^*Ax, x \rangle = \langle AA^*x, x \rangle,$$

that is, $\langle Bx, x \rangle = 0$, where $B = A^*A - AA^*$. Since B is self-adjoint, we see by Problem 6.2.3 that $\langle Bx, x \rangle = 0$ for all $x \in H$ iff $B = 0$, that is, $A^*A = AA^*$ which means that A is normal.

(b) Let $B = \alpha A + \beta A^*$. Then,

$$\begin{aligned}BB^* &= (\alpha A + \beta A^*)(\bar{\alpha}A^* + \bar{\beta}A) \\ &= |\alpha|^2 AA^* + \alpha\bar{\beta}A^2 + \bar{\alpha}\beta(A^*)^2 + |\beta|^2 A^*A, \\ B^*B &= (\bar{\alpha}A^* + \bar{\beta}A)(\alpha A + \beta A^*) \\ &= |\alpha|^2 A^*A + \bar{\alpha}\beta(A^*)^2 + \alpha\bar{\beta}A^2 + |\beta|^2 AA^*.\end{aligned}$$

If $|\alpha| = |\beta|$ then $BB^* = B^*B$, that is, B is normal.

(c) For any $A \in BL(H)$, we have

$$\|A^2\| = \|A \cdot A\| \leq \|A\| \|A\| = \|A\|^2.$$

Let A be normal. For all $x \in H$,

$$\|Ax\|^2 = \langle Ax, Ax \rangle = \langle A^*Ax, x \rangle \leq \|A^*Ax\| \|x\|.$$

But $\|A^*(Ax)\| = \|A(Ax)\|$ by (a). Hence,

$$\begin{aligned}\|Ax\|^2 &\leq \|A^2x\| \|x\| \leq \|A^2\| \|x\|^2, \\ \|Ax\| &\leq \|A^2\|^{1/2} \|x\|.\end{aligned}$$

This implies that $\|A\| \leq \|A^2\|^{1/2}$, that is, $\|A\|^2 \leq \|A^2\|$. This proves (c).

(d) Let A be normal. Then, by (a),

$$A(x) = 0 \iff A^*(x) = 0;$$

hence $Z(A) = Z(A^*)$. Thus,

$$R(A^*)^\perp = R(A)^\perp \tag{15}$$

by Problem 6.1.15(c). Since $R(A)$ and $R(A^*)$ are subspaces, it follows from Problem 5.3.5(b) that

$$R(A)^{\perp\perp} = \overline{R(A)}, \quad R(A^*)^{\perp\perp} = \overline{R(A^*)}.$$

So (d) follows from (15). This completes the proof. ■

PROBLEM 6.2.19 Let H be a Hilbert space of dimension at least two. Show that there are operators on H which are not normal.

Solution Let E be an orthonormal basis for H . Then E contains at least two distinct elements u_1, u_2 . Let $Y = \text{span}\{u_1, u_2\}$ and $Z = Y^\perp$. Then every $x \in H$ can be uniquely expressed as $x = y + z$ with $y \in Y, z \in Z$, and hence as

$$x = au_1 + bu_2 + z, \quad a, b \in \mathbb{K}, z \in Z. \quad (16)$$

Let

$$A(u_1) = u_2, \quad A(u_2) = 2u_1, \quad A(z) = 0 \quad \text{for } z \in Z, \quad (17)$$

$$B(u_1) = 2u_2, \quad B(u_2) = u_1, \quad B(z) = 0 \quad \text{for } z \in Z. \quad (18)$$

These uniquely determine linear maps A, B on H . If x is given by (16), then

$$\|x\|^2 = |a|^2 + |b|^2 + \|z\|^2 \geq |a|^2 + |b|^2$$

by the Pythagoras theorem. Hence, by (17),

$$\|Ax\|^2 = \|au_2 + 2bu_1\|^2 = |a|^2 + |2b|^2 \leq 4\|x\|^2.$$

Thus A is bounded with $\|A\| \leq 4$. Similarly, B defined by (18), is bounded. Let x be given by (16) and

$$y = cu_1 + du_2 + z_1, \quad c, d \in \mathbb{K}, z_1 \in Z.$$

Then,

$$\langle Ax, y \rangle = \langle au_2 + 2bu_1, cu_1 + du_2 + z_1 \rangle.$$

Since u_1, u_2, z_1 are orthogonal, we get

$$\langle Ax, y \rangle = ad + 2bc.$$

Similarly,

$$\langle x, By \rangle = \langle au_1 + bu_2 + z, 2cu_2 + du_1 \rangle = ad + 2bc.$$

Thus we see that $\langle Ax, y \rangle = \langle x, By \rangle$ for all $x, y \in H$, that is, $B = A^*$. Since

$$AB(u_1) = A(2u_2) = 4u_1, \quad BA(u_1) = B(u_2) = u_1,$$

we see that $AB \neq BA$, that is, $AA^* \neq A^*A$. Hence A is not normal. This completes the proof. ■

PROBLEM 6.2.20 Let H be a Hilbert space and W be the set of all normal operators on H . Prove the following:

- (a) W is a closed subset of $BL(H)$.
- (b) W cannot be a proper subspace of $BL(H)$.
- (c) If the dimension of H is at least two, then W cannot be a subspace of $BL(H)$.

Solution (a) Let $A \in \overline{W}$. Then there exist $A_n \in W$ such that $A_n \rightarrow A$ in the norm of $BL(H)$. So $A_n A_n^* = A_n^* A_n$, and hence

$$\begin{aligned} \|AA^* - A^*A\| &\leq \|AA^* - A_n A_n^*\| + \|A_n A_n^* - A_n A_n^*\| + \|A_n A_n^* - A_n^* A_n\| \\ &\quad + \|A_n^* A_n - A_n^* A\| + \|A_n^* A - A^* A\| \\ &= \|(A - A_n)A^*\| + \|A_n(A^* - A_n^*)\| + 0 \\ &\quad + \|A_n^*(A_n - A)\| + \|(A_n^* - A^*)A\| \\ &\leq \|A - A_n\| \|A\| + \|A_n\| \|A - A_n\| + \|A_n\| \|A_n - A\| \\ &\quad + \|A_n - A\| \|A\|, \end{aligned}$$

where we have used the fact that

$$\|ST\| \leq \|S\| \|T\|, \quad \|T^*\| = \|T\|.$$

Since $\|A_n - A\| \rightarrow 0$ and $\|A_n\| \rightarrow \|A\|$ as $n \rightarrow \infty$, we must have

$$AA^* - A^*A = 0,$$

that is, $A \in W$. Thus W is closed.

(b) Suppose that $W \neq BL(H)$. Then, there is $T \in BL(H)$ such that T is not normal. Let

$$A = \frac{1}{2}(T + T^*), \quad B = \frac{1}{2}(T - T^*)$$

Then $A^* = A$ and $B^* = -B$. So A and B are normal whereas $A + B = T$ is not normal. Thus, $A, B \in W$ but $A + B \notin W$. Therefore, W cannot be subspace unless $W = BL(H)$.

(c) If the dimension of H is at least two, then $W \neq BL(H)$ by Problem 6.2.19. So the result follows from (b). This completes the proof. ■

PROBLEM 6.2.21 Let H be a Hilbert space. Show that if A and B are normal operators on H and $AB^* = B^*A$, then $A + B$ and AB are normal. Show also that $A, B, A + B$ and AB can be normal even though $AB^* \neq B^*A$.

Solution Let A and B be normal and $AB^* = B^*A$. Then, $(AB^*)^* = (B^*A)^*$, that is, $BA^* = A^*B$. Hence,

$$\begin{aligned} (A + B)(A + B)^* &= (A + B)(A^* + B^*) = AA^* + AB^* + BA^* + BB^*, \\ (A + B)^*(A + B) &= (A^* + B^*)(A + B) = A^*A + A^*B + B^*A + B^*B \\ &= AA^* + BA^* + AB^* + BB^*. \end{aligned}$$

These give

$$(A + B)(A + B)^* = (A + B)^*(A + B).$$

Again,

$$(AB)(AB)^* = ABB^*A^* = AB^*BA^*,$$

$$\begin{aligned}(AB)^*(AB) &= B^*A^*AB = B^*AA^*B \\ &= AB^*BA^*.\end{aligned}$$

Therefore, $(AB)(AB)^* = (AB)^*(AB)$, that is, AB is normal. This proves the first part.

For the second part we consider two self-adjoint operators A and B such that $BA = -AB \neq 0$. For instance, we can take $H = \mathbb{K}^2$ and let A and B be represented by the 2×2 matrices

$$M_1 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

respectively. Since these matrices are real and symmetric, the operators A and B are self-adjoint by Problem 6.1.10. Since

$$M_1M_2 = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix} = -M_2M_1,$$

we see that $AB = -BA \neq 0$.

Therefore, $AB^* - B^*A = AB - BA = 2AB \neq 0$. Thus $A + B$, being self-adjoint, is normal. Since $(AB)^* = B^*A^* = BA = -AB$, we have

$$(AB)^*(AB) = -(AB)^2 = (AB)(AB)^*.$$

Thus AB also is normal. This completes the proof. ■

PROBLEM 6.2.22 Let $k(s, t)$ be a square-integrable function on the unit square $[0, 1] \times [0, 1]$. For $x \in L^2[0, 1]$ let

$$(Ax)(s) = \int_0^1 k(s, t)x(t) dt, \quad 0 \leq s \leq 1.$$

Show that this defines a bounded operator A on $L^2[0, 1]$. Show also that:

- (a) A self-adjoint if $k(s, t) = \overline{k(t, s)}$ for all (s, t)
- (b) A is normal if

$$\int_0^1 k(s, u)\overline{k(t, u)} du = \int_0^1 \overline{k(u, s)} k(u, t) du \quad (19)$$

for all (s, t) .

Solution For $x \in L^2[0, 1]$, $0 \leq s \leq 1$, we have

$$|(Ax)(s)| \leq \int_0^1 |k(s, t)||x(t)| dt \leq \left(\int_0^1 |k(s, t)|^2 dt \right)^{1/2} \left(\int_0^1 |x(t)|^2 dt \right)^{1/2}.$$

Hence,

$$\|Ax\|^2 = \int_0^1 |Ax(s)|^2 ds \leq \int_0^1 ds \int_0^1 |k(s, t)|^2 dt \|x\|^2 = \|k\|_2^2 \|x\|^2.$$

This first of all shows that $A(x) \in L^2[0, 1]$. It also shows that A is bounded and $\|A\| \leq \|k\|_2$, the L^2 -norm of k on $[0, 1] \times [0, 1]$. Since A is easily seen to be linear, the first part is complete.

For the second part, let

$$k_1(s, t) = \overline{k(t, s)}, \quad k_2(s, t) = |k(s, t)|, \quad k_3(s, t) = |k_1(s, t)|$$

and, for $x \in L^2[0, 1]$,

$$B_i(x)(s) = \int_0^1 k_i(s, t)x(t) dt, \quad 0 \leq s \leq 1, i = 1, 2, 3. \quad (20)$$

By the same argument as above, we see that B_1, B_2 and B_3 are bounded linear operators on $H = L^2[0, 1]$. Exactly as in the second part of solution of Problem 6.1.12, we see that

$$\langle x, B_1 y \rangle = \langle Ax, y \rangle \quad \text{for all } x, y \in L^2[0, 1].$$

Thus, $B_1 = A^*$. So if $k(s, t) = k_1(s, t)$, then $A = B_1 = A^*$, that is, A self-adjoint. This proves (a).

For (b), suppose that (19) holds for all s, t . For $x \in L^2[0, 1]$, we have, by (20),

$$\begin{aligned} AB_1(x)(s) &= \int_0^1 k(s, u)B_1(x)(u) du \\ &= \int_0^1 k(s, u) \left\{ \int_0^1 k_1(u, t)x(t) dt \right\} du. \end{aligned}$$

We wish to change the order of integration. Now,

$$\begin{aligned} \int_0^1 du \int_0^1 |k(s, u)k_1(u, t)x(t)| dt &= \int_0^1 k_2(s, u) du \int_0^1 k_3(u, t)|x(t)| dt \\ &= \int_0^1 k_2(s, u) du B_3(|x|)(u) \\ &= B_2(B_3|x|)(s), \end{aligned}$$

by (20). If this is finite, we can change the order of integration by Fubini's theorem. But $B_2B_3(|x|)$ exists in $L^2[0, 1]$. So, $B_2B_3(|x|)(s)$ is finite for almost all $s \in [0, 1]$; for such s , we get

$$AB_1(x)(s) = \int_0^1 \left\{ \int_0^1 k(s, u)k_1(u, t) du \right\} x(t) dt \quad (21)$$

Similarly,

$$B_1A(x)(s) = \int_0^1 k_1(s, u)A(x)(u) du$$

$$\begin{aligned} &= \int_0^1 k_1(s, u) \left\{ \int_0^1 k(u, t) x(t) dt \right\} du \\ &= \int_0^1 \left\{ \int_0^1 k_1(s, u) k(u, t) du \right\} x(t) dt \end{aligned} \quad (22)$$

for almost all $s \in [0, 1]$. So, if (19) holds, then combining (21) and (22), we see that $AB_1(x)(s) = B_1A(x)(s)$ for almost all $s \in [0, 1]$. This means that $AB_1(x) = B_1A(x)$. This, being true for all $x \in L^2[0, 1]$, implies that $AB_1 = B_1A$, that is, $AA^* = A^*A$. So A is normal. ■

7

Spectral Theory in Hilbert Spaces

This chapter is devoted to spectral theory of bounded operators on Hilbert spaces. In Section 7.1 we consider spectral properties in general. In the case of compact self-adjoint operators, the eigenvalue and eigenvectors play a significant role in the so-called spectral representation of the operator. This concept is introduced in Section 7.2.

7.1 The Spectrum and Numerical Range

In Section 4.1 we introduced the concept of spectral values of operators on normed spaces. These concepts for operators on Hilbert spaces are recalled here.

Let H be a Hilbert space and $A \in BL(H)$. A scalar $k \in \mathbb{K}$ is said to be a *spectral value* of A if $A - kI$ is not invertible in $BL(H)$. We say that k is an *eigenvalue* of A if $A - kI$ is not one-one, and k is an *approximate eigenvalue* of A if $A - kI$ is not bounded below, that is, there is no $\alpha > 0$ such that

$$\alpha\|x\| \leq \|Ax - kx\| \quad \text{for all } x \in H.$$

The *spectrum* $\sigma(A)$ of A is the set of all spectral values of A ; the *eigen-spectrum* $\sigma_e(A)$ is the set of all eigenvalues of A . The set of all approximate eigenvalues of A is denoted by $\sigma_a(A)$. The set

$$W(A) = \{\langle Ax, x \rangle : x \in H, \|x\| = 1\}$$

is called the *numerical range* of A .

The *eigenspace* of A corresponding to k is the space $Z(A - kI)$. If k is an eigenvalue, the corresponding eigenspace contains nonzero vectors; in that

case, the nonzero vectors in $Z(A - kI)$ are called *eigenvectors* corresponding to k .

Let A be self-adjoint. Then, $\langle Ax, x \rangle = \langle x, Ax \rangle$ is real for all $x \in H$. Let

$$m_A = \inf_{\|x\|=1} \langle Ax, x \rangle, \quad M_A = \sup_{\|x\|=1} \langle Ax, x \rangle.$$

Then m_A and M_A are spectral values of A , and

$$\|A\| = \max\{|m_A|, |M_A|\} = \sup_{\|x\|=1} |\langle Ax, x \rangle|.$$

[See Problems 7.1.3 and 7.1.4.]

Let H be a Hilbert space of finite dimension n . Let $u_1, \dots, u_n$ be an orthonormal basis for H . An operator $A \in BL(H)$ is represented by the matrix $M = (k_{ij})$, where $k_{ij} = \langle Au_j, u_i \rangle$. If I is the $n \times n$ unit matrix, then

$$p(t) = \det(M - tI)$$

is a polynomial in t of degree n . Hence, $p(t)$ has n roots in $\mathbb{C}$. The spectral values of A are the roots of this polynomial; these roots are also eigenvalues of A . The polynomial $p(t)$ is called the *characteristic polynomial* of A . If A is self-adjoint, then the roots of $p(t)$ are real.

EXERCISES

1. If I is the identity operator on a Hilbert space H , find $\sigma(I)$ and $W(I)$.
2. Let $H = \mathbb{R}^2$ and A be represented by the matrix $M = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$. Find $\sigma(A)$, $W(A)$, m_A , M_A and $\|A\|$.
3. Let $H = \mathbb{R}^3$ and A be represented by the matrix

$$M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Find $\sigma(A)$, m_A , M_A and $\|A\|$.

4. Let $H = \mathbb{R}^2$ and A be represented by the matrix $M = \begin{pmatrix} 3 & 2 \\ -1 & 0 \end{pmatrix}$. Find the eigenvalues and the corresponding eigenspaces Y_1, Y_2 for A . Show that every $x \in H$ is expressible as $x = y_1 + y_2$ with $y_i \in Y_i$. Is this expression unique?
5. Let $H = \mathbb{R}^2$ and A be represented by the matrix $M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Show that H does not have a basis consisting of eigenvectors of A .

6. Let $H = \mathbb{R}^3$ and $A \in BL(H)$ be represented by the matrix

$$M = \begin{pmatrix} 3 & 2 & 1 \\ -1 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Find the eigenvalues and the corresponding eigenspaces.

PROBLEMS

PROBLEM 7.1.1 Let $A \in BL(H)$, where H is a Hilbert space. Prove the following:

- (a) $\sigma(A^*) = \{k : \bar{k} \in \sigma(A)\}$.
- (b) $\sigma(A^*) = \sigma_e(A^*) \cup \{k : \bar{k} \in \sigma_a(A)\}$.
- (c) $\sigma(A^*) = \sigma_a(A^*) \cup \{k : \bar{k} \in \sigma_e(A)\}$.

Solution (a) For any $k \in \mathbb{K}$ the operator $A - kI$ is invertible in $BL(H)$ iff

$$(A - kI)B = I = B(A - kI) \tag{1}$$

for some $B \in BL(H)$. But (1) holds iff

$$B^*(A - kI)^* = I = (A - kI)^* B^*,$$

that is,

$$B^*(A^* - \bar{k}I) = I = (A^* - \bar{k}I)B^*.$$

Therefore, we see that $A - kI$ is invertible in $BL(H)$ iff $A^* - \bar{k}I$ is invertible in $BL(H)$. Thus,

$$k \notin \sigma(A) \iff \bar{k} \notin \sigma(A^*)$$

This gives (a).

(b) Since $\sigma_e(A^*) \subset \sigma(A^*)$ and

$$\bar{k} \in \sigma_a(A) \implies \bar{k} \in \sigma(A) \implies k \in \sigma(A^*)$$

by (a), we see that

$$\sigma_e(A^*) \cup \{k : \bar{k} \in \sigma_a(A)\} \subset \sigma(A^*). \tag{2}$$

To prove the reverse inclusion suppose that $k \notin \sigma_e(A^*)$ and $\bar{k} \notin \sigma_a(A)$. Then the zero space $Z(A^* - kI)$ is $\{0\}$ and $A - \bar{k}I$ is bounded below. In Problem 6.1.15(c) we proved that $Z(T) = R(T^*)^\perp$. Using this with $T = A^* - kI$, we see that

$$R(A - \bar{k}I)^\perp = Z(A^* - kI) = \{0\}.$$

Consequently,

$$H = \{0\}^\perp = R(A - \bar{k}I)^{\perp\perp} = \overline{R(A - \bar{k}I)}, \quad (3)$$

where the last equality follows from Problem 5.3.5(b) with $M = R(A - \bar{k}I)$. Let y be any element of H . Because of (3) there exist y_n such that $y_n \rightarrow y$ and $y_n = (A - \bar{k}I)(x_n)$. Since $A - \bar{k}I$ is bounded below there is $\alpha > 0$ such that

$$\alpha\|x\| \leq \|Ax - \bar{k}x\| \quad \text{for } x \in H \quad (4)$$

Hence,

$$\alpha\|x_n - x_m\| \leq \|A(x_n - x_m) - \bar{k}(x_n - x_m)\| = \|y_n - y_m\|$$

which shows that $\{x_n\}$ is a Cauchy sequence in H . So $\{x_n\}$ converges to some $x \in H$. Therefore,

$$\begin{aligned} y_n &= A(x_n) - \bar{k}x_n \rightarrow Ax - \bar{k}x, \\ y &= Ax - \bar{k}x \in R(A - \bar{k}I). \end{aligned}$$

Thus we have proved that $A - \bar{k}I$ is onto. Since it is one-one by (4), it has an inverse B which is linear. From (4) we also have

$$\alpha\|Bx\| \leq \|(A - \bar{k}I)Bx\| = \|x\|.$$

This shows that $(A - \bar{k}I)^{-1} = B \in BL(H)$. Hence, $\bar{k} \notin \sigma(A)$ and so $k \notin \sigma(A^*)$, by (a). This combined with (2) proves (b).

(c) Changing A to A^* in (b), we get

$$\sigma(A) = \sigma_e(A) \cup \{k : \bar{k} \in \sigma_a(A^*)\}.$$

Hence, $\bar{k} \in \sigma(A)$ iff $\bar{k} \in \sigma_e(A)$ or $k \in \sigma_a(A^*)$, that is,

$$k \in \{k : \bar{k} \in \sigma_e(A)\} \cup \sigma_a(A^*).$$

Since $k \in \sigma(A^*)$ iff $\bar{k} \in \sigma(A)$ by (a), the proof is complete. ■

PROBLEM 7.1.2 Let A be a normal operator on a Hilbert space H . Prove that

- (a) $Z(A - kI) = Z(A^* - \bar{k}I)$ for $k \in \mathbb{K}$,
- (b) $\sigma(A) = \sigma_a(A)$,
- (c) eigenvectors corresponding to distinct eigenvalues are orthogonal.

Solution (a) It can be seen that $A - kI$ is normal since A is normal. If T is normal then $\|Tx\| = \|T^*x\|$ for all x by Problem 6.2.18(a). Hence,

$$\|(A - kI)x\| = \|(A - kI)^*x\| = \|(A^* - \bar{k}I)x\|$$

for all x . So,

$$(A - kI)(x) = 0 \iff (A^* - \bar{k}I)(x) = 0.$$

This gives (a).

(b) It is clear that $\sigma_a(A) \subset \sigma(A)$. For the reverse inclusion, let $k \in \sigma(A)$. Then, $\bar{k} \in \sigma(A^*)$ by Problem 7.1.1(a). So by Problem 7.1.1(b),

$$\bar{k} \in \sigma_e(A^*) \quad \text{or} \quad k \in \sigma_a(A).$$

If $\bar{k} \in \sigma_e(A^*)$, then $Z(A^* - \bar{k}I) \neq \{0\}$, and so, by (a), $Z(A - kI) \neq \{0\}$, that is, $k \in \sigma_e(A)$. But $\sigma_e(A) \subset \sigma_a(A)$. It follows that $k \in \sigma_a(A)$ in any case. This proves (b).

(c) Let x_1 and x_2 be eigenvectors corresponding to two distinct eigenvalues k_1 and k_2 respectively. Then,

$$x_1 \in Z(A - k_1I), \quad x_2 \in Z(A - k_2I) = Z(A^* - \bar{k}_2I).$$

Hence, $Ax_1 - k_1x_1 = 0$ and $A^*x_2 - \bar{k}_2x_2 = 0$. Therefore,

$$\begin{aligned} k_1 \langle x_1, x_2 \rangle &= \langle k_1x_1, x_2 \rangle = \langle Ax_1, x_2 \rangle \\ &= \langle x_1, A^*x_2 \rangle = \langle x_1, \bar{k}_2x_2 \rangle \\ &= k_2 \langle x_1, x_2 \rangle. \end{aligned}$$

Since $k_1 \neq k_2$, this implies that $\langle x_1, x_2 \rangle = 0$. This gives (c) and completes the proof. ■

PROBLEM 7.1.3 Let A be a bounded self-adjoint operator on a Hilbert space H and

$$m_A = \inf_{\|x\|=1} \langle Ax, x \rangle, \quad M_A = \sup_{\|x\|=1} \langle Ax, x \rangle.$$

Prove that

$$\|A\| = \sup_{\|x\|=1} |\langle Ax, x \rangle| = \max(|m_A|, |M_A|).$$

Solution Since the result is trivial when $A = 0$, we may suppose that A is nonzero. Write $\langle A(z), z \rangle = \phi(z)$ for $z \in H$ and let $K = \sup_{\|x\|=1} |\phi(x)|$. Since $\phi(kz) = |k|^2\phi(z)$ for all z , we see that if $z \neq 0$ and $x = z/\|z\|$, then $|\phi(z)| = \|z\|^2|\phi(x)|$. Thus,

$$|\phi(z)| \leq K\|z\|^2 \quad \text{for all } z \in H. \tag{5}$$

Since

$$|\phi(x)| = |\langle Ax, x \rangle| \leq \|Ax\|\|x\| \leq \|A\|\|x\|^2 = \|A\| \quad \text{for } \|x\| = 1,$$

we first observe that $K \leq \|A\|$.

To prove the reverse inequality, let $x \in H$ with $\|x\| = 1$. Suppose that $\|A(x)\| \neq 0$, $v = \alpha x$ and $w = (1/\alpha)A(x)$, where $\alpha = \sqrt{\|A(x)\|}$. Then,

$$\|v\|^2 = \alpha^2 \quad \text{and} \quad \|w\|^2 = \alpha^{-2}\|A(x)\|^2 = \alpha^2$$

and hence,

$$\|v\|^2 = \|w\|^2 = \|A(x)\|. \quad (6)$$

Further,

$$\begin{aligned} \langle Av + Aw, v + w \rangle &= \langle Av, v \rangle + \langle Av, w \rangle + \langle Aw, v \rangle + \langle Aw, w \rangle, \\ \phi(v + w) &= \phi(v) + \langle Av, w \rangle + \langle Aw, v \rangle + \phi(w). \end{aligned}$$

Similarly,

$$\phi(v - w) = \phi(v) - \langle Av, w \rangle - \langle Aw, v \rangle + \phi(w).$$

But

$$\langle Aw, v \rangle = \langle w, Av \rangle = \langle \alpha^{-1}A(x), \alpha A(x) \rangle = \|A(x)\|^2$$

and

$$\langle Av, w \rangle = \langle \alpha A(x), \alpha^{-1}A(x) \rangle = \|A(x)\|^2.$$

Thus,

$$\begin{aligned} \phi(v + w) - \phi(v - w) &= 2\|A(x)\|^2 + 2\|A(x)\|^2, \\ 4\|A(x)\|^2 &\leq |\phi(v + w)| + |\phi(v - w)|. \end{aligned}$$

Using (5) and (6), we get

$$\begin{aligned} 4\|A(x)\|^2 &\leq K \{ \|v + w\|^2 + \|v - w\|^2 \} = K \{ 2\|v\|^2 + 2\|w\|^2 \} \\ &= K \{ 2\|A(x)\| + 2\|A(x)\| \}. \end{aligned}$$

This shows that $\|A(x)\|^2 \leq K\|A(x)\|$, that is, $\|A(x)\| \leq K$. This holds whenever $\|x\| = 1$, whether $\|A(x)\|$ is zero or not. Consequently, $\|A\| \leq K$. This proves that $\|A\| = K$.

To prove that $K = \max(|m_A|, |M_A|)$, note that $-\phi(x) \leq -m_A \leq |m_A|$ and $\phi(x) \leq |M_A|$ for $\|x\| = 1$. Since $\phi(x)$ is real, this implies that $|\phi(x)| \leq \max(|m_A|, |M_A|)$. Thus, $K \leq \max(|m_A|, |M_A|)$. On the other hand, there exist x_n such that $\|x_n\| = 1$ and $\phi(x_n) \rightarrow M_A$ and so, $|\phi(x_n)| \rightarrow |M_A|$. Similarly, there exist y_n such that $\|y_n\| = 1$ and $|\phi(y_n)| \rightarrow |m_A|$. It follows that $K \geq \max(|m_A|, |M_A|)$. This proves the second part. ■

PROBLEM 7.1.4 Let A be a bounded self-adjoint operator on a Hilbert space H and m_A, M_A be as in Problem 7.1.3. Prove that m_A and M_A are spectral values of A . Deduce that $\|A\|$ or $-\|A\|$ is a spectral value of A .

Solution Since the result is trivial when $A = 0$, we may suppose that A is nonzero. First consider a special case in which $m_A = 0 < M_A$ and write M instead of M_A . Thus, $M = \sup_{\|x\|=1} |\langle Ax, x \rangle| = \|A\|$ by Problem 7.1.3. There exist sequences $\{k_n\}$, $\{x_n\}$ such that

$$k_n = \langle Ax_n, x_n \rangle, \quad \|x_n\| = 1, \quad k_n \rightarrow M.$$

Since $\|Ax_n - Mx_n\|^2 = \langle Ax_n - Mx_n, Ax_n - Mx_n \rangle$, we see that

$$\|Ax_n - Mx_n\|^2 = \langle Ax_n, Ax_n \rangle - M\langle Ax_n, x_n \rangle - M\langle x_n, Ax_n \rangle + M^2\langle x_n, x_n \rangle.$$

Since $\langle x_n, Ax_n \rangle = \langle Ax_n, x_n \rangle$, this gives

$$\|Ax_n - Mx_n\|^2 = \|Ax_n\|^2 - 2M\langle Ax_n, x_n \rangle + M^2$$

But $\|Ax_n\| \leq \|A\| = M$ and $\langle Ax_n, x_n \rangle = k_n$.

It follows that

$$\|Ax_n - Mx_n\|^2 \leq 2M^2 - 2k_nM \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

This shows that $M \in \sigma_a(A) \subset \sigma(A)$.

Generally, it is observed that if $B = A - \lambda I$, then

$$k \in W(A) \Leftrightarrow k - \lambda \in W(B) \quad \text{and} \quad k \in \sigma(A) \Leftrightarrow k - \lambda \in \sigma(B).$$

Hence, $m_B = m_A - \lambda$ and $M_B = M_A - \lambda$. Thus, if $B = A - m_A I$, then $m_B = 0 < M_A - m_A = M_B$. By the special case already proved, we see that $M_B \in \sigma(B)$ and so, $M_A = M_B + m_A \in \sigma(A)$.

Finally, let $B = -A$. Then,

$$k \in W(A) \Leftrightarrow -k \in W(B) \quad \text{and} \quad k \in \sigma(A) \Leftrightarrow -k \in \sigma(B).$$

Hence, $M_B = -m_A$. In this way, we get $-m_A \in \sigma(B)$ and so, $m_A \in \sigma(A)$.

For the last part, since $\|A\| = \max(|m_A|, |M_A|)$ by Problem 7.1.3, we find that there is a spectral value k , namely, m_A or M_A such that $|k| = \|A\| \neq 0$. Since the spectral values of A are real, we see that $k = \|A\|$ or $-\|A\|$. This completes the proof. ■

PROBLEM 7.1.5 Let $A \in BL(H)$ where H is a Hilbert space and, $W(A)$ be the numerical range of A . Prove the following:

- (a) $\sigma(A) \subset \overline{W(A)}$.
- (b) A is self-adjoint $\implies \sigma(A) \subset \mathbb{R}$.
- (c) converse of (b) is false.
- (d) Let A be self-adjoint; then A is positive iff its spectrum consists of nonnegative real values only.

Solution (a) Let $k \in \sigma(A)$. Then $\bar{k} \in \sigma(A^*)$ by Problem 7.1.1(a), and hence $\bar{k} \in \sigma_e(A^*)$ or $k \in \sigma_a(A)$ by Problem 7.1.1(b). If $\bar{k} \in \sigma_e(A^*)$, then there is x such that

$$A^*(x) = \bar{k}x, \quad \|x\| = 1;$$

hence

$$k = k\langle x, x \rangle = \langle x, \bar{k}x \rangle = \langle x, A^*x \rangle = \langle Ax, x \rangle \in W(A).$$

Let $k \in \sigma_a(A)$. Then there exist x_n with $\|x_n\| = 1$ and $\|Ax_n - kx_n\| \rightarrow 0$ as $n \rightarrow \infty$. Let $k_n = \langle Ax_n, x_n \rangle$. Then $k_n \in W(A)$, and

$$k_n - k = \langle Ax_n, x_n \rangle - k\langle x_n, x_n \rangle = \langle Ax_n - kx_n, x_n \rangle.$$

So,

$$|k_n - k| \leq \|Ax_n - kx_n\| \|x_n\| \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Hence, $k_n \rightarrow k, k \in \overline{W(A)}$. This gives (a).

(b) Let A be self-adjoint. Then,

$$\langle Ax, x \rangle = \langle x, Ax \rangle = \overline{\langle Ax, x \rangle};$$

hence $\langle Ax, x \rangle$ is real for all x . Thus, $W(A) \subset \mathbb{R}$. But $\mathbb{R}$ is a closed subset of $\mathbb{K}$. So, $\overline{W(A)} \subset \mathbb{R}$; and $\sigma(A) \subset \mathbb{R}$ by (a).

(c) Let $H = \mathbb{C}^2$ and A be represented by the 2×2 matrix $M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

Then A^* is represented by the conjugate of the transpose of M , which is different from M . So A is not self-adjoint. The spectral values of A are the roots of the characteristic equation, namely,

$$\begin{vmatrix} 1-t & 1 \\ 0 & 1-t \end{vmatrix} = 0,$$

giving the solution $t = 1$. So, $\sigma(A) = \{1\} \subset \mathbb{R}$.

(d) Let A be positive. Then, $\langle Ax, x \rangle \geq 0$ for all $x \in H$. So $W(A) \subset [0, \infty)$. But $[0, \infty)$ is a closed subset of $\mathbb{K}$. Therefore, by (a),

$$\sigma(A) \subset \overline{W(A)} \subset [0, \infty).$$

Conversely, suppose that $\sigma(A) \subset [0, \infty)$. If $\inf \{\langle Ax, x \rangle : \|x\| = 1\} = m_A$, then m_A is a spectral value, and so $m_A \geq 0$. If $x \neq 0$ and $\alpha = 1/(\|x\|)$, then $\|\alpha x\| = 1$, and so

$$0 \leq m_A \leq \langle A(\alpha x), \alpha x \rangle = \alpha^2 \langle Ax, x \rangle.$$

It follows that $\langle Ax, x \rangle \geq 0$ for all $x \in H$, that is, A is positive. This completes the proof. ■

PROBLEM 7.1.6 Let A be a unitary operator on a Hilbert space H . If $\sigma(A)$ and $W(A)$ are its spectrum and numerical range, show that

(a) $W(A) \subset \{k : |k| \leq 1\}$,

(b) $\sigma(A) \subset \{k : |k| = 1\}$.

Solution (a) Let $k \in W(A)$. Then $k = \langle Ax, x \rangle$ for some x with $\|x\| = 1$. Since

$$\|Ax\|^2 = \langle Ax, Ax \rangle = \langle x, A^*Ax \rangle = \langle x, x \rangle = 1,$$

we get

$$|k| = |\langle Ax, x \rangle| \leq \|Ax\| \|x\| = 1.$$

This gives (a).

(b) Let $k \in \sigma(A)$. Then $k \in \overline{W(A)}$ by Problem 7.1.5(a). Hence, $|k| \leq 1$ by (a). Since $AA^* = I = A^*A$, we see that $A^{-1} = A^*$ exists in $BL(H)$. So $k \neq 0$. Since

$$A - kI = A - kAA^* = -kA(A^* - \frac{1}{k}I),$$

we see that if $A^* - (1/k)I$ is invertible in $BL(H)$ so is $A - kI$. But $A - kI$ is not invertible since $k \in \sigma(A)$. So $A^* - (1/k)I$ is not invertible, that is, $1/k \in \sigma(A^*)$. Therefore, $1/k \in \overline{W(A^*)}$. Since A^* is unitary, we can use (a) with A^* instead of A . This gives

$$|\frac{1}{k}| \leq 1, \quad |k| \geq 1.$$

Thus, $|k| = 1$, and the proof is complete. ■

PROBLEM 7.1.7 Let $A \in BL(H)$, where H is a Hilbert space. Prove that A has an eigenvalue λ with $|\lambda| = \|A\|$ iff there is $x \in H$ with $\|x\| = 1$ and $|\langle Ax, x \rangle| = \|A\|$.

Solution Suppose that A has an eigenvalue λ with $|\lambda| = \|A\|$. Then, there exists $x \in H$ with $Ax = \lambda x$ and $\|x\| = 1$. For this x , we have

$$\begin{aligned} \langle Ax, x \rangle &= \langle \lambda x, x \rangle = \lambda \langle x, x \rangle = \lambda, \\ |\langle Ax, x \rangle| &= |\lambda| = \|A\|. \end{aligned}$$

Conversely, suppose that

$$|\langle Ax, x \rangle| = \|A\| \quad \text{for some } x \in H \text{ with } \|x\| = 1.$$

Let $\lambda = \langle Ax, x \rangle$. Then,

$$\langle Ax - \lambda x, Ax - \lambda x \rangle = \langle Ax, Ax \rangle - \langle Ax, \lambda x \rangle - \langle \lambda x, Ax \rangle + \langle \lambda x, \lambda x \rangle.$$

Hence,

$$\begin{aligned} \|Ax - \lambda x\|^2 &= \|Ax\|^2 - \bar{\lambda}\lambda - \lambda\bar{\lambda} + \lambda\bar{\lambda} = \|Ax\|^2 - |\lambda|^2 \\ &\leq \|A\|^2\|x\|^2 - \|A\|^2 = 0. \end{aligned}$$

Thus, $Ax = \lambda x$, and λ is an eigenvalue. This completes the proof. ■

PROBLEM 7.1.8 Let H be a complex Hilbert space and A be a normal operator on H . Prove that if $\sigma(A) = \{0\}$, then $A = 0$. Show also that this may be false in the following cases:

- (i) A is not normal.
- (ii) H is a real Hilbert space.

Solution Since H is a complex Banach space, the spectral radius formula holds:

$$\lim_{n \rightarrow \infty} \|A^n\|^{1/n} = \sup \{|k| : k \in \sigma(A)\}. \quad (7)$$

Since A is normal, $\|A^2\| = \|A\|^2$ by Problem 6.2.18(c). Since A^2 also is normal, we get $\|A^4\| = \|A\|^4$. By induction,

$$\|A^n\|^{1/n} = \|A\| \quad \text{for } n = 2^j, j = 1, 2, \dots \quad (8)$$

The formula (7) asserts the existence of the limit. So the limit must be $\|A\|$ by (8). Hence if $\sigma(A) = \{0\}$, the right-hand side of (7) is 0. Therefore, $\|A\| = 0$, that is, $A = 0$.

For the second part, first consider case (i). Let $H = \mathbb{C}^2$ and A be the linear map on H represented by the matrix $M = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Its spectral values are the roots of the characteristic equation

$$\begin{vmatrix} 0-t & 1 \\ 0 & 0-t \end{vmatrix} = 0,$$

that is, $t^2 = 0$. Hence, $\sigma(A) = \{0\}$ even though A is not zero.

For case (ii), let $\mathbb{K} = \mathbb{R}$, $H = \mathbb{R}^3$ and $A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear map represented by the matrix

$$M_1 = \begin{pmatrix} 0 & 1 & 1 \\ -1 & 0 & 1 \\ -1 & -1 & 0 \end{pmatrix}.$$

Then A^* is represented by the transpose of M_1 , which is $-M_1$. Hence, A^*A and AA^* are both represented by $-M_1^2$. Thus A is normal.

The spectral values of A are the roots of the characteristic equation

$$\begin{vmatrix} 0-t & 1 & 1 \\ -1 & 0-t & 1 \\ -1 & -1 & 0-t \end{vmatrix} = 0.$$

This reduces to $t(t^2 + 3) = 0$. It has only one root, namely 0, in the scalar field $\mathbb{R}$. So, $\sigma(A) = \{0\}$. Thus A is a nonzero normal operator with $\sigma(A) = 0$. This completes the proof. ■

PROBLEM 7.1.9 Show that a unitary operator on a Hilbert space need not have an eigenvalue.

Solution Let $H = l^2$, e_n be the n th unit vector in l^2 , and

$$v_{n-1} = e_{2n-1}, \quad v_{-n} = e_{2n}$$

for $n = 1, 2, \dots$. Then, $\{v_j : j = 0, \pm 1, \pm 2, \dots\}$ is an orthonormal basis for H . Since

$$\sum_{j=-\infty}^{\infty} |\langle x, v_j \rangle|^2 = \|x\|^2 < \infty \quad \text{for } x \in H,$$

we see by the Riesz-Fischer theorem that

$$A(x) = \sum_{j=-\infty}^{\infty} \langle x, v_j \rangle v_{j+1} \quad (9)$$

exists for each $x \in H$. It is clear that A is linear, and

$$\|Ax\|^2 = \sum_{j=-\infty}^{\infty} |\langle Ax, v_{j+1} \rangle|^2 = \sum_{j=-\infty}^{\infty} |\langle x, v_j \rangle|^2 = \|x\|^2.$$

This implies that $A \in BL(H)$.

For all $x \in H$ and integers m , we have

$$\langle x, A^*v_m \rangle = \langle Ax, v_m \rangle = \langle x, v_{m-1} \rangle$$

by (9). It follows that $A^*v_m = v_{m-1}$ for all integers m . From (9) we also see that $A(v_m) = v_{m+1}$. Hence,

$$A^*Av_m = A^*v_{m+1} = v_m$$

$$AA^*v_m = Av_{m-1} = v_m.$$

Since $\{v_m : m = 0, \pm 1, \pm 2, \dots\}$ is an orthonormal basis, it follows that

$$A^*A(x) = x = AA^*(x) \quad \text{for } x \in H$$

Thus A is unitary.

Suppose that A has an eigenvalue k . Then, $Ax = kx$ for some $x \neq 0$. So $\langle x, v_j \rangle \neq 0$ for some j . Further, for all integers m ,

$$k\langle x, v_m \rangle = \langle kx, v_m \rangle = \langle Ax, v_m \rangle = \langle x, v_{m-1} \rangle$$

by (9). Repeated application of this shows that

$$k^r \langle x, v_j \rangle = \langle x, v_{j-r} \rangle,$$

$$k^r \langle x, v_{j+r} \rangle = \langle x, v_j \rangle \neq 0$$

for $r = 1, 2, \dots$. The last relation implies that $k \neq 0$. Hence,

$$|\langle x, v_j \rangle|^2 \sum_{r=1}^{\infty} |k|^{2r} = \sum_{r=1}^{\infty} |\langle x, v_{j-r} \rangle|^2 \leq \|x\|^2 < \infty, \quad (10)$$

$$|\langle x, v_j \rangle|^2 \sum_{r=1}^{\infty} |k|^{-2r} = \sum_{r=1}^{\infty} |\langle x, v_{j+r} \rangle|^2 \leq \|x\|^2 < \infty, \quad (11)$$

using Bessel's inequality. Since $\langle x, v_j \rangle \neq 0$, (10) implies that $|k| < 1$ and (11) implies that $|k| > 1$. This contradiction shows that A cannot have an eigenvalue. This completes the proof. ■

PROBLEM 7.1.10 Suppose that $\{u_n\}$ is a total orthonormal sequence in a separable Hilbert space H . Let $A \in BL(H)$ and, for some $\lambda \in \mathbb{K}$,

$$A(u_n) = \lambda u_n - u_{n+1} \quad \text{for } n = 1, 2, \dots$$

Find $\sigma(A)$.

Solution Let $B = \lambda I - A$. Then, $B \in BL(H)$, and

$$B(u_n) = \lambda u_n - A(u_n) = u_{n+1}, \quad n = 1, 2, \dots$$

Since $\{u_n\}$ is total and B is linear and continuous we have, for all $x \in H$,

$$x = \sum_{n=1}^{\infty} \langle x, u_n \rangle u_n, \quad (12)$$

$$B(x) = \sum_{n=1}^{\infty} \langle x, u_n \rangle B(u_n) = \sum_{n=1}^{\infty} \langle x, u_n \rangle u_{n+1}. \quad (13)$$

Parseval's formula gives

$$\|Bx\|^2 = \sum_{n=1}^{\infty} |\langle x, u_n \rangle|^2 = \|x\|^2.$$

Hence, $\|B\| = 1$. Therefore,

$$k \in \sigma(B) \implies |k| \leq \|B\| = 1,$$

as in any Banach space (see Problem 4.2.1).

Next suppose that $k \notin \sigma(B)$. Then, $(B - kI)^{-1}$ exists in $BL(H)$. If $x = (B - kI)^{-1}(u_1)$, then, from (12) and (13),

$$\begin{aligned} u_1 &= Bx - kx = \sum_{n=1}^{\infty} \langle x, u_n \rangle u_{n+1} - k \sum_{n=1}^{\infty} \langle x, u_n \rangle u_n \\ &= -k \langle x, u_1 \rangle u_1 + \sum_{n=2}^{\infty} (-k \langle x, u_n \rangle + \langle x, u_{n-1} \rangle) u_n. \end{aligned}$$

This implies that

$$-k \langle x, u_1 \rangle = 1, \quad \langle x, u_{n-1} \rangle = k \langle x, u_n \rangle$$

for $n = 2, 3, \dots$. Hence,

$$k^n \langle x, u_n \rangle = -1 \quad \text{for } n \geq 1$$

by induction. Therefore, $k \neq 0$, and

$$\sum_{n=1}^{\infty} |k|^{-2n} = \sum_{n=1}^{\infty} |\langle x, u_n \rangle|^2 = \|x\|^2 < \infty.$$

This is possible only if $|k|^{-1} < 1$, that is, $|k| > 1$.

Thus we have proved that

$$k \notin \sigma(B) \iff |k| > 1.$$

Hence,

$$A - \mu I = \lambda I - B - \mu I = -(B - (\mu - \lambda)I)$$

is invertible iff $|\mu - \lambda| > 1$. Therefore,

$$\sigma(A) = \{\mu \in \mathbb{K} : |\mu - \lambda| \leq 1\}. \quad \blacksquare$$

PROBLEM 7.1.11 Let $A \in BL(H)$, where H is a Hilbert space. Show that λ is an approximate eigenvalue of A iff there is a sequence $\{B_n\}$ in $BL(H)$ such that $\|B_n\| = 1$ for all n and $\|(A - \lambda I)B_n\| \rightarrow 0$ as $n \rightarrow \infty$.

Solution Suppose that there is a sequence $\{B_n\}$ in $BL(H)$ such that $\|B_n\| = 1$ and $\|(A - \lambda I)B_n\| \rightarrow 0$. If $A - \lambda I$ is bounded below, then there is $\alpha > 0$ such that

$$\alpha\|x\| \leq \|(A - \lambda I)x\| \quad \text{for all } x \in H. \quad (14)$$

For each n we can find $y_n \in H$ such that

$$\|y_n\| = 1, \quad \|B_n(y_n)\| > \frac{1}{2}\|B_n\| = \frac{1}{2}.$$

Using (14), we get

$$\frac{1}{2}\alpha < \alpha\|B_n(y_n)\| \leq \|(A - \lambda I)B_n y_n\| \leq \|(A - \lambda I)B_n\|.$$

Since the last member in the above inequalities tends to zero as n tends to ∞ , this contradicts the fact that $\alpha > 0$. Therefore, $A - \lambda I$ cannot be bounded below, that is, λ is an approximate eigenvalue.

Conversely, suppose that λ is an approximate eigenvalue of A . Then there is a sequence $\{x_n\}$ in H with $\|x_n\| = 1$ and $\|Ax_n - \lambda x_n\| \rightarrow 0$. Choose $u \in H$ with $\|u\| = 1$, and define $B_n : H \rightarrow H$ by

$$B_n(x) = \langle x, u \rangle x_n, \quad n = 1, 2, \dots$$

Then B_n is linear, and

$$\begin{aligned} \|B_n x\| &= |\langle x, u \rangle| \|x_n\| \leq \|x\| \|u\| \|x_n\| = \|x\| \quad \text{for all } x, \\ \|B_n u\| &= \|\langle u, u \rangle x_n\| = 1. \end{aligned}$$

It follows that $\|B_n\| = 1$. Again, for all $x \in H$, we have

$$\begin{aligned} (A - \lambda I)B_n(x) &= (A - \lambda I)\langle x, u \rangle x_n, \\ \|(A - \lambda I)B_n(x)\| &= |\langle x, u \rangle| \|(A - \lambda I)x_n\| \\ &\leq \|x\| \|u\| \|Ax_n - \lambda x_n\|. \end{aligned}$$

Hence,

$$\|(A - \lambda I)B_n\| \leq \|Ax_n - \lambda x_n\| \|u\| \rightarrow 0.$$

Thus, $\|(A - \lambda I)B_n\| \rightarrow 0$ and $\|B_n\| = 1$. This completes the proof. $\blacksquare$

PROBLEM 7.1.12 Let H be a separable Hilbert space, $\{u_n\}$ be an orthonormal basis for H , and $\{k_n\}$ be a bounded sequence of scalars. Show that

$$A(x) = \sum_{n=1}^{\infty} k_n \langle x, u_n \rangle u_n, \quad x \in H,$$

defines a normal operator on H . [Such an operator is called a *diagonal operator*.] Find the eigenvalues and spectrum of A .

Solution Let $|k_n| \leq \alpha$. Then,

$$\sum_{n=1}^{\infty} |k_n \langle x, u_n \rangle|^2 \leq \alpha^2 \sum_{n=1}^{\infty} |\langle x, u_n \rangle|^2 \leq \alpha^2 \|x\|^2.$$

So the Riesz-Fischer theorem implies the existence of $A(x)$ for each $x \in H$. It is clear that A is linear. Moreover,

$$\langle Ax, u_n \rangle = k_n \langle x, u_n \rangle, \quad n = 1, 2, \dots \quad (15)$$

and hence by Parseval's formula,

$$\|Ax\|^2 = \sum_{n=1}^{\infty} |\langle Ax, u_n \rangle|^2 = \sum_{n=1}^{\infty} |k_n \langle x, u_n \rangle|^2 \leq \alpha^2 \|x\|^2.$$

So, $\|Ax\| \leq \alpha \|x\|$, for all $x \in H$. Thus A is bounded.

Since $\{u_n\}$ is total, we have

$$x = \sum_{n=1}^{\infty} \langle x, u_n \rangle u_n, \quad x \in H.$$

Hence, for any $k \in \mathbb{K}$, $x \in H$,

$$Ax - kx = \sum_{n=1}^{\infty} (k_n - k) \langle x, u_n \rangle u_n. \quad (16)$$

Thus,

$$\langle Ax - kx, u_m \rangle = (k_m - k) \langle x, u_m \rangle, \quad m = 1, 2, \dots,$$

and $Ax - kx = 0$ iff $\langle Ax - kx, u_m \rangle = 0$ for all $m \geq 1$, that is,

$$(k_m - k) \langle x, u_m \rangle = 0 \quad \text{for all } m \geq 1. \quad (17)$$

This holds if $k = k_n$ and $x = u_n$ for some n . Hence, $Au_n = k_n u_n$; and so each k_n is an eigenvalue. On the other hand, if $k \neq k_n$ for any n , then (17) holds only if $\langle x, u_m \rangle = 0$ for all m , that is, $x = 0$. So, $Ax - kx = 0$ only if $x = 0$. Thus k is not an eigenvalue if $k \neq k_n$ for any n . Hence,

$$\sigma_e(A) = \{k_n : n \geq 1\}$$

Next suppose that k is not an approximate eigenvalue of A . Then, $A - kI$ is bounded below. So, there is $\beta > 0$ such that

$$\beta\|x\| \leq \|Ax - kx\| \quad \text{for all } x \in H.$$

Taking $x = u_m$ in (16), this gives

$$\beta \leq \|Au_m - ku_m\| = \|(k_m - k)u_m\| = |k_m - k|. \quad (18)$$

Let $V = \{k_n : n = 1, 2, \dots\}$. Since $\beta > 0$, (18) implies that $k \notin \overline{V}$, the closure of V . Thus we have proved that

$$\overline{V} \subset \sigma_a(A). \quad (19)$$

On the other hand, suppose that $k \notin \overline{V}$. Then there exists $\beta > 0$ such that $|k_n - k| > \beta$ for all $n \geq 1$. Let $\alpha_n = (k_n - k)^{-1}$. Then $\{\alpha_n\}$ is bounded. Hence, as before,

$$B(x) = \sum_{n=1}^{\infty} \alpha_n \langle x, u_n \rangle u_n, \quad x \in H,$$

defines an element in $BL(H)$. Moreover,

$$\langle Bx, u_n \rangle = \alpha_n \langle x, u_n \rangle, \quad n \geq 1.$$

Therefore, from (16),

$$\begin{aligned} AB(x) - kB(x) &= \sum_{n=1}^{\infty} (k_n - k) \langle Bx, u_n \rangle u_n \\ &= \sum_{n=1}^{\infty} (k_n - k) \alpha_n \langle x, u_n \rangle u_n = \sum_{n=1}^{\infty} \langle x, u_n \rangle u_n = x. \end{aligned}$$

Similarly, since

$$\langle Ax - kx, u_n \rangle = (k_n - k) \langle x, u_n \rangle$$

from (16), we get

$$B(Ax - kx) = \sum_{n=1}^{\infty} \alpha_n (k_n - k) \langle x, u_n \rangle u_n = x.$$

Hence, $(A - kI)B = I = B(A - kI)$. This shows that $k \notin \sigma(A)$. Therefore,

$$\sigma(A) \subset \overline{V}.$$

Since $\sigma_a(A) \subset \sigma(A)$, this together with (19) implies that

$$\sigma(A) = \overline{V} = \text{closure of } \{k_n : n \geq 1\}.$$

Finally, since $\langle Ax - k_n x, u_n \rangle = 0$ by (15), we see that

$$\langle Ax - k_n x, Au_n \rangle = \langle Ax - k_n x, k_n u_n \rangle = 0,$$

and hence

$$\langle A^* Ax, u_n \rangle = \langle Ax, Au_n \rangle = \langle k_n x, Au_n \rangle$$

for all $n \geq 1, x \in H$. Since $\{u_n\}$ is total, we get

$$A^* A(x) = \sum_{n=1}^{\infty} \langle A^* Ax, u_n \rangle u_n = \sum_{n=1}^{\infty} \langle k_n x, Au_n \rangle u_n.$$

Again, since

$$AA^*(x) = \sum_{n=1}^{\infty} k_n \langle A^* x, u_n \rangle u_n = \sum_{n=1}^{\infty} k_n \langle x, Au_n \rangle u_n,$$

we see that $AA^*(x) = A^* A(x)$ for all $x \in H$. Thus A is normal. ■

PROBLEM 7.1.13 Let H be a separable Hilbert space and $A \in BL(H)$. Show that A is a diagonal operator with respect to some orthonormal basis for H iff A is normal and H is the closure of the span of its eigenvectors.

Solution Suppose that A is a diagonal operator with respect to an orthonormal basis $\{u_n\}$. This means that there is a sequence $\{k_n\}$ of scalars such that

$$A(x) = \sum k_n \langle x, u_n \rangle u_n, \quad x \in H. \quad (20)$$

Then, exactly as in the last part of the solution of Problem 7.1.12, we see that

$$AA^*(x) = \sum k_n \langle x, Au_n \rangle u_n = A^* A(x).$$

Thus A is normal. Since $A(u_n) = k_n u_n$ by (20), each k_n is an eigenvalue and u_n an eigenvector corresponding to k_n . So the span of eigenvectors contains the span of $\{u_n\}$. But $\overline{\text{span}}\{u_n\} = H$ by Problem 5.2.14 since $\{u_n\}$ is an orthonormal basis for H . Therefore, H is the closure of the span of eigenvectors.

Conversely, suppose that A is a normal operator and $H = \overline{\text{span}}(E)$ where E is the set of eigenvectors of A . Since A is normal, eigenvectors corresponding to distinct eigenvalues are orthogonal. Since H is separable, every orthonormal set in it is countable. Let $k_1, k_2, \dots$ be the distinct elements of $\sigma_e(A)$, and $Y_n = Z(A - k_n I)$ be the eigenspace corresponding to k_n . Then Y_n is a Hilbert space and is nonzero. Let $\{u_{ni} : i = 1, 2, \dots\}$ be an orthonormal basis for Y_n . Then, $\{u_{ni} : i \geq 1, n \geq 1\}$ is an orthonormal set in H .

Suppose that $\langle x, u_{ni} \rangle = 0$ for all n, i . Let y be any eigenvector of A . Then y must correspond to some eigenvalue k_n , that is, $y \in Y_n$ for some n . So,

$$y = \sum_{i \geq 1} \langle y, u_{ni} \rangle u_{ni}.$$

By the linearity and continuity of the inner product, we get

$$\langle y, x \rangle = \sum_i \langle y, u_{ni} \rangle \langle u_{ni}, x \rangle = 0.$$

This shows that $\langle x, y \rangle = 0$ for all $y \in E$, that is, $x \in E^\perp$. But,

$$E^{\perp\perp} = \overline{\text{span}}(E), \quad E^{\perp\perp\perp} = E^\perp$$

by Problems 5.3.5 and 5.3.2. Hence,

$$E^{\perp\perp} = H, \quad E^\perp = E^{\perp\perp\perp} = H^\perp = \{0\};$$

thus, $x = 0$. This shows that $\{u_{ni} : n \geq 1, i \geq 1\}$ is an orthonormal basis consisting of eigenvectors. Therefore, for any $x \in H$,

$$\begin{aligned} x &= \sum_{n,i} \langle x, u_{ni} \rangle u_{ni} \\ A(x) &= \sum_{n,i} \langle x, u_{ni} \rangle A(u_{ni}) \end{aligned}$$

by the linearity and continuity of A . However, $A(u_{ni}) = k_n u_{ni}$. So,

$$A(x) = \sum_{n,i} k_n \langle x, u_{ni} \rangle u_{ni}.$$

This shows that A is a diagonal operator with respect to the orthonormal basis $\{u_{ni}\}$. This completes the proof. ■

PROBLEM 7.1.14 Let H be a Hilbert space, $\{u_n\}$ be a countable orthonormal set, not necessarily total, in H , $\{k_n\}$ be a bounded sequence of scalars, and E denote the set $\{k_n : n = 1, 2, \dots\}$. For $x \in H$, let

$$A(x) = \sum_{n \geq 1} k_n \langle x, u_n \rangle u_n. \tag{21}$$

Prove the following:

(a) $A \in BL(H)$ and

$$E \subset \sigma_e(A) \subset E \cup \{0\}.$$

(b) $\overline{E} \subset \sigma(A) \subset \overline{E} \cup \{0\}$.

(c) If $k \notin \sigma(A)$, then the inverse B of $A - kI$ is given by

$$\begin{aligned} B(x) &= \sum \frac{\langle x, u_n \rangle}{k_n} u_n && \text{if } k = 0, \\ kB(x) &= -x + \sum \frac{k_n}{k_n - k} \langle x, u_n \rangle u_n && \text{if } k \neq 0. \end{aligned}$$

Solution (a) Let $Y = \overline{\text{span}}\{u_n\}$. Then Y is a closed subspace of H , and $\{u_n\}$ is an orthonormal basis for the Hilbert space Y . Exactly as in the solution of Problem 7.1.12, we see that $A(x)$ exists for each $x \in H$. It is clear that A is linear, $A(x) \in Y$ and $\langle Ax, u_n \rangle = k_n \langle x, u_n \rangle$ for $n \geq 1, x \in H$. So,

$$\|Ax\|^2 = \sum_n |\langle Ax, u_n \rangle|^2 = \sum_n |k_n \langle x, u_n \rangle|^2 \leq \lambda^2 \|x\|^2$$

if $|k_n| \leq \lambda$. Thus $A \in BL(H)$.

From (21) we see that $A(u_n) = k_n u_n$. Hence each k_n is an eigenvalue, that is,

$$E \subset \sigma_e(A).$$

Let $k \notin E \cup \{0\}$ and $Ax = kx$. Then, $x = (1/k)A(x) \in Y$. Hence,

$$x = \sum \langle x, u_n \rangle u_n$$

since $\{u_n\}$ is an orthonormal basis for Y . So, by (21),

$$k \sum \langle x, u_n \rangle u_n = kx = A(x) = \sum k_n \langle x, u_n \rangle u_n.$$

This implies that

$$k \langle x, u_n \rangle = k_n \langle x, u_n \rangle \quad \text{for all } n \geq 1.$$

But $k \neq k_n$ for any n . So, $\langle x, u_n \rangle = 0$ for all n , and hence $x = 0$. Thus, k is not an eigenvalue. It follows that

$$\sigma_e(A) \subset E \cup \{0\}.$$

(b) Let $k \notin \sigma_a(A)$. Then, $A - kI$ is bounded below, that is, there is $\alpha > 0$ such that

$$\alpha \|x\| \leq \|Ax - kx\| \quad \text{for } x \in H.$$

Since $A(u_n) = k_n u_n$ by (21), this gives

$$\alpha \|u_n\| \leq \|k_n u_n - k u_n\| = |k_n - k|.$$

Hence, $|k_n - k| \geq \alpha > 0$ for $n = 1, 2, \dots$. This implies that $k \notin \overline{E}$. Thus we see that

$$\overline{E} \subset \sigma_a(A) \subset \sigma(A). \quad (22)$$

Let $k \notin \overline{E}$ and $k \neq 0$. Then there exists $\beta > 0$ such that $|k - k_n| > \beta$ for all n . If $|k_n| \leq \lambda$, then

$$\left| \frac{k_n}{k_n - k} \right| \leq \frac{\lambda}{\beta} \quad \text{for all } n.$$

Hence, as in (a) above,

$$T(x) = \sum_{n \geq 1} \frac{k_n}{k_n - k} \langle x, u_n \rangle u_n, \quad x \in H \quad (23)$$

defines an operator $T \in BL(H)$. Moreover,

$$\langle Tx, u_n \rangle = \frac{k_n}{k_n - k} \langle x, u_n \rangle,$$

for $n \geq 1, x \in H$. Hence,

$$\begin{aligned} AT(x) &= \sum k_n \langle Tx, u_n \rangle u_n = \sum \frac{k_n^2}{k_n - k} \langle x, u_n \rangle u_n, \\ TA(x) &= \sum \frac{k_n}{k_n - k} \langle Ax, u_n \rangle u_n = \sum \frac{k_n^2}{k_n - k} \langle x, u_n \rangle u_n \end{aligned}$$

since $\langle Ax, u_n \rangle = k_n \langle x, u_n \rangle$ by (21). Thus,

$$\begin{aligned} A(x) + kT(x) &= \sum \left(k_n + \frac{kk_n}{k_n - k} \right) \langle x, u_n \rangle u_n \\ &= \sum \frac{k_n^2}{k_n - k} \langle x, u_n \rangle u_n = AT(x) = TA(x). \end{aligned}$$

Hence, $A + kT = AT = TA$. Thus we see that

$$\begin{aligned} (A - kI)(T - I) &= AT - A - kT + kI = kI, \\ (T - I)(A - kI) &= TA - kT - A + kI = kI. \end{aligned} \quad (24)$$

This implies that $A - kI$ is invertible in $BL(H)$ if $k \notin \overline{E}$ and $k \neq 0$. Therefore,

$$\sigma(A) \subset \overline{E} \cup \{0\}.$$

This combined with (22) proves (b).

(c) Let $k \notin \sigma(A)$. Then $k \notin \overline{E}$ by (b). If $k \neq 0$ then (24) shows that $(A - kI)^{-1} = (1/k)(T - I)$; hence, by (23),

$$k(A - kI)^{-1}(x) = T(x) - x = -x + \sum \frac{k_n}{k_n - k} \langle x, u_n \rangle u_n.$$

Suppose that $k = 0$. Then A^{-1} exists in $BL(H)$. Let x be any element in H . Then, $x = y + z$, where $y \in Y, z \in Y^\perp$. So, $\langle z, u_n \rangle = 0$ for all n ; hence, $A(z) = 0$ by (21) and $z = A^{-1}A(z) = 0$. Thus, $x = y \in Y$. This shows that $Y = H$, and hence $\{u_n\}$ is an orthonormal basis for H . Since $0 = k \notin \overline{E}$, we see that $\{1/k_n\}$ is bounded. So, as in (a),

$$B(x) = \sum \frac{1}{k_n} \langle x, u_n \rangle u_n, \quad x \in H,$$

defines an element in $BL(H)$. Also, for $x \in H$,

$$BA(x) = \sum \frac{1}{k_n} \langle Ax, u_n \rangle u_n = \sum \frac{1}{k_n} k_n \langle x, u_n \rangle u_n = x,$$

since $\{u_n\}$ is an orthonormal basis for H . Hence,

$$B(x) = BAA^{-1}(x) = A^{-1}(x), \quad x \in H,$$

that is, $B = A^{-1}$. This completes the proof. ■

PROBLEM 7.1.15 Let A be an operator on a Hilbert space H . A subspace F of H is said to be *invariant* for A if $A(F) \subset F$. [For example, the range $R(A)$ is invariant for A .] If A is normal, E is a set of eigenvectors of A , $F_1 = E^\perp$ and F_2 is the closure of the span of E , show that both F_1 and F_2 are closed invariant subspaces for A .

Solution By Problem 5.3.1 we see that $E^\perp$ is a closed subspace of H . Let $x \in E^\perp$. If $v \in E$, then $\langle x, v \rangle = 0$ and v is an eigenvector. So, $Av = kv$ for some $k \in \mathbb{K}$. Since A is normal it follows from Problem 7.1.2(a) that $A^*v = \bar{k}v$. Hence,

$$\langle Ax, v \rangle = \langle x, A^*v \rangle = \langle x, \bar{k}v \rangle = k\langle x, v \rangle = 0.$$

This shows that $A(x) \in E^\perp$. Therefore, $A(E^\perp) \subset E^\perp$, that is, $F_1 = E^\perp$ is invariant for A .

Let $E_1 = \text{span}(E)$ and $x \in E_1$. Then,

$$x = c_1v_1 + \cdots + c_nv_n, \quad v_i \in E.$$

Since v_i is an eigenvector, $A(v_i) = k_iv_i$ for some $k_i \in \mathbb{K}$. Hence,

$$A(x) = \sum_{i=1}^n c_i A(v_i) = \sum_{i=1}^n c_i k_i v_i \in E_1.$$

Therefore, $A(x) \in E_1$ for all $x \in E_1$. Let $x \in F_2 = \overline{E_1}$. Then there exist x_n in E_1 such that $x_n \rightarrow x$. So $A(x_n) \rightarrow A(x)$. Thus $A(x) \in \overline{E_1}$ since $A(x_n) \in E_1$. This proves that $A(x) \in F_2$ for all $x \in F_2$, that is, $A(F_2) \subset F_2$. Hence F_2 is invariant for A . This completes the proof. ■

PROBLEM 7.1.16 Let H be a complex Hilbert space, W be the set of all self-adjoint operators in $BL(H)$, and W_1 be the set of all unitary operators B in $BL(H)$ such that $1 \notin \sigma(B)$. For $A \in W$, write

$$U(A) = (A - iI)(A + iI)^{-1}.$$

Show that U is a bijective transform of W onto W_1 , its inverse being given by

$$U^{-1}(B) = i(I + B)(I - B)^{-1}, \quad B \in W_1.$$

[$U(A)$ is called the *Cayley transform* of A .]

Solution If A is self-adjoint then its spectral values are real, by Problem 7.1.5(b), and hence $A - iI$ and $A + iI$ are invertible in $BL(H)$. Let $B = U(A)$. Then, B is invertible and

$$\begin{aligned} B^{-1} &= (A + iI)(A - iI)^{-1} = (A - iI + 2iI)(A - iI)^{-1} \\ &= I + 2i(A - iI)^{-1} = (A - iI)^{-1}(A - iI + 2iI) \\ &= (A - iI)^{-1}(A + iI). \end{aligned}$$

Since $B(A + iI) = A - iI$ and A is self-adjoint, we have

$$(A + iI)^* B^* = (A - iI)^*.$$

Thus,

$$\begin{aligned} (A - iI)B^* &= A + iI, \\ B^* &= (A - iI)^{-1}(A + iI) = B^{-1}. \end{aligned}$$

This shows that B is unitary. Again,

$$\begin{aligned} B &= (A + iI - 2iI)(A + iI)^{-1} = I - 2i(A + iI)^{-1}, \\ I - B &= 2i(A + iI)^{-1}. \end{aligned}$$

This shows that $I - B$ is invertible in $BL(H)$; hence $1 \notin \sigma(B)$. Therefore, $B \in W_1$.

Again, let $A \in W$ and $B = U(A)$. Then,

$$\begin{aligned} B(A + iI) &= A - iI, \\ iB + iI &= A - BA = A(I - B), \\ i(B + I)(I - B)^{-1} &= A. \end{aligned}$$

Thus, if $A_1, A_2 \in W$ and $U(A_1) = U(A_2) = B$, then

$$A_1 = i(I + B)(I - B)^{-1} = A_2.$$

This shows that U is one-one.

Let B be any element in W_1 . Then, $BB^* = I = B^*B$ and $I - B$ is invertible. Let

$$A = i(I + B)(I - B)^{-1}. \quad (25)$$

Then,

$$A(I - B) = i(I + B). \quad (26)$$

From this we get

$$\begin{aligned} (I - B^*)A^* &= -i(I + B^*), \\ (B - BB^*)A^* &= -i(B + BB^*), \quad (B - I)A^* = -i(B + I), \\ A^* &= i(I + B)(I - B)^{-1}. \end{aligned}$$

Hence, $A^* = A$ by (25). So $A \in W$. Again, from (26),

$$A - iI = iB + AB = (iI + A)B.$$

Taking adjoints, we get

$$A + iI = B^*(-iI + A) = B^{-1}(A - iI).$$

Hence,

$$I = B^{-1}(A - iI)(A + iI)^{-1} = B^{-1}U(A),$$

which shows that $B = U(A)$. This shows that U is onto. From (25) we also see that

$$U^{-1}(B) = A = i(I + B)(I - B)^{-1}.$$

This completes the proof. ■

PROBLEM 7.1.17 Let $H = L^2[0, 1]$, where the scalar field $\mathbb{K}$ is $\mathbb{R}$. For $x \in H$, let

$$A(x)(s) = \int_0^1 s^2 t^2 x(t) dt, \quad 0 \leq s \leq 1.$$

Show that this defines a self-adjoint operator $A \in BL(H)$. Find m_A and M_A .

Solution Let $k(s, t) = s^2 t^2$. Then,

$$\begin{aligned} \int_0^1 |k(s, t)| dt &= \int_0^1 s^2 t^2 dt = \frac{s^2}{3} \leq \frac{1}{3}, \\ \int_0^1 |k(s, t)| ds &\leq \frac{1}{3}. \end{aligned}$$

Hence, by Problem 6.1.12, we see that $A \in BL(H)$ and that

$$A^*(x)(s) = \int_0^1 \overline{k(t, s)} x(t) dt$$

Since $\overline{k(t, s)} = t^2 s^2 = k(s, t)$, we see that $A^* = A$, that is, A is self-adjoint.

For $x \in L^2[0, 1]$, we have

$$\begin{aligned} \langle Ax, x \rangle &= \int_0^1 A(x)(s)x(s) ds \\ &= \int_0^1 \left\{ \int_0^1 s^2 t^2 x(t) dt \right\} x(s) ds \\ &= \int_0^1 s^2 x(s) ds \int_0^1 t^2 x(t) dt = \left(\int_0^1 t^2 x(t) dt \right)^2. \end{aligned} \quad (27)$$

This shows that $\langle Ax, x \rangle \geq 0$ for all $x \in H$. If $x(t) = b(1 + at)$, then

$$\int_0^1 t^2 x(t) dt = b \int_0^1 (t^2 + at^3) dt = b \left(\frac{1}{3} + \frac{a}{4} \right); \quad (28)$$

$$\int_0^1 |x(t)|^2 dt = b^2 \int_0^1 (1 + 2at + a^2 t^2) dt = b^2 \left(1 + a + \frac{a^2}{3} \right). \quad (29)$$

If $a = -4/3$ and $b^2(1 + a + a^2/3) = 1$, that is, $b = \sqrt{27/7}$, then we see from (27), (28) and (29), that

$$\langle Ax, x \rangle = 0, \quad \|x\| = 1.$$

Thus $0 \in W(A)$, and so $m_A = 0$. Again,

$$\left| \int_0^1 t^2 x(t) dt \right| \leq \left(\int_0^1 t^4 dt \right)^{1/2} \left(\int_0^1 |x(t)|^2 dt \right)^{1/2} = 5^{-1/2} \|x\|.$$

This combined with (27) shows that $\langle Ax, x \rangle \leq 1/5$ for all x with $\|x\| = 1$. If $x(t) = \sqrt{5} t^2$, then

$$\begin{aligned} \int_0^1 t^2 x(t) dt &= \sqrt{5} \int_0^1 t^4 dt = 5^{-1/2}, \\ \int_0^1 (x(t))^2 dt &= 5 \int_0^1 t^4 dt = 1. \end{aligned}$$

Therefore, $\langle Ax, x \rangle = 1/5, \|x\| = 1$. This shows that $M_A = 1/5$. ■

PROBLEM 7.1.18 Let $A \in BL(H)$, where H is a Hilbert space. If A is self-adjoint and invertible, show that

$$\|A^{-1}\| \geq \max \left\{ \frac{1}{|m_A|}, \frac{1}{|M_A|} \right\}.$$

Show by means of an example that strict inequality can occur here.

Solution Let $B = A^{-1}$. Then $AB = I$, and hence

$$B^* A^* = I, \quad B^* A = I, \quad B^* = A^{-1} = B.$$

Thus B also is self-adjoint. If $k \neq 0$, then

$$A - kI = kA \left(\frac{1}{k} I - B \right).$$

This shows that $k \in \sigma(A)$ iff $1/k \in \sigma(B)$. Since m_A and M_A are spectral values of A , we see that $1/m_A$ and $1/M_A$ are spectral values of B . Since

$$\sigma(B) \subset \overline{W(B)} = [m_B, M_B],$$

it follows that

$$m_B \leq \frac{1}{m_A} \leq M_B, \tag{30}$$

$$m_B \leq \frac{1}{M_A} \leq M_B. \tag{31}$$

Since $A = B^{-1}$, we can interchange A and B in this and get

$$m_A \leq \frac{1}{m_B} \leq M_A, \quad (32)$$

$$m_A \leq \frac{1}{M_B} \leq M_A. \quad (33)$$

We see from (30) that if $m_B > 0$, then $m_A > 0$, and from (32) that if $m_A > 0$ then $m_B > 0$. Similarly, from (31) and (33) we see that $M_A < 0$ iff $M_B < 0$. Suppose that $m_A > 0$ or $M_A < 0$. Then, m_A, M_A, m_B and M_B are of the same sign. Multiplying the inequality in (33) by M_B/m_A and comparing the result with the second inequality in (30), we see that

$$M_B = \frac{1}{m_A}.$$

Similarly, from (31) and (32), we get

$$m_B = \frac{1}{M_A}.$$

Hence, if $m_A > 0$ or $M_A < 0$,

$$\|B\| = \max \{|m_B|, |M_B|\} = \max \left\{ \frac{1}{|M_A|}, \frac{1}{|m_A|} \right\}.$$

Next consider the case $m_A < 0 < M_A$. Then,

$$m_B \leq \frac{1}{m_A} < 0 < \frac{1}{M_A} \leq M_B$$

from (30) and (31). Therefore,

$$\begin{aligned} \|B\| &\geq |m_B| \geq \frac{1}{|m_A|}, \quad \|B\| \geq M_B \geq \frac{1}{M_A}, \\ \|B\| &\geq \max \left\{ \frac{1}{|m_A|}, \frac{1}{|M_A|} \right\}. \end{aligned}$$

This proves the first part.

For the second part let $H = \mathbb{R}^3$, $\mathbb{K} = \mathbb{R}$ and A be represented by the matrix

$$M_1 = \begin{pmatrix} 3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Since M_1 is symmetric and $\det(M_1) \neq 0$, we see that A is self-adjoint and invertible. The inverse B is represented by the matrix

$$M_2 = \begin{pmatrix} 1/3 & 0 & 0 \\ 0 & -1/2 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

If $x \in \mathbb{R}^3$, then

$$\begin{aligned} A(x) &= (3x(1), -2x(2), -x(3)), \\ \langle Ax, x \rangle &= 3(x(1))^2 - 2(x(2))^2 - (x(3))^2. \end{aligned}$$

Hence,

$$\begin{aligned} \langle Ax, x \rangle &\leq 3(x(1))^2 \leq 3\|x\|^2 \\ \langle Ax, x \rangle + 2\|x\|^2 &= 5(x(1))^2 + (x(3))^2 \geq 0 \end{aligned}$$

for all $x \in \mathbb{R}^3$. So if $\|x\| = 1$, then

$$\langle Ax, x \rangle \leq 3 \tag{34}$$

$$\langle Ax, x \rangle + 2 \geq 0 \tag{35}$$

If $x = (1, 0, 0)$, we have equality in (34), and if $x = (0, 1, 0)$, we have equality in (35). Thus we see that $m_A = -2, M_A = 3$. On the other hand, if $x = (0, 0, 1)$, then $\|x\| = 1, B(x) = (0, 0, -1)$, and

$$1 = \|Bx\| \leq \|B\| \|x\| = \|B\|.$$

So $\|B\| \geq 1$, whereas

$$\max \left\{ \frac{1}{|m_A|}, \frac{1}{|M_A|} \right\} = \frac{1}{2}.$$

This completes the proof. ■

PROBLEM 7.1.19 Let $A \in BL(H)$, where H is a Hilbert space. Prove that its numerical range $W(A)$ is a convex set.

Solution It is clear that $W(A)$ is nonempty if $H \neq \{0\}$. Let k_1 and k_2 be two distinct elements of $W(A)$. Then,

$$k_1 = \langle Ax_1, x_1 \rangle, \quad k_2 = \langle Ax_2, x_2 \rangle,$$

where $\|x_1\| = \|x_2\| = 1$. Let

$$\begin{aligned} \alpha &= \frac{1}{k_1 - k_2} (\langle Ax_1, x_2 \rangle - k_2 \langle x_1, x_2 \rangle), \\ \beta &= \frac{1}{k_1 - k_2} (\langle Ax_2, x_1 \rangle - k_2 \langle x_2, x_1 \rangle), \\ \gamma &= \pm \frac{\alpha - \bar{\beta}}{|\alpha - \bar{\beta}|} \quad \text{if } \alpha \neq \bar{\beta} \end{aligned}$$

and $\gamma = \pm 1$ if $\alpha = \bar{\beta}$, where the sign ‘+’ or ‘-’ is so chosen that

$$a = \operatorname{Re} \gamma \langle x_2, x_1 \rangle \geq 0.$$

For real s, t , let $x = tx_1 + s\gamma x_2$, and

$$g(s, t) = \frac{1}{k_1 - k_2} (\langle Ax, x \rangle - k_2 \langle x, x \rangle).$$

Then,

$$\begin{aligned} \langle x, x \rangle &= \langle tx_1 + s\gamma x_2, tx_1 + s\gamma x_2 \rangle \\ &= t^2 \|x_1\|^2 + ts\bar{\gamma} \langle x_1, x_2 \rangle + st\gamma \langle x_2, x_1 \rangle + s^2 |\gamma|^2 \|x_2\|^2 \\ &= t^2 + 2ast + s^2 \end{aligned} \quad (36)$$

Further,

$$\begin{aligned} \langle Ax, x \rangle &= \langle tAx_1 + s\gamma Ax_2, tx_1 + s\gamma x_2 \rangle \\ &= t^2 \langle Ax_1, x_1 \rangle + ts\bar{\gamma} \langle Ax_1, x_2 \rangle + st\gamma \langle Ax_2, x_1 \rangle + s^2 |\gamma|^2 \langle Ax_2, x_2 \rangle \\ &= t^2 k_1 + ts\bar{\gamma} \langle Ax_1, x_2 \rangle + st\gamma \langle Ax_2, x_1 \rangle + s^2 k_2. \end{aligned}$$

From these we get

$$\begin{aligned} (k_1 - k_2)g(s, t) &= \langle Ax, x \rangle - k_2 \langle x, x \rangle \\ &= t^2(k_1 - k_2) + st\bar{\gamma}\alpha(k_1 - k_2) + st\gamma\beta(k_1 - k_2). \end{aligned}$$

Hence,

$$g(s, t) = t^2 + stb, \quad (37)$$

where $b = \alpha\bar{\gamma} + \beta\gamma$. Using the values of γ we see that if $\alpha \neq \bar{\beta}$, then

$$\pm b|\alpha - \bar{\beta}| = \alpha(\bar{\alpha} - \beta) + \beta(\alpha - \bar{\beta}) \in \mathbb{R},$$

and if $\alpha = \bar{\beta}$, then

$$\pm b = \alpha + \beta = \bar{\beta} + \beta \in \mathbb{R}.$$

Thus we see that, in any case, b is real. Therefore, $g(s, t)$ is real for real s, t . For $0 \leq t \leq 1$, choose s by

$$s = -at + \sqrt{a^2 t^2 + 1 - t^2}. \quad (38)$$

Then, $\langle x, x \rangle = 1$ by (36). Moreover, $g(s, t)$ becomes a real continuous function $h(t)$ for $0 \leq t \leq 1$. If $t = 1$, then $s = 0$, since $a \geq 0$, and so $h(1) = 1$ by (37). Further, $h(0) = 0$. Hence for any λ with $0 < \lambda < 1$, there is t with $0 < t < 1$ and $h(t) = \lambda$. With the corresponding value of s given by (38), we get a point $x \in H$ such that $\langle x, x \rangle = 1$, and

$$\frac{1}{k_1 - k_2} (\langle Ax, x \rangle - k_2 \langle x, x \rangle) = g(s, t) = h(t) = \lambda.$$

Hence,

$$\langle Ax, x \rangle - k_2 = (k_1 - k_2)\lambda,$$

that is, $\lambda k_1 + (1 - \lambda)k_2 = \langle Ax, x \rangle \in W(A)$. This completes the proof. ■

PROBLEM 7.1.20 Let $A \in BL(H)$, where H is a Hilbert space, and $W(A)$ be the numerical range of A . Prove the following:

- (a) $W(A) = W(UAU^{-1})$ for every unitary operator U on H .
- (b) If $W(A)$ contains more than one point then the derived set of $W(A)$ is the same as its closure.

Solution (a) Let U be unitary and $B = UAU^{-1}$. Let $k \in W(A)$. Then $k = \langle Ax, x \rangle$ with $\|x\| = 1$. If $y = Ux$, then $By = UAU^{-1}y = UAx$, and so

$$\langle By, y \rangle = \langle UAx, Ux \rangle = \langle Ax, U^*Ux \rangle = \langle Ax, x \rangle = k.$$

Thus, $k = \langle By, y \rangle$ and $\|y\| = \|Ux\| = \|x\| = 1$. Hence, $k \in W(B)$. Since U^{-1} also is unitary we see, similarly, that if $k \in W(B)$ then $k \in W(U^{-1}BU) = W(A)$. Thus, $W(A) = W(B)$.

(b) Since a limit point of $W(A)$ is necessarily a point of its closure $\overline{W(A)}$, it is enough if we prove that every point in the closure is a limit point. Let $k \in \overline{W(A)}$ and $\epsilon > 0$ be given. There exists $k_1 \in W(A)$ such that $|k - k_1| < \epsilon/2$. Since $W(A)$ contains more one point we can choose k_2 in $W(A)$ different from k_1 . Let

$$k(t) = tk_1 + (1 - t)k_2, \quad 0 \leq t \leq 1.$$

Since $W(A)$ is a convex set by Problem 7.1.19, we see that $k(t) \in W(A)$ for $0 \leq t \leq 1$. Since $k(t)$ is a continuous function of t and $k(t) \rightarrow k_1$ as $t \rightarrow 1$, we can choose t with $0 < t < 1$ and such that

$$|k(t) - k_1| < \frac{\epsilon}{2}, \quad k(t) \neq k.$$

So, $|k(t) - k| \leq |k(t) - k_1| + |k_1 - k| < \epsilon$. Since $k(t) \in W(A)$ and $k(t) \neq k$, this shows that k is a limit point of $W(A)$. This completes the proof. ■

PROBLEM 7.1.21 Let $A \in BL(H)$, where H is a finite dimensional Hilbert space. If

- (i) A is self-adjoint or
- (ii) A is normal and the scalar field $\mathbb{K}$ is $\mathbb{C}$,

show that there exist scalars $t_1, \dots, t_m$ and subspaces $Y_1, \dots, Y_m$, which are pairwise orthogonal, such that, for all $x \in H$,

$$\begin{aligned} x &= y_1 + \dots + y_m, \quad y_i \in Y_i, \\ A(x) &= t_1 y_1 + \dots + t_m y_m. \end{aligned}$$

Solution Suppose that M is the matrix representing A w.r.t. some orthonormal basis, and $t_1, \dots, t_m$ are the distinct roots in $\mathbb{C}$ of the characteristic polynomial $p(t) = \det(M - tI)$. If $t_i \in \mathbb{K}$, then t_i is an eigenvalue of A .

Suppose that $\mathbb{K} = \mathbb{R}$. Then A is self-adjoint and so M is a real symmetric matrix. If M is of order $n \times n$, then it represents a linear map $T : \mathbb{C}^n \rightarrow \mathbb{C}^n$ which also is self-adjoint. The roots of $p(t)$ are eigenvalues of T and hence real. Consequently, $t_i \in \mathbb{R}$ for all i . Thus, if (i) or (ii) holds then $t_1, \dots, t_m$ lie in the scalar field $\mathbb{K}$ and so are precisely the distinct eigenvalues of A . This shows, in particular, that A has eigenvalues.

Let $Y_i = Z(A - t_i I)$ be the eigenspace corresponding to t_i . Since A is normal in both the cases (i) and (ii), we see that the eigenvectors corresponding to distinct eigenvalues are orthogonal by Problem 7.1.2(c). Thus, $Y_i \perp Y_j$ for $i \neq j$. Let $Y = Y_1 + \dots + Y_m$, that is,

$$Y = \{y \in H : y = y_1 + \dots + y_m, y_i \in Y_i\}.$$

We shall prove that $Y = H$.

Let $x \in Y^\perp$ and $y \in Y$. Then $y = y_1 + \dots + y_m$ with $y_i \in Y_i$. It can be seen that $Y_i \subset Y$ and so $\langle x, y_i \rangle = 0$. Since A is normal, we see from Problem 7.1.2(a) that $y_i \in Z(A^* - \bar{t}_i I)$, that is, $A^*(y_i) = \bar{t}_i y_i$. Hence,

$$\begin{aligned}\langle Ax, y_i \rangle &= \langle x, A^* y_i \rangle = \langle x, \bar{t}_i y_i \rangle = 0, \\ \langle A^* x, y_i \rangle &= \langle x, A y_i \rangle = \langle x, t_i y_i \rangle = 0.\end{aligned}$$

It follows that $\langle Ax, y \rangle = 0$ and $\langle A^* x, y \rangle = 0$. This holds for any $y \in Y$. Thus, $Ax \in Y^\perp$ and $A^* x \in Y^\perp$ for any $x \in Y^\perp$.

Let $H_1 = Y^\perp$. Then A and A^* map H_1 into itself. Let B and C denote the restrictions to H_1 of A and A^* respectively. Then $B, C \in BL(H_1)$. For all $x, y \in H_1$,

$$\langle Bx, y \rangle = \langle Ax, y \rangle = \langle x, A^* y \rangle = \langle x, Cy \rangle.$$

Thus, $C = B^*$. Hence if A is self-adjoint or normal so is B . Suppose that $H_1 \neq \{0\}$. We proved that A has eigenvalues. Applying this result with H changed to H_1 and A changed to B , we see that B has eigenvalues. If k is an eigenvalue of B , then $Ax = Bx = kx$ for some $x \neq 0, x \in H_1$, hence k is an eigenvalue of A . Therefore, $k = t_i$ and $x \in Y_i \subset Y$. This is not possible since $x \in H_1 = Y^\perp$ and $x \neq 0$. This shows that $Y^\perp = H_1 = \{0\}$, and hence, $Y = H$.

We proved that every $x \in H$ can be expressed as

$$x = y_1 + \dots + y_m, \quad y_i \in Y_i.$$

Since $A(y_i) = t_i y_i$, we get

$$A(x) = t_1 y_1 + \dots + t_m y_m.$$

This completes the proof. ■

PROBLEM 7.1.22 Let A be a normal operator on a finite dimensional Hilbert space H over $\mathbb{C}$. Show that $A^* = p(A)$, where p is some polynomial with complex coefficients. Deduce that if an operator B commutes with A then it commutes with A^* .

Solution By Problem 7.1.21, we see that there exist subspaces $Y_1, \dots, Y_m$ which are mutually orthogonal such that every $x \in H$ can be expressed as

$$x = y_1 + \dots + y_m, \quad y_i \in Y_i, \quad (39)$$

and there exist scalars $t_1, \dots, t_m$ such that

$$A(x) = t_1 y_1 + \dots + t_m y_m. \quad (40)$$

Since Y_i are orthogonal, we see that the expression of x given by (39) is unique. It follows from (40) that

$$A(y_i) = t_i y_i, \quad i = 1, \dots, m. \quad (41)$$

Using Problem 7.1.2(a), we have

$$A^*(y_i) = \overline{t_i} y_i, \quad i = 1, \dots, m. \quad (42)$$

Repeated application of (41) gives

$$A^r(y_i) = t_i^r y_i, \quad i = 1, \dots, m, \quad r \geq 1.$$

It follows that if $p(t)$ is any polynomial in t , then

$$p(A)(y_i) = p(t_i) y_i, \quad i = 1, \dots, m. \quad (43)$$

Let

$$p(t) = \sum_{j=1}^m l_j(t) \overline{t_j},$$

where

$$l_j(t) = \prod_{i \neq j} \frac{t - t_i}{t_j - t_i}.$$

This is the Lagrange interpolating polynomial. It can be seen that

$$l_j(t_i) = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j, \end{cases}$$

and hence $p(t_i) = \overline{t_i}$. Using (43) with this polynomial, we get

$$p(A)(y_i) = \overline{t_i} y_i = A^*(y_i)$$

by (42). Hence from (39),

$$p(A)(x) = \sum_{i=1}^m p(A)(y_i) = \sum_{i=1}^m A^*(y_i) = A^*(x).$$

This is true for any $x \in H$. Thus $p(A) = A^*$.

Finally, if $AB = BA$, then $A^r B = B A^r$ for $r = 0, 1, 2, \dots$, and hence, $p(A)B = B p(A)$, that is, $A^* B = B A^*$. This completes the proof. ■

7.2 Compact Self-Adjoint Operators and Spectral Theorem

Let H be a Hilbert space. Recall that a linear map $A : H \rightarrow H$ is *compact* or *completely continuous* if for every bounded sequence $\{x_n\}$ in H the sequence $\{Ax_n\}$ has a subsequence converging in H .

If $\{A_n\}$ is a sequence of compact operators on H and $A_n \rightarrow A$ in $BL(H)$, then A is compact. If $A \in BL(H)$ and the range of A is of finite dimension, then A is compact.

For a nonzero compact self-adjoint operator A on a Hilbert space H we have the following theorem:

There is a finite or infinite sequence $\{\lambda_n\}$ of nonzero real numbers with $|\lambda_1| \geq |\lambda_2| \geq \dots$, and a corresponding finite or infinite orthonormal sequence $\{u_n\}$ in H such that

$$A(x) = \sum_{n \geq 1} \lambda_n \langle x, u_n \rangle u_n, \quad x \in H.$$

Moreover,

- (i) if $\{\lambda_n\}$ is an infinite sequence then $\lambda_n \rightarrow 0$,
- (ii) $\lambda_1, \lambda_2, \dots$ are precisely the nonzero eigenvalues of A , and
- (iii) u_n is an eigenvector corresponding to λ_n .

Let see Problem 7.2.6, $k(s, t)$ be a square integrable function on $[0, 1] \times [0, 1]$. For $x \in L^2[0, 1]$, let

$$A(x)(s) = \int_0^1 k(s, t)x(t) dt, \quad 0 \leq s \leq 1.$$

In Problem 4.1.16 we found that A is a compact operator on $H = L^2[0, 1]$. By Problem 6.2.22, A is self-adjoint if $k(s, t) = \overline{k(t, s)}$.

An operator $P \in BL(H)$ is called an *orthogonal projection* or *projection* on H if (i) it is idempotent, that is, $P^2 = P$, and (ii) its zero space $Z(P)$ and range space $R(P)$ are orthogonal.

Suppose that A is a nonzero compact self-adjoint operator on a Hilbert space H , and $t_1, t_2, \dots$ are the distinct nonzero eigenvalues of A with $|t_1| \geq |t_2| \geq \dots$. Let $t_0 = 0$ and P_j be the orthogonal projection whose range $R(P_j)$ is the eigenspace $Z(A - t_j I)$ of $t_j, j = 0, 1, 2, \dots$. Then,

$$x = \sum_{n \geq 0} P_n(x), \quad x \in H, \tag{1}$$

where the sum exists in H for each x , and

$$A = \sum_{n \geq 1} t_n P_n, \tag{2}$$

where the sum exists in $BL(H)$. Moreover, $R(P_j)$ is finite dimensional for $j \geq 1$ and $P_i P_j = 0$ for $i \neq j$.

This result is known as *spectral theorem*. See Problem 7.2.22. It gives the spectral decomposition of A . If the sum in (2) is finite, then it is equivalent to

$$A(x) = \sum_{n \geq 1} t_n P_n(x) \quad \text{for } x \in H. \quad (3)$$

If the sum is infinite, then (2) and (3) are different. However, (2) implies (3). This follows from the fact that if $A_n \rightarrow A$ in $BL(H)$, then

$$\|A_n(x) - A(x)\| \leq \|A_n - A\| \|x\| \rightarrow 0, \quad x \in H.$$

In Problem 7.1.21 we have given a finite dimensional version of this. There we have proved this even for normal operators assuming that the scalar field $\mathbb{K}$ is $\mathbb{C}$.

EXERCISES

1. Let $H = \mathbb{R}^2$ and $A \in BL(H)$ be represented by the matrix $M = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$.

Find mutually orthogonal subspaces Y_1, Y_2 and scalars t_1, t_2 such that

$$\begin{aligned} x &= y_1 + y_2, \quad y_i \in Y_i, \\ Ax &= t_1 y_1 + t_2 y_2. \end{aligned}$$

2. Let $H = \mathbb{R}^2$ and $A \in BL(H)$ be represented by the matrix $M = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$.

Find mutually orthogonal subspaces Y_1, Y_2 and scalars t_1, t_2 such that

$$\begin{aligned} x &= y_1 + y_2, \quad y_i \in Y_i, \\ Ax &= t_1 y_1 + t_2 y_2. \end{aligned}$$

3. Let $H = \mathbb{R}^2$ and $A \in BL(H)$ be represented by the matrix $M = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$.

Find scalars t_1, t_2 and orthogonal projections P_1, P_2 such that

$$\begin{aligned} x &= P_1(x) + P_2(x), \\ Ax &= t_1 P_1(x) + t_2 P_2(x). \end{aligned}$$

for all $x \in H$.

4. Let $H = \mathbb{R}^3$ and $A \in BL(H)$ be represented by the matrix

$$M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Find scalars t_1, t_2, t_3 and orthogonal projections P_1, P_2, P_3 such that

$$\begin{aligned}x &= P_1(x) + P_2(x) + P_3(x), \\Ax &= t_1 P_1(x) + t_2 P_2(x) + t_3 P_3(x)\end{aligned}$$

for all $x \in H$.

5. Let $H = \mathbb{R}^3$ and $A \in BL(H)$ be represented by the matrix

$$M = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Find scalars t_1, t_2, t_3 and orthogonal projections P_1, P_2, P_3 such that

$$\begin{aligned}x &= P_1(x) + P_2(x) + P_3(x), \\Ax &= t_1 P_1(x) + t_2 P_2(x) + t_3 P_3(x)\end{aligned}$$

for all $x \in H$.

6. Let $H = \mathbb{C}^3$ and $A \in BL(H)$ be represented by the matrix

$$M = \begin{pmatrix} 0 & 1 & 1 \\ -1 & 0 & 1 \\ -1 & -1 & 0 \end{pmatrix}.$$

Find scalars t_1, t_2, t_3 and orthogonal projections P_1, P_2, P_3 such that

$$\begin{aligned}x &= P_1(x) + P_2(x) + P_3(x), \\Ax &= t_1 P_1(x) + t_2 P_2(x) + t_3 P_3(x)\end{aligned}$$

for all $x \in H$.

PROBLEMS

PROBLEM 7.2.1 Let $A \in BL(H)$, where H is a Hilbert space. Prove that

- (a) A is compact iff A^*A is compact,
- (b) if A is compact then A^* is compact.

Solution (a) Suppose that A is compact. Let $\{x_n\}$ be a bounded sequence in H , $y_n = A(x_n)$ and $z_n = A^*A(x_n) = A^*(y_n)$. Then, $\{y_n\}$ has a subsequence $\{y_{n_j}\}$ converging to some $y \in H$. So,

$$z_{n_j} = A^*(y_{n_j}) \rightarrow A^*(y)$$

Therefore, $\{z_{n_j}\}$ is a convergent subsequence of $\{A^*A(x_n)\}$. This shows that A^*A is compact.

Conversely, suppose that A^*A is compact. Let $\{x_n\}$, $\{y_n\}$, $\{z_n\}$ be as above. Then, $\{z_n\}$ has a convergent subsequence $\{z_{n_j}\}$. Suppose that $\|x_n\| \leq \alpha$ for all n . Then for all n, m , we have

$$\begin{aligned}\langle y_n - y_m, y_n - y_m \rangle &= \langle A(x_n - x_m), A(x_n - x_m) \rangle \\ &= \langle x_n - x_m, A^*A(x_n - x_m) \rangle \\ &= \langle x_n - x_m, z_n - z_m \rangle.\end{aligned}$$

Hence,

$$\|y_n - y_m\|^2 \leq \|x_n - x_m\| \|z_n - z_m\| \leq 2\alpha \|z_n - z_m\|.$$

Since $\{z_{n_j}\}$ is a Cauchy sequence, this implies that $\{y_{n_j}\}$ also is a Cauchy sequence. Thus $\{y_{n_j}\}$ converges in H . This implies that A is compact.

(b) Suppose that A is compact. Let $\{x_n\}$ be a bounded sequence in H , and $y_n = A^*(x_n)$. Then, $\{y_n\}$ is a bounded sequence since A^* is bounded. Hence, $\{Ay_n\}$ has a convergent subsequence $\{Ay_{n_j}\}$, that is, $\{AA^*x_n\}$ has a convergent subsequence $\{AA^*x_{n_j}\}$. This shows that AA^* is compact. But $AA^* = (A^*)^*A^*$. It follows that A^* is compact by (a). This completes the proof. ■

PROBLEM 7.2.2 Let H be a complex Hilbert space, and A be a compact normal operator on H . Show that there is $x \in H$ such that

$$|\langle Ax, x \rangle| = \|A\|, \quad \|x\| = 1.$$

Solution It is known that the spectrum $\sigma(A)$ of a bounded operator on a Banach space is a compact subset of the scalar field. Since $|k|$ is a continuous function of k , we see that

$$\sup \{|k| : k \in \sigma(A)\} = |\lambda|$$

for some $\lambda \in \sigma(A)$. That is, $r_\sigma(A) = |\lambda|$. Since H is a complex Banach space, the spectral radius formula, namely,

$$\lim_{n \rightarrow \infty} \|A^n\|^{1/n} = r_\sigma(A),$$

holds. Since A is normal, we have $\|A^2\| = \|A\|^2$. Since $A^2, A^4, \dots$ also are normal, it follows that

$$\|A^n\| = \|A\|^n \quad \text{for } n = 2^j, j = 1, 2, \dots$$

Hence, $\|A^n\|^{1/n} = \|A\|$ for these values of n . Letting $n \rightarrow \infty$ through these values, we see that $r_\sigma(A) = \|A\|$. Thus, there is $\lambda \in \sigma(A)$ with $|\lambda| = \|A\|$. If $\lambda = 0$, then $A = 0$ and so the result is trivially true. So let $\lambda \neq 0$.

By Problem 7.1.2(b) we see that λ is an approximate eigenvalue of A . Hence there is a sequence $\{x_n\}$ such that $\|x_n\| = 1$ and $\|Ax_n - \lambda x_n\| \rightarrow$

0. Since A is compact, $\{Ax_n\}$ has a convergent subsequence $\{Ax_{n_j}\}$. If $A(x_{n_j}) \rightarrow y$, then

$$\lambda x_{n_j} = (\lambda x_{n_j} - Ax_{n_j}) + A(x_{n_j}) \rightarrow y. \quad (4)$$

Therefore, $\lambda A(x_{n_j}) = A(\lambda x_{n_j}) \rightarrow A(y)$. Consequently, $\lambda y = A(y)$. Since $\|\lambda x_{n_j}\| = |\lambda|$ for all j , we see from (4) that $\|y\| = |\lambda| \neq 0$. Hence λ is an eigenvalue of A . Thus there is $x \in H$ with $\|x\| = 1$ and $A(x) = \lambda x$. For this x ,

$$|\langle Ax, x \rangle| = \langle \lambda x, x \rangle = |\lambda| = \|A\|.$$

This completes the proof. ■

PROBLEM 7.2.3 Let $S, T \in BL(H)$, where H is a Hilbert space. If T is compact and

$$(i) S^*S \leq T^*T \quad \text{or} \quad (ii) SS^* \leq TT^*,$$

then show that S is compact.

Solution When we write $A \leq B$, it is implied that A and B are self-adjoint and $\langle Ax, x \rangle \leq \langle Bx, x \rangle$ for all x . Let $S^*S \leq T^*T$ and $\{x_n\}$ be a bounded sequence in H . Then, for all x ,

$$\begin{aligned} \langle Sx, Sx \rangle &= \langle S^*Sx, x \rangle \leq \langle T^*Tx, x \rangle = \langle Tx, Tx \rangle, \\ \|Sx\|^2 &\leq \|Tx\|^2. \end{aligned}$$

Hence for all n, m ,

$$\|S(x_n - x_m)\| \leq \|T(x_n - x_m)\|. \quad (5)$$

Since T is compact, $\{Tx_n\}$ has a convergent subsequence $\{T(x_{n_j})\}$. So (5) shows that $\{S(x_{n_j})\}$ is a Cauchy sequence in H ; hence it converges. This shows that S is compact if (i) is true.

Next suppose that (ii) holds. Since T^* is compact by Problem 7.2.1(b), we see from what has been proved above that S^* is compact and hence S is compact. This completes the proof. ■

PROBLEM 7.2.4 Let A be a compact operator on a Hilbert space, and $\{u_n\}$ be an infinite orthonormal sequence in H . Show that $A(u_n) \rightarrow 0$ in H .

Solution Suppose that $\{Au_n\}$ does not tend to zero. Then there is $\epsilon > 0$ and a subsequence $\{Au_{n_r}\}$ such that $\|Au_{n_r}\| \geq \epsilon$ for all r . Let $v_r = u_{n_r}$. Since $\|v_r\| = 1$ and A is compact, $\{Av_r\}$ has a convergent subsequence $\{A(v_{r_i})\}$. Let $w_i = v_{r_i}$ and $A(w_i) \rightarrow x$. Then,

$$\langle A(w_i), y \rangle \rightarrow \langle x, y \rangle \quad \text{for } y \in H.$$

But,

$$\sum_i |\langle Aw_i, y \rangle|^2 = \sum_i |\langle w_i, A^*y \rangle|^2 \leq \|A^*y\|^2$$

by Bessel's inequality. Since $\|A^*y\| < \infty$, it follows that $\langle Aw_i, y \rangle \rightarrow 0$ as $i \rightarrow \infty$. Hence, $\langle x, y \rangle = 0$ for any $y \in H$. Thus, $x = 0$ and so $A(w_i) \rightarrow 0$. Since this contradicts the fact that

$$\|Aw_i\| = \|A(v_{r_i})\| \geq \epsilon \quad \text{for all } i,$$

the proof is complete. ■

PROBLEM 7.2.5 Let A be a compact operator on a Hilbert space H and $\delta > 0$. Prove the following:

- (a) Any set of linearly independent eigenvectors of A corresponding to eigenvalues k with $|k| > \delta$ is finite.
- (b) The eigenspace corresponding to a nonzero eigenvalue of A is finite dimensional.
- (c) A has only a countable number of distinct eigenvalues.

Solution (a) Suppose that there is an infinite sequence of linearly independent vectors x_n corresponding to eigenvalues k_n with $|k_n| > \delta$. Let $\{u_n\}$ be the orthonormal sequence constructed out of $\{x_n\}$ using the Gram-Schmidt process. Let

$$F_n = \text{span}\{x_1, \dots, x_n\} = \text{span}\{u_1, \dots, u_n\}.$$

If $x = c_1x_1 + \dots + c_nx_n$ is any element of F_n , then

$$\begin{aligned} A(x) &= \sum_{j=1}^n c_j A(x_j) = \sum_{j=1}^n c_j k_j x_j \in F_n, \\ A(x) - k_n x &= \sum_{j=1}^n c_j (k_j - k_n) x_j = \sum_{j=1}^{n-1} c_j (k_j - k_n) x_j \in F_{n-1}. \end{aligned}$$

Let $n > m$ and $y = A(u_n) - k_n u_n - A(u_m)$. Since $u_m \in F_m \subset F_{n-1}$, we see that

$$A(u_m) \in F_{n-1}, \quad A(u_n) - k_n u_n \in F_{n-1},$$

and hence $y \in F_{n-1}$. It follows that $\langle y, k_n u_n \rangle = 0$, and so

$$\|Au_n - Au_m\|^2 = \|y + k_n u_n\|^2 = \|y\|^2 + \|k_n u_n\|^2 \geq |k_n|^2 > \delta^2.$$

Thus, $\|Au_n - Au_m\| > \delta$ whenever $n > m$. This shows that $\{A(u_n)\}$ cannot have a convergent subsequence. Since $\{u_n\}$ is bounded, this contradicts the compactness of A and completes the proof of (a).

(b) If the eigenspace corresponding to a nonzero eigenvalue k is infinite dimensional, then it contains an infinite number of linearly independent vectors; this contradicts (a) if $0 < \delta < |k|$.

(c) Let there be an infinite number of distinct eigenvalues $k_1, k_2, \dots$ whose absolute values are greater than δ . Let $A(x_n) = k_n x_n, x_n \neq 0$. It can

be seen that eigenvectors corresponding to distinct eigenvalues are linearly independent. This contradicts (a). It follows that for $m = 1, 2, \dots$, the set

$$E_m = \left\{ k \in \sigma_e(A) : |k| > \frac{1}{m} \right\}$$

is finite. Since

$$\sigma_e(A) \subset \{0\} \cup \bigcup_{m=1}^{\infty} E_m,$$

we see that $\sigma_e(A)$ is countable. This completes the proof. ■

PROBLEM 7.2.6 Let A be a nonzero compact self-adjoint operator on a Hilbert space H . Prove that there is a finite or infinite sequence $\{\lambda_n\}$ of nonzero real numbers with $|\lambda_1| \geq |\lambda_2| \geq \dots$ and a corresponding finite or infinite orthonormal sequence $\{u_n\}$ in H such that

$$A(x) = \sum_{n \geq 1} \lambda_n \langle x, u_n \rangle u_n, \quad x \in H.$$

Also show that

- (a) If $\{\lambda_n\}$ is an infinite sequence then $\lambda_n \rightarrow 0$ as $n \rightarrow \infty$.
- (b) Any λ_n is repeated only a finite number of times in the sequence.
- (c) $\lambda_1, \lambda_2, \dots$ are precisely the nonzero eigenvalues of A .

Solution Since A is compact, its nonzero spectral values are eigenvalues. So, by Problem 7.1.6, A has a nonzero eigenvalue λ_1 such that $|\lambda_1| = \|A\| > 0$. Let u_1 be an eigenvector corresponding to λ_1 with $\|u_1\| = 1$, $F_1 = \text{span}\{u_1\}$ and $H_1 = F_1^\perp$. Then, H_1 is a Hilbert space. Since $Au_1 = \lambda_1 u_1$ and $\langle Ax, u_1 \rangle = \langle x, Au_1 \rangle$, we see that

$$\langle Ax, u_1 \rangle = \lambda_1 \langle x, u_1 \rangle = 0 \quad \text{for } x \in F_1^\perp.$$

This shows that $Ax \in F_1^\perp$ for $x \in F_1^\perp$. Thus, if A_1 is the restriction of A to H_1 , then it is a bounded operator on H_1 ; it is self-adjoint and compact. If A_1 is nonzero, then repeat the argument with A, H replaced by A_1, H_1 .

More generally, assume that we have already found an orthonormal set $\{u_1, \dots, u_m\}$ and real numbers λ_j with $A(u_j) = \lambda_j u_j$ and $|\lambda_1| \geq \dots \geq |\lambda_m| > 0$. Let $F_m = \text{span}\{u_1, \dots, u_m\}$ and $H_m = F_m^\perp$. Then, H_m is a Hilbert space. As before,

$$\langle Ax, u_j \rangle = \langle x, Au_j \rangle = \lambda_j \langle x, u_j \rangle = 0 \quad \text{for all } x \in F_m^\perp.$$

This shows that if A_m is the restriction of A to H_m , then it is a bounded operator on H_m ; it is self-adjoint and compact.

Suppose that A_m is nonzero. As it is compact and self-adjoint, we can repeat argument given in the beginning with A, H replaced by A_m, H_m . Let $H_0 = H$ and $A_0 = A$. We see that A_m has a nonzero eigenvalue λ_{m+1} such

that $|\lambda_{m+1}| = \|A_m\| > 0$. Let $u_{m+1} \in H_m$ be an eigenvector corresponding to λ_{m+1} with $\|u_{m+1}\| = 1$. Then, u_{m+1} is orthogonal to u_j for $1 \leq j \leq m$. Thus, $\{u_1, \dots, u_{m+1}\}$ is an orthonormal set and $A(u_j) = \lambda_j u_j$ for $1 \leq j \leq m+1$. Since $H_m \subset H_{m-1}$, we see that $|\lambda_{m+1}| = \|A_m\| \leq \|A_{m-1}\| = \lambda_m$. The procedure can be repeated until we arrive at a stage where A_m is identically zero.

First suppose that some A_m is identically zero. For each $x \in H$, we have $x = y + z$, where $y \in F_m$ and $z \in H_m$. Then,

$$y = \sum_{j=1}^m k_j u_j, \quad A(y) = \sum_{j=1}^m k_j A(u_j) \quad A(z) = A_m(z) = 0.$$

Further, since $u_j \perp u_i$ for $i \neq j$ and $u_j \perp H_m$, we get

$$\langle x, u_j \rangle = \langle y, u_j \rangle + \langle z, u_j \rangle = \left\langle \sum_{i=1}^m k_i u_i, u_j \right\rangle + 0 = k_j.$$

This shows that

$$A(y) = \sum_{j=1}^m \langle x, u_j \rangle A(u_j) = \sum_{j=1}^m \langle x, u_j \rangle \lambda_j u_j.$$

This gives the desired expression for A if A_m is the zero operator.

Now, suppose that every A_m is nonzero. We get an infinite orthonormal set $\{u_n\}$ and a sequence $\{\lambda_n\}$ of real numbers with $A(u_n) = \lambda_n u_n$ and $|\lambda_n| \geq |\lambda_{n+1}|$ for all n . So, $\delta = \lim_{n \rightarrow \infty} |\lambda_n|$ exists and $|\lambda_n| \geq \delta$ for all n . So, for $n \neq m$,

$$\|A(u_n) - A(u_m)\|^2 = \|\lambda_n u_n - \lambda_m u_m\|^2 = |\lambda_n|^2 \|u_n\|^2 + |\lambda_m|^2 \|u_m\|^2,$$

since $\langle u_n, u_m \rangle = 0$. Thus, $\|A(u_n) - A(u_m)\|^2 \geq 2\delta^2$ for $n \neq m$. If $\delta > 0$, then this gives a contradiction, since $\{u_n\}$ is bounded and A is compact. This proves that $\lambda_n \rightarrow 0$ as $n \rightarrow \infty$. This, in particular, shows that no λ_n is repeated an infinite number of times in the sequence.

Let $Y = \text{span}\{u_n\}$, $F_0 = \overline{Y}$, the closure of Y and $H_0 = F_0^\perp$. For every $x \in H$, we have $x = y + z$, where $y \in F_0$ and $z \in H_0$. Since $F_n \subset Y \subset F_0$ for all n , we see that $H_0 = F_0^\perp \subset F_n^\perp = H_n$. Hence,

$$z \in H_n, \quad \|A(z)\| = \|A_n(z)\| \leq \|A_n\| \|z\| = |\lambda_{n+1}| \|z\|$$

for all n . Since $\lambda_n \rightarrow 0$, this shows that $A(z) = 0$. Further, $\langle x, u_n \rangle = \langle y, u_n \rangle + \langle z, u_n \rangle = \langle y, u_n \rangle$. Again, by Problem 5.2.7,

$$y = \sum_{n=1}^{\infty} \langle y, u_n \rangle u_n = \sum_{n=1}^{\infty} \langle x, u_n \rangle u_n, \quad A(y) = \sum_{n=1}^{\infty} \langle x, u_n \rangle A(u_n).$$

Now, $A(u_n) = \lambda_n u_n$. Since $A(z) = 0$, we get $A(x) = A(y) + 0$. This gives

$$A(x) = A(y) = \sum_{n=1}^{\infty} \lambda_n \langle x, u_n \rangle u_n.$$

It is to be proved that A has no nonzero eigenvalues other than λ_n . Let λ be an eigenvalue and u be a corresponding eigenvector of norm 1. Then,

$$\lambda u = A(u) = \sum_{n \geq 1} \lambda_n \langle u, u_n \rangle u_n.$$

If $\lambda \neq \lambda_n$, then $\langle u, u_n \rangle = 0$ as eigenvectors corresponding to distinct eigenvalues are orthogonal. Hence, if $\lambda \neq \lambda_n$ for any n , then $\lambda u = 0$, $\lambda = 0$. This completes the proof. ■

PROBLEM 7.2.7 Let A be a nonzero compact operator on a Hilbert space H . Prove the following:

- (a) If A is normal then every nonzero spectral value of A is an eigenvalue.
- (b) If A is self-adjoint then $\|A\|$ or $-\|A\|$ is an eigenvalue of A .
- (c) If A is not self-adjoint we can have $|\lambda| < \|A\|$ for every spectral value λ of A .
- (d) If A is self-adjoint then its nonzero eigenvalues $\lambda_1, \lambda_2, \dots$, can be arranged so that

$$\|A\| = |\lambda_1| \geq |\lambda_2| \geq \dots$$

Solution (a) Let A be a compact normal operator in $BL(H)$, and λ be a nonzero spectral value of A . Then λ is an approximate eigenvalue by Problem 7.1.2(b). So there is a sequence $\{x_n\}$ with $\|x_n\| = 1$ for all n and $\|Ax_n - \lambda x_n\| \rightarrow 0$. As in the last part of the solution of Problem 7.2.2, we see that λ is an eigenvalue of A .

(b) Let A be a nonzero compact self-adjoint operator in $BL(H)$. Then the bounds m_A and M_A of the numerical range $W(A)$ are spectral values of A , and moreover,

$$\|A\| = \max\{|m_A|, |M_A|\}.$$

So there is a spectral value λ , namely, m_A or M_A , such that $|\lambda| = \|A\| \neq 0$. Since the spectral values of A are real, we see that $\lambda = \|A\|$ or $-\|A\|$, and λ is an eigenvalue by (a).

(c) Let $H = \mathbb{R}^2$ and $A \in BL(H)$ be represented by the matrix

$$M = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}.$$

Using Problem 1.2.13 we see that $\|A\| = (1/2)(\sqrt{10} + \sqrt{2})$. The eigenvalues, being the roots of the characteristic equation, are 1, 2. Thus, $|\lambda| < \|A\|$ for each eigenvalue λ .

(d) As in the solution of Problem 7.2.5(c), we see that for each $\delta > 0$ there exist only a finite number of eigenvalues k such that $|k| > \delta$. It follows that the nonzero eigenvalues of A can be written as $\lambda_1, \lambda_2, \dots$, where $|\lambda_1| \geq |\lambda_2| \geq \dots$. Since

$$\sigma(A) \subset \overline{W(A)} = [m_A, M_A]$$

and $\|A\| = \max\{|m_A|, |M_A|\}$, we see that $|\lambda_n| \leq \|A\|$ for all n . By (b) there is an eigenvalue λ_1 with $|\lambda_1| = \|A\|$. This proves (d). ■

PROBLEM 7.2.8 Let A be a nonzero compact operator on a Hilbert space H . Prove that there exists a finite or infinite nondecreasing sequence $\{\alpha_n\}$ of positive numbers and orthonormal sequences $\{u_n\}$ and $\{v_n\}$ such that

$$A(x) = \sum_{n \geq 1} \alpha_n \langle x, u_n \rangle v_n, \quad x \in H, \quad (6)$$

$$A^*(x) = \sum_{n \geq 1} \alpha_n \langle x, v_n \rangle u_n, \quad x \in H. \quad (7)$$

Solution Since A is compact, we see that A^*A is a compact self-adjoint positive operator. So its nonzero eigenvalues are positive and hence can be written as $\alpha_1^2, \alpha_2^2, \dots$, where $\alpha_n \geq \alpha_{n+1} > 0$ for all n and $\alpha_n \rightarrow 0$ if $\{\alpha_n\}$ is infinite. Thus, by the theorem stated in the beginning of this section,

$$A^*A(x) = \sum \alpha_n^2 \langle x, u_n \rangle u_n, \quad x \in H, \quad (8)$$

where $\{u_n\}$ is the orthonormal sequence of eigenvectors corresponding to $\{\alpha_n^2\}$. Let $v_n = A(u_n/\alpha_n)$,

$$A_m(x) = \sum_{n=1}^m \alpha_n \langle x, u_n \rangle v_n, \quad x \in H, \quad (9)$$

and $B_m = A - A_m$. Then, $A(u_n) = \alpha_n v_n$, and

$$\begin{aligned} \alpha_j \alpha_n \langle v_j, v_n \rangle &= \langle Au_j, Au_n \rangle = \langle u_j, A^* Au_n \rangle = \langle u_j, \alpha_n^2 u_n \rangle, \\ \alpha_j \langle v_j, v_n \rangle &= \alpha_n \langle u_j, u_n \rangle. \end{aligned}$$

This shows that $\{v_n\}$ is an orthonormal sequence. Further,

$$\alpha_n \langle Ax, v_n \rangle = \langle Ax, Au_n \rangle = \langle A^* Ax, u_n \rangle = \alpha_n^2 \langle x, u_n \rangle$$

from (8). From (9) we see that

$$\alpha_n \langle x, u_n \rangle = \langle A_m x, v_n \rangle \quad \text{for } n \leq m.$$

Thus we get

$$\langle B_m x, v_n \rangle = \langle Ax - A_m x, v_n \rangle = \langle Ax, v_n \rangle - \langle A_m x, v_n \rangle = 0$$

for $1 \leq n \leq m$, and

$$\langle x, A^* v_n \rangle = \langle Ax, v_n \rangle = \langle x, \alpha_n u_n \rangle$$

for all $n \geq 1$. Since this is true for all $x \in H$, we must have $A^*(v_n) = \alpha_n u_n$ for all $n \geq 1$. Thus for all $x, y \in H$,

$$\langle A_m^* B_m x, y \rangle = \langle B_m x, A_m y \rangle = \langle B_m x, \sum_{n=1}^m \alpha_n \langle y, u_n \rangle v_n \rangle = 0.$$

This implies that $A_m^* B_m(x) = 0$ for all $x \in H$. Hence,

$$\begin{aligned} B_m^* B_m(x) &= (A^* - A_m^*) B_m(x) = A^* B_m(x) \\ &= A^* A(x) - A^* A_m(x) \\ &= \sum_{n \geq 1} \alpha_n^2 \langle x, u_n \rangle u_n - \sum_{n=1}^m \alpha_n \langle x, u_n \rangle \alpha_n u_n. \end{aligned} \quad (10)$$

Since $\alpha_n \leq \alpha_{n+1}$ for all $n > m$ and $\{u_n\}$ is orthonormal this gives

$$\begin{aligned} \|B_m^* B_m(x)\|^2 &= \left\| \sum_{n > m} \alpha_n^2 \langle x, u_n \rangle u_n \right\|^2 \\ &= \sum_{n > m} \alpha_n^4 |\langle x, u_n \rangle|^2 \leq \alpha_{m+1}^4 \|x\|^2 \end{aligned}$$

by Bessel's inequality. Thus,

$$\begin{aligned} \langle B_m x, B_m x \rangle &= \langle B_m^* B_m x, x \rangle \leq \|B_m^* B_m x\| \|x\| \leq \alpha_{m+1}^2 \|x\|^2, \\ \|B_m(x)\| &\leq \alpha_{m+1} \|x\|, \quad x \in H. \end{aligned}$$

If $\{\alpha_n\}$ is infinite, then $\alpha_{m+1} \rightarrow 0$ as $m \rightarrow \infty$, and so

$$\|A(x) - A_m(x)\| = \|B_m(x)\| \rightarrow 0.$$

This gives (6). If $\{\alpha_n\}$ contains only a finite number of terms $\alpha_1, \dots, \alpha_N$, then we define A_m, B_m for $1 \leq m \leq N$ only. In this case, (10) with $m = N$ shows that $B_N^* B_N(x) = 0$. Therefore, exactly as above, $\langle B_N(x), B_N(x) \rangle = \langle B_N^* B_N x, x \rangle = 0$ for all $x \in H$. This implies that $A(x) = A_N(x)$. Thus, (6) is proved in this case also.

For (7) we have, for $x, y \in H$,

$$\begin{aligned} \langle x, A^* y \rangle &= \langle Ax, y \rangle = \left\langle \sum \alpha_n \langle x, u_n \rangle v_n, y \right\rangle \\ &= \sum \alpha_n \langle x, u_n \rangle \langle v_n, y \rangle = \left\langle x, \sum \alpha_n \langle y, v_n \rangle u_n \right\rangle. \end{aligned}$$

Therefore,

$$A^*(y) = \sum \alpha_n \langle y, v_n \rangle u_n, \quad y \in H.$$

This completes the proof. ■

PROBLEM 7.2.9 Let $\{u_\alpha : \alpha \in L\}$ be an orthonormal basis for a Hilbert space H . Suppose that $A \in BL(H)$ and

$$\sum_{\alpha \in L} \|A(u_\alpha)\|^2 < \infty. \quad (11)$$

Prove the following:

- (a) $\sum_{\alpha \in L} \|A^*(u_\alpha)\|^2 = \sum_{\alpha \in L} \|A(u_\alpha)\|^2$.
 (b) $\sum_{i \in J} \|A(v_i)\|^2 = \sum_{\alpha \in L} \|A(u_\alpha)\|^2$ for any other orthonormal basis

$$\{v_i : i \in J\} \quad \text{for } H.$$

- (c) A is compact.

[An operator A for which (11) holds is called a *Hilbert-Schmidt operator*.]

Solution (a) Note that if (11) holds then all but a countable number of terms in it must be zero. For each $\beta \in L$, Parseval's formula for $\|A^*(u_\beta)\|$ gives

$$\|A^*u_\beta\|^2 = \sum_{\alpha} |\langle A^*u_\beta, u_\alpha \rangle|^2 = \sum_{\alpha} |\langle u_\beta, Au_\alpha \rangle|^2.$$

Hence,

$$\begin{aligned} \sum_{\beta \in L} \|A^*(u_\beta)\|^2 &= \sum_{\beta} \sum_{\alpha} |\langle u_\beta, Au_\alpha \rangle|^2 \\ &= \sum_{\alpha} \sum_{\beta} |\langle Au_\alpha, u_\beta \rangle|^2 = \sum_{\alpha} \|Au_\alpha\|^2, \end{aligned}$$

using Parseval's formula for $A(u_\alpha)$. The change of order of summation is valid since all the terms are non-negative.

- (b) Exactly as above,

$$\begin{aligned} \sum_{i \in J} \|Av_i\|^2 &= \sum_i \sum_{\alpha} |\langle Av_i, u_\alpha \rangle|^2 = \sum_i \sum_{\alpha} |\langle v_i, A^*u_\alpha \rangle|^2 \\ &= \sum_{\alpha} \sum_i |\langle A^*u_\alpha, v_i \rangle|^2 \\ &= \sum_{\alpha} \|A^*u_\alpha\|^2 = \sum_{\alpha} \|Au_\alpha\|^2. \end{aligned}$$

Here the last equality is given by (a) and the previous one by Parseval's formula as $\{v_i\}$ is an orthonormal basis.

- (c) Since

$$\sum_{\alpha} \|A^*u_\alpha\|^2 = \sum_{\alpha} \|Au_\alpha\|^2 < \infty,$$

the set $E = \{u_\alpha : A^*(u_\alpha) \neq 0\}$ is countable. Let $E = \{w_n\}$. Then, for all $x \in H$,

$$\begin{aligned} A(x) &= \sum_{\alpha} \langle Ax, u_\alpha \rangle u_\alpha \\ &= \sum_{\alpha} \langle x, A^* u_\alpha \rangle u_\alpha = \sum_{n \geq 1} \langle x, A^* w_n \rangle w_n. \end{aligned} \quad (12)$$

If E contains only a finite number of terms, then the range $R(A)$ is $\text{span}(E)$, and hence finite dimensional. So A is compact in this case.

Suppose that E is infinite. For $m = 1, 2, \dots$, let

$$A_m(x) = \sum_{n=1}^m \langle x, A^* w_n \rangle w_n, \quad x \in H, \quad (13)$$

and $B_m = A - A_m$. Then A_m has a finite dimensional range and hence compact. From (12) and (13) we get, using the orthonormality of $\{w_n\}$,

$$\begin{aligned} \|B_m(x)\|^2 &= \left\| \sum_{n=m+1}^{\infty} \langle x, A^* w_n \rangle w_n \right\|^2 = \sum_{n=m+1}^{\infty} \|\langle x, A^* w_n \rangle w_n\|^2 \\ &= \sum_{n=m+1}^{\infty} |\langle x, A^* w_n \rangle|^2 \leq \sum_{n=m+1}^{\infty} \|x\|^2 \|A^*(w_n)\|^2. \end{aligned}$$

This is true for all $x \in H$. Hence,

$$\|B_m\| \leq \left(\sum_{n=m+1}^{\infty} \|A^* w_n\|^2 \right)^{1/2}.$$

This tends to zero as $m \rightarrow \infty$. This shows that $A_m \rightarrow A$. Therefore, A is compact. This completes the proof. ■

PROBLEM 7.2.10 Let H be an infinite dimensional Hilbert space, $\{u_n\}$ be an orthonormal basis for H , $\{v_n\}$ be some orthonormal sequence in H and $\{k_n\}$ be a sequence of scalars. Prove the following:

(a) If $\{k_n\}$ is bounded, then

$$A(x) = \sum_{n=1}^{\infty} k_n \langle x, u_n \rangle v_n, \quad x \in H,$$

defines an operator A in $BL(H)$.

(b) A is compact iff $k_n \rightarrow 0$.

(c) A is a Hilbert-Schmidt operator iff $\sum |k_n|^2 < \infty$.

Solution (a) If $|k_n| \leq \alpha$, then

$$\sum_{n=1}^{\infty} |k_n \langle x, u_n \rangle|^2 \leq \alpha^2 \sum_{n=1}^{\infty} |\langle x, u_n \rangle|^2 = \alpha^2 \|x\|^2.$$

This, together with the Riesz-Fischer theorem, shows that $A(x)$ exists for each $x \in H$. The orthonormality of $\{v_n\}$ implies that

$$\|Ax\|^2 = \sum_{n=1}^{\infty} \|k_n \langle x, u_n \rangle v_n\|^2 \leq \alpha^2 \|x\|^2.$$

Since A is clearly linear, we see that $A \in BL(H)$.

(b) First suppose that $k_n \rightarrow 0$. For $m = 1, 2, \dots$, let

$$\alpha_m = \sup_{n \geq m} |k_n|,$$

$$A_m(x) = \sum_{n=1}^m k_n \langle x, u_n \rangle v_n, \quad x \in H$$

and $B_m = A - A_m$. Then A_m is compact since its range is finite dimensional, and further,

$$B_m(x) = \sum_{n=m+1}^{\infty} k_n \langle x, u_n \rangle v_n, \quad x \in H.$$

Since $\{v_n\}$ is orthonormal, this gives

$$\begin{aligned} \|B_m(x)\|^2 &= \sum_{n=m+1}^{\infty} \|k_n \langle x, u_n \rangle v_n\|^2 \\ &\leq \alpha_m^2 \sum_{n=m+1}^{\infty} |\langle x, u_n \rangle|^2 \leq \alpha_m^2 \|x\|^2. \end{aligned}$$

This implies that $\|B_m\| \leq \alpha_m \rightarrow 0$ as $m \rightarrow \infty$. Thus, $A_m \rightarrow A$, and so A is compact.

For the converse suppose that $\{k_n\}$ does not to zero. Then, for some $\epsilon > 0$, we have $|k_n| \geq \epsilon$ for an infinite number of values of n , say, for $n = n_1, n_2, \dots$. The definition of $A(x)$ shows that $A(u_m) = k_m v_m$ for $m = 1, 2, \dots$. Therefore,

$$\begin{aligned} \|A(u_{n_i}) - A(u_{n_j})\|^2 &= \|k_{n_i} v_{n_i} - k_{n_j} v_{n_j}\|^2 \\ &= |k_{n_i}|^2 + |k_{n_j}|^2 \geq 2\epsilon^2. \end{aligned}$$

This is true for all $i, j, i \neq j$. This implies that $\{A(u_{n_i})\}$ cannot have a Cauchy subsequence; so it cannot have a convergent subsequence. Thus A cannot be compact since $\{u_{n_i}\}$ is a bounded sequence. This proves (b).

(c) Since $A(u_n) = k_n v_n$, we have

$$\sum_{n=1}^{\infty} \|A(u_n)\|^2 = \sum_{n=1}^{\infty} \|k_n v_n\|^2 = \sum_{n=1}^{\infty} |k_n|^2.$$

Since $\{u_n\}$ is an orthonormal basis, this gives (c). ■

PROBLEM 7.2.11 Show that the Jacobi matrix given in Problem 6.1.1 represents a compact operator on l^2 with respect to the standard basis iff $a_n \rightarrow 0$, $b_n \rightarrow 0$ and $c_n \rightarrow 0$.

Solution Let A be represented by the Jacobi matrix

$$M = \begin{bmatrix} a_1 & b_1 & 0 & 0 & 0 & \dots \\ c_2 & a_2 & b_2 & 0 & 0 & \dots \\ 0 & c_3 & a_3 & b_3 & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$$

If e_n is the n th unit vector, then

$$\begin{aligned} A(e_n) &= \sum_{i=1}^{\infty} \langle A e_n, e_i \rangle e_i = \sum_{i=1}^{\infty} k_{in} e_i \\ &= b_{n-1} e_{n-1} + a_n e_n + c_{n+1} e_{n+1}, \end{aligned}$$

where $b_0 = 0$, $e_0 = 0$. Hence,

$$\|A e_n\|^2 = |b_{n-1}|^2 + |a_n|^2 + |c_{n+1}|^2.$$

If A is compact then $A(e_n) \rightarrow 0$, by Problem 7.2.4; this implies that $b_n \rightarrow 0$, $a_n \rightarrow 0$ and $c_n \rightarrow 0$.

Conversely, suppose that $b_n \rightarrow 0$, $a_n \rightarrow 0$ and $c_n \rightarrow 0$. Let

$$\begin{aligned} B(x) &= \sum_{n=1}^{\infty} b_{n-1} \langle x, e_n \rangle e_{n-1}, \quad x \in H, \\ C(x) &= \sum_{n=1}^{\infty} a_n \langle x, e_n \rangle e_n, \quad x \in H, \\ D(x) &= \sum_{n=1}^{\infty} c_{n+1} \langle x, e_n \rangle e_{n+1}, \quad x \in H. \end{aligned}$$

Then B, C and D are compact operators on l^2 by Problem 7.2.10(b). We see from Problem 6.1.1 that M represents an element $A \in BL(H)$. Further,

$$\begin{aligned} B(e_m) + C(e_m) + D(e_m) &= b_{m-1} e_{m-1} + a_m e_m + c_{m+1} e_{m+1} \\ &= A(e_m). \end{aligned}$$

It follows that $A(x) = B(x) + C(x) + D(x)$ for all x . Hence A is compact. This completes the proof. ■

PROBLEM 7.2.12 Let $k(s, t)$ be a continuous scalar-valued function on the unit square $[0, 1] \times [0, 1]$, which is not identically zero, and is such that $k(s, t) = \overline{k(t, s)}$ for all s, t in $[0, 1]$. Let A be defined on $L^2[0, 1]$ by

$$A(x)(s) = \int_0^1 k(s, t)x(t) dt, \quad 0 \leq s \leq 1, \quad x \in L^2[0, 1].$$

Show that there is a sequence $\{\lambda_n\}$ of nonzero real numbers and an orthonormal sequence $\{u_n\}$ of continuous functions on $[0, 1]$ such that, for $x \in L^2[0, 1]$,

$$A(x)(s) = \sum_{n \geq 1} \lambda_n \langle x, u_n \rangle u_n(s),$$

where the series, if infinite, converges uniformly for $0 \leq s \leq 1$. Show also that $\sum |\lambda_n|^2 < \infty$.

Solution Let $H = L^2[0, 1]$. It is known from Problem 4.1.15 that A is a compact operator on H . It can be seen that A is nonzero, and by Problem 6.2.22 we see that A is self-adjoint. Let $\{\lambda_n\}$ be the sequence, finite or infinite, of nonzero eigenvalues of A with $|\lambda_n| \geq |\lambda_{n+1}|$, and $\{v_n\}$ be the corresponding orthonormal sequence of eigenvectors. Then,

$$A(x) = \sum_{n \geq 1} \lambda_n \langle x, v_n \rangle v_n. \quad (14)$$

Here the sum exists and equality holds as elements of H .

Since $k(s, t)$ is continuous and hence uniformly continuous on $[0, 1] \times [0, 1]$, for each $\epsilon > 0$, there is $\delta > 0$ such that

$$|k(s_1, t) - k(s_2, t)| < \epsilon \quad \text{for } |s_1 - s_2| < \delta, \quad 0 \leq t \leq 1.$$

Hence for any $x \in H$ and for $|s_1 - s_2| < \delta$,

$$\begin{aligned} |Ax(s_1) - Ax(s_2)| &= \left| \int_0^1 [k(s_1, t) - k(s_2, t)]x(t) dt \right| \\ &\leq \int_0^1 |k(s_1, t) - k(s_2, t)| |x(t)| dt \\ &\leq \left(\int_0^1 \epsilon^2 dt \right)^{1/2} \left(\int_0^1 |x(t)|^2 dt \right)^{1/2} = \epsilon \|x\|. \end{aligned}$$

This shows that $A(x)$ is a continuous function on $[0, 1]$ for $x \in H$.

Let x be any element of H . Since

$$\sum_n |\langle x, v_n \rangle|^2 \leq \|x\|^2 < \infty,$$

the Riesz-Fischer theorem implies that

$$y = \sum_n \langle x, v_n \rangle v_n \quad (15)$$

exists in H . Let

$$y_m = \sum_{n=1}^m \langle x, v_n \rangle v_n, \quad m = 1, 2, \dots \quad (16)$$

First suppose that the sequence $\{v_n\}$ is infinite. Then, $y_m \rightarrow y$ in H . Let $|k(s, t)| \leq \alpha$ for all $s, t \in [0, 1]$. For $0 \leq s \leq 1$, we get

$$\begin{aligned} |A(y_m)(s) - A(y)(s)| &\leq \int_0^1 |k(s, t)| |y_m(t) - y(t)| dt \\ &\leq \left(\int_0^1 |k(s, t)|^2 dt \right)^{1/2} \left(\int_0^1 |y_m(t) - y(t)|^2 dt \right)^{1/2} \\ &\leq \alpha \|y_m - y\|. \end{aligned}$$

This shows that $A(y_m)(s) \rightarrow A(y)(s)$, as $m \rightarrow \infty$, uniformly for $0 \leq s \leq 1$.

From (14), we first see that $A(v_n) = \lambda_n v_n$, and hence

$$A(x) = \sum_{n=1}^{\infty} \langle x, v_n \rangle A(v_n) = A(y)$$

by (14) and (15). So $A(x) - A(y)$ is the zero element of $L^2[0, 1]$. But $A(x) - A(y) = A(x - y)$ is a continuous function on $[0, 1]$; hence it must be identically zero in $[0, 1]$. Therefore, $A(x)(s) = A(y)(s)$ for $0 \leq s \leq 1$. Thus, $A(y_m)(s) \rightarrow A(x)(s)$ as $m \rightarrow \infty$, uniformly for $0 \leq s \leq 1$.

Let $u_n = (1/\lambda_n)A(v_n)$. Then u_n is a continuous function on $[0, 1]$. Since $A(v_n) = \lambda_n v_n$, we see that u_n and v_n represent the same element in H . So $\{u_n\}$ is an orthonormal sequence in $L^2[0, 1]$, and $\langle x, v_n \rangle = \langle x, u_n \rangle$. Hence from (16),

$$A(y_m) = \sum_{n=1}^m \langle x, v_n \rangle A(v_n) = \sum_{n=1}^m \langle x, u_n \rangle \lambda_n u_n.$$

This equality holds as elements of H . However, the first and last members represent continuous functions on $[0, 1]$. Hence, exactly as above,

$$A(y_m)(s) = \sum_{n=1}^m \lambda_n \langle x, u_n \rangle u_n(s).$$

Thus,

$$\sum_{n=1}^m \lambda_n \langle x, u_n \rangle u_n(s) \rightarrow A(x)(s)$$

as $m \rightarrow \infty$, uniformly for $0 \leq s \leq 1$.

Next suppose that $\{v_n\}$ contains only a finite number of terms. Let them be $v_1, \dots, v_N$. If $u_n = (1/\lambda_n)A(v_n)$ as above, then (14) becomes

$$A(x) = \sum_{n=1}^N \lambda_n \langle x, v_n \rangle v_n = \sum_{n=1}^N \lambda_n \langle x, u_n \rangle u_n.$$

This implies that

$$A(x)(s) = \sum_{n=1}^N \lambda_n \langle x, u_n \rangle u_n(s).$$

It remains to prove that $\sum |\lambda_n|^2 < \infty$. Let $k_s(t) = k(s, t)$ for $0 \leq t \leq 1$. Since $\overline{k_s}$ is continuous on $[0, 1]$, it lies in $L^2[0, 1]$. So,

$$A(v_n)(s) = \int_0^1 k(s, t) v_n(t) dt = \langle v_n, \overline{k_s} \rangle.$$

Therefore,

$$\begin{aligned} \sum_{n \geq 1} \|A(v_n)\|^2 &= \sum_n \int_0^1 |A(v_n)(s)|^2 ds = \int_0^1 \sum_n |A(v_n)(s)|^2 ds \\ &= \sum_n |\langle \overline{k_s}, v_n \rangle|^2 \leq \int_0^1 \|\overline{k_s}\|^2 ds \end{aligned}$$

using Bessel's inequality. If $|k(s, t)| \leq \alpha$, then

$$\int_0^1 \|\overline{k_s}\|^2 ds = \int_0^1 ds \int_0^1 |\overline{k(s, t)}|^2 dt \leq \alpha^2.$$

Since $A(v_n) = \lambda_n v_n$, we get

$$\sum_n |\lambda_n|^2 = \sum_n \|A(v_n)\|^2 \leq \alpha^2.$$

This completes the proof. ■

PROBLEM 7.2.13 Let $z \in L^2(-\pi, \pi]$ and extend z to $\mathbb{R}$ as a 2π -periodic function. For x in $L^2[-\pi, \pi]$, let

$$A(x)(s) = \frac{1}{2\pi} \int_{-\pi}^{\pi} z(s-t)x(t) dt.$$

Prove the following:

- (a) If $\sum_{n=-\infty}^{\infty} c_n e^{int}$ is the Fourier series of z then for all $x \in L^2[-\pi, \pi]$,

$$A(x)(s) = \sum_{n=-\infty}^{\infty} c_n \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} x(t) e^{-int} dt \right) e^{ins},$$

the series converging uniformly and absolutely on $[-\pi, \pi]$.

- (b) A is compact.
 (c) The eigenvalues of A are given by the Fourier coefficients c_n of z , and the corresponding eigenfunctions are e^{ins} , $n = 0, \pm 1, \pm 2, \dots$

Solution (a) Let $u_n(t) = (2\pi)^{-1/2}e^{int}$, $n = 0, \pm 1, \pm 2, \dots$. Then $\{u_n\}$ is an orthonormal basis for $H = L^2[-\pi, \pi]$. For $s \in [-\pi, \pi]$, let $z_s(t) = z(s - t)$. Then z_s belongs to $L^2[-\pi, \pi]$, and so

$$\begin{aligned} A(x)(s) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} z_s(t)x(t) dt = \frac{1}{2\pi} \langle x, \overline{z_s} \rangle \\ &= \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} \langle x, u_n \rangle \langle u_n, \overline{z_s} \rangle \end{aligned} \quad (17)$$

by the formula given in Problem 5.2.14(c). But,

$$\begin{aligned} \langle u_n, \overline{z_s} \rangle &= \int_{-\pi}^{\pi} (2\pi)^{-1/2} e^{int} z(s - t) dt \\ &= (2\pi)^{-1/2} e^{ins} \int_{-\pi}^{\pi} e^{-in(s-t)} z(s - t) dt \\ &= (2\pi)^{1/2} e^{ins} c_n, \end{aligned}$$

where

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} z(t) e^{-int} dt = (2\pi)^{-1/2} \langle z, u_n \rangle.$$

Let

$$\hat{x}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} x(t) e^{-int} dt = (2\pi)^{-1/2} \langle x, u_n \rangle.$$

Thus (17) becomes

$$Ax(s) = \sum_{n=-\infty}^{\infty} c_n \hat{x}(n) e^{ins} \quad (18)$$

for $s \in [-\pi, \pi]$. Since

$$|c_n \hat{x}(n) e^{ins}| = |c_n| |\hat{x}(n)|,$$

$$\begin{aligned} \sum |c_n| |\hat{x}(n)| &= \frac{1}{2\pi} \sum |\langle z, u_n \rangle| |\langle x, u_n \rangle| \\ &\leq \frac{1}{2\pi} \left(\sum |\langle z, u_n \rangle|^2 \right)^{1/2} \left(\sum |\langle x, u_n \rangle|^2 \right)^{1/2} \\ &\leq \frac{1}{2\pi} \|z\| \|x\|, \end{aligned}$$

we see that the series in (18) converges uniformly and absolutely for all s .

(b) Since the series in (18) converges uniformly, it converges in the norm of $H = L^2[-\pi, \pi]$. So (18) gives

$$A(x) = \sum_{n=-\infty}^{\infty} c_n \langle x, u_n \rangle u_n, \quad x \in H \quad (19)$$

as elements of H . Since $\sum |c_n|^2 < \infty$, we see that $c_n \rightarrow 0$ as $|n| \rightarrow \infty$, and hence A is compact by Problem 7.2.10(b).

(c) From what has been obtained in Problem 7.1.12, we see that if A is given by (19), then

$$\sigma_e(A) = \{c_n : n = 0, \pm 1, \pm 2, \dots\}.$$

From (19) we also see that $A(u_n) = c_n u_n$. Thus u_n is an eigenvector corresponding to c_n . This completes the proof. ■

PROBLEM 7.2.14 Let H be the space $L^2[0, 1]$ over $\mathbb{K} = \mathbb{R}$, $k(s, t) = \min(s, t)$ for $0 \leq s, t \leq 1$ and $\mu \neq 0$. Show that the integral equation

$$x(s) - \mu \int_0^1 k(s, t)x(t) dt = y(s), \quad x, y \in H,$$

has a unique solution if $\mu \neq (n - \frac{1}{2})^2 \pi^2$ for $n = 1, 2, \dots$. Find the solution in that case.

Solution For $x \in H$, let $k(s, t) = \min(s, t)$ and

$$A(x)(s) = \int_0^1 k(s, t)x(t) dt, \quad 0 \leq s \leq 1.$$

Then k is continuous on $[0, 1] \times [0, 1]$ and $k(s, t) = k(t, s)$. So A is a compact self-adjoint operator in $BL(H)$, as observed in the beginning of the solution of Problem 7.2.12.

For $x \in L^2[0, 1]$ let $y = A(x)$. Then,

$$y(s) = \int_0^1 k(s, t)x(t) dt = \int_0^s tx(t) dt + s \int_s^1 x(t) dt.$$

The last two integrals are absolutely continuous functions of s . Hence, y is absolutely continuous and, for almost all s ,

$$y'(s) = sx(s) + s(-x(s)) + \int_s^1 x(t) dt = \int_s^1 x(t) dt.$$

Hence y' is absolutely continuous and $y''(s) = -x(s)$ a.e. Since $\mathbb{K} = \mathbb{R}$, we get

$$\begin{aligned} \langle Ax, x \rangle &= \int_0^1 y(s)x(s) ds = - \int_0^1 y(s)y''(s) ds \\ &= - \left[y(s)y'(s) \right]_0^1 + \int_0^1 (y'(s))^2 ds. \end{aligned}$$

As $y(0) = 0 = y'(1)$, this shows that $\langle Ax, x \rangle \geq 0$ for all $x \in H$.

Let λ be a nonzero eigenvalue of A . Then $\lambda > 0$ since A is positive. If $A(x) = \lambda x$, $x \neq 0$ let $y = A(x)$. Then $y = -\lambda y''$. The solution of this differential equation is

$$y(s) = c_1 \sin(\alpha s) + c_2 \cos(\alpha s)$$

where $\alpha = \lambda^{-1/2}$. Since $y(0) = 0 = y'(1)$,

$$c_2 = 0, \quad c_1 \alpha \cos \alpha = 0.$$

Hence, $\alpha = (n - \frac{1}{2})\pi$ and $\lambda = (n - \frac{1}{2})^{-2}\pi^{-2}$ with $y(s) = c_1 \sin((n - \frac{1}{2})\pi s)$. For $n = 1, 2, \dots$, let

$$\lambda_n = \left(n - \frac{1}{2}\right)^{-2} \pi^{-2}, \quad u_n(s) = \sqrt{2} \sin\left((n - \frac{1}{2})\pi s\right). \quad (20)$$

Then $\|u_n\| = 1$, the nonzero eigenvalues of A are λ_n and the corresponding eigenvectors are cu_n . Consequently,

$$A(x) = \sum_{n=1}^{\infty} \lambda_n \langle x, u_n \rangle u_n, \quad x \in H, \quad (21)$$

where λ_n, u_n are given by (20).

The given integral equation can be written as $x(s) - \mu A(x)(s) = y(s)$, that is,

$$Ax - \frac{1}{\mu}x = -\frac{1}{\mu}y, \quad x \in H. \quad (22)$$

Since $\mu \neq (n - \frac{1}{2})^2\pi^2$, we see that $1/\mu$ is not an eigenvalue of A . Since A is compact and self-adjoint its nonzero spectral values are eigenvalues, as in the solution of Problem 7.2.2. This implies that $1/\mu$ is not a spectral value of A . Thus $B = (A - (1/\mu)I)^{-1}$ exists in $BL(H)$, and hence (22) has a unique solution $x = -(1/\mu)B(y)$.

By Problem 7.1.14(c) with $k = 1/\mu$ and $k_n = \lambda_n$, we see that if A is given by (21), then $B = (A - (1/\mu)I)^{-1}$ is given by

$$\frac{1}{\mu}B(x) = -x + \sum_{n=1}^{\infty} \frac{\mu\lambda_n}{\mu\lambda_n - 1} \langle x, u_n \rangle u_n, \quad x \in H.$$

Hence the solution is given by

$$x = -\frac{1}{\mu}B(y) = y + \sum_{n=1}^{\infty} \frac{\mu\lambda_n}{1 - \mu\lambda_n} \langle y, u_n \rangle u_n.$$

By (20) we see that

$$\langle y, u_n \rangle u_n(s) = 2c_n \sin\left((n - \frac{1}{2})\pi s\right),$$

where

$$c_n = \frac{1}{\sqrt{2}} \langle y, u_n \rangle = \int_0^1 y(t) \sin \left(\left(n - \frac{1}{2} \right) \pi t \right) dt. \quad (23)$$

Since

$$\frac{\mu \lambda_n}{1 - \mu \lambda_n} = \frac{\mu}{\left(n - \frac{1}{2} \right)^2 \pi^2 - \mu},$$

the solution is

$$x(s) = y(s) + 2\mu \sum_{n=1}^{\infty} \frac{c_n}{\left(n - \frac{1}{2} \right)^2 \pi^2 - \mu} \sin \left(\left(n - \frac{1}{2} \right) \pi s \right)$$

for almost all $s \in [0, 1]$, where c_n is given by (23). ■

PROBLEM 7.2.15 Let $P \in BL(H)$ where H is a Hilbert space. Show that

- (a) P is an orthogonal projection iff $P = P^*P$,
- (b) orthogonal projections are self-adjoint.

Solution Suppose that P is an orthogonal projection. For $x \in H$ we have

$$P(x - Px) = Px - P^2x = Px - Px = 0,$$

and hence $x - Px \in Z(P)$. Since $Z(P) \perp R(P)$, we see that, for all $x, y \in H$,

$$\langle x - Px, Py \rangle = 0, \quad \langle x, Py \rangle = \langle Px, Py \rangle.$$

Interchanging x and y , we get

$$\langle y, Px \rangle = \langle Py, Px \rangle = \overline{\langle x, Py \rangle}.$$

Thus,

$$\langle Px, y \rangle = \langle x, Py \rangle \quad \text{for all } x, y \in H.$$

This shows that $P^* = P$. This readily gives (b). Further, $P^*P = P^2 = P$.

On the other hand, suppose that $P = P^*P$. Then, $P^* = (P^*P)^* = P^*P = P$, and hence $P = P^*P = P^2$. Thus P is self-adjoint and idempotent. If Px is any element of $R(P)$ and $y \in Z(P)$, then

$$\langle Px, y \rangle = \langle x, P^*y \rangle = \langle x, Py \rangle = \langle x, 0 \rangle = 0.$$

This shows that $R(P) \perp Z(P)$. Therefore, P is an orthogonal projection and the proof is complete. ■

PROBLEM 7.2.16 Let $P \in BL(H)$, where H is a Hilbert space. Show that P is an orthogonal projection iff P is idempotent and $\|P\| \leq 1$.

Solution Suppose that P is an orthogonal projection. Then $P = P^*P$ by Problem 7.2.15 and P is idempotent. Hence, for all $x \in H$,

$$\begin{aligned}\langle Px, Px \rangle &= \langle P^*Px, x \rangle = \langle Px, x \rangle, \\ \|Px\|^2 &= \langle Px, x \rangle \leq \|Px\| \|x\|, \\ \|Px\| &\leq \|x\|.\end{aligned}$$

Thus, $\|P\| \leq 1$.

Conversely, suppose that P is idempotent and $\|P\| \leq 1$. Let $x \in H$, and $y \in Z(P)$ with $y \neq 0$. Let

$$k = \frac{\langle Px, y \rangle}{\|y\|^2}, \quad z = Px - ky.$$

Then $Pz = P^2x - kPy = Px$, since $Py = 0$. Hence,

$$\begin{aligned}\langle Px - ky, Px - ky \rangle &= \langle Px, Px \rangle - \bar{k}\langle Px, y \rangle - k\langle y, Px \rangle + k\bar{k}\langle y, y \rangle \\ &= \|Px\|^2 - \bar{k}k\|y\|^2 - k\bar{k}\|y\|^2 + k\bar{k}\|y\|^2, \\ \|z\|^2 &= \langle Px - ky, Px - ky \rangle = \|Px\|^2 - |k|^2\|y\|^2.\end{aligned}$$

However, $\|Pz\| \leq \|P\| \|z\| \leq \|z\|$. So,

$$|k|^2\|y\|^2 = \|Pz\|^2 - \|z\|^2 \leq 0$$

which implies that $k = 0$, that is, $\langle Px, y \rangle = 0$. Since this is true even when $y = 0$, we see that $Px \perp y$ for all $x \in H$ and $y \in Z(P)$. Thus, $R(P) \perp Z(P)$, and P is an orthogonal projection. This completes the proof. ■

PROBLEM 7.2.17 Let $P \in BL(H)$, where H is a Hilbert space, and

$$\langle Px, x \rangle = \|Px\|^2 \quad \text{for all } x \in H. \quad (24)$$

Show that if

- (i) P is self-adjoint, or
- (ii) the scalar field $\mathbb{K}$ is $\mathbb{C}$,

then P is an orthogonal projection. Show also that if the scalar field is $\mathbb{R}$ then (24) alone does not imply that P is an orthogonal projection.

Solution Let $A = P^*P - P$. From (24) we have

$$\begin{aligned}\langle P^*Px, x \rangle &= \langle Px, Px \rangle = \langle Px, x \rangle, \quad x \in H, \\ \langle Ax, x \rangle &= 0, \quad x \in H.\end{aligned} \quad (25)$$

If P is self-adjoint then A is self-adjoint, and hence (25) implies that $A = 0$ by Problem 6.2.3. If $\mathbb{K} = \mathbb{C}$, then (25) together with Problem 6.2.4(a)

implies that A is self-adjoint. Thus in both cases (i) and (ii), we have $A = 0$, that is, $P^*P = P$. Hence P is an orthogonal projection by Problem 7.2.15. This proves the first statement.

To prove the second statement, let $H = \mathbb{R}^2$ with the scalar field $\mathbb{R}$. Let $P : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by the matrix $M = \begin{pmatrix} 1/2 & 1/2 \\ -1/2 & 1/2 \end{pmatrix}$. If $y = Px$, then

$$y(1) = \frac{1}{2}x(1) + \frac{1}{2}x(2), \quad y(2) = -\frac{1}{2}x(1) + \frac{1}{2}x(2).$$

So,

$$\begin{aligned} \langle Px, x \rangle &= y(1)x(1) + y(2)x(2) = \frac{1}{2}(x(1))^2 + \frac{1}{2}(x(2))^2, \\ \|Px\|^2 &= (y(1))^2 + (y(2))^2 = \frac{1}{2}(x(1))^2 + \frac{1}{2}(x(2))^2. \end{aligned}$$

Thus, $\langle Px, x \rangle = \|Px\|^2$ for all $x \in \mathbb{R}^2$. If $Px = y$ and $P^2x = Py = z$, then

$$\begin{aligned} z(1) &= \frac{1}{2}y(1) + \frac{1}{2}y(2) = \frac{1}{2}x(2) \\ z(2) &= -\frac{1}{2}y(1) + \frac{1}{2}y(2) = -\frac{1}{2}x(1). \end{aligned}$$

This shows that $P^2 \neq P$ and so P is not a projection. This completes the proof. ■

PROBLEM 7.2.18 Let H be a Hilbert space and P_1, P_2 be orthogonal projections on H . Prove the following:

- (a) P_1P_2 is an orthogonal projection iff $P_1P_2 = P_2P_1$.
- (b) $P_1 + P_2$ is an orthogonal projection iff $P_1P_2 = 0 = P_2P_1$.
- (c) $P_1 + P_2$ and $P_1 - P_2$ are orthogonal projectons iff $P_2 = 0$.

Solution (a) Orthogonal projections are self-adjoint and idempotent. So, if P_1P_2 is an orthogonal projection, then

$$P_1P_2 = (P_1P_2)^* = P_2^*P_1^* = P_2P_1.$$

Conversely, if $P_1P_2 = P_2P_1$, then

$$\begin{aligned} (P_1P_2)^*(P_1P_2) &= (P_2^*P_1^*)(P_1P_2) = P_2P_1P_1P_2 \\ &= P_2P_1P_2 = P_1P_2P_2 = P_1P_2. \end{aligned}$$

So P_1P_2 is an orthogonal projection by Problem 7.2.15.

(b) By Problem 7.2.15, we see that $P_1 + P_2$ is an orthogonal projection iff

$$\begin{aligned} P_1 + P_2 &= (P_1 + P_2)^*(P_1 + P_2) = (P_1^* + P_2^*)(P_1 + P_2) \\ &= (P_1 + P_2)(P_1 + P_2) = P_1^2 + P_1P_2 + P_2P_1 + P_2^2 \\ &= P_1 + P_1P_2 + P_2P_1 + P_2. \end{aligned}$$

Hence, $P_1 + P_2$ is an orthogonal projection iff

$$P_1 P_2 + P_2 P_1 = 0. \quad (26)$$

If $P_1 P_2 = 0 = P_2 P_1$, then this is true. Conversely, suppose that (26) holds. Then, multiplying (26) on the left by P_2 , we get

$$P_2 P_1 P_2 + P_2^2 P_1 = 0. \quad (27)$$

Multiplying (27) on the right by P_2 , we get

$$\begin{aligned} P_2 P_1 P_2^2 + P_2^2 P_1 P_2 &= 0, \\ P_2 P_1 P_2 + P_2 P_1 P_2 &= 0. \end{aligned}$$

Thus, $P_2 P_1 P_2 = 0$, and so $P_2 P_1 = 0$ by (27) and $P_1 P_2 = 0$ by (26).

(c) If $P_2 = 0$, then $P_1 + P_2 = P_1 = P_1 - P_2$ is an orthogonal projection. Conversely, suppose that $P_1 + P_2$ and $P_1 - P_2$ are orthogonal projections. Then, $P_1 P_2 = 0 = P_2 P_1$ by (b). Also,

$$\begin{aligned} P_1 - P_2 &= (P_1 - P_2)(P_1 - P_2) = P_1^2 - P_1 P_2 - P_2 P_1 + P_2^2 \\ &= P_1 + P_2; \end{aligned}$$

hence $P_2 = 0$. This completes the proof. ■

PROBLEM 7.2.19 Let P_1 and P_2 be orthogonal projections of a Hilbert space H and $P = P_1 + P_2 - P_1 P_2$. Show that

- (a) P is an orthogonal projection iff $P_1 P_2 = P_2 P_1$,
- (b) $R(P) = R(P_1) + R(P_2)$ if $P_1 P_2 = P_2 P_1$.

Solution (a) If P is an orthogonal projection, then $P = P^*$, and so

$$P_1 + P_2 - P_1 P_2 = P_1^* + P_2^* - P_2^* P_1^* = P_1 + P_2 - P_2 P_1.$$

Therefore, $P_1 P_2 = P_2 P_1$.

Conversely, suppose that $P_1 P_2 = P_2 P_1$. Let $P_3 = P_2 - P_2 P_1$. Then,

$$\begin{aligned} P_3 P_1 &= P_2 P_1 - P_2 P_1^2 = 0, \\ P_1 P_3 &= P_1 P_2 - P_1 P_2 P_1 = P_1 P_2 - P_1^2 P_2 = 0, \\ P_2 P_3 &= P_2^2 - P_2^2 P_1 = P_3. \end{aligned}$$

From these we see that

$$\begin{aligned} P_3^* P_3 &= (P_2^* - P_1^* P_2^*) P_3 = P_2 P_3 - P_1 P_2 P_3 \\ &= P_3 - P_1 P_3 = P_3. \end{aligned}$$

Hence, P_3 is an orthogonal projection by Problem 7.2.15. So, $P = P_1 + P_3$ is an orthogonal projection by Problem 7.2.18(b) since $P_1 P_3 = 0 = P_3 P_1$.

(b) Let $P_1P_2 = P_2P_1$. For all $x \in H$, we have

$$P(x) = P_1(x) + P_2(x) - P_1P_2(x) = P_1(y) + P_2(x),$$

where $y = x - P_2(x)$. This shows that $R(P) \subset R(P_1) + R(P_2)$.

On the other hand, let $y_1 = P_1(x_1)$ and $y_2 = P_2(x_2)$. Then,

$$\begin{aligned} Py_1 &= P_1y_1 + P_2y_1 - P_1P_2y_1 \\ &= P_1(P_1x_1) + P_2(P_1x_1) - P_2P_1(P_1x_1) \\ &= P_1x_1 + P_2P_1x_1 - P_2P_1x_1 = P_1x_1, \\ Py_2 &= P_1(P_2x_2) + P_2(P_2x_2) - P_1P_2(P_2x_2) \\ &= P_1P_2x_2 + P_2x_2 - P_1P_2x_2 = P_2x_2. \end{aligned}$$

Therefore,

$$y_1 + y_2 = P_1x_1 + P_2x_2 = P(y_1 + y_2) \in R(P).$$

This shows that $R(P_1) + R(P_2) \subset R(P)$. Thus, $R(P) = R(P_1) + R(P_2)$. This completes the proof. ■

PROBLEM 7.2.20 Let H be a Hilbert space and $P \in BL(H)$ be an orthogonal projection. Show that

- (a) $R(P) = Z(P)^\perp$,
- (b) if Q is an orthogonal projection of $Y = R(P)$, then QP is an orthogonal projection of H .

Solution (a) If P is an orthogonal projection, then $R(P) \perp Z(P)$. So, $R(P) \subset Z(P)^\perp$. Let $y \in Z(P)^\perp$. Then,

$$Py \in R(P) \subset Z(P)^\perp,$$

and so $y - Py \in Z(P)^\perp$. But

$$P(y - Py) = Py - P^2y = 0.$$

Therefore, $y - Py \in Z(P)$. This implies that $y - Py = 0$, and hence $y = Py \in R(P)$. This shows that $Z(P)^\perp \subset R(P)$; and (a) is proved.

(b) Let $E = QP$. It is clear that E maps H into itself and that E is linear and continuous. If $y \in R(P)$, then $y = Px$, and so

$$Py = P^2x = Px = y.$$

Let $x \in H$ and $z = Ex = QPx$. Then $z \in R(Q) \subset Y$. Thus, $Pz = z$ since $z \in Y = R(P)$. Similarly, $Qz = z$. Hence,

$$E^2x = Ez = QPz = Qz = z = Ex.$$

This shows that $E^2 = E$.

Let $x \in Z(E)$ and $y \in R(E)$. Then $PQy = Py = y$ since $y \in R(Q)$ and $y \in R(P)$. So

$$\begin{aligned}\langle x, y \rangle &= \langle x, PQy \rangle = \langle Px, Qy \rangle \\ &= \langle QPx, y \rangle = \langle Ex, y \rangle,\end{aligned}$$

where we have used first the fact that P is self-adjoint on H and then that Q is self-adjoint on Y . Since $Ex = 0$, it follows that $\langle x, y \rangle = 0$. This shows that $Z(E) \perp R(E)$. Thus, E is an orthogonal projection on H , and the proof is complete. ■

PROBLEM 7.2.21 Let $A \in BL(H)$, where H is a finite dimensional Hilbert space. Suppose that $P_1, \dots, P_m$ are nonzero orthogonal projections such that

$$P_i P_j = 0 \quad \text{for } i \neq j, \quad (28)$$

$$I = P_1 + P_2 + \dots + P_m, \quad (29)$$

and $k_1, \dots, k_m$ are distinct scalars such that

$$A = k_1 P_1 + \dots + k_m P_m. \quad (30)$$

Show that $k_1, \dots, k_m$ are the distinct eigenvalues of A and $P_1, \dots, P_m$ are the orthogonal projections onto the corresponding eigenspaces.

Solution Let $Y_j = Z(A - k_j I)$, the zero space of $A - k_j I$. If $y \in R(P_j)$, then $y = P_j(x)$ for some $x \in H$, and hence, using (30) and (28),

$$A(y) = \sum_{i=1}^m k_i P_i(y) = \sum_{i=1}^m k_i P_i P_j(x) = k_j P_j^2(x) = k_j y,$$

which implies that $y \in Y_j$.

Conversely, suppose that $y \in Y_j$. Then, $A(y) = k_j y$ and

$$y = \sum_{i=1}^m P_i(y) \quad (31)$$

$$A(y) = \sum_{i=1}^m k_i P_i(y)$$

by (29) and (30). Hence

$$0 = A(y) - k_j y = \sum_{i=1}^m (k_i - k_j) P_i(y).$$

Since $P_r P_i = 0$ for $i \neq r$, we see that

$$0 = P_r(0) = \sum_{i=1}^m (k_i - k_j) P_r P_i(y) = (k_r - k_j) P_r^2(y).$$

Since $k_r - k_j \neq 0$ for $r \neq j$, it follows that $P_r(y) = 0$ if $r \neq j$. Hence from (31),

$$y = P_j(y) \in R(P_j).$$

We have thus proved that $R(P_j) = Y_j$.

Since P_j is nonzero, $Y_j = R(P_j)$ contains nonzero vectors. This implies that k_j is an eigenvalue and Y_j is the corresponding eigenspace. It remains to prove that A has no eigenvalue other than $k_1, \dots, k_m$. For this let λ be different from each k_j and $Ax = \lambda x$. Then, as above,

$$Ax - \lambda x = \sum_{i=1}^m k_i P_i(x) - \lambda \sum_{i=1}^m P_i(x) = \sum_{i=1}^m (k_i - \lambda) P_i(x),$$

and so,

$$0 = P_r(Ax - \lambda x) = \sum_{i=1}^m (k_i - \lambda) P_r P_i(x) = (k_r - \lambda) P_r^2(x).$$

Hence, $P_r(x) = 0$ since $k_r \neq \lambda$. This is true for each r . Therefore, by (29),

$$x = \sum_{r=1}^m P_r(x) = 0.$$

This shows that λ cannot be an eigenvalue, and completes the proof. ■

PROBLEM 7.2.22 Let A be a nonzero compact self-adjoint operator on a Hilbert space H and $t_1, t_2, \dots$ be the distinct nonzero eigenvalues of A with $|t_1| \geq |t_2| \geq \dots$. Let $t_0 = 0$ and P_j be the orthogonal projection whose range $R(P_j)$ is the eigenspace $Z(A - t_j I)$ of $t_j, j = 0, 1, 2, \dots$. Prove that

$$x = P_0(x) + \sum_{n \geq 1} P_n(x), \quad x \in H,$$

where the sum exists in H for each x and

$$A = \sum_{n \geq 1} t_n P_n,$$

where the sum exists in $BL(H)$. Also show that $R(P_j)$ is finite dimensional for $j \geq 1$ and $P_i P_j = 0$ for $i \neq j$. [This result is the spectral theorem for compact self-adjoint operators.]

Solution By Problem 7.2.6, there is a finite or infinite orthonormal sequence $\{u_n\}$ in H and real numbers λ_n such that $A(u_n) = \lambda_n u_n$, $|\lambda_n| \geq |\lambda_{n+1}| > 0$ for all n and

$$A(x) = \sum_{n \geq 1} \lambda_n \langle x, u_n \rangle u_n, \quad x \in H. \quad (32)$$

Here, the sequence $\{\lambda_n\}$ is precisely

$$t_1, \dots, t_1, t_2, \dots, t_2, t_3, \dots,$$

where each t_i is some λ_n and is repeated only a finite number of times. If sequence $\{\lambda_n\}$ is infinite, then $\lambda_n \rightarrow 0$ as $n \rightarrow \infty$. Consequently, if the sequence $\{t_i\}$ is infinite, then $t_i \rightarrow 0$ as $i \rightarrow \infty$. The eigenvectors u_n can be correspondingly renamed as u_{ij} so that $A(u_{ij}) = t_i u_{ij}$. We can rewrite (32) as follows:

$$A(x) = \sum_{i \geq 1} t_i \sum_{j \geq 1} \langle x, u_{ij} \rangle u_{ij}, \quad x \in H. \quad (33)$$

If $x \in H$ and $i, j \geq 1$, then $P_i x \in \text{span}\{u_{i1}, u_{i2}, \dots\}$ and $P_j x \in \text{span}\{u_{j1}, u_{j2}, \dots\}$. Since eigenvectors corresponding to distinct eigenvalues are orthogonal, we see that $R(P_i) \perp R(P_j)$ for $i \neq j$. It can be seen that this is true for all $i, j \geq 0$. Consequently, if x is any element in H , then

$$P_j x \in R(P_j) \subset R(P_i)^\perp = Z(P_i), \quad P_i(P_j x) = 0.$$

This shows that $P_i P_j = 0$ for $i \neq j$.

For each $i \geq 1$, the eigenspace $Z_i = Z(A - t_i I) = R(P_i)$ of t_i , is finite dimensional and the set $\{u_{i1}, u_{i2}, \dots\}$ is an orthonormal basis for this. Let $\{u_{0\alpha}\}$ be an orthonormal basis for $R(P_0) = Z(A)$. Clearly, $R(P_i)$ are closed subspaces and hence, are Hilbert spaces themselves for $i \geq 0$. Further, $P_i x \in R(P_i)$ for $i \geq 0$, $x \in H$. Hence, they have the Fourier expansions

$$P_i x = \sum_{j \geq 1} \langle P_i x, u_{ij} \rangle u_{ij} \quad \text{for } i \geq 1,$$

$$P_0 x = \sum_{\alpha} \langle P_0 x, u_{0\alpha} \rangle u_{0\alpha}$$

in the sense that the sums exist in H and the equality holds. Since

$$\langle P_i x, u \rangle = \langle x, P_i u \rangle = \langle x, u \rangle \quad \text{if } u \in R(P_i),$$

we get

$$P_i(x) = \sum_{j \geq 1} \langle x, u_{ij} \rangle u_{ij} \quad \text{for } i \geq 1, \quad P_0(x) = \sum_{\alpha} \langle x, u_{0\alpha} \rangle u_{0\alpha}$$

Combining this with (33), we get

$$A(x) = \sum_{i \geq 1} t_i P_i(x), \quad x \in H$$

in the sense that the sum exists in H and is equal to $A(x)$.

Consider a set S consisting of vectors u_{ij} , $i, j \geq 1$ and $u_{0\alpha}$. This is an orthonormal set of eigenvectors. Suppose that $x \in H$ is orthogonal to each element in S . Then, $\langle x, u_{ij} \rangle = 0$ for all $i, j \geq 1$ and hence, from (33), we

see that $A(x) = 0$, that is, $x \in Z(A)$. But $\langle x, u_{0\alpha} \rangle = 0$ for each α and $\{u_{0\alpha}\}$ is an orthonormal basis for $Z(A)$. This implies that $x = 0$. Thus, if x is orthogonal to all elements in S , then $x = 0$. This shows that S is an orthonormal basis for H . So, any $x \in H$ has the Fourier expansion

$$x = \sum_{\alpha} \langle x, u_{0\alpha} \rangle u_{0\alpha} + \sum_{i \geq 1} \sum_{j \geq 1} \langle x, u_{ij} \rangle u_{ij}$$

which readily gives

$$x = P_0(x) + \sum_{i \geq 1} P_i(x) \quad \text{for each } x \in H.$$

It is to be proved that $A = \sum_{i \geq 1} t_i P_i$ as elements in $BL(H)$. This is obvious if there are only a finite number of terms in the sum. So, let $\{t_i\}$ be an infinite sequence. For $x \in H$ and $n = 1, 2, \dots$, let

$$A_n(x) = \sum_{i=1}^n t_i P_i(x) \quad \text{and} \quad x_n = P_0(x) + \sum_{i=1}^n P_i(x).$$

If $i \neq j$, then $P_i(x) \perp R(P_j)$ as $R(P_i) \perp R(P_j)$. Hence, distinct terms in these sums are orthogonal. By Pythagoras theorem, we get $\|x_n\|^2 = \sum_{i=0}^n \|P_i(x)\|^2$ and for $m > n$,

$$\|A_m(x) - A_n(x)\|^2 = \left\| \sum_{i=n+1}^m t_i P_i(x) \right\|^2 = \sum_{i=n+1}^m |t_i|^2 \|P_i(x)\|^2.$$

Since $\|x_n\| \rightarrow \|x\|$, we see that $\sum_{i=0}^{\infty} \|P_i(x)\|^2 = \|x\|^2$. Moreover, $|t_i| \leq |t_{n+1}|$ for $i > n$. This shows that

$$\|A_m(x) - A_n(x)\|^2 \leq |t_{n+1}|^2 \sum_{i=n+1}^{\infty} \|P_i(x)\|^2 \leq |t_{n+1}|^2 \|x\|^2.$$

Letting $m \rightarrow \infty$, we get

$$\|A(x) - A_n(x)\|^2 \leq |t_{n+1}|^2 \|x\|^2, \quad \|A(x) - A_n(x)\| \leq |t_{n+1}| \|x\|.$$

This is true for all $x \in H$. Hence, $\|A - A_n\| \leq |t_{n+1}|$. Since $t_n \rightarrow 0$ as $n \rightarrow \infty$, this shows that $A_n \rightarrow A$ in $BL(H)$. This completes the proof. ■

PROBLEM 7.2.23 Let H be a Hilbert space, $\{u_n\}$ be an infinite orthonormal basis for H , and $F_n = \text{span}\{u_1, \dots, u_n\}$ for $n = 1, 2, \dots$. If P_n is the orthogonal projection onto F_n , show that

- (a) $P_n(x) \rightarrow x$ for each $x \in H$,
- (b) $\|P_n - I\|$ does not tend to zero.

Solution (a) Since $\{u_1, \dots, u_n\}$ is an orthonormal basis for $F_n = R(P_n)$, we have, for each $x \in H$,

$$P_n(x) = k_1 u_1 + \dots + k_n u_n. \quad (34)$$

So for $1 \leq j \leq n$,

$$k_j = \langle P_n x, u_j \rangle = \langle x, P_n^* u_j \rangle.$$

But $P_n^*(u_j) = P_n(u_j) = u_j$ since $u_j \in F_n = R(P_n)$. Thus, $k_j = \langle x, u_j \rangle$, and hence from (34),

$$P_n(x) = \sum_{j=1}^n k_j u_j = \sum_{j=1}^n \langle x, u_j \rangle u_j. \quad (35)$$

Since $\{u_n\}$ is an orthonormal basis, we have the Fourier expansion

$$x = \sum_{n=1}^{\infty} \langle x, u_n \rangle u_n, \quad x \in H,$$

the series being convergent in H . The n -th partial sum of this is $P_n(x)$ by (35). So, $P_n(x) \rightarrow x$ in H .

(b) Since $P_n(u_{n+1}) = 0$ by (35), we see that

$$\|(P_n - I)u_{n+1}\| = \|-u_{n+1}\| = 1,$$

$$1 \leq \|P_n - I\| \|u_{n+1}\| = \|P_n - I\|.$$

This shows that $\|P_n - I\|$ cannot tend to zero, and thus the proof is complete. ■

PROBLEM 7.2.24 Let H be a complex Hilbert space and A be a compact normal operator on H . Show that there is a finite or infinite sequence $\{c_n\}$ of nonzero complex numbers with $|c_1| \geq |c_2| \geq \dots$, and a sequence $\{P_n\}$ of orthogonal projections such that $P_n P_m = 0$ for $n \neq m$ and

$$A(x) = \sum_{n \geq 1} c_n P_n(x) \quad \text{for } x \in H.$$

Solution Note that a bounded sequence $\{x_n\}$ in H has a subsequence $\{x_{n_j}\}$ such that $\{A(x_{n_j})\}$ converges, and so $\{A^* A(x_{n_j})\}$ converges. Thus we see that $A^* A$ is compact, self-adjoint and nonzero.

Let t be a nonzero eigenvalue of $A^* A$ and $Y = Z(A^* A - tI)$ be the corresponding eigenspace. Then, Y is finite dimensional since $A^* A$ is compact. As A is normal and $A^* A(y) = ty$ for all $y \in Y$, we get

$$A^* A(Ay) = AA^* Ay = At y = tAy,$$

$$A^* A(A^* y) = A^* A^* Ay = A^* t y = tA^* y.$$

This shows that Ay and $A^* y$ belong to Y for all $y \in Y$. Let B and C denote the restrictions to Y of A and A^* respectively. We see that $C = B^*$ and that

B is a normal operator on the finite dimensional Hilbert space Y . There exist mutually orthogonal subspaces $M_1, \dots, M_m$ of Y and scalars $k_1, \dots, k_m$ such that, for all $x \in Y$,

$$\begin{aligned} x &= y_1 + \dots + y_m, \quad y_j \in M_j \\ B(x) &= k_1 y_1 + \dots + k_m y_m. \end{aligned}$$

Let Q_j be the orthogonal projection of H onto M_j . Then Q_j is zero on M_i for $i \neq j$ and $Q_j(y) = y$ for $y \in M_j$. Hence,

$$\begin{aligned} Q_1 x + \dots + Q_m x &= x, \quad x \in Y \\ k_1 Q_1 x + \dots + k_m Q_m x &= B(x), \quad x \in Y. \end{aligned} \quad (36)$$

Using Problem 7.2.21 with $H = Y$, we can see that $k_1, \dots, k_m$ are the distinct eigenvalues of B and $M_j = R(Q_j)$ is the eigenspace corresponding to k_j . Since B is normal, we have by Problem 7.1.2(a),

$$Z(B^* - \overline{k_j}I) = Z(B - k_j I) = M_j.$$

Choose x in Y with $Q_j(x) \neq 0$. Since $Q_j(x) \in M_j \subset Y$, we have

$$AQ_j(x) = BQ_j(x) = k_j Q_j(x).$$

Hence k_j is an eigenvalue of A . Since $A^*Ay = ty$ for $y \in Y$, we also get

$$\begin{aligned} tQ_j(x) &= A^*AQ_j(x) = k_j A^*Q_j(x) \\ &= k_j B^*Q_j(x) = k_j \overline{k_j} Q_j(x). \end{aligned}$$

Therefore, $|k_j|^2 = t$. This is true for $1 \leq j \leq m$.

Let $t_1, t_2, \dots$ be the distinct nonzero eigenvalues of A^*A with $|t_1| \geq |t_2| \geq \dots$. Let $Y_i = Z(A^*A - t_i I)$ and B_i be the restriction of A to Y_i . The spectral theorem for compact self-adjoint operators shows that if S_i is the orthogonal projection onto Y_i and S_0 is the orthogonal projection onto $Y_0 = Z(A^*A)$, then for all $x \in H$,

$$x = S_0(x) + \sum_{i \geq 1} S_i(x), \quad (37)$$

$$A(x) = \sum_{i \geq 1} t_i S_i(x).$$

Apply the result found above with $t = t_i$. Then, Y and B become Y_i and B_i . Corresponding to (36), we have

$$B_i(x) = \sum_{j=1}^{m_i} k_{ij} Q_{ij}(x), \quad x \in Y_i, \quad (38)$$

where $k_{i1}, k_{i2}, \dots$ are distinct eigenvalues of A with

$$|k_{ij}|^2 = t_i \quad (39)$$

and Q_{ij} are orthogonal projections of Y_i . Moreover, if M_{ij} is the range of Q_{ij} , then $M_{i1}, M_{i2}, \dots$ are mutually orthogonal subspaces of Y_i .

If $x \in H$ and $i \geq 1$, then $S_i(x) \in Y_i$, and so by (38),

$$AS_i(x) = BS_i(x) = \sum_{j=1}^{m_i} k_{ij} Q_{ij} S_i(x) = \sum_{j=1}^{m_i} k_{ij} E_{ij}(x),$$

where $E_{ij} = Q_{ij} S_i$. If $x \in Y_0 = Z(A^*A)$, then

$$\langle Ax, Ax \rangle = \langle A^*Ax, x \rangle = 0,$$

that is, $Ax = 0$. Thus, $AS_0(x) = 0$ for all $x \in H$. Hence, by (37) and (40),

$$A(x) = AS_0(x) + \sum_{i \geq 1} AS_i(x) = \sum_{i \geq 1} \sum_{j=1}^{m_i} k_{ij} E_{ij}(x) \quad (40)$$

for all $x \in H$. From Problem 7.2.20(b), we see that $E_{ij} = Q_{ij} S_i$ is an orthogonal projection of H . Denoting the complex numbers,

$$k_{11}, k_{12}, \dots, k_{21}, \dots$$

by $c_1, c_2, \dots$ and the projections

$$E_{11}, E_{12}, \dots, E_{21}, \dots$$

by $P_1, P_2, \dots$ respectively, we can write (40) as

$$A(x) = \sum_{n \geq 1} c_n P_n(x), \quad x \in H.$$

Since $|t_1| \geq |t_2| \geq \dots$, it follows from (39) that $|c_1| \geq |c_2| \geq \dots$. We also have

$$R(E_{ij}) \subset R(Q_{ij}) = M_{ij} \subset Y_i.$$

The spaces $M_{i1}, M_{i2}, \dots$ are mutually orthogonal for the same i . Further, Y_i also are mutually orthogonal as they are eigenspaces corresponding to distinct eigenvalues $t_1, t_2, \dots$ of A^*A . It follows that the ranges of E_{ij} , $i \geq 1, j \geq 1$, are mutually orthogonal. So, $R(P_n) \perp R(P_m)$ for $n \neq m$. Hence,

$$\langle P_n P_m x, y \rangle = \langle P_m x, P_n y \rangle = 0$$

for all $n \neq m$ and $x, y \in H$. This implies that $P_n P_m = 0$ for $n \neq m$. Thus, the proof is complete. ■

PROBLEM 7.2.25 Let $A \in BL(H)$, where H is a Hilbert space. Suppose that there is a sequence $\{c_n\}$ of nonzero scalars and a sequence $\{P_n\}$ of nonzero orthogonal projections such that $P_n P_m = 0$ for $n \neq m$, and

$$A(x) = \sum_{n=1}^{\infty} c_n P_n(x), \quad x \in H. \quad (41)$$

If $c_n \rightarrow 0$ and $R(P_n)$ is finite dimensional for each n , prove the following:

- (a) A is normal and compact.
- (b) $\{c_n\}$ are the distinct nonzero eigenvalues of A .
- (c) $R(P_n)$ is the eigenspace corresponding to c_n .

Solution (a) Given that the series in (41) converges in H for each x in H . Since

$$\langle P_n x, P_m x \rangle = \langle x, P_n P_m x \rangle = 0$$

for $n \neq m$, we see that the terms in (41) are mutually orthogonal. So (41) implies that

$$\sum_{n=1}^{\infty} \|c_n P_n x\|^2 = \|Ax\|^2 < \infty. \quad (42)$$

Let

$$B(x) = \sum_{n=1}^{\infty} \overline{c_n} P_n(x), \quad x \in H. \quad (43)$$

Since the terms in (43) are mutually orthogonal, we see using Problem 5.2.5 that this sum exists in H because of (42). Further,

$$\|Bx\|^2 = \sum_{n=1}^{\infty} \|\overline{c_n} P_n(x)\|^2 = \|Ax\|^2.$$

It is clear that B is linear. Thus, $B \in BL(H)$. Moreover, for $x, y \in H$,

$$\begin{aligned} \langle Ax, y \rangle &= \left\langle \sum_{n=1}^{\infty} c_n P_n x, y \right\rangle = \sum_{n=1}^{\infty} c_n \langle P_n x, y \rangle \\ &= \sum_{n=1}^{\infty} c_n \langle x, P_n y \rangle = \langle x, \sum_{n=1}^{\infty} \overline{c_n} P_n y \rangle \\ &= \langle x, By \rangle. \end{aligned}$$

Hence, $B = A^*$. Since

$$\|Ax\| = \|Bx\| = \|A^*x\| \quad \text{for all } x \in H,$$

we see that A is normal by Problem 6.2.18(a).

Let $Y_i = R(P_i)$ and $\{v_{ij} : 1 \leq j \leq m_i\}$ be an orthonormal basis for Y_i . Then, for all $x \in H$,

$$\langle P_i x, v_{ij} \rangle = \langle x, P_i(v_{ij}) \rangle = \langle x, v_{ij} \rangle,$$

and so,

$$\|P_i x\|^2 = \sum_j |\langle P_i x, v_{ij} \rangle|^2 = \sum_j |\langle x, v_{ij} \rangle|^2.$$

Since $Y_i = R(P_i)$ are mutually orthogonal, we see that

$$\{v_{ij} : i \geq 1, 1 \leq j \leq m_i\}$$

is an orthonormal set in H . So Bessel's inequality gives, for all $x \in H$,

$$\sum_{i=1}^{\infty} \|P_i x\|^2 = \sum_{i=1}^{\infty} \sum_j |\langle x, v_{ij} \rangle|^2 \leq \|x\|^2. \quad (44)$$

Let

$$\alpha_n = \sup_{i \geq n} |c_i|, \quad A_n = \sum_{i=1}^n c_i P_i.$$

Then $\alpha_n \rightarrow 0$ as $n \rightarrow \infty$, and by (41),

$$A(x) - A_n(x) = \sum_{i=n+1}^{\infty} c_i P_i(x), \quad x \in H.$$

Since the terms in this sum are mutually orthogonal, we get, for all $x \in H$,

$$\begin{aligned} \|(A - A_n)x\|^2 &= \sum_{i=n+1}^{\infty} \|c_i P_i x\|^2 \\ &\leq \alpha_n^2 \sum_{i=n+1}^{\infty} \|P_i x\|^2 \leq \alpha_n^2 \|x\|^2 \end{aligned}$$

by (44). It follows that $\|A - A_n\| \leq \alpha_n \rightarrow 0$. Hence, $A_n \rightarrow A$ in $BL(H)$. But the range of A_n is spanned by a finite number of vectors $v_{ij}, 1 \leq i \leq n, 1 \leq j \leq m_i$. So each A_n is compact. Thus A is compact.

(b) Let $y \in Y_j = R(P_j)$. Then, $y = P_j(x), x \in H$; hence from (41),

$$A(y) = \sum_{n=1}^{\infty} c_n P_n P_j(x) = c_j P_j(x) = c_j y.$$

Since Y_j contains nonzero vectors, this shows that c_j is an eigenvalue of A . It also shows that

$$Y_j \subset Z(A - c_j I) \quad (45)$$

Next suppose that $Ay = \lambda y$. Using (41), we get

$$\lambda P_i(y) = P_i(Ay) = \sum_{n=1}^{\infty} c_n P_i P_n(y) = c_i P_i(y). \quad (46)$$

First suppose that $\lambda \neq c_i$ for any i . Then (46) implies that $P_i(y) = 0$ for all y . Hence, by (41), $\lambda y = A(y) = 0$. So, $y = 0$ if $\lambda \neq 0$. This shows that A does not have an eigenvalue λ with $\lambda \neq 0, \lambda \neq c_i$ for any i . This proves (b).

(c) Use (46) with $\lambda = c_j$. We see that if $A(y) = c_j y$, then $P_i(y) = 0$ for $i \neq j$; hence by (41),

$$c_j y = A(y) = c_j P_j(y) \in R(P_j) = Y_j.$$

Thus, $y \in Y_j$ as $c_j \neq 0$. Since this is true whenever $y \in Z(A - c_j I)$, it follows from (45) that

$$Y_j = Z(A - c_j I).$$

This completes the proof. ■

PROBLEM 7.2.26 Show that if H is a Hilbert space, then every compact operator on H is the limit, in the norm of $BL(H)$, of a sequence of bounded operators with finite dimensional ranges.

Solution Let A be a nonzero compact operator on a Hilbert space H . As observed in the beginning of the solution of Problem 7.2.24, we see that A^*A is compact. It is nonzero and self-adjoint. Let $t_1, t_2, \dots$ be the distinct nonzero eigenvalues of A^*A with $|t_1| \geq |t_2| \geq \dots, t_0 = 0$ and P_j be the orthogonal projection of H onto $Y_j = Z(A^*A - t_j I)$ for $j = 0, 1, 2, \dots$. Then the spectral theorem shows that

$$x = \sum_{i \geq 0} P_i(x), \quad x \in H, \quad (47)$$

the series being convergent in H . Moreover, $P_i P_j = 0$ for $i \neq j$, and for $i \geq 1$, the eigenspace $Y_i = R(P_i)$ has a finite basis $\{v_{ij} : 1 \leq j \leq m_i\}$. Hence for all $x \in H$, we have

$$\begin{aligned} \langle P_i x, v_{ij} \rangle &= \langle x, P_i(v_{ij}) \rangle = \langle x, v_{ij} \rangle, \quad i \geq 1, j \geq 1, \\ P_i(x) &= \sum_{j=1}^{m_i} \langle P_i x, v_{ij} \rangle v_{ij} = \sum_{j=1}^{m_i} \langle x, v_{ij} \rangle v_{ij}, \quad i \geq 1, \\ x - P_0(x) &= \sum_{i \geq 1} \sum_{j=1}^{m_i} \langle x, v_{ij} \rangle v_{ij}, \end{aligned}$$

where the last step is obtained from (47). Let $v_{11}, v_{12}, \dots, v_{21}, \dots$ be denoted by $u_1, u_2, \dots$ respectively. Thus, we can write this as

$$x - P_0(x) = \sum_{n \geq 1} \langle x, u_n \rangle u_n, \quad x \in H. \quad (48)$$

If $i \neq j$, then for all $x, y \in H$,

$$\langle P_i x, P_j y \rangle = \langle x, P_i P_j y \rangle = 0$$

since $P_i P_j = 0$. This shows that $Y_i = R(P_i)$ are mutually orthogonal. It follows that $\{u_n\}$ is orthonormal. Since $P_0(x) \in Y_0$ and $Y_0 = Z(A^*A)$, we see that

$$\langle AP_0 x, AP_0 x \rangle = \langle A^* AP_0 x, P_0 x \rangle = 0,$$

and hence $AP_0(x) = 0$ for all $x \in H$. So, by (48),

$$A(x) = \sum_{n \geq 1} \langle x, u_n \rangle A(u_n), \quad x \in H. \quad (49)$$

First suppose that A^*A has only a finite number of nonzero eigenvalues t_i . Then the sum in (47) is finite. Consequently, the sum in (49) is finite. So $R(A)$ is spanned by $A(u_1), \dots, A(u_m)$ for some m , and hence the result holds in this case.

Next suppose that $\{t_n\}$ is infinite, and let

$$A_n(x) = \sum_{j=1}^n \langle x, u_j \rangle A(u_j), \quad x \in H.$$

Then $\{u_n\}$ is infinite, and from (49),

$$A(x) - A_n(x) = \sum_{j=n+1}^{\infty} \langle x, u_j \rangle A(u_j), \quad x \in H. \quad (50)$$

Each u_n belongs to some Y_i , and hence $A^*A(u_n) = t_i u_n$ for some i . Thus, for $n \neq m$,

$$\langle Au_n, Au_m \rangle = \langle A^* Au_n, u_m \rangle = t_i \langle u_n, u_m \rangle = 0$$

since $\{u_n\}$ is orthonormal. Hence the terms in (50) are mutually orthogonal. It follows that

$$\|Ax - A_n x\|^2 = \sum_{j=n+1}^{\infty} \|\langle x, u_j \rangle A(u_j)\|^2.$$

The compactness of A implies, by Problem 7.2.4, that $A(u_n) \rightarrow 0$. Consequently, if $\alpha_n = \sup_{j > n} \|Au_j\|$, then $\alpha_n \rightarrow 0$, and

$$\|(A - A_n)x\|^2 \leq \sum_{j=n+1}^{\infty} |\langle x, u_j \rangle|^2 \alpha_n^2 \leq \alpha_n^2 \|x\|^2$$

by Bessel's inequality. Since this is true for all $x \in H$, we get

$$\|A - A_n\| \leq \alpha_n \rightarrow 0.$$

Therefore, $A_n \rightarrow A$ in $BL(H)$. Since $R(A_n)$ is finite dimensional, the proof is complete. ■

PROBLEM 7.2.27 Let A be a compact operator on a Hilbert space H . Show that if A commutes with AA^* then A is normal. Show also this may not be true if A not compact.

Solution Let $\{x_n\}$ be a bounded sequence in H and $y_n = A^*(x_n)$. Then, $\{y_n\}$ is bounded since A^* is bounded. Hence, $\{A(y_n)\}$ has a convergent subsequence. Thus, $\{AA^*(x_n)\}$ has a convergent subsequence. This shows that AA^* is compact. Let A be nonzero. Then AA^* is nonzero, compact and self-adjoint. Let $t_1, t_2, \dots$ be the distinct nonzero eigenvalues of AA^* , with $|t_1| \geq |t_2| \geq \dots$. Let $t_0 = 0$ and P_i be the orthogonal projection of H onto $Y_i = Z(AA^* - t_i I)$ for $i = 0, 1, 2, \dots$. Then the spectral theorem shows that

$$x = \sum_{i \geq 0} P_i(x), \quad x \in H, \quad (51)$$

the series being convergent in H . Moreover, $P_i P_j = 0$ for $i \neq j$ and $Y_i = R(P_i)$ is finite dimensional for $i \geq 1$.

Let $x \in Y_i$. Then, $AA^*(x) = t_i x$. Given that $A(AA^*) = (AA^*)A$. Taking adjoints, $AA^*A^* = A^*AA^*$. Hence,

$$\begin{aligned} AA^*(Ax) &= AAA^*x = A(t_i x) = t_i(Ax), \\ AA^*(A^*x) &= A^*AA^*x = A^*(t_i x) = t_i(A^*x). \end{aligned}$$

These show that Ax and A^*x belong to Y_i for all $x \in Y_i$. This is true for $i = 0, 1, 2, \dots$.

Let $i \geq 1$, and B_i and C_i denote the restrictions to Y_i of A and A^* , respectively. Then B_i and C_i are in $BL(Y_i)$, and for all $x \in Y_i$,

$$B_i C_i(x) = B_i A^*x = AA^*x = t_i x. \quad (52)$$

Since $t_i \neq 0$, (52) shows that B_i is onto Y_i . Consequently, B_i is one-one since Y_i is finite dimensional for $i \geq 1$. Thus, B_i^{-1} exists and (52) shows that $B_i C_i = t_i I$. Hence,

$$C_i = t_i B_i^{-1}, \quad C_i B_i = t_i I.$$

Thus for all $x \in Y_i, i \geq 1$, we have

$$A^*Ax = A^*B_i x = C_i B_i x = t_i x = AA^*x.$$

Since $P_i(x) \in Y_i$ for all $x \in H, i \geq 1$, we have

$$A^*AP_i(x) = AA^*P_i(x) \quad \text{for all } x \in H, i \geq 1. \quad (53)$$

Let $x \in Y_0 = Z(AA^*)$ and $y = A(x)$. Then, $y \in Y_0$ and so,

$$\langle A^*y, A^*y \rangle = \langle AA^*y, y \rangle = 0.$$

Hence,

$$A^*Ax = A^*y = 0 = AA^*x, \quad x \in Y_0.$$

Since $P_0(x) \in Y_0$ for all $x \in H$, we get

$$A^*AP_0(x) = AA^*P_0(x), \quad x \in H. \quad (54)$$

Using (51), we see that

$$\begin{aligned} A^*A(x) &= \sum_{i \geq 0} A^*AP_i(x), \quad x \in H, \\ AA^*(x) &= \sum_{i \geq 0} AA^*P_i(x), \quad x \in H. \end{aligned}$$

Hence by (53) and (54),

$$A^*Ax = AA^*x \quad \text{for all } x \in H.$$

Thus A is normal.

For the second part, let H be an infinite dimensional separable Hilbert space and $\{u_n\}$ be an orthonormal basis for H . Let

$$T(x) = \sum_{n=1}^{\infty} \langle x, u_{n+1} \rangle u_n.$$

Then T is clearly linear and

$$\|Tx\|^2 = \sum_{n=1}^{\infty} |\langle x, u_{n+1} \rangle|^2 \leq \|x\|^2.$$

Hence $T \in BL(H)$. As in Problem 6.1.13, we see that $TT^* = I \neq T^*T$. Thus T commutes with TT^* , but T is not normal. This completes the proof. $\blacksquare$

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